

**NIST Technical Note
NIST TN 2126-upd1**

Heats of Combustion and Related Properties of Pure Substances

Donald R. Burgess, Jr.
Anthony P. Hamins

This publication is available free of charge from:
<https://doi.org/10.6028/NIST.TN.2126-upd1>

NIST Technical Note
NIST TN 2126-upd1

Heats of Combustion and Related Properties of Pure Substances

Donald R. Burgess, Jr.
Chemical Sciences Division
Material Measurement Laboratory

Anthony P. Hamins
Fire Research Division
Engineering Laboratory

This publication is available free of charge from:
<https://doi.org/10.6028/NIST.TN.2126-upd1>

published December 2023
updated February 2026



U.S. Department of Commerce
Gina M. Raimondo, Secretary

National Institute of Standards and Technology
Laurie E. Locascio, NIST Director and Under Secretary of Commerce for Standards and Technology

Certain equipment, instruments, software, or materials, commercial or non-commercial, are identified in this paper in order to specify the experimental procedure adequately. Such identification does not imply recommendation or endorsement of any product or service by NIST, nor does it imply that the materials or equipment identified are necessarily the best available for the purpose.

NIST Technical Series Policies

[Copyright, Use, and Licensing Statements](#)

[NIST Technical Series Publication Identifier Syntax](#)

Publication History

Approved by the NIST Editorial Review Board on 2022-11-18

Supersedes NIST Technical Note 2126 (December 2023) <https://doi.org/10.6028/NIST.TN.2126>

How to Cite this NIST Technical Series Publication

Burgess, DR, Jr., Hamins, AP (2026) Heats of Combustion and Related Properties of Pure Substances. (National Institute of Standards and Technology, Gaithersburg, MD), NIST Technical Note (TN) NIST TN 2126-upd1
<https://doi.org/10.6028/NIST.TN.2126-upd1>

ORCID iDs

Donald R. Burgess, Jr.: 0000-0002-3428-6540

Anthony P. Hamins: 0000-0001-7531-4256

Contact Information

dburgess@nist.gov

Abstract

In this work, we compiled and evaluated various properties pertaining to the heats of combustion of about 420 compounds of physical, chemical, and engineering interest. This includes normal boiling points, melting points, latent heats (enthalpies) of vaporization, heats (enthalpies) of fusion, heats (enthalpies) of formation and combustion, and heat capacities (specific heats).

Keywords

Boiling points; combustion; enthalpies; heat capacities; heats of combustion; heats of formation; heats of vaporization; melting points.

Contents

1. Introduction	1
1.1. Overview	1
1.2. List of Compound Names	1
1.3. Compound Information	5
1.4. Organization of Compounds by Classes	11
1.5. References for Introduction.....	13
2. Normal Melting Points T_m and Normal Boiling Points T_b ($P = 1 \text{ atm} = 101.325 \text{ kPa}$)	14
2.1. Overview	14
2.2. Normal Melting and Normal Boiling Points (compilation).....	17
Table 2-1. Normal Melting and Normal Boiling Points	17
2.3. Recommended Melting and Boiling Points.....	81
Table 2-2. Recommended Normal Melting and Normal Boiling Points	81
Table 2-2(1). Small Molecules	81
Table 2-2(2). Hydrocabons	82
Table 2-2(3). Aromatic Hydrocarbons.....	86
Table 2-2(4). Oxidized Hydrocarbons.....	88
Table 2-2(5). Nitrogen-Substituted Hydrocarbons	92
Table 2-2(6). Sulfur-Substituted Hydrocarbons	94
Table 2-2(7). Halogenated Hydrocarbons	96
Table 2-2(8). Siloxanes	99
Table 2-3. Notations for Methods.....	100
2.4. References for Normal Melting and Normal Boiling Points.....	101
3. Latent Enthalpies of Vaporization $\Delta_v H(T_b)$ and Enthalpies of Fusion $\Delta_{fus} H(T_m)$	158
3.1. Overview	158
Table 3-1. Notation used for Tables 3-3 and 3-4 transition temperatures and enthalpies of transition	160
Table 3-2. Notation for methods used for transition temperatures and enthalpies of transition ..	161
3.2. Transition Temperatures T_{trs} and Enthalpies of Transition $\Delta_{trs} H$ (compilation).....	162
Table 3-3. Transition temperatures T_{trs} and enthalpies of transition $\Delta_{trs} H$	162
3.3. Recommended Transition Temperatures T_{trs} and Enthalpies of Transition $\Delta_{trs} H$	196
3.3.1. Data Tables.....	196

Table 3-4. Recommended transition temperatures T_{trs} and enthalpies of transition $\Delta_{\text{trs}}H$	196
Table 3-4(1). Small Molecules	196
Table 3-4(2). Hydrocarbons.....	197
Table 3-4(3). Aromatic Hydrocarbons.....	200
Table 3-4(4). Oxidized Hydrocarbons.....	202
Table 3-4(5). Nitrogen-Substituted Hydrocarbons	205
Table 3-4(6). Sulfur-Substituted Hydrocarbons	207
Table 3-4(7). Halogenated Hydrocarbons	208
Table 3-4(8). Siloxanes	211
3.3.2. Discussion of Uncertainties in Heats of Vaporization	212
Table 3-4(9). Derived uncertainties in EOS boiling points	215
Table 3-4(10). Analytical expression for derived uncertainties in heats of vaporization	218
Table 3-4(11). Legend for column headings in Tables 3-4(10) and 3-4(11)	219
Figure 3-1. Correlation between uncertainty in latent enthalpy of vaporization $U(\Delta_v H)$ and uncertainty in boiling point $U(T_b)$	220
Figure 3-2. Relative deviation in uncertainty $U_{\Delta T}(\Delta_v H)/\Delta_v H$ in enthalpy of vaporization from correlated value	221
Figure 3-3. Relative deviation in uncertainty $U_{\Delta T}(\Delta_v H)/\Delta_v H$ in enthalpy of vaporization from correlated value (considering both enthalpic and entropic contributions).....	222
3.3.3. References for Heats of Vaporization and Fusion.....	223
4. Heats of Combustion and Formation	249
4.1. Overview	249
Table 4-1. Notations for column headings.....	250
Table 4-2. Notations for phase column.....	251
Table 4-3. Notations for selected values column.....	252
Table 4-4. Notations for methods column	252
Table 4-5. Notations for revised values contained in column "R"	254
4.2. Heats of Dilution.....	255
Table 4-6. Compounds and references for aqueous dilutions for HF.....	256
Table 4-7. Compounds and references for aqueous dilution for HCl	257
4.3. Heats of Combustion, Formation, and Transition (compilation)	258
Table 4-8. Heats of Combustion, Formation, and Transition (compilation)	258
4.4. Recommended Heats of Combustion, Formation, and Transition	324

Table 4-9. Recommended Heats of Combustion, Formation, and Transition	325
Table 4-9(1). Small Molecules	325
Table 4-9(2). Hydrocarbons.....	327
Table 4-9(3). Aromatic Hydrocarbons.....	333
Table 4-9(4). Oxidized Hydrocarbons.....	335
Table 4-9(5). Nitrogen-Substituted Hydrocarbons	340
Table 4-9(6). Sulfur-Substituted Hydrocarbons	343
Table 4-9(7). Halogenated Hydrocarbons	344
Table 4-10. Recommended Gross and Net Heats of Combustion	350
4.5. References for Heats of Combustion, Formation, and Transition	370
5. Heat Capacities.....	401
5.1. Overview.....	401
5.2. Heat Capacities (compilation)	402
Table 5-1. Heat capacities in gas, liquid, and solid phases	402
5.3. Recommended Heat Capacities.....	426
Table 5-2. Recommended heat capacities in gas, liquid, and solid phases	426
Table 5-2(1). Small Molecules	426
Table 5-2(2). Hydrocarbons.....	427
Table 5-2(3). Aromatic Hydrocarbons.....	431
Table 5-2(4). Oxidized Hydrocarbons.....	433
Table 5-2(5). Nitrogen-Substituted Hydrocarbons	437
Table 5-2(6). Sulfur-Substituted Hydrocarbons	439
Table 5-2(7). Halogenated Hydrocarbons	440
5.4. References for Heat Capacities	444

1. Introduction

1.1. Overview

In this work, we compiled and evaluated various properties pertaining to the heats of combustion of about 400 compounds of physical, chemical, and engineering interest. The information provided here are for hydrocarbons, oxidized hydrocarbons, nitrogen-substituted hydrocarbons, sulfur-substituted hydrocarbons, and halogenated hydrocarbons. The data provided here includes normal boiling points, melting points, latent heats (enthalpies) of vaporization, heats (enthalpies) of fusion, heats (enthalpies) of formation and combustion, and heat capacities (specific heats).

In this Introduction, we first provide an index into the recommended values in Section 2 (Melting and Boiling Points), Section 3 (Heats of Transition), Section 4 (Heats of Combustion, Formation, and Transition), and Section 5 (Heat Capacities). Next in Table 1.1, an alphabetical list of all the compound names (and alternate names) along with the preferred names. For the most part, we used IUPAC standard nomenclature for chemical names, but in some cases used common names (*e.g.*, isobutane vs. 2-methyl propane or acetylene vs. ethyne). We then provide in Table 1.2, information about the compound: the compound name, Chemical Abstracts Registry Number (CASNO), Molecular Formula, and Molecular Weight (MW). In Table 1.3, we provide a list of the compounds along with the chemical classes for each.

In each of the sections, we provide the data for the compounds ordered first by primary/secondary compound class (*e.g.*, Hydrocarbons/Alkanes, Alkenes, etc. or Oxidized Hydrocarbons/Alcohols, Ethers, etc.) and then ordered by molecular formula. The data were taken from many individual sources in the literature, reviews and evaluations, and from the NIST databases NIST REFPROP [2018LEM], NIST TRC TDE,[2019DIK] and the NIST Chemistry WebBook[2021LIN]. At the end of each section, there is a list of the original source references. We provide uncertainties (*unc*) from the original sources as reported (where available). Our recommended uncertainties are generally expanded uncertainties U (2σ , 95 % confidence level), while many are standard uncertainties u (1σ , 68 % confidence level. The inconsistency in the uncertainties (1σ vs. 2σ) is because the uncertainties in the literature are mixed and often not explicitly stated. The recommended values provided here were selected from reliable original sources (they are not weighted averages) and then an uncertainty was assigned either from the original source (1σ or 2σ), from the range of values (2σ), or from our estimate (generally 2σ). This is an extensive compilation, however, for some of the compounds and quantities, it is not comprehensive, but there is an adequate selection of values for all compounds from the literature.

This is a revised version of this document which was originally published in December 2023. We have corrected a number of errors, including adding missing references and inconsistencies in sub-section headings. We have also updated a number of the normal boiling points providing more exact values. The authors thank Karen De Lannoye at NIST for her “eagle eye” in identifying many of the errors in the original document

1.2. List of Compound Names

Name	Preferred Name
(1-methylethyl)benzene	isopropylbenzene
(1-methylpropyl)benzene	<i>sec</i> -butylbenzene
1,1'-bicyclohexyl	1,1'-bicyclohexyl
1,1-dichloroethene	1,1-dichloroethene
1,1-dichloroethylene	1,1-dichloroethene
1,1-difluoroethene	1,1-difluoroethene
1,1-difluoroethylene	1,1-difluoroethene
1,1-dimethyl-1-propanethiol	1,1-dimethyl-1-propanethiol
1,2,3,4-tetrahydronaphthalene	tetralin
1,2-butadiene	1,2-butadiene
1,2-dichloroethene, (<i>E</i>)-	1,2-dichloroethene, (<i>E</i>)-
1,2-dichloroethene, (<i>Z</i>)-	1,2-dichloroethene, (<i>Z</i>)-
1,2-dichloroethylene, (<i>E</i>)-	1,2-dichloroethene, (<i>E</i>)-
1,2-dichloroethylene, (<i>Z</i>)-	1,2-dichloroethene, (<i>Z</i>)-
1,2-difluoroethene, (<i>E</i>)-	1,2-difluoroethene, (<i>E</i>)-
1,2-difluoroethene, (<i>Z</i>)-	1,2-difluoroethene, (<i>Z</i>)-
1,2-difluoroethylene, (<i>E</i>)-	1,2-difluoroethene, (<i>E</i>)-
1,2-difluoroethylene, (<i>Z</i>)-	1,2-difluoroethene, (<i>Z</i>)-
1,2-dimethylbenzene	1,2-dimethylbenzene
1,2-ethanediol	1,2-ethanediol
1,2-pentadiene	1,2-pentadiene
1,3,5,7-tetraazaadamantane	methenamine
1,3,5-triazine-2,4,6-triamine	1,3,5-triazine-2,4,6-triamine
1,3-dimethylbenzene	1,3-dimethylbenzene
1,4-dimethylbenzene	1,4-dimethylbenzene
1-methyl-4-isopropylbenzene	<i>p</i> -isopropyltoluene
1-oxacyclopent-4-ene-2,5-dione	2,5-furandione
1-propanol	1-propanol
1-propenal	1-propenal
1-propylamine	1-propylamine
2,2-dimethylbutane	neohexane
2,2-dimethylhexane	neooctane

2,2-dimethylpentane	2,2-dimethylpentane
2,2-dimethylpropane	2,2-dimethylpropane
2,2-dimethylpropanethiol	neopentanethiol
2,5-furandione	2,5-furandione
2-bornanone	D-camphor
2-isopropyltoluene	2-isopropyltoluene
2-methyl-1,3-butadiene	2-methyl-1,3-butadiene
2-methyl-1-butanol	2-methyl-1-butanol
2-methyl-1-propanol	isobutanol
2-methyl-2-butanethiol	1,1-dimethyl-1-propanethiol
2-methyl-2-butanol	2-methyl-2-butanol
2-methyl-2-butene	isopentene
2-methyl-2-pentanethiol	<i>tert</i> -pentanethiol
2-methylbutane	isopentane
2-methylhexane	2-methylhexane
2-methyloctane	2-methyloctane
2-methylpentane	isohexane
2-methylpentane	2-methylpentane
2-methylpropan-2-ol	<i>tert</i> -butanol
2-methylpropane	isobutane
2-methylpropene	isobutene
2-methylpropylene	isobutene
2-propanol	isopropanol
2-propenal	2-propenal
3-isopropyltoluene	3-isopropyltoluene
3-methyl-1-butanethiol	isopentanethiol
3-methyl-1-butanol	isopentanol
3-methylphenol	3-methylphenol
4-isopropyltoluene	4-isopropyltoluene
acetaldehyde	acetaldehyde
acetone	acetone
acetonitrile	ethane nitrile
acetylene	acetylene

acrolein	2-propenal
allene	allene
aminobenzene	aniline
aniline	aniline
benzocyclohexane	benzocyclohexane
benzocyclopentane	benzocyclopentane
bicyclohexane	1,1'-bicyclohexyl
bromoethene	bromoethene
bromoethylene	bromoethene
bromomethane	bromomethane
butadiyne	butadiyne
butanenitrile	butanenitrile
butanoic acid	butanoic acid
butanone	butanone
butenyne	butenyne
butylamine	<i>n</i> -butylamine
carbon tetrachloride	tetrachloromethane
carbonyl dichloride	phosgene
carbonyl fluoride	carbonyl fluoride
chloroethylene	chloroethene
chloroacetylene	chloroacetylene
chloroethane	chloroethane
chloroethene	chloroethene
chloroethyne	chloroacetylene
chloroform	trichloromethane
chloromethane	chloromethane
chlorotrifluoroethene	chlorotrifluoroethene
chlorotrifluoroethylene	chlorotrifluoroethene
<i>cis</i> -decahydronaphthalene	<i>cis</i> -decalin
<i>cis</i> -decalin	<i>cis</i> -decalin
cumene	isopropylbenzene
cyanoethene	propenenitrile
cyanomethane	ethane nitrile
D-camphor	D-camphor
diacetylene	butadiyne
dichloroacetylene	dichloroacetylene

dichloroethyne	dichloroacetylene
dichloromethane	dichloromethane
diethyl ether	diethyl ether
diisopropyl ether	diisopropyl ether
dimethyl ether	dimethyl ether
dimethyl ketone	acetone
ethanol	ethanol
ethane nitrile	ethane nitrile
ethane	ethane
ethenyl acetate	vinyl acetate
ethenyl bromide	bromoethene
ethenyl ethanoate	vinyl acetate
ethenylbenzene	ethenylbenzene
ethyl 2-propenoate	ethyl 2-propenoate
ethyl acetate	ethyl acetate
ethyl acrylate	ethyl 2-propenoate
ethyl bromide	bromomethane
ethyl chloride	chloroethane
ethyl cyanide	propanenitrile
ethyl ethanoate	ethyl acetate
ethyl ether	diethyl ether
ethyl fluoride	fluoroethane
ethyl formate	ethyl formate
ethyl methanoate	ethyl formate
ethylallene	1,2-pentadiene
ethylene	ethene
ethylene glycol	1,2-ethanediol
ethylene oxide	oxirane
ethyne	acetylene
fluoroethane	fluoroethane
fluoroethene	fluoroethene
fluoroethylene	fluoroethylene
fluoromethane	fluoromethane
formyl fluoride	carbonyl fluoride
furan	furan
glycerin	glycerin

glycerol	glycerin
hexachloroethane	hexachloroethane
hexamethylenetetramine	methenamine
hydrazoic acid	hydrogen azide
hydrogen azide	hydrogen azide
indane	benzocyclopentane
isoamyl alcohol	2-methyl-1-butanol
isobutane	isobutane
isobutanol	isobutanol
Isobutene	isobutene
Isoheptane	2-methylhexane
Isohexane	isohexane
Isononane	2-methyloctane
isooctane	isooctane
isopentane	isopentane
isopentanethiol	isopentanethiol
isopentanol	isopentanol
isopentene	isopentene
isoprene	2-methyl-1,3-butadiene
isopropanol	isopropanol
isopropylbenzene	isopropylbenzene
isopropylether	diisopropyl ether
maleic anhydride	2,5-furandione
<i>m</i> -cresol	3-methylphenol
<i>m</i> -cymene	3-isopropyltoluene
melamine	1,3,5-triazine-2,4,6-triamine
methenamine	methenamine
methyl benzene	toluene
methyl chloride	chloromethane
methyl cyanide	ethane nitrile
methyl ether	dimethyl ether
methyl ethyl ketone	butanone
methyl fluoride	fluoromethane
methyl formate	methyl formate
methyl methacrylate	methyl 2-methylpropenoate
methyl 2-methylpropenoate	methyl 2-methylpropenoate

methyl methonate	methyl formate
methylacetylene	propyne
methylallene	1,2-butadiene
methylene chloride	dichloromethane
methylketene	1-propenal
methyloxacyclopropane	methyloxacyclopropane
methyloxirane	methyloxacyclopropane
<i>m</i> -isopropyltoluene	3-isopropyltoluene
<i>m</i> -xylene	1,3-dimethylbenzene
<i>N,N</i> -dimethylaniline	<i>N,N</i> -dimethylaniline
<i>n</i> -butylamine	<i>n</i> -butylamine
neoheptane	2,2-dimethylpentane
neohexane	2,2-dimethylbutane
neohexane	neohexane
neononane	2,2-dimethylheptane
neooctane	neooctane
neopentane	2,2-dimethylpropane
neopentanethiol	neopentanethiol
<i>n</i> -propanol	1-propanol
<i>n</i> -propyl cyanide	butanenitrile
<i>n</i> -propylacetate	<i>n</i> -propylacetate
<i>n</i> -propylamine	1-propylamine
<i>o</i> -cymene	2-isopropyltoluene
<i>o</i> -isopropyltoluene	2-isopropyltoluene
oxacyclobutane	oxacyclobutane
oxacyclohexane	oxacyclohexane
oxacyclopentadiene	Furan
oxacyclopropane	oxirane
oxane	oxacyclohexane
oxetane	oxacyclobutane
oxirane	oxirane
<i>o</i> -xylene	1,2-dimethylbenzene
<i>p</i> -cymene	4-isopropyltoluene
perchloroethane	hexachloroethane
phosgene	phosgene
<i>p</i> -isopropyltoluene	4-isopropyltoluene

propadiene	allene
propenenitrile	propenenitrile
propyl acetate	<i>n</i> -propylacetate
propyne	propyne
<i>p</i> -xylene	1,4-dimethylbenzene
<i>sec</i> -butylbenzene	<i>sec</i> -butylbenzene
styrene	ethenylbenzene
<i>tert</i> -butanol	<i>tert</i> -butanol
<i>tert</i> -pentanethiol	<i>tert</i> -pentanethiol
<i>tert</i> -pentanol	2-methyl-2-butanol
tetrachloroethene	tetrachloroethene
tetrachloroethylene	tetrachloroethene
tetrachloromethane	tetrachloromethane
tetrafluoroethene	tetrafluoroethene
tetrafluoroethylene	tetrafluoroethene
tetrahydropyran	oxacyclohexane
tetralin	benzocyclohexane

thiacyclopropane	thiacyclopropane
thiirane	thiacyclopropane
toluene	toluene
<i>trans</i> -decahydronaphthalene	<i>trans</i> -decalin
<i>trans</i> -decalin	<i>trans</i> -decalin
trichloroethene	trichloroethene
trichloroethylene	trichloroethene
trichloromethane	trichloromethane
trifluoroethene	trifluoroethene
trifluoroethylene	trifluoroethene
vinyl acetate	vinyl acetate
vinyl bromide	bromoethene
vinyl chloride	chloroethene
vinyl ethanoate	vinyl acetate
vinylacetylene	butenyne
xylidene	<i>N,N</i> -dimethylaniline

1.3. Compound Information

Compound	CASNO	Formula	MW
1,1,1,2-tetrachloroethane	106-46-7	C ₂ H ₂ Cl ₄	167.85
1,1,1,2-tetrafluoroethane	811-97-2	C ₂ H ₂ F ₄	102.03
1,1,1-trichloroethane	71-55-6	C ₂ H ₃ Cl ₃	133.40
1,1,1-trifluoroethane	420-46-2	C ₂ H ₃ F ₃	84.04
1,1,2,2-tetrachloroethane	79-34-5	C ₂ H ₂ Cl ₄	167.85
1,1,2,2-tetrafluoroethane	359-35-3	C ₂ H ₂ F ₄	102.03
1,1,2-trichloroethane	79-00-5	C ₂ H ₃ Cl ₃	133.40
1,1,2-trifluoroethane	430-66-0	C ₂ H ₃ F ₃	84.04
1,1'-bicyclohexyl	92-51-3	C ₁₂ H ₂₂	166.30
1,1-dichloro-1-fluoroethane	1717-00-6	C ₂ H ₃ Cl ₂ F	116.95
1,1-dichloroethane	75-34-3	C ₂ H ₄ Cl ₂	98.96
1,1-dichloroethene	75-35-4	C ₂ H ₂ Cl ₂	96.94
1,1-difluoroethane	75-37-6	C ₂ H ₄ F ₂	66.05
1,1-difluoroethene	75-38-7	C ₂ H ₂ F ₂	64.03
1,1-dimethyl-1-propanethiol	1679-09-0	C ₅ H ₁₂ S	104.21
1,1-dimethylcyclopentane	1638-26-2	C ₇ H ₁₄	98.19
1,1-dimethylhydrazine	57-14-7	C ₂ H ₈ N ₂	60.10
1,2,3-trimethylbenzene	526-73-8	C ₉ H ₁₂	120.19
1,2,4-trimethylbenzene	95-63-6	C ₉ H ₁₂	120.19
1,2-butadiene	590-19-2	C ₄ H ₆	54.09
1,2-dibromoethane	106-93-4	C ₂ H ₄ Br ₂	187.86
1,2-dichloro-1-fluoroethane	430-57-9	C ₂ H ₃ Cl ₂ F	116.95
1,2-dichloro-1-fluoroethene	430-58-0	C ₂ HCl ₂ F	114.93
1,2-dichloroethane	107-06-2	C ₂ H ₄ Cl ₂	98.96
1,2-dichloroethene, (E)-	156-60-5	C ₂ H ₂ Cl ₂	96.94
1,2-dichloroethene, (Z)-	156-59-2	C ₂ H ₂ Cl ₂	96.94
1,2-diethylbenzene	135-01-3	C ₁₀ H ₁₄	134.22
1,2-difluoroethane	624-72-6	C ₂ H ₄ F ₂	66.05
1,2-difluoroethene, (E)-	1630-78-0	C ₂ H ₂ F ₂	64.03
1,2-difluoroethene, (Z)-	1630-77-9	C ₂ H ₂ F ₂	64.03
1,2-dimethylbenzene	95-47-6	C ₈ H ₁₀	106.17
1,2-dimethylcyclohexene	1674-10-8	C ₈ H ₁₄	110.20
1,2-ethanediol	107-21-1	C ₂ H ₆ O ₂	62.07

1,2-pentadiene	591-95-7	C ₅ H ₈	68.12
1,3,5-triazine-2,4,6-triamine	108-78-1	C ₃ H ₆ N ₆	126.12
1,3,5-trimethylbenzene	108-67-8	C ₉ H ₁₂	120.19
1,3,5-trioxane	110-88-3	C ₃ H ₆ O ₃	90.08
1,3-butadiene	106-99-0	C ₄ H ₆	54.09
1,3-cyclohexadiene	592-57-4	C ₆ H ₈	80.13
1,3-cyclopentadiene	542-92-7	C ₅ H ₆	66.10
1,3-diethylbenzene	141-93-5	C ₁₀ H ₁₄	134.22
1,3-diethylcyclohexane	1678-99-5	C ₁₀ H ₂₀	140.27
1,3-dimethylbenzene	108-38-3	C ₈ H ₁₀	106.17
1,3-dioxane	505-22-6	C ₄ H ₈ O ₂	88.11
1,3-pentadiene	504-60-9	C ₅ H ₈	68.12
1,4-cyclohexadiene	628-41-1	C ₆ H ₈	80.13
1,4-diethylbenzene	105-05-5	C ₁₀ H ₁₄	134.22
1,4-dimethylbenzene	106-42-3	C ₈ H ₁₀	106.17
1,4-dioxane	123-91-1	C ₄ H ₈ O ₂	88.11
1,4-pentadiene	591-93-5	C ₅ H ₈	68.12
1,5-cyclooctadiene	111-78-4	C ₈ H ₁₂	108.18
1-butanethiol	109-79-5	C ₄ H ₁₀ S	90.19
1-butanol	71-36-3	C ₄ H ₁₀ O	74.12
1-butene	106-98-9	C ₄ H ₈	56.11
1-butyne	107-00-6	C ₄ H ₆	54.09
1-chloro-1,1-difluoroethane	75-68-3	C ₂ H ₃ ClF ₂	100.50
1-chloro-1-fluoroethane	1615-75-4	C ₂ H ₄ ClF	82.50
1-chloro-2,2-difluoroethane	338-65-8	C ₂ H ₃ ClF ₂	100.50
1-decene	872-05-9	C ₁₀ H ₂₀	140.27
1-ethylcyclohexene	1453-24-3	C ₈ H ₁₄	110.20
1-ethylcyclopentene	2146-38-5	C ₇ H ₁₂	96.17
1-heptanethiol	1639-09-4	C ₇ H ₁₆ S	132.27
1-heptene	592-76-7	C ₇ H ₁₄	98.19
1-heptyne	628-71-7	C ₇ H ₁₂	96.17
1-hexanethiol	111-31-9	C ₆ H ₁₄ S	118.24
1-hexene	592-41-6	C ₆ H ₁₂	84.16
1-hexyne	693-02-7	C ₆ H ₁₀	82.14

1-isobutanethiol	513-44-0	C ₄ H ₁₀ S	90.19
1-methyl-2-ethylbenzene	611-14-3	C ₉ H ₁₂	120.19
1-methyl-3-ethylbenzene	620-14-4	C ₉ H ₁₂	120.19
1-methyl-4-ethylbenzene	622-96-8	C ₉ H ₁₂	120.19
1-methylcyclohexene	591-49-1	C ₇ H ₁₂	96.17
1-methylnaphthalene	90-12-0	C ₁₁ H ₁₀	142.20
1-methylpyrrole	96-54-8	C ₅ H ₇ N	81.12
1-nonene	124-11-8	C ₉ H ₁₈	126.24
1-octene	111-66-0	C ₈ H ₁₆	112.21
1-octyne	629-05-0	C ₁₁ H ₁₄	146.23
1-pentanethiol	110-66-7	C ₅ H ₁₂ S	104.21
1-pentanol	71-41-0	C ₅ H ₁₂ O	88.15
1-penten-3-yne	646-05-9	C ₅ H ₆	66.10
1-pentene	109-67-1	C ₅ H ₁₀	70.13
1-pentyne	627-19-0	C ₅ H ₈	68.12
1-propanethiol	107-03-9	C ₃ H ₈ S	76.16
1-propanol	71-23-8	C ₃ H ₈ O	60.10
1-propenal	6004-44-0	C ₃ H ₄ O	56.06
1-propylamine	107-10-8	C ₃ H ₉ N	59.11
2,2,3-trimethylbutane	464-06-2	C ₇ H ₁₆	100.20
2,2,4-trimethylpentane	540-84-1	C ₈ H ₁₈	114.23
2,2-dimethylhexane	590-73-8	C ₈ H ₁₈	114.23
2,3-dimethylbutane	79-29-8	C ₆ H ₁₄	86.18
2,3-dimethylpentane	565-59-3	C ₇ H ₁₆	100.20
2,4-diisocyanatotoluene	584-84-9	C ₉ H ₆ N ₂ O ₂	174.16
2,4-dimethylpentane	108-08-7	C ₇ H ₁₆	100.20
2,5-furandione	108-31-6	C ₄ H ₂ O ₃	98.06
2-butanethiol	513-53-1	C ₄ H ₁₀ S	90.19
2-butanol	78-92-2	C ₄ H ₁₀ O	74.12
2-butene, (E)-	624-64-6	C ₄ H ₈	56.11
2-butene, (Z)-	590-18-1	C ₄ H ₈	56.11
2-butyne	503-17-3	C ₄ H ₆	54.09
2-chloro-1,1,1-trifluoroethane	75-88-7	C ₂ H ₂ ClF ₃	118.49
2-hexyne	764-35-2	C ₆ H ₁₀	82.14
2-isopropyltoluene	527-84-4	C ₁₀ H ₁₄	134.22
2-methoxyethanol	109-86-4	C ₃ H ₈ O ₂	76.09
2-methyl-1,3-butadiene	78-79-5	C ₅ H ₈	68.12

2-methyl-1-butanol	137-32-6	C ₅ H ₁₂ O	88.15
2-methyl-1-butene	563-46-2	C ₅ H ₁₀	70.13
2-methyl-2-butanol	75-85-4	C ₅ H ₁₂ O	88.15
2-methyl-2-butene	513-35-9	C ₅ H ₁₀	70.13
2-methylcyclohexanone	583-60-8	C ₇ H ₁₂ O	112.17
2-methylfuran	534-22-5	C ₅ H ₆ O	82.10
2-methylheptane	592-27-8	C ₈ H ₁₈	114.23
2-methylnaphthalene	91-57-6	C ₁₁ H ₁₀	142.20
2-methylphenol	95-48-7	C ₇ H ₈ O	108.14
2-methylpyridine	109-06-8	C ₆ H ₇ N	93.13
2-pentanol	6032-29-7	C ₅ H ₁₂ O	88.15
2-pentanone	107-87-9	C ₅ H ₁₀ O	86.13
2-pentene, (E)-	646-04-8	C ₅ H ₁₀	70.13
2-pentene, (Z)-	627-20-3	C ₅ H ₁₀	70.13
2-pentyne	627-21-4	C ₅ H ₈	68.12
2-propenal	107-02-8	C ₃ H ₄ O	56.06
3,3-dimethylpentane	562-49-2	C ₇ H ₁₆	100.20
3-ethylcyclopentene	694-35-9	C ₇ H ₁₂	96.17
3-hexyne	928-49-4	C ₆ H ₁₀	82.14
3-isopropyltoluene	535-77-3	C ₁₀ H ₁₄	134.22
3-methyl-1-butene	563-45-1	C ₅ H ₁₀	70.13
3-methyl-2-butanol	598-75-4	C ₅ H ₁₂ O	88.15
3-methylcyclohexanone	591-24-2	C ₇ H ₁₂ O	112.17
3-methylcyclopentene	1120-62-3	C ₆ H ₁₀	82.14
3-methylhexane	589-34-4	C ₇ H ₁₆	100.20
3-methylpentane	96-14-0	C ₆ H ₁₄	86.18
3-methylphenol	108-39-4	C ₇ H ₈ O	108.14
3-pentanol	584-02-1	C ₅ H ₁₂ O	88.15
3-pentanone	96-22-0	C ₅ H ₁₀ O	86.13
3-penten-1-yne	2206-23-7	C ₅ H ₆	66.10
4-isopropyltoluene	99-87-6	C ₁₀ H ₁₄	134.22
4-methylcyclohexanone	589-92-4	C ₇ H ₁₂ O	112.17
4-methylphenol	106-44-5	C ₇ H ₈ O	108.14
acenaphthene	83-32-9	C ₁₂ H ₁₀	154.21
acenaphthylene	208-96-8	C ₁₂ H ₈	152.19
acetaldehyde	75-07-0	C ₂ H ₄ O	44.05
acetic acid	64-19-7	C ₂ H ₄ O ₂	60.05

acetic anhydride	108-24-7	C ₄ H ₆ O ₃	102.09
acetone	67-64-1	C ₃ H ₆ O	58.08
acetylene	74-86-2	C ₂ H ₂	26.04
allene	463-49-0	C ₃ H ₄	40.06
alpha-D-glucose	492-62-6	C ₆ H ₁₂ O ₆	180.16
ammonia	7664-41-7	H ₃ N	17.03
ammonium perchlorate	7790-98-9	ClH ₄ NO ₄	117.49
aniline	62-53-3	C ₆ H ₇ N	93.13
anthracene	120-12-7	C ₁₄ H ₁₀	178.23
azacyclobutane	503-29-7	C ₃ H ₇ N	57.09
azacyclopentane	123-75-1	C ₄ H ₉ N	71.12
azacyclopropane	151-56-4	C ₂ H ₅ N	43.07
benzaldehyde	100-52-7	C ₇ H ₆ O	106.12
benzene	71-43-2	C ₆ H ₆	78.11
benzenethiol	108-98-5	C ₆ H ₆ S	110.18
benzoic acid	65-85-0	C ₇ H ₆ O ₂	122.12
benzyl alcohol	100-51-6	C ₇ H ₈ O	108.14
biphenylene	259-79-0	C ₁₂ H ₈	152.19
bromochlorodifluoromethane	353-59-3	CB ₂ ClF ₂	165.36
bromochlorofluoromethane	593-98-6	CHBrClF	147.37
bromochloromethane	74-97-5	CH ₂ BrCl	129.38
bromodichlorofluoromethane	353-58-2	CB ₂ Cl ₂ F	181.82
bromodichloromethane	75-27-4	CHBrCl ₂	163.83
bromodifluoromethane	1511-62-2	CHBrF ₂	130.92
bromoethane	74-96-4	C ₂ H ₅ Br	108.97
bromoethene	593-60-2	C ₂ H ₃ Br	106.95
bromofluoromethane	373-52-4	CH ₂ BrF	112.93
bromomethane	74-83-9	CH ₃ Br	94.94
bromotrichloromethane	75-62-7	CBrCl ₃	198.27
bromotrifluoromethane	75-63-8	CBrF ₃	148.91
butadiyne	460-12-8	C ₄ H ₂	50.06
butanal	123-72-8	C ₄ H ₈ O	72.11
butanenitrile	109-74-0	C ₄ H ₇ N	69.11
butanoic acid	107-92-6	C ₄ H ₈ O ₂	88.11
butanoic anhydride	106-31-0	C ₈ H ₁₄ O ₃	158.19
butanone	78-93-3	C ₄ H ₈ O	72.11
butenyne	689-97-4	C ₄ H ₄	52.07

carbon	7440-44-0	C	12.01
carbon dioxide	124-38-9	CO ₂	44.01
carbon disulfide	75-15-0	CS ₂	76.14
carbon monoxide	630-08-0	CO	28.01
carbonyl difluoride	353-50-4	CF ₂ O	66.01
carbonyl fluoride	1493-02-3	CHFO	48.02
chlorobenzene	108-90-7	C ₆ H ₅ Cl	112.56
chlorodifluoromethane	75-45-6	CHClF ₂	86.47
chloroethane	75-00-3	C ₂ H ₅ Cl	64.51
chloroethene	75-01-4	C ₂ H ₃ Cl	62.50
chloroacetylene	593-63-5	C ₂ HCl	60.48
chlorofluoromethane	593-70-4	CH ₂ ClF	68.48
chloromethane	74-87-3	CH ₃ Cl	50.49
chlorotrifluoroethene	79-38-9	C ₂ ClF ₃	116.47
chlorotrifluoromethane	75-72-9	CClF ₃	104.46
cis-decalin	493-01-6	C ₁₀ H ₁₈	138.25
coronene	191-07-1	C ₂₄ H ₁₂	300.35
cyanogen	460-19-5	C ₂ N ₂	52.03
cyclobutadiene	1120-53-2	C ₄ H ₄	52.07
cyclobutane	287-23-0	C ₄ H ₈	56.11
cyclobutanone	1191-95-3	C ₄ H ₆ O	70.09
cyclobutene	822-35-5	C ₄ H ₆	54.09
cycloheptane	291-64-5	C ₇ H ₁₄	98.19
cycloheptene	628-92-2	C ₇ H ₁₂	96.17
cyclohexane	110-82-7	C ₆ H ₁₂	84.16
cyclohexanone	108-94-1	C ₆ H ₁₀ O	98.14
cyclohexene	110-83-8	C ₆ H ₁₀	82.14
cyclohexylamine	108-91-8	C ₆ H ₁₃ N	99.17
cyclopentane	287-92-3	C ₅ H ₁₀	70.13
cyclopentanone	120-92-3	C ₅ H ₈ O	84.12
cyclopentene	142-29-0	C ₅ H ₈	68.12
cyclopropane	75-19-4	C ₃ H ₆	42.08
cyclopropanone	5009-27-8	C ₃ H ₄ O	56.06
cyclopropene	2781-85-3	C ₃ H ₄	40.06
D-camphor	76-22-2	C ₁₀ H ₁₆ O	152.23
diborane	19287-45-7	B ₂ H ₆	16.86
dibromochlorofluoromethane	353-55-9	CB ₂ ClF	226.27

dibromochloromethane	124-48-1	CHBr ₂ Cl	208.28
dibromochloromethane	124-48-1	CHBr ₂ Cl	208.28
dibromodichloromethane	594-18-3	CBBr ₂ Cl ₂	242.72
dibromodifluoromethane	75-61-6	CBBr ₂ F ₂	209.82
dibromofluoromethane	1868-53-7	CHBr ₂ F	191.83
dibromomethane	74-95-3	CH ₂ Br ₂	173.83
dichlorodifluoromethane	75-71-8	CCl ₂ F ₂	120.91
dichloroacetylene	7572-29-4	C ₂ Cl ₂	94.93
dichlorofluoromethane	75-43-4	CHCl ₂ F	102.92
dichloromethane	75-09-2	CH ₂ Cl ₂	84.93
diethyl ether	60-29-7	C ₄ H ₁₀ O	74.12
diethyl sulfide	352-93-2	C ₄ H ₁₀ S	90.19
diethylamine	109-89-7	C ₄ H ₁₁ N	73.14
difluorochloromethane	75-45-6	CHClF ₂	86.47
difluoroethyne	689-99-6	C ₂ F ₂	62.02
difluoromethane	75-10-5	CH ₂ F ₂	52.02
diisopropyl ether	108-20-3	C ₆ H ₁₄ O	102.17
dimethyl disulfide	624-92-0	C ₂ H ₆ S ₂	94.20
dimethyl ether	115-10-6	C ₂ H ₆ O	46.07
dimethyl sulfide	75-18-3	C ₂ H ₆ S	62.13
dimethyl sulfoxide	67-68-5	C ₂ H ₆ OS	78.13
dimethylamine	124-40-3	C ₂ H ₇ N	45.08
di- <i>n</i> -butyl sulfide	544-40-1	C ₈ H ₁₈ S	146.29
di- <i>n</i> -propyl ether	111-43-3	C ₆ H ₁₄ O	102.17
di- <i>n</i> -propyl sulfide	111-47-7	C ₆ H ₁₄ S	118.24
ethane	74-84-0	C ₂ H ₆	30.07
ethanenitrile	75-05-8	C ₂ H ₃ N	41.05
ethanethiol	75-08-1	C ₂ H ₆ S	62.13
ethanol	64-17-5	C ₂ H ₆ O	46.07
ethene	74-85-1	C ₂ H ₄	28.05
ethenylbenzene	100-42-5	C ₈ H ₈	104.15
ethyl acetate	141-78-6	C ₄ H ₈ O ₂	88.11
ethyl propenoate	140-88-5	C ₅ H ₈ O ₂	100.12
ethyl formate	109-94-4	C ₄ H ₆ O ₂	86.09
ethyl isopropyl sulfide	5145-99-3	C ₅ H ₁₂ S	104.21
ethyl methyl ether	540-67-0	C ₃ H ₈ O	60.10
ethyl methyl sulfide	624-89-5	C ₃ H ₈ S	76.16

ethyl propyl sulfide	4110-50-3	C ₅ H ₁₂ S	104.21
ethylamine	75-04-7	C ₂ H ₇ N	45.08
ethylbenzene	100-41-4	C ₈ H ₁₀	106.17
ethylcyclohexane	1678-91-7	C ₈ H ₁₆	112.21
ethylcyclopentane	1640-89-7	C ₇ H ₁₄	98.19
fluoranthene	206-44-0	C ₁₆ H ₁₀	202.25
fluorene	86-73-7	C ₁₃ H ₁₀	166.22
fluoroethane	353-36-6	C ₂ H ₅ F	48.06
fluoroethene	75-02-5	C ₂ H ₃ F ₃	84.04
fluoroethyne	2713-09-9	C ₂ HF	44.03
fluoromethane	593-53-3	CH ₃ F	34.03
formaldehyde	50-00-0	CH ₂ O	30.03
formic acid	64-18-6	CH ₂ O ₂	46.03
furan	110-00-9	C ₄ H ₄ O	68.07
glycerin	56-81-5	C ₃ H ₈ O ₃	92.09
graphite	7782-42-5	C	12.01
hexachloroethane	67-72-1	C ₂ Cl ₆	236.74
hexafluoroethane	76-16-4	C ₂ F ₆	138.01
hexamethyldisiloxane	107-46-0	C ₆ H ₁₈ OSi ₂	162.38
hexanenitrile	628-73-9	C ₆ H ₁₁ N	97.16
hydrazine	302-01-2	H ₄ N ₂	32.05
hydrogen	1333-74-0	H ₂	2.02
hydrogen azide	7782-79-8	HN ₃	43.03
hydrogen cyanide	74-90-8	CHN	27.03
hydrogen peroxide	7722-84-1	H ₂ O ₂	34.01
hydrogen sulfide	7783-06-4	H ₂ S	34.08
indane	496-11-7	C ₉ H ₁₀	118.18
indene	95-13-6	C ₉ H ₈	116.16
iodomethane	74-88-4	CH ₃ I	141.94
isobutanal	78-84-2	C ₄ H ₈ O	72.11
isobutane	75-28-5	C ₄ H ₁₀	58.12
isobutanol	78-83-1	C ₄ H ₁₀ O	74.12
isobutene	115-11-7	C ₄ H ₈	56.11
isobutyl ethyl ether	628-81-9	C ₆ H ₁₄ O	102.17
isobutylamine	78-81-9	C ₄ H ₁₁ N	73.14
Isobutylbenzene	538-93-2	C ₁₀ H ₁₄	134.22
isobutylcyclohexane	1678-98-4	C ₁₀ H ₂₀	140.27

isoheptane	591-76-4	C ₇ H ₁₆	100.20
isohexane	107-83-5	C ₆ H ₁₄	86.18
isononane	3221-61-2	C ₉ H ₂₀	128.26
isopentane	78-78-4	C ₅ H ₁₂	72.15
isopentanethiol	541-31-1	C ₅ H ₁₂ S	104.21
isopentanol	123-51-3	C ₅ H ₁₂ O	88.15
isopropanethiol	75-33-2	C ₃ H ₈ S	76.16
isopropanol	67-63-0	C ₃ H ₈ O	60.10
isopropyl methyl ether	598-53-8	C ₄ H ₁₀ O	74.12
isopropyl methyl sulfide	1551-21-9	C ₄ H ₁₀ S	90.19
isopropylamine	75-31-0	C ₃ H ₉ N	59.11
isopropylbenzene	98-82-8	C ₉ H ₁₂	120.19
isopropylcyclohexane	696-29-7	C ₉ H ₁₈	126.24
isopropylcyclopentane	3875-51-2	C ₈ H ₁₆	112.21
methane	74-82-8	CH ₄	16.04
methanethiol	74-93-1	CH ₄ S	48.11
methanol	67-56-1	CH ₄ O	32.04
methenamine	100-97-0	C ₆ H ₁₂ N ₄	140.19
methyl acetate	79-20-9	C ₃ H ₆ O ₂	74.08
methyl butanoate	623-42-7	C ₅ H ₁₀ O ₂	102.13
methyl formate	107-31-3	C ₄ H ₆ O ₂	86.09
methyl isopropyl sulfide	1551-21-9	C ₄ H ₁₀ S	90.19
methyl 2-methylpropenoate	80-62-6	C ₅ H ₈ O ₂	100.12
methyl nitrate	598-58-3	CH ₃ NO ₃	77.04
methyl propanoate	554-12-1	C ₄ H ₈ O ₂	88.11
methyl propyl sulfide	3877-15-4	C ₄ H ₁₀ S	90.19
methylamine	74-89-5	CH ₅ N	31.06
methylcyclobutane	598-61-8	C ₅ H ₁₀	70.13
methylcyclohexane	108-87-2	C ₇ H ₁₄	98.19
methylcyclopentane	96-37-7	C ₆ H ₁₂	84.16
methylenecyclobutane	1120-56-5	C ₅ H ₈	68.12
methylenecyclohexane	1192-37-6	C ₇ H ₁₂	96.17
methylethylamine	624-78-2	C ₃ H ₉ N	59.11
methyloxacyclopropane	75-56-9	C ₃ H ₆ O	58.08
<i>N,N</i> -dimethylaniline	121-69-7	C ₈ H ₁₁ N	121.18
naphthacene	92-24-0	C ₁₈ H ₁₂	228.29
naphthalene	91-20-3	C ₁₀ H ₈	128.17

<i>n</i> -butane	106-97-8	C ₄ H ₁₀	58.12
<i>n</i> -butanoic acid	107-92-6	C ₄ H ₈ O	88.11
<i>n</i> -butyl methyl ether	628-28-4	C ₅ H ₁₂ O	88.15
<i>n</i> -butylamine	109-73-9	C ₄ H ₁₁ N	73.14
<i>n</i> -butylbenzene	104-51-8	C ₁₀ H ₁₄	134.22
<i>n</i> -decane	124-18-5	C ₁₀ H ₂₂	142.28
<i>n</i> -dodecane	112-40-3	C ₁₂ H ₂₆	170.33
neoheptane	590-35-2	C ₇ H ₁₆	100.20
neohexane	75-83-2	C ₆ H ₁₄	86.18
neononane	1071-26-7	C ₉ H ₂₀	128.26
neopentane	463-82-1	C ₅ H ₁₂	72.15
neopentanethiol	1679-08-9	C ₅ H ₁₂ S	104.21
<i>n</i> -heptane	142-82-5	C ₇ H ₁₆	100.20
<i>n</i> -hexadecane	544-76-3	C ₁₆ H ₃₄	226.44
<i>n</i> -hexane	110-54-3	C ₆ H ₁₄	86.18
<i>n</i> -hexanoic acid	142-62-	C ₆ H ₁₂ O ₂	116.16
nitrobenzene	98-95-3	C ₆ H ₅ NO ₂	123.11
nitrogen	7727-37-9	N ₂	28.01
nitroglycerin	55-63-0	C ₃ H ₅ N ₃ O ₉	227.09
nitromethane	75-52-5	CH ₃ NO ₂	61.04
nitrous oxide	10024-97-2	N ₂ O	44.01
<i>n</i> -nonane	111-84-2	C ₉ H ₂₀	128.26
<i>n</i> -octane	111-65-9	C ₈ H ₁₈	114.23
<i>n</i> -pentane	109-66-0	C ₅ H ₁₂	72.15
<i>n</i> -pentanoic acid	109-52-4	C ₅ H ₁₀ O	86.13
<i>n</i> -propyl methyl ether	557-17-5	C ₄ H ₁₀ O	74.12
<i>n</i> -propylacetate	109-60-4	C ₅ H ₁₀ O ₂	102.13
<i>n</i> -propylbenzene	103-65-1	C ₉ H ₁₂	120.19
<i>n</i> -propylcyclopentane	2040-96-2	C ₈ H ₁₆	112.21
<i>n</i> -undecane	1120-21-4	C ₁₁ H ₂₄	156.31
octamethylcyclotetrasiloxane	556-67-2	C ₈ H ₂₄ O ₄ Si ₄	296.62
oxacyclobutane	503-30-0	C ₃ H ₆ O	58.08
oxacyclohexane	142-68-7	C ₅ H ₁₀ O	86.13
oxacyclopentane	109-99-9	C ₄ H ₈ O	72.11
oxirane	75-21-8	C ₂ H ₄ O	44.05
oxygen	7782-44-7	O ₂	32.00
pentachloroethane	76-01-7	C ₂ HCl ₅	202.29

pentachlorophenol	87-86-5	C ₆ HCl ₅ O	266.34
pentafluoroethane	354-33-6	C ₂ HF ₅	120.02
pentanenitrile	110-59-8	C ₅ H ₉ N	83.13
perylene	198-55-0	C ₂₀ H ₁₂	252.31
phenanthrene	85-01-8	C ₁₄ H ₁₀	178.23
phenol	108-95-2	C ₆ H ₆ O	94.11
phosgene	75-44-5	CCl ₂ O	98.92
piperidine	110-89-4	C ₅ H ₁₁ N	85.15
propanal	123-38-6	C ₃ H ₆ O	58.08
propane	74-98-6	C ₃ H ₈	44.10
propanenitrile	107-12-0	C ₃ H ₅ N	55.08
propanoic acid	79-09-4	C ₃ H ₆ O ₂	74.08
propanoic anhydride	123-62-6	C ₆ H ₁₀ O ₃	130.14
propene	115-07-1	C ₃ H ₆	42.08
propenenitrile	107-13-1	C ₃ H ₃ N	53.06
propylcyclohexane	1678-92-8	C ₉ H ₁₈	126.24
propyne	74-99-7	C ₃ H ₄	40.06
pyrene	129-00-0	C ₁₆ H ₁₀	202.25
pyrrole	109-97-7	C ₄ H ₅ N	67.09
sec-butylbenzene	135-98-8	C ₁₀ H ₁₄	134.22
sucrose	57-50-1	C ₁₂ H ₂₂ O ₁₁	342.30
sulfur dioxide	7446-09-5	O ₂ S	64.06
tert-butanol	75-65-0	C ₄ H ₁₀ O	74.12
tert-butyl methyl ether	1634-04-4	C ₅ H ₁₂ O	88.15
tert-butylamine	75-64-9	C ₄ H ₁₁ N	73.14
tert-butylbenzene	98-06-6	C ₁₀ H ₁₄	134.22
tert-butylcyclohexane	3178-22-1	C ₁₀ H ₂₀	140.27
tert-pentanethiol	1679-09-0	C ₅ H ₁₂ S	104.21
tetrabromomethane	558-13-4	CBr ₄	331.63

tetrachloroethene	127-18-4	C ₂ Cl ₄	165.83
tetrachloromethane	56-23-5	CCl ₄	153.82
tetrafluoroethene	116-14-3	C ₂ F ₄	100.02
tetrafluoromethane	75-73-0	CF ₄	88.00
tetralin	119-64-2	C ₁₀ H ₁₂	132.20
tetranitromethane	509-14-8	CN ₄ O ₈	196.03
thiacyclobutane	287-27-4	C ₃ H ₆ S	74.14
thiacyclohexane	1613-51-0	C ₅ H ₁₀ S	102.20
thiacyclopentane	110-01-0	C ₄ H ₈ S	88.17
thiacyclopropane	420-12-2	C ₂ H ₄ S	60.12
thiophene	110-02-1	C ₄ H ₄ S	84.14
toluene	108-88-3	C ₇ H ₈	92.14
trans-decalin	493-02-7	C ₁₀ H ₁₈	138.25
tribromochloromethane	594-15-0	CB ₃ Cl	287.18
tribromofluoromethane	353-54-8	CB ₃ F	270.72
tribromomethane	75-25-2	CHBr ₃	252.73
trichloroethene	79-01-6	C ₂ HCl ₃	131.39
trichlorofluoromethane	75-69-4	CCl ₃ F	137.37
trichloromethane	67-66-3	CHCl ₃	119.38
triethanolamine	102-71-6	C ₆ H ₁₅ NO ₃	149.19
triethylamine	121-44-8	C ₆ H ₁₅ N	101.19
trifluoroethene	359-11-5	C ₂ HF ₃	82.02
trifluoromethane	75-46-7	CHF ₃	70.01
trimethylamine	75-50-3	C ₃ H ₉ N	59.11
trinitromethane	517-25-9	CHN ₃ O ₆	151.04
trinitrotoluene	118-96-7	C ₇ H ₅ N ₃ O ₆	227.13
urea	57-13-6	CH ₄ N ₂ O	60.06
vinyl acetate	108-05-4	C ₄ H ₆ O ₂	86.09
water	7732-18-5	H ₂ O	18.02

1.4. Organization of Compounds by Classes

In the tables in this document, the compounds are arranged first by primary and secondary chemical class and then by molecular formula.

Primary and Secondary Chemical Class

1. Small Molecules

- a. Hydrogen/Oxygen
- b. Nitrogen/Hydrogen
- c. Nitrogen/Oxygen
- d. Sulfur/Hydrogen/Oxygen
- e. Boron/Hydrogen
- f. Hydrogen Halides

2. Hydrocarbons

- a. Carbon
- b. Alkanes
- c. Alkenes
- d. Alkadienes
- e. Alkynes
- f. Alkenynes
- g. Alkadiynes
- h. Cycloalkanes
- i. Cycloalkenes
- j. Cycloalkadienes
- k. Bicycloalkanes

3. Aromatic Hydrocarbons

- a. Alkylbenzenes
- b. Alkenylbenzenes
- c. Benzocycloalkanes
- d. Benzocycloalkenes
- e. Alkyl naphthalenes
- f. Polycyclic Aromatics

4. Oxidized Hydrocarbons

- a. Carbon Oxides
- b. Alkanols
- c. Glycols
- d. Sugars
- e. Phenols
- f. Alkyl ethers
- g. Aldehydes
- h. Alkenals
- i. Ketones
- j. Carboxylic Acids
- k. Alkyl Esters
- l. Alkenyl Esters
- m. Alkanoic Anhydrides
- n. Oxacycloalkanes
- o. Oxacycloalkadienes
- p. Oxacycloketones
- q. Dioxanes and Trioxanes
- r. Furandione and D-Camphor
- s. Aromatic Alcohols
- t. Aromatic Aldehydes and Acids

5. Nitrogen-Substituted Hydrocarbons

- a. Hydrogen Cyanides
- b. Alkylamines
- c. Cycloalkylamines
- d. Azacycloalkanes
- e. Methenamines
- f. Azacycloalkadienes
- g. Alkanenitriles
- h. Alkenenitriles
- i. Cyanogens
- j. Alkylhydrazines
- k. Melamines
- l. Pyridines
- m. Anilines
- n. Oxidized Amines
- o. Alkyl Nitrates and Nitroalkanes
- p. Isocyanates
- q. Nitro Aromatics

6. Sulfur-Substituted Hydrocarbons

- a. Alkanethiols
- b. Alkyl Sulfides
- c. Thiacycloalkanes
- d. Thiacycloalkenes
- e. Aromatic Thiols
- f. Alkyl Sulfoxides

7. Halogenated Hydrocarbons

- a. Fluoromethanes
- b. Fluoroethanes
- c. Fluoroethenes
- d. Chloromethanes
- e. Chloroethanes
- f. Chloroethenes
- g. Chloroethynes

- h. Chloroaromatics
- i. Bromomethanes
- j. Bromoethanes
- k. Bromoethenes
- l. Iodomethanes
- m. Chlorofluoromethanes
- n. Chlorofluoroethanes
- o. Chlorofluoroethenes

- p. Bromofluoroalkanes
- q. Bromochloroalkanes
- r. Bromochlorofluoroalkanes
- s. Carbonyl Halides
- t. Chlorophenols

8. Siloxanes

1.5. References for Introduction

- [2018LEM] Lemmon, E. W.; Bell, I. H.; Huber, M. L.; McLinden, M. O.; Reference Fluid Thermodynamic and Transport Properties (REFPROP), NIST Standard Reference Database 23, Vers. 10.0, 2018
- [2019DIK] Diky, V.; Chirico, R. D.; Frenkel, M.; Bazyleva, A.; Magee, J. W.; Paulechka, E.; Kazakov, A.; Lemmon, E. W.; Muzny, C. D.; Smolyanitsky, A. Y.; Townsend, S.; Kroenlein, K.; ThermoData Engine (TDE), Vers. 10.4.2, NIST Standard Reference Database 103b, (2019)
- [2021LIN] Linstrom P. J.; W.G. Mallard, Eds., NIST Chemistry WebBook, NIST Standard Reference Database Number 69, National Institute of Standards and Technology, Gaithersburg MD, 20899, <https://doi.org/10.18434/T4D303>, (accessed July 22, 2021)

2. Normal Melting Points T_m and Normal Boiling Points T_b ($P = 1 \text{ atm} = 101.325 \text{ kPa}$)

2.1. Overview

In this section, we compiled, evaluated, and recommended normal melting points T_m and normal boiling points T_b ($P = 1 \text{ atm} = 101.325 \text{ kPa}$) and of hydrocarbons, substituted hydrocarbons, and a few other compounds of physical, chemical, and engineering interest. (A few triple points and sublimation points are included). The substituted hydrocarbons include different classes of oxygen-, nitrogen-, sulfur-, and halogen-substituted hydrocarbons. We compiled about 2200 values from the literature for normal melting points and normal melting points for about 420 compounds from about 870 sources in the literature. From these data, we recommend about 100 melting points and about 400 boiling points for the substances. All of these data are provided in Table 2-1. Although this is an extensive compilation, for some compounds, it is not comprehensive. All compounds, however, have a good selection of values from the literature, sufficient to provide an evaluated recommendation. The boiling points provided here were derived from direct boiling point, liquid-vapor equilibrium, vapor pressure determinations – and also from equation-of-state (EOS) formulations. The most accurate normal boiling points come from NIST TRC REFPROP[2018LEM] and NIST TRC TDE,[2019DIK] although for some compounds we did select other values. All values that were compiled are given in Table 2-1, while only the recommended values are provided in Table 2-2. All sources of data for the boiling and melting points that were compiled from the literature are given in the references.

We provide uncertainties from the original sources where available. In many cases, we have assigned estimated uncertainties where the original sources did not provide them, or we considered them to be unreasonable (such as they were providing precision not accuracy). The recommended values of boiling points provided here were selected from reliable original sources (not weighted averages). The uncertainties given here for the recommended values of the boiling points are generally expanded uncertainties at the 95 % confidence level (2σ , Type B).[1994TAY] The uncertainty column is identified with “unc”, since many workers did not identify whether the uncertainties were 1σ or 2σ . All our recommended values are, in general, 2σ . The highest quality data are those derived from equation of state (EOS) formulations with about one half of these (excluding outliers) having very low expanded uncertainties of (0.01 to 0.06) K, and the other half having still low expanded uncertainties of (0.06 to 0.16) K. The expanded uncertainties (2σ) for the EOS predicted normal boiling points were derived from the reported expanded uncertainties (2σ) in pressure and the temperature dependence of the pressure, $(1/P) (\partial P/\partial T)$. Vapor pressure determinations of boiling points are relatively accurate – but about half as accurate as those derived from EOS formulations. In many cases, we were able to assign an estimated average uncertainty to each set of data from the measurements of individual research groups by comparing the deviations of their reported boiling points to other data in the literature, as well as considering reported uncertainties, when provided in the original sources (these estimated uncertainties are generally (0.07, 0.10, or 0.15) K. We estimated that about 45 % of these sets of boiling points had deviations of about (0.05 to 0.10) K, about 35 % had deviations of (0.10 to 0.25) K, about 15 % had deviations of (0.25 to 0.6) K, and about 5 % had deviations >0.6 K. For boiling points derived from other vapor pressure measurements, where there was insufficient data for sampling a set from a particular

research group, we assigned a generic expanded uncertainty of $U = 0.15$ K, except in a few cases where we believed that it was higher. Vapor-liquid equilibrium determinations of boiling points are also relatively accurate – we estimated that most of the boiling points had expanded uncertainties on the order of $U = (0.15 \text{ to } 0.25)$ K – roughly a median value. For direct determinations of boiling points, our analysis suggests that about 50 % of the values had expanded uncertainties of $U = (0.05 \text{ to } 0.20)$ K, 20 % with $U = (0.2 \text{ to } 0.5)$ K, 20 % with $U = (0.5 \text{ to } 1.0)$ K, and 10 % with $U > 1$ K. Where the boiling points were derived from direct boiling point measurements, we assigned expanded uncertainties based on the range of values (Type B uncertainties)[1994TAY] from many different determinations.

We also provide suggested melting points for compounds above 273.15 K (0 °C). These are important for determining enthalpies of combustion for some compounds which require enthalpies of melting or sublimation. We categorized the methods that were used to determine the boiling points and melting points reported in each of the sources, as best we could.

The overall procedure was that we first compiled values from the literature taking many from the NIST Chemistry Webbook, [2021LIN] and then selected recommended values for each compound. We then used the NIST TRC Thermodata Engine (TDE) [2021DIK] to independently screen our selections and make corrections as needed. In most cases, the values from TDE agreed with our selections within the uncertainties. A fraction of the uncertainties assigned by TDE (a semi-automated evaluation method) were either too small or too large relative to our assigned uncertainties. We investigated these differences and made corrections as needed. We then also used NIST TRC REFPROP[2018LEM] to screen our selections for boiling points – and some of our recommended values are selected from REFPROP, since they are derived from EOS formulations. In general, however, we attempted to provide the original values from the literature, rather than from evaluated sources (including from REFPROP) to provide traceability to the original measurements. We found that in many cases, the values from the evaluated sources are very nearly identical to those in the original sources.

The columns labeled “Phase” and “TRS” indicate the phase and the type of the transition point, respectively. For compounds that are in the gas phase at room temperature, we have designated these as Phase=“gas” and transition point trs=“BP” to reflect this. For example, H₂ becomes a gas at T=20.369 K and a liquid below that temperature. We have also included triple points (trs=“TP”) for just a few compounds where the melting and triple points do not differ by much. The column labeled “SEL” indicates that data is a selected/recommended value. Those labeled “SEL” are the selected value at standard state (e.g., [*liq* → *gas*], where compound is a liquid at standard state), while those labeled “SEL1” or “SEL2” are for other states (e.g., [*sol* → *liquid*], melting [*sol* → *liquid*], or sublimation [*sol* → *gas*]). The column labeled “Method” provides an indication of the method used to determine the boiling point or melting point. A legend for the “Method” column is provided in Table 2-3.

We provide recommended values for normal melting and normal boiling points on the International Temperature Scale of 1990 (ITS-90).[1990PRE] Recommended values based on measurements before 1990 (IPTS-48 and IPTS-68 scales) were (mostly) corrected to the ITS-90 scale.[1992GOL][1994RUS] Generally, the corrections were less than 0.03 K, since the temperatures for the substances in this set of largely hydrocarbons and substituted hydrocarbons are on the order

of just (200 to 500) K, and much less than uncertainties on the order of (0.10 to 0.20) K. Only at much higher temperatures (>800 K) do these corrections become greater than 0.1 K.

2.2. Normal Melting and Normal Boiling Points (compilation)

All values that were compiled are given in Table 2-1, while only the recommended values are provided in Table 2-2, with all references for melting and boiling point data provided in the last subsection 2.4. Values in **bold** are updated values from the original version of this document. Note: e-H₂ is equilibrium H₂ which is essentially all (99.8 %) parahydrogen (isomer with antiparallel proton spins) at 20 K. At room temperature, normal H₂ is 75 % orthohydrogen (isomer with parallel proton spins) and 25 % parahydrogen.

Table 2-1. Normal Melting and Normal Boiling Points

Formula	Compound	Phase	SEL	trs	T_{trs}/K	unc/K	Method	Reference
SMALL MOLECULES								
Hydrogen/Oxygen								
e-H ₂	hydrogen	gas	SEL	BP	20.271	0.003	EOS	2009LEA
				BP	20.271	0.001	VP	1977ANC
				BP	20.270	0.002	VP	1963BAR
				BP	20.272	0.002	VP	1948WOO
n-H ₂		gas	SEL	BP	20.369	0.003	EOS	2009LEA
o-H ₂		gas	SEL	BP	20.38	0.003	EOS	2009LEA
O ₂	oxygen	gas	SEL	BP	90.188	0.029	EOS	1991STE
				BP	90.188	0.029	EOS-rev	2018LEM
H ₂ O	water	liq	SEL	BP	373.124	0.014	EOS	2002WAG
		sol	SEL1	MP	273.153	0.001	EOS	2006FEI
			SEL2	TP	273.16	0.01	DEF	1990PRE
		liq		BP	373.124	0.014	EOS-rev	2018LEM
H ₂ O ₂	hydrogen peroxide	liq	SEL	BP	426.31	0.05	VLE rev	2004MAN
		sol	SEL1	MP	272.699	0.008	SLE	1951FOL
			SEL2	TP	272.75	0.02	TP	1954GIG
		liq		BP	426.2	1.2	BP rev	2019DIK
				BP	423.4		VLE Extrap	1952SCA
		sol		MP	273.2		MP	1954GIG
				MP	272.3		MP	1928CUT
				MP	271.5		MP	1920MAA
				TP	272.74	0.05	review	2019DIK
Nitrogen/Hydrogen								
N ₂	nitrogen	gas	SEL	BP	77.355	0.007	EOS	2000SPA
				BP	77.355	0.007	EOS-rev	2018LEM
				BP	77.355	0.015	VP	1997NOW

			BP	77.356	0.004	EOS	1986JAC	
			BP	77.355	0.004	VP	1973WAG	
			BP	77.359	0.015	VP	1970WEB	
			BP	77.338	0.006	VP	1966MOU	
			BP	77.364	0.01	VP	1961STR	
			BP	77.361	0.008	VP	1954ARM	
H3N	ammonia	gas	SEL	BP	239.823	0.020	EOS	2018GAO
			BP	239.823	0.020	EOS-rev	2018LEM	
			BP	239.8	0.3	VP rev	1947STU	
H4N2	hydrazine	liq	SEL	BP	386.7	0.9	BP	1989TIP
			BP	387.227	0.015	BP rev	2019DIK	
			BP	386.8	1.2	BP rev	2012KRO	
			BP	387.3		BP	1934HIE	
HN3	hydrogen azide	liq	SEL	BP	308.9	1.5	BP rev	2007LID
Nitrogen/Oxygen								
N2O	nitrous oxide	gas	SEL	BP	184.75	0.07	cal	1974ATA
			BP	184.68	0.07	EOS	2006LEM	
			BP	184.68	0.07	EOS-rev	2018LEM	
			BP	184.74	0.02	VP	1945HOG	
ClH4NO4	ammonium perchlorate	sol	SEL	dec	595.9	4.0	DTA	2004LIU
			SEL	TP	513.2	1.0	TP rev	2019DIK
Sulfur/Hydrogen/Oxygen								
H2S	hydrogen sulfide	gas	SEL	BP	212.85	0.10	EOS	2006LEM
			BP	212.85	0.10	EOS-rev	2018LEM	
			BP	212.86	0.26	BP rev	2012KRO	
			BP	218.9			2004SAK	
			BP	212.87	0.10	EOS	1983GOO	
			BP	212.60	0.18	VP rev	1947STU	
O2S	sulfur dioxide	gas	SEL	BP	263.14	0.09	EOS	2016GAO
			BP	263.14	0.09	EOS-rev	2018LEM	
Boron/Hydrogen								
B2H6	diborane	gas	SEL	BP	180.6	0.1	VP	1959PAR
			BP	180.80	0.19	BP rev	2019DIK	
			BP	180.80	0.19	VLE	2012KRO	
			BP	180.7		VP	1961DIT	
			BP	180.6			1956WIR	
			BP	180.7			1956LAN	
Hydrogen Halides								

ClH	hydrogen chloride	gas		BP	188.173	0.088	EOS	2018THO
				BP	188.173	0.088	EOS-rev	2018LEM
				BP	188.29	0.1	VP	1986HEN
				BP	188.19	0.1	VP	1981NUN
HYDROCARBONS								
Carbon								
C	graphite	sol	SEL	MP	4800	100	MP rev	2005SAV
				MP	4800	100	MP	2004BAS
				MP	4800	200	MP	2003KOR
				MP	4660		MP	2001KER
				MP	4750	150	MP	2000BAS
				MP	4800	150	MP	1998MUS
				MP	4900	200	MP	1993POT
				MP	4330	50	MP	1990CEZ
				MP	4700		MP	1988SHE
Alkanes								
CH ₄	methane	gas	SEL	BP	111.667	0.013	EOS	1991SET
				BP	111.667	0.013	EOS-rev	2018LEM
				BP	111.7			2016HUR
				BP	111.67	0.22	BP rev	2012KRO
				BP	111.65	0.02		1976PAV
				BP	111.70	0.08	VP	1972PRY
C ₂ H ₆	ethane	gas	SEL	BP	184.566	0.004	EOS	2006BUC
				BP	184.566	0.004	EOS-rev	2018LEM
				BP	184.6		BP comp	2016HUR
				BP	184.57	0.17	BP rev	2012KRO
				BP	184.6	0.2		1971STR
				BP	184.5	0.1		1964ZIE
				BP	184.10	0.15		1937WIT
				BP	184.9	0.5	VP	1928COF
				BP	184.48	0.10	VP	1926LOO
C ₃ H ₈	propane	gas	SEL	BP	231.039	0.009	EOS	2009LEM
				BP	231.039	0.009	EOS-rev	2018LEM
				BP	231.1		BP comp	2016HUR
				BP	231.036	0.066	BP rev	2012KRO
				BP	231.07	0.08		1977GOO
				BP	231.1			1971STR
				BP	231.1		review	1948ROS

C ₄ H ₁₀	isobutane	gas	SEL	BP	231.02	0.15	VP	1938KEM
				BP	228.7	0.5	VP	1928COF
				BP	261.40	0.26	EOS	2006BUC2
				BP	261.40	0.26	EOS-rev	2018LEM
				BP	261.4		BP comp	2016HUR
				BP	261.4	0.4	BP rev	2012KRO
				BP	261.13	0.15	VLE	1976STE
				BP	261.0	0.3	VLE	1973DAS
				BP	261.4	0.2	VLE	1967GIL
				BP	261.41	0.15	VP	1940AST1
	<i>n</i> -butane	gas	SEL	BP	263.0	0.5	VP	1928COF
				BP	272.66	0.12	EOS	2006BUC2
				BP	272.66	0.12	EOS-rev	2018LEM
				BP	272.7		BP comp	2016HUR
				BP	272.66	0.17	BP rev	2012KRO
				BP	272.64	0.08	VP	1982HAY
				BP	272.3	0.3	VLE	1973DAS2
				BP	272.7		VLE	1967GIL
				BP	272.77	0.15	VP	1940AST2
				BP	272.7	0.5	VP	1928COF
C ₅ H ₁₂	isopentane	liq	SEL	BP	300.99	0.07	VP	1945WIL
				BP	301.0	0.7	EOS-rev	2018LEM
				BP	300.98	0.15	BP rev	2012KRO
				BP	301.0	0.7	EOS	2006LEM
				BP	301.0	0.2	GLC	1957DES
				BP	301.00	0.18	VP rev	1947STU
				BP	301.05	0.15	BP distill	1947FEN
	neopentane	gas	SEL	BP	282.650	0.028	EOS	2006LEM
				BP	282.665	0.019	BP rev	2019DIK
				BP	282.650	0.028	EOS-rev	2018LEM
				BP	282.655	0.093	BP rev	2012KRO
				BP	282.65	0.03	EOS	2006LEM
				BP	282.64	0.04	VP	1975HOP
				BP	282.65	0.06	VP ebul	1974OSB
				BP	282.7	0.2	GLC	1957DES
	<i>n</i> -pentane	liq	SEL	BP	309.20	0.07	VP	1945WIL
				BP	309.21	0.12	EOS	2022THO
				BP	309.3		VLE	2021TSU

C ₆ H ₁₄	2,3-dimethylbutane	liq	SEL	BP	309.21	0.12	EOS-rev	2018LEM
				BP	309.2		BP comp	2016HUR
				BP	309.21	0.21	BP rev	2012KRO
				BP	309.20	0.06	VP ebul	1974OSB
				BP	309.2	0.2	GLC	1957DES
				BP	309.20	0.15	BP rev	1947FEN
	3-methylpentane	liq	SEL	BP	331.11	0.07	VP	1945WIL
				BP	331.12	0.09	BP rev	2019DIK
				BP	331.18	0.06	EOS-rev	2018LEM
				BP	331.2		VLE	1992HIA
				BP	331.3		VLE	1970KIR
				BP	331.0		VLE	1967MCC
				BP	331.4			1948RUS
				BP	331.10	0.15	BP distill	1947FEN
				BP	331.2		VLE	1946KAY
	isohehexane	liq	SEL	BP	336.41	0.07	VP	1945WIL
				BP	336.40	0.06	BP rev	2019DIK
				BP	336.402	0.055	BP rev	2012KRO
				BP	336.4	0.2	GLC	1957DES
				BP	336.30	0.15	BP distill	1947FEN
	neohexane	liq	SEL	BP	336.5	0.4	BP rev	1935LES
				BP	333.40	0.07	VP	1945WIL
				BP	333.39	0.03	BP rev	2019DIK
				BP	333.36	0.06	EOS-rev	2018LEM
				BP	333.36	0.21	BP rev	2012KRO
				BP	333.36	0.06	EOS	2006LEM
	<i>n</i> -hexane	liq	SEL	BP	334.4	0.3	VLE	1999LOR
				BP	333.40	0.15	BP distill	1947FEN
				BP	322.87	0.07	VP	1945WIL
				BP	322.87	0.05	BP rev	2019DIK
				BP	322.85	0.13	EOS-rev	2018LEM
				BP	332.9	0.2	GLC	1957DES
				BP	322.80	0.15	BP distill	1947FEN
				BP	341.86	0.05	VP	1945WIL
				BP	341.87	0.13	EOS	2022THO2
				BP	341.87	0.13	EOS-rev	2018LEM
				BP	341.9		BP comp	2016HUR
				BP	341.87	0.15	BP rev	2012KRO

C ₇ H ₁₆	2,2,3-trimethylbutane	liq	SEL	BP	342.0	0.1	VLE	1998FEN
				BP	341.85	0.15	BP distill	1947FEN
				BP	341.9		BP rev	1935LES
				BP	354.01	0.10	VP	1949FOR
				BP	353.99	0.09	BP rev	2019DIK
				BP	354.0	0.2	GLC	1957DES
				BP	354.2		VLE	1949DEN
				BP	353.99	0.15	BP distill	1947FEN
				BP	354.0			1943MCA
				BP	353.9		BP rev	1922TIM
	2,3-dimethylpentane	liq	SEL	BP	362.91	0.10	VP	1949FOR
				BP	362.9	0.4	BP rev	2019DIK
				BP	363.0		VLE	1999MON
				BP	363.3		VLE	1960KYL
				BP	362.9	0.5	VLE	1974OSB
				BP	363.0	0.2	GLC	1957DES
				BP	362.87	0.15	BP distill	1947FEN
	2,4-dimethylpentane	liq	SEL	BP	353.7	0.3	VLE	1976TRI
				BP	353.6	0.3	BP rev	2019DIK
	3,3-dimethylpentane	liq	SEL	BP	359.19	0.05	VP	1945WIL
				BP	359.19	0.08	BP rev	2019DIK
				BP	359.21	0.10	VP	1949FOR
				BP	359.25	0.15	BP distill	1947FEN
	3-methylhexane	liq	SEL	BP	364.97	0.10	VP	1949FOR
				BP	364.9	0.4	BP rev	2019DIK
				BP	365.00	0.15	BP rev	1947FEN
	isoheptane	liq	SEL	BP	363.18	0.10	VP	1949FOR
				BP	363.1	0.4	BP rev	2019DIK
				BP	363.20	0.15	BP distill	1947FEN
	neoheptane	liq	SEL	BP	352.33	0.07	VP	1945WIL
				BP	352.33	0.06	BP rev	2019DIK
				BP	352.25	0.15	BP distill	1947FEN
				BP	352.1		BP rev	1935LES
	<i>n</i> -heptane	liq	SEL	BP	371.55	0.07	VP	1945WIL
				BP	371.55	0.07	EOS-rev	2018LEM
				BP	371.6		BP comp	2016HUR
				BP	371.53	0.29	BP rev	2012KRO
				BP	371.5	0.1	VLE	1995WIS

C ₈ H ₁₈	2,2,4-trimethylpentane	liq	SEL	BP	371.51	0.07	BP distill BP rev	1954DOU
				BP	371.55	0.15		1947FEN
				BP	371.6			1935LES
				BP	372.36	0.07		1945WIL
				BP	372.36	0.14		2018LEM
				BP	390.9			2016HUR
	2,2-dimethylhexane	liq	SEL	BP	372.4	0.1	BP comp BP rev	2012KRO
				BP	372.4	0.2		1995HIA2
				BP	379.97	0.07	VLE BP rev	1945WIL
				BP	379.967	0.014		2019DIK
				BP	379.99	0.15	GLC	1957DES
				BP	380.2			1951AUD
				BP	379.7		VP rev BP distill BP	1948BUC
				BP	379.77	0.18		1947STU
				BP	380.05	0.15		1947FEN
				BP	379.97	0.10		1946FOR
				BP	380.3			1945HER
	2-methylheptane	liq		BP	390.77	0.07	VP BP rev	1945WIL
				BP	390.78	0.09		2019DIK
				BP	390.8	0.2	BP rev BP distill	1957DES
				BP	390.90	0.15		1947FEN
	3-methylheptane	liq		BP	390.4		BP rev VP	1935LES
				BP	392.05	0.07		1945WIL
				BP	392.04	0.08	BP rev VP	2019DIK
	<i>n</i> -octane	liq	SEL	BP	398.79	0.07		1945WIL
				BP	398.79	0.14	EOS-rev BP comp	2018LEM
				BP	398.8			2016HUR
				BP	398.9		VLE BP rev	2013FER
				BP	398.77	0.28		2012KRO
				BP	399.2		VLE VLE	2007RED
				BP	399.2			2006SUR
				BP	398.8		VLE VLE	2000HIA
				BP	398.6			1999WIS
				BP	398.8	0.2	VLE VLE	1995HIA2
				BP	398.8			1995HIA
				BP	398.7		VLE VLE	1986PAU
				BP	398.8			1982RED
				BP	398.75	0.20	VLE	1976RAD

C ₉ H ₂₀	isononane	liq	SEL	BP	398.75	0.15	BP distill	1947FEN
				BP	398.6		BP rev	1935LES
				BP	416.5	0.3	VP	1948ROS
				BP	416.4	1.3	BP rev	2019DIK
				BP	416.4	0.2	BP rev	1950MEA
				BP	416.41	0.15	BP distill	1947FEN
				BP	416.0		BP rev	1935LES
				BP	403.63	0.15	BP rev	1947FEN
				BP	405.6	2.2	BP rev	2019DIK
				BP	423.93	0.07	VP	1945WIL
	neononane	liq	SEL	BP	423.91	0.15	EOS-rev	2018LEM
				BP	424.0		BP comp	2016HUR
				BP	423.91	0.17	BP rev	2012KRO
				BP	423.91	0.15	EOS	2006LEM
				BP	423.82	0.25		1982RED
				BP	423.95	0.10	VP	1949FOR
				BP	447.27	0.07	VP	1945WIL
				BP	447.27	0.16	EOS	2006LEM
				BP	447.27	0.16	EOS-rev	2018LEM
				BP	447.27	0.14	BP rev	2012KRO
C ₁₀ H ₂₂	<i>n</i> -decane	liq	SEL	BP	447.3	0.1	VP	1988GIE
				BP	447.3	0.2	GLC	1957DES
				BP	447.4	0.3		1955HAN
				BP	447.2	0.2	BP rev	1947FEN
				BP	469.04	0.05	VP	1955CAM
				BP	469.04	0.19	BP rev	2019DIK
				BP	469.26	0.10	VP	1996VIT
				BP	469.05	0.15	BP rev	1947FEN
				BP	489.48	0.05	VLE	1996DEJ
				BP	489.4	0.3	BP rev	2019DIK
C ₁₁ H ₂₄	<i>n</i> -undecane	liq	SEL	BP	489.44	0.08	EOS-rev	2018LEM
				BP	489.44	0.08	EOS	2004LEM
				BP	489.41	0.15	BP rev	1947FEN
				BP	489.44	0.15	VP	1945WIL
				BP	263.6		EOS-rev	2018LEM
				BP	559.9	0.2	VP	1954CAM
				BP	469.05	0.15	BP rev	1947FEN
				BP	469.26	0.10	VP	1996VIT
				BP	469.04	0.19	BP rev	2019DIK
				BP	469.04	0.05	VP	1955CAM
C ₁₂ H ₂₆	<i>n</i> -dodecane	liq	SEL	BP	469.05	0.15	BP rev	1947FEN
				BP	489.48	0.05	VLE	1996DEJ
				BP	489.4	0.3	BP rev	2019DIK
				BP	489.44	0.08	EOS-rev	2018LEM
				BP	489.44	0.08	EOS	2004LEM
				BP	489.41	0.15	BP rev	1947FEN
				BP	489.44	0.15	VP	1945WIL
				BP	263.6		EOS-rev	2018LEM
				BP	559.9	0.2	VP	1954CAM
				BP	469.05	0.15	BP rev	1947FEN
C ₁₆ H ₃₄	<i>n</i> -hexadecane	liq	SEL	BP	469.05	0.15	BP rev	1947FEN
				BP	489.48	0.05	VLE	1996DEJ
				BP	489.4	0.3	BP rev	2019DIK
				BP	489.44	0.08	EOS-rev	2018LEM
				BP	489.44	0.08	EOS	2004LEM
				BP	489.41	0.15	BP rev	1947FEN
				BP	489.44	0.15	VP	1945WIL
				BP	263.6		EOS-rev	2018LEM
				BP	559.9	0.2	VP	1954CAM
				BP	469.05	0.15	BP rev	1947FEN

		sol	SEL1	MP	291.27	0.05	MP	1986SNO
			SEL2	TP	291.3			2018LEM
		liq		BP	560.1	1.5	BP rev	2019DIK
				BP	560.5	0.6	VP	2014SII
				BP	560.2			1989CRC
				BP	559	2		1950MUM
				BP	559.2		BP rev	1949BOO
Alkenes								
C ₂ H ₄	ethene	gas	SEL	BP	169.38	0.02	EOS	2000SMU
				BP	169.38	0.02	EOS-rev	2018LEM
				BP	169.4		BP comp	2016HUR
				BP	169.38	0.12	BP rev	2012KRO
				BP	169.35	0.02		1986JAH
				BP	169.37	0.03	VP	1950MIC
				BP	169.9	0.2	VP	1940LAM
				BP	169.3	0.5	VP	1928COF
C ₃ H ₆	propene	gas	SEL	BP	225.53	0.04	EOS-rev	2018LEM
				BP	225.5		BP comp	2016HUR
				BP	225.5	0.4	BP rev	2012KRO
				BP	225.5	0.4		1946MOR
				BP	225.9	0.2	VP	1940LAM
				BP	225.38	0.15	VP	1939POW
				BP	226.2	0.5	VP	1928COF
C ₄ H ₈	1-butene	gas	SEL	BP	266.87	0.07	VP	1940LAM
				BP	266.82	0.29	BP rev	2019DIK
				BP	266.84	0.06	EOS-rev	2018LEM
				BP	267.0		BP comp	2016HUR
				BP	266.84	0.24	BP rev	2012KRO
				BP	266.9	0.15	VLE	1976STE
				BP	266.9	0.2		1949FOE
				BP	266.9		BP	1946MOR
				BP	267.24	0.11	VP	1940LAM
				BP	266.6		BP	1930LUC
				BP	267.4	0.5	VP	1928COF
	2-butene, (E)-	gas	SEL	BP	274.02	0.05	VP	1945GUT
				BP	274.03	0.11	BP rev	2019DIK
				BP	274.03	0.08	EOS-rev	2018LEM
				BP	274.03	0.09	BP rev	2012KRO

				BP	274.03	0.08	EOS	2005LEM
				BP	274.44	0.07	VP	1940LAM
				BP	274.11	0.15		1937CAR
				BP	274.2	0.5	VP	1928COF
2-butene, (Z)-	gas	SEL	BP	276.87	0.08	VP	1944SCO	
			BP	276.83	0.10	BP rev	2019DIK	
			BP	276.87	0.05	EOS-rev	2018LEM	
			BP	276.87	0.08	BP rev	2012KRO	
			BP	276.87	0.05	EOS	2005LEM	
			BP	276.9	0.5		1949FOE	
			BP	277.07	0.14	VP	1940LAM	
			isobutene	gas	SEL	BP	266.04	0.15
BP	266.17	0.29				BP rev	2019DIK	
BP	266.15	0.13				EOS-rev	2018LEM	
BP	266.2	0.2				BP rev	2012KRO	
BP	266.15	0.13				EOS	2005LEM	
BP	266.4	0.5					1949FOE	
BP	266.6	0.5				VP	1928COF	
C ₅ H ₁₀	1-pentene	liq				SEL	BP	303.11
			BP	303.10	0.07		BP rev	2019DIK
			BP	303.10	0.01		EOS-rev	2018LEM
			BP	303.2			BP comp	2016HUR
			BP	303.11	0.07		BP rev	2012KRO
			BP	303.22	0.04			1951DAY
			BP	303.12	0.06		VP	1950FOR
			BP	303.15	0.15		BP rev	1947FEN
2-methyl-1-butene	liq	SEL	BP	304.29	0.13	VP	1949SCO	
			BP	304.25	0.21	BP rev	2019DIK	
			BP	304.3	0.2	BP rev	2012KRO	
			BP	304.3	0.2	GLC	1957DES	
			BP	304.20	0.15	BP rev	1947FEN	
2-methyl-2-butene	liq	SEL	BP	311.70	0.15	VP ebul	1949SCO	
			BP	311.70	0.28	BP rev	2019DIK	
			BP	311.7	0.2	GLC	1957DES	
			BP	311.9	0.2	VP	1940LAM	
2-pentene, (E)-	liq	SEL	BP	309.48	0.08	VP	1950SCO2	
			BP	309.48	0.03	BP rev	2019DIK	
			BP	309.476	0.036	BP rev	2012KRO	

				BP	309.0				1948THO
				BP	309.5			VLE	1946GER
				BP	309.0			VLE	1941EWE
				BP	309.4				1937CAR
2-pentene, (Z)-	liq	SEL	BP		310.07	0.08		VP ebul	1950SCO2
					310.07	0.03		BP rev	2019DIK
					310.076	0.035		BP rev	2012KRO
					310.0				1951HOF
					310.2				1948THO
					309.7				1937CAR
3-methyl-1-butene	gas	SEL	BP		293.20	0.08		VP ebul	1950SCO2
					293.21	0.04		BP rev	2019DIK
					293.201	0.034		BP rev	2012KRO
					293.2	0.5			1946GER
C ₆ H ₁₂ 1-hexene	liq	SEL	BP		336.64	0.06		VP	1950FOR
					336.60	0.08		BP rev	2019DIK
					336.7			BP comp	2016HUR
					336.58	0.08		BP rev	2012KRO
					336.7	0.2		VLE	1966SUR
					336.6	0.2		GLC	1957DES
C ₇ H ₁₄ 1-heptene	liq	SEL	BP		366.71	0.06		VP	1950FOR
					366.77	0.09		BP rev	2019DIK
					366.8			BP comp	2016HUR
					366.76	0.07		BP rev	2012KRO
					366.7	0.8			1965KAA
					366.80	0.15		BP rev	1947FEN
C ₈ H ₁₆ 1-octene	liq	SEL	BP		363.4				1935MUL
					394.43	0.06		VP	1950FOR
					394.42	0.10		BP rev	2019DIK
					394.5			BP comp	2016HUR
					394.4	0.1		BP rev	2012KRO
					394.4	0.2		GLC	1957DES
C ₉ H ₁₈ 1-nonene	liq	SEL	BP		394.40	0.15		BP rev	1947FEN
					420.02	0.06		VP	1950FOR
					420.01	0.06		BP rev	2019DIK
					418.1				1957LOE
C ₁₀ H ₂₀ 1-decene	liq	SEL	BP		418.5				1931WIL
					443.72	0.06		VP	1950FOR

				BP	443.72	0.09	BP rev	2019DIK
				BP	443.8			1950MEA
				BP	443.8			1935MUL
Alkadienes								
C ₃ H ₄	allene	gas	SEL	BP	238.3	0.5	VP rev	1947STU
				BP	238.2	0.6	BP rev	2019DIK
				BP	240.9	0.3	EOS-rev	2018LEM
				BP	238.6		BP comp	2016HUR
				BP	239.0	0.5	BP rev	2012KRO
				BP	237.7			1956BLO
				BP	238.7	0.4		1954POM
				BP	240.0	0.2	VP	1940LAM
				BP	238.15	0.15	VP	1930LIV
				BP	241.2			1905LES
C ₄ H ₆	1,2-butadiene	gas	SEL	BP	284.1	0.4		1946STR
				BP	284.00	0.17	BP rev	2019DIK
				BP	284.0		BP comp	2016HUR
				BP	284.1	0.4	eval	2012KRO
				BP	284.0		eval	1985MAJ
				BP	291.9	1.0	VP rev	1947STU
				BP	283.6			1947AST
				BP	283.5			1931HUR
	1,3-butadiene	gas	SEL	BP	268.44	0.15	VLE	1976STE
				BP	268.59	0.19	BP rev	2019DIK
				BP	268.8		BP comp	2016HUR
				BP	268.60	0.17	BP rev	2012KRO
				BP	268.7	0.4		1949FOE
				BP	268.62	0.15	VP	1933HEI
C ₅ H ₈	1,2-pentadiene	liq	SEL	BP	318.01	0.06	VP	1950FOR
				BP	318	2		2021FIS
				BP	317.99	0.03	BP rev	2019DIK
				BP	318.1		BP comp	2016HUR
				BP	317.6	2.1	BP rev	2012KRO
				BP	318.80	0.06	VP ebul	1969OSB
	1,3-pentadiene	liq	SEL	BP	315.3	0.5	BP distill	1935MUL
				BP	315.4	2.0	BP rev	2019DIK
				BP	315.3	1.7	BP rev	2012KRO
				BP	314.9	1.5	BP distill	1956BUS

1,4-pentadiene			liq	SEL	BP	315.4	0.5	VP rev	1947STU
					BP	299.12	0.15	VP	1950FOR
					BP	299.10	0.03	BP rev	2019DIK
					BP	299.10	0.04	BP rev	2012KRO
					BP	298.93	0.06	VP extrap	1969OSB
					BP	299.3	0.6		1949ELS
					BP	298.90	0.12	VP	1940LAM
2-methyl-1,3-butadiene			liq	SEL	BP	307.19	0.15	VLE	1985HOW
					BP	307.20	0.06	BP rev	2019DIK
					BP	307.20	0.06	BP rev	2012KRO
					BP	307.26	0.08	VP	1964GUB
					BP	307.30	0.15		1963LEV
					BP	307.0	0.6		1952GIL
					BP	307.2	0.5		1949FOE
Alkynes									
C ₂ H ₂	acetylene	gas	SEL	BP	188.44	0.10	VP extrap	1956AMB	
				BP	188.26	0.15	EOS-rev	2018LEM	
	sol	SEL1	TP	191.75	0.15	EOS-rev	2018LEM		
			BP	188.8	0.5	VP extrap	2019DIK		
	gas			BP	188.5		sub rev	2013HAY	
				BP	189.0		BP rev	1982BUC	
				BP	189.1		sub VP	1952ROB	
				BP	189.1	0.5	sub VP	1950CLA	
				BP	189.6		sub VP	1921MAA	
				BP	189.1		sub VP	1915BUR	
				TP	192.0		sub rev	2019DIK	
	C ₃ H ₄	propyne	gas	SEL	BP	249.81	0.15	VP	1967VAN
BP					249.92	0.16	BP rev	2019DIK	
BP					250		EOS-rev	2018LEM	
BP					249.9		BP comp	2016HUR	
BP					248	15	BP rev	2012KRO	
BP					250		EOS	1992POL	
BP					249.9	1.0		1950RIC	
BP					245.4		VP	1921MAA	
BP					249.7			1905LES	
sol		SEL1	TP	170.5		EOS-rev	2018LEM		
			TP	170.5		EOS	1992POL		
C ₄ H ₆	1-butyne	gas	SEL	BP	281.22	0.10	VP	1950AST	

			BP	281.21	0.06	BP rev	2019DIK
			BP	281.23	0.05	EOS-rev	2018LEM
			BP	281.22	0.14	BP rev	2012KRO
			BP	281.9		VLE	1952ROB
			BP	281.3	0.3		1947BOO
			BP	281.3		VP rev	1947API
			BP	281.3			1946STR
2-butyne	liq	SEL	BP	300.1	0.2		1954POM2
			BP	300.106	0.012	BP rev	2019DIK
			BP	300.9	0.7	BP rev	2012KRO
			BP	300.1		BP	1945HEN
			BP	299.8			1939CON
C ₅ H ₈ 1-pentyne	liq	SEL	BP	313.5	0.3	VLE	2021TSU
			BP	313.0	1.3	BP rev	2019DIK
			BP	313	1	BP rev	2012KRO
			BP	313.4			1954POM2
			BP	313.4		BP	1945HEN
2-pentyne	liq	SEL	BP	329.3	0.2		1954POM2
			BP	328.9	1.7	BP rev	2019DIK
			BP	328.9	1.3	BP rev	2012KRO
			BP	329.2			1954MIL
			BP	329.2		BP	1945HEN
C ₆ H ₁₀ 1-hexyne	liq	SEL	BP	344.5	0.2	VLE	2021TSU
			BP	344.32	0.16	BP rev	2019DIK
			BP	344.28	0.15	BP rev	2012KRO
			BP	344.5			1954POM2
			BP	344.6		BP	1945HEN
2-hexyne	liq	SEL	BP	357.6	0.2	VLE	2021TSU
			BP	357.25	0.17	BP rev	2019DIK
			BP	357.5	0.2	BP rev	2012KRO
			BP	357.7			1954POM2
3-hexyne	liq	SEL	BP	354.6	0.3		1954POM2
			BP	354.07	0.21	BP rev	2019DIK
			BP	354.7	0.5	BP rev	2012KRO
			BP	356.3	0.5	VP	1965RON
			BP	354.4			1955LEM
C ₇ H ₁₂ 1-heptyne	liq	SEL	BP	354.4			1954MIL
			BP	372.9	0.3	BP	1945HEN

				BP	372.93	0.16	BP rev	2019DIK
				BP	372.7		VLE	1948HEN
C ₈ H ₁₄	1-octyne	liq	SEL	BP	399.4	0.3	BP	1945HEN
				BP	399.38	0.06	BP rev	2019DIK
Alkenynes								
C ₄ H ₄	butenyne	gas	SEL	BP	276.2	2.0		1954HEN
				BP	279.2	1.3	BP rev	2019DIK
				BP	278.3		BP comp	2016HUR
				BP	279.2	1.0	BP rev	2012KRO
				BP	278.5	0.5	VP rev	1947STU
C ₅ H ₆	1-penten-3-yne	liq	SEL	BP	332	4		1933JAC
				BP	332	4		2019DIK
	3-penten-1-yne	liq	SEL	BP	317.1	3.0		1954POM2
				BP	319	5	BP est	2019DIK
				BP	319	4	BP rev	2012KRO
				BP	320.2			1957PET
Alkadiynes								
C ₄ H ₂	butadiyne	gas	SEL	BP	283.7	1.0		1954POM2
				BP	282.9	1.0	BP rev	2019DIK
				BP	283.5		BP comp	2016HUR
				BP	284.1	2.8	BP rev	2012KRO
				BP	282.5	0.5	VP rev	1947STU
Cycloalkanes								
C ₃ H ₆	cyclopropane	gas	SEL	BP	240.33	0.12	VP	1946RUE1
C ₃ H ₆	cyclopropane	sol	SEL1	MP	145.55	0.14	heat	1946RUE1
				BP	240.30	0.08	BP rev	2019DIK
				BP	241.7	0.4	EOS-rev	2018LEM
				BP	240.3		BP comp	2016HUR
				BP	242	7	BP rev	2012KRO
				BP	241.7	0.4	EOS	1992POL
				BP	239.8	1.0	VP extrap	1962HAS
				BP	241.7	1.5	VP	1955GRA
				BP	239.7		VP	1947STU
				BP	240.5	0.5		1939GRO
				BP	240.3			1936HAS
				BP	240.1		VP	1934KOH
C ₄ H ₈	cyclobutane	gas	SEL	BP	285.6	0.2	VP	1953RAT
				BP	285.7	0.4	BP rev	2019DIK

C ₅ H ₁₀	cyclopentane	liq	SEL	BP	286.1		BP comp	2016HUR
				BP	285.7	0.4	BP rev	2012KRO
				BP	285.3			1953RAT
				BP	285.7			1952KAA
				BP	285.7	0.4		1949ROB
				BP	284.5			1949CAS
				BP	322.39	0.07	VP	1945WIL
				BP	322.40	0.06	EOS-rev	2018LEM
				BP	322.5		BP comp	2016HUR
			SEL	BP	322.40	0.06	EOS	2015GED
				BP	322.40	0.09	BP rev	2012KRO
				BP	322.4		VLE	1978SEV
				BP	322.4	0.2	GLC	1957DES
				BP	322.5		VLE	1956MYE
				BP	322.40	0.15		1949FOE
				BP	322.5		VLE	1949DEN
				BP	322.35	0.15		1947FEN
				BP	322.39	0.07	VP	1945WIL
	methylcyclobutane	liq	SEL	BP	310.3	1	BP distil	1953PIN2
				BP	310	4	BP rev	2019DIK
				BP	310	3	BP rev	2012KRO
				BP	309.5			1937GRO
C ₆ H ₁₂	cyclohexane	liq	SEL	BP	353.86	0.07	VP	1945WIL
				BP	353.865	0.017	EOS-rev	2018LEM
				BP	353.9		BP comp	2016HUR
				BP	353.87	0.03	EOS	2014ZHO
				BP	353.89	0.27	BP rev	2012KRO
				BP	353.90	0.15	VLE	1998FEN
				BP	353.8		VLE	1995WIS
				BP	354.0		VLE	1994TOJ
				BP	353.7		VLE	1989RAM
				BP	353.9		VLE	1989MAT
				BP	354.04	0.15	VP	1974KER
				BP	363.82	0.03	VLE	1973MAT
				BP	353.90	0.15	VLE	1965MAR
				BP	353.9	0.2	GLC	1957DES
				BP	353.85	0.15	BP rev	1947FEN
	methylcyclopentane	liq	SEL	BP	344.94	0.07	VP	1945WIL

C ₇ H ₁₄	1,1-dimethylcyclopentane	liq	SEL	BP	344.94	0.06	BP rev	2019DIK
				BP	344.937	0.061	BP rev	2012KRO
				BP	344.75	0.04	VP rev	1979KUD
				BP	344.97	0.15	VLE	1970WEA
				BP	345.0		VLE	1962KAE
				BP	345.0		VLE	1961BEL
				BP	345.0	0.2	GLC	1957DES
				BP	345.00	0.15		1949FOE
				BP	345.0		VLE	1949DEN
				BP	344.95	0.15	BP rev	1947FEN
				BP	344.8		BP	1945CRA
				BP	345.0		BP rev	1935LES
				BP	360.97	0.10	VP	1949FOR
				BP	361.0	0.3	BP rev	2019DIK
	cycloheptane	liq	SEL	BP	361.1		VLE	1949DEN
				BP	360.65	0.15	BP rev	1947FEN
				BP	391.92	0.2	VP	1956FIN
				BP	391.94	0.08	BP rev	2019DIK
				BP	391.7		BP rev	1989WEA
				BP	393.0			1986FIS
				BP	391.5			1976MEY
				BP	392.6			1965MUE
				BP	392.1	0.2	GLC	1957DES
				BP	392.0		VP rev	1954API
				BP	391.6			1952KAA
				BP	391.3		BP rev	1935LES
	ethylcyclopentane	liq	SEL	BP	376.59	0.10	VP	1949FOR
				BP	376.6	0.3	BP rev	2019DIK
				BP	376.60	0.15	BP rev	1947FEN
				BP	376.9		BP	1945CRA
	methylcyclohexane	liq	SEL	BP	374.06	0.07	VP	1945WIL
				BP	374.04	0.09	BP rev	2019DIK
				BP	374.01	0.11	EOS-rev	2018LEM
				BP	374.01	0.11	BP rev	2012KRO
				BP	374.12	0.15	VLE	1991AIZ
				BP	374.1		VLE	1977WIS
				BP	373.9			1974RAO
				BP	373.9		VLE	1973WAD

C ₈ H ₁₆		liq	SEL	BP	374.1		VLE	1973NAG
				BP	374.1		VLE	1973CHI
				BP	374.0		VLE	1969RAJ
				BP	374.1	0.2	GLC	1957DES
				BP	374.0		BP rev	1935LES
	ethylcyclohexane	liq	SEL	BP	404.91	0.07	VP	1945WIL
				BP	404.92	0.03	BP rev	2019DIK
				BP	405.0	0.15	BP rev	1935LES
	isopropylcyclopentane	liq	SEL	BP	399.55	0.06	VP	1949FOR
				BP	399.544	0.022	BP rev	2019DIK
				BP	399.70	0.15		1949FOE
				BP	399.55	0.15	BP rev	1947FEN
				BP	399.6		BP	1945CRA
	n-propylcyclopentane	liq	SEL	BP	404.08	0.06	VP	1949FOR
				BP	404.07	0.07	BP rev	2019DIK
				BP	404.1	0.2	GLC	1957DES
				BP	404.1			1956EPS
				BP	404.20	0.15		1949FOE
				BP	404.1		BP rev	1949BOO2
				BP	404.15	0.15	BP rev	1947FEN
				BP	404.0		BP	1945CRA
C ₉ H ₁₈	isopropylcyclohexane	liq	SEL	BP	427.70	0.10	VP	1949FOR
				BP	427.62	0.20	BP rev	2019DIK
				BP	427.8		BP rev	1945SMI
				BP	427.7		VLE	1938SMI
	propylcyclohexane	liq	SEL	BP	429.86	0.10	VP	1949FOR
				BP	429.85	0.08	BP rev	2019DIK
				BP	429.7		VLE	1950CHI
				BP	429.8		BP rev	1946BOO
				BP	429.9		BP rev	1945SMI
	1,3-diethylcyclohexane	liq	SEL	BP	445.7	1.5		1962ALL
				BP	447.1	2.4	BP rev	2019DIK
				BP	447.2		BP	1933SIG
C ₁₀ H ₂₀	isobutylcyclohexane	liq	SEL	BP	444.46	0.07	VP	1949FOR
				BP	444.46	0.13	BP rev	2019DIK
				BP	444.4			1949BOO2
	tert-butylcyclohexane	liq	SEL	BP	444.73	0.10	VP	1949FOR

				BP	444.71	0.16	BP rev	2019DIK
				BP	444.9		BP rev	1945SMI
				BP	444.7		BP rev	1944BOO
Cycloalkenes								
C ₃ H ₄	cyclopropene	gas	SEL	BP	237.6	1.5	BP	1941SCH
				BP	237.7	1.3	BP rev	2019DIK
				BP	237.6	1.3	BP rev	2012KRO
C ₄ H ₆	cyclobutene	gas	SEL	BP	275.6	0.2	VP	1941HEI
				BP	275.70	0.20	BP rev	2019DIK
				BP	275.7	0.5	EOS-rev	2018LEM
				BP	275.2		BP rev	1989WEA
C ₅ H ₈	cyclopentene	liq	SEL	BP	317.37	0.06	VP	1950FOR
				BP	317.4	0.1	VLE	2021TSU
				BP	317.37	0.04	BP rev	2019DIK
				BP	317.37	0.05	BP rev	2012KRO
				BP	317.6			1955TIM
				BP	317.40	0.15		1949FOE
				BP	317.6			1948ROB
				BP	317.2			1938CAR
				BP	317.5		VP	1937MAE
	methylenecyclobutane	liq	SEL	BP	315.33	0.15	BP rev	1947FEN
				BP	315	5	BP est	2019DIK
				BP	315	4	BP rev	2012KRO
				BP	315.0	0.5		1955CRI
				BP	314.9			1949ROB
C ₆ H ₁₀	3-methylcyclopentene	liq	SEL	BP	338	1	BP	1945CRA
				BP	338.3	2.2	BP rev	2019DIK
				BP	338.1			1948ADK
				BP	338.35	0.15	BP rev	1947FEN
	cyclohexene	liq	SEL	BP	356.10	0.04	VP	1973MEY
				BP	356.07	0.08	BP rev	2019DIK
				BP	356.0		BP comp	2016HUR
				BP	353.89	0.27	BP rev	2012KRO
				BP	356.2		VLE	1989MAT
				BP	356.1	0.2	GLC	1957DES
				BP	356.2			1955TIM
				BP	356.13	0.06	VP	1950FOR

C ₇ H ₁₂	1-ethylcyclopentene	liq	SEL	BP	379.5	0.2	VP rev BP rev	1946API
				BP	379.31	0.08		2019DIK
				BP	379.6			1965SAM
				BP	379.6	0.2		1959PIN
				BP	379.7			1958TUR
				BP	380.5			1952VAN
				BP	377			1948ADK
	1-methylcyclohexene	liq	SEL	BP	383.40	0.15	BP rev VP BP rev VLE	1935LES
				BP	383.35	0.12		1960CAM
				BP	383.3			2019DIK
				BP	383.1			1999BEV
				BP	382.7			1965SAM
				BP	382.7			1958TUR
	3-ethylcyclopentene	liq	SEL	BP	371.3		BP BP rev	1933SIG
				BP	371.0	3.3		1945CRA
	cycloheptene	liq	SEL	BP	386.1	0.5	GLC BP rev GLC	2019DIK
				BP	386.4	3.5		1958TUR
				BP	388.1	0.2		2019DIK
				BP	385.6			1957DES
				BP	386.1			1956TRA
				BP	385.6			1954DOE
C ₈ H ₁₄	methylenecyclohexane	liq	SEL	BP	376.55	0.04	VP BP rev BP rev BP BP	1973MEY
				BP	376.553	0.024		2019DIK
				BP	376.55	0.03		2012KRO
				BP	375.7			1958TUR
				BP	375.4			1956COP
				BP	376			1956CHI
				BP	356			1952STE
	1,2-dimethylcyclohexene	liq	SEL	BP	409	2	BP BP est BP	1952GOE
				BP	410	3		1938CHI
				BP	408.9			2019DIK
				BP				1933SIG
				BP				1960CAM
				BP				2019DIK
	1-ethylcyclohexene	liq	SEL	BP	410.09	0.05	VP BP rev BP BP	1965SAM
				BP	410.126	0.026		1962COP
				BP	408.4			1958TUR
				BP	409.9			
				BP	409	1		

				BP	409.4		BP	1933SIG
Cycloalkadienes								
C ₄ H ₄	cyclobutadiene	gas	SEL	BP	286	12	BP est	2019DIK
C ₅ H ₆	1,3-cyclopentadiene	liq	SEL	BP	313.4	0.2	VP	1965HUL
				BP	313.34	0.07	BP rev	2019DIK
				BP	313.6	1.3	BP rev	2012KRO
				BP	314.7	0.5	VLE	1967LES
				BP	314.2	0.5	GLC	1957DES
				BP	313.9			1949MUR
				BP	313.2			1941BRO
				BP	313.3		BP	1936KIS
C ₆ H ₈	1,3-cyclohexadiene	liq	SEL	BP	353.462	0.005	VP	1989STE
				BP	353.475	0.022	BP rev	2019DIK
				BP	353.48	0.05	BP rev	2012KRO
				BP	353.47	0.04	VP	1973MEY
				BP	354.2	0.2	GLC	1957DES
				BP	353			1951BLO
				BP	353.7			1940WHI
	1,4-cyclohexadiene	liq	SEL	BP	362.653	0.005		1989STE
				BP	362.664	0.021	BP rev	2019DIK
				BP	362.67	0.04	BP rev	2012KRO
				BP	362.7		VP rev	1962API
C ₈ H ₁₂	1,5-cyclooctadiene	liq	SEL	BP	421.8	2.0		1985TAS
				BP	424.0		VLE	2021TSU
				BP	418.8	2.7	BP est	2019DIK
Bicycloalkanes								
C ₁₀ H ₁₈	<i>cis</i> -decalin	liq	SEL	BP	468.90	0.10	VP	1955CAM
				BP	468.94	0.14	BP rev	2019DIK
				BP	468.97	0.15	BP rev	2012KRO
				BP	468.70	0.15		1949FOE
				BP	468.9			1949BOO
	<i>trans</i> -decalin	liq	SEL	BP	460.40	0.10	VP	1955CAM
				BP	460.41	0.21	BP rev	2019DIK
				BP	460.41	0.25	BP rev	2012KRO
				BP	462.90	0.15		1949FOE
C ₁₂ H ₂₂	cyclohexylcyclohexane	liq	SEL	BP	511.1	0.3	VP	1980WIE
		sol	SEL1	MP	277.2	0.5	MP	1983O'R
		liq		BP	511.2			2021EPA

	BP	510.89	0.25	BP rev	2019DIK
	BP	509.2		BP comp	2016HUR
	BP	510.92	0.25	BP rev	2012KRO
	BP	512.2	0.3	BP	1963MEA
	BP	512.2	0.4	VLE	1961JES
	BP	511.35	0.15	VP	1955MYE
	BP	512.0	0.3		1951WIS
	BP	510.2		BP	1933SIG

AROMATIC HYDROCARBONS								
Alkylbenzenes								
C ₆ H ₆	benzene	liq	SEL	BP	353.23	0.07	VP	1945WIL
		sol	SEL1	MP	278.67	0.02	MP	1996LIN
		liq		BP	353.22	0.13	EOS-rev	2018LEM
				BP	353.2		BP comp	2016HUR
				BP	353.22	0.13	EOS	2012THO
				BP	353.22	0.08	BP rev	2012KRO
				BP	353.3	0.2	VLE	1991DIC
				BP	353.3	0.6	VP isoten	1971EON
				BP	353.2		VLE	1960KYL
				BP	353.25	0.15	BP rev	1947FEN
				BP	353.4		BP rev	1935LES
		sol		MP	278.64	0.04	SLE	1986SNO
				MP	278.8	0.2	MP	1979MAH
				MP	278.6	0.1	MP	1970STO
		C ₇ H ₈	toluene	liq	SEL	BP	383.75	0.07
sol	SEL1			MP	178.2	0.2	MP	1976RAD
liq				BP	383.75	0.04	EOS-rev	2018LEM
				BP	383.8		BP comp	2016HUR
				BP	383.75	0.04	EOS	2006LEM
				BP	383.8		VLE	1999TRE
				BP	383.8	0.2	VLE	1991DIC
				BP	383.8		VLE	1991AIZ
				BP	383.80	0.15	BP rev	1947FEN
				BP	383.7		BP rev	1935LES
C ₈ H ₁₀	1,2-dimethylbenzene	liq	SEL	BP	417.55	0.07	VP	1945WIL
				BP	417.54	0.04	BP rev	2019DIK
				BP	417.52	0.19	EOS-rev	2018LEM
				BP	417.52	0.19	EOS	2012ZHO
				BP	417.40	0.15	VP extrap	1967AMB
				BP	417.9	0.2		1955TIM
				BP	417.55	0.15	BP rev	1947FEN
				BP	417.6		BP rev	1935LES
	1,3-dimethylbenzene	liq	SEL	BP	412.23	0.07	VP	1945WIL
				BP	412.23	0.06	BP rev	2019DIK
				BP	412.21	0.07	EOS-rev	2018LEM
				BP	412.21	0.07	EOS	2012ZHO

			BP	412.23	0.07	BP rev	2012KRO
			BP	412.2	0.2	VLE	1991DIC
			BP	411.11	0.15	VP extrap	1967AMB
			BP	412.3	0.2	GLC	1957DES
			BP	412.30	0.15	BP rev	1947FEN
			BP	412.4		BP rev	1935LES
			SEL BP	411.48	0.07	VP	1945WIL
			BP	411.47	0.07	EOS-rev	2018LEM
			BP	411.47	0.07	EOS	2012ZHO
			BP	411.467	0.064	BP rev	2012KRO
1,4-dimethylbenzene	liq	SEL	BP	411.4	0.2	VLE	1990WIS
			BP	411.53	0.06	VP ebul	1974OSB
			BP	411.11	0.15	VP extrap	1967AMB
			BP	411.55	0.15	BP rev	1947FEN
			BP	411.6		BP rev	1935LES
			SEL BP	409.32	0.07	VP	1945WIL
			BP	409.32	0.09	BP rev	2019DIK
			BP	409.31	0.11	EOS-rev	2018LEM
			BP	409.3		BP comp	2016HUR
			BP	409.31	0.11	EOS	2012ZHO
ethylbenzene	liq	SEL	BP	409.15	0.20	VLE	1991DIC
			BP	408.96	0.15	VP	1967AMB
			BP	409.40	0.15	BP rev	1947FEN
			BP	409.3		BP rev	1935LES
			SEL BP	449.22	0.15	VP	1967AMB
			BP	449.235	0.093	BP rev	2019DIK
			BP	449.1			1956FIE
			BP	449.3		BP	1954SCH
			BP	449.23	0.10	VP	1949FOR
			BP	449.2			1948REY
C ₉ H ₁₂ 1,2,3-trimethylbenzene	liq	SEL	BP	449.25	0.15	BP rev	1947FEN
			SEL BP	442.49	0.15	VP	1967AMB
			BP	442.52	0.18	BP rev	2019DIK
			BP	442.5		VLE	1979KUR
			BP	442.50	0.10	VP	1949FOR
			BP	442.35	0.15	BP rev	1947FEN
			SEL BP	437.86	0.15	VP	1967AMB
			BP	437.85	0.14	BP rev	2019DIK
1,2,4-trimethylbenzene	liq	SEL					
1,3,5-trimethylbenzene	liq	SEL					

			BP	437.9				1951CHI
			BP	437.87	0.10	VP		1949FOR
			BP	437.85	0.15	BP rev		1947FEN
			BP	437.9				1946BOO
			BP	437.8				1943LEC
			BP	437.9		VP		1940STU
			BP	437.8				1933MAI
			BP	437.8		BP rev		1921TIM
2-ethyltoluene	liq	SEL	BP	438.24	0.15	VP		1949FOR
			BP	438.29	0.14	BP rev		2019DIK
			BP	438.1				1956FIE
			BP	438.1				1951KUT
			BP	438.30	0.15	BP rev		1947FEN
3-ethyltoluene	liq	SEL	BP	434.44	0.10	VP		1949FOR
			BP	434.44	0.16	BP rev		2019DIK
			BP	434.55	0.15	BP rev		1947FEN
4-ethyltoluene	liq	SEL	BP	435.13	0.10	VP		1949FOR
			BP	435.13	0.26	BP rev		2019DIK
			BP	435.10	0.15	BP rev		1947FEN
isopropylbenzene	liq	SEL	BP	425.53	0.07	VP		1945WIL
			BP	425.53	0.14	BP rev		2019DIK
			BP	425.6		BP comp		2016HUR
			BP	425.54	0.20	VLE		1989CEP
			BP	425.54	0.15	VP		1967AMB
			BP	425.55	0.15	BP rev		1947FEN
<i>n</i> -propylbenzene	liq	SEL	BP	432.35	0.07	VP		1945WIL
			BP	432.36	0.07	BP rev		2019DIK
			BP	433.1		VLE		1988PAU
			BP	432.4		VLE		1968KES
			BP	432.37	0.15	VP		1967AMB
			BP	432.6				1950BRY
			BP	432.32	0.10	VP		1949FOR
			BP	432.40	0.15			1949FOE
			BP	432.40	0.15	BP rev		1947FEN
			BP	432.2		BP rev		1945SMI
C ₁₀ H ₁₄ 1,2-diethylbenzene	liq	SEL	BP	456.57	0.10	VP		1949FOR
			BP	456.57	0.14	BP rev		2019DIK
			BP	456.2				1952BER

			BP	456.5			1949BIR
			BP	454.35	0.15	BP rev	1947FEN
			BP	456.7			1946KAR
1,3-diethylbenzene	liq	SEL	BP	454.25	0.15	VP	1949FOR
			BP	454.24	0.17	BP rev	2019DIK
			BP	454.4		BP	1949BIR
			BP	454.40	0.15	BP rev	1947FEN
			BP	454.8			1946GIB
1,4-diethylbenzene	liq	SEL	BP	456.90	0.10	VP	1949FOR
			BP	456.89	0.17	BP rev	2019DIK
			BP	456.90	0.15	VP	1967AMB
			BP	456.9			1960MAI
			BP	456.6			1952BER
			BP	456.90	0.15	BP rev	1947FEN
			BP	457			1946KAR
2-isopropyltoluene	liq	SEL	BP	451.45	0.20	VP ebul	1959MCD
			BP	451.4	1.0	BP rev	2019DIK
			BP	451.3	1.4	BP rev	2012KRO
			BP	451.3	0.3		1950BOO
			BP	451.50	0.15	BP distill	1947FEN
3-isopropyltoluene	liq	SEL	BP	448.05	0.12	VP ebul	1959MCD
			BP	448.1	1.1	BP rev	2019DIK
			BP	448.2	0.4	BP rev	2012KRO
			BP	448.4	0.4		1949BIR
			BP	448.34	0.15	BP distill	1947FEN
4-isopropyltoluene	liq	SEL	BP	450.26	0.06	VP ebul	1959MCD
			BP	450.28	0.08	BP rev	2019DIK
			BP	450	4	BP rev	2012KRO
			BP	450.3			2010HAY
			BP	450.30	0.15	BP rev	1947FEN
Isobutylbenzene	liq	SEL	BP	445.95	0.15	BP rev	1947FEN
			BP	445.88	0.14	BP rev	2019DIK
<i>n</i> -butylbenzene	liq	SEL	BP	456.42	0.10	VP	1949FOR
			BP	456.42	0.08	BP rev	2019DIK
			BP	456.42	0.15	VP	1967AMB
			BP	456.5		BP	1949BIR
			BP	456.25	0.15	BP rev	1947FEN
sec-butylbenzene	liq	SEL	BP	446.30	0.15	BP rev	1947FEN

				BP	446.44	0.03	BP rev	2019DIK
tert-butylbenzene		liq	SEL	BP	442.27	0.15	VP	1949FOR
				BP	442.26	0.03	BP rev	2019DIK
				BP	442.3		BP	1955RUS
				BP	442			1954CAD
				BP	442.30	0.15		1949FOE
				BP	442.6		BP	1949BIR
				BP	442.1			1947LEG
				BP	442.25	0.15	BP rev	1947FEN
				BP	442			1946RUE2
				BP	442.5		BP rev	1945SMI
				BP	442			1945GRE
				Alkenylbenzenes				
C ₈ H ₈	ethenylbenzene	liq	SEL	BP	418.5	0.3		1950BOO
				BP	418.5	0.4	BP rev	2019DIK
				BP	418.3		BP comp	2016HUR
				BP	418.4	0.9	BP rev	2012KRO
				BP	418.8	0.3	VLE	1955JUN
				BP	418.30	0.15	VP	1955DRE
Benzocycloalkanes								
C ₉ H ₁₀	benzocyclopentane	liq	SEL	BP	450.938	0.002	VP ebul	2016CHI
				BP	450.96	0.11	BP rev	2019DIK
				BP	450.9		VP	1978OSB
				BP	451.0	0.2	BP rev	1947FEN
				BP	451.0		BP rev	1944BOO
		sol	SEL1	TP	221.786	0.01	MP	2016CHI
C ₁₀ H ₁₂	benzocyclohexane	liq	SEL	BP	480.6	0.5		1953KAR
				BP	480.3	0.5	BP rev	2019DIK
				BP	480.2		BP comp	2016HUR
				BP	480.3	0.4	BP rev	2012KRO
				BP	480.2			1955HAM
				BP	480.5	0.2	VP	1922HER
Benzocycloalkenes								
C ₉ H ₈	benzocyclopentene	liq	SEL	BP	455.72	0.15	BP rev	1947FEN
				BP	455.72	0.23	BP rev	2019DIK
				BP	455.4		GLC	1967COO
				BP	454.7	0.5	VP rev	1947STU
Alkyl naphthalenes								

C ₁₀ H ₈	naphthalene	sol liq	SEL	MP	353.4	0.3	DSC	1980KRA
			SEL1	BP	491.13	0.10	EOS	1993CHI
				BP	491.1	0.0	BP rev	2019DIK
				BP	491.1		BP comp	2016HUR
				BP	491.137	0.024	BP rev	2012KRO
				BP	491.5		VP ebul	1990AMB
				BP	491.7			1981WIL
				BP	490.73	0.15	VP	1968FOW
				BP	491.10	0.15	VP	1967AMB
				BP	491.11	0.10	VP	1955CAM
		sol		BP	491.2	0.3		1943LEC
				BP	491.0	0.4	VP isoten	1923MOR
				MP	353.3	0.5	VLE	1986SCH
				MP	353.43	0.02	rev	1984SZA
				MP	353.8	0.2	MP	1980RAD
C ₁₁ H ₁₀	1-methylnaphthalene	liq	SEL	BP	517.56	0.10	VP	1955CAM
		sol	SEL1	MP	242.65	0.15	MP	1954HIP
		liq		BP	517.78	0.17	BP rev	2019DIK
				BP	517.9		BP comp	2016HUR
				BP	517.78	0.21	BP rev	2012KRO
				BP	518.0	0.5	GLC	1967COO
	2-methylnaphthalene			BP	516.15	0.15	VP	1955MYE
		sol		MP	242.52	0.06	rev	1955STR
		sol	SEL	MP	307.3	0.4	MP	1958RAM
		liq	SEL1	BP	514.10	0.10	VP	1981WIE
				BP	514.23	0.17	BP rev	2019DIK
				BP	514.2			2013NIK
Polycyclic Aromatics				BP	514.24	0.21	BP rev	2012KRO
				BP	514.20	0.10	VP rev	1955CAM
				BP	514.3		VLE	1949OTH
				BP	514.3			1940MAI
		sol		MP	307.7	0.2	MP	1954HIP
	acenaphthylene	sol	SEL	MP	366.4	1.0	MP	1965BOY
		liq	SEL1	BP	551	5	BP est	2019DIK
	biphenylene			BP	553.2		BP rev	2014HAY
		sol	SEL	MP	384.4	0.5	MP	1974YOK
		liq	SEL1	BP	548.5	3.0	BP est	2019DIK

C ₁₂ H ₁₀	acenaphthene	sol liq	SEL	MP	364.0	1.0	MP	1990DON
			SEL1	BP	550.4	0.4	VP isoten	1923MOR
				BP	550.6	0.3	BP rev	2019DIK
C ₁₃ H ₁₀	fluorene	liq	SEL	BP	570.2	1.0	VP	1982SIV
		sol	SEL	MP	388	2	MP SLE	1989SED
		liq		BP	570.7	0.7	BP rev	2019DIK
				BP	569.6			1943LEC
				BP	570.2	0.4	VP isoten	1923MOR
C ₁₄ H ₁₀	anthracene	sol	SEL	MP	489.5	0.3	MP	1988COO
		liq	SEL1	BP	614.9	0.4	VP isoten	1923MOR
				BP	614.093	0.011	BP rev	2019DIK
				BP	613.1			1931BUR
	phenanthrene	sol	SEL	MP	372.7	0.3	MP	1990DON
		liq	SEL1	BP	609.00	0.06	VP ebul	1975OSB
				BP	611.7	0.8	BP rev	2019DIK
				BP	605.3			1981WIL
				BP	611.5	0.4	VP isoten	1923MOR
		sol		MP	372.8	0.3	MP	1980RAM
C ₁₆ H ₁₀	fluoranthene	sol	SEL	MP	381.0	0.3	MP	1973CAS
		liq	SEL1	BP	660	5	VP	1955TSY
				BP	647	6	BP est	2019DIK
	pyrene	sol	SEL	MP	424.3	0.6	MP	1987JUD
		liq	SEL1	BP	668	5	VP	1955TSY
				BP	662	4	BP est	2019DIK
C ₁₈ H ₁₂	naphthacene	sol	SEL	MP	613	2	MP	1952INO
		liq	SEL1	BP	716	2	BP	1967WAK
				BP	714	5	BP est	2019DIK
C ₂₀ H ₁₂	perylene	sol		MP	550.9	2.3	TP	2019DIK
C ₂₄ H ₁₂	coronene	sol	SEL	MP	710.5	0.5	MP	1980SMI
		liq	SEL1	BP	798.2		BP rev	1989WEA

OXIDIZED HYDROCARBONS								
Carbon Oxides								
CO ₂	carbon dioxide	sol	SEL	Sub	194.686	0.003	sub rev	1996BED
				Sub	194.685	0.003	EOS-rev	2018LEM
				Sub	194.685	0.003	EOS	1996SPA
				Sub	194.685	0.003	sub VP	1966BAR
				Sub	194.686	0.006	sub VP	1963LOV
				Sub	194.7	0.1	sub VP	1956AMB
				Sub	194.7	0.1	sub VP	1937GIA
				Sub	194.7	0.1	sub VP	1933MEY
CO	carbon monoxide	gas sol gas sol	SEL SEL1	BP	81.65	0.26	EOS	1979BRO
				TP	68.2		EOS	2006LEM
				BP	81.6		EOS-rev	2018LEM
				BP	81.6		EOS	2006LEM
				BP	81.61	0.05	VP	1932CLA
				TP	68.2		EOS-rev	2018LEM
Alkanols								
CH ₄ O	methanol	liq	SEL	BP	337.63	0.12	EOS	1993DER
				BP	337.63	0.12	EOS-rev	2018LEM
				BP	337.9		BP comp	2016HUR
				BP	337.63	0.16	BP rev	2012KRO
				BP	337.4		VLE	1998BLA
				BP	337.65	0.15	VLE	1998ARC
				BP	337.7		VLE	1997COM
				BP	337.5		VLE	1995RED
				BP	337.7		VLE	1992HIA
				BP	337.7		VLE	1991KAT
				BP	337.9		VLE	1991CAB
				BP	337.6		VLE	1990GUL
				BP	337.8		VLE	1988WIS
				BP	337.7		VLE	1978OCH
				BP	337.71	0.03	VLE	1973MAT
				BP	337.7		VLE	1970KIR
				BP	337.70	0.15	VP	1970AMB
				BP	337.4			1970AMB
C ₂ H ₆ O	ethanol	liq	SEL	BP	351.44	0.10	VLE	1998ARC
				BP	351.6	0.5	EOS-rev	2018LEM
				BP	351.4		BP comp	2016HUR

C ₃ H ₈ O	1-propanol	liq	SEL	BP	351.6	0.5	EOS	2014SCH
				BP	351.39	0.19	BP rev	2012KRO
				BP	351.5		VLE	1997COM
				BP	351.4		VLE	1995HIA
				BP	351.5		VLE	1991KAT
				BP	351.5		VLE	1985DRI
				BP	351.5		VLE	1979THO
				BP	351.45	0.15	VP	1970AMB
				BP	351.5		VLE	1960COP
				BP	351.45	0.15	VLE	1949KRE
				BP	370.3	0.2	VLE	1995HIA
				BP	370.11	0.23	BP rev	2019DIK
				BP	370.4		BP comp	2016HUR
				BP	370.19	0.15	BP rev	2012KRO
				BP	370.36	0.15	VLE	2010GAO
				BP	370.30	0.15	VP	1970AMB
	isopropanol	liq	SEL	BP	355.37	0.08	VP	1963BID
				BP	355.39	0.18	BP rev	2019DIK
				BP	353.5		BP comp	2016HUR
				BP	356.36	0.13	BP rev	2012KRO
				BP	355.7		VLE	1990GUL
				BP	355.5		VLE	1985RAO2
				BP	355.6		VLE	1985DRI
				BP	355.0		VLE	1980YOS
				BP	355.5		VLE	1971NAG
				BP	355.5		VLE	1966KAT
				BP	355.42	0.15	VLE	1960TER
				BP	390.8	0.2	VLE	1996GON
C ₄ H ₁₀ O	1-butanol	liq	SEL	BP	390.6	0.3	BP rev	2019DIK
				BP	390.79	0.16	BP rev	2012KRO
				BP	390.8		VLE	1995WIS
				BP	390.7		VLE	1995RED
				BP	390.8		VLE	1992TAN
				BP	390.8		VLE	1991CAB
				BP	390.9		VP	1988GIE
				BP	390.9		VLE	1988AGU
				BP	390.8		VLE	1987KRI
				BP	390.8		VLE	1987KRI

				BP	390.8		VLE	1978TAM			
				BP	390.9		VLE	1970KAT			
				BP	391.02	0.15	VP	1969KEM			
				BP	391.4	0.2	VLE	1965HES			
				BP	390.88	0.12	VP	1963BID			
2-butanol	liq	SEL	BP	372.64	0.14	VP	1963BID				
			BP	372.4	0.6	BP rev	2019DIK				
			BP	372.56	0.18	BP rev	2012KRO				
			BP	372.6		VLE	2011LLA				
			BP	372.7		VLE	1998FEN				
			BP	372.6		VLE	1987KRI				
			BP	372.4		VLE	1985RAO				
			BP	372.6			1960SKI				
isobutanol	liq	SEL	BP	372.5		VLE	1953HEL				
			BP	380.90	0.15	VLE	1995RED				
			BP	380.84	0.10	BP rev	2019DIK				
			BP	380.98	0.03	BP rev	2012KRO				
			BP	381.0		VLE	1988AGU				
			BP	380.9		VLE	1987KRI				
			BP	381.1		VLE	1981RAV				
			BP	380.9		VLE	1975WIS				
<i>tert</i> -butanol	sol liq	SEL SEL1	MP	298.87	0.10	MP	1988KIM				
			BP	355.47	0.15	VP	1963BID				
			BP	355.51	0.06	BP rev	2019DIK				
			BP	355.49	0.09	BP rev	2012KRO				
			BP	355.6		VLE	1999LOR				
			BP	355.3			1995RED				
			BP	355.7		VLE	1988WIS3				
			BP	355.6		VLE	1988AGU				
			BP	355.6		VLE	1987KRI				
			BP	355.5		VLE	1977GOV				
			BP	355.47	0.15	VP	1969BRO				
			BP	355.49	0.15	VP	1963BEY				
			BP	355.4			1956STU				
			BP	355.6			1956OMU				
			C ₅ H ₁₂ O	1-pentanol	liq	SEL	BP	410.9	0.3	VLE	1991YOS

			BP	411.0	0.9	BP rev	2019DIK
			BP	411.11	0.26	BP rev	2012KRO
			BP	411.1			1995RED
			BP	410.2		VLE	1988WIS
			BP	411			1982KAR
			BP	410.8		VLE	1976TAM
			BP	411.2			1974RAO
			BP	411.2		VLE	1972MIL
			BP	410.9			1970AMB
2-methyl-1-butanol	liq	SEL	BP	402.14	0.17	VP ebul	2010CEN
			BP	402.01	0.22	BP rev	2019DIK
			BP	402.15	0.07	BP rev	2012KRO
			BP	401.8	0.5	VLE	1953MCK
			BP	403.8	0.2	VP	1935BUT
2-methyl-2-butanol	liq	SEL	BP	375.30	0.17	VP ebul	2010CEN
			BP	375.22	0.26	BP rev	2019DIK
2-pentanol	liq	SEL	BP	392.48	0.17	VP ebul	2010CEN
			BP	392.39	0.22	BP rev	2019DIK
			BP	392.25	0.17	VLE	2012MAR
			BP	392.22	0.17	VLE	2009GIE
			BP	393.02	0.25	VLE	1984RAO
			BP	392.33	0.17	VP ebul	1963THO
			BP	392.29	0.15	VP	1935BUT
3-methyl-2-butanol	liq	SEL	BP	385.03	0.17	VP ebul	2010CEN
			BP	384.63	0.27	BP rev	2019DIK
			BP	385.6		VLE	1953MCK
			BP	384.6			1937GIN
			BP	386.1		VLE	1930LEC
3-pentanol	liq	SEL	BP	386.6			1916WIL
			BP	388.62	0.17	VP ebul	2010CEN
isopentanol	liq	SEL	BP	388.57	0.24	BP rev	2019DIK
			BP	404.73	0.17	VP ebul	2010CEN
			BP	404.57	0.27	BP rev	2019DIK
			BP	405.2		BP comp	2016HUR
			BP	403.95	0.13	BP rev	2012KRO
			BP	404.55	0.06	VLE	2010CEP
			BP	403.5		VLE	1992TAN
			BP	403.9		VP	1986MAR

				BP	403.8		VLE	1985RAO
				BP	404.9		VLE	1981RAV
				BP	405.2		VLE	1972MIL
				BP	405.2		VLE	1965RED
				BP	405		VLE	1960TER
Glycols								
C ₂ H ₆ O ₂	1,2-ethanediol	liq	SEL	BP	470.35	0.08	VLE	2013QIA
				BP	470.5	0.9	BP rev	2019DIK
				BP	470.3	0.2	EOS-rev	2018LEM
				BP	470.3	0.3	VLE	2013PLA
				BP	470.36	0.10	VP	2012KAM
				BP	470.4	0.4	VLE	2004NAG
				BP	470.5	0.5	VLE	1982DIZ
				BP	470.50	0.03	VP	1981MAR
				BP	470.44	0.06	VP	1981AMB
				BP	470.4	0.5	VLE	1971SOK
				BP	470.38	0.25	VLE	1959KOG
				BP	470.73	0.25	VP rev	1952JON
				BP	470.4	0.5	VLE	1949OTH
				BP	471.0	0.5	BP rev	1935TIM
				BP	470.4	0.5	BP	1931SMY
C ₃ H ₈ O ₂	1,2-propanediol	liq	SEL	BP	461.31	0.06	VLE	2011MAT
				BP	461.2	0.8	EOS	2021EIS
				BP	461.5	0.4	BP rev	2019DIK
				BP	461.2		VLE	2013ZHA
				BP	460.43	0.23	EOS	2002STE
				BP	461.2	0.4	VP comp	1999DEA
				BP	461.0	0.4	VP	1955TIM
				BP	461.4	0.4	VP comp	1954MAR
				BP	459.8	1.0	VP	1952WES
				BP	459.2	1.5	VP	1950CLE
				BP	460.6	0.6	VP	1935SCH
2-methoxyethanol		liq	SEL	BP	397.54	0.15	VLE	2012RED
				BP	397.47	0.04	BP rev	2019DIK
				BP	397.41	0.07	BP rev	2012KRO
				BP	397.6		VP	2012BEJ
				BP	397.4		VLE	2010MAR
				BP	397.5	0.4	VLE	1995RED

				BP	397.6	0.4	VLE	1995KRI
				BP	397.7		VLE	1979CHA
				BP	397.4			1970QUI
				BP	397.29	0.15	VP	1956PIC
Sugars								
C ₃ H ₈ O ₃	glycerin	liq	SEL	BP	563.03	0.10	BP	2016ANI
		sol	SEL1	MP	291.1	0.5	MP	1958COS
C ₆ H ₁₀ O ₅	cellulose	sol	SEL	MP	408.2		MP	2020POD
C ₆ H ₁₂ O ₆	alpha-D-glucose	sol	SEL	MP	421.8	0.6	MP	2004HUR
		gas	SEL1	BP	706	50	est	2019DIK
		sol		MP	414		MP	1996DOM
				MP	423.2		MP	1983RAE
	fructose	sol	SEL	MP	376.7	0.3	MP	2004HUR
				MP	378.2		MP	1995FAN
C ₁₂ H ₂₂ O ₁₁	sucrose	sol	SEL	MP	456	4	MP	2004HUR
				MP	459	3	dec	2021EPA
				MP	461.2		MP	2001GLO
				MP	458.2		MP	1983RAE
				MP	462	3	MP	1950KOF
Phenols								
C ₇ H ₈ O	2-methylphenol	liq		BP	464.1		VP	1958BID
		sol	SEL	MP	304.12	0.10	MP	1960AND
		liq	SEL1	BP	464.14	0.10	VP rev	1949DRE2
				BP	464.15	0.05	BP rev	2019DIK
				BP	464.3			1947LEC
				BP	464.30	0.18	VP	1947GOL
				BP	463.9			1944PAR
		sol		MP	464.1	0.3	MP	1949DRE
	4-methylphenol	sol	SEL	MP	307.84	0.20	MP	1960AND
		liq	SEL1	BP	474.99	0.10	VP rev	1949DRE2
				BP	475.10	0.28	BP rev	2019DIK
				BP	475		VLE	1968TAY
				BP	474.90	0.18	VP	1947GOL
Alkyl ethers								
C ₂ H ₆ O	dimethyl ether	gas	SEL	BP	248.37	0.09	EOS	2011WU
				BP	248.37	0.09	EOS-rev	2018LEM
				BP	248.34	0.15	VP	1941KEN
C ₃ H ₈ O	ethyl methyl ether	gas	SEL	BP	279.7	0.5		1941IPA

C ₄ H ₁₀ O	diethyl ether	liq	SEL	BP	280.59	0.19	VP rev	2019DIK	
				BP	280.7			1947STU	
				BP	281.2			1924BER	
				BP	307.55	0.15	VP	1972AMB	
				BP	307.59	0.03	BP rev	2019DIK	
	isopropyl methyl ether	liq	SEL	BP	307.7		VLE	2011GAO	
				BP	307.63	0.15	VP	1922TAY	
				BP	303.90	0.02		1976AMB	
				BP	303.917	0.014		2019DIK	
				BP	311.8			1976AMB	
	<i>n</i> -propyl methyl ether	liq	SEL	BP	311.70	0.03	VP rev	2019DIK	
				BP	312.2			1947STU	
				BP	343.29	0.02		VLE	2007BOU
				BP	343.3	0.0		2019DIK	
				BP	343.3			1976AMB	
	<i>tert</i> -butyl methyl ether	liq	SEL	BP	328.20	0.15		VLE	1999MON
				BP	328.23	0.13	BP rev	2019DIK	
				BP	341.6	0.3	VLE	1996RES	
				BP	341.58	0.14	BP rev	2019DIK	
				BP	341.4	0.3	VLE	1999MON	
C ₆ H ₁₄ O	diisopropyl ether	liq	SEL	BP	341.49	0.15	VP	1969CID	
				BP	363.22	0.04	VP	1973MEY	
				BP	363.22	0.03	BP rev	2019DIK	
				BP	364.2		VLE	1972MAR	
				BP	359.50	0.15	VP	1969CID	
	di- <i>n</i> -propyl ether	liq	SEL	BP	363.9			1950SPU	
				BP	354.2	1.0			1932NOR
				BP	354.2	1.9			2019DIK
				BP	354.2				
				BP	354.2				
Aldehydes									
CH ₂ O	formaldehyde	gas	SEL	BP	254.1	0.2	VP	1935SPE	
				BP	254.0	0.4	BP rev	2019DIK	
				BP	253.9		BP comp	2016HUR	
				BP	253.4	0.6	BP rev	2012KRO	
				BP	254.05	0.15	BP rev	1977REI	
C ₂ H ₄ O	acetaldehyde	gas	SEL	BP	293.5	0.2	VLE	1950COL	
				BP	293.8	0.5	BP rev	2019DIK	
				BP	294.2		BP comp	2016HUR	
				BP	293.6	0.8	BP rev	2012KRO	
				BP	293.6				

C ₃ H ₆ O	propanal	liq	SEL	BP	293.5	0.3	VLE	1963BUL				
				BP	293.9	0.3		1955SHI				
				BP	293.9	0.6		1950CHR				
				BP	321.19	0.15	VLE	2010GAO				
				BP	321.15	0.08	BP rev	2019DIK				
				BP	321.12	0.03	VLE	1984ENG				
				BP	321.17	0.05		1977SVO				
				BP	321.17	0.03	VLE	1977BOU				
				BP	321.19	0.03	VLE	1973MAT				
				BP	321.20	0.03		1972COU				
C ₄ H ₈ O	butanal	liq	SEL	BP	321.15	0.03	VLE	1953MCK				
				BP	321.1			1962TJE				
				BP	348.0	0.4	VLE	1985HSU				
				BP	347.95	0.15	BP rev	2019DIK				
				BP	348.3	0.5	BP rev	2012KRO				
				BP	347.9	0.3	VLE	1984ENG				
				BP	348.33	0.25	VP	1959SEP				
				BP	347.4	0.5		1955BRO				
				BP	347.7	0.5	VLE	1953MCK				
				BP	348.4	0.5		1943HUR				
	isobutanal	liq	SEL	BP	348.0	0.3		1938DOL				
				BP	347.8	0.5		1938COO				
				BP	348.9	0.5		1922TIM				
				BP	337.26	0.25	VP	1959SEP				
				BP	337.19	0.11	BP rev	2019DIK				
				BP	337.25	0.14	BP rev	2012KRO				
				BP	337.3	0.5		1941HEA				
				Alkenals								
				C ₃ H ₄ O	1-propenal	gas	SEL	BP	264		BP est	2021CHE
					2-propenal	liq	SEL	BP	325.4	0.3		1979MAR
BP	325.46	0.10	BP rev					2019DIK				
BP	325.7		BP comp					2016HUR				
BP	325.48	0.11	BP rev					2012KRO				
BP	325.71	0.18	VP rev					1947STU				
Ketones												
C ₃ H ₆ O	acetone	liq	SEL	BP	329.22	0.29	EOS	2006LEM				
				BP	329.22	0.29	EOS-rev	2018LEM				
				BP	329.2	0.2	VLE	1999TRE				

				BP	329.5		VLE	1990GUL
				BP	329.2		VP	1986TAK
				BP	329.42	0.15	VP	1974AMB
				BP	329.0		VLE	1970WIL
				BP	329.5			1955KOC
C ₄ H ₈ O	butanone	liq	SEL	BP	352.72	0.03	VLE	1973MAT
				BP	352.71	0.02	BP rev	2019DIK
				BP	352.7		VLE	2011LLA
				BP	352.7		VLE	2002DEL
				BP	352.8		VLE	1994YAH
				BP	352.63	0.15	VP	1961NIC
C ₅ H ₁₀ O	2-pentanone	liq	SEL	BP	375.42	0.15	VLE	1973SCH
				BP	375.38	0.05		2019DIK
				BP	375.1			2003AUC
				BP	375.3		VP	1975AMB
				BP	376.0			1966MEY
	3-pentanone	liq	SEL	BP	375.08	0.06	VP	1965COL
				BP	375.08	0.06	BP rev	2019DIK
				BP	376.41	0.10	VP rev	1949DRE2
Carboxylic Acids								
CH ₂ O ₂	formic acid	liq	SEL	BP	374.1	0.4	VLE	2011BAS
		sol	SEL1	MP	281.5	0.2	MP	1981KOH
		liq		BP	374.35	0.24	BP rev	2019DIK
				BP	374.4		VLE	2007INC
				BP	373.7		VLE	1990KAT
				BP	374.8		VP	1987AMB2
				BP	373.6			1973LAK
				BP	373.8		VLE	1960CON
				BP	373.7			1952WAR
				BP	373.9		VP rev	1949DRE2
				BP	374.0		VLE	1949ALP
C ₂ H ₄ O ₂	acetic acid	liq	SEL	BP	391.20	0.15	VLE	1991SAY
		sol	SEL1	MP	289.48	0.16	MP	1959MCD
		liq		BP	390.97	0.05	BP rev	2019DIK
				BP	391.1		VLE rev	1990HUI2
				BP	391.1		VLE	1988KAT
				BP	391.3		VLE	1968BAR
				BP	390.99	0.08	VP ebul	1959MCD

C ₃ H ₆ O ₂	propanoic acid	liq	SEL	BP	414.0	0.3	VLE	1982WIS
				BP	414.04	0.10	BP rev	2019DIK
				BP	413.99	0.15	VLE	2010GAO
				BP	414.24	0.10	VP rev	1949DRE2
C ₄ H ₈ O ₂	<i>n</i> -butanoic acid	liq	SEL	BP	436.41	0.10	VP rev	1949DRE2
				BP	436.4	0.4	BP rev	2019DIK
				BP	436.7		VLE	1976RAD
				BP	437.2		VLE	1954PUR
C ₅ H ₁₀ O ₂	<i>n</i> -pentanoic acid	liq	SEL	BP	459.1	0.3	VP	1964ADR
				BP	458.5	0.6		2019DIK
				BP	459.2			2012GUN
				BP	459.0			1987AMB2
				BP	459.4			1955HAN2
				BP	459.0			1955GEN
C ₆ H ₁₂ O ₂	<i>n</i> -hexanoic acid	liq	SEL	BP	478.2	0.5		1949PAT
				BP	477.7	0.8		2019DIK
				BP	477.3			1958BAK
				BP	478.2			1952JAM
				BP	478.2			1938VAN
				BP	478.1			1933KOH
Alkyl Esters								
C ₂ H ₄ O ₂	methyl formate	liq	SEL	BP	305.1	0.3	VP	1928NEL
				BP	304.76	0.08	BP rev	2019DIK
C ₃ H ₆ O ₂	ethyl formate	liq	SEL	BP	327.29	0.05	VLE	1998SOT
				BP	327.23	0.04	BP eval	2019DIK
	methyl acetate	liq	SEL	BP	330.11	0.15	VLE	2016MAT
				BP	330.2		VLE	2019JIA
				BP	329.81	0.06		2019DIK
				BP	330.1		VLE	2018SHA
				BP	330.0		VLE	1969HUD
C ₄ H ₈ O ₂	ethyl acetate	liq	SEL	BP	350.21	0.15	VP	1965POL
				BP	350.15	0.05	BP rev	2019DIK
	methyl propanoate	liq	SEL	BP	352.75	0.25	VLE	2014RIO
				BP	352.40	0.15		2019DIK
				BP	351.7			1989SUS
				BP	352.85	0.25		1977SVO
				BP	352.9			1959TIM
C ₅ H ₁₀ O ₂	methyl butanoate	liq	SEL	BP	375.74	0.25	VLE	2012RIO

				BP	375.36	0.12	VLE VLE	2019DIK
				BP	375.8			2012AGA
				BP	375.0			2004ORT
				BP	375.5			2003ORT
				BP	373.7			1956SUM
				BP	375.8			1927LEC
				BP	375.9			1910YOU
<i>n</i> -propylacetate	liq	SEL	BP	374.7	0.4	VLE VP	1977SVO	
			BP	374.28	0.07		2019DIK	
			BP	374.3			2015SUS	
			BP	374.3			2013FER	
			BP	374.6			1965POL	
			BP	374.9			1959TIM	
Alkenyl Esters								
C ₄ H ₆ O ₂	vinyl acetate	liq	SEL	BP	345.3	0.4	VLE	1965SWA
				BP	345.73	0.09	BP rev	2019DIK
				BP	345.88	0.15	VLE	1963CAP
C ₅ H ₈ O ₂	ethyl propenoate	liq	SEL	BP	372.59	0.15	VLE	2013LOM
				BP	372.0	0.5	BP rev	2019DIK
				BP	370.2		VP rev	1987DAU
				BP	372.6		VLE	1973FIL
				BP	372.69	0.15	VP rev	1947STU
methyl 2-methylpropenoate	liq	SEL	BP	369.7	0.5	VP	1956BRO	
			BP	373.76	0.12	BP rev	2019DIK	
Alkanoic Anhydrides								
C ₄ H ₆ O ₃	acetic anhydride	liq	SEL	BP	412.62	0.15	VP VP VLE	1976RAD
				BP	412.65	0.05		2019DIK
				BP	412.4			1987AMB
				BP	411.76	0.08		1959MCD
				BP	412.5			1950NEL
C ₆ H ₁₀ O ₃	propanoic anhydride	liq	SEL	BP	440.3	1.5		1961WRI
				BP	440.8	0.8		2019DIK
				BP	438.2			1958LEV
C ₈ H ₁₄ O ₃	butanoic anhydride	liq	SEL	BP	471.4	2.0	BP comp	1921TIM2
				BP	471.2	2.5		2019DIK
				BP	471.4			1961WRI
				BP	471.4			1958LEV
Oxacycloalkanes								

C ₂ H ₄ O	oxacyclopropane	gas	SEL	BP	283.58	0.04	VP ebul	1959MCD
				BP	283.59	0.04	BP rev	2019DIK
				BP	283.66	0.13	EOS-rev	2018LEM
				BP	283.66	0.13	EOS	2015THO
				BP	283.65	0.15	VLE	1950COL
C ₃ H ₆ O	methyloxacyclopropane	liq	SEL	BP	307.36	0.08	VP	1959MCD
				BP	307.34	0.07	BP rev	2019DIK
				BP	307.27	0.14	EOS-rev	2018LEM
				BP	307.4		VLE	1989MUH
				BP	307.56	0.25	VLE	1985KOL
				BP	307.2		VP	1966BOT
				BP	307.9			1956PRI
				BP	308.2			1937MOU
				BP	308.0			1934ALL
	oxacyclobutane	liq	SEL	BP	320.73	0.10	VP	1980OSB
				BP	320.73	0.02		2019DIK
				BP	321.6			1960CHA
				BP	320.8			1934ALL
C ₄ H ₈ O	oxacyclopentane	liq	SEL	BP	339.15	0.15	VLE	1998WIS
				BP	339.14	0.06		2019DIK
				BP	339.1			1996KRI
				BP	338.9		VLE	1975SAD
				BP	339.1			1946BOO
C ₅ H ₁₀ O	oxacyclohexane	liq	SEL	BP	361.36	0.20	VLE	2012MEJ
				BP	361.32	0.14		2019DIK
				BP	361.2			2005GIN
				BP	361.2		VLE	2000ROD
				BP	361.1			1964MOE
				BP	361.2			1957KIR
				BP	361.2			1934ALL
Oxacycloalkadienes								
C ₄ H ₄ O	oxacyclopentadiene	liq	SEL	BP	304.51	0.08	VP	1952GUT2
				BP	304.50	0.07	BP rev	2019DIK
C ₅ H ₆ O	2-methyloxacyclopentadiene	liq	SEL	BP	338.4	0.5	VP isoten	1971EON
				BP	337.1	0.4	BP rev	2019DIK
Oxacycloketones								
C ₃ H ₄ O	cyclopropanone	liq	SEL	BP	333	18	BP est	2019DIK
C ₄ H ₆ O	cyclobutanone	liq	SEL	BP	372.9	0.5		1972WOL

				BP	371.95	0.04	BP rev	2019DIK
				BP	372.9	0.5	BP	1972WOL
				BP	372.2			1949ROB
				BP	367.8	0.2	VP	1942BEN
C ₅ H ₈ O	cyclopentanone	liq	SEL	BP	403.9	0.3	BP	1972WOL
				BP	403.68	0.07	BP rev	2019DIK
				BP	403.8		BP	1937TIM
C ₆ H ₁₀ O	cyclohexanone	liq	SEL	BP	428.57	0.15	VP	1987AMB
				BP	428.54	0.10	BP rev	2019DIK
				BP	428.54	0.10	VP	1973MEY
				BP	428.6	0.3	BP	1965GRO
C ₇ H ₁₂ O	2-methylcyclohexanone	liq	SEL	BP	438.0	0.5	VLE	1993BUR
				BP	437.56	0.05	BP rev	2019DIK
				BP	438.2	0.5	BP	1940WHI
	3-methylcyclohexanone	liq	SEL	BP	442.7	0.5	BP	1940WHI
				BP	442.41	0.25	BP rev	2019DIK
	4-methylcyclohexanone	liq	SEL	BP	444.6	0.5	BP	1940WHI
				BP	444.1	0.3	BP	2019DIK
Dioxanes and Trioxanes								
C ₃ H ₆ O ₃	1,3,5-trioxane	sol	SEL	MP	335.7	2.0	MP	1963LEV
		liq	SEL1	BP	387.2	0.4	VLE	1966SER
		sol		MP	333.4	2.0	TP	1988VAN
C ₄ H ₈ O ₂	1,3-dioxane	liq	SEL	BP	378.1	0.5	BP	1967PIH
				BP	377.9	2.8	BP est	2019DIK
				BP	377.6	0.5	BP	1972RAK
	1,4-dioxane	liq	SEL	BP	374.3	0.3	VLE	1956MCC
				BP	374.4	0.4	BP rev	2019DIK
				BP	374.44	0.15	VP	1938CRE
Furandione and D-Camphor								
C ₁₀ H ₁₆ O	D-Camphor	sol	SEL	MP	452	10	MP	1992AND
				MP	463.4		MP	1990DON
				MP	452.0		MP	1979MJO
C ₄ H ₂ O ₃	2,5-furandione	sol	SEL	MP	325.3	0.5	MP	1983DEW
		liq	SEL1	BP	478	5	VP rev	1947STU
				BP	476	11	BP est	2019DIK
				BP	473	5	VLE	2002GU
Aromatic Alcohols								
C ₆ H ₆ O	phenol	sol	SEL	MP	314.05	0.15	MP	1960AND

		liq	SEL1	BP	454.91	0.10	VP rev	1949DRE2
				BP	454.96	0.17	BP rev	2019DIK
				BP	455.19	0.18	VP	1947GOL
C ₇ H ₈ O	3-methylphenol	liq	SEL	BP	475.50	0.20	VLE	1989CEP2
		sol	SEL1	MP	285.37	0.20	MP	1960AND
		liq		BP	475.4		BP rev	2019DIK
				BP	475.28	0.18	VP	1947GOL
				BP	475.9		BP rev	1937TIM
				BP	475.3		VLE	1921KEN
		sol		MP	285.1		MP	1937TIM
	benzyl alcohol	liq	SEL	BP	478.5	0.3	VLE	1998SEN
				BP	478.3	0.9	BP rev	2019DIK
				BP	478.6		VLE	1994RES
				BP	478.49	0.07	VP	1990AMB2
				BP	478.2	0.3	GLC	1977KUC
				BP	477.9			1967MAR
				BP	478.45	0.10	VP rev	1949DRE2
				BP	478.4		VLE	1947LEC2
				BP	478.9		BP rev	1922TIM
Aromatic Aldehydes and Acids								
C ₇ H ₆ O	benzaldehyde	liq	SEL	BP	451.15	0.20	VLE	1981RAO
				BP	452.4		VLE	1958LEV
				BP	451.68	0.18	VP rev	1947STU
				BP	452.4		VLE	1943LEC
				BP	452.2		VLE	1941EWE2
C ₇ H ₆ O ₂	benzoic acid	sol	SEL	MP	395.47	0.02	MP	1989HAL
		liq	SEL1	BP	523.5	0.5	VP	1931BUR
				BP	523.5	0.5	VP rev	1947STU
				BP	523.2	0.8	BP rev	2019DIK
				BP	522.2		BP rev	1989WEA
				BP	523.3	0.7		1930MAR
				BP	522.6		VP	1927KLO

NITROGEN-SUBSTITUTED HYDROCARBONS								
Hydrogen Cyanides								
CHN	hydrogen cyanide	liq	SEL	BP	298.8	0.3	VP	1926SIN
				BP	298.78	0.04	BP rev	2019DIK
Alkylamines								
CH ₅ N	methylamine	gas	SEL	BP	266.84	0.15	VP	1937AST
				BP	266.76	0.19	BP rev	2019DIK
C ₂ H ₇ N	ethylamine	gas	SEL	BP	289.79	0.18	VP rev	1947STU
				BP	289.80	0.09	BP rev	2019DIK
C ₃ H ₉ N	<i>n</i> -propylamine	liq	SEL	BP	320.37	0.06	VP ebul	1968OSB
				BP	320.36	0.05	BP rev	2019DIK
				BP	321.7	0.5	VP rev	1947STU
	isopropylamine	liq	SEL	BP	304.92	0.06	VP ebul	1968OSB
				BP	304.91	0.04	BP rev	2019DIK
C ₄ H ₁₁ N	isobutylamine	liq	SEL	BP	341.1	0.2	VP VP VP	1977RAD
				BP	340.69	0.05		2019DIK
				BP	340.7			1980OSB
				BP	340.9			1979MAJ
				BP	341.1			1962HOL
				BP	340.9			1959COS
				BP	340.1			1951BAR
				BP	341.1			1949SCH
				BP	340.8		VP	1935SIM
				BP	341.2			1921TIM2
	<i>n</i> -butylamine	liq	SEL	BP	350.1	0.3	VLE	1971LET
				BP	350.17	0.13	BP rev	2019DIK
	<i>tert</i> -butylamine	liq	SEL	BP	317.19	0.06	VP ebul	1968OSB
				BP	317.18	0.04	BP rev	2019DIK
C ₂ H ₇ N	dimethylamine	gas	SEL	BP	280.02	0.15	VP	1939AST
				BP	280.3	0.4	BP rev	2019DIK
C ₃ H ₉ N	methylethylamine	liq	SEL	BP	305.6	0.5	VLE	1985WOL
				BP	305.8	0.6	BP rev	2019DIK
C ₄ H ₁₁ N	diethylamine	liq	SEL	BP	328.48	0.15	VLE VLE	1971LET
				BP	328.48	0.13		2019DIK
				BP	328.6			1967NAK
				BP	328.5			1967HUM
				BP	328.5			1959COS
C ₃ H ₉ N	trimethylamine	gas	SEL	BP	276.00	0.15	VP	1944AST

C ₆ H ₁₅ N	triethylamine	liq	SEL	BP	276.07	0.24	BP rev	2019DIK
				BP	361.93	0.25	VLE	1962BIT
				BP	362.2	0.3	BP rev	2019DIK
Cycloalkylamines								
C ₆ H ₁₃ N	cyclohexylamine	liq	SEL	BP	406.98	0.15	VLE	1960NOV
				BP	406.8	0.3	BP rev	2019DIK
				BP	407.4			1940WHI
				BP	407.0			1939LEW
Azacycloalkanes								
C ₂ H ₅ N	azacyclopropane	liq	SEL	BP	328.6	1.0		1956SEA
				BP	327.4	0.5		2019DIK
C ₃ H ₇ N	azacyclobutane	liq	SEL	BP	333.5	1.5		2019DIK
				BP	336			1982BUC
C ₄ H ₉ N	azacyclopentane	liq	SEL	BP	359.71	0.15	VP	1959MCC
				BP	359.71	0.07	BP rev	2019DIK
C ₅ H ₁₁ N	azacyclohexane	liq	SEL	BP	379.37	0.06	VP ebul	1968OSB
				BP	379.34	0.13	BP rev	2019DIK
Methenamines								
C ₆ H ₁₂ N ₄	methenamine	sol	SEL	MP	359	10	sub	2002VER
Azacycloalkadienes								
C ₄ H ₅ N	azacyclopentadiene	liq	SEL	BP	402.93	0.06	VP ebul	1968OSB
				BP	402.90	0.05	BP rev	2019DIK
				BP	399.5	0.6	VP isoten	1971EON
C ₅ H ₇ N	1-methylazacyclopentadiene	liq	SEL	BP	385.90	0.06	VP ebul	1968OSB
				BP	385.87	0.04	BP rev	2019DIK
				BP	384.7	0.6	VP isoten	1971EON
Alkanenitriles								
C ₂ H ₃ N	ethanenitrile	liq	SEL	BP	355.05	0.15	VP	1974DOJ
				BP	354.74	0.03	BP rev	2019DIK
C ₃ H ₅ N	propanenitrile	liq	SEL	BP	370.7	0.3	VP rev	1949DRE2
				BP	370.41	0.29	BP rev	2019DIK
C ₄ H ₇ N	butanenitrile	liq	SEL	BP	390.77	0.04	VP	1973MEY
				BP	390.74	0.04	BP rev	2019DIK
C ₅ H ₉ N	pentanenitrile	liq	SEL	BP	414.42	0.10	VP rev	1949DRE2
				BP	414.39	0.17	BP rev	2019DIK
C ₆ H ₁₁ N	hexanenitrile	liq	SEL	BP	436.61	0.04	VP	1973MEY
				BP	436.59	0.05	BP rev	2019DIK
				BP	436.3			1961WRI

				BP	436.88	0.10	VP rev	1949DRE2
Alkenenitriles								
C ₃ H ₃ N	propenenitrile	liq	SEL	BP	351.1	0.5	VLE	1964SEV
				BP	350.7	0.9	BP rev	2019DIK
Cyanogens								
C ₂ N ₂	cyanogen	gas	SEL	BP	251.92	0.20	VP	1939RUE
				BP	251.954	0.002	BP rev	2019DIK
				BP	252.0		BP rev	1985MAJ
				BP	252.18	0.18	VP rev	1947STU
				BP	251.8		VP	1925PER
Alkylhydrazines								
C ₂ H ₈ N ₂	1,1-dimethylhydrazine	liq	SEL	BP	335.4	0.3	VLE	1992FER
				BP	335.2	0.7	BP rev	2019DIK
				BP	337.1		BP rev	1989WEA
				BP	336.2		VP	1985MAJ
				BP	334.3		VP	1970KOB
				BP	335.3		VLE	1956CAR
				BP	335.57	0.20	VP	1953AST
Melamines								
C ₃ H ₆ N ₆	melamine	sol	SEL	MP	627		MP	2005LID
Pyridines								
C ₅ H ₅ N	pyridine	liq	SEL	BP	388.36	0.06	BP rev	2019DIK
				BP	388.46	0.10		1967FRI
				BP	388.36	0.06		1957MCC
C ₆ H ₇ N	2-methylpyridine	liq	SEL	BP	402.52	0.08	VP	1963SCO
				BP	402.53	0.04	BP rev	2019DIK
Anilines								
C ₆ H ₇ N	aniline	liq	SEL	BP	457.1	0.2	VLE	1979KUR
				BP	457.2	0.5	BP rev	2019DIK
				BP	457.7		VLE	1992GUP
				BP	457.0		VLE	1965DES
				BP	457.6			1962STE
				BP	457.1		VLE	1962STA
				BP	456.99	0.05		1962HAT
				BP	457.3	0.3	VLE	1959ELL
				BP	457.5		VLE	1950KOR
				BP	457.1			1949DRE
				BP	457.6		BP rev	1926TIM

C ₈ H ₁₁ N	N,N-dimethylaniline	liq	SEL	BP	466.5	0.4	BP	1976RAD
		sol	SEL1	MP	275.7	0.3	MP	1976RAD
		liq		BP	466.7	1.0	BP rev	2019DIK
			BP	466.5	0.5	VP rev	1947STU	
			BP	466.5	0.4	BP	1946LEC	
			BP	466.6	0.5	VP	1925NEL	
Oxidized Amines								
CH ₄ N ₂ O	urea	sol	SEL	MP	406.2	0.5	TP	1972ZOR
				MP	405.6	0.4	TP	2019DIK
				MP	406.5	0.5	TP	2011RAI
				MP	405.8	0.5	TP	2010ZEN
				MP	406.5	0.5	TP	1987DEL
C ₆ H ₁₅ NO ₃	triethanolamine	liq	SEL	BP	608	5	VP extrap	1959MCD
		sol	SEL1	MP	294.7	0.5	MP	1959MCD
		liq		BP	605	8	BP rev	2019DIK
			BP	623.2		BP rev	2014HAY	
			BP	608.6		BP rev	2013ONE	
Alkyl Nitrates and Nitroalkanes								
CH ₃ NO ₃	methyl nitrate	liq	SEL	BP	338.0		VLE	1947LEC
				BP	338	3	BP rev	2019DIK
C ₃ H ₅ N ₃ O ₉	nitroglycerin	sol	SEL	MP	286.7		MP	2001ONE
CN ₄ O ₈	tetranitromethane	liq	SEL	BP	397.9		VP	1952EDW
		sol	SEL1	MP	287.1	0.5	MP	1955TIM
		liq		BP	399.2		BP rev	1985MAJ
CHN ₃ O ₆	trinitromethane	sol	SEL	MP	292	3	MP	1939LEW
CH ₃ NO ₂	nitromethane	liq	SEL	BP	374.34	0.08	VP	1954MCC
				BP	374.32	0.09	BP rev	2019DIK
Isocyanates								
C ₉ H ₆ N ₂ O ₂	2,4-diisocyanatotoluene	liq	SEL	BP	524.8	2.0	VP	1975FRE
		sol	SEL1	MP	293.7	1.0	MP	2020FIS
		liq		BP	524		BP rev	2010HAY
			BP	529		VP	1990DAU	
Nitro Aromatics								
C ₆ H ₅ NO ₂	nitrobenzene	liq	SEL	BP	483.81	0.15	VLE	1952BRO
		sol	SEL1	MP	278.89	0.20	MP	1954WIT
		liq		BP	483.82	0.29	BP rev	2019DIK
		sol		MP	279.2	1.0	MP	1965MAL
C ₇ H ₅ N ₃ O ₆	trinitrotoluene	sol	SEL	MP	355.1	1.0	MP	1990HUI

SULFUR-SUBSTITUTED HYDROCARBONS								
Alkanethiols								
CH ₄ S	methanethiol	gas	SEL	BP	279.05	0.10		1960MOR
				BP	279.10	0.06	BP rev	2019DIK
				BP	279.10	0.06	BP rev	2012KRO
				BP	279.05	0.15	BP ebul	1960HAI
				BP	279.10	0.15	VP ebul	1942RUS
C ₂ H ₆ S	ethanethiol	liq	SEL	BP	308.15	0.15	VP ebul	1966OSB
				BP	308.14	0.15	BP rev	2019DIK
				BP	308.14	0.15	BP rev	2012KRO
				BP	308.19	0.15		1953BIR
				BP	308.19	0.15	VLE	1949DEN
C ₃ H ₈ S	1-propanethiol	liq	SEL	BP	340.87	0.15	VP	1956PEN
				BP	340.84	0.15	BP rev	2019DIK
				BP	340.95	0.15	BP ebul	1960HAI
				BP	340.97	0.15	VLE	1949DEN
	isopropanethiol	liq	SEL	BP	325.71	0.15	VP ebul	1966OSB
		BP		325.69	0.15	BP rev	2019DIK	
		BP		325.80	0.15	BP ebul	1956HAI	
C ₄ H ₁₀ S	1-butanethiol	liq	SEL	BP	371.56	0.15	VP ebul	1966OSB
				BP	371.59	0.15	BP rev	2019DIK
				BP	371.20	0.15	BP ebul	1956HAI
	1-isobutanethiol	liq	SEL	BP	361.64	0.15	VP ebul	1966OSB
		BP		361.63	0.15	BP rev	2019DIK	
		BP		361.65	0.15	BP ebul	1960HAI	
	2-butanethiol	liq	SEL	BP	358.13	0.15	VP ebul	1966OSB
		BP		358.12	0.15	BP rev	2019DIK	
		BP		358.20	0.15	BP ebul	1956HAI	
C ₅ H ₁₂ S	1,1-dimethyl-1-propanethiol	liq		BP	372.35	0.15	BP ebul	1960HAI
			SEL	BP	372.28	0.15	VP ebul	1966OSB
				BP	372.28	0.15	BP rev	2019DIK
	1-pentanethiol	liq	SEL	BP	399.79	0.15	VP ebul	1966OSB
		BP		399.77	0.15	BP rev	2019DIK	
	isopentanethiol	liq	SEL	BP	391.47	0.15	VP	1966OSB
				BP	391.47	0.01		2019DIK
	neopentanethiol	liq	SEL	BP	376.81	0.15	VP	1966OSB
				BP	376.74	0.09		2019DIK
				BP	376.9			1969LAN

	<i>tert</i> -pentanethiol	liq	SEL	BP	398.16	0.15	VP	1966OSB
				BP	398.15	0.02		2019DIK
C ₆ H ₁₄ S	1-hexanethiol	liq	SEL	BP	425.81	0.15	VP ebul	1966OSB
				BP	425.80	0.15	BP rev	2019DIK
				BP	425.75	0.15	BP ebul	1960HAI
C ₇ H ₁₆ S	1-heptanethiol	liq	SEL	BP	450.07	0.15	VP ebul	1966OSB
				BP	450.06	0.15	BP rev	2019DIK
				BP	450.05	0.15	BP ebul	1960HAI
Alkyl Sulfides								
CS ₂	carbon disulfide	liq	SEL	BP	319.37	0.15	VP	1962WAD
				BP	319.35	0.15	BP rev	2019DIK
				BP	319.37	0.15	BP rev	2012KRO
				BP	319.37	0.15		1972BOU
				BP	318.35	0.15		1949FEW
C ₂ H ₆ S ₂	dimethyl disulfide	liq	SEL	BP	382.89	0.15	VP	1950SCO
				BP	382.86	0.15	BP rev	2019DIK
				BP	382.89	0.15	VP	1952WHI
C ₂ H ₆ S	dimethyl sulfide	liq	SEL	BP	310.45	0.15	VP ebul	1942OSB
				BP	310.47	0.15	BP rev	2019DIK
				BP	310.42	0.15	BP rev	2012KRO
				BP	310.43	0.15	GLC	1957DES
				BP	310.55	0.15	BP ebul	1956HAI
				BP	310.43	0.15	VP	1952WHI
C ₃ H ₈ S	ethyl methyl sulfide	liq	SEL	BP	339.80	0.15	VP ebul	1952WHI
				BP	339.76	0.15	BP rev	2019DIK
C ₄ H ₁₀ S	diethyl sulfide	liq	SEL	BP	365.25	0.15	VP ebul	1966OSB
				BP	365.21	0.15	BP rev	2019DIK
				BP	365.21	0.15	GLC	1957DES
				BP	365.21	0.15	VP	1952WHI
	isopropyl methyl sulfide	liq	SEL	BP	358.0	0.2		1956HAI
				BP	357.88	0.05		2019DIK
				BP	357.8		VP	1952WHI
				BP	358.0			1951MCA
	methyl isopropyl sulfide	liq		BP	357.88	0.15	BP rev	2019DIK
				BP	358.0	0.2	BP ebul	1956HAI
				BP	357.97	0.15	VP	1952WHI
	methyl propyl sulfide	liq	SEL	BP	368.69	0.15	VP	1952WHI
				BP	368.66	0.15	BP rev	2019DIK

C ₅ H ₁₂ S	ethyl isopropyl sulfide	liq	SEL	BP	368.80	0.15	BP ebul	1956HAI
				BP	380.4	1.0	VP	1952WHI
				BP	380.50	0.02		2019DIK
				BP	380.5			1951MCA
				BP	376.1			1933ING
	ethyl propyl sulfide	liq	SEL	BP	391.65	0.15	VP	1952WHI
				BP	391.62	0.15	BP rev	2019DIK
				BP	391.65	0.15	GLC	1957DES
C ₆ H ₁₄ S	di- <i>n</i> -propyl sulfide	liq	SEL	BP	415.98	0.15	VP	1952WHI
				BP	415.96	0.15	BP rev	2019DIK
				BP	415.85	0.15	BP ebul	1960HAI
C ₈ H ₁₈ S	di- <i>n</i> -butyl sulfide	liq	SEL	BP	462.06	0.15	VP	1952WHI
				BP	462.06	0.15	BP rev	2019DIK
				BP	461.95	0.15	BP ebul	1960HAI
C ₂ H ₄ S	thiacyclopropane	liq	SEL	BP	328.1	0.4	VP	1952GUT
				BP	328.06	0.01		2019DIK
				BP	328.9			1947LEC
Thiacycloalkanes								
C ₃ H ₆ S	thiacyclobutane	liq	SEL	BP	368.11	0.15	BP rev	2019DIK
C ₄ H ₈ S	thiacyclopentane	liq	SEL	BP	394.27	0.15	VP ebul	1966OSB
				BP	394.26	0.15	BP rev	2019DIK
				BP	393.42	0.15	VP isoten	1971EON
				BP	394.26	0.15		1962DAV
				BP	394.60	0.15	VP	1952WHI
C ₅ H ₁₀ S	thiacyclohexane	liq	SEL	BP	368.11	0.15	BP rev	2019DIK
Thiacycloalkenes								
C ₄ H ₄ S	thiacyclopentadiene	liq	SEL	BP	357.31	0.15	VP	1949WAD
				BP	357.24	0.15	BP rev	2019DIK
				BP	356.52	0.15	VP isoten	1971EON
				BP	357.25	0.15	GLC	1957DES
				BP	357.25	0.15	VP	1952WHI
Aromatic Thiols								
C ₆ H ₆ S	benzenethiol	liq	SEL	BP	442.29	0.15	VP ebul	1966OSB
				BP	442.25	0.15	BP rev	2019DIK
				BP	442.25	0.15	BP ebul	1960HAI
Alkyl Sulfoxides								
C ₂ H ₆ OS	dimethyl sulfoxide	liq	SEL	BP	463.86	0.20	VP	1972JAK
		sol	SEL1	MP	291.6	0.3	MP	1948DOU

liq	BP	464.37	0.22	BP rev	2019DIK
-----	----	--------	------	--------	---------

HALOGEN-SUBSTITUTED HYDROCARBONS								
Fluoromethanes								
CH ₃ F	fluoromethane	gas	SEL	BP	194.80	0.06	VP	1948MIC
				BP	194.89	0.09	BP rev	2019DIK
				BP	194.836	0.023	EOS-rev	2018LEM
				BP	194.836	0.023	EOS	2006LEM
				BP	195.2			1991PAP
				BP	194.8	1.6	VP	1998HOL
				BP	194.3	1.0		1989FON
				BP	194.86	0.05	VP	1983OI
				BP	195.2			1968FIL
				BP	194.7		BP	1940GRO
CH ₂ F ₂	difluoromethane	gas	SEL	BP	221.499	0.008	EOS	1997TIL
				BP	221.499	0.008	EOS-rev	2018LEM
				BP	221.5		BP rev	2010SYN
				BP	221.4			1997BAR
				BP	221.7	1.0		1994WID
				BP	221.48	0.05		1993WEB
				BP	221.42	0.05		1987KAN
				BP	221.45	0.17		1973LOS
				BP	221.46	0.15	VP	1968MAL
				BP	221.2			1968FIL
				BP	221.6			1964CRO
				BP	221.6			1939HEN
CHF ₃	trifluoromethane	gas	SEL	BP	191.132	0.017	EOS-rev	2003PEN
				BP	191.132	0.017	EOS-rev	2018LEM
				BP	188.8		BP rev	2010SYN
				BP	191.15	0.04	VP fit	1982POP
				BP	191.1			1962VAL
				BP	191.0			1964CRO
				BP	189.0			1962VAL
				BP	190.98	0.15	VP	1962VAL
				BP	191.0	0.5		1959HOU
				BP	191.19	0.05		1971STR
				BP	191.0			1956THO
				BP	189.0			2003PEN
CF ₄	tetrafluoromethane	gas	SEL	BP	145.24	0.15	VP	1979LOB
				BP	145.20	0.04	BP rev	2019DIK

				BP	145.11	0.07	EOS-rev	2018LEM
				BP	145.21	0.15		2010YU
				BP	145.6	1.0		2007ZHU
				BP	145.11	0.07	EOS-rev VP	1990PLA
				BP	145.56	0.20		1972REG
				BP	145.0			1971STR
				BP	145.1	0.15		1969SMI
				BP	145.2			1968FIL
				BP	145.16	0.20		1968BLA
				BP	145.13	0.25		1967SIM
				BP	145.1			1967GIL
				BP	145.1			1964CRO
				BP	145.34	0.20		1960CHA
				BP	145.1		1956THO	
				BP	145.4		1933THO	
BP	145.15	0.15	VP	1933MEN				
Fluoroethanes								
C ₂ H ₅ F	fluoroethane	gas	SEL	BP	235.69	0.15	VP	1975VID
				BP	235.55	0.27	BP rev	2019DIK
				BP	235.61	0.04	EOS-rev	2018LEM
				BP	235.1	0.5		2018HE
				BP	235.61	0.04	EOS	2016QI
				BP	236.1	0.3	BP rev	2010SYN
				BP	235.7	0.4	VLE extrap	2006HAN
				BP	235.7	0.4	VP	2005CHE
				BP	235.9	0.3	VLE	2001KUL
				BP	235.63	0.15	VLE	1998BEY
				BP	236.0			1990PCR
				BP	235.5	0.3		1989WEA
				BP	235.5	0.3	BP	1936GRO
C ₂ H ₄ F ₂	1,1-difluoroethane	gas	SEL	BP	249.13	0.05	EOS	1996OUT
				BP	249.13	0.05	EOS-rev	2018LEM
				BP	248.5		BP rev	2010SYN
				BP	249.1		VP ebul	1993SIL
				BP	246.7	1.0	VP rev	1947STU
				BP	248.5		BP	1936HEN
1,2-difluoroethane		liq gas	SEL	BP	304	10		1955EDG
				BP	287	9	BP est	2019DIK

C ₂ H ₃ F ₃	1,1,1-trifluoroethane	liq		BP	293.0		BP rev	2012KRO
				BP	303.9			2010SYN
				BP	304.5			1960KLA
		gas		BP	283.7		BP	1936HEN
		gas	SEL	BP	225.91	0.04	EOS	2000LEM
				BP	225.92	0.18	BP rev	2019DIK
				BP	225.91	0.04	EOS-rev	2018LEM
				BP	225.6		BP rev	2010SYN
				BP	226.2		VLE	2001KUL
				BP	226		BP rev	1990PCR
	1,1,2-trifluoroethane	gas	SEL	BP	225.8	0.4	VP	1944RUS
				BP	226.5		BP	1936HEN
				BP	226.4			1933SWA
				BP	276.8	0.3	BP rev	2019DIK
				BP	278.2		BP rev	2010SYN
				BP	276.1			1940YOU
				BP	278.2	1.0	BP	1936HEN
				BP	247.076	0.009	EOS	1994TIL
				BP	247.07	0.22	BP rev	2019DIK
				BP	247.076	0.009	EOS-rev	2018LEM
C ₂ H ₂ F ₄	1,1,1,2-tetrafluoroethane	gas	SEL	BP	247.1	0.5	VLE	2011DON
				BP	246.7		BP rev	2010SYN
				BP	247.0	0.1	VLE	1996BLA
				BP	247.076	0.004	EOS	1994TIL
				BP	247.02	0.22	VLE	1992GOO
				BP	246.6		BP rev	1990PCR
				BP	247.15	0.25	VLE	1990NIS
				BP	248.5		quote	1963EDG
				BP	253.23	0.15	VP	1991MAE
				BP	253.44	0.12	BP rev	2019DIK
	1,1,2,2-tetrafluoroethane	gas	SEL	BP	253.7		BP rev	2010SYN
				BP	250.15	0.08	VP rev	1979KUD
				BP	250.2			1959EIS
				BP	250.2		VP	1940YOU
				BP	225.06	0.04	EOS	2005LEM
				BP	225.06	0.04	EOS-rev	2018LEM
				BP	225.3	0.7	VP	2001KUL
				BP	224.6		BP	1999TAK
C ₂ HF ₅	pentafluoroethane	gas	SEL	BP	225.06	0.04	EOS	2005LEM
				BP	225.06	0.04	EOS-rev	2018LEM
				BP	225.3	0.7	VP	2001KUL
				BP	224.6		BP	1999TAK

C ₂ F ₆	hexafluoroethane	gas	SEL	BP	225.1	0.2	BP	1997BAR
				BP	225.05	0.07	VP	1994WEB
				BP	224.83	0.23	VP	1992WIL
				BP	224.7	1.0	BP	1968FIL
				BP	225.2	1.0	BP	1952CUC
				BP	195.06	0.06	EOS	2006LEM
				BP	195.06	0.06	BP rev	2019DIK
				BP	195.06	0.06	EOS-rev	2018LEM
C ₂ F ₆	hexafluoroethane	gas	SEL	BP	194.87	0.20	VP	1948PAC
Fluoroethenes								
C ₂ H ₃ F	fluoroethene	gas	SEL	BP	201.7	0.5	VP rev	1947STU
				BP	201	14	BP est	2019DIK
C ₂ H ₂ F ₂	1,1-difluoroethene	gas	SEL	BP	187.6	1.2	BP rev	2019DIK
				BP	190.2		BP rev	2010SYN
				BP	190.2		BP rev	1989WEA
	1,2-difluoroethene, (E)-	gas	SEL	BP	221	20	BP est	2019DIK
				BP	231		BP rev	2010SYN
	1,2-difluoroethene, (Z)-	gas	SEL	BP	220	21	BP rev	2019DIK
				BP	253		BP rev	2010SYN
C ₂ HF ₃	trifluoroethene	gas	SEL	BP	211.7	1.5	BP	1961CRA
				BP	211.8	0.6	BP rev	2019DIK
				BP	220.2		BP comp	2001HOR
				BP	222		BP rev	1990PCR
				BP	212.7		BP	1953PRO
C ₂ F ₄	tetrafluoroethene	gas	SEL	BP	197.51	0.25	VP	1953FUR
				BP	197.57	0.04	BP rev	2019DIK
				BP	199.2		BP	1953LAZ
				BP	196.7		BP	1953ATK
				BP	196.9		VP	1946REN
				BP	196.8		VP	1933RUF
Chloromethanes								
CH ₃ Cl	chloromethane	gas	SEL	BP	248.99	0.15	VLE	1948GAN
				BP	249.01	0.07	BP rev	2019DIK
				BP	249.0	0.3	VP	1988HOL
				BP	249.1			1966GAL
				BP	249.31	0.15	VP	1947BEE
				BP	249.4			1943MCG
CH ₂ Cl ₂	dichloromethane	liq	SEL	BP	312.92	0.15	VLE	1960MUE

					BP	312.96	0.04	BP rev	2019DIK
					BP	312.9		VLE	1994KRI
					BP	312.9			1972BOU
					BP	312.9			1950MUM
					BP	313.34	0.25	VLE	1948GAN
CHCl ₃	trichloromethane	liq	SEL	BP	334.42	0.18		VP rev	1947STU
				BP	334.31	0.05		BP rev	2019DIK
CCl ₄	tetrachloromethane	liq	SEL	BP	349.88	0.05		VP ebul	1959HIL
				BP	349.86	0.07		BP rev	2019DIK
				BP	349.7			BP rev	2010SYN
Chloroethanes									
C ₂ H ₅ Cl	chloroethane	gas	SEL	BP	285.40	0.15		VP	1948GOR
				BP	285.45	0.06		BP rev	2019DIK
C ₂ H ₄ Cl ₂	1,1-dichloroethane	liq	SEL	BP	330.42	0.15		VP	1956LI
				BP	330.35	0.03		BP rev	2019DIK
				BP	330.2			VLE	1999IGO
				BP	330.4			VLE	1988WIS2
				BP	330.3			VLE	1985KOV
				BP	330.4			VLE	1951NEW
1,2-dichloroethane		liq	SEL	BP	356.70	0.20		VLE	1996KRI
				BP	356.58	0.15		BP rev	2019DIK
				BP	356.65	0.03		EOS-rev	2018LEM
				BP	356.58	0.08		VP	2008AMI
				BP	356.7			VLE	1989MAT
				BP	356.6			VLE	1986PAU2
				BP	356.61	0.13		VP	1951SIE
				BP	356.62	0.08		VP	1943E.W
1,1,1-trichloroethane		liq		BP	356.98	0.15		VP	1929PEA
			SEL	BP	347.35	0.20		BP ebul	1985RED
				BP	347.22	0.07		BP rev	2019DIK
				BP	347.4			VLE	2007RAD
				BP	347.0			VLE	2005KUM
				BP	347.0	0.3		VLE	1996KRI
				BP	346.7			VLE	1995COM
1,1,2-trichloroethane		liq		BP	347.2			VLE	1982VIS
				BP	341.0			VP extrap	1944RUB
			SEL	BP	386.82	0.10		VP rev	1949DRE2
				BP	386.74	0.08		BP rev	2019DIK

				BP	386.8			1967HAN	
				BP	387.2		BP rev	1955TIM	
C ₂ H ₂ Cl ₄	1,1,1,2-tetrachloroethane	liq	SEL	BP	403.3	0.3	VP rev	1949DRE2	
				BP	403.33	0.05	BP rev	2019DIK	
				BP	403.0	0.4	VLE	1985KOV	
1,1,2,2-tetrachloroethane			liq	SEL	BP	422.3	1.0	VP	1950MAT
				BP	419.1	0.6	BP rev	2019DIK	
				BP	419.25	0.15	VLE	1996KRI	
				BP	419.4		BP ebul	1985RED	
C ₂ HCl ₅	pentachloroethane	liq	SEL	BP	435.1	0.3	VLE	1988MIH	
				BP	435	5	BP rev	2019DIK	
				BP	433.6		BP rev	1943MCG	
				BP	435.1		BP	1941EWE	
				BP	434.6			1935EAR	
				BP	441.6	2.0	BP	1930NEL	
				BP	435.1		VLE	1926LEC	
C ₂ Cl ₆	hexachloroethane	sol	SEL	MP	460.0	2.0	MP	1962SHA	
		liq		BP	463.0	0.5	VP rev	1947STU	
				BP	458.2		VP extrap	2019DIK	
			sol	MP	458.9	0.3	MP	1956MIL	
Chloroethenes									
C ₂ H ₃ Cl	chloroethene	gas	SEL	BP	259.36	0.09	VP ebul	1959MCD	
				BP	259.40	0.05	BP rev	2019DIK	
				BP	259.44	0.07	EOS-rev	2018LEM	
C ₂ H ₂ Cl ₂	1,1-dichloroethene	liq	SEL	BP	304.71	0.05	VP ebul	1959HIL2	
				BP	304.69	0.04	BP rev	2019DIK	
				BP	304.55	0.15	VLE	1985KOV2	
				BP	304.8		VLE	1981EFR	
1,2-dichloroethene, (E)-			liq	SEL	BP	320.73	0.25	VP	1947KET
				BP	320.8	0.6	BP rev	2019DIK	
				BP	320.8		VLE	1985MAC	
				BP	320.5		VLE	1985KOV2	
				BP	321.4		VLE	1951NEW	
				BP	321.9		VLE	1951FLO	
1,2-dichloroethene, (Z)-			liq	SEL	BP	333.6	0.5	VP	1947KET
				BP	333.32	0.21	BP rev	2019DIK	
				BP	333.2			1951AND	
				BP	333.5		BP	1941MAR	

C ₂ HCl ₃	trichloroethene	liq	SEL	BP	360.5	0.4	VP isoten	1944MCD
				BP	359.84	0.11	BP rev	2019DIK
C ₂ Cl ₄	tetrachloroethene	liq	SEL	BP	394.29	0.15	VLE	1970POL
				BP	394.25	0.10	BP rev	2019DIK
Chloroacetylenes								
C ₂ HCl	chloroacetylene	gas	SEL	BP	238	10	BP rev	1987STE
				BP	243		explod	2021EPA
				BP	243	9	predict	2021CHE
C ₂ Cl ₂	dichloroacetylene	sol	SEL	explod	306	10	explod	2021EPA
		liq		BP	365	15	BP rev	2012KRO
				BP	306		BP rev	1999LEW
				BP	306	10		1967SMI
Chloroaromatics								
C ₆ H ₅ Cl	chlorobenzene	liq	SEL	BP	405.21	0.25	EOS-rev	2018LEM
				BP	404.7	0.3	BP eval	2019DIK
				BP	404.6	0.5	VLE	1995KRI
				BP	405.2	0.5	VLE	1990JOS
				BP	404.9	0.5	VLE	1989RED
				BP	405.0	0.5	VLE	1988AGU
Bromomethanes								
CH ₃ Br	bromomethane	gas	SEL	BP	276.70	0.15	VP	1947BEE
				BP	276.66	0.15	BP rev	2019DIK
				BP	276.7	0.1	VP rev	1979KUD
CH ₂ Br ₂	dibromomethane	liq	SEL	BP	370.1	0.1	VP rev	1979KUD
				BP	370.1	0.5	BP rev	2019DIK
				BP	371.7	1.0	VP rev	1947STU
CHBr ₃	tribromomethane	liq	SEL	BP	422.36	0.05	VP rev	1979KUD
				BP	422.37	0.23	BP rev	2019DIK
				BP	421.83	0.25	VLE	1941SIM
CBr ₄	tetrabromomethane	sol	SEL	MP	367.6	2.0	MP	1970SIL
		liq	SEL1	BP	462.6	0.6	BP rev	1979KUD
				BP	462.7		BP comp	2021EPA
				BP	462.6	1.0	VP rev	1947STU
Bromoethanes								
C ₂ H ₅ Br	bromoethane	liq	SEL	BP	311.54	0.19	VP	1930ZMA
				BP	311.5	0.4	BP rev	2019DIK
C ₂ H ₄ Br ₂	1,2-dibromoethane	liq	SEL	BP	404.56	0.10	VP rev	1949DRE
				BP	404.58	0.06	BP rev	2019DIK

Bromoethenes								
C ₂ H ₃ Br	bromoethene	liq	SEL	BP	306	18	BP est	2019DIK
		gas		BP	288.6	0.5	VP	1937GUY
Iodoalkanes								
CH ₃ I	iodomethane	liq	SEL	BP	315.58	0.02	VP rev	1979KUD
				BP	315.58	0.09	BP rev	2019DIK
Chlorofluoromethanes								
CH ₂ ClF	chlorofluoromethane	gas	SEL	BP	264.1	1.0	BP rev	1979KUD
				BP	260	13	BP est	2019DIK
				BP	264.1		BP rev	2010SYN
				BP	264			1990PCR
CHClF ₂	chlorodifluoromethane	gas	SEL	BP	232.32	0.02	VP rev	1979KUD
				BP	232.34	0.03	BP rev	2019DIK
				BP	232.4		BP rev	2010SYN
				BP	232.28	0.15	VP	1964KLE
CHCl ₂ F	dichlorofluoromethane	gas	SEL	BP	282.05	0.10	VP rev	1979KUD
				BP	282.01	0.09	BP rev	2019DIK
				BP	282.0	0.8	EOS-rev	2018LEM
				BP	282.0	0.8	EOS	1990PLA
				BP	282.1		BP rev	2010SYN
				BP	282.3	0.2	VP rev	1947STU
CHClF ₂	difluorochloromethane	gas	SEL	BP	232.32	0.12	VP rev	1979KUD
				BP	232.34	0.03	BP rev	2019DIK
CClF ₃	chlorotrifluoromethane	gas	SEL	BP	191.7	0.1	VP rev	1979KUD
				BP	191.75	0.04	BP rev	2019DIK
				BP	191.67	0.02	EOS-rev	2018LEM
				BP	191.8		BP rev	2010SYN
				BP	191.67	0.02	EOS	2000MAG
				BP	191.91	0.18	VP rev	1947STU
CCl ₂ F ₂	dichlorodifluoromethane	gas	SEL	BP	243.38	0.04	VP rev	1979KUD
				BP	243.42	0.05	BP rev	2019DIK
				BP	243.40	0.05	EOS-rev	2018LEM
				BP	243.4		BP rev	2010SYN
				BP	243.40	0.05	EOS	1992MAR
				BP	243.29	0.18	VP rev	1947STU
CCl ₃ F	trichlorofluoromethane	gas	SEL	BP	296.78	0.02	VP rev	1979KUD
				BP	296.84	0.06	BP rev	2019DIK
				BP	296.86	0.06	EOS-rev	2018LEM

				BP	297.0		BP rev	2010SYN
				BP	296.858	0.056	EOS	1992JAC
				BP	296.78	0.06	VP ebul	1941OSB
				BP	296.86	0.10	VP	1940BEN
Chlorofluoroethanes								
C ₂ H ₄ ClF	1-chloro-1-fluoroethane	gas	SEL	BP	289.3	1.5	BP	1936HEN
				BP	289.3	1.5	BP est	2019DIK
				BP	289.3		BP rev	2010SYN
C ₂ H ₃ Cl ₂ F	1,1-dichloro-1-fluoroethane	liq	SEL	BP	305.26	0.25	VP	1991MAE
				BP	305.222	0.024	BP rev	2019DIK
				BP	305.2		BP rev	2010SYN
				BP	304.7		VP	1997DUA
				BP	305.60	0.15	VP	1993DEF
				BP	305.4		VP ebul	1992WEB
				BP	305.10	0.25	BP	1943HEN
				BP	304.9		BP	1936HEN
				1,2-dichloro-1-fluoroethane		liq	SEL	BP
			BP	346.9	1.9	BP est	2019DIK	
			BP	349.2		BP rev	2010SYN	
C ₂ H ₃ ClF ₂	1-chloro-1,1-difluoroethane	gas	SEL	BP	264.04	0.25	VP ebul	1993SIL
				BP	264.04	0.24	BP rev	2019DIK
				BP	264.0		BP rev	2010SYN
				BP	263.6		BP	1936HEN
1-chloro-2,2-difluoroethane		liq	SEL	BP	308.3	0.2	BP	1936HEN
			BP	309.2		BP rev	2010SYN	
C ₂ H ₂ ClF ₃	2-chloro-1,1,1-trifluoroethane	gas	SEL	BP	279.3	0.2	BP	1936HEN
				BP	279.19	0.22	BP rev	2019DIK
				BP	280.1		BP rev	2010SYN
Chlorofluoroethenes								
C ₂ HCl ₂ F	1,2-dichloro-1-fluoroethene	liq	SEL	BP	305.7	2.5	BP rev	2010SYN
				BP	308		BP rev	1990PCR
C ₂ ClF ₃	chlorotrifluoroethene	gas	SEL	BP	244.8	0.1	VP	1951OLI
				BP	244.85	0.19	BP rev	2019DIK
				BP	246			1990PCR
				BP	244.7		BP rev	1990ALD
				BP	246.0			1982ERA
Bromofluoromethanes								
CH ₂ BrF	bromofluoromethane	gas	SEL	BP	290.7	2.0	BP rev	1979KUD

		liq		BP	310	12	BP est	2019DIK
				BP	310.0		BP rev	2012KRO
		gas		BP	290.7		BP rev	2010SYN
				BP	292.2		VP rev	1999DYK
CHBrF ₂	bromodifluoromethane	gas	SEL	BP	257.5	0.2	VP rev	1979KUD
				BP	257.5	0.4	BP rev	2019DIK
				BP	258.7		BP rev	2010SYN
CHBr ₂ F	dibromodifluoromethane	liq	SEL	BP	337	2	BP rev	1979KUD
				BP	351	13	BP est	2019DIK
				BP	338.1		BP rev	2010SYN
CBrF ₃	bromotrifluoromethane	gas	SEL	BP	215.30	0.20	VP ebul	1959MCD
				BP	215.32	0.08	BP rev	2019DIK
				BP	215.4		BP rev	2010SYN
				BP	215.3	0.3	VP rev	1979KUD
CBr ₂ F ₂	dibromodifluoromethane	gas	SEL	BP	295.93	0.04	VP rev	1979KUD
				BP	295.94	0.04	BP rev	2019DIK
				BP	297.7		BP rev	2010SYN
				BP	295.94	0.05	VP ebul	1959MCD
CBr ₃ F	tribromodifluoromethane	liq	SEL	BP	379.6	1.0	BP rev	1979KUD
				BP	404	17	BP est	2019DIK
				BP	379.7		BP rev	2010SYN
Bromochloromethanes								
CH ₂ BrCl	bromochloromethane	liq	SEL	BP	341.19	0.08	VP rev	1979KUD
				BP	341.12	0.04	BP rev	2019DIK
				BP	341.22	0.05	VP ebul	1959MCD
CHBrCl ₂	bromodichloromethane	liq	SEL	BP	363.3	0.4	BP rev	1979KUD
				BP	363	3	BP est	2019DIK
				BP	362.7		BP rev	2010SYN
CHBr ₂ Cl	dibromochloromethane	liq	SEL	BP	393	2	BP rev	1979KUD
				BP	405	13	BP est	2019DIK
CBrCl ₃	bromotrichloromethane	liq	SEL	BP	378.1	1.0	VP rev	1979KUD
				BP	378	3	BP est	2019DIK
				BP	376	4	BP rev	2012KRO
				BP	376.7		BP	1950ZAK
CBr ₂ Cl ₂	dibromodichloromethane	liq	SEL	BP	393	2	BP rev	1979KUD
				BP	423.2			2021EPA
				BP	414	22	BP est	2019DIK
				BP	423.4		BP rev	2014HAY

CBr ₃ Cl	tribromochloromethane	liq		BP	329.9		BP rev	1955VAR
			SEL	BP	433	1	BP rev	1979KUD
				BP	452	18	BP est	2019DIK
Bromochlorofluoromethanes								
CHBrClF	bromochlorofluoromethane	liq	SEL	BP	309	2	BP rev	1979KUD
				BP	313	16	BP est	2019DIK
				BP	309.7		BP	1952HAS
				BP	309.3	1.0	BP	1942BER
CBrClF ₂	bromochlorodifluoromethane	gas	SEL	BP	269.14	0.04	VP rev	1979KUD
				BP	269.24	0.03	BP rev	2019DIK
				BP	270.7		BP rev	2010SYN
				BP	269.19	0.15	VP	1960GLE
CBrCl ₂ F	bromodichlorofluoromethane	liq	SEL	BP	323	15	VP	1979KUD
				BP	338	20	BP est	2019DIK
				BP	325		BP rev	2010SYN
				BP	323.6			1957MIL
CBr ₂ ClF	dibromochlorofluoromethane	liq	SEL	BP	353	2	BP rev	1979KUD
				BP	376	19	BP est	2019DIK
				BP	353		BP rev	2010SYN
				BP	353.2	1.0	BP	1952HAS
Carbonyl Halides								
CHFO	carbonyl fluoride	gas	SEL	BP	245.2	2.0	VP	1964FIS
CF ₂ O	carbonyl difluoride	gas	SEL	BP	188.62	0.20	VP	1968PAC
CCl ₂ O	phosgene	gas	SEL	BP	280.66	0.10	VP	1948GIA
				BP	280.66	0.17	BP rev	2019DIK
Chlorophenols								
C ₆ HCl ₅ O	pentachlorophenol	liq	SEL	BP	583	3	VP	1947STU
		sol	SEL1	MP	462.8	1.0	MP	1959MCD
		liq		BP	588	19	BP est	2019DIK
		sol		MP	463.8	1.0	MP	1990DON

Siloxanes								
C ₆ H ₁₈ OSi ₂	hexamethyldisiloxane	liq	SEL	BP	373.64	0.10	VP	1961SCO
C ₈ H ₂₄ O ₄ Si ₄	octamethylcyclotetrasiloxane	liq	SEL	BP	448.9	0.6	EOS	2016THO
		sol	SEL1	MP	290.7	1.0	MP	1968MAR
		liq		BP	446.2	2.0	VLE	1969MYE
				BP	448.0	1.0	VP	1954OST

2.3. Recommended Melting and Boiling Points

Table 2-2. Recommended Normal Melting and Normal Boiling Points

Table 2-2(1). Small Molecules

Formula	Compound	Phase	SEL	trs	T_{trs}/K	unc/K	Method	Reference
Hydrogen/Oxygen								
H ₂	hydrogen	gas	SEL	BP	20.369	0.014	EOS	2009LEA
O ₂	oxygen	gas	SEL	BP	90.188	0.029	EOS	1991STE
H ₂ O	water	liq	SEL	BP	373.124	0.014	EOS	2002WAG
			SEL1	MP	273.153	0.001	EOS	2006FEI
		sol	SEL2	TP	273.16	0.01	DEF	1990PRE
			H ₂ O ₂	hydrogen peroxide	liq	SEL	BP	426.31
sol	SEL1	MP			272.699	0.008	SLE	1951FOL
	SEL2	TP			272.75	0.02	TP	1954GIG

N ₂	nitrogen	gas	SEL	BP	77.355	0.007	EOS	2000SPA
H ₃ N	ammonia	gas	SEL	BP	239.823	0.020	EOS	2018GAO
H ₄ N ₂	hydrazine	liq	SEL	BP	386.7	0.9	BP	1989TIP
HN ₃	hydrogen azide	liq	SEL	BP	308.9	1.5	BP rev	2007LID
Nitrogen/Oxygen								
N ₂ O	nitrous oxide	gas	SEL	BP	184.75	0.07	cal	1974ATA
ClH ₄ NO ₄	ammonium perchlorate	sol	SEL	dec	595.9	4.0	DTA	2004LIU
			SEL	TP	513.2	1.0	TP rev	2019DIK
Sulfur/Hydrogen/Oxygen								
H ₂ S	hydrogen sulfide	gas	SEL	BP	212.85	0.10	EOS	2006LEM
O ₂ S	sulfur dioxide	gas	SEL	BP	263.14	0.09	EOS	2016GAO
Boron/Hydrogen								
B ₂ H ₆	diborane	gas	SEL	BP	180.6	0.1	VP	1959PAR
Hydrogen Halides								
ClH	hydrogen chloride	gas		BP	188.173	0.088	EOS	2018THO

Table 2-2(2). Hydrocabons

Formula	Compound	Phase	SEL	trs	T_{trs}/K	unc/K	Method	Reference
Carbon								
C	graphite	sol	SEL	MP	4800	100	MP rev	2005SAV
Alkanes								
CH ₄	methane	gas	SEL	BP	111.667	0.013	EOS	1991SET
C ₂ H ₆	ethane	gas	SEL	BP	184.566	0.004	EOS	2006BUC
C ₃ H ₈	propane	gas	SEL	BP	231.039	0.009	EOS	2009LEM
C ₄ H ₁₀	isobutane	gas	SEL	BP	261.40	0.26	EOS	2006BUC2
	<i>n</i> -butane	gas	SEL	BP	272.66	0.12	EOS	2006BUC2
C ₅ H ₁₂	isopentane	liq	SEL	BP	300.99	0.07	VP	1945WIL
	neopentane	gas	SEL	BP	282.650	0.028	EOS	2006LEM
	<i>n</i> -pentane	liq	SEL	BP	309.20	0.07	VP	1945WIL
C ₆ H ₁₄	2,3-dimethylbutane	liq	SEL	BP	331.09	0.07	VP	1945WIL
	3-methylpentane	liq	SEL	BP	336.41	0.07	VP	1945WIL
	isohexane	liq	SEL	BP	333.40	0.07	VP	1945WIL
	neohexane	liq	SEL	BP	322.87	0.07	VP	1945WIL
	<i>n</i> -hexane	liq	SEL	BP	341.86	0.05	VP	1945WIL
C ₇ H ₁₆	2,2,3-trimethylbutane	liq	SEL	BP	354.01	0.10	VP	1949FOR
	2,3-dimethylpentane	liq	SEL	BP	362.91	0.10	VP	1949FOR
	2,4-dimethylpentane	liq	SEL	BP	353.7	0.3	VLE	1976TRI
	3,3-dimethylpentane	liq	SEL	BP	359.19	0.05	VP	1945WIL
	3-methylhexane	liq	SEL	BP	364.97	0.10	VP	1949FOR
	isoheptane	liq	SEL	BP	363.18	0.10	VP	1949FOR
	neoheptane	liq	SEL	BP	352.33	0.07	VP	1945WIL
	<i>n</i> -heptane	liq	SEL	BP	371.55	0.07	VP	1945WIL
C ₈ H ₁₈	2,2,4-trimethylpentane	liq	SEL	BP	372.36	0.07	VP	1945WIL
	2,2-dimethylhexane	liq	SEL	BP	379.97	0.07	VP	1945WIL
	2-methylheptane	liq	SEL	BP	390.77	0.07	VP	1945WIL
	3-methylheptane	liq	SEL	BP	392.05	0.07	VP	1945WIL
	<i>n</i> -octane	liq	SEL	BP	398.79	0.14	EOS	2018LEM
C ₉ H ₂₀	isononane	liq	SEL	BP	416.5	0.3	VP	1948ROS
	neononane	liq	SEL	BP	403.63	0.15	BP rev	1947FEN
	<i>n</i> -nonane	liq	SEL	BP	423.93	0.07	VP	1945WIL
C ₁₀ H ₂₂	<i>n</i> -decane	liq	SEL	BP	447.27	0.07	VP	1945WIL

C ₁₁ H ₂₄	<i>n</i> -undecane	liq	SEL	BP	469.05	0.15	BP rev	1947FEN
C ₁₂ H ₂₆	<i>n</i> -dodecane	liq	SEL	BP	489.48	0.05	VLE	1996DEJ
		sol	SEL1	MP	263.5	0.2	MP	1983OTT
			SEL2	TP	263.6		EOS	2004LEM
C ₁₆ H ₃₄	<i>n</i> -hexadecane	liq	SEL	BP	559.9	0.2	VP	1954CAM
		sol	SEL1	MP	291.3	0.1	MP	1986SNO
		liq		BP	559.9	0.2	EOS-rev	2018LEM
		sol	SEL2	TP	291.3		TP rev	2018LEM
Alkenes								
C ₂ H ₄	ethene	gas	SEL	BP	169.38	0.02	EOS	2000SMU
C ₃ H ₆	propene	gas	SEL	BP	225.53	0.04	EOS-rev	2018LEM
C ₄ H ₈	1-butene	gas	SEL	BP	266.87	0.07	VP	1940LAM
	2-butene, (E)-	gas	SEL	BP	274.02	0.15	VP	1945GUT
	2-butene, (Z)-	gas	SEL	BP	276.87	0.08	VP	1944SCO
	isobutene	gas	SEL	BP	266.04	0.15	VP	1940LAM
	1-pentene	liq	SEL	BP	303.11	0.08	VP	1949SCO
C ₅ H ₁₀	2-methyl-1-butene	liq	SEL	BP	304.29	0.13	VP	1949SCO
	2-methyl-2-butene	liq	SEL	BP	311.70	0.15	VP	1949SCO
	2-pentene, (E)-	liq	SEL	BP	309.48	0.08	VP	1950SCO2
	2-pentene, (Z)-	liq	SEL	BP	310.07	0.08	VP ebul	1950SCO2
	3-methyl-1-butene	gas	SEL	BP	293.20	0.08	VP	1950SCO2
C ₆ H ₁₂	1-hexene	liq	SEL	BP	336.64	0.06	VP	1950FOR
C ₇ H ₁₄	1-heptene	liq	SEL	BP	366.71	0.06	VP	1950FOR
C ₈ H ₁₆	1-octene	liq	SEL	BP	394.43	0.06	VP	1950FOR
C ₉ H ₁₈	1-nonene	liq	SEL	BP	420.02	0.06	VP	1950FOR
C ₁₀ H ₂₀	1-decene	liq	SEL	BP	443.72	0.06	VP	1950FOR
Alkadienes								
C ₃ H ₄	allene	gas	SEL	BP	238.3	0.5	VP rev	1947STU
C ₄ H ₆	1,2-butadiene	gas	SEL	BP	284.1	0.4		1946STR
	1,3-butadiene	gas	SEL	BP	268.44	0.15	VLE	1976STE
C ₅ H ₈	1,2-pentadiene	liq	SEL	BP	318.01	0.06	VP	1950FOR
	1,3-pentadiene	liq	SEL	BP	315.3	0.5	BP distill	1935MUL
	1,4-pentadiene	liq	SEL	BP	299.12	0.15	VP	1950FOR
	2-methyl-1,3-butadiene	liq	SEL	BP	307.19	0.15	VLE	1985HOW
Alkynes								
C ₂ H ₂	acetylene	gas	SEL	BP	188.44	0.10	VP extrap	1956AMB
		sol	SEL1	TP	191.75	0.15	EOS-rev	2018LEM
C ₃ H ₄	propyne	gas	SEL	BP	249.80	0.15	VP	1967VAN

C ₄ H ₆	1-butyne	gas	SEL	BP	281.22	0.10	VP	1950AST
	2-butyne	liq	SEL	BP	300.1	0.2		1954POM2
C ₅ H ₈	1-pentyne	liq	SEL	BP	313.5	0.3	VLE	2021TSU
	2-pentyne	liq	SEL	BP	329.3	0.2		1954POM2
C ₆ H ₁₀	1-hexyne	liq	SEL	BP	344.5	0.2	VLE	2021TSU
	2-hexyne	liq	SEL	BP	357.6	0.2	VLE	2021TSU
	3-hexyne	liq	SEL	BP	354.6	0.3		1954POM2
C ₇ H ₁₂	1-heptyne	liq	SEL	BP	372.9	0.3	BP	1945HEN
C ₈ H ₁₄	1-octyne	liq	SEL	BP	399.4	0.3	BP	1945HEN
Alkenynes								
C ₄ H ₄	butenyne	gas	SEL	BP	276.2	2.0		1954HEN
C ₅ H ₆	1-penten-3-yne	liq	SEL	BP	332	4		1933JAC
	3-penten-1-yne	liq	SEL	BP	317.1	3.0		1954POM2
Alkadiynes								
C ₄ H ₂	butadiyne	gas	SEL	BP	283.7	1.0		1954POM2
Cycloalkanes								
C ₃ H ₆	cyclopropane	gas	SEL	BP	240.33	0.12	VP	1946RUE1
		sol	SEL1	MP	145.55	0.14	heat	1946RUE1
C ₄ H ₈	cyclobutane	gas	SEL	BP	285.6	0.2	VP	1953RAT
C ₅ H ₁₀	cyclopentane	liq	SEL	BP	322.39	0.07	VP	1945WIL
	methylcyclobutane	liq	SEL	BP	310.3	1	BP distil	1953PIN2
C ₆ H ₁₂	cyclohexane	liq	SEL	BP	353.86	0.07	VP	1945WIL
	methylcyclopentane	liq	SEL	BP	344.94	0.07	VP	1945WIL
C ₇ H ₁₄	1,1-dimethylcyclopentane	liq	SEL	BP	360.97	0.10	VP	1949FOR
	cycloheptane	liq	SEL	BP	391.92	0.2	VP	1956FIN
	ethylcyclopentane	liq	SEL	BP	376.59	0.10	VP	1949FOR
	methylcyclohexane	liq	SEL	BP	374.06	0.07	VP	1945WIL
C ₈ H ₁₆	ethylcyclohexane	liq	SEL	BP	404.91	0.07	VP	1945WIL
	isopropylcyclopentane	liq	SEL	BP	399.55	0.06	VP	1949FOR
	<i>n</i> -propylcyclopentane	liq	SEL	BP	404.08	0.06	VP	1949FOR
C ₉ H ₁₈	isopropylcyclohexane	liq	SEL	BP	427.70	0.10	VP	1949FOR
	propylcyclohexane	liq	SEL	BP	429.86	0.10	VP	1949FOR
C ₁₀ H ₂₀	1,3-diethylcyclohexane	liq	SEL	BP	445.7	1.5		1962ALL
	isobutylcyclohexane	liq	SEL	BP	444.46	0.07	VP	1949FOR
	<i>tert</i> -butylcyclohexane	liq	SEL	BP	444.73	0.10	VP	1949FOR
Cycloalkenes								
C ₃ H ₄	cyclopropene	gas	SEL	BP	237.6	1.5	BP	1941SCH
C ₄ H ₆	cyclobutene	gas	SEL	BP	275.6	0.2	VP	1941HEI

C ₅ H ₈	cyclopentene	liq	SEL	BP	317.37	0.06	VP	1950FOR
	methylenecyclobutane	liq	SEL	BP	315.33	0.15	BP rev	1947FEN
C ₆ H ₁₀	3-methylcyclopentene	liq	SEL	BP	338	1	BP	1945CRA
	cyclohexene	liq	SEL	BP	356.08	0.04	VP	1973MEY
C ₇ H ₁₂	1-ethylcyclopentene	liq	SEL	BP	279.5	0.2	VP rev	1946API
	1-methylcyclohexene	liq	SEL	BP	383.40	0.15	VP	1960CAM
	3-ethylcyclopentene	liq	SEL	BP	371.3		BP	1945CRA
	cycloheptene	liq	SEL	BP	386.1	0.5	GLC	1958TUR
C ₈ H ₁₄	methylenecyclohexane	liq	SEL	BP	376.55	0.04	VP	1973MEY
	1,2-dimethylcyclohexene	liq	SEL	BP	409	2	BP	1938CHI
	1-ethylcyclohexene	liq	SEL	BP	410.09	0.05	VP	1960CAM
Cycloalkadienes								
C ₄ H ₄	cyclobutadiene	gas	SEL	BP	286	12	BP est	2019DIK
C ₅ H ₆	1,3-cyclopentadiene	liq	SEL	BP	313.4	0.2	VP	1965HUL
C ₆ H ₈	1,3-cyclohexadiene	liq	SEL	BP	353.462	0.005	VP	1989STE
	1,4-cyclohexadiene	liq	SEL	BP	362.653	0.005	VP	1989STE
C ₈ H ₁₂	1,5-cyclooctadiene	liq	SEL	BP	421.8	2.0		1985TAS
Bicycloalkanes								
C ₁₀ H ₁₈	<i>cis</i> -decalin	liq	SEL	BP	468.90	0.10	VP	1955CAM
	<i>trans</i> -decalin	liq	SEL	BP	460.40	0.10	VP	1955CAM
C ₁₂ H ₂₂	cyclohexylcyclohexane	liq	SEL	BP	511.1	0.3	VP	1980WIE
		sol	SEL1	MP	277.2	0.5	MP	1983O'R

Table 2-2(3). Aromatic Hydrocarbons

Formula	Compound	Phase	SEL	trs	T_{trs}/K	unc/K	Method	Reference
Alkylbenzenes								
C ₆ H ₆	benzene	liq	SEL	BP	353.23	0.07	VP	1945WIL
		sol	SEL1	MP	278.67	0.02	MP	1996LIN
C ₇ H ₈	toluene	liq	SEL	BP	383.75	0.07	VP	1945WIL
		sol	SEL1	MP	178.2	0.2	MP	1976RAD
C ₈ H ₁₀	1,2-dimethylbenzene	liq	SEL	BP	417.55	0.07	VP	1945WIL
	1,3-dimethylbenzene	liq	SEL	BP	412.23	0.07	VP	1945WIL
	1,4-dimethylbenzene	liq	SEL	BP	411.48	0.07	VP	1945WIL
	ethylbenzene	liq	SEL	BP	409.32	0.07	VP	1945WIL
C ₉ H ₁₂	1,2,3-trimethylbenzene	liq	SEL	BP	449.22	0.15	VP	1967AMB
	1,2,4-trimethylbenzene	liq	SEL	BP	442.49	0.15	VP	1967AMB
	1,3,5-trimethylbenzene	liq	SEL	BP	437.86	0.15	VP	1967AMB
	2-ethyltoluene	liq	SEL	BP	438.24	0.15	VP	1949FOR
	3-ethyltoluene	liq	SEL	BP	434.44	0.10	VP	1949FOR
	4-ethyltoluene	liq	SEL	BP	435.13	0.10	VP	1949FOR
	isopropylbenzene	liq	SEL	BP	425.53	0.07	VP	1945WIL
	<i>n</i> -propylbenzene	liq	SEL	BP	432.35	0.07	VP	1945WIL
C ₁₀ H ₁₄	1,2-diethylbenzene	liq	SEL	BP	456.57	0.10	VP	1949FOR
	1,3-diethylbenzene	liq	SEL	BP	454.25	0.15	VP	1949FOR
	1,4-diethylbenzene	liq	SEL	BP	456.90	0.10	VP	1949FOR
	2-isopropyltoluene	liq	SEL	BP	451.45	0.20	VP ebul	1959MCD
	3-isopropyltoluene	liq	SEL	BP	448.05	0.12	VP ebul	1959MCD
	4-isopropyltoluene	liq	SEL	BP	450.26	0.06	VP ebul	1959MCD
	Isobutylbenzene	liq	SEL	BP	445.95	0.15	BP rev	1947FEN
	<i>n</i> -butylbenzene	liq	SEL	BP	456.42	0.10	VP	1949FOR
	sec-butylbenzene	liq	SEL	BP	446.30	0.15	BP rev	1947FEN
	<i>tert</i> -butylbenzene	liq	SEL	BP	442.27	0.15	VP	1949FOR
Alkenylbenzenes								
C ₈ H ₈	ethenylbenzene	liq	SEL	BP	418.5	0.3		1950BOO
Benzocycloalkanes								
C ₉ H ₁₀	benzocyclopentane	liq	SEL	BP	450.938	0.002	VP ebul	2016CHI
C ₉ H ₁₀	benzocyclopentane	sol	SEL1	TP	221.786	0.01	VP ebul	2016CHI
C ₁₀ H ₁₂	benzocyclohexane	liq	SEL	BP	480.6	0.5		1953KAR
Benzocycloalkenes								
C ₉ H ₈	benzocyclopentene	liq	SEL	BP	455.72	0.15	BP rev	1947FEN

Alkylnaphthalenes								
C ₁₀ H ₈	naphthalene	sol	SEL	MP	353.4	0.3	DSC	1980KRA
		liq	SEL1	BP	491.13	0.10	EOS	1993CHI
C ₁₁ H ₁₀	1-methylnaphthalene	liq	SEL	BP	517.56	0.10	VP	1981WIE
		sol	SEL1	MP	242.65	0.15	MP	1954HIP
	2-methylnaphthalene	sol	SEL	MP	307.3	0.4	MP	1958RAM
		liq	SEL1	BP	514.10	0.10	VP	1981WIE
Polycyclic Aromatics								
C ₁₂ H ₈	acenaphthylene	sol	SEL	MP	366.4	1.0	MP	1965BOY
		liq	SEL1	BP	551	5	BP est	2019DIK
	biphenylene	sol	SEL	MP	384.4	0.5	MP	1974YOK
		liq	SEL1	BP	548.5	3.0	BP est	2019DIK
C ₁₂ H ₁₀	acenaphthene	sol	SEL	MP	364.0	1.0	MP	1990DON
		liq	SEL1	BP	550.4	0.4	VP isoten	1923MOR
C ₁₃ H ₁₀	fluorene	liq	SEL	BP	570.2	1.0	VP	1982SIV
		sol	SEL	MP	388	2	MP SLE	1989SED
C ₁₄ H ₁₀	anthracene	sol	SEL	MP	489.5	0.3	MP	1988COO
		liq	SEL1	BP	614.9	0.4	VP isoten	1923MOR
	phenanthrene	sol	SEL	MP	372.7	0.3	MP	1990DON
		liq	SEL1	BP	609.00	0.06	VP ebul	1975OSB
C ₁₆ H ₁₀	fluoranthene	sol	SEL	MP	381.0	0.3	MP	1973CAS
		liq	SEL1	BP	660	5	VP	1955TSY
	pyrene	sol	SEL	MP	424.3	0.6	MP	1987JUD
		liq	SEL1	BP	668	5	VP	1955TSY
C ₁₈ H ₁₂	naphthacene	sol	SEL	MP	613	2	MP	1952INO
		liq	SEL1	BP	716	2	BP	1967WAK
C ₂₀ H ₁₂	perylene	sol		MP	550.9	2.3	TP	2019DIK
C ₂₄ H ₁₂	coronene	sol	SEL	MP	710.5	0.5	MP	1980SMI
		liq	SEL1	BP	798.2		BP rev	1989WEA

Table 2-2(4). Oxidized Hydrocarbons

Formula	Compound	Phase	SEL	trs	T_{trs}/K	unc/K	Method	Reference
Carbon Oxides								
CO ₂	carbon dioxide	sol	SEL	Sub	194.686	0.003	sub rev	1996BED
CO	carbon monoxide	gas	SEL	BP	81.65	0.26	EOS	1979BRO
		sol	SEL1	TP	68.2		EOS	2006LEM
Alkanols								
CH ₄ O	methanol	liq	SEL	BP	337.63	0.12	EOS	1993DER
C ₂ H ₆ O	ethanol	liq	SEL	BP	351.44	0.10	VLE	1998ARC
C ₃ H ₈ O	1-propanol	liq	SEL	BP	370.3	0.2	VLE	1995HIA
	isopropanol	liq	SEL	BP	355.37	0.08	VP	1963BID
C ₄ H ₁₀ O	1-butanol	liq	SEL	BP	390.8	0.2	VLE	1996GON
	2-butanol	liq	SEL	BP	372.64	0.14	VP	1963BID
	isobutanol	liq	SEL	BP	380.90	0.15	VLE	1995RED
	<i>tert</i> -butanol	sol	SEL	MP	298.87	0.10	MP	1988KIM
		liq	SEL1	BP	355.47	0.15	VP	1963BID
C ₅ H ₁₂ O	1-pentanol	liq	SEL	BP	410.9	0.3	VLE	1991YOS
	2-methyl-1-butanol	liq	SEL	BP	402.14	0.17	VP ebul	2010CEN
	2-methyl-2-butanol	liq	SEL	BP	375.30	0.17	VP ebul	2010CEN
	2-pentanol	liq	SEL	BP	392.48	0.17	VP ebul	2010CEN
	3-methyl-2-butanol	liq	SEL	BP	385.03	0.17	VP ebul	2010CEN
	3-pentanol	liq	SEL	BP	388.62	0.17	VP ebul	2010CEN
	isopentanol	liq	SEL	BP	404.73	0.17	VP ebul	2010CEN
Glycols								
C ₂ H ₆ O ₂	1,2-ethanediol	liq	SEL	BP	470.35	0.08	VLE	2013QIA
C ₃ H ₈ O ₂	1,2-propanediol	liq	SEL	BP	461.31	0.06	VLE	2011MAT
	2-methoxyethanol	liq	SEL	BP	397.54	0.15	VLE	2012RED
Sugars								
C ₃ H ₈ O ₃	glycerin	liq	SEL	BP	563.03	0.10	BP	2016ANI
		sol	SEL1	MP	291.1	0.5	MP	1958COS
C ₆ H ₁₀ O ₅	cellulose	sol		MP	408.2		MP	2020POD
C ₆ H ₁₂ O ₆	alpha-D-glucose	sol	SEL	MP	421.8	0.6	MP	2004HUR
		gas	SEL1	BP	706	50	est	2019DIK
	fructose	sol	SEL	MP	376.7	0.3	MP	2004HUR
C ₁₂ H ₂₂ O ₁₁	sucrose	sol	SEL	MP	456	4	MP	2004HUR
Phenols								
C ₇ H ₈ O	2-methylphenol	liq		BP	464.1		VP	1958BID

		sol	SEL	MP	304.12	0.10	MP	1960AND
		liq	SEL1	BP	464.14	0.10	VP rev	1949DRE2
4-methylphenol		sol	SEL	MP	307.82	0.20	MP	1960AND
		liq	SEL1	BP	474.99	0.10	VP rev	1949DRE2
Alkyl ethers								
C ₂ H ₆ O	dimethyl ether	gas	SEL	BP	248.37	0.09	EOS	2011WU
C ₃ H ₈ O	ethyl methyl ether	gas	SEL	BP	279.7	0.5		1941IPA
C ₄ H ₁₀ O	diethyl ether	liq	SEL	BP	307.55	0.15	VP	1972AMB
	isopropyl methyl ether	liq	SEL	BP	303.90	0.02		1976AMB
	<i>n</i> -propyl methyl ether	liq	SEL	BP	311.8			1976AMB
C ₅ H ₁₂ O	<i>n</i> -butyl methyl ether	liq	SEL	BP	343.29	0.02	VLE	2007BOU
	<i>tert</i> -butyl methyl ether	liq	SEL	BP	328.20	0.15	VLE	1999MON
C ₆ H ₁₄ O	diisopropyl ether	liq	SEL	BP	341.6	0.3	VLE	1996RES
	di- <i>n</i> -propyl ether	liq	SEL	BP	363.22	0.04	VP	1973MEY
	isobutyl ethyl ether	liq	SEL	BP	354.2	1.0		1932NOR
Aldehydes								
CH ₂ O	formaldehyde	gas	SEL	BP	254.1	0.2	VP	1935SPE
C ₂ H ₄ O	acetaldehyde	gas	SEL	BP	293.5	0.2	VLE	1950COL
C ₃ H ₆ O	propanal	liq	SEL	BP	321.19	0.15	VLE	2010GAO
C ₄ H ₈ O	butanal	liq	SEL	BP	348.0	0.4	VLE	1985HSU
	isobutanal	liq	SEL	BP	337.26	0.25	VP	1959SEP
Alkenals								
C ₃ H ₄ O	1-propenal	gas	SEL	BP	264		BP est	2021CHE
	2-propenal	liq	SEL	BP	325.4	0.3		1979MAR
Ketones								
C ₃ H ₆ O	acetone	liq	SEL	BP	329.22	0.29	EOS	2006LEM
C ₄ H ₈ O	butanone	liq	SEL	BP	352.72	0.03	VLE	1973MAT
C ₅ H ₁₀ O	2-pentanone	liq	SEL	BP	375.42	0.15	VLE	1973SCH
	3-pentanone	liq	SEL	BP	375.08	0.06	VP	1965COL
Carboxylic Acids								
CH ₂ O ₂	formic acid	liq	SEL	BP	374.1	0.4	VLE	2011BAS
		sol	SEL1	MP	281.5	0.2	MP	1981KOH
C ₂ H ₄ O ₂	acetic acid	liq	SEL	BP	391.20	0.15	VLE	1991SAY
		sol	SEL1	MP	289.48	0.16	MP	1959MCD
C ₃ H ₆ O ₂	propanoic acid	liq	SEL	BP	414.0	0.3	VLE	1982WIS
C ₄ H ₈ O ₂	<i>n</i> -butanoic acid	liq	SEL	BP	436.42	0.10	VP rev	1949DRE2
C ₅ H ₁₀ O ₂	<i>n</i> -pentanoic acid	liq	SEL	BP	459.1	0.3		1964ADR
C ₆ H ₁₂ O ₂	<i>n</i> -hexanoic acid	liq	SEL	BP	478.2	0.5		1949PAT

Alkyl Esters								
C ₂ H ₄ O ₂	methyl formate	liq	SEL	BP	305.1	0.3	VP	1928NEL
C ₃ H ₆ O ₂	ethyl formate	liq	SEL	BP	327.29	0.05	VLE	1998SOT
	methyl acetate	liq	SEL	BP	330.11	0.15	VLE	2016MAT
C ₄ H ₈ O ₂	ethyl acetate	liq	SEL	BP	350.21	0.15	VP	1965POL
	methyl propanoate	liq	SEL	BP	352.75	0.25		2014RIO
C ₅ H ₁₀ O ₂	methyl butanoate	liq	SEL	BP	375.74	0.25	VLE	2012RIO
	<i>n</i> -propylacetate	liq	SEL	BP	374.7	0.4		1977SVO
Alkenyl Esters								
C ₄ H ₆ O ₂	vinyl acetate	liq	SEL	BP	345.3	0.4	VLE	1965SWA
C ₅ H ₈ O ₂	ethyl propenoate	liq	SEL	BP	372.59	0.15	VLE	2013LOM
	methyl 2-methylpropenoate	liq	SEL	BP	369.7	0.5	VP	1956BRO
Alkanoic Anhydrides								
C ₄ H ₆ O ₃	acetic anhydride	liq	SEL	BP	412.62	0.15		1976RAD
C ₆ H ₁₀ O ₃	propanoic anhydride	liq	SEL	BP	440.3	1.5		1961WRI
C ₈ H ₁₄ O ₃	butanoic anhydride	liq	SEL	BP	471.4	2.0		1921TIM2
Oxacycloalkanes								
C ₂ H ₄ O	oxacyclopropane	gas	SEL	BP	283.58	0.04	VP ebul	1959MCD
C ₃ H ₆ O	methyloxacyclopropane	liq	SEL	BP	307.36	0.08	VP ebul	1959MCD
	oxacyclobutane	liq	SEL	BP	320.73	0.10	VP	1980OSB
C ₄ H ₈ O	oxacyclopentane	liq	SEL	BP	339.15	0.15	VLE	1998WIS
C ₅ H ₁₀ O	oxacyclohexane	liq	SEL	BP	361.36	0.20	VLE	2012MEJ
Oxacycloalkadienes								
C ₄ H ₄ O	oxacyclopentadiene	liq	SEL	BP	304.49	0.08	VP	1952GUT2
C ₅ H ₆ O	2-methyloxacyclopentadiene	liq	SEL	BP	338.4	0.5	VP isoten	1971EON
Oxacycloketones								
C ₃ H ₄ O	cyclopropanone	liq	SEL	BP	333	18	BP est	2019DIK
C ₄ H ₆ O	cyclobutanone	liq	SEL	BP	372.9	0.5		1972WOL
C ₅ H ₈ O	cyclopentanone	liq	SEL	BP	403.9	0.3	BP	1972WOL
C ₆ H ₁₀ O	cyclohexanone	liq	SEL	BP	428.57	0.15	VP	1987AMB
C ₇ H ₁₂ O	2-methylcyclohexanone	liq	SEL	BP	438.0	0.5	VLE	1993BUR
	3-methylcyclohexanone	liq	SEL	BP	442.7	0.5	BP	1940WHI
	4-methylcyclohexanone	liq	SEL	BP	444.6	0.5	BP	1940WHI
Dioxanes and Trioxanes								
C ₃ H ₆ O ₃	1,3,5-trioxane	sol	SEL	MP	335.7	2.0	MP	1963LEV
		liq	SEL1	BP	387.2	0.4	VLE	1966SER
C ₄ H ₈ O ₂	1,3-dioxane	liq	SEL	BP	378.1	0.5	BP	1967PIH

1,4-dioxane	liq	SEL	BP	374.3	0.3	VLE	1956MCC
Furandione and D-Camphor							
C ₁₀ H ₁₆ O	D-camphor	sol	SEL	MP	452	10	MP
C ₄ H ₂ O ₃	2,5-furandione	sol	SEL	MP	325.3	0.5	MP
		liq	SEL1	BP	478	5	VP rev
Aromatic Alcohols							
C ₆ H ₆ O	phenol	sol	SEL	MP	314.03	0.15	MP
		liq	SEL1	BP	454.91	0.10	VP rev
C ₇ H ₈ O	3-methylphenol	liq	SEL	BP	475.5	0.2	VLE
		sol	SEL1	MP	285.36	0.20	MP
	benzyl alcohol	liq	SEL	BP	478.5	0.3	VLE
Aromatic Aldehydes and Acids							
C ₇ H ₆ O	benzaldehyde	liq	SEL	BP	451.1	0.2	VLE
C ₇ H ₆ O ₂	benzoic acid	sol	SEL	MP	395.478	0.010	MP
		liq	SEL1	BP	523.5	0.5	VP rev

Table 2-2(5). Nitrogen-Substituted Hydrocarbons

Formula	Compound	Phase	SEL	trs	T_{trs}/K	unc/K	Method	Reference
Hydrogen Cyanides								
CHN	hydrogen cyanide	liq	SEL	BP	298.8	0.3	VP	1926SIN
Alkylamines								
CH ₅ N	methylamine	gas	SEL	BP	266.84	0.15	VP	1937AST
C ₂ H ₇ N	ethylamine	gas	SEL	BP	289.79	0.18	VP rev	1947STU
C ₃ H ₉ N	<i>n</i> -propylamine	liq	SEL	BP	320.37	0.06	VP ebul	1968OSB
	isopropylamine	liq	SEL	BP	304.92	0.06	VP ebul	1968OSB
C ₄ H ₁₁ N	isobutylamine	liq	SEL	BP	341.1	0.2		1977RAD
	<i>n</i> -butylamine	liq	SEL	BP	350.1	0.3	VLE	1971LET
	<i>tert</i> -butylamine	liq	SEL	BP	317.19	0.06	VP ebul	1968OSB
C ₂ H ₇ N	dimethylamine	gas	SEL	BP	280.02	0.15	VP	1939AST
C ₃ H ₉ N	methylethylamine	liq	SEL	BP	305.6	0.5	VLE	1985WOL
C ₄ H ₁₁ N	diethylamine	liq	SEL	BP	328.48	0.15		1971LET
C ₃ H ₉ N	trimethylamine	gas	SEL	BP	276.00	0.15	VP	1944AST
C ₆ H ₁₅ N	triethylamine	liq	SEL	BP	361.93	0.25	VLE	1962BIT
Cycloalkylamines								
C ₆ H ₁₃ N	cyclohexylamine	liq	SEL	BP	406.98	0.15	VLE	1960NOV
Azacycloalkanes								
C ₂ H ₅ N	azacyclopropane	liq	SEL	BP	328.6	1.0		1956SEA
C ₃ H ₇ N	azacyclobutane	liq	SEL	BP	333.5	1.5		2019DIK
C ₄ H ₉ N	azacyclopentane	liq	SEL	BP	359.71	0.15	VP	1959MCC
C ₅ H ₁₁ N	azacyclohexane	liq	SEL	BP	379.37	0.06	VP ebul	1968OSB
Methenamines								
C ₆ H ₁₂ N ₄	methenamine	sol	SEL	MP	359	10	sub	2002VER
Azacycloalkadienes								
C ₄ H ₅ N	azacyclopentadiene	liq	SEL	BP	402.93	0.06	VP ebul	1968OSB
C ₅ H ₇ N	1-methylazacyclopentadiene	liq	SEL	BP	385.90	0.06	VP ebul	1968OSB
Alkanenitriles								
C ₂ H ₃ N	ethanenitrile	liq	SEL	BP	355.05	0.15	VP	1974DOJ
C ₃ H ₅ N	propanenitrile	liq	SEL	BP	370.7	0.3	VP rev	1949DRE2
C ₄ H ₇ N	butanenitrile	liq	SEL	BP	390.77	0.04	VP	1973MEY
C ₅ H ₉ N	pentanenitrile	liq	SEL	BP	414.42	0.10	VP rev	1949DRE2
C ₆ H ₁₁ N	hexanenitrile	liq	SEL	BP	436.61	0.04	VP	1973MEY
Alkenenitriles								

C ₃ H ₃ N	propenenitrile	liq	SEL	BP	351.1	0.5	VLE	1964SEV
Cyanogens								
C ₂ N ₂	cyanogen	gas	SEL	BP	251.92	0.20	VP	1939RUE
Alkylhydrazines								
C ₂ H ₈ N ₂	1,1-dimethylhydrazine	liq	SEL	BP	335.4	0.3	VLE	1992FER
Melamines								
C ₃ H ₆ N ₆	melamine	sol	SEL	MP	627		MP	2005LID
Pyridines								
C ₅ H ₅ N	pyridine	liq	SEL	BP	388.36	0.06	eval	2019DIK
C ₆ H ₇ N	2-methylpyridine	liq	SEL	BP	402.52	0.08	VP	1963SCO
Anilines								
C ₆ H ₇ N	aniline	liq	SEL	BP	457.1	0.2	VLE	1979KUR
C ₈ H ₁₁ N	N,N-dimethylaniline	liq	SEL	BP	466.5	0.4	BP	1976RAD
		sol	SEL1	MP	275.7	0.3	MP	1976RAD
Oxidized Amines								
CH ₄ N ₂ O	urea	sol	SEL	MP	406.2	0.5	TP	1972ZOR
C ₆ H ₁₅ NO ₃	triethanolamine	liq	SEL	BP	608	5	VP extrap	1959MCD
		sol	SEL1	MP	294.7	0.5	MP	1959MCD
Alkyl Nitrates and Nitroalkanes								
CH ₃ NO ₃	methyl nitrate	liq	SEL	BP	338.0		VLE	1947LEC
C ₃ H ₅ N ₃ O ₉	nitroglycerin	sol	SEL	MP	286.7		MP	2001ONE
CN ₄ O ₈	tetranitromethane	liq	SEL	BP	397.9		VP	1952EDW
		sol	SEL1	MP	287.1	0.5	MP	1955TIM
CHN ₃ O ₆	trinitromethane	sol	SEL	MP	292	3	MP	1939LEW
CH ₃ NO ₂	nitromethane	liq	SEL	BP	374.34	0.08	VP	1954MCC
Isocyanates								
C ₉ H ₆ N ₂ O ₂	2,4-diisocyanatotoluene	liq	SEL	BP	524.8	2.0	VP	1975FRE
		sol	SEL1	MP	293.7	1.0	MP	2020FIS
Nitro Aromatics								
C ₆ H ₅ NO ₂	nitrobenzene	liq	SEL	BP	483.81	0.15	VLE	1952BRO
		sol	SEL1	MP	278.89	0.20	MP	1954WIT
C ₇ H ₅ N ₃ O ₆	trinitrotoluene	sol	SEL	MP	355.1	1.0	MP	1990HUI

Table 2-2(6). Sulfur-Substituted Hydrocarbons

Formula	Compound	Phase	SEL	trs	T_{trs}/K	unc/K	Method	Reference
Alkanethiols								
CH ₄ S	methanethiol	gas	SEL	BP	279.05	0.10		1960MOR
C ₂ H ₆ S	ethanethiol	liq	SEL	BP	308.15	0.15	VP ebul	1966OSB
C ₃ H ₈ S	1-propanethiol	liq	SEL	BP	340.87	0.15	VP	1956PEN
	isopropanethiol	liq	SEL	BP	325.71	0.15	VP ebul	1966OSB
C ₄ H ₁₀ S	1-butanethiol	liq	SEL	BP	371.56	0.15	VP ebul	1966OSB
	1-isobutanethiol	liq	SEL	BP	361.64	0.15	VP ebul	1966OSB
	2-butanethiol	liq	SEL	BP	358.13	0.15	VP ebul	1966OSB
C ₅ H ₁₂ S	1,1-dimethyl-1-propanethiol	liq	SEL	BP	372.35	0.15	BP ebul	1960HAI
				BP	372.28	0.15	VP ebul	1966OSB
	1-pentanethiol	liq	SEL	BP	399.79	0.15	VP ebul	1966OSB
	isopentanethiol	liq	SEL	BP	391.47	0.15	VP	1966OSB
	neopentanethiol	liq	SEL	BP	376.81	0.15	VP	1966OSB
	<i>tert</i> -pentanethiol	liq	SEL	BP	398.16	0.15	VP	1966OSB
C ₆ H ₁₄ S	1-hexanethiol	liq	SEL	BP	425.81	0.15	VP ebul	1966OSB
C ₇ H ₁₆ S	1-heptanethiol	liq	SEL	BP	450.07	0.15	VP ebul	1966OSB
Alkyl Sulfides								
CS ₂	carbon disulfide	liq	SEL	BP	319.37	0.15	VP	1962WAD
C ₂ H ₆ S ₂	dimethyl disulfide	liq	SEL	BP	382.89	0.15	VP	1950SCO
C ₂ H ₆ S	dimethyl sulfide	liq	SEL	BP	310.45	0.15	VP ebul	1942OSB
C ₃ H ₈ S	ethyl methyl sulfide	liq	SEL	BP	339.80	0.15	VP ebul	1952WHI
C ₄ H ₁₀ S	diethyl sulfide	liq	SEL	BP	365.25	0.15	VP ebul	1966OSB
	isopropyl methyl sulfide	liq	SEL	BP	358.0	0.2		1956HAI
	methyl propyl sulfide	liq	SEL	BP	368.69	0.15	VP	1952WHI
C ₅ H ₁₂ S	ethyl isopropyl sulfide	liq	SEL	BP	380.4	1.0	VP	1952WHI
	ethyl propyl sulfide	liq	SEL	BP	391.65	0.15	VP	1952WHI
C ₆ H ₁₄ S	di- <i>n</i> -propyl sulfide	liq	SEL	BP	415.98	0.15	VP	1952WHI
C ₈ H ₁₈ S	di- <i>n</i> -butyl sulfide	liq	SEL	BP	462.06	0.15	VP	1952WHI
C ₂ H ₄ S	thiacyclopropane	liq	SEL	BP	328.1	0.4	VP	1952GUT
Thiacycloalkanes								
C ₃ H ₆ S	thiacyclobutane	liq	SEL	BP	368.11	0.15	BP rev	2019DIK
C ₄ H ₈ S	thiacyclopentane	liq	SEL	BP	394.27	0.15	VP ebul	1966OSB
C ₅ H ₁₀ S	thiacyclohexane	liq	SEL	BP	368.11	0.15	BP rev	2019DIK
Thiacycloalkenes								
C ₄ H ₄ S	thiacyclopentadiene	liq	SEL	BP	357.31	0.15	VP	1949WAD

Aromatic Thiols								
C ₆ H ₆ S	benzenethiol	liq	SEL	BP	442.29	0.15	VP ebul	1966OSB
Alkyl Sulfoxides								
C ₂ H ₆ OS	dimethyl sulfoxide	liq	SEL	BP	463.86	0.20	VP	1972JAK
		sol	SEL1	MP	291.6	0.3	MP	1948DOU

Table 2-2(7). Halogenated Hydrocarbons

Formula	Compound	Phase	SEL	trs	T_{trs}/K	unc/K	Method	Reference
Fluoromethanes								
CH ₃ F	fluoromethane	gas	SEL	BP	194.8		EOS	2006LEM
CH ₂ F ₂	difluoromethane	gas	SEL	BP	221.499	0.008	EOS	1997TIL
CHF ₃	trifluoromethane	gas	SEL	BP	191.3		EOS	2003PEN
CF ₄	tetrafluoromethane	gas	SEL	BP	145.24	0.15	VP	1979LOB
Fluoroethanes								
C ₂ H ₅ F	fluoroethane	gas	SEL	BP	235.69	0.15	VP	1975VID
C ₂ H ₄ F ₂	1,1-difluoroethane	gas	SEL	BP	249.13	0.05	EOS	1996OUT
	1,2-difluoroethane	liq	SEL	BP	304	10		1955EDG
C ₂ H ₃ F ₃	1,1,1-trifluoroethane	gas	SEL	BP	225.91	0.04	EOS	2000LEM
			SEL	BP	278.2	1.0	BP	1936HEN
C ₂ H ₂ F ₄	1,1,1,2-tetrafluoroethane	gas	SEL	BP	247.076	0.009	EOS	1994TIL
	1,1,2,2-tetrafluoroethane	gas	SEL	BP	253.23	0.15	VP	1991MAE
C ₂ HF ₅	pentafluoroethane	gas	SEL	BP	225.06	0.04	EOS	2005LEM
C ₂ F ₆	hexafluoroethane	gas	SEL	BP	195.06	0.06	EOS	2006LEM
Fluoroethenes								
C ₂ H ₃ F	fluoroethene	gas	SEL	BP	201.7	0.5	VP rev	1947STU
C ₂ H ₂ F ₂	1,1-difluoroethene	gas	SEL	BP	187.6	1.2	BP rev	2019DIK
	1,2-difluoroethene, (E)-	gas	SEL	BP	221	20	BP est	2019DIK
	1,2-difluoroethene, (Z)-	gas	SEL	BP	220	21	BP rev	2019DIK
C ₂ HF ₃	trifluoroethene	gas	SEL	BP	211.7	1.5	BP	1961CRA
C ₂ F ₄	tetrafluoroethene	gas	SEL	BP	197.51	0.25	VP	1953FUR
Chloromethanes								
CH ₃ Cl	chloromethane	gas	SEL	BP	248.99	0.15	VLE	1948GAN
CH ₂ Cl ₂	dichloromethane	liq	SEL	BP	312.92	0.15	VLE	1960MUE
CHCl ₃	trichloromethane	liq	SEL	BP	334.42	0.18	VP rev	1947STU
CCl ₄	tetrachloromethane	liq	SEL	BP	349.88	0.05	VP ebul	1959HIL
Chloroethanes								
C ₂ H ₅ Cl	chloroethane	gas	SEL	BP	285.40	0.15	VP	1948GOR
C ₂ H ₄ Cl ₂	1,1-dichloroethane	liq	SEL	BP	330.42	0.15	VP	1956LI
	1,2-dichloroethane	liq	SEL	BP	356.70	0.20	VLE	1996KRI
C ₂ H ₃ Cl ₃	1,1,1-trichloroethane	liq	SEL	BP	347.35	0.20	BP ebul	1985RED
	1,1,2-trichloroethane	liq	SEL	BP	386.82	0.10	VP rev	1949DRE2
C ₂ H ₂ Cl ₄	1,1,1,2-tetrachloroethane	liq	SEL	BP	403.3	0.3	VP rev	1949DRE2
	1,1,2,2-tetrachloroethane	liq	SEL	BP	422.3	1.0	VP	1950MAT

C ₂ HCl ₅	pentachloroethane	liq	SEL	BP	435.1	0.3	VLE	1988MIH
C ₂ Cl ₆	hexachloroethane	sol	SEL	MP	460.2	2.0	MP	1962SHA
		liq	SEL1	BP	463.0	0.5	VP rev	1947STU
Chloroethenes								
C ₂ H ₃ Cl	chloroethene	gas	SEL	BP	259.36	0.09	VP ebul	1959MCD
C ₂ H ₂ Cl ₂	1,1-dichloroethene	liq	SEL	BP	304.71	0.05	VP ebul	1959HIL2
	1,2-dichloroethene, (E)-	liq	SEL	BP	320.73	0.25	VP	1947KET
	1,2-dichloroethene, (Z)-	liq	SEL	BP	333.6	0.5	VP	1947KET
C ₂ HCl ₃	trichloroethene	liq	SEL	BP	360.5	0.4	VP isoten	1944MCD
C ₂ Cl ₄	tetrachloroethene	liq	SEL	BP	394.29	0.15	VLE	1970POL
Chloroethynes								
C ₂ HCl	chloroacetylene	gas	SEL	BP	238	10	BP rev	1987STE
C ₂ Cl ₂	dichloroacetylene	sol	SEL	explod	306	10	explod	2021EPA
Chloroaromatics								
C ₆ H ₅ Cl	chlorobenzene	liq	SEL	BP	405.21	0.25	EOS-rev	2018LEM
Bromomethanes								
CH ₃ Br	bromomethane	gas	SEL	BP	276.70	0.15	VP	1947BEE
CH ₂ Br ₂	dibromomethane	liq	SEL	BP	370.1	0.1	VP rev	1979KUD
CHBr ₃	tribromomethane	liq	SEL	BP	422.36	0.05	VP rev	1979KUD
CBr ₄	tetrabromomethane	sol	SEL	MP	367.6	2.0	MP	1970SIL
		liq	SEL1	BP	462.6	0.6	BP rev	1979KUD
CH ₃ Br	bromomethane	gas	SEL	BP	276.70	0.15	VP	1947BEE
Bromoethanes								
C ₂ H ₅ Br	bromoethane	liq	SEL	BP	311.54	0.19	VP	1930ZMA
C ₂ H ₄ Br ₂	1,2-dibromoethane	liq	SEL	BP	404.56	0.10	VP rev	1949DRE
Bromoethenes								
C ₂ H ₃ Br	bromoethene	liq	SEL	BP	306	18	BP est	2019DIK
Iodomethanes								
CH ₃ I	iodomethane	liq	SEL	BP	315.58	0.02	VP rev	1979KUD
Chlorofluoromethanes								
CH ₂ ClF	chlorofluoromethane	gas	SEL	BP	264.1	1.0	BP rev	1979KUD
CHClF ₂	chlorodifluoromethane	gas	SEL	BP	232.32	0.02	VP rev	1979KUD
CHCl ₂ F	dichlorofluoromethane	gas	SEL	BP	282.05	0.10	VP rev	1979KUD
CHClF ₂	difluorochloromethane	gas	SEL	BP	232.32	0.12	VP rev	1979KUD
CClF ₃	chlorotrifluoromethane	gas	SEL	BP	191.7	0.1	VP rev	1979KUD
CCl ₂ F ₂	dichlorodifluoromethane	gas	SEL	BP	243.38	0.04	VP rev	1979KUD
CCl ₃ F	trichlorofluoromethane	gas	SEL	BP	296.78	0.02	VP rev	1979KUD

Chlorofluoroethanes									
C ₂ H ₄ ClF	1-chloro-1-fluoroethane	gas	SEL	BP	289.3	1.5	BP	1936HEN	
C ₂ H ₃ Cl ₂ F	1,1-dichloro-1-fluoroethane	liq	SEL	BP	305.26	0.25	VP	1991MAE	
	1,2-dichloro-1-fluoroethane	liq	SEL	BP	347.0	0.2	BP	1936HEN	
C ₂ H ₃ ClF ₂	1-chloro-1,1-difluoroethane	gas	SEL	BP	264.04	0.25	VP ebul	1993SIL	
	1-chloro-2,2-difluoroethane	liq	SEL	BP	308.3	0.2	BP	1936HEN	
C ₂ H ₂ ClF ₃	2-chloro-1,1,1-trifluoroethane	gas	SEL	BP	279.3	0.2	VP	1936HEN	
Chlorofluoroethenes									
C ₂ HCl ₂ F	1,2-dichloro-1-fluoroethene	liq	SEL	BP	305.7	2.5	BP rev	2010SYN	
C ₂ ClF ₃	chlorotrifluoroethene	gas	SEL	BP	244.8	0.1	VP	1951OLI	
Bromofluoromethanes									
CH ₂ BrF	bromofluoromethane	gas	SEL	BP	290.7	2.0	BP rev	1979KUD	
CHBrF ₂	bromodifluoromethane	gas	SEL	BP	257.5	0.2	VP rev	1979KUD	
CHBr ₂ F	dibromofluoromethane	liq	SEL	BP	337	2	BP rev	1979KUD	
CBrF ₃	bromotrifluoromethane	gas	SEL	BP	215.30	0.20	VP ebul	1959MCD	
CBr ₂ F ₂	dibromodifluoromethane	gas	SEL	BP	295.93	0.04	VP rev	1979KUD	
CBr ₃ F	tribromofluoromethane	liq	SEL	BP	379.6	1.0	BP rev	1979KUD	
Bromochloromethanes									
CH ₂ BrCl	bromochloromethane	liq	SEL	BP	341.21	0.08	VP rev	1979KUD	
CHBrCl ₂	bromodichloromethane	liq	SEL	BP	363.3	0.4	BP rev	1979KUD	
CHBr ₂ Cl	dibromochloromethane	liq	SEL	BP	393	2	BP rev	1979KUD	
CBrCl ₃	bromotrichloromethane	liq	SEL	BP	378.1	1.0	VP rev	1979KUD	
CBr ₂ Cl ₂	dibromodichloromethane	liq	SEL	BP	393	2	BP rev	1979KUD	
CBr ₃ Cl	tribromochloromethane	liq	SEL	BP	433	1	BP rev	1979KUD	
Bromochlorofluoromethanes									
CHBrClF	bromochlorofluoromethane	liq	SEL	BP	309	2	BP rev	1979KUD	
CBrClF ₂	bromochlorodifluoromethane	gas	SEL	BP	269.14	0.04	VP rev	1979KUD	
CBrCl ₂ F	bromodichlorofluoromethane	liq	SEL	BP	323	15	VP	1979KUD	
CBr ₂ ClF	dibromochlorofluoromethane	liq	SEL	BP	353	2	BP rev	1979KUD	
Carbonyl Halides									
CHFO	carbonyl fluoride	gas	SEL	BP	245.2	2.0	VP	1964FIS	
CF ₂ O	carbonyl difluoride	gas	SEL	BP	188.62	0.20	VP	1968PAC	
CCl ₂ O	phosgene	gas	SEL	BP	280.66	0.10	VP	1948GIA	
Chlorophenols									
C ₆ HCl ₅ O	pentachlorophenol	liq	SEL	BP	583	3	VP	1947STU	
		sol	SEL1	MP	462.8	1.0	MP	1959MCD	

Table 2-2(8). Siloxanes

Formula	Compound	Phase	SEL	trs	T_{trs}/K	unc/K	Method	Reference
Siloxanes								
C ₆ H ₁₈ OSi ₂	hexamethyldisiloxane	liq	SEL	BP	373.64	0.10	VP	1961SCO
C ₈ H ₂₄ O ₄ Si ₄	octamethylcyclotetrasiloxane	liq	SEL	BP	448.9	0.6	EOS	2016THO
		sol	SEL1	MP	290.7	1.0	MP	1968MAR

Table 2-3. Notations for Methods

Method	Description	Method	Description
BP	Boiling point (unspec. method)	predict	Predicted
BP comp	BP compilation	quote	Quoted value
BP distill	BP distillation method	rev	Review
BP ebul	BP ebulliometry method	sub	Sublimation (unspec. method)
BP est	Estimated BP	sub rev	Sublimation review
BP eval	Evaluated BP	sub SVE	Sublimation SVE
BP rev	BP review	sub VP	Sublimation VP
dec	Decomposition	sub, EOS	Sublimation EOS
DSC	Differential Scanning Calorimetry	TP	Triple point (unspec. method)
DTA	Differential Thermal Analysis	VLE	Vapor-Liquid Equilibrium (multicomponent)
EOS	Equation-of-State	VLE extrap	VLE extrapolation
est	Estimated	VLE rev	VLE review
eval	Evaluated	VP	Vapor pressure method
explod	Explodes	VP ebul	Ebulliometry method
GLC	Gas-Liquid Chromatography	VP extrap	VP extrapolation
MP	Melting point (unspec. method)	VP isoten	VP isoteniscope method
MP rev	MP review	VP rev	VP review
MP SLE	MP Solid-Liquid Equilibrium		

2.4. References for Normal Melting and Normal Boiling Points

- [1834CLA] Clapeyron, M. C.; *Mémoire sur la puissance motrice de la chaleur*, J. l'École Polytech. 3, 153 (1834).
- [1905LES] Lespieau, R.; Chavanne, G.; Liquefaction of allene and allylene. *Comptes Rendus*, 1905, 140, 1035
- [1910YOU] Young, S., The Internal Heat of Vaporization constants of thirty pure substances, *Sci. Proc. R. Dublin Soc.*, 1910, 12, 374.
- [1915BUR] Burrell, G. A.; Robertson, I. W.; The Vapor Pressures of Acetylene, Ammonia and Isobutane at Temperatures Below their Normal Boiling Points. *J. Am. Chem. Soc.*, 1915, 37, 2482
- [1916WIL] Willcox, M.; Brunel, R.F., The reversible replacement of alcohols in aldehydealcoholates. *J. Am. Chem. Soc.*, 1916, 38, 1821
- [1920MAA] Maass, O.; Hatcher, W. H.; The properties of pure hydrogen peroxide. I., *J. Am. Chem. Soc.*, 1920, 42, 2548
- [1921KEN] Kendall, J.; Beaver, J.J., Compound Formation in Phenol-cresol Mixtures, *J. Am. Chem. Soc.*, 1921, 43, 1853
- [1921MAA] Maass, O.; Wright, C.H., Some physical properties of hydrocarbons containing two and three carbon atoms, *J. Am. Chem. Soc.*, 1921, 43, 1098
- [1921TIM] Timmermans, J., The Freezing Points of Organic Substances IV. New Exp. Determinations, *Bull. Soc. Chim. Belg.*, 1921, 30, 62.
- [1921TIM2] Timmermans, J.; Matlaar, J.F., Freezing points of organic substances VI. New experimental determinations., *Bull. Soc. Chim. Belg.*, 1921, 30, 213
- [1922HER] Herz, W.; Schuftan, P., Physikalisch-chemische Untersuchungen an Tetralin und Dekalin, *Z. Phys. Chem. (Frankfurt/Main)*, 1922, 101, 269
- [1922TAY] Taylor, Robert S.; Smith, Leighton B., The vapor pressures, densities and some derived quantities for ether at low temperatures, *J. Am. Chem. Soc.*, 1922, 44, 2450
- [1922TIM] Timmermans, J., Investigation of the Freezing Point of Organic Substances VII, *Bull. Soc. Chim. Belg.*, 1922, 31, 389
- [1923MOR] Mortimer, F. Spencer.; Murphy, Ray V., The Vapor Pressures of Some Substances Found in Coal Tar., *Ind. Eng. Chem.*, 1923, 15, 1140
- [1924BER] Berthoud, A.; Brum, R.; Physical Properties of Some Organic Compounds. *J. Chim. Phys. Phys.-Chim. Biol.*, 1924, 21, 143
- [1925NEL] Nelson, O.A.; Wales, H., Vapor Pressures and Boiling Points of Mono- and Dimethylanilines and Mono- and Diethylanilines, *J. Am. Chem. Soc.*, 1925, 47, 867

- [1925PER] Perry, J. H.; Bardwell, D. C.; The vapor pressures of solid and liquid cyanogen. J. Am. Chem. Soc., 1925, 47, 2629
- [1926LEC] Lecat, M.; New binary azeotropes: 3rd list. Ann. Soc. Sci. Bruxelles, Ser. B, 1926, 45, 284
- [1926LOO] Loomis, A.G.; Walters, J.E., The vapor pressure of ethane near the normal boiling point 1, J. Am. Chem. Soc., 1926, 48, 2051
- [1926SIN] Sinozaki, Heima; Hara, Ryoaburo; Mitsukuri, Shinroku, The vapour pressures of hydrogen cyanide, Bull. Chem. Soc. Jpn., 1926, 1, 59
- [1926TIM] Timmermans, J.; Martin, F., The Work of Intl. Bur. of Physical-Chemical Standards - Methods and Apparatus, J. Chim. Phys. Phys.-Chim. Biol., 1926, 23, 733
- [1927KLO] Klosky, S.; Woo, L. P. L.; Flanigan, R. J.; The vapor pressure curve of benzoic acid. J. Am. Chem. Soc., 1927, 49, 1280
- [1927LEC] Lecat, M., New binary azeotropes: 6th list, Ann. Soc. Sci. Bruxelles, Ser. B, 1927, 47, 63
- [1928COF] Coffin, C.C.; Maass, O., The Preparation and Physical Properties of α , β - and γ -Butylene and Normal and Isobutane, J. Am. Chem. Soc., 1928, 50, 1427
- [1928CUT] Cuthbertson, A.C.; Matheson, G.L.; Maass, O., The Freezing Point and Density of Pure Hydrogen Peroxide, J. Am. Chem. Soc., 1928, 50, 1120
- [1928NEL] Nelson, O.A., Vapor Pressures of Fumigants: II---Methyl, Ethyl, n-Propyl, Isopropyl, n-Butyl, Secondary Butyl and Isobutyl Formates 1, Ind. Eng. Chem., 1928, 20, 1382
- [1929PEA] Pearce, J.N.; Peters, P.E., The Vapor Pressure of Ethylene Chloride Between -30° and 100°, J. Phys. Chem., 1929, 33, 873
- [1930LEC] Lecat, M., Azeotropism in Binary Mixtures Containing an Alcohol Mixed with an amine, a nitro derivative, an ether or water., Z. Anorg. Allg. Chem., 1930, 186, 119
- [1930LIV] Livingston, R.; Heisig, G. B.; The vapor pressure of allene as a function of temperature, J. Am. Chem. Soc, 1930, 52, 2409
- [1930LUC] Lucas, H. J. ; Dillon, R. T. ; Young, W. G.; Qualitative estimation of the composition of butene mixtures by distillation methods. J. Am. Chem. Soc., 1930, 52, 1949
- [1930MAR] Marti, F. B.; Methods and equipment used at the bureau of physico-chemical standards: 3 Physico-chemical properties of some solid organic compounds at normal temperatures. Bull. Soc. Chim. Belg., 1930, 39, 590
- [1930NEL] Nelson, O.A., Vapor Pressures of Fumigants, Ind. Eng. Chem., 1930, 22, 971
- [1930ZMA] Zmaczynski, M.A., Recherches Ebullioscopiques et Tonometriques Comparatives de 8 Substances Organiques Etalons, J. Chim. Phys. Phys. Chim. Biol., 1930, 27, 503

- [1931BUR] Burriel, F.; Physico-Chemical Study of Some Solid Organic Compounds at Ordinary Temperatures, and Their Correlation with Temperature. *An. R. Soc. Esp. Fis. Quim.*, 1931, 29, 89
- [1931HUR] Hurd, C.D.; Meinert, R.N., Synthesis and pyrolysis of methylallene and ethylacetylene, *J. Am. Chem. Soc.*, 1931, 53, 289
- [1931SMY] Smyth, C.P.; Walls, W.S., Electric moment and molecular structure: IV the glycols, *J. Am. Chem. Soc.*, 1931, 53, 2115
- [1931WIL] Wilkinson, R., CCCCXXII.—Studies in the olefin series. Part I. The synthesis of Δ^1 -olefins. *J. Chem. Soc.*, 1931, 3057
- [1932CLA] Clayton, J.O.; Giaque, W.F., The Heat Capacity and Entropy of Carbon Monoxide. Heat of Vaporization Vapor Pressure of Solid and Liquid. Free Energy to 5000 K from Spectroscopic Data, *J. Am. Chem. Soc.*, 1932, 54, 2610
- [1932NOR] Norris, J. F.; Rigby, G. W.; Reactivity of Atoms and Groups in Organic Compounds XII. The Preparation and Properties of Mixed Aliphatic Ethers with Special Reference to those Containing the Tert-Butyl Radical. *J. Am. Chem. Soc.*, 1932, 54, 2088
- [1933HEI] Heisig, G.B., Action of Radon on Some Unsaturated Hydrocarbons. III. Vinylacetylene and Butadiene, *J. Am. Chem. Soc.*, 1933, 55, 2304
- [1933ING] Ingold; Jessop; Kuriyam; Mandour, Influence of Poles and Polar Linkages on the Course Pursued by Elimination Reactions XVIII. Thermal Decomposition of Sulphonium Hydroxides, *J. Chem. Soc.*, 1933, 3455
- [1933JAC] Jacobson, R. A.; Carothers, W. H.; Acetylene polymers and their derivatives: VII alpha-alkyl-beta-vinylacetylenes. *J. Am. Chem. Soc.*, 1933, 55, 1622
- [1933KOH] Kohlraush, K.W.F.; Koppl, F.; Pongratz, A., *Z. Phys. Chem., Abt. B*, 1933, 21, 242.
- [1933MAI] Mair, B.J.; Schickanz, S.T., The isolation of mesitylene, pseudocumene, and hemimellitene from an Oklahoma petroleum, *J. Res. Natl. Bur. Stand.*, 1933, 11, 665
- [1933MEN] Menzel, Walter; Mohry, Friedrich, Die Dampfdrucke des CF₄ und NF₃ und der Tripelpunkt des CF₄, *Z. Anorg. Allg. Chem.*, 1933, 210, 257
- [1933MEY] Meyers, C. H.; Van Dusen, M. S.; The Vapor Pressures of Liquid and Solid Carbon Dioxide., *J. Res. Natl. Bur. Stand.*, 1933, 10, 381
- [1933RUF] Ruff, O.; Bretschneider, O.; The Preparation of Hexafluoroethane and Tetrafluoroethene from Tetrafluoromethane. *Z. Anorg. Allg. Chem.*, 1933, 210, 173
- [1933SIG] Signaigo, F.K.; Cramer, P.L., The preparation of some mono- and dialkylcyclohexanes, *J. Am. Chem. Soc.*, 1933, 55, 3326

- [1933SWA] Swarts, F.; Comptes Rendus 167, 1261 (1933)
- [1933THO] Thornton, N. V.; Burg, A. B.; Schlesinger, H. I.; The Behaviour of Dichlorodifluoromethane and of Chlorotrifluoromethane in the Electric Discharge, J. Am. Chem. Soc., 1933, 55, 3177
- [1934ALL] Allen, J.S.; Hibbert, H., Studies on reactions relating to carbohydrates and polysaccharides: XVII. the oxygen valence angle and the structure of glucose and related compounds, J. Am. Chem. Soc., 1934, 56, 1398
- [1934HIE] Hieber, W; Woerner, A.; Thermochemical measurements of complex-forming amines and alcohols. Z. Elektrochem. Angew. Phys. Chem., 1934, 40, 252
- [1934KOH] Kohlraush, K. W. F.; Koppl, F. Z. Phys. Chem. Abt. B, 1934, 26, 209
- [1935BUT] Butler, J.A.V.; Ramchandani, C.N.; Thomson, D.W., The Solubility of Non-Electrolytes. Part 1. The Free Energy of Hydration of Some Aliphatic Alcohols, J. Chem. Soc., 1935, 280
- [1935EAR] Earp, D. P.; Glasstone, S.; Dielectric polarisation and molecular-compound formation in solution. J. Chem. Soc., 1935, 1709
- [1935LES] Leslie, R.T.; White, J.D., Present Status of the Isolation and Identification of The Volatile Hydrocarbons in a Midcontinent Petroleum, J. Res. Natl. Bur. Stand., 1935, 15, 211
- [1935MUL] Mulliken, S.P.; Wakeman, R.L.; Gerry, H.T., The preparation of certain alkenes, alkadienes and alkynes, J. Am. Chem. Soc., 1935, 57, 1605
- [1935SCH] Schierholtz, O.J.; Staples, M.L., Vapor Pressure of Certain Glycols, J. Am. Chem. Soc., 1935, 57, 2709
- [1935SIM] Simon, A.; Huter, J.; Vapor Pressure Curves, Melting Point and Chemical Constants of Dimethyl, Trimethyl- and Isobutylamines. Z. Elektrochem., 1935, 41, 28
- [1935SPE] Spence, R.; Wild, W., The Vapour-Pressure Curve of Formaldehyde, and Some Related Data, J. Chem. Soc., 1935, 138, 506
- [1935TIM] Timmermans, J.; Hennaut-Roland, M., Physical constants of 20 organic compounds. VII., J. Chim. Phys. Phys.-Chim. Biol., 1935, 32, 501
- [1936GRO] Grosse, A.V., Refractive Indices at Low Temperatures, J. Am. Chem. Soc., 1937, 59, 2739
- [1936HAS] Hass, E.B.; McBee, E.T.; Hinds, G.E.; Glusenkamp, E.W., Synthesis of cyclopropane, Ind. Eng. Chem., 1936, 28, 1178
- [1936HEN] Henne, A.L.; Renoll, M.W., Fluoroethanes and Fluoroethylenes. V. J. Am. Chem. Soc., 1936, 58, 889
- [1936KIS] Kistiakowsky, G. B.; Ruhoff, J. R.; Smith, H. A.; Vaughan, W. E.; Heats of Organic Reactions IV. Hydrogenation of Some Dienes and of Benzene. J. Am. Chem. Soc., 1936, 58, 146

- [1937AST] Aston, J.G.; Siller, C.W.; Messerly, G.H., Heat capacities and entropies of organic compounds. III. Methylamine from 12 K to the boiling point. Heat of vaporization and vapor pressure. The entropy from molecular data, J. Am. Chem. Soc., 1937, 59, 1743
- [1937CAR] Carr, E.R.; Stucklen, H., The Ultraviolet Absorption Spectra of the Isomers of Butene-2 and Pentene-2, J. Am. Chem. Soc., 1937, 59, 2138.
- [1937GIA] Giauque, W.F.; Egan, C.J., Carbon Dioxide. The Heat Capacity and Vapor Pressure of the Solid. The Heat of Sublimation. Thermodynamic and Spectroscopic Values of the Entropy, J. Chem. Phys., 1937, 5, 45
- [1937GIN] Ginnings, P.M.; Baum, R., Aqueous Solubilities of the Isomeric Pentanols, J. Am. Chem. Soc., 1937, 59, 1111
- [1937GRO] Grosse, A.V.; Ipatieff, V.N., The reaction of cycloparaffins with aromatic hydrocarbons: decycloalkylation, J. Org. Chem., 1937, 447
- [1937GUY] Guyer, A.; Schutze, H.; Weidenmann, M., Uber einige thermische Eigenschaften des Vinylbromids, Helv. Chim. Acta, 1937, 20, 936
- [1937MAE] Maess, L.; Mueffling, L., The vapor pressure of several hydrocarbons and ketones, Angew. Chem., 1937, 50, 759
- [1937MOU] Moureu, H.; Dode, M., Heats of Formation of Ethylene Oxide, of Ethandiol and of Several Homologs, Bull. Soc. Chim. Fr., 1937, 637
- [1937TIM] Timmermans, J.; Hennaut-Roland, M., Works from International Bureau at Physical-Chemical Standards. VIII. Physical constants of 20 organic compounds, J. Chim. Phys. Phys.-Chim. Biol., 1937, 34, 693
- [1937WIT] Witt, R.K.; Kemp, J.D., The heat capacity of ethane from 15 K to the boiling point. The heat of fusion and the heat of vaporization., J. Am. Chem. Soc., 1937, 59, 273
- [1938CAR] Carr, E.P.; Stucklen, H., The ultraviolet absorbtion of simple hydrocarbons: IV unsaturated cyclic hydrocarbons in the schumann region, J. Chem. Phys., 1938, 6, 55
- [1938CHI] Chiurdoglu, G.; Bull. Soc. Chim. Belg., 1938, 47, 241
- [1938COO] Coomber, D.I.; Partington, J.R., Studies in dielectric polarisation. Part XXIII the dipole moments of some aliphatic and aromatic aldehydes and of anthrone, J. Chem. Soc., 1938, 1444.
- [1938CRE] Crenshaw, J.L.; Cope, A.C.; Finkelstein, N.; Rogan, R., The Dioxanates of the Mercuric Halides, J. Am. Chem. Soc., 1938, 60, 2308
- [1938DOL] Dolliver, M.A.; Gresham, T.L.; Kistiakowsky, G.B.; Smith, E.A.; Vaughan, W.E., Heats of Organic Reactions VI. Heats of Hydrogenation of Some Oxygen-Containing Compounds, J. Am. Chem. Soc., 1938, 60, 440
- [1938KEM] Kemp, J.D.; Egan, C.J., Hindered rotation of the methyl groups in propane. The heat capacity, vapor pressure, heats of fusion and vaporization of propane. Entropy and density of the gas, J. Am. Chem. Soc., 1938, 60, 1521

- [1938SMI] Smittenberg, J.; Hoog, H.; Henkes, R.A., Freezing Points of A Number of Pure Hydrocarbons of the Gasoline Boiling Range and of Some of their Binary Mixtures, J. Am. Chem. Soc., 1938, 60, 17
- [1938VAN] van Romburgh, P., Recl. Trav. Chim. Pays-Bas, 1938, 57, 494
- [1939AST] Aston, J.G.; Eidinoff, M.L.; Forster, W.S., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of dimethylamine, J. Am. Chem. Soc., 1939, 61, 1539
- [1939CON] Conn, J.B.; Kistiakowsky, G.B.; Smith, E.A., Heats of organic reactions. VIII. Some further hydrogenations, including those of some acetylenes., J. Am. Chem. Soc., 1939, 61, 1868
- [1939GRO] Grosse, A.V.; Linn, C.B., Refraction Data on Liquid C3-Hydrocarbons, J. Am. Chem. Soc., 1939, 61, 751
- [1939HEN] Henne, A. L.; Renoll, M. W.; Leicester, H. M.; J. Am. Chem. Soc., 1939, 61, 938
- [1939LEW] Lewis, G.L.; Smyth, C.P., The Dipole Moments and Structures of Ozone, Silicobromoform and Dichlorogermane, J. Am. Chem. Soc., 1939, 61, 3067
- [1939POW] Powell, T.M.; Giauque, W.F., Propylene. The heat capacity, vapor pressure, heats of fusion and vaporization. The third law of thermodynamics and orientation equilibrium in the solid, J. Am. Chem. Soc., 1939, 61, 2366
- [1939RUE] Ruehrwein, R. A.; Giauque, W. F.; The entropy of cyanogen. Heat capacity and vapor pressure of solid and liquid. Heats of fusion and vaporization. J. Am. Chem. Soc., 1939, 61, 2940
- [1940AST1] Aston, J.G.; Kennedy, R.M.; Schumann, S.C., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of isobutane, J. Am. Chem. Soc., 1940, 62, 2059
- [1940AST2] Aston, J.G.; Messerly, G.H., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of n-butane, J. Am. Chem. Soc., 1940, 62, 1917
- [1940BEN] Benning, A.F.; McHarness, R.C., Thermodynamic Properties of Fluorochloromethanes and -Ethan's Vapor Pressure of Three Fluorochloromethanes and Trifluorotrichloroethane, Ind. Eng. Chem., 1940, 32, 497
- [1940GRO] Grosse, A. V.; Wackher, R. C.; Linn, C. B.; Physical Properties of the Alkyl Fluorides and A Comparison of the Alkyl Fluorides with the Other Alkyl Halides and with the Alkyls of the Elements of Period II J. Phys. Chem., 1940, 44, 275
- [1940LAM] Lamb, Arthur B.; Roper, Edwin E., The Vapor Pressures of Certain Unsaturated Hydrocarbons, J. Am. Chem. Soc., 1940, 62, 806
- [1940MAI] Mair, B.J.; Streiff, A.J., Separation of the aromatic hydrocarbons, and the isolation of n-dodecane, naphthalene, 1-methylnaphthalene, and 2-methylnaphthalene from the kerosene fraction of petroleum, J. Res. Natl. Bur. Stand., 1940, 24, 395
- [1940STU] Stuckey, J.M.; Saylor, J.H., Vapor Pressures of Organic Compounds I., J. Am. Chem. Soc., 1940, 62, 2922

- [1940WHI] White, A.H.; Bishop, W.S., Dielectric Evidence of Molecular Rotation in the Crystals of Certain Non-aromatic Compounds, J. Am. Chem. Soc., 1940, 62, 8
- [1940YOU] Young, D. W. S.; Fukuhara, N.; Bigelow, L. A.; The Action of Elementary Fluorine Upon Organic Compounds VIII. The Influence of Dilution on the Vapor Phase Fluorination of Ethane. J. Am. Chem. Soc., 1940, 62, 1171
- [1941BRO] Brooks, B.T.; Wilbert, G., The Reaction of Cyclopentadiene and Ketene, J. Am. Chem. Soc., 1941, 63, 870
- [1941EWE] Ewell, R.H.; Hardy, P.E., Isomerization Equilibria among the Branched Chain Pentenes, J. Am. Chem. Soc., 1941, 63, 3460
- [1941EWE2] Ewell, R.H.; Welch, L.M., Maximum Boiling Mixtures of chloroparaffins with Donor Liquids, J. Am. Chem. Soc., 1941, 63, 2475.
- [1941HEA] Hearne. G.; Tamele, M.; Converse, W., Derivatives of Allylic Chloride, Ind. Eng. Chem., 1941, 33, 805
- [1941HEI] Heisig, G.B., The Preparation and the Vapor Pressures of Cyclobutene and Cyclobutane, J. Am. Chem. Soc., 1941, 63, 1698
- [1941IPA] Ipatieff, V. N.; Burwell, R. L.; The Catalytic Preparation and Interconversion of Simple and Mixed Ethers. J. Am. Chem. Soc., 1941, 63, 969
- [1941KEN] Kennedy, R.M.; Sagenkahn, M.; Aston, J.G., The heat capacity and entropy, heats of fusion and vaporization, and the vapor pressure of dimethyl ether. The density of gaseous dimethyl ether, J. Am. Chem. Soc., 1941, 63, 2267
- [1941MAR] Maryott, A.A.; Hobbs, M.E.; Gross, P.M., Bond Moment Additivity and the Electric Moments of Some Halogenated Hydrocarbons, J. Am. Chem. Soc., 1941, 63, 659
- [1941OSB] Osborne, D.W.; Garner, C.S.; Doescher, R.N.; Yost, D.M., The heat capacity, entropy, heats of fusion and vaporization and vapor pressure of fluorotrichloromethane, J. Am. Chem. Soc., 1941, 63, 3496
- [1941SCH] Schlatter, M. J.; The preparation of cyclopropene. J. Am. Chem. Soc., 1941, 63, 1733
- [1941SIM] Simnikov, I.P., Increase of Enthalpy During Formation of Azeotropic Mixtures, Zh. Obshch. Khim., 1941, 14, 483
- [1942BEN] Benson, S.W.; Kistiakowsky, G.B., The Photochemical Decomposition of Cyclic Ketones, J. Am. Chem. Soc., 1942, 64, 80
- [1942BER] Berry, K. L.; Sturtevant, J. M.; Fluorochlorobromomethane, J. Am. Chem. Soc., 1942, 64, 1599
- [1942OSB] Osborne, D.W.; Doescher, R.N.; Yost, D.M., The heat capacity, heats of fusion and vaporization, vapor pressure and entropy of dimethyl sulfide, J. Am. Chem. Soc., 1942, 64, 169
- [1942RUS] Russell, H., Jr.; Osborne, D.W.; Yost, D.M., The heat capacity, entropy, heats of fusion, transition, and vaporization and vapor pressures of methyl mercaptan, J. Am. Chem. Soc., 1942, 64, 165

- [1943HEN] Henne, A.L.; Plueddeman, E.P., The Addition of Hydrogen Fluoride to Haloolefins, J. Am. Chem. Soc., 1943, 65, 1271
- [1943HUR] Hurdis, E.C.; Smyth, C.P., J. Am. Chem. Soc., 1943, 65, 89.
- [1943LEC] Lecat, M., Azeotropes of Ethyl Urethane and other Azeotropes, C. R. Hebd. Seances Acad. Sci., 1943, 217, 273
- [1943MCA] McArdle, E.H.; Robertson, A.E., Viscosity of Solutions in Branched-Chain Paraffins. Ind. Eng. Chem., Anal. Ed., 1943, 15, 484
- [1943MCG] McGovern, E. W.,; Ind. Eng. Chem., 1943, 35, 1230
- [1943MCG] McGovern, E. W.; Chlorohydrocarbon Solvents. Ind. Eng. Chem., 1943, 35, 1230
- [1944AST] Aston, J.G.; Sagenkahn, M.L.; Szasa, G.J.; Moessen, G.W.; Zuhr, H.F., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of trimethylamine. The entropy from spectroscopic and molecular data, J. Am. Chem. Soc., 1944, 66, 1171
- [1944BOO] Boord, C.E.; Perilstein, W.L.; Greenlee, K.W., , Am. Pet. Inst. Hydrocarbon Res. Proj., Sixth Annu. Ref., Ohio State Univ., Aug. 31, 1944
- [1944MCD] McDonald, H.J., The vapor pressure and heat of vaporization of trichloroethylene, J. Phys. Chem., 1944, 48, 47
- [1944PAR] Pardee, W. A.; Weinrich, W.; Physical properties of alkylated phenols. Ind. Eng. Chem., 1944, 36, 595
- [1944RUB] Rubin, T.R.; Levedahl, B.H.; Yost, D.M., The heat capacity, heat of transition, vaporization, vapor pressure and entropy of 1,1,1-trichloroethane, J. Am. Chem. Soc., 1944, 66, 279
- [1944RUS] Russell, H., Jr.; Golding, D.R.V.; Yost, D.M., The heat capacity, heats of transition, fusion and vaporization, vapor pressure and entropy of 1,1,1-trifluoroethane, J. Am. Chem. Soc., 1944, 66, 16
- [1944SCO] Scott, R.B.; Ferguson, W.J.; Brickwedde, F.G., Thermodynamic properties of cis-2-butene from 15° to 1,500 K, J. Res. Natl. Bur. Stand., 1944, 33, 1
- [1945CRA] Crane, G.; Boord, C.E.; Henne, A.L., The Preparation of Cyclopentenenes and Cyclopentanes I., J. Am. Chem. Soc., 1945, 67, 1237
- [1945GRE] Greensfelder, B.S.; Voge, H.H.; Good, G.M., Ind. Eng. Chem., 1945, 37, 1168
- [1945GUT] Guttman, L.; Pitzer, K.S., trans-2-Butene. The heat capacity, heats of fusion and vaporization, and vapor pressure. The entropy and barrier to internal rotation, J. Am. Chem. Soc., 1945, 67, 324
- [1945HEN] Henne, A.L.; Greenlee, K.W., Preparation and Physical Constants of Acetylenic Compounds, J. Am. Chem. Soc., 1945, 67, 484
- [1945HER] Herington, E.F.G.; Rideal, E.K., The Catalytic Cyclization of Aliphatic Hydrocarbons II., Proc. R. Soc. London, A, 1945, 184, 447

- [1945HOG] Hoge, H. J.; "Vapor pressure, latent heat of vaporization, and triple-point temperature of nitrous oxide," J. Res. Natl. Bur. Stand., 1945, 34, 281-93
- [1945SMI] Smith, H.A.; Pennekamp, E.F.H., The Catalytic Hydrogenation of the Benzene Nucleus II. The Hydrogenation of Benzene and Mono-alkylbenzenes, J. Am. Chem. Soc., 1945, 67, 276
- [1945WIL] Williamham, C.B.; Taylor, W.J.; Pignocco, J.M.; Rossini, F.D., Vapor Pressures and Boiling Points of Some Paraffin, Alkylcyclopentane, Alkylcyclohexane, and Alkylbenzene Hydrocarbons, J. Res. Natl. Bur. Stand., 1945, 35, 219
- [1946API] Am. Pet. Inst. Res. Proj. 45, Ohio State Univ., 1946.
- [1946BOO] Boord, C.E.; Greenlee, K.W.; Perilstein, W.L., The Synthesis, Purification and Prop. of Hydrocarbons of Low Mol. Weight, Am. Pet. Inst. Res. Proj. 45, Eighth Annu. Rep., Ohio State Univ., June 30, 1946
- [1946FOR] Forziati, A. F.; Glasgow, A. R.; Willingham, C. B.; Rossini, F. D.; Specification and Properties of 29 Paraffin, 4 Alkylcyclopentanes, 10 Alkylcyclohexanes and 8 Alkylbenzene Hydrocarbons. J. Res. Natl. Bur. Stand., 1946, 36, 129
- [1946GER] Gerding, H.; van der Vet, A.P., Raman spectra of the seven isomeric pentenes and quantitative analysis of two mixtures by means of the Raman effect, Recl. Trav. Chim. Pays-Bas, 1946, 64, 257
- [1946GIB] Gibbons, L.C.; Thompson, J.F.; Reynolds, T.W.; Wright, J.I.; Chanan, H.H.; Lamberti, J.M.; Hipsher, H.F.; Karabinos, J.V., The Purification and Physical Constants of Aromatic Compounds, J. Am. Chem. Soc., 1946, 68, 1130
- [1946KAR] Karabinos, J.V.; Serijan, K.T.; Gibbons, L.C., ortho- and para-diethylbenzenes, J. Am. Chem. Soc., 1946, 68, 2107
- [1946KAY] Kay, W.B., The Vapor Pressures and Saturated Liquid and Vapor Densities of the Isomeric Hexanes, J. Am. Chem. Soc., 1946, 68, 1336
- [1946LEC] Lecat, M.; Azeotropes of Amines. Ann. Soc. Sci. Bruxelles, Ser. 1, 1946, 60, 169
- [1946MOR] Morrell, C. E.; Paltz, W. J.; Packie, J. W.; Asbury, W. C.; Brown, C. L.; Trans. Am. Inst. Chem. Eng., 1946, 42, 473
- [1946REN] Renfrew, M. M.; Lewis, E. E.; Polytetrafluoroethylene: Heat-resistane, Chemically Inert Plastic. Ind. Eng. Chem., 1946, 38, 870
- [1946RUE1] Ruehrwein, R.A.; Powell, T.M., The Heat Capacity, Vapor Pressure, Heats of Fusion and Vaporization of Cyclopropane. Entropy and Density of the Gas 1, J. Am. Chem. Soc., 1946, 68, 1063
- [1946RUE2] Rueggeberg, W.H.; Cushing, M.L.; Cook, W.A., Condensation of Aliphatic Alcohols with Aryl Hydrocarbons in the Presence of Chlorosulfuric Acid, J. Am. Chem. Soc., 1946, 68, 191

- [1946SLA] Slabey, V.A., The Reduction of Pentaerythrityl Tetrabromide in Ethanol with Zinc. Isolation of Spiropentane. J. Am. Chem. Soc., 1946, 68, 1335
- [1946STR] Streiff, A.J.; Murphy, E.T.; Cahill, J.C.; Soule, L.F.; Sedlak, V.A.; Willingham, C.B.; Rossini, F.D., , Am. Pet. Inst. Res. Proj. 6, Natl. Bur. Stand., 1946.
- [1947API] Am. Pet. Inst. Res. Proj. 45, Ohio State Univ., 1947
- [1947AST] Aston, J. G.; Szasz, G. J.; The Thermodynamics of Butadiene-1,2 from Calorimetric and Spectroscopic Data. J. Am. Chem. Soc., 1947, 69, 3108
- [1947BEE] Beersmans, J.; Jungers, J.C., Synthese et Etude des Chlorure, Bromure et Iodure de Deuteromethyle, Bull. Soc. Chim. Belg., 1947, 56, 238
- [1947BOO] Boord, C.E.; Greenlee, K.W.; Perilstein, W.L.; Derfer, J.M., The Synthesis, Purification and Prop. of Hydrocarbons of Low Mol. Wt., Am. Pet. Inst. Res. Proj. 45, Ninth Annu. Rep., Ohio State Univ., June 30, 1947
- [1947FEN] Fenske, M.R.; Braun, W.G.; Wiegand, R.V.; Quiggle, D.; McCormick, R.H.; Rank, D.H., Raman Spectra of Hydrocarbons, Anal. Chem., 1947, 19, 700
- [1947GOL] Goldblum, K.B.; Martin, R.W.; Young, R.B., Vapor Pressure Data for Phenols, Ind. Eng. Chem., 1947, 39, 474
- [1947KET] Ketelaar, J.A.A.; van Velden, P.F.; Zalm, P., Vapour Pressures and Heats of Vaporization of cis- and trans-1,2-Dichloroethene, Recl. Trav. Chim. Pays-Bas, 1947, 66, 721
- [1947LEC] Lecat, M., Orthobaric Azeotropes of Sulfides, Bull. Cl. Sci., Acad. R. Belg., 1947, 33, 160
- [1947LEC2] Lecat, M., Some azeotropes of which one constituent is heterocyclic nitrogen, Ann. Soc. Sci. Bruxelles, Ser. 1, 1947, 61, 73.
- [1947LEG] Legge, D.I., Alkylbenzenes from Benzene and Isobutene. I. Preparation, Identification of Several Fractions. J. Am. Chem. Soc., 1947, 69, 2079
- [1947STU] Stull, Daniel R., Vapor Pressure of Pure Substances. Organic and Inorganic Compounds, Ind. Eng. Chem., 1947, 39, 517
- [1948ADK] Adkins, H.; Roebuck, A. K.; Isomerization of Cyclohexenes to Cyclopentenenes: Oxidation of These Alkenes to Glycols and Acids. J. Am. Chem. Soc., 1948, 70, 4041
- [1948BUC] Buck, F.R.; Elsner, B.B.; Forbes, E.J.; Morrell, S.H.; Smith, J.C.; Wallsgrove, E.R., The Use of Zinc Alkyls in the Preparation of Hydrocarbons with a Quaternary Carbon Atom, J. Inst. Pet., 1948, 34, 339
- [1948DOU] Douglas, T.B., Vapor Pressure of Methyl Sulfoxide from 20 to 50 deg. Calculation of the Heat of Vaporization, J. Am. Chem. Soc., 1948, 70, 2001.
- [1948GAN] Ganeff, J.M.; Jungers, J.C., Tensions de Vapeur du Systeme CH₃Cl - CH₂Cl₂, Bull. Soc. Chim. Belg., 1948, 57, 82

- [1948GIA] Giauque, W.F.; Jones, W.M., Carbonyl chloride. Entropy. Heat capacity. Vapor pressure. Heats of fusion and vaporization. Comments on solid sulfur dioxide structure, J. Am. Chem. Soc., 1948, 70, 120
- [1948GOR] Gordon, J.; Giauque, W.F., The entropy of ethyl chloride. Heat capacity from 18 to 287 K. Vapor pressure. Heats of fusion and vaporization, J. Am. Chem. Soc., 1948, 70, 1506
- [1948HEN] Hennion, G.F.; Groebner, J.J., Some Azeotropes of Alkylacetylenes and Ethyl Alcohol, J. Am. Chem. Soc., 1948, 70, 426
- [1948MIC] Michels, A.; Wassenaar, T., Vapour pressure of methylfluoride, Physica, 1948, 14, 104
- [1948PAC] Pace, E.L.; Aston, J.G., The thermodynamics of hexafluoroethane from calorimetric and spectroscopic data, J. Am. Chem. Soc., 1948, 70, 566
- [1948REY] Reynolds, T.W.; Ebersole, E.R.; Lamberti, J.M.; Chanan, H.H.; Ordin, P.M., Ind. Eng. Chem., 1948, 40, 1751
- [1948ROB] Robey, R.F.; Wiese, H.K., Chemical Analysis of Refinery C5 Hydrocarbon Fractions, Anal. Chem., 1948, 20, 926
- [1948ROS] Rossini, F.D., Pure compounds from petroleum purification and purity of hydrocarbons, Anal. Chem., 1948, 20, 110
- [1948RUS] Rust, F.F.; Seubold, F.H.; Vaughan, W.E., Dec. of Di-t-Alkyl Peroxides. II. React. of the Resultant Free Radical, J. Am. Chem. Soc., 1948, 70, 95
- [1948THO] Thornton, V.; Herald, A.E., Anal. Chem., 1948, 20, 9
- [1948WOO] H.W. Woolley, R.B. Scott, F.G. Brickwedde; "Compilation of thermal properties of hydrogen in its various isotopic and ortho-para modifications," J. Res. NBS 41, 379-475 (1948).
- [1949ALP] Alpert, N.; Elving, P.J., Vapor-liquid equilibria in binary systems ethylene dichloride-toluene and formic acid-acetic acid, Ind. Eng. Chem., 1949, 41, 2864
- [1949BIR] Birch, S.F.; Dean, R.A.; Fidler, F.A.; Lowry, R.A., The preparation of the C(10) monocyclic aromatic hydrocarbons, J. Am. Chem. Soc., 1949, 71, 1362
- [1949BOO] Boord, C.E.; Greenlee, K.W.; Perilstein, W.L.; Derfer, J.M., The Synthesis, Purification and Prop. of Hydrocarbons of Low Mol. Wt., Am. Pet. Inst. Res. Proj. 45, Eleventh Annu. Rep., Ohio State Univ., 1949.
- [1949BOO2] Boord, C.E.; Henne, A.L.; Greenlee, K.W.; Perilstein, W.L.; Derfer, J.M., The Grignard Reagent in hydrocarbon synthesis, Ind. Eng. Chem., 1949, 41, 609
- [1949CAS] Cason, J.; Way, R.L., The preparation of cyclobutane, J. Org. Chem., 1949, 14, 31
- [1949DEN] Denyer, R.L.; Fidler, F.A.; Lowry, R.A., Azeotrope Formation Between Thiols and Hydrocarbons, Ind. Eng. Chem., 1949, 41, 2727
- [1949DRE] Dreisbach, R.R.; Martin, R.A., Physical Data on Some Organic Compounds, Ind. Eng. Chem., 1949, 41, 2875

- [1949DRE2] Dreisbach, R.R.; Shrader, S.A., Vapor Pressure--Temperature Data on Some Organic Compounds, Ind. Eng. Chem., 1949, 41, 2879
- [1949ELS] Elsner, B.B.; Wallsgrove, E.R., J. Inst. Pet., 1949, 35, 259
- [1949FEW] Few, A.V.; Smith, J.W., The Solvent Effect on the Dipole Moment of Aniline, J. Chem. Soc., 1949, 1949, 3057
- [1949FOE] Foehr, E.G.; Fenske, M.R., Magneto-optic Rotation of Hydrocarbons, Ind. Eng. Chem., 1949, 41, 1956
- [1949FOR] Forziati, A.F.; Norris, W.R.; Rossini, F.D., Vapor Pressures and Boiling Points of Sixty API-NBS Hydrocarbons, J. Res. Natl. Bur. Stand., 1949, 43, 555
- [1949KRE] Kretschmer, Carl B.; Wiebe, Richard., Liquid-Vapor Equilibrium of Ethanol--Toluene Solutions, J. Am. Chem. Soc., 1949, 71, 1793
- [1949MUR] Murphy, M.T.; Duggan, A.C., Pyrolysis of Butadiene, J. Am. Chem. Soc., 1949, 71, 3347.
- [1949OTH] Othmer, D.F.; Savitt, S.A.; Krasner, A.; Goldberg, A.M.; Markowitz, D., Composition of vapors from boiling binary solutions. Systems with m- or p-cresol as one component., Ind. Eng. Chem., 1949, 41, 572
- [1949PAT] Patschky, A.; Liermeier, F., Z. Anal. Chem., 1949, 129, 62
- [1949ROB] Roberts, J.D.; Sauer, C.W., Small-Ring Compounds. III. Synthesis of Cyclobutanone, Cyclobutanol, Cyclobutene and Cyclobutane. J. Am. Chem. Soc., 1949, 71, 3925
- [1949SCH] Schuerch, C.; Huntress, E. H.; The Schmidt Reaction II. A New Rearrangement Observed during Degradation of Triethylacetic Acid. J. Am. Chem. Soc., 1949, 71, 2238
- [1949SCO] Scott, D.W.; Waddington, G.; Smith, J.C.; Huffman, H.M., Thermodynamic properties of three isomeric pentenes, J. Am. Chem. Soc., 1949, 71, 2767
- [1949WAD] Waddington, G.; Knowlton, J.W.; Scott, D.W.; Oliver, G.D.; Todd, S.S.; Hubbard, W.N.; Smith, J.C.; Huffman, H.M., Thermodynamic properties of thiophene, J. Am. Chem. Soc., 1949, 71, 797
- [1950AST] Aston, J.G.; Mastrangelo, S.V.R.; Moessen, G.W., The thermodynamics of 1-butyne from calorimetric and spectroscopic data, J. Am. Chem. Soc., 1950, 72, 5287
- [1950BOO] Boord, C.E.; Greenlee, K.W.; Derfer, J.M., The Synthesis, Purification and Prop. of Hydrocarbons of Low Mol. Wt., Am. Pet. Inst. Res. Proj. 45, Twelfth Annu. Rep., Ohio State Univ., 1950.
- [1950BRY] Bryce-Smith, D.; Turner, E.E., J. Chem. Soc., 1950, 1975
- [1950CHI] Chiurdoglu, G.; Fierens, P.J.C.; Henkert, C., Equilibrium of Isomerization of Cyclic Hydrocarbons I. Isomerization by Aluminum Chloride of Cyclopentane and Cyclohexane Hydrocarbons, C₈H₁₆, Bull. Soc. Chim. Belg., 1950, 59, 140
- [1950CHR] Christensen, L. D.; Smith, J.O., Pressure-enthalpy diagram for acetaldehyde, Ind. Eng. Chem., 1950, 42, 2128

- [1950CLA] Clark, A. M.; Din, F.; Equilibria Between Solid, Liquid, and Gaseous Phases at Low Temperature binary systems acetylene - carbon dioxide, acetylene - ethylene and acetylene - ethane. *Trans. Faraday Soc.*, 1950, 46, 901
- [1950CLE] Clendenning, K.A.; Macdonald, F.J.; Wright, D.E., Effects of position isomerism on the physical properties of glycols, *Can. J. Res., Sect. B*, 1950, 28, 608
- [1950COL] Coles, K.F.; Popper, Felix, Vapor-Liquid Equilibria. Ethylene Oxide - Acetaldehyde and Ethylene Oxide - Water Systems, *Ind. Eng. Chem.*, 1950, 42, 1434
- [1950FOR] Forziati, A. F.; Camin, D. L.; Rossini, F. D.; Density, Refractive Index, Boiling Point, and Vapor Pressure of Eight Monoolefin (1-Alkene), Six Pentadiene, and Two Cyclomonoolefin Hydrocarbons. *J. Res. Natl. Bur. Stand.*, 1950, 45, 406
- [1950KOF] Kofler, L.; Sitte, H., Melting Point Determination of Substances with Melt with Decomposition, *Monatsh. Chem.*, 1950, 81, 619
- [1950KOR] Kortuem, G.; Moegling, D.; Woerner, F., The Efficiency of Circulation Apparatuses for Determining Boiling and Vapor Pressure Diagrams of Binary Liquids., *Chem.-Ing.-Tech.*, 1950, 22, 453
- [1950MAT] Matthews, J.B.; Sumner, J.F.; Moelwyn-Hughes, E.A., The Vapour Pressures of Certain Liquids, *Trans. Faraday Soc.*, 1950, 46, 797
- [1950MEA] Mears, T.W.; Fookson, A.; Pomerantz, P.; Rich, E.H.; Dussinger, C.S.; Howard, F.L., Syntheses and Properties of Two Olefins, Six Paraffins, and Their Intermediates, *J. Res. Natl. Bur. Stand.*, 1950, 44, 299
- [1950MIC] Michels, A.; Wassenaar, T., The vapour pressure of ethylene, *Physica*, 1950, 16, 221
- [1950MUM] Mumford, S.A.; Phillips, J.W.C., 19. The Physical Properties of Some Aliphatic Compounds, *J. Chem. Soc.*, 1950, 75
- [1950NEL] Nelson, P. A.; Markham, A. E.; Vapor-liquid equilibrium of pyridine-acetic anhydride solutions. *J. Am. Chem. Soc.*, 1950, 72, 2417
- [1950RIC] Rice, F.O.; Wall, L.A., The pyrolysis of isobutylene, *J. Am. Chem. Soc.*, 1950, 72, 3967
- [1950SCO] Scott, D.W.; Finke, H.L.; Gross, M.E.; Guthrie, G.B.; Huffman, H.M., 2,3-Dithiabutane: low temperature heat capacity, heat of fusion, heat of vaporization, vapor pressure, entropy and thermodynamic functions, *J. Am. Chem. Soc.*, 1950, 72, 2424
- [1950SCO2] Scott, D.W.; Waddington, G., Vapor Pressure of cis-2-Pentene, trans-2-Pentene and 3-Methyl-1-butene, *J. Am. Chem. Soc.*, 1950, 72, 4310
- [1950SPU] Spurr, R.A.; Zeitlin, H., The dipole moments of some aliphatic ethers, *J. Am. Chem. Soc.*, 1950, 72, 4832
- [1950ZAK] Zakharkin, L.I., Bromotrichloromethane, *Sint. Org. Soedin.*, 1950-52, 1952, 18
- [1951AND] Andrews, L.J.; Keefer, R.M., The Argentation of Organic Iodides, *J. Am. Chem. Soc.*, 1951, 73, 5733

- [1951AUD] Audsley, A.; Goss, F.R., Atom Polarisation. Part II. Relationships Between Molecular Dimensions and Boiling Points for Branched-Chain Paraffins, *J. Chem. Soc.*, 1951, 120
- [1951BAR] Barcelo, J. R.; Le Fevre, R. J. W.; Smythe, B. M.; Variations of dipole moment with state for n-propyl and n-butylamines with notes on the apparent moments of in solutions of certain other amines.. *Trans. Faraday Soc.*, 1951, 47, 357
- [1951BLO] Blomquist, A.T.; Kwiatek, J., The Addition of Ketene to Cyclic Conjugated Dienes, *J. Am. Chem. Soc.*, 1951, 73, 2098
- [1951CHI] Chiurdoglu, G., Determination of cis-trans isomerism in cyclanes by means of the chiurdoglu rule: II pairs of trisubstituted stereoisomers, *Bull. Soc. Chim. Belg.*, 1951, 60, 39
- [1951DAY] Day, H.O.; Felsing, W.A., The Compresibility of 1-Pentene, *J. Am. Chem. Soc.*, 1951, 73, 4839
- [1951FLO] Flom, D. G.; Alpert, N.; Elving, P. J.; Systems involving cis- or trans-dichloroethylene and an acetal or ether. *Ind. Eng. Chem.*, 1951, 43, 1178
- [1951FOL] Foley, W. T.; Giguere, P. A.; Hydrogen peroxide and its analogues. II. Phase equilibrium in the system hydrogen peroxide-water, *Can. J. Chem.*, 1951, 29, 123
- [1951HOF] Hoff, M.C.; Greenlee, K.W.; Boord, C.E., The Quantitavtive Inversion of cis- and trans-Alkene Isomers: A New Synthesis of Normal cis-Alkenes, *J. Am. Chem. Soc.*, 1951, 73, 3329
- [1951KUT] Kutz, W. M.; Nickels, J. E.; McGovern, J. J.; Corson, B. B.; *J. Org. Chem.*, 1951, 16, 699
- [1951MCA] Mcallan, D.T.; Cullum, T.V.; Dean, R.A.; Fidler, F.A., The Preparation and Properties of Sulfur Compounds Related to Petroleum I. The Dialkyl Sulfides and Disulfides, *J. Am. Chem. Soc.*, 1951, 73, 3627
- [1951NEW] Newitt, D.M.; Weale, K.E., Pressure-volume-temperature relations in liquids and liquid mixtures: II the compression isotherms of some organic liquids up to 1000 atmospheres, and the compressions of some aqueous and nonaqueous, *J. Chem. Soc.*, 1951, 3092
- [1951OLI] Oliver, G. D.; Grisard, J. W.; Cunningham, C. W.; Thermodynamic Properties and P-V-T Relations of Chlorotrifluoroethylene. *J. Am. Chem. Soc.*, 1951, 73, 5719
- [1951SIE] Sieg, L.; J.L. Cruetzen, and W. Jost, *Z. Phys. Chem. (Leipzig)*, 1951, 198, 263
- [1951WIS] Wise, P. H.; Serijan, K. T.; Goodman, I. A.; *Nat. Advis. Comm. Aeronaut., Tech. Rep. Nat. Advis. Comm. Aeronaut.*, 1951, 1003
- [1952BER] Bergmann, E. D.; Resnick, C.; *J. Org. Chem.*, 1952, 17, 1291
- [1952BRO] Brown, I., Liquid-Vapour Equilibria. III. The Systems Benzene-n-Heptane, n-Hexane-Chlorobenzene, and cycloHexane-Nitrobenzene, *Aust. J. Sci. Res. Ser. A*, 1952, 530

- [1952CUC] Cuculo, J.A.; Bigelow, L.A., The Action of elementary Fluorine upon Organic Compounds. XVII. The Direct Fluorination of Acetonitrile, *J. Am. Chem. Soc.*, 1952, 74, 710
- [1952EDW] Edwards, G., The Vapour Pressure of Tetranitromethane, *Trans. Faraday Soc.*, 1952, 48, 513
- [1952GIL] Gilbert, R.D.; Williams, L.W., *J. Am. Chem. Soc.*, 1952, 74, 4114
- [1952GOE] Goering, H.L.; Abell, P.I.; Aycok, B.F., *J. Am. Chem. Soc.*, 1952, 74, 3588
- [1952GUT] Guthrie, G. B.; Scott, D. W.; Waddington, G.; Thiacyclopropane (ethylene sulfide): Infrared Spectrum, Vapor Pressure, and some Thermodynamic Properties. *J. Am. Chem. Soc.*, 1952, 74, 2795
- [1952GUT2] Guthrie, G.B., Jr.; Scott, D.W.; Hubbard, W.N.; Katz, C.; McCullough, J.P.; Gross, M.E.; Williamson, K.D.; Waddington, G., Thermodynamic properties of furan, *J. Am. Chem. Soc.*, 1952, 74, 4662
- [1952HAS] Haszeldine, R.N.; *J. Chem. Soc.*, 1952, 4259
- [1952INO] Inokuchi, Hiroo; Shiba, Sukekuni; Handa, Takashi; Akamatu, Hideo, Heats of Sublimation of Condensed Polynuclear Aromatic Hydrocarbons, *Bull. Chem. Soc. Jpn.*, 1952, 25, 299
- [1952JAM] James, A.T.; Martin, A.J.P., *Biochem. J.*, 1952, 50, 679
- [1952JON] Jones, W.S.; Tamplin, W.S., Chapter 3. Physical Properties of Ethylene Glycol in Glycols, George O. Curme, Jr., ed(s)., Reinhold Publishing Corporation, New York, 1952, 27
- [1952KAA] Kaarsemaker, S.; Coops, J., Thermal quantities of some cycloparaffins. Part III. Results of measurements, *Rec. Trav. Chim. Pays/Bas*, 1952, 71, 261
- [1952ROB] Robey, R. F.; Hudson, B. E.; Wiese, H. K.; Determination of individual acetylenes in butadiene and in c4 hydrocarbon fractions. *Anal. Chem.*, 1952, 24, 1080
- [1952SCA] Scatchard, G.; G. M. Kavanagh, L. B. Ticknor, Vapor-Liquid Equilibrium. VIII. Hydrogen Peroxide—Water Mixtures. *J. Am. Chem. Soc.* 1952, 74, 3715
- [1952STE] Stevens, H.C.; Grummitt, O., The Stereochemistry of the Reactions of cis- and trans-2-Chlorocyclohexanol with Thionyl Chloride, *J. Am. Chem. Soc.*, 1952, 74, 4876
- [1952VAN] Van derbij, J.R.; Kooijman, E.C., Properties of a number of cyclic olefins, *Recl. Trav. Chim. Pays-Bas*, 1952, 71, 837
- [1952WAR] Waring, W., Thermodynamic Properties of Formic Acid, *Chem. Rev.*, 1952, 51, 171.
- [1952WES] West, P.W.; Senise, P.; Burkhalter, T.S., *Anal. Chem.*, 1952, 24, 1250
- [1952WHI] White, P.T.; Barnard-Smith, D.G.; Fidler, F.A., Vapor Pressure--Temperature Relationships of Sulfur Compounds Related to Petroleum, *Ind. Eng. Chem.*, 1952, 44, 1430
- [1953AST] Aston, J.G.; Wood, J.L.; Zolki, T.P., The thermodynamic properties and configuration of unsymmetrical dimethylhydrazine, *J. Am. Chem. Soc.*, 1953, 75, 6202

- [1953ATK] Atkinson, B.; Trenwith, A. B.; The Thermal Decomposition of Tetrafluoroethylene. J. Chem. Soc., 1953, 2082
- [1953BIR] Birch, S.F., Sulphur Compounds in Petroleum, J. Inst. Pet., 1953, 39, 185.
- [1953FUR] Furukawa, G.T.; McCoskey, R.E.; Reilly, M.L., Heat capacity, heats of fusion and vaporization, and vapor pressure of tetrafluoroethylene, J. Res. Natl. Bur. Stand., 1953, 51, 69
- [1953HEL] Hellwig, L.R.; Van Winkle, M., Vapour-liquid equilibria for ethyl alcohol binary systems., Ind. Eng. Chem., 1953, 45, 624
- [1953KAR] Karo, W.; McLaughlin, R.L.; Hipsher, H.F., Dicyclic Hydrocarbons VI. 1,2,3,5-tetrahydronaphthalene and 1-Alkyl-1,2,3, 4-tetrahydronaphthalenes, J. Am. Chem. Soc., 1953, 75, 3233
- [1953LAZ] LaZerte, J.D.; Hals, L. J.; Reid, T.S.; Smith, G. H.; Pyrolyses of the Salts of the Perfluoro Carboxylic Acids. J. Am. Chem. Soc., 1953, 75, 4525
- [1953MCK] McKenna, F.E.; Tartar, H.V.; Lingafelter, E.C., Studies of hemiacetal formation in alcohol - aldehyde systems: II refraction studies, J. Am. Chem. Soc., 1953, 75, 604
- [1953OWE] Owen, L. N.; Saharia, G. .S.; 526. Alicyclic glycols. Part XII. Derivatives of cycloheptane-1: 2-diol. . Chem. Soc., 1953, 2582
- [1953PIN] Pines, H.; Huntsman, W.D.; Ipatieff, V.N., Isomerization Accompanying Alkylation. IX. The Reaction of Benzene with n- and Isopropylcyclopropane. The Synthesis of n- and Isopropylcyclopropane, J. Am. Chem. Soc., 1953, 75, 2311
- [1953PIN2] Pines, H.; Huntsman, W.D.; Ipatieff, V.N., "Isomerization of Saturated Hydrocarbons. XI.1 The Isomerization of Alkylcyclopropanes and Alkylcyclobutanes in the Presence of Aluminum Halide," J. Am. Chem. Soc. 1953, 75, 10, 2315–2317
- [1953PRO] Prober, M.; The Synthesis and Polymerization of Some Fluorinated Styrenes. J. Am. Chem. Soc., 1953, 968
- [1953RAT] Rathjens, G.W., Jr.; Gwinn, W.D., Heat capacities and entropy of cyclobutane, J. Am. Chem. Soc., 1953, 75, 5629
- [1954API] Am. Pet. Inst. Res. Proj. 45, Tech. Rep. 13, Ohio State Univ., 1954
- [1954ARM] Armstrong, G. T.; "The Vapor Pressure of Nitrogen," J. Res. Natl. Bur. Stand., 1954, 53, 263
- [1954CAD] Cadogan, J.I.G.; Hey, D.H.; Williams, G.H., Homolytic Aromatic Substitution V. Partial Rate Factors for the Phenylation of Biphenyl, J. Chem. Soc., 1954, 3352
- [1954CAM] Camin, David L.; Forziati, Alphonse F.; Rossini, Frederick D., Physical Properties of n-Hexadecane, n-Decylcyclopentane, n-Decylcyclohexane, 1-Hexadecene and n-Decylbenzene, J. Phys. Chem., 1954, 58, 440
- [1954DOE] Doering, W. von E.; Knox, L. H.; J. Am. Chem. Soc., 1954, 76, 3203

- [1954DOU] Douglas, T. B.; Furukawa, G. T.; McCloskey, R. E.; Ball, A. F.; Calorimetric Properties of Normal heptane From 0° to 520 K. J. Res. Natl. Bur. Stand., 1954, 53, 139
- [1954GIG] Giguere, P. A.; Liu, I. D.; Dugdale, J. S.; Morrison, J. A.; Hydrogen peroxide: the low temperature heat capacity of the solid and the third law entropy, Can. J. Chem., 1954, 32, 117
- [1954HEN] Hennion, G.F.; Price, C.C.; McKeon, T.F., Jr., The Preparation of Vinylacetylene, J. Am. Chem. Soc., 1954, 76, 5160.
- [1954HIP] Hipsher, H.F.; Wise, P.H., Dicyclic Hydrocarbons VIII. 1-Alkyl-naphthalenes and Some of Their Tetrahydro Derivatives, J. Am. Chem. Soc., 1954, 76, 1747
- [1954MAR] C. Marsden, Solvents and Allied Substances Manual (Cleaver-Hume Press LTD, London, 1954)
- [1954MCC] McCullough, J.P.; Scott, D.W.; Pennington, R.E.; Hossenlopp, I.A.; Waddington, Guy, Nitromethane: The Vapor Heat Capacity, Heat of Vaporization, Vapor Pressure and Gas Imperfection; the Chemical Thermodynamic Properties from 0 to 1500°K., J. Am. Chem. Soc., 1954, 76, 4791
- [1954MIL] Miller, H.M.; Greenlee, K.W.; Derfer, J.M.; Boord, C.E., J. Org. Chem., 1954, 19, 1882
- [1954OST] Osthoff, Robert C.; Grubb, W.T., Physical Properties of Organosilicon Compounds. III. Thermodynamic Properties of Octamethylcyclotetrasiloxane, J. Am. Chem. Soc., 1954, 76, 399
- [1954POM] Pomerantz, P.; Fookson, A.; Mears, T.W.; Rothberg, S.; Howard, F.L., Synthesis and Physical Properties of Several Aliphatic and Alicyclic Hydrocarbons, J. Res. Natl. Bur. Stand., 1954, 52, 59
- [1954POM2] Pomerantz, P.; Fookson, A.; Mears, T.W.; Rothberg, S.; Howard, F.L., Synthesis and Physical Properties of Several Acetylenic Hydrocarbons, J. Res. Natl. Bur. Stand., 1954, 52, 51
- [1954PUR] Purnell, J.H.; Bowden, S.T., Miscibility Phenomena in Ternary Systems Containing Diphenyl Ether Water, and an Aliphatic Alcohol, J. Chem. Soc., 1954, 539
- [1954SCH] Schlatter, M.J., J.; t-Alkyl Groups. II. Use of the t-Butyl Group as a Blocking Group in the Synthesis of 1,2,3-Trialkylbenzenes. J. Am. Chem. Soc., 1954, 76, 4952.
- [1954WIT] Witschonke, C.R., Freezing point and purity data for some organic compounds, Anal. Chem., 1954, 26, 562
- [1955BRO] Brown, H.C.; Nakagawa, M., Steric effects of elimination reactions: III the effect of the steric requirements of alkyl substitutes upon the extent and direction of unimolecular elimination in the solvolysis of secondary alkyl, J. Am. Chem. Soc., 1955, 77, 3614.
- [1955CAM] Camin, David L.; Rossini, Frederick D., Physical Properties of Fourteen API Research Hydrocarbons, C 9 to C 15, J. Phys. Chem., 1955, 59, 1173
- [1955CRI] Criegee, R.; Kerckow, A.; Zinke, H., Chem. Ber., 1955, 88, 1878

- [1955DRE] Dreyer, R.; Martin, W.; von Weber, U., Die Settigungsampfdrucke von Benzol, Toluol, ethylbenzol, Styrol, Cumol und Brombenzol zwischen 10 und 760 Torr, J. Prakt. Chem., 1955, 1, 324
- [1955EDG] Edgell, W. F.; Parts, L.; J. Am. Chem., 1955, 77, 4899
- [1955GEN] Gensler, W.J.; Mahadevan, A.P., Preparation and Structure of Tridecadiyne-5,8, J. Am. Chem. Soc., 1955, 77, 3076
- [1955GRA] Gragson, J. T.; Greenlee, K. W.; Derfer, J. M.; Boord, C. E. Cyclopropane Hydrocarbons from .gamma.-Bromo Ethers J. Org. Chem., 1955, 20, 275
- [1955HAM] Hammond, B.R.; Stokes, R.H., Diffusion in Binary Liquid Mixtures. Part 2 The Diffusion of Carbon Tetrachloride in Some Organic Solvents at 25 deg, Trans. Faraday Soc., 1955, 51, 1641
- [1955HAN] Hansen, R.S.; Hansen, R.D., J. Phys. Chem., 1955, 59, 496
- [1955HAN2] Hansen, R.S.; Miller, F.A.; Christian, S.D., Activity coefficients of components in the systems water - acetic acid, water - propionic acid and water - n-butyric acid at 25 C, J. Phys. Chem., 1955, 59, 391
- [1955JUN] Junghans, W.; Von Weber, U., The Binary System Ethylbenzene-Styrene , 90 and 120 C by an entrainment method and heats of mixing at 20 C, J. Prakt. Chem., 1955, 2, 265.
- [1955KOC] Kochi, J.K., J. Am. Chem. Soc., 1955, 77, 5274
- [1955LEM] Lemaire, H.; Lucas, H. J. ; The Mercury-catalyzed Addition of Acetic Acid to 3-Hexyne. J. Am. Chem. Soc., 1955, 77, 939
- [1955MYE] Myers, H. S.; Fenske, M. R.; Measurement and Correlation of Vapor Pressures Data for High Boiling Hydrocarbons. Ind. Eng. Chem., 1955, 47, 1652
- [1955NIE] Nielsen, J.R.; Classenn, H.H.; Moran, N.B., Infrared and Raman Spectra of Fluorinated Ethanes VIII. Pentafluoroethane, J. Chem. Phys., 1955, 23, 329
- [1955RUS] Russell, G.A.; Brown, H.C., The Competitive Halogenation of Cyclohexane and Aalkyl Hydrocarbons Evidence as to the Nature of the Transition States in Halogenation Reactions, J. Am. Chem. Soc., 1955, 77, 4578
- [1955SHI] Shiner, V.J.; Boskin, M.J.; Smith, M.L., The Effects of γ -Methyl Substitution on the Rates of the Bimolecular Displacement and Elimination Reactions of Alkyl Halides, J. Am. Chem. Soc., 1955, 77, 5525
- [1955STR] Streiff, A.J.; Hulme, A.R.; Cowie, P.A.; Krouskop, N.C.; Rossini, F.D., Purification, Purity, and Freezing Points of Sixty-four American Petroleum Institute Standard and Research Hydrocarbons, Anal. Chem., 1955, 27, 411
- [1955TIM] Timmermans, J.; Hennaut-Roland, M., Work of the International Bureau of Physical-Chemical Standards. IX. The Physical Constants of Twenty Organic Compounds, J. Chim. Phys. Phys.-Chim. Biol., 1955, 52, 223
- [1955TSY] Tsyapkina, O.Y., Study of Vacuum Pressure Influence on Efficiency of Separation of Some Polynuclear Compounds of Coal Tar Rectifications, Zh. Prikl. Khim. (Moscow), 1955, 28, 185

- [1955VAR] Varshni, Y. P.; Mohan, H.; J. Indian Chem. Soc., 1955, 32, 211
- [1956AMB] Ambrose, D., The Vapour Pressures and Critical Temperatures of Acetylene and Carbon Dioxide, Trans. Faraday Soc., 1956, 52, 772
- [1956BLO] Blomquist, A., T.; Verdol, J. A.; THERMAL DIMERIZATION OF ALLENE TO 1,2-DIMETHYLENECYCLOBUTANE. J. Am. Chem. Soc., 1956, 78, 109
- [1956BRO] Brockhaus, Von A.; Jenckel, E., Über die kinetik des thermischen abbaues von polymethacrylsäuremethylester., Makromol. Chem., 1956, 18, 262
- [1956BUS] Bussert, J.F.; Greenlee, K.W.; Derfer, J.M.; Boord, C.E., Synthesis of the Geometrical Isomers of 1,2,3-Trimethylcyclohexane, J. Am. Chem. Soc., 1956, 78, 6076.
- [1956CAR] Carleton, L. T.; Phase equilibria in dimethylhydrazine-water system. Ind. Eng. Chem. Chem. Eng. Data Ser., 1956, 1, 21
- [1956CHI] Chiurdoglu, G.; Laune, J.; Poelmans, M., Acyclic compounds with Quarternary Carbon IV. Conjugation with 1-Ethoxycaronylcyclopropanespirocyclanes, Bull Soc. Chim. Belges, 1956, 65, 257
- [1956COP] Cope, A.C.; Bumgardner, C.L., Cyclic Polyolefins XL. cis-cis and cis-trans-1,3-Cyclooctadiene from Cycloocten-3-yl dimethylamine, J. Am. Chem. Soc., 1956, 78, 2812.
- [1956EPS] Epstein, M.B.; Willingham, C.B.; Mair, B.J.; Rossini, F.D., Hydrocarbons in the 126 to 132 deg Fraction of Petroleum, Anal. Chem., 1956, 28, 1924
- [1956FIE] Field, F. H.; Hastings, S. H.; Determination of Unsaturated Hydrocarbons by Low-voltage Mass Spectrometry. Anal. Chem., 1956, 28, 1248
- [1956FIN] Finke, H. L.; Scott, D. W.; Gross, M. E.; Messerly, J. F.; Waddington, G.; Cycloheptane, Cyclooctane and 1,3,5-Cycloheptatriene. Low Temperature Thermal Properties, Vapor Pressure and Derived Chemical Thermodynamic Properties. J. Am. Chem. Soc., 1956, 78, 5469
- [1956HAI] Haines, W.E.; Helm, R.V.; Cook, G.L.; Ball, J.S., Purification and Properties of Ten Organic Sulfur Compounds-Second Series, J. Phys. Chem., 1956, 60, 549
- [1956LAN] Landolt, H. H.; Bornstein, R.; "Zahlenwerte und Function aus Physik, Chemie, Astronomie, Geophysik and Technik," 3 Teil, II Band, Springer-Verlag, Berlin, 1956
- [1956LI] Li, J.C.M.; Pitzer, K.S., The thermodynamic properties of 1,1-dichloroethane: heat capacities from 14 to 294 K., heats of fusion and vaporization, vapor pressure and entropy of the ideal gas. The barrier to internal rotation, J. Am. Chem. Soc., 1956, 78, 1077
- [1956MCC] McCormack, J.B.; Walkup, J.H.; Rush, R.I., Boiling Point Comp. Diagram for the Sys. 1,4-Dioxane-n-Butyl Alcohol alcohol, J. Phys. Chem., 1956, 60, 826.

- [1956MIL] Miller, W.T.; Dittman, A.L., The Mechanism of Fluorination I. Fluorine Sensitized Oxidation of Trichloro- and Tetrachloroethylene, *J. Am. Chem. Soc.*, 1956, 78, 2793
- [1956MYE] Myers, H.S., Binary mixtures of naphthenes and aromatics vapor-liquid equilibrium data, *Ind. Eng. Chem.*, 1956, 48, 1104
- [1956OMU] Omura, I.; Higasi, K.; Baba, H., Ionization potentials of some organic molecules. II. Aliphatic compounds, *Bull. Chem. Soc. Japan*, 1956, 29, 504
- [1956PEN] Pennington, R.E.; Scott, D.W.; Finke, H.L.; McCullough, J.P.; Messerly, J.F.; Hossenlopp, I.A.; Waddington, G., The chemical thermodynamic properties and rotational tautomerism of 1-propanethiol, *J. Am. Chem. Soc.*, 1956, 78, 3266
- [1956PIC] Pick, J.; Fried, V.; Hala, E.; Vilim, O., Der Dampfdruck des Athylenglykolmonomethylathers und des Athylenglykolmonoathylathers, *Collect. Czech. Chem. Commun.*, 1956, 21, 260
- [1956PRI] Price, C.C.; Osgan, M., The Polymerization of Isopropylene Oxide, *J. Am. Chem. Soc.*, 1956, 78, 4787
- [1956SEA] Searles, S.; Tamres, M.; Block, F.; Quarterman, L. A.; Hydrogen Bonding and Basicity of Cyclic Imines. *J. Am. Chem. Soc.*, 1956, 78, 4917
- [1956STU] Stutchbury, J.E., Compressions of Organic Liquids and Their Mixtures with Water, *Aust. J. Chem.*, 1956, 9, 536
- [1956SUM] Sumrell, G.; Ham, G.E., *J. Am. Chem. Soc.*, 1956, 78, 5573
- [1956THO] Thorp, N.; Scott, R. L.; Fluorocarbon solutions at low temperatures. I. The liquid mixtures CF₄-CHF₃, CF₄-CH₄, CF₄-Kr, CH₄-Kr., *J. Phys. Chem.*, 1956, 60, 670
- [1956TRA] Traynham, J. G.; Sehnert, M. F.; Ring Size and Reactivity of Cyclic Olefins: Complexation with Aqueous Silver Ion, *J. Am. Chem. Soc.*, 1956, 78, 4024
- [1956WIR] Wirth, H.E., Palmer, E.D., *J. Phys. Chem.* 60, 911 (1956)
- [1957DES] Desty, D.H.; Whyman, B.H.F., Application of Gas-Liquid Chromatography to Analysis of Liquid Petroleum Fractions, *Anal. Chem.*, 1957, 29, 320
- [1957KIR] Kirmann, A.; Hamaide, N., New Preparation of Oxacyclanes, *Bull. Soc. Chim. Fr.*, 1957, 1957, 789
- [1957LOE] Loev, B.; Massengale, J. T.; Notes - Cation Exchange Resins as Catalysts in the Alkylation of Phenols. *J. Org. Chem.*, 1957, 22, 988
- [1957MCC] McCullough, J. P.; Douslin, D. R.; Messerly, J. F.; Hossenlopp, I. A.; Kincheloe, T. C.; Waddington, G.; Pyridine: Experimental and Calculated Chemical Thermodynamic Properties Between 0 and 1500 K; A Revised Vibrational Assignment, *J. Am. Chem. Soc.*, 1957, 79, 4289
- [1957MIL] Miller, W. T.; Bergman, E., Jr.; Fainberg, A. H.; *J. Am. Chem. Soc.* 79, 4159 (1957)

- [1957PET] Petrov, A.A.; Kolesova, V.A.; Porfir'eva, Y.I., Conjugated Systems. LXXVIII Raman Spectra and Reactivity of Vinylacetylenic Hydrocarbons, *Zh. Obshch. Khim.*, 1957, 27, 2081
- [1958BAK] Bak, E.; Geankoplis, C. J.; Distribution coefficients of fatty acids between water and methyl isobutyl ketone. *Ind. Eng. Chem. Chem. Eng. Data Ser.*, 1958, 3, 256
- [1958BID] Biddiscombe, D. P.; Martin, J. F.; Vapour pressures of phenol and the cresols. *Trans. Faraday Soc.*, 1958, 54, 1316
- [1958COS] Costello, J.M.; Bowden, S.T., The Temperature Variation of Orthobaric Density Difference in Liquid-Vapor Systems III. Alcohols, *Recl. Trav. Chim. Pays-Bas*, 1958, 77, 36
- [1958LEV] Levi, L.; Nicholls, R.V.V., Formation of Styrene by Pyrolysis of Aromatic Heterocyclic Aldehyde- Aliphatic Acid Anhydride Mixtures on Morden Bentonite, *Ind. Eng. Chem.*, 1958, 50, 1005
- [1958RAM] Rampolla, R.W.; Smyth, C.P., *J. Am. Chem. Soc.*, 1958, 80, 1057
- [1958TUR] Turner, R.B.; Garner, R.H., Heats of Hydrogenation V. Relative Stabilities in Certain Exocyclic- Endocyclic Olefin Pairs, *J. Am. Chem. Soc.*, 1958, 80, 1424
- [1959COS] Costello, J.M.; Bowden, S.T., The Temperature Variation of Orthobaric Density Difference in Liquid-Vapour Systems. V. Amines, *Recl. Trav. Chim. Pays-Bas*, 1959, 78, 391
- [1959EIS] Eiseman, B.J., Why Refrigerant 22 should be favored for Absorption refrigeration, *ASHRAE J.*, 1959, 1, 45
- [1959ELL] Ellis, S.R.M.; Razavipour, M., Vapour-liquid equilibria at reduced pressures: n-butanol-ethyl benzene; aniline-p-cymene, *Chem. Eng. Sci.*, 1959, 11, 99
- [1959HIL] Hildenbrand, D.L.; McDonald, R.A., The Heat of Vaporization and Vapor Pressure of Carbon Tetrachloride; the Entropy from Calorimetric Data, *J. Phys. Chem.*, 1959, 63, 1521
- [1959HIL2] Hildenbrand, D. L.; McDonald, R. A.; Kramer, W. R.; Stull, D. R.; Thermodyn. and spectroscopic study of vinylidene chloride. I. Thermodyn. properties of the solid, liquid, and ideal gas. *J. Chem. Phys.*, 1959, 30, 930
- [1959HOU] Hou, Y.-C.; Martin, J. J.; Physical and Thermodynamic properties of trifluoromethane, *AIChE J.*, 1959, 5, 125
- [1959KOG] Kogan, V. B.; Fridman, V. M.; Romanova, T. G.; Separation of mixtures of alcohols and hydrocarbons by extraction, *J. Appl. Chem. USSR (Engl. Transl.)*, 1959, 32, 864
- [1959MCC] McCullough, J.P.; Douslin, D.R.; Hubbard, W.N.; Todd, S.S.; Messerly, J.F.; Hossenlopp, I.A.; Frow, F.R.; Dawson, J.P.; Waddington, G., Pyrrolidine: Chemical thermodynamic properties between 0 and 1500°K; effect of pseudorotation; and an unusual thermal anomaly in the liquid state, *J. Am. Chem. Soc.*, 1959, 81, 5884
- [1959MCD] McDonald, R.A.; Shrader, S.A.; Stull, D.R., Vapor Pressures and Freezing Points of Thirty Pure Organic Compounds., *J. Chem. Eng. Data*, 1959, 4, 311

- [1959PAR] Paridon, L. J.; MacWood, G. E.; Vapor Pressure of Diborane. J. Phys. Chem. 1959, 63, 1997
- [1959PIN] Pinchas, S.; Shabtai, J.; Herling, J.; Gil-Av, E.; The Preparation and Infrared Spectra of the Three Ethylcyclopentenes, Methylene-, and ethyldenecyclopentane. J. Inst. Pet., 1959, 45, 311
- [1959SEP] Seprakova, M.; Paulech, J.; Dykyj, J., Dampfdruck der Butyraldehyde, Chem. Zvesti, 1959, 13, 313
- [1959SKI] Skinner, H.A.; Snelson, A., Heats of Hydrogenation. Part 3, Trans. Faraday Soc., 1959, 55, 404
- [1959TIM] Timmermans, J.; Hennaut-Roland, M., Work of the International Bureau of Physico-Chemical Properties physical constants of twenty organic compounds, J. Chim. Phys. Phys.-Chim. Biol., 1959, 56, 984
- [1960AND] Andon, R.J.L.; Biddiscombe, D.P.; Cox, J.D.; Handley, R.; Harrop, D.; Herington, E.F.G.; Martin, J.F., Thermodynamic properties of organic oxygen compounds. Part I. Preparation and physical properties of pure phenol, cresols, and xylenols, J. Chem. Soc., 1960, 5246
- [1960CAM] Camin, D. L.; Rossini, F. D.; Physical properties of 16 selected C7 and C8 alkene hydrocarbons. J. Chem. Eng. Data, 1960, 5, 368
- [1960CHA] Chan, S.I.; Zinn, J.; Fernandez, J.; Gwinn, W.D., Trimethylene Oxide. I. Microwave Spectrum, Dipole Moment, and Double Minimum Vibration, J. Chem. Phys., 1960, 33, 1643
- [1960CHA2] Chari, N. C.; Thermodynamic Properties of Carbon Tetrafluoride. Ph.D. Thesis, University of Michigan, Ann Arbor, MI (1960)
- [1960CON] Conti, J.J.; Othmer, D.F.; Gilmont, R., Composition of Vapors from Boiling Binary Solutions. Systems Containing Formic Acid, Acetic Acid, Water and Chloroform, J. Chem. Eng. Data, 1960, 5, 301
- [1960COP] Copp, J.L.; Findlay, T.J.V., Thermodynamics of Binary Systems Containing Amines. Part 4., Trans. Faraday Soc., 1960, 56, 13
- [1960GLE] Glew, D.N., The gas hydrate of bromochlorodifluoromethane, Can. J. Chem., 1960, 38, 208
- [1960HAI] Haines, W. E.; Helm. R. V.; Cook, G. L.; Ball, J. S.; Purification and Properties of Ten Organic Sulfur Compounds, J. Chem. Eng. Data, 1960, 5, 112
- [1960KLA] Klaeboe, P.; Nielsen, J. R.; Infrared and Raman Spectra of Fluorinated Ethanes XIII. 1,2-Difluoroethane. J. Chem. Phys., 1960, 33, 1764
- [1960KYL] Kyle, B.G.; Tetlow, N.J., Perfluorotributylamine as an Extractive Distillation Solvent for a Hydrocarbon Mixture, J. Chem. Eng. Data, 1960, 5, 275
- [1960MAI] Mair, B. J.; Krouskop, N. C.; Rossini, F. D.; Seventeen Aromatic Hydrocarbons Separated from the 180deg to 200deg C. Fraction of Petroleum. J. Chem. Eng. Data, 1960, 5, 186
- [1960MOR] Morris, J.C.; Lanum, W.J.; Helm, R.V.; Haines, W.E.; Cook, G.L.; Ball, J.S., Purification and Properties of Ten Organic Sulfur Compounds, J. Chem. Eng. Data, 1960, 5, 112

- [1960MUE] Mueller, C.R.; Ignatowski, A.J., Equilibrium and Transport Properties of the Carbon Tetrachloride-Methylene Chloride System, J. Chem. Phys., 1960, 32, 1430
- [1960NOV] Novák, J.; Matous, J.; Pick, J., Gleichgewicht flüssigkeit-dampf XXIV. Gleichgewicht flüssigkeit-dampf im system cyclohexylamin-cyclohexanol-anilin, Collect. Czech. Chem. Commun., 1960, 25, 2405
- [1960SKI] Skinner, H.A.; Snelson, A., The Heats of Combustion of the Four Isomeric Butyl Alcohols, Trans. Faraday Soc., 1960, 56, 1776
- [1960TER] Terry, T.D.; Kepner, R.E.; Dinsmore, W., Binary Systems Containing Isoamyl Alcohol or Active Amyl Alcohol, J. Chem. Eng. Data, 1960, 5, 403
- [1961BEL] Belknap, R.C.; Weber, J.H., Vapor-Liquid Equilibrium at Atmospheric Pressure. Systems Containing Ethyl Alcohol, n-Hexane, Benzene, and Methylcyclopentane, J. Chem. Eng. Data, 1961, 6, 485
- [1961CRA] Craig, N. C.; Entemann, E. A.; Thermodynamics of cis-trans isomerizations, the 1,2-difluoroethylenes. J. Am. Chem. Soc., 1961, 83, 3047
- [1961DIT] Ditter, J.F.; Perrine, J.C.; Shapiro, I., Vapor Pressure of Deuterodiborane, J. Chem. Eng. Data, 1961, 6, 271
- [1961JES] Jessup, R.S.; Stanley, C.L., Heats and volumes of mixing in several C(12) hydrocarbon systems, J. Chem. Eng. Data, 1961, 6, 368
- [1961NIC] Nickerson, J.K.; Kobe, K.A.; McKetta, John J., The thermodynamic properties of the methyl ketone series, J. Phys. Chem., 1961, 65, 1037
- [1961SCO] Scott, D.W.; Messerly, J.F.; Todd, S.S.; Guthrie, G.B.; Hossenlopp, I.A.; Moore, R.T.; Osborn, A.; Berg, W.T.; McCullough, J.P., Hexamethyldisiloxane: chemical thermodynamic properties and internal rotation about the siloxane linkage, J. Phys. Chem., 1961, 65, 1320
- [1961STR] T. R. Strobridge, "The thermodynamic properties of nitrogen from 64 to 300 K between 0.1 and 200 atmospheres," Natl. Bur. Stand. Tech. Note No. 129, 1961
- [1961WRI] Wright, F. J.; Influence of temperature on viscosity of nonassociated liquids. J. Chem. Eng. Data, 1961, 6, 454
- [1962ALL] Allinger, N.L.; Hu, S., Conformational Analysis XXI. The Ethyl Group, J. Am. Chem. Soc., 1962, 84, 370
- [1962API] Am. Pet. Inst. Res. Proj. 58A, Annu. Rep., 1962
- [1962BIT] Bittrich, H.J.; Kauer, E., Zur Thermodynamik des Systems Diathylamin-Triathylamin. I. Das Flüssigkeits-Dampf-Gleichgewicht, Z. Phys. Chem. (Frankfurt/Main), 1962, 219, 224
- [1962BRO] Brown, I.; Smith, F., Aust. J. Chem., 1962, 15, 3
- [1962COP] Cope, A. C.; Youngquist, M. J.; Proximity Effects XXV. Pyrolysis of Cyclooctyl and Cyclodecyl Esters. J. Am. Chem. Soc., 1962, 84, 2411

- [1962DAV] Davies, J.V.; Sunner, S., Heats of Combustion and Formation of Thiolane and the Thiolenes, *Acta Chem. Scand.*, 1962, 16, 1870
- [1962HAS] Haselden, G. G.; Snowden, P. Equilibrium Properties of the Carbon Dioxide+ Propylene and Carbon Dioxide + Cyclopropane Systems at Low Temperatures *Trans. Faraday Soc.*, 1962, 58, 1515
- [1962HAT] Hatton, W.E.; Hildenbrand, D.L.; Sinke, G.C.; Stull, D.R., Chemical thermodynamic properties of aniline, *J. Chem. Eng. Data*, 1962, 7, 229
- [1962HOL] Holmberg, K. E.; Isotopic Effects in Systems Containing an Amine and Carbon Dioxide. *Acta Chem. Scand.*, 1962, 16, 2117
- [1962KAE] Kaes, G.L.; Weber, J.H., Vapor-Liquid Equilibrium at Atmospheric Pressure. Systems Containing Ethyl Alcohol, n-Hexane, Benzene, and Methylcyclopentane, *J. Chem. Eng. Data*, 1962, 7, 344
- [1962SHA] Sharpe, A.N.; Walker, S., Molecular interaction: polarisation studies of mixtures of pyridine with polyhalogenated hydrocarbons, *J. Chem. Soc.*, 1962, 157
- [1962STA] Stadnicki, J.S., Tonometric Investigations of Binary and Ternary Azeotropes. IV. Binary System Aniline-n-Undecane, *Bull. Acad. Pol. Sci., Ser. Sci. Chim.*, 1962, 10, 299
- [1962STE] Stephenson, R.W.; Van Winkle, M., Modification of Relative Volatilities by Addition of Solvent, *J. Chem. Eng. Data*, 1962, 7, 510.
- [1962TJE] Tjebbes, J., *Acta Chem. Scand.*, 1962, 16, 953
- [1962VAL] Valentine, R.H.; Brodale, G.E.; Giaque, W.F., Trifluoromethane: entropy, low temperature heat capacity, heats of fusion and vaporization, and vapor pressure, *J. Phys. Chem.*, 1962, 66, 392
- [1962WAD] Waddington, Guy; Smith, J.C.; Williamson, K.D.; Scott, D.W., Carbon disulfide as a reference substance for vapor-flow calorimetry; The chemical thermodynamic properties, *J. Phys. Chem.*, 1962, 66, 1074
- [1963BAR] C. R. Barber and A. Horsford, ".The determination of the boiling and triple points of equilibrium hydrogen and its vapour pressure-temperature relation," *Br. J. Appl. Phys.* 14, 920 (1963).
- [1963BEY] Beynon, Eugene T.; McKetta, John J., The thermodynamic properties of 2-methyl-2-propanol, *J. Phys. Chem.*, 1963, 67, 2761
- [1963BID] Biddiscombe, D.P.; Collerson, R.R.; Handley, R.; Herington, E.F.G.; Martin, J.F.; Sprake, C.H.S., Thermodynamic Properties of Organic Oxygen Compounds. Part 8. Purification and Vapor Pressures of the Propyl and Butyl Alcohols, *J. Chem. Soc.*, 1963, 1954
- [1963BUL] Bull, S.Sh.; Seregrenaja, I.I.; Tsherbakora, P.R., Isothermic Equilibrium of Liquid-Steam in System Water-Acetoaldehyde, *Khim. Prom. (Moscow)*, 1963, 7, 507

- [1963CAP] Capková, A.; Fried, V., Gleichgewicht flüssigkeit-dampf XXX. System vinylacetat-essigsäure, Collect. Czech. Chem. Commun., 1963, 28, 2235
- [1963EDG] Edgell, W.F.; Riethof, T.R.; Ward, C., J.; The infrared and Raman spectra of CF₃CH₂I, CF₃CH₂Br, and CF₃CH₂F. Mol. Spectrosc., 1963, 11, 92
- [1963LEV] Le Fevre, R.J.W.; Sundaram, A.; Pierens, R.K., Molecular Polarisability: the Anisotropy of the Carbon-Oxygen Link, J. Chem. Soc., 1963, 479
- [1963LOV] Lovejoy, D. R.; Some Boiling and Triple Points Below 0° C, Nature, 1963, 197, 353
- [1963MEA] Mears, T.W.; Stanley, C.L.; Compere, E.L.; Howard, F.L., Synthesis, purification, and physical properties of seven twelve-carbon hydrocarbons, J. Res. Natl. Bur. Stand., Sect. A, 1963, 67, 475.
- [1963SCO] Scott, D.W.; Hubbard, W.N.; Messerly, J.F.; Todd, S.S.; Hossenlopp, I.A.; Good, W.D.; Douslin, D.R.; McCullough, J.P., Chemical Thermodynamic Properties and Internal Rotation of Methylpyridines I. 2-Methylpyridine, J. Phys. Chem., 1963, 67, 680
- [1963THO] Thomas, L. H.; Meatyard, R.; Viscosity and Molecular Association. IV. Association of Monohydric Alcohols and Some Hindered Phenols. J. Chem. Soc., 1963, 1986
- [1964ADR] Adriaanse, N.; Dekker, H.; Coops, J.; Some Physical Constants of Normal, Saturated Fatty Acids and Their Methyl Esters. Recl. Trav. Chim. Pays-Bas, 1964, 83, 557
- [1964CRO] Croll, I. M.; Scott, R. L.; Fluorocarbon Solutions at Low Temperatures IV. The Liquid Mixtures CH₄ + CClF₃, CH₂F₂ + CClF₃, CHF₃ + CClF₃, CF₄ + CClF₃, C₂H₆ + CClF₃, C₂H₆ + CF₄, and CHF₃ + CF₄ J. Phys. Chem., 1964, 68, 3853
- [1964FIS] Fischer, G.; Buchanan, AS, Vapour pressures of formyl fluoride and carbonyl chlorofluoride, Aust. J. Chem., 1964, 17, 481
- [1964GUB] Gubkov, A.N.; Fermor, N.A.; Smirnov, N.I., Vapor Pressure of Mono-Poly Systems, Zh. Prikl. Khim. (Leningrad), 1964, 37, 2204
- [1964KLE] Kletskii, A.V., Dependence of Vapor Pressure Freon - 22, Inzh. Fiz. Zh., 1964, 7, 40
- [1964MOE] Moelwyn-Hughes, E.A.; Thorpe, P.L., The physical and thermodynamic properties of some associated solutions, Proc. R. Soc. London, A, 1964, 277, 423
- [1964SEV] Sevrugova, N.N.; Sokorskii, V.A.; Zhavoronkov, N.M., Phase of Equilibrium for Mixtures Acrylonitrile-Acetonitrile, Zh. Prikl. Khim. (Leningrad), 1964, 37, 1989
- [1964ZIE] Ziegler, W.T.; Kirk, B.S.; Mullins, J.C.; Berquist, A.R., Calculation of the vap. press. & heats of vaporization & sublimation of liqs. & solids below 1 atms. prss. VII ethane, Ga. Inst. Technol., Eng. Exp. Stn., Proj. A-764, Tech. Rep. No. 1, 1964

- [1965BOY] Boyd, R.H.; Christensen, R.L.; Pua, R., The Heats of Combustion of Acenaphthene, Acenaphthylene, and Fluoranthene. Strain and Delocalization in Bridged Naphthalenes, *J. Am. Chem. Soc.*, 1965, 87, 3554
- [1965COL] Collerson, R.R.; Counsell, J.F.; Handley, R.; Martin, J.F.; Sprake, C.H.S., Thermodynamic Properties of Organic Oxygen Compounds. Part XV. Purification and Vapour Pressures of Some Ketones and Ethers, *J. Chem. Soc.*, 1965, 3697
- [1965DES] Deshpande, D.D.; Pandya, M.V., Thermodynamics of Binary Solutions Part 1.--Heats of Mixing of Aniline in Benzene, Carbontetrachloride and Chlorobenzene, *Trans. Faraday Soc.*, 1965, 61, 1858
- [1965GRO] Grove, E.L.; Walden, G.E., Variation of Dielectric Constant with Temperature for SOme Five- and Six-Carbon Ketones, *J. Chem. Eng. Data*, 1965, 10, 98
- [1965HES] Hessel, D.; Geiseler, G., Uber die Druckabhangigkeit des heteroazeotropen Systems n-Butanol/Wasser, *Z. Phys. Chem. (Leipzig)*, 1965, 229, 199
- [1965HUL] Hull, H. S.; Reid, A. F.; Turnbull, A. G.; Vapour Pressures of Cyclopentadiene and Bis(cyclopentadienyl)magnesium. *Aust. J. Chem.*, 1965, 18, 249
- [1965KAA] Kaal. T.A.; Ordyan, M.B.; Eidus, Y.T., Synthesis of Carboxylic Acid Derivatives Uner Conditions of Acid Catalysis From Carbon Monoxide, Olefines, and Acylating Agents, *Zh. Org. Khim.*, 1965, 1, 1187
- [1965MAL] Mallikarjun, S.; Hill, N.E., Temperature dependence of viscosity and dielectric relaxation time in simple polar liquids, *Trans. Faraday Soc.*, 1965, 61, 1389
- [1965MAR] Marinichev, A.N.; Susarev, M.P., Study of a Liquid-Vapor Equilibrium in the Systems of Acetone/Methanol and Acetone/Cyclohexane at Temperatures 35, 45, 55 C and Pressure 760 Torr, *Zh. Prikl. Khim. (Moscow)*, 1965, 38, 378
- [1965MUE] Mueller, E.; Padeken, H. G.; Salamon, M.; Fiedler, G.; Direct Photooxidation of Cycloalkanes. *Chem. Ber.*, 1965, 98, 1893
- [1965POL] Polák, J.; Mertl, I., Saturated vapour pressure of methyl acetate, ethyl acetate, n-propyl acetate, methyl propionate, and ethyl propionate, *Collect. Czech. Chem. Commun.*, 1965, 30, 3526
- [1965RED] Reddy, M.S.; Rao, C.V., Isothermal and Isobaric Vapor-Liquid Equilibrium Data System: Toluene- Isoamyl Alcohol, *J. Chem. Eng. Data*, 1965, 10, 309
- [1965RON] Rondeau, R.E.; Harrah, L.A., Melting Point and Vapor Pressures of 3-Hexyne., *J. Chem. Eng. Data*, 1965, 10, 84
- [1965SAM] Samartsev, M. A.; Ogloblin, K.A.; Interaction of nitrosyl Chloride with unsaturate compounds. XVII. cis-trans - isomerism of aliphatic chloronitroso compounds. *J. Org. Chem. USSR (Engl. Transl.)*, 1965, 1, 29
- [1965SWA] Swamy, P.A.; Van Winkle, Matthew, Vapor-Liquid Equilibria at 760 mm Mercury for the System Vinyl Acetate-2,4-Dimethyl Pentane., *J. Chem. Eng. Data*, 1965, 10, 214

- [1966BAR] Barber, C. R.; The sublimation temperature of carbon dioxide, *Brit. J. Appl. Phys*, 1966, 17, 391
- [1966BOT] Bott, T. R.; Sadler, H. N.; Vapor pressure of propylene oxide.. *J. Chem. Eng. Data*, 1966, 11, 25
- [1966GAL] Gallant, R. W.; Physical properties of hydrocarbons: V chlorinated methanes.. *Hydrocarbon Process.*, 1966, 45, 161
- [1966KAT] Katti, P.K.; Prakash, O., Viscosities of Binary Mixtures of Carbon Tetrachloride with Methanol and Isopropyl Alcohol, *J. Chem. Eng. Data*, 1966, 11, 46
- [1966MEY] Meyer, E. F.; Wagner, R. E.; Cohesive Energy in Polar Organic Liquids. *J. Phys. Chem.*, 1966, 70, 3162
- [1966MOU] Moussa, M. R.; Muijlwijk, R.; van Dijk, H.; "The vapor pressure of liquid nitrogen," *Physica*, 1966, 32, 900-12
- [1966OSB] Osborn, A. G.; Douslin, D. R.; Vapor Pressure Relations of 36 Sulfur Compounds Present in Petroleum. *J. Chem. Eng. Data*, 1966, 11, 502
- [1966SER] Serebrennaya, I. I.; Byk, S. Sh.; Liquid-vapor equilibrium in the system trioxane + benzene and trioxane + methylene chloride at atmospheric pressure. *Zh. Prikl. Khim.*, 1966, 39, 1441
- [1966SUR] Suryanarayana, Y.S.; Van Winkle, M., Solvent effect on relative volatility: n-hexane + 1-hexene system., *J. Chem. Eng. Data*, 1966, 11, 7
- [1967AMB] Ambrose, D.; Broderick, B.E.; Townsend, R., The Vapour Pressures Above the Normal Boiling Point and the Critical Pressures of Some Aromatic Hydrocarbons, *J. Chem. Soc. A*., 1967, 633
- [1967COO] Cooper, A. R.; Crowne, C. W. P.; Farrell, P. G.; Gas-Liquid Chromatographic Studies of Electron-Donor-Acceptor Systems. *Trans. Faraday Soc.*, 1967, 63, 447
- [1967FRI] Fried, V.; Gallant, P.; Schneier, G.B., Vapor-liquid equilibrium in the system pyridine-tetrachloroethylene, *J. Chem. Eng. Data*, 1967, 12, 504
- [1967GIL] Gilmour, J.B.; Zwicker, J.O.; Katz, J.; Scott, R.L., Fluorocarbon Solutions at Low Temperatures V. The Liquid Mixtures C₂H₆ + C₂F₆, C₃H₈ + C₂F₆, CH₄ + C₃F₈, C₂H₆ + C₃F₈, C₃H₈ + C₃F₈, n-C₄H₁₀ + C₃F₈, i-C₄H₁₀ + C₃F₈, C₃H₈ + n-C₄F₁₀, n-C₆H₁₄ + n-C₄F₁₀, *J. Phys. Chem.*, 1967, 71, 3259
- [1967HAN] Hanson, D.O.; Van Winkle, M., Alteration of the Relative Volatility of n-Hexane-1-Hexene by Oxygenated and Chlorinated Solvents, *J. Chem. Eng. Data*, 1967, 12, 319.
- [1967HUM] Humphrey, J.L.; Van Winkle, M., Vapor-Liquid Equilibria at 60 deg C for n-Hexane-Alkyl Amines and 1-Hexene-Alkyl Amines, *J. Chem. Eng. Data*, 1967, 12, 526
- [1967LES] Lesteva, T.M.; Ogoradnikov, S.K.; Morozova, A., Kinetics of Dimerisation of Cyclopentadiene and Equilibrium of Liquid-Steam in System of Isoprene-Cyclopentadiene, *Zh. Prikl. Khim. (Leningrad)*, 1967, 40, 891
- [1967MAR] Marks, G.W.; Acoustic velocity with relation to chemical constitution in alcohols. *J. Acoust. Soc. Am.*, 1967, 41, 103

- [1967MCC] McConnell, C.G.; Van Winkle, M., Vapor-Liquid Equilibria in the Acetone-2,3-Dimethylbutane and Chloroform-2,3-Dimethylbutane Systems, *J. Chem. Eng. Data*, 1967, 12, 430
- [1967NAK] Nakanishi, K.; Shirai, H.; Minamiyama, T., Vapor-Liquid Equilibria of Binary Systems Containing Alcohols (Methanol with Aliphatic Amines), *J. Chem. Eng. Data*, 1967, 12, 591
- [1967PIH] Pihlaja, K.; Heikkila, J.; Heats of Combustion of 1,3-Dioxane and Its Methyl Derivatives. *Acta Chem. Scand.*, 1967, 21, 2390
- [1967SIM] Simon, M.; Knobler, C. M.; Duncan, A. G.; *Cryogenics*, 1967, 7, 138
- [1967SMI] Smirnov, K. M.; A. P. Tomilov, and A. I. Shchekotikhin, *Russ. Chem. Rev., Engl. Trans.* 36, 326, 1967
- [1967VAN] van Hook, W.A., Vapor Pressures of the Methylacetylenes, H₃CCCH, H₃CCCD, D₃CCCH, and D₃CCCD, *J. Chem. Phys.*, 1967, 46, 1907
- [1967WAK] Wakayama, N.; Inokuchi, H.; Heats of Sublimation of Polycyclic Aromatic Hydrocarbons and Their Molecular Packings. *Bull. Chem. Soc. Jpn.*, 1967, 40, 2267
- [1968BAR] Baradarajan, A.; Satyanarayana, M., Vapor - Liquid Equilibria of the Benzene - Cyclohexane - Acetic Acid System at Atmospheric Pressure, *J. Chem. Eng. Data*, 1968, 13, 148
- [1968BLA] Blagoi, Yu. P.; Butko, A. E.; Mikhailenko, S. A.; Yakuba, V. V.; Velocity of sound in liquid propene, carbon tetrafluoride, and ammonia. *Zh. Fiz. Khim.*, 1968, 42, 1075
- [1968FIL] Filatov, A. S.; Makarov, S. P.; Yakubovich, A. Ya.; Halogenation of 1,1,1-Trifluoroazomethane *Zh. Obshch. Khim.*, 1968, 38, 33
- [1968FIL2] Filatov, A.S.; Makarov, S.P.; Yakubovich, A.Ya., Some Chemical Properties of Pentafluoroazomethane, *Zh. Obshch. Khim.*, 1968, 38, 35
- [1968FOW] Fowler, Lewis.; Trump, Walter N.; Vogler, Carl E., Vapor pressure of naphthalene. Measurements between 40.deg. and 180.deg., *J. Chem. Eng. Data*, 1968, 13, 209
- [1968KES] Kesselman, W.D.; Hollenbach, G.E.; Myers, A.L.; Humphreys, A.E., Vapor-Liquid Equilibrium Data for Benzene-Alkylbenzene Systems, *J. Chem. Eng. Data*, 1968, 13, 34
- [1968MAL] Malbrunot, P. F.; Meunier, P. A.; Scatena, G. M.; Mears, W. H.; Murphy, K. P.; Sinka, J. V.; Pressure-Volume-Temperature Behavior of Difluoromethane *J. Chem. Eng. Data*, 1968, 13, 16
- [1968MAR] Marsh, K.N., Thermodynamics of Octamethylcyclotetrasiloxane mixtures, *Trans. Faraday Soc.*, 1968, 64, 883
- [1968OSB] Osborn, A. G.; Douslin, D. R.; Vapor Pressure Relations of 13 Nitrogen Compounds Related to Petroleum, *J. Chem. Eng. Data*, 1968, 13, 534
- [1968PAC] Pace, E.L.; Reno, M.A., Thermodynamic Properties of Carbonyl Fluoride from 12 K to its Boiling Point. The Entropy from Molecular and Spectroscopic Data, *J. Chem. Phys.*, 1968, 48, 1231

- [1968TAY] Taylor, Z.L.; Wingard, R.E., Vapor-Liquid Equilibrium in the n-Octane-p-Cresol System, J. Chem. Eng. Data, 1968, 13, 301
- [1969BRO] Brown, I.; Fock, W.; Smith, F., The thermodynamic properties of solutions of normal and branched alcohols in benzene and n-hexane, J. Chem. Thermodyn., 1969, 1, 273
- [1969CID] Cidlinský, J.; Polák, J., Saturated vapour pressures of some ethers, Collect. Czech. Chem. Commun., 1969, 34, 1317
- [1969HUD] Hudson, J. W.; Van Winkle, M.; Multicomponent vapor-liquid equilibria in systems of mixed positive and negative deviations. J. Chem. Eng. Data, 1969, 14, 310
- [1969KEE] Keenan, A. G.; Siegmund, R. F.; Thermal decomposition of ammonium perchlorate. Q. Rev. Chem. Soc., 1969, 23, 430
- [1969KEM] Kemme, Herbert R.; Kreps, Saul I., Vapor pressure of primary n-alkyl chlorides and alcohols, J. Chem. Eng. Data, 1969, 14, 98
- [1969LAN] Lanum, W.J.; Morris, J.C., Physical Properties of Some Sulfur and Nitrogen Compounds, J. Chem. Eng. Data, 1969, 14, 93
- [1969MYE] Myers, R.S.; Clever, H.L., Surface Tension of Octamethylcyclotetrasiloxane and Hexamethyldisilazane and Their Solutions with Carbon Tetrachloride and n-Hexadecane, J. Chem. Eng. Data, 1969, 14, 161
- [1969OSB] Osborn, Ann G.; Douslin, Donald R., Vapor pressure relations for the seven pentadienes, J. Chem. Eng. Data, 1969, 14, 208
- [1969RAJ] Raju, B.N.; Rao, D.P., Isobaric vapor-liquid equilibrium data for the systems 1-propanol- methylcyclohexane and methylcyclohexane-1- butanol, J. Chem. Eng. Data, 1969, 14, 283
- [1969SMI] Smith, J. H.; Pace, E. L.; The thermodynamic properties of carbon tetrafluoride from 12 K to its boiling point. The significance of the parameter .nu. J. Phys. Chem., 1969, 73, 4232
- [1970AMB] Ambrose, D.; Sprake, C.H.S., Thermodynamic properties of organic oxygen compounds XXV. Vapour pressures and normal boiling temperatures of aliphatic alcohols, J. Chem. Thermodyn., 1970, 2, 631
- [1970KAT] Kato, M.; Konishi, H.; Hirata, M., New apparatus for isobaric dew and bubble point method methanol + water, ethyl acetate + ethanol, water + 1-butanol, and ethyl acetate + water systems, J. Chem. Eng. Data, 1970, 15, 435
- [1970KIR] Kirby, C.E.; Van Winkle, M., Vapor-liquid equilibria 2,3-dimethylbutane + methanol and 2,3- dimethylbutane + methanol + chloroform systems, J. Chem. Eng. Data, 1970, 15, 177
- [1970KOB] Kobe, K. A.; Mathews, J. F.; Critical Properties and Vapor Pressures of Some Organic Nitrogen and Oxygen Compounds. J. Chem. Eng. Data, 1970, 15, 182

- [1970POL] Polak, Jiri; Murakami, Sachio; Lam, V.T.; Benson, George C., Excess enthalpy, volume, and Gibbs free energy of cyclopentane-tetrachloroethylene mixtures at 25.deg., J. Chem. Eng. Data, 1970, 15, 323
- [1970QUI] Quitzsch, K.; Hofmann, H.P.; Hering, D.; Salzer, R.; Geiseler, G., Z. Phys. Chem. (Leipzig), 1970, 243, 321
- [1970SIL] Silver, L.; Rudman, R., Polymorphism of the Crystalline Methylchloromethane Compounds. A Differential Scanning Calorimetric Study, J. Phys. Chem., 1970, 74, 3134
- [1970STO] Stokes, R.H.; Levien, B.J.; Marsh, K.N., A continuous dilution dilatometer. The excess volume for the system cyclohexane + benzene, J. Chem. Thermodyn., 1970, 43
- [1970WEA] Weatherford, R.M.; Van Winkle, M., Vapor-Liquid Equilibria of the Quinary System Hexane, Methylcyclopentane, Cyclohexane, Benzene, and Toluene, J. Chem. Eng. Data, 1970, 15, 386
- [1970WEB] Weber, L. A.; "Some vapor pressure and P,V,T data on nitrogen in the range 65 to 140 K," J. Chem. Thermodyn., 1970, 2, 839-846
- [1970WIL] Willock, J.M.; Van Winkle, M., Binary and Ternary Vapor-Liquid Equilibria of Methano-Acetone-2,3-Dimethylbutane, J. Chem. Eng. Data, 1970, 15, 281
- [1971EON] Eon, C.; Pommier, C.; Guiochon, G., Vapor pressures and second virial coefficients of some five-membered heterocyclic derivatives, J. Chem. Eng. Data, 1971, 16, 408
- [1971LET] Letcher, T.M.; Bayles, J.W., Thermodynamics of some Binary Liquid Mixtures Containing Aliphatic Amines, J. Chem. Eng. Data, 1971, 16, 266
- [1971NAG] Nagata, I.; Ohta, T., Isobaric vapor-liquid equilibria for three binary and two ternary systems, J. Chem. Eng. Data, 1971, 16, 164
- [1971SOK] Sokolov, N. M.; Tsygankova, L. N.; Zhavoronkov, N. M.; The liquid-vapour equilibrium in the water-ethylene glycol and water - 1, 2 - propylene glycol systems at various pressures, Teor. Osn. Khim. Tekhnol., 1971, 5, 900
- [1971STR] Streng, A.G., Miscibility and Compatibility of Some Liquid and Solidified Gases at Low Temperature, J. Chem. Eng. Data, 1971, 16, 357
- [1972AMB] Ambrose, D.; Sprake, C.H.S.; Townsend, R., Thermodynamic properties of organic oxygen compounds XXIX. The vapour pressure of diethyl ether, J. Chem. Thermodyn., 1972, 4, 247
- [1972BOU] Boublik, T.; Aim, K., Heats of Vaporization of Simple Non-spherical Molecule Compounds, Collect. Czech. Chem. Commun., 1972, 37, 3513
- [1972COU] Counsell, J. F.; Lee, D. A.; Thermodynamic Properties of Organic Oxygen Compounds 30. Vapor Heat Capacity and Enthalpy of Vaporization of Propanal. J. Chem. Thermodyn., 1972, 4, 915
- [1972JAK] Jakli, G.; van Hook, W.A., The Vapor Pressures of Dimethyl Sulfoxide and Hexadeuterodimethyl Sulfoxide from about 313 to 453 K, J. Chem. Thermodyn., 1972, 4, 857

- [1972MAR] Maripuri, V.O.; Ratcliff, G.A., Isothermal Vapor-Liquid Equilibria in Binary Mixtures Containing Alkanes and Ethers, *J. Chem. Eng. Data*, 1972, 17, 454
- [1972MIL] Miller, K.J.; Huang, H.-S., Vapor-liquid equilibrium for binary systems 2-butanone with 2-butanol, 1-pentanol, and isoamyl alcohol, *J. Chem. Eng. Data*, 1972, 17, 77
- [1972PRY] Prydz, R.; Goodwin, R.D., Experimental Melting and Vapor Pressures of Methane, *J. Chem. Thermodyn.*, 1972, 4, 127
- [1972RAK] Rakhmankulov, D. L.; Isagulyants, V. I.; Zlotskii, S. S.; *J. Appl. Chem. USSR (Engl. Transl.)*, 1972, 45, 2907
- [1972REG] Regnier, J., Tension de Vapeur de L'Ethane Entre 80 et 135 K, *J. Chim. Phys.*, 1972, 69, 942
- [1972WOL] Wolf, G., Thermochemical studies of cyclic ketones, *Helv. Chim. Acta*, 1972, 55, 1446
- [1972ZOR] Zordan, T.A.; Hurkot, D.G.; Peterson, M.; Hepler, L.G., Enthalpies and entropies of melting from differential scanning calorimetry and freezing point depressions: urea, methylurea, 1, 1-dimethylurea, 1, 3-dimethylurea, tetramethylurea, and thiourea, *Thermochim. Acta*, 1972, 5, 21.
- [1973CAS] Casellato, F.; Vecchi, C.; Girell, A., Differential calorimetric study of polycyclic aromatic hydrocarbons, *Thermochim. Acta*, 1973, 6, 361
- [1973CHI] Chinikamala, A.; Houth, G.N.; Taylor, Z.L., Vapor-Liquid Equilibria of Binary Systems Containing Selected Hydrocarbons with Perfluorobenzene, *J. Chem. Eng. Data*, 1973, 18, 322
- [1973DAS] Das, T.R.; Reed, C.O., Jr.; Eubank, P.T., PVT Surface and Thermodynamic Properties of Isobutane, *J. Chem. Eng. Data*, 1973, 18, 253
- [1973DAS2] Das, T.R.; Reed, C.O., Jr.; Eubank, P.T., PVT Surface and Thermodynamic Properties of n-Butane, *J. Chem. Eng. Data*, 1973, 18, 244
- [1973FIL] Filippov, N. A.; Reikhsfel'd, V. O.; Liquid-vapor equilibrium in the systems ethyl acrylate + n-butanol, n-butanol + n-butyl acrylate, and ethyl acrylate + n-butanol + n-butyl acrylate. *Zh. Prikl. Khim.*, 1973, 46, 2041
- [1973LAK] Lakhanpal, M. L.; Kapoor, K. K.; Lal, G.; Mandal, H. G.; Thermochemical studies of some partially miscible liquid pairs of formic acid with benzene, toluene, and p-xylene, and of acetic anhydride and furfural with cyclohexane. *Indian J. Chem.*, 1973, 11, 471
- [1973LOS] Los, L. V.; Saturation pressure of Freon 22 vapor *Kholod. Tekh. Tekhnol.*, 1973, 80
- [1973MAT] Matsunaga, I.; Katayama, T.; Vapor-liquid equilibria for three binary mixtures of propionaldehyde. *J. Chem. Eng. Jpn.*, 1973, 6, 397
- [1973MEY] Meyer, Edwin F.; Hotz, Roger D., High-precision vapor-pressure data for eight organic compounds, *J. Chem. Eng. Data*, 1973, 18, 359

- [1973NAG] Nagata, I.; Ohta, T.; Uchiyama, Y., Excess gibbs free energies for binary systems isopropanol with benzene, cyclohexane, methycyclohexane, J. Chem. Eng. Data, 1973, 18, 54
- [1973SCH] Scheller, W.A.; Narasimha Rao, S.V., Isothermal Vapor-Liquid Equilibrium Data for System Heptane-2-Pentanone at 90 C, J. Chem. Eng. Data, 1973, 18, 223
- [1973WAD] Wade, J.C.; Taylor, Z.L., Vapor-liquid equilibrium in perfluorobenzene-benzene-methylcyclohexane system., J. Chem. Eng. Data, 1973, 18, 424
- [1973WAG] Wagner, W.; "New vapour pressure measurements for argon and nitrogen and a new method for establishing rational vapour pressure equations," Cryogenics, 1973, 13, 470-82
- [1974AMB] Ambrose, D.; Sprake, C.H.S.; Townsend, R., Thermodynamic properties of organic oxygen compounds XXXIII. The vapour pressure of acetone, J. Chem. Thermodyn., 1974, 6, 693
- [1974ATA] Atake, T.; Chihara, H., "A New Condensed Gas Calorimeter. Thermodynamic Properties of Solid and Liquid Dinitrogen Oxide," Bull. Chem. Soc. Jpn., 1974, 47, 2126-2136
- [1974DOJ] Dojcansky, J.; Heinrich, J., Saturated Vapour Pressure of Acetonitrile, Chem. Zvesti, 1974, 28, 157
- [1974KER] Kerns, W.J.; Anthony, R.G.; Eubank, P.T., Volumetric Properties of Cyclohexane Vapor, AIChE Symp. Ser., 1974, 70, 140, 14
- [1974OSB] Osborn, Ann G.; Douslin, Donald R., Vapor-pressure relations for 15 hydrocarbons, J. Chem. Eng. Data, 1974, 19, 114
- [1974RAO] Rao, M.V.P.; Naidu, P.R., Excess volumes of binary mixtures of alcohols in methylcyclohexane, Can. J. Chem., 1974, 52, 788
- [1974YOK] Yokozeki, A.; Wilcox, C.F.; Bauer, S.H., Biphenylene. Internuclear Distances and Their Root Mean Square Amplitudes of Vibration, J. Am. Chem. Soc., 1974, 96, 1026
- [1975AMB] Ambrose, D.; Ellender, J. H.; Lees, E. B.; Sprake, C. H. S.; Townsend, R.; Thermodynamic Properties of Organic Oxygen Compounds XXXVII. Vapor Pressure of Some Aliphatic Ketones. J. Chem. Thermodyn., 1975, 7, 453
- [1975FRE] Frensdorff, H.K.; Adams, R.K., Vapor pressure of 2,4-tolylene diisocyanate, J. Chem. Eng. Data, 1975, 20, 13
- [1975HOP] Hopfner, A.; Parekh, N.; Horner, Ch.; Abdel-Hamid, A., Der Dampfdruck-Isotopie-Effekt von deuterierten Neopentanen, Ber. Bunsen-Ges. Phys. Chem., 1975, 79, 216
- [1975OSB] Osborn, Ann G.; Douslin, Donald R., Vapor pressures and derived enthalpies of vaporization for some condensed-ring hydrocarbons, J. Chem. Eng. Data, 1975, 20, 229
- [1975SAD] Sada, E.; Morisue, T.; Miyahara, K., Salt Effects on Vapor-Liquid Equilibrium of Tetrahydrofuran-Water System, J. Chem. Eng. Data, 1975, 20, 283
- [1975VID] Vidaurri, F.C., Ethyl Fluoride Vapor Pressure, J. Chem. Eng. Data, 1975, 20, 349

- [1975WIS] Wisniak, J.; Tamir, A., Vapor-liquid equilibria of sec-butanol + isobutanol, sec-butanol + tert- butanol, and isobutanol + tert-butanol systems, J. Chem. Eng. Data, 1975, 20, 388
- [1976AMB] Ambrose, D.; Ellender, J. H.; Sprake, C. H. S.; Townsend, R.; Thermodynamic properties of organic oxygen compounds. XLIII. Vapour pressures of some ethers. J. Chem. Thermodyn., 1976, 8, 165
- [1976MEY] Meyer, E. F.; Hotz, C. A.; Cohesive Energies in Polar Organic Liquids. 3. Cyclic Ketones. J. Chem. Eng. Data, 1976, 21, 274
- [1976PAV] Pavese, F.; Cagna, G.; Ferri, D., The Triple Point of Pure Methane in Proc. Int. Cryog. Eng. Conf. 6, 1976
- [1976RAD] Radwan, M. H.; Hanna, A. A.; Binary Azeotropes Containing Butyric Acid. J. Chem. Eng. Data, 1976, 21, 285
- [1976STE] Steele, K. Polling, B. E.; Manley, D. B.; Vapor pressures for the system 1-butene, isobutane, and 1,3-butadiene. J. Chem. Eng. Data, 1976, 21, 399
- [1976TAM] Tamir, A.; Wisniak, J., Binary Vapor-Liquid Equilibria of Some Amyl Alcohols, J. Chem. Eng. Data, 1976, 21, 182
- [1976TRI] Tripathi, R.P.; Krishna, S.; Gulati, I.B., Isobaric Binary Vapor-Liquid Equilibria in Cyclohexane-tert-Butyl Alcohol and 2,4-Dimethylpentane-tert-Butyl Alcohol Systems, J. Chem. Eng. Data, 1976, 21, 44
- [1977ANC] Ancsin, J.; "Thermometric Fixed Points of Hydrogen," Metrologia 13, 79 (1977)
- [1977BOU] Bougard, J.; Jadot, R., Isothermal Vapor-Liquid Equilibria for the System 1,2-Dichloropropane- Propanal, J. Chem. Eng. Data, 1977, 22, 88
- [1977GOO] Goodwin, R.D., , Natl. Bur. Stand., Tech. Rep.] NBSIR 77-860, 1977
- [1977GOV] Govindaswamy, S.; Andiappan, A.N.; Lakshmanan, S.M., Vapor-liquid equilibria of the binary and ternary systems containing n-hexane(1) + benzene(2) + tert-butyl alcohol(3) at 760 mm Hg pressure, J. Chem. Eng. Data, 1977, 22, 264
- [1977KUC] Kuchhal, R. K.; Mallik, K. L.; Gupta, P. L.; Studies on thermodynamics of solution by gas chromatography: solubility measurements of hydrocarbons in cycloalkanols. Can. J. Chem., 1977, 55, 1273
- [1977RAD] Radecki, A.; Kaczmarek, B.; Phase Equilibria in Ternary Systems of Hexamethyldisiloxane-Butylamines- Water. J. Chem. Eng. Data, 1977, 22, 168
- [1977REI] Reid, R. C.; Prausnitz, J. M.; Sherwood, T. K. The Properties of Gases and Liquids, 3rd ed.; McGraw-Hill Book Company: New York, 1977
- [1977SVO] Svoboda, V.; Vesely, F.; Holub, R.; Pick, J., Heats of vaporization of alkyl acetates and propionates, Collect. Czech. Chem. Commun., 1977, 42, 943
- [1977WIS] Wisniak, J.; Tamir, A., Thermodynamic Properties of the Systems Methylcyclohexane-Toluene, n-Heptane-Toluene, and Methylcyclohexane-Toluene-n-Heptane, J. Chem. Eng. Data, 1977, 22, 391

- [1978CAL] Calado, J. C. G.; Staveley, L. A. K.; The triple-point temperature and pressure of hydrogen bromide, and the vapour pressure of the liquid to about 205 K, *J. Chem. Thermodyn.*, 1978, 10, 897
- [1978OCH] Ochi, K.; Lu, B.C.Y., Determination and Correlation of Binary Vapor-Liquid Equilibrium Data, *Fluid Phase Equilib.*, 1978, 1, 185
- [1978OSB] Osborn, A. G.; Scott, D. W.; Vapor-pressure and enthalpy of vaporization of indan and five methyl- substituted indans. *J. Chem. Thermodyn.*, 1978, 10, 619
- [1978SEV] Sevcik, M.; Boublik, T.; Biros, J., Excess thermodynamic functions in the cyclopentane(1) + chloroform(2) system at 298.15 K, *Collect. Czech. Chem. Commun.*, 1978, 43, 2242
- [1978TAM] Tamir, A.; Wisniak, J., Vapor-liquid equilibria in the ternary systems tert-butyl alcohol + diethyl ketone + methyl isobutyl ketone and diethyl ketone + methyl isobutyl ketone + n-butyl alcohol and their binaries, *J. Chem. Eng. Data*, 1978, 23, 293
- [1979BRO] Brown, G. N.; W.T. Ziegler, *Adv. Cryo. Eng.*, 1979, 25, 662
- [1979CHA] Chandak, B.S.; Nageshwar, G.D.; Mene, P.S., Excess gibbs free energy, enthalpy, and volume and viscosity of acetone-methyl cellosolve mixtures, *J. Chem. Eng. Data*, 1979, 24, 267.
- [1979KUD] Kudchadker, A.P.; Kudchadker, S.A.; Shukla, R.P.; Patnaik, P.R., Vapor pressures and boiling points of selected halomethanes, *J. Phys. Chem. Ref. Data*, 1979, 8, 499
- [1979KUR] Kurtyka, Z.M.; Kurtyka, E.A., Vapor-Liquid Equilibria in the Systems Aniline-Pseudocumene, o-Toluidine-n-Decane, and m-Xylidine-n-Dodecane at Atmos. Press., *J. Chem. Eng. Data*, 1979, 24, 15
- [1979LOB] Lobo, L. Q.; Staveley, L. A. K.; The vapour pressure of tetrafluoromethane. *Cryogenics*, 1979, 19, 335
- [1979MAH] Maheshwari, R.C.; Suri, S.K.; Tewari, U.S., Excess Volumes of Mixing of Binary Mixtures of Arsenic Tribromide with Benzene, Cyclohexane, and Carbon Tetrachloride at 303.15, 308.15 and 313.15 K, *J. Chem. Eng. Data*, 1979, 24, 237
- [1979MAJ] Majer, V.; Svoboda, V.; Koubek, J.; Pick, J.; Enthalpy data of liquids. XVI. Temperature dependence of heats of vaporization, saturated vapor pressures, and cohesive energies for a group of amines. *Collect. Czech. Chem. Commun.*, 1979, 44, 3521
- [1979MAR] Markovnik, V.S.; Sachek, A.I.; Peshchenko, A.D.; Shvaro, O.V.; Andreevskii, D.N.; Olizarevich, N.M., *Termodin. Org. Soedin.*, 1979, 107
- [1979MJO] Mjojo, C.C., Order-disorder phenomena: part 2. order-disorder phase equilibria in d-and l-systems of camphor and related compounds., *J. Chem. Soc., Faraday Trans. 2*, 1979, 75, 692
- [1979THO] Thorat, R.T.; Nageshwar, G.D.; Mene, P.S., Excess thermodynamic properties of the ternary system ethyl acetate + ethanol + 2-ethoxyethanol, *J. Chem. Eng. Data*, 1979, 24, 270

- [1980KRA] Krajewska, A.; Pigon, K., Phase diagrams in the binary systems of 2,4,7-trinitrofluoren-9-one with aromatic and heteroaromatic compounds., *Thermochim. Acta*, 1980, 41, 187
- [1980OSB] Osborn, A. G.; Scott, D. W.; Vapor Pressure of 17 Miscellaneous Organic Compounds. *J. Chem. Thermodyn.*, 1980, 12, 429
- [1980RAD] Radomska, M.; Radomski, R., Calorimetric Studies of Binary Systems of 1,3,5-Trinitrobenzene with Naphthalene, Anthracene, and Carbazole I. Phase Transitions and Heat Capacities of the Pure Components and Charge Transfer Complex, *Thermochim. Acta*, 1980, 40, 405
- [1980RAM] Rama Varma, K.T.; Kumaran, M.K.; Narayanan, P.; Seetharaman, T.S., Solid-Liquid Equilibria in Mixtures of Biphenyl + p-Nitrophenol, + m-Nitrophenol, + o-Nitrophenol, and Phenanthrene + p-Nitrophenol at Atmospheric Pressure, *J. Chem. Thermodyn.*, 1980, 12, 615
- [1980SMI] Smith, G.W., Phase behavior of some condensed polycyclic aromatics, *Mol. Cryst. Liq. Cryst.*, 1980, 64, 15
- [1980WIE] Wieczorek, S. A.; Kobayashi, R.; Vapor pressure measurements of diphenylmethane, thianaphthene, and bicyclohexyl at elevated temperatures. *J. Chem. Eng. Data*, 1980, 25, 302
- [1980YOS] Yoshikawa, Y.; Takagi, A.; Kato, M., Indirect Determination of Vapor-Liquid Equilibria by a Small Ebulliometer Tetrahydrofuran-Alcohol Binary Systems, *J. Chem. Eng. Data*, 1980, 25, 344
- [1981AMB] Ambrose, D.; Hall, D. J. Thermodynamic Properties of Organic Oxygen Compounds L. The Vapor Pressures of 1,2-Ethanediol(Ethylene Glycol) and Bis(2-hydroxyethyl) Ether (Diethylene Glycol) *J. Chem. Thermodyn.*, 1981, 13, 61
- [1981EFR] Efremov, R. V.; Kolesnikov, V. A.; Danov, S. M.; Kormushechkin, V. D.; Liquid-vapor equilibrium in systems containing vinylidene chloride and other monomers. *Zh. Prikl. Khim.*, 1981, 54, 422
- [1981KOH] Kohler, F.; Atrops, H.; Kalali, H.; Liebermann, E.; Wilhelm, E.; Ratkovics, F.; Salamon, T., Molecular Interactions in Mixtures of Carboxylic Acids with Amines 1. Melting Curves and Viscosities, *J. Phys. Chem.*, 1981, 85, 2520
- [1981MAR] Markovnik, V.S.; Peshchenko, A.D.; Sachek, A.I.; Shvaro, O.V.; Andreyevskii, D.N. Vapor pressure of some glycols *Termodin. Org. Soedin.*, 1981, 54
- [1981NUN] Nunes da Ponte, M.; Staveley, L. A. K.; "The equation of state and thermodynamic properties of liquid hydrogen chloride," *J. Chem. Thermodyn.*, 1981, 13, 179-186
- [1981RAO] Rao, K.V.; Ravi Prasad, A.; Chiranjivi, C., Isobaric Binary Vapor-Liquid Equilibria for the System Allyl Alcohol-Toluene and Toluene-Benzaldehyde, *J. Chem. Eng. Data*, 1981, 26, 413
- [1981RAV] Ravi Prasad, A.; Rao, K.V.; Chiranjivi, C., Binary Vapor-Liquid Equilibria of Iso Alcohol-Tetrachloroethene Systems, *J. Chem. Eng. Data*, 1981, 26, 211

- [1981WIE] Wieczorek, S. A.; Kobayashi, R.; Vapor pressure measurements of 1-methylnaphthalene, 2-methylnaphthalene and 9,10-dihydrophenanthrene at elevated temperatures. *J. Chem. Eng. Data*, 1981, 26, 8
- [1981WIL] Wilson, G. M.; Johnston, R. H.; Hwang, S.-C.; Tsonopoulos, C.; Volatility of Coal Liquids at High Temperatures and Pressures. *Ind. Eng. Chem. Proc. Des. Dev.*, 1981, 20, 94
- [1982BUC] Buckingham, J.; Donaghy, S.M., *Dictionary of Organic Compounds: Fifth Edition*, Chapman and Hall, New York, 1982
- [1982DIZ] Dizechi, M.; Marschell, E., Viscosity of Some Binary and Ternary Liquid Mixtures, *J. Chem. Eng. Data*, 1982, 27, 358
- [1982ERA] Erastov, P. A.; Kolesov, V. P.; Enthalpy of Formation of Trifluorochloroethene. *J. Chem. Thermodyn.*, 1982, 14, 103
- [1982HAY] Haynes, W. M.; Goodwin, R. D.; Thermophysical Properties of Normal Butane from 135 to 700 K at Pressures to 70 MPa. *NBS Monograph 169* (1982)
- [1982KAR] Karunakar, J.; Reddy, K.D.; Rao, M.V.P., Excess volumes of a homologous series of aliphatic alcohols with benzonitrile, *J. Chem. Eng. Data*, 1982, 27, 346
- [1982POP] Popowicz, A.; Oi, T.; Shulman, J.; Ishida, T.; Vapor pressure isotope effects in liquid fluoroform *J. Chem. Phys.*, 1982, 76, 3732
- [1982RED] Reddy, S.S.; Reddy, K.D.; Rao, M.V.P., Excess Volumes of Homologous Series of Aliphatic Hydrocarbons with Chlorobenzene, Nitrobenzene, and Benzonitrile, *J. Chem. Eng. Data*, 1982, 27, 173
- [1982SIV] Sivaraman, A.; Kobayashi, R.; Investigation of vapor pressures and heats of vaporization of condensed aromatic compounds at elevated temperatures. *J. Chem. Eng. Data*, 1982, 27, 264
- [1982VIS] Viswanath, D. S.; Rao, M. V.; Isobaric vapor-liquid equilibria of the p-xylene + 1,1,1- trichloroethane system.. *J. Chem. Eng. Data*, 1982, 27, 414
- [1982WIS] Wisniak, J.; Tamir, A., Vapor-Liquid Equilibrium in the Systems Propyl Bromide-Acetic Acid, Propyl Bromide-Propionic Acid, and Propyl Bromide-Acetic Acid- Propionic Acid, *J. Chem. Eng. Data*, 1982, 27, 430
- [1983DEW] De Wit, H.G.M.; Offringa, J.C.A.; De Kruif, C.G.; van Miltenburg, J.C., Thermodynamic properties of molecular organic crystals containing nitrogen, oxygen and sulfur. III. molar heat capacities measured by DSC. scanning calorimetry., *Thermochim. Acta*, 1983, 65, 43
- [1983GOO] Goodwin, R.D., Hydrogen sulfide provisional thermophysical properties from 188 to 700K at pressures to 75 MPa, Report, NBSIR-83-1694; NTIS No. PB84-122704, 177 pp., 1983.
- [1983OI] Oi, T.; Shulman, J.; Popowicz, A.; Ishida, Y.; Vapor pressure isotope effects in liquid methyl fluoride *J. Phys. Chem.*, 1983, 87, 3153

- [1983O'R] O'Rourke, D.F.; Mraw, S.C., Heat capacities and enthalpies of fusion of dibenzothiophene (220 to 560 K) and of biphenyl, cyclohexylbenzene, and cyclohexylcyclohexane (220 to 475 K). Enthalpies and temperatures of three transitions in solid cyclohexylcyclohexane, *J. Chem. Thermodyn.*, 1983, 15, 489
- [1983OTT] Ott, J.B.; Goates, J.R., (Solid + liquid) Phase Equilibria in Binary Mixtures Containing Benzene, a Cycloalkane, an n-Alkane, or Tetrachloromethane. An Equation for Representing (Solid + liquid) Phase Equilibria, *J. Chem. Thermodyn.*, 1983, 15, 267
- [1983RAE] Raemy, A.; Schweizer, T. F. J. *Thermal Anal.* 1983, 28, 95–108
- [1984ENG] Eng, R.; Sandler, S.I., Vapor-liquid equilibria for three aldehyde/hydrocarbon mixtures, *J. Chem. Eng. Data*, 1984, 29, 156
- [1984RAO] Rao, K. V.; Prasad, A. R.; Measurement, Correlation and Prediction of Binary/Vapor - Liquid Equilibria For Alcohol - Tetrachloroethene Systems. *Can. J. Chem. Eng.*, 1984, 62, 142
- [1984SZA] Szafranski, A.M., Solid-Liquid Equilibrium, *Int. DATA Ser., Sel. Data Mixtures, Ser. A*, 1984, No. 1, 40
- [1985DRI] Dribika, M.M.; Rashed, I.G.; Biddulph, M.W., Vapor-Liquid Equilibria in the Ternary System Methyl Alcohol-Ethyl Alcohol-Isopropyl Alcohol at Atmospheric Pressure, *J. Chem. Eng. Data*, 1985, 30, 146
- [1985HOW] Howat, C. S.; Swift, G. W.; Vapor-liquid phase equilibria for the binaries isoprene-dicyclopentadiene and n-pentane-dicyclopentadiene at 313 and 323 K. *J. Chem. Eng. Data*, 1985, 30, 281
- [1985HSU] Hsu, F.-J.; Lazar, J.M.; Engelhardt, C.E.; Tapper, M.B.; Birnbaum, C.S.; Fried, V.; Zudkevitch, D., Isothermal vapor-liquid equilibria for the n-butanal/toluene system, *AIChE Symp. Ser.*, 1985, 81, 130
- [1985KOL] Kolodziej, E. J.; Jin, Z. L.; Greenkorn, R. A.; Chao, K.-C.; Vapor-liquid equilibrium in binary mixtures of propylene oxide with 1,2-dichloropropane and with tert-butanol. *AIChE Symp. Ser.*, 1985, 81, 80
- [1985KOV] Kovac, A.; Svoboda, J.; Ondrus, I., Vapour-liquid equilibrium of some binary systems containing 1,1,2-trichloroethane, *Chem. Zvesti*, 1985, 39, 737
- [1985KOV2] Kovac, A.; Svoboda, J.; Ondrus, I., Vapor-liquid equilibrium of some binary systems containing 1,1-dichloroethylene, *Chem. Zvesti*, 1985, 39, 729
- [1985MAC] Machat, V.; Boublik, T.; Vapour-liquid equilibrium and excess molar volume of (trans-1,2-dichloroethene + cyclopentane) at 293.15 K. *J. Chem. Thermodyn.*, 1985, 17, 887
- [1985MAJ] Majer, V.; Svoboda, V., *Enthalpies of Vaporization of Organic Compounds: A Critical Review and Data Compilation*, Blackwell Scientific Publications, Oxford, 1985
- [1985RAO] Rao, K.P.C.; Reddy, K.S.; Excess volumes and excess isentropic compressibilities of binary mixtures of N,N-dimethylformamide with branched alcohols at 303.15 K, *Thermochim. Acta*, 1985, 91, 321

- [1985RAO2] Rao, Y.V.C.; Basu, R., Vapor-liquid equilibria for the 2-propanol-toluene system at 760 and 550 Torr, J. Chem. Eng. Data, 1985, 30, 361.
- [1985RED] Reddy, K.D.; Rao, M.V.P., Excess volumes of nitrobenzene with the alkyl-substituted chloroethanes at 303.15 and 313.15 K., J. Chem. Eng. Data, 1985, 30, 36
- [1985TAS] Taskinen, E.; Nummelin, K.; Relative thermodynamic stabilities of the isomeric cyclooctadienes. Acta Chem. Scand., 1985, B39, 791
- [1985WOL] Wolff, H.; Schiller, O.; The vapour pressure behaviour and the association of n-methylethylamine, diethylamine, and their n-deuterioanalogues in mixtures with n-hexane. Fluid Phase Equilib., 1985, 22, 185
- [1986FIS] Fischer, J.; Weiss, A.; Transport properties of liquids. V. Self-diffusion, viscosity, and mass density of ellipsoidal shaped molecules in the pure liquid phase. Ber. Bunsen-Ges. Phys. Chem., 1986, 90, 896
- [1986JAC] Jacobsen, R.T.; Stewart, R.B.; Jahangiri, M., Thermodynamic Properties of Nitrogen from the Freezing Line to 2000 K at Pressures to 1000 MPa, J. Phys. Chem. Ref. Data, 1986, 15, 735-909
- [1986JAH] Jahangiri, M.; Jacobsen, R.T.; Stewart, R.B.; McCarty, R.D., Thermodynamic Properties of Ethylene from the Freezing Line to 450 K at Pressures to 260 MPa, J. Phys. Chem. Ref. Data, 1986, 15, 593
- [1986MAR] Markovnik, V.S.; Shvaro, O.V.; Al'khimovich, V.M.; Sachek, A.I., Temperature dependence of vapor pressure, of heats of formation, combustion, and vaporization of 3-methyl-1-butanol, Termodin. Org. Soedin., 1986, 1986, 68
- [1986PAU] Paul, H.; Krug, J.; Knapp, H., Measurements of VLE, hE and vE for Binary Mixtures of n-Alkanes with n-Alkylbenzenes, Thermochim. Acta, 1986, 108, 9
- [1986PAU2] Paul, H.-I.; Krug, J.; Gutsche, B.; Knapp, H., Measurements of VLE, hE, and vE for Binary Mixtures of Dibutyl Ether with 1-Chloropentane, 1,2-Dichloroethane, and 1,1,1-Trichloroethane trichloroethane, J. Chem. Eng. Data, 1986, 31, 448
- [1986SCH] Schmitt, W.J.; Reid, R.C., Solubility of Monofunctional Organic Solids in Chemical Diverse Supercritical Fluids, J. Chem. Eng. Data, 1986, 31, 204
- [1986SNO] Snow, R.L.; Ott, J.B.; Goates, J.R.; Marsh, K.N.; O'Shea, S.; Stokes, R.H., (Solid + liquid) and (vapor + liquid) phase equilibria and excess enthalpies for (benzene + n-tetradecane), (benzene + n-hexadecane), (cyclohexane + n-tetradecane), and (cyclohexane + n-hexadecane), J. Chem. Thermodyn., 1986, 18, 107
- [1986TAK] Takagi, S.; Ishikawa, M.; Kumada, M.; Kimura, T.; Fujishiro, R., Vapor Pressure and Thermodynamic Properties of Hexamethyldisilane at 305-387 K, Thermochim. Acta, 1986, 109, 55.
- [1987AMB] Ambrose, D.; Ghiassee, N. B.; Vapor Pressure, Critical Temperature and Critical Pressure of Acetic Anhydride. J. Chem. Thermodyn., 1987, 19, 911

- [1987AMB2] Ambrose, D.; Ghasseer, N. B.; Vapor Pressures and Critical Temperatures and Critical Pressures of Some Alkanoic Acids: C1 to C10. *J. Chem. Thermodyn.*, 1987, 19, 505
- [1987DAU] Daubert, T. E.; Jalowka, J. W.; Goren, V.; Vapor pressure of 22 pure industrial chemicals. *AIChE Symp. Ser.*, 1987, 83, 128
- [1987DEL] DellaGatta, G.; Ferro, D., Enthalpies of fusion and solid-to-solid transition, entropies of fusion for urea and twelve alkylurea, *Thermochim. Acta*, 1987, 122, 143
- [1987JUD] Judy, C.L.; Pontikos, N.M.; Acree, W.E., Solubility of Pyrene in Binary Solvent Mixtures Containing Cyclohexane, *J. Chem. Eng. Data*, 1987, 32, 60.
- [1987KAN] Kanungo, A.; Oi, T.; Popowicz, A.; Ishida, T.; Vapor pressure isotope effects in liquid methylene difluoride *J. Phys. Chem.*, 1987, 91, 4198
- [1987KRI] Krishnaiah, A.; Choudary, N.V., Activity Coefficients and Excess Gibbs Free Energies of 1,2-Dichloroethane with Isomeric Butanols, *J. Chem. Eng. Data*, 1987, 32, 196
- [1987STE] Stephenson, Richard M.; Malanowski, Stanislaw, *Handbook of the Thermodynamics of Organic Compounds*, 1987
- [1988AGU] Aguirre-Ode, F.; Koo, J., Activity Coefficients in Thiophene-Butanol Systems Near the Freezing Point of Thiophene, *J. Chem. Eng. Data*, 1988, 33, 310
- [1988COO] Coon, J.E.; Sediawan, W.B.; Auwaerter, J.E.; McLaughlin, E., Solubilities of families of heterocyclic polynuclear aromatics in organic solvents and their mixtures, *J. Solution Chem.*, 1988, 17, 519
- [1988GIE] Gierycz, P.; Gregorowicz, J.; Malanowski, S., Vapor Pressures and Excess Gibbs Energies of (Butan-1-ol+n-octane or n-Decane) at 373.15 and 383.15 K, *J. Chem. Thermodyn.*, 1988, 20, 385
- [1988HOL] Holldorff, H.; Knapp, H.; Vapor pressures of n-butane, dimethyl ether, methyl chloride, methanal and the vapor-liquid equilibrium of dimethyl ether - methanol. *Fluid Phase Equilib.*, 1988, 40, 113
- [1988KAT] Kato, M., Vapor-Liquid Equilibrium Measurements for Binary Systems of Acetic Acid with Ethyl Acetate and Vinyl Acetate by the Dew-Bubble Point Temperature Method, *J. Chem. Eng. Data*, 1988, 33, 499
- [1988KIM] Kim, E.S.; Marsh, K.N., Excess Volumes for 2-Methyl-2-propanol + Water at 5 K Intervals from 303.15 to 323.15 K, *J. Chem. Eng. Data*, 1988, 33, 288
- [1988MIH] Mihaila, G.; Popa, A.; Cruceanu, M.; Popovici, E.; Ciugureanu, C.; Echilibrul lichid-vapori in sisteme binare cu derivapi clorurapi ai acetilenei. *Rev. Chim. (Bucharest)*, 1988, 39, 687
- [1988PAU] Paul, H.-I.; Krug, J.; Knapp, H., Measurements of VLE, vE, and hE for binary mixtures of 1-chlorohexane with three n-alkylbenzenes: toluene, ethylbenzene, n-propylbenzene, *J. Chem. Eng. Data*, 1988, 33, 453

- [1988SHE] Sheindlin MA, Senchenko VN. Experimental investigation of graphite thermodynamical properties in the vicinity of melting point. *Sov. Phys. Dokl*, 1988, 33, 142–5
- [1988VAN] van Bommel, M.J.; van Miltenburg, J.C.; Schuijff, A., Heat-capacity measurements and thermodynamic functions of 1,3,5-triazine and 1,3,5-trioxane, *J. Chem. Thermodyn.*, 1988, 20, 397
- [1988WIS] Wisniak, J.; Tamir, A., Association Effects in the Methanol-1-Pentanol System, *J. Chem. Eng. Data*, 1988, 33, 432
- [1988WIS2] Wisniak, J.; Tamir, A., Vapor-Liquid Equilibria at 760 mmHg for the System 1,1-Dichloroethane- Propyl Bromide, *J. Chem. Eng. Data*, 1988, 33, 108
- [1988WIS3] Wisniak, J.; Tamir, A., Vapor-Liquid Equilibria at 760 mmHg in the Systems Propyl Bromide- tert-Butyl Alcohol and Propyl Bromide-p-Xylene, *J. Chem. Eng. Data*, 1988, 33, 106
- [1989CEP] Cepeda, E.; Gonzalez, C.; Resa, J.M., Isobaric vapor-Liquid Equilibrium for the Cumene-Phenol System, *J. Chem. Eng. Data*, 1989, 34, 270
- [1989CEP2] Cepeda, E.; Gonzalez, C.; Resa, J.M.; Ortiz de Salido, C., Isobaric Vapor-Liquid Equilibrium for the Cumene-m-Cresol and Phenol- m-Cresol Systems at 13.33 kPa, *J. Chem. Eng. Data*, 1989, 34, 429
- [1989CRC] CRC Handbook of Data on Organic Compounds, 2nd Edition, Weast,R.C and Grasselli, J.G., ed(s)., CRC Press, Inc., Boca Raton, FL, 1989
- [1989FON] Fonseca, I. M. A.; Lobo, L. Q.; Thermodynamics of liquid mixtures of xenon and methyl fluoride *Fluid Phase Equilib.*, 1989, 47, 249
- [1989HAL] Halawa, M.M.; Georgiuss, R.S.; El-Sabban, S.; Ammar, M.M., Study of the freezing temperature of benzoic acid as a standard reference material in thermometry, *J. Mater. Sci.*, 1989, 24, 2619
- [1989MAT] Mato, F.; Gonzalez, G.; Arroyo, F.J., Vapor-Liquid Equilibria at 760 mmHg in the Binary Systems Cyclohexene-1,2-Dichloroethane and Cyclohexane-1,2-Dichloroethane, *J. Chem. Eng. Data*, 1989, 34, 179
- [1989MUH] Muhlenbruch, M.; Figurski, G.; The vapor-liquid equilibrium of the binary systems acetaldehyde-propylene oxide, acetaldehyde-methylformate, and propylene oxide-propan-2-ol at 101.325 kPa. *Z. Phys. Chem. (Leipzig)*, 1989, 270, 305
- [1989RAM] Ramanjaneyulu, K.; Surendranath, K.N.; Krishnaiah, A., Excess volumes of binary mixtures of trichloroethylene with some aliphatic, alicyclic and aromatic hydrocarbons, *Fluid Phase Equilib.*, 1989, 44, 357
- [1989RED] Reddy, B.S.; Rao, B.V.S., Vapor-liquid equilibrium data of tert-butyl alcohol-chlorobenzene at 101.3 kPa, *J. Chem. Eng. Data*, 1989, 34, 26
- [1989SED] Sediawan, W.B.; Gupta, S.; McLaughlin, E., Solid-Liquid Phase Diagrams of Binary Aromatic Hydrocarbon Mixtures from Calorimetric Studies, *J. Chem. Eng. Data*, 1989, 34, 223

- [1989STE] Steele, W. V.; Chirico, R. D.; Nguyen, A.; Hossenlopp, I. A.; Smith, N. K.; Determination of some pure compound ideal-gas enthalpies of formation. *AIChE Symp. Ser.*, 1989, 85, 140
- [1989SUS] Susial, P.; Ortega, J.; de Alfonso, C.; Alonso, C., Vapor-Liquid Equilibrium Measurements for Methyl Propanoate-Ethanol and Methyl Propanoate-Propan-1-ol at 101.32 kPa, *J. Chem. Eng. Data*, 1989, 34, 247
- [1989TIP] Tipton, T.; Stone, D.A.; KuBulat, K.; Person, W.B., Experimental and theoretical studies of the infrared spectra of hydrazines: N₂H₄, N₂H₃D, N₂H₂D₂, N₂HD₃, and N₂D₄, *J. Phys. Chem.*, 1989, 93, 2917
- [1989WEA] Weast, R.C; Grasselli, J.G., ed(s)., *CRC Handbook of Data on Organic Compounds*, 2nd Edition, CRC Press, Inc., Boca Raton, FL, 1989
- [1990ALD] Aldrich Chemical Company Inc., *Catalog Handbook of Fine Chemicals*, Aldrich Chemical Company, Inc., Milwaukee WI, 1990
- [1990AMB] Ambrose, D.; Ewing, M. B.; Ghiassee, N. B.; Sanchez Ochoa, J. C.; The ebulliometric method of vapor-pressure measurement: vapor pressures of benzene, hexafluorobenzene, and naphthalene. *J. Chem. Thermodyn.*, 1990, 22, 589
- [1990AMB2] Ambrose, D.; Ghiassee, N. B.; Vapor pressures, critical temperatures, and critical pressures of benzyl alcohol, octan-2-ol, and 2-ethylhexan-1-ol. *J. Chem. Thermodyn.*, 1990, 22, 307
- [1990CEZ] Cezairliyan, A., Miiller, A.P. Measurement of the radiance temperature (at 655 nm) of melting graphite near its triple point by a pulse-heating technique. *Int. J. Thermophys.*, 1990, 11, 643–651
- [1990DAU] Daubert, T. E.; Hutchison, G.; Vapor pressure of 18 pure industrial chemicals. *AIChE Symp. Ser.*, 1990, 86, 93
- [1990DON] Donnelly, J.R.; Drewes, L.A.; Johnson, R.L.; Munslow, W.D.; Knapp, K.K.; Sovocool, G.W., Purity and heat of fusion data for environmental standards as determined by differential scanning calorimetry, *Thermochim. Acta*, 1990, 167, 155
- [1990GUL] Gultekin, N., Vapor-Liquid Equilibria at 1 atm. for Ternary and Quarternary Systems Composed of Acetone, Methanol, 2-Propanol, and 1-Propanol, *J. Chem. Eng. Data*, 1990, 35, 132
- [1990HUI] Hu, R.; Li, J.; Liang, Y.; Wu, S.; Sun, L.; Wang, Y., Thermal behavior of 2,2,2-trinitroethyl-4,4,4-trinitrobutyrate, *J. Therm. Anal.*, 1990, 36, 1155
- [1990HUI2] Hui, F.; Shuxian, Y.; Zhongming, M.; Qi, W.; Genghua, C.; Shi-Jun, H., Studies on Isothermal Vapor-Liquid Equilibrium for Associating Systems of Acetic Acid - Propylene Glycol Monomethyl Ether in Experimental Results for DIPPR 1990-91 Projects on Phase Equilibria and Pure Component Properties, *DIPPR Data Series No. 2*, 1994

- [1990JOS] Joshi, S.S.; Aminabhavi, T.M.; Shukla, S.S., Densities and Shear Viscosities of Anisole with Nitrobenzene, Chlorobenzene Carbon Tetrachloride, 1,2-Dichloroethane and Cyclohexane from 25 to 40 C, J. Chem. Eng. Data, 1990, 35, 247
- [1990KAT] Kato, M.; Yoshikawa, H.; Yamaguchi, M.; Vapor-liquid equilibrium measurements of three binary systems made of formic acid, acetic acid and propionic acid by the dew-bubble point temperature method. Fluid Phase Equilib., 1990, 54, 47
- [1990NIS] Nishiumi, H.; Yokoyama, T.; Vapor-Liquid equilibrium for the system of R134a-R22, Proc. Symp. Thermophys. Prop., 11th, 1990, Tokyo, Japan, p 95,
- [1990PCR] PCR Inc., Research Chemicals Catalog 1990-1991, PCR Inc., Gainesville, FL, 1990
- [1990PLA] Platzter, B., Polt, A., and Maurer, G., "Thermophysical Properties of Refrigerants," Berlin, Springer-Verlag, 1990
- [1990PRE] H. Preston-Thomas, The International Temperature Scale of 1990 (ITS-90), Metrologia 27, 3-10 (1990)
- [1990WIS] Wisniak, J.; Tamir, A., Isobaric Vapor-Liquid Equilibria in the Systems Propyl Bromide-Methyl Ethyl Ketone, Methyl Ethyl Ketone-o-Xylene, and Vinyl Acetate-Methyl Methacrylate, J. Chem. Eng. Data, 1990, 35, 147.
- [1991AIZ] Aizawa, K.; Kato, M., Vapor-liquid equilibrium determination by total pressure measurements for three binary systems made of 1,2-dimethoxyethane with toluene, methylcyclohexane, or (trifluoromethyl)benzene, J. Chem. Eng. Data, 1991, 36, 159
- [1991CAB] Cabezas, J.L.; Beltran, S.; Coca, J., Isobaric vapor-liquid equilibrium data for the binary systems 1,2-dimethoxyethane + alcohols, J. Chem. Eng. Data, 1991, 36, 184
- [1991DIC] Di Cave, S.; Mazzarotta, B., Isobaric Vapor-Liquid Equilibria for the Binary Systems Formed by Acetonitrile and Aromatic Hydrocarbons, J. Chem. Eng. Data, 1991, 36, 293
- [1991KAT] Kato, M.; Tanaka, H.; Yoshikawa, H., Vapor-liquid equilibria for four binary systems made of alcohol and cyclohexylamine at 100 kPa., J. Chem. Eng. Data, 1991, 36, 387
- [1991MAE] Maezawa, Y.; Sato, H. Watanabe, K.; Liquid Densities and Vapor Pressures of 1,1,2,2-Tetrafluoroethane (HFC 134) and 1,1 -Dichloro-1 -fluoroethane (HCFC 141b). J. Chem. Eng. Data, 1991, 36, 141
- [1991PAP] Papousek, D.; Tesar, R.; Pracna, P.; Civis, S.; Winnewisser, M.; Belov, S. P.; Tret'yakov, M. Y.; High-resolution Fourier transform and submillimeter-wave study of the .nu.6 band of methyl-12C fluoride J. Mol. Spectrosc., 1991, 147, 279
- [1991SAY] Sayar, A.A.; Tatli, B.; Dramur, U., Liquid-Liquid Equilibria of the Water + Acetic Acid + Cyclohexyl Acetate Ternary, J. Chem. Eng. Data, 1991, 36, 378

- [1991SET] Setzmann, U. and Wagner, W., A New Equation of State and Tables of Thermodynamic Properties for Methane Covering the Range from the Melting Line to 625 K at Pressures up to 1000 MPa, J. Phys. Chem. Ref. Data, 1991, 20, 1061
- [1991STE] Stewart, R.B., Jacobsen, R.T, and Wagner, W., Thermodynamic Properties of Oxygen from the Triple Point to 300 K with Pressures to 80 MPa, J. Phys. Chem. Ref. Data, 1991, 20, 917
- [1991YOS] Yoshikawa, H.; Kato, M., Vapor-liquid equilibria for three binary systems made of methyl ethyl ketone with acetone, methyl acetate, or N-amyl alcohol, J. Chem. Eng. Data, 1991, 36, 57
- [1992AND] Andersson, O.; Ross, R.G., Thermal conductivity of (+)- and (±)-camphor at pressures up to 0.5 GPa and temperatures down to 40 K, Mol. Phys., 1992, 76, 433
- [1992FER] Ferriol, M.; Laachach, A.; Cohen-Adad, M. T.; Getzen, F. W.; Jorat, L.; Noyel, G.; Huck, J.; Bureau, J. C.; Vapor-liquid equilibria in the binary systems water-methylhydrazine and water-1,1-dimethylhydrazine. Thermodynamic modeling in relation to the structure of the liquid phase. Fluid Phase Equilib., 1992, 71, 287
- [1992GOL] R. N. Goldberg and R. D. Weir, "Conversion of temperatures and thermodynamic properties to the basis of the International Temperature Scale of 1990," Pure Appl. Chem. 64, 1545-1562 (1992).
- [1992GOO] Goodwin, A. R. H.; Defibaugh, D. R.; Weber, L. A. The vapor pressure of 1,1,1,2-tetrafluoroethane (R134a) and chlorodifluoromethane (R22) Int. J. Thermophys., 1992, 13, 837
- [1992GUP] Gupta, S.K.; Nanoti, S.M.; Rawat, B.S., Vapor-Liquid Equilibria for the Systems Hexane-Aniline and Hexane-Benzene-Aniline at Atmospheric Pressure, J. Chem. Eng. Data, 1992, 37, 162
- [1992HIA] Hiaki, T.; Yamato, K.; Kojima, K., Vapor-liquid equilibria of 2,3-dimethylbutane + methanol or ethanol at 101.3 kPa, J. Chem. Eng. Data, 1992, 37, 203
- [1992JAC] Jacobsen, R.T, Penoncello, S.G., and Lemmon, E.W., "A Fundamental Equation for Trichlorofluoromethane (R-11)," Fluid Phase Equilib., 1992, 80, 45
- [1992MAR] Marx, V., Pruss, A., and Wagner, W., "Neue Zustandsgleichungen fuer R 12, R 22, R 11 und R 113. Beschreibung des thermodynamischen Zustandsverhaltens bei Temperaturen bis 525 K und Druecken bis 200 MPa," Duesseldorf, VDI Verlag, Series 19 (Waermetechnik/Kaeltetechnik), No. 57, 1992
- [1992POL] Polt, A., Platzer, B., and Maurer, G., "Parameter der Thermischen Zustandsgleichung von Bender fuer 14 Mehratomige Reine Stoffe," Chem. Tech. (Leipzig), 1992, 44, 216
- [1992TAN] Tanaka, H.; Muramatsu, T.; Kato, M., Isobaric Vapor-Liquid Equilibria for Three Binary Systems of 2-Butanone with 3-Methyl-1-butanol, 1-Butanol, or 2-Butanol, J. Chem. Eng. Data, 1992, 37, 164
- [1992WEB] Weber, L. A.; Ebulliometric measurement of the vapor pressures of R123 and R141b. Fluid Phase Equilib., 1992, 80, 141

- [1992WIL] Wilson, L. C.; Wilding, W. V.; Wilson, G. M.; Rowley, R. L.; Felix, V. M.; Chisolm-Carter, T.; Thermophysical properties of HFC-125, *Fluid Phase Equilib.*, 1992, 80, 167
- [1993BUR] Burguet, M. C.; Monton, J. B.; Sanchotello, M.; Vazquez, M. I.; Vapor-Liquid Equilibria of Cyclohexanone + Cyclohexanol and Cyclohexanone + 2-Methylcyclohexanone Systems at 4.00 and 26.66 kPa. *J. Chem. Eng. Data*, 1993, 38, 328
- [1993CHI] Chirico, R. D.; Knipmeyer, S. E.; Nguyen, A.; Steele, W. V.; The thermodynamic properties to the temperature 700 K of naphthalene and of 2,7-dimethylnaphthalene. *J. Chem. Thermodyn.*, 1993, 25, 1461
- [1993DEF] Defibaugh, D. R.; Goodwin, A. R. H.; Morrison, G.; Weber, L. A.; Thermodynamic properties of 1,1-dichloro-1-fluoroethane (R141b). *Fluid Phase Equilib.*, 1993, 85, 271
- [1993DER] de Reuck, K.M. and Craven, R.J.B., *Methanol*, International Thermodynamic Tables of the Fluid State-12, IUPAC, Blackwell Scientific Publications, London, 1993.
- [1993POT] Pottlacher G, Hixson RS, Melnitzky S, Kaschnitz E, Winkler MA, Jager H, Thermophysical properties of POCOAXF-5Q graphite up to melting. *Thermochim. Acta*, 1993, 218, 183
- [1993SIL] Silva, A. M.; Weber, L. A.; Ebulliometric Measurement of the Vapor Pressure of 1-Chloro-1,1-difluoroethane and 1,1-Difluoroethane. *J. Chem. Eng. Data*, 1993, 38, 644
- [1993WEB] Weber, L. A.; Goodwin, A. R. H.; Ebulliometric measurement of the vapor pressure of difluoromethane *J. Chem. Eng. Data*, 1993, 38, 254
- [1994KRI] Krishnaiah, A.; Viswanath, D.S.; Toghiani, H., Isothermal Vapor-Liquid Equilibria of the System Isopropyl Acetate + Cyclopropane in Experimental Results for DIPPR 1990-91 Projects on Phase Equilibria and Pure Component Properties, DIPPR Data Series No. 2, 1994
- [1994RES] Resa, J. M.; Quintana, T.; Cepeda, E.; Isobaric vapor-liquid equilibrium of the systems cumene + benzyl alcohol and phenol + benzyl alcohol at 10 kPa. *J. Chem. Eng. Data*, 1994, 39, 98
- [1994RUS] R. L. Rusby, R. P. Hudson and M. Durieux, "Revised values for (t₉₀-t₆₈) from 630°C to 1064°C," *Metrologia* 31, 149-153 (1994).
- [1994TAY] Taylor, B. N.; Kuyatt, C.; Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results, NIST Technical Note 1297 (1994). <https://doi.org/10.6028/NIST.TN1297> (accessed July 22, 2021)
- [1994TIL] Tillner-Roth, R. and Baehr, H.D.; "An International Standard Formulation of the Thermodynamic Properties of 1,1,1,2-Tetrafluoroethane (HFC-134a) for Temperatures from 170 K to 455 K at Pressures up to 70 MPa," *J. Phys. Chem. Ref. Data*, 1994, 23, 657
- [1994TOJ] Tojo, J.; Ramallo, A.V.; Orge, B., Vapor-Liquid Equilibria for the Ternary System Methyl Ethyl Ketone + Cyclohexane + Heptane at 101.325 kPa, *J. Chem. Eng. Data*, 1994, 39, 480

- [1994WEB] Weber, L. A.; Silva, A. M.; Measurements of the Vapor Pressures of Difluoromethane, 1-Chloro-1,2,2,2-tetrafluoroethane, and Pentafluoroethane, J. Chem. Eng. Data, 1994, 39, 808
- [1994WID] Widiatmo, J. V.; Sato, H.; Watanabe, K.; Saturated-liquid densities and vapor pressures of 1,1,1-trifluoroethane, difluoromethane, and pentafluoroethane J. Chem. Eng. Data, 1994, 39, 304
- [1994YAH] Yahiaoui, A.; Gonzalez, J. A.; Ait-Kaci, A.; Jose, J.; Kehiaian, H. V.; Isothermal vapor-liquid equilibria for the 2-butanone + benzene + n-octane system. Fluid Phase Equilib., 1994, 98, 179
- [1995COM] Comelli, F.; Francesconi, R.; Isothermal Vapor-Liquid Equilibria for 2,2,4-Trimethylpentane + Five Chlorohydrocarbon Binary Systems at 343.15 K. J. Chem. Eng. Data, 1995, 40, 21
- [1995FAN] Fan, J.; Angell, C. A. Thermochim. Acta, 1995, 266, 9–30
- [1995HIA] Hiaki, T.; Takahashi, K.; Tsuji, T.; Hongo, M.; Kojima, K., Vapor-Liquid Equilibria of Ethanol + Octane at 343.15 K and 1-Propanol + Octane at 358.15 K, J. Chem. Eng. Data, 1995, 40, 271
- [1995HIA2] Hiaki, T.; Takahashi, K.; Tsuji, T.; Hongo, M.; Kojima, K., Vapor-Liquid Equilibria of 1-Propanol or 2-Propanol with Octane at 101.3 kPa, J. Chem. Eng. Data, 1995, 40, 274
- [1995KRI] Krishnan, K.M.; Rambabu, K.; Venkateswarlu, P.; Raman, G.K., A Study on Mixing Properties of Binary Mixtures of 2-Methoxyethanol with Aromatic Hydrocarbons, J. Chem. Eng. Data, 1995, 40, 132
- [1995RED] Reddy, V.K.; Rambabu, K.; Devarajulu, T.; Krishnaiah, A., Volume of Mixing, Speed of Sound, and Viscosity of Methyl Cellosolve with Aliphatic Alcohols at 308.15 K, J. Chem. Eng. Data, 1995, 40, 124
- [1995WIS] Wisniak, J.; Apelblat, A.; Zabicky, J.; Feingold, I., Isobaric Vapor-Liquid Equilibria in the Binary Systems of 1-Bromopropane with Cyclohexane, Heptane, and 1-Butanol, J. Chem. Eng. Data, 1995, 40, 120
- [1996BED] R E Bedford, G Bonnier, H Maas, and F Pavese, Recommended values of temperature on the International Temperature Scale of 1990 for a selected set of secondary reference points, Metrologia, 1996, 33, 133
- [1996BLA] Blanke, W.; Weiss, R. Investigation of the (p, V, T) behavior of the fluids R124 and R134a at 120 -300 K and at pressures of up to 12 Mpa, PTB-Ber. W - Phys.-Tech. Bundesanst., PTB-W-64, 1996, 1-25
- [1996DEJ] Dejoz, A.; Gonzalez-Alfaro, V.; Miguel, P. J.; Vazquez, M. I.; "Isobaric Vapor-Liquid Equilibria for Binary Systems Composed of Octane, Decane, and Dodecane at 20 kPa," J. Chem. Eng. Data, 1996, 41, 93-96
- [1996DOM] Domalski, Eugene S.; Hearing, Elizabeth D., Heat Capacities and Entropies of Organic Compounds in the Condensed Phase. Volume III, J. Phys. Chem. Ref. Data, 1996, 25, 1
- [1996GON] Gonzalez, E.; Ortega, J., Densities and Isobaric Vapor-Liquid Equilibria for the Mixtures Formed by Four Butyl Esters and 1-Butanol, J. Chem. Eng. Data, 1996, 41, 53
- [1996KRI] Krishnaiah, A.; Surendranath, K.N., Densities, Speeds of Sound, and Viscosities of Mixtures of Oxolane with Chloroethanes and Chloroethenes, J. Chem. Eng. Data, 1996, 41, 1012

- [1996LIN] Linek, J.; Wichterle, I.; Marsh, K.N., Vapor-liquid Equilibria for N-Methyl-3-pyrrolidone + Benzene, + Toluene, and + Methylcyclohexane, J. Chem. Eng. data, 1996, 41, 1212
- [1996OUT] Outcalt, S.L. and McLinden, M.O., A Modified Benedict-Webb-Rubin Equation of State for the Thermodynamic Properties of R152a (1,1-Difluoroethane), J. Phys. Chem. Ref. Data, 1996, 25, 605
- [1996RES] Resa, J.M.; Echebarria, S.; Betolaza, M.A.; Ruiz, A.; Moradillo, B., Isobaric Vapor-Liquid Equilibria of 3-Pentanone with Acetone and Isopropyl Ether at 101.3 kPa, J. Chem. Eng. Data, 1996, 41, 63
- [1996SPA] Span, R. and Wagner, W.; A New Equation of State for Carbon Dioxide Covering the Fluid Region from the Triple-Point Temperature to 1100 K at Pressures up to 800 MPa. J. Phys. Chem. Ref. Data, 1996, 25, 1509
- [1997BAR] Barley, M. H.; Morrison, J. D.; O'Donnel, A.; Parker, I. B.; Petherbridge, S.; Wheelhouse, R. W.; Vapour-liquid equilibrium data for binary mixtures of some new refrigerants Fluid Phase Equilib., 1997, 140, 183
- [1997COM] Comelli, F.; Francesconi, R., Isothermal Vapor-Liquid Equilibria Measurements, Excess Molar Enthalpies, and Excess Molar Volumes of Dimethyl Carbonate + Methanol, + Ethanol, and + Propan-1-ol at 313.15 K, J. Chem. Eng. Data, 1997, 42, 705
- [1997DUA] Duarte-Garza, H. A.; Hwang, C.-A.; Kellerman, S. A.; Miller, R. C.; Hall, K. R.; Holste, J. C.; Marsh, K. N.; Gammon, B. E.; Vapor pressure, vapor density, and liquid density for 1,1-dichloro-1-fluoroethane(R-141b). J. Chem. Eng. Data, 1997, 42, 497
- [1997NOW] Nowak, P.; Kleinrahm, R.; Wagner, W.; "Measurement and correlation of the (p, rho, T) relation of nitrogen. II. Saturated-liquid and saturated-vapour densities and vapour pressures along the entire coexistence curve," J. Chem. Thermodyn., 1997, 29, 1157-1174
- [1997TIL] Tillner-Roth, R. and Yokozeki, A., An International Standard Equation of State for Difluoromethane (R-32) for Temperatures from the Triple Point at 136.34 K to 435 K and Pressures up to 70 MPa, J. Phys. Chem. Ref. Data, 1997, 26, 1273
- [1998ARC] Arce, A.; Martinez-Ageitos, J.; Soto, A., Vapor-Liquid Equilibria at 101.32 kPa of the Ternary Systems 2-Methoxy-2-methylpropane + Methanol + Water and 2-Methoxy-2-methylpropane + Ethanol + Water, J. Chem. Eng. Data, 1998, 43, 708
- [1998BEY] Beyerlein, A. L.; DesMarteau, D. D.; Kul, I.; Zhao, G.; Properties of novel fluorinated compounds and their mixtures as alternative refrigerants Fluid Phase Equilib., 1998, 287, 150
- [1998BLA] Blanco, A.M.; Ortega, J., Densities and Vapor-Liquid Equilibrium Values for Binary Mixtures Composed of Methanol + an Ethyl Ester at 141.3 kPa with Application of an Extended Correlation Equation for Isobaric VLE Data, J. Chem. Eng. Data, 1998, 43, 638

- [1998FEN] Feng, L.-C.; Chou, C.-H.; Tang, M.; Chen, Y.P., Vapor-Liquid Equilibria of Binary Mixtures 2-Butanol + Butyl Acetate, Hexane + Butyl Acetate, and Cyclohexane + 2-Butanol at 101.3 kPa, J. Chem. Eng. Data, 1998, 43, 658
- [1998HOL] Holcomb, C.D.; Magee, J. W.; Scott, J. L.; Outcalt, S. L.; Haynes, W. M.; Selected Thermodynamic Properties for Mixtures of R-32 (difluoromethane), R-125 (pentafluoroethane), R-134a (1,1,1,2-tetrafluoroethane), R-143a (1,1,1-trifluoroethane), R-41 (fluoromethane), R-290 (propane), and R-744 (carbon dioxide), NIST Technical Note 1397, Boulder CO, 1998
- [1998MUS] Musella M, Ronchi C, Brykin M, Sheindlin M. The molten state of graphite: an experimental study. J Appl Phys, 1998, 84, 2530–7
- [1998SEN] Senol, A.; Vapor-Liquid Equilibria on the Systems Ethyl Ethanoate + 2-Methyl-2-butanol, 2-Methyl-1-Propanol + 3-Methyl-1-butanol, and Cyclohexanol + Benzyl Alcohol at 101.32 kPa. J. Chem. Eng. Data, 1998, 43, 763
- [1998SOT] Soto, A.; Hernandez, P.; Ortega, J.; Experimental VLE at 101.32 kPa in binary systems composed of ethyl methanoate and alkan-1-ols or alkan-2-ols and treatment of data using a correlation with temperature-dependent parameters. Fluid Phase Equilib., 1998, 146, 351
- [1998WIS] Wisniak, J.; Fishman, E.; Shaulitch, R., Isobaric Vapor-Liquid Equilibria in the Systems 2-Butanone + Heptane and 3-Butanone + Oxolane, J. Chem. Eng. Data, 1998, 43, 537
- [1999BEV] Bevia, F. R.; Torres, M. J. F.; Vapor-liquid equilibrium for the binary system 1-methylcyclohexene + heptane at 101.3 kPa. ELDATA: The International Electronic Journal of Physico-Chemical Data, 1999, 5, 143
- [1999DEA] Dean, J. A.; Lange's Handbook of Chemistry (McGraw-Hill, Inc., New York, 1999)
- [1999DYK] Dykyj, J.; Svoboda, J.; Wilhoit, R.C.; Frenkel, M.; Hall, K.R.; Vapor Pressure and Antoine Coefficients for Hydrocarbons and Sulfur-, Selenium-, Tellurium-, and Halogen-Containing Organic Compounds. Vol. IV, Springer, Berlin, 1999
- [1999IGO] Igoudjilene, O.; Ait-Kaci, A.; Jose, J.; Static measurements of the total vapor pressure of binary mixtures of butan-2-one with 1,1,1-trichloroethane or 1,1-dichloroethane in the temperature range from 263 K to 343 K. ELDATA: The International Electronic Journal of Physico-Chemical Data, 1999, 5, 165
- [1999LEW] Lewis, R.J. Sax's Dangerous Properties of Industrial Materials. 10th ed. Volume 2, New York, John Wiley & Sons, 1999
- [1999LOR] Loras, S.; Aucejo, A.; Munoz, R.; Ordonez, L.M., Phase Equilibria in the Systems 2-Methyl-2-propanol + Methyl 1,1-Dimethylpropyl Ether and 2-Methylpentane + 2-Methyl-2-propanol + Methyl 1,1-Dimethylpropyl Ether, J. Chem. Eng. Data, 1999, 44, 1163

- [1999MON] Monton, J.B.; Torre, J.d.I.; Burguet, M.C.; Munoz, R.; Loras, S., Isobaric Vapor-Liquid Equilibrium in the Systems 2,3-Dimethylpentane + Methyl 1,1-dimethylethyl Ether, + Diisopropyl Ether and + Methyl 1,1-Dimethylpropyl Ether, J. Chem. Eng. Data, 1999, 44, 1158
- [1999TAK] Takagi, T.; T. Sakura, T. Tsuji, M. Hongo, Bubble point pressure for binary mixtures of difluoromethane with pentafluoroethane and 1,1,1,2-tetrafluoroethane. Fluid Phase Equilib. 162 (1999) 171–179
- [1999TRE] Treszczanowicz, T.; Kasprzycka-Guttman, T.; Treszczanowicz, A.J., Distribution Coefficient of β -Carotene in Systems Composed of Hexane, Water, and 2-Propanone and Toluene, Water and 2-Propanone, J. Chem. Eng. Data, 1999, 44, 970
- [1999WIS] Wisniak, J.; Galindo, G.; Reich, R.; Segura, H.; Isobaric vapor - liquid equilibrium in the systems ethyl 1,1-dimethylethyl ether + 2,2,4- trimethyl pentane and + octane. Phys. Chem. Liq., 1999, 37, 649
- [2000BAS] Basharin A Yu, Fortov VE. Parameters of SLV triple point of graphite. Abstract, 14th Symposium on Thermophysical Properties. (June 25–30 2000, Boulder, USA). Compiled and edited by Haynes WM and Stevenson BA. National Institute of Standards and Technology, 2000, p. 159
- [2000HIA] Hiaki, T.; Tatsuhana, K.; Isobaric vapor-liquid equilibria for binary and ternary systems composed of 2-methoxy-2-mehtylpropane, ethanol, 2-methyl-2-propanol, and octane at 101.3 kPa. J. Chem. Eng. Data, 2000, 45, 564
- [2000KUI] Kul, I.; DesMarteau, D. D.; Beyerlein, A. L.; Vapor-liquid equilibria of novel chemicals and their mixtures as R-22 alternatives Fluid Phase Equilib., 2000, 173, 263
- [2000LEM] Lemmon, E.W. and Jacobsen, R.T, An International Standard Formulation for the Thermodynamic Properties of 1,1,1-Trifluoroethane (HFC-143a) for Temperatures from 161 to 450 K and Pressures to 50 MPa, J. Phys. Chem. Ref. Data, 2000, 29, 521
- [2000MAG] Magee, J.W., Outcalt, S.L., and Ely, J.F., "Molar Heat Capacity C(v), Vapor Pressure, and (p, rho, T) Measurements from 92 to 350 K at Pressures to 35 MPa and a New Equation of State for Chlorotrifluoromethane (R13)," Int. J. Thermophys., 2000, 21, 1097
- [2000ROD] Rodriguez, S.; Artigas, H.; Lafuente, C.; Mainar, A. M.; Royo, F. M.; Isobaric vapour-liquid equilibrium of binary mixtures of some cyclic ethers with chlorocyclohexane at 40.0 and 101.3 kPa. Thermochim. Acta, 2000, 362, 153
- [2000SMU] Smukala, J., Span, R., and Wagner, W., A New Equation of State for Ethylene Covering the Fluid Region for Temperatures from the Melting Line to 450 K at Pressures up to 300 MPa, J. Phys. Chem. Ref. Data, 2000, 29, 1053

- [2000SPA] Span, R., Lemmon, E.W., Jacobsen, R.T, Wagner, W., and Yokozeki, A., A Reference Equation of State for the Thermodynamic Properties of Nitrogen for Temperatures from 63.151 to 1000 K and Pressures to 2200 MPa, J. Phys. Chem. Ref. Data, 2000, 29, 1361
- [2001GLO] Gloria, H.; Sievert, D. J. Agric. Food Chem. 2001, 49, 2433–2436
- [2001HOR] Horvath, A. L.; Boiling points of halogenated organic compounds. Chemosphere, 2001, 44, 897
- [2001KER] Kerley GI, Chhabildas L. Multicomponent-Multiphase Equation of State for Carbon. Sandia report, SAND2001-2619, Sandia National Laboratories, 2001
- [2001KUL] Kul, I.; DesMarteau, D. D.; Beyerlein, A. L.; Vapor-liquid equilibria for CF₃OCF₂H/fluorinated ethane and CF₃SF₅/fluorinated ethane mixtures as potential R22 alternatives, Fluid Phase Equilib., 2001, 185, 241
- [2001ONE] O'Neil, M.J. (ed.). The Merck Index - An Encyclopedia of Chemicals, Drugs, and Biologicals. 13th Edition, Whitehouse Station, NJ: Merck and Co., Inc., 2001
- [2002DEL] del Mar Olaya, M. ; Gomis, V.; Molto, J. ; Ruiz, F.; Isobaric Vapor-Liquid Equilibria for the Binary System 2-Propanol + Methyl Ethyl Ketone and for the Ternary System 1-Butanol + 2-Propanol + Methyl Ethyl Ketone at 101.3 kPa. J. Chem. Eng. Data, 2002, 47, 65
- [2002GU] Gu, F.; Wang, L.; Wu, Z.; Vapor-Liquid and Liquid-Liquid Equilibrium for Octane + Maleic Anhydride System. J. Chem. Eng. Data, 2002, 47, 643
- [2002STE] Steele, W. V.; R. D. Chirico, S. E. Knipmeyer, and A. Nguyen, J. Chem. Eng. Data, 2002, 47, 689
- [2002VER] Verevkin, Sergey P., Relationships among strain energies of mono- and poly-cyclic cyclohexanoid molecules and strain of their component rings, J. Chem. Thermodyn., 2002, 34, 263
- [2002WAG] Wagner, W. and Pruss, A., The IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use, J. Phys. Chem. Ref. Data, 2002, 31, 387
- [2003AUC] Aucejo, A.; Gabaldon, C.; Loras, S.; Marzal, P.; Sancho-Tello, M.; Phase Equilibria in the Binary and Ternary Systems Composed of Diethyl ketone, 2-Pentanone and 3-Pentanol at 101.3 kPa. J. Chem. Eng. Data, 2003, 48, 1128
- [2003KOR] Korobenko VN, Savvatimskiy AI, Blackbody design for high temperature (1800 to 5500 K) of metals and carbon in liquid states under fast heating. In: Ripple DC, editor. Temperature: Its measurement and control in science and industry. Conference Proceedings, vol. 7. Melville, NY, American Institute of Physics, 2003. p. 783–8
- [2003ORT] Ortega, J.; Espiau, F.; Tojo, J.; Canosa, J.; Rodriguez, A.; Isobaric Vapor-Liquid Equilibria and Excess Properties for the Binary Systems of Methyl Esters + Heptane. J. Chem. Eng. Data, 2003, 48, 1183

- [2003PEN] Penoncello, S.G., Lemmon, E.W., Jacobsen, R.T, and Shan, Z., "A Fundamental Equation for Trifluoromethane (R-23)," J. Phys. Chem. Ref. Data, 2003, 32, 1473
- [2004BAS] Basharin A Yu, Brykin MV, Marin M Yu, Pakhomov IS, Sitnikov SF. Methods of increasing the measurement accuracy during the experimental determination of the melting point of graphite. High Temp., 2004, 42, 64–71
- [2004HUR] Hurtt, M.; Pitkanen, I.; Knuutinen, J.; Melting behaviour of DD-sucrose, DD-glucose and DD-fructose, Carbohydrate Res., 2004, 339, 2267
- [2004LEM] Lemmon, E.W. and Huber, M.L., "Thermodynamic Properties of n-Dodecane," Energy Fuels, 2004, 18, 960
- [2004LIU] Liu, L.; Li, F.; Tan, L.; Ming, L.; Yi, Y. Effects of Nanometer Ni, Cu, Al and NiCu Powders on the Thermal Decomposition of Ammonium Perchlorate. *Propellants, Explos., Pyrotech.* **2004**, 29, 34– 38.
- [2004MAN] Manatt, S. L.; Manatt, M. R. R. On the Analyses of Mixture Vapor Pressure Data: The Hydrogen Peroxide/Water System and Its Excess Thermodynamic Functions, Chem. Eur. J., 2004, 6540
- [2004NAG] Nagashima, A.; Yoshii, S.; Matsuda, H.; Ochi, K.; Measurement and Correlation of Excess Molar Enthalpies for Ethylene Glycol + Alkanol Systems at the Temperatures (298.15, 308.15, and 323.15) K, J. Chem. Eng. Data, 2004, 49, 286
- [2004ORT] Ortega, J.; Espiau, F.; Postigo, M.; Excess Properties and Isobaric Vapor-Liquid Equilibria for Binary Mixtures of Methyl Esters + tert-Butanol. J. Chem. Eng. Data, 2004, 49, 1602
- [2004SAK] Sakoda, N.; Uematsu, M., A Thermodynamic Property Model for Fluid Phase Hydrogen Sulfide, Int. J. Thermophys., 2004, 25, 709
- [2005CHE] Chen, Q.; Hong, R.-h.; Chen, G.-m.; Vapor Pressure Measurements of Ethyl Fluoride J. Chem. Eng. Data, 2005, 50, 1586
- [2005GIN] Giner, B.; Lopez, M. C.; Cea, P.; Lafuente, C.; Royo, F. M.; Study of tetrahydropyran-chlorobutane VLE using the gamma-phi and phi-phi approaches. Fluid Phase Equilib., 2005, 232, 50
- [2005KUM] Kumari, S. R.; Venkateswarlu, P.; Prabhakar, G.; Excess volumes and speeds of sound of mixtures of 1,2-dibromoethane with chlorinated ethanes and ethenes at 303.15 K. J. Chem. Thermodyn., 2005, 37, 737
- [2005LEM] Lemmon, E.W. and Jacobsen, R.T, "A New Functional Form and New Fitting Techniques for Equations of State with Application to Pentafluoroethane (HFC-125)," J. Phys. Chem. Ref. Data, 2005, 34, 69
- [2005LID] Lide, D.R.; CRC Handbook of Chemistry and Physics 86th Edition 2005-2006. CRC Press, Taylor & Francis, Boca Raton, FL, 2005
- [2005SAV] Savvatimskiy, A. I.; Measurements of the melting point of graphite and the properties of liquid carbon (a review for 1963–2003). Carbon, 2005, 43, 1115

- [2006BUC] Bückner, D. and Wagner, W., A Reference Equation of State for the Thermodynamic Properties of Ethane for Temperatures from the Melting Line to 675 K and Pressures up to 900 MPa, J. Phys. Chem. Ref. Data, 2006, 35, 205
- [2006BUC2] Bückner, D. and Wagner, W., Reference Equations of State for the Thermodynamic Properties of Fluid Phase n-Butane and Isobutane, J. Phys. Chem. Ref. Data, 2006, 35, 929
- [2006FEI] Feistel, R.; Wagner, W.; A New Equation of State for H₂O Ice Ih, J. Phys. Chem. Ref. Data, 2006, 35, 1021
- [2006HAN] Han, X. H.; Chen, G. M.; Li, C. S.; Qiao, X. G.; Cui, X. L.; Wang, Q.; Isothermal Vapor-Liquid Equilibrium of (Pentafluoroethane + Fluoroethane) at Temperatures between (265.15 and 303.15) K Obtained with a Recirculating Still J. Chem. Eng. Data, 2006, 51, 1232
- [2006LEM] Lemmon, E.W. and Span, R., Short Fundamental Equations of State for 20 Industrial Fluids, J. Chem. Eng. Data, 2006, 51, 785
- [2006RED] Reddy, KVNS; Reddy, GS, Krishnaiah, A.; Excess volumes, densities, speeds of sound and viscosities for the binary systems of diisopropyl ether with hydrocarbons at 303.15 K. Thermochim. Acta, 2006, 440, 43
- [2006SUR] Suresh Reddy, K. V. N.; Sankara Reddy, G.; Krishnaiah, A. Excess volumes, densities, speeds of sound and viscosities for the binary systems of diisopropyl ether with hydrocarbons at 303.15K Thermochim. Acta, 2006, 440, 43-50
- [2007BOU] Boukais-Belaribi, G.; Belaribi, B. F.; Lohmann, J.; Jose, J.; Isothermal vapour liquid equilibria and excess molar enthalpies of hex-2-yne + methyl butyl ether and hex-3-yne + dibutyl ether mixtures. Fluid Phase Equilib., 2007, 262, 180
- [2007INC] Ince, E.; Kirbaslar, S. I.; Sahin, S.; Liquid-Liquid Equilibria for Ternary Systems of Water + Formic Acid + Dibasic Esters. J. Chem. Eng. Data, 2007, 52, 1889
- [2007LID] Lide, D.R. CRC Handbook of Chemistry and Physics 88th Edition 2007-2008. CRC Press, Taylor & Francis, Boca Raton, FL 2007
- [2007RAD] Radhamma, M.; Liao, W.-C.; Venkatesu, P.; Rao, M. V. P.; Lee, M. J.; Lin, H. M.; Excess molar enthalpies of dimethylsulfoxide with chloroethanes and chloroethenes at 298.15 K. Thermochim. Acta, 2007, 465, 1
- [2007RED] Reddy, V. N. S.; Rao, P. S.; Krishnaiah, A.; Experimental and theoretical values of sound speeds and viscosities for the binary systems of MTBE with hydrocarbons. J. Mol. Liq., 2007, 135, 14
- [2007ZHU] Zhu, H.-B.; Gong, M.-Q.; Zhang, Y.; Wu, J.-F.; Vapor-Liquid Equilibrium Data of the Methane + Tetrafluoromethane System at Temperatures from (159.61 to 178.93) K J. Chem. Eng. Data, 2007, 52, 463
- [2008AMI] Amireche-Ziar, F.; G. Boukais-Belaribi, A. Jakob, I. Mokbel, and F.B. Belaribi, Fluid Phase Equilib. 2008, 268, 39
- [2009GIE] Gierycz, P.; Kosowski, A.; Swietlik, R.; Vapor-Liquid Equilibria in Binary Systems Formed by Cyclohexane with Alcohols. J. Chem. Eng. Data, 2009, 54, 2996

- [2009LEA] Leachman, J.W., Jacobsen, R.T, Penoncello, S.G., and Lemmon, E.W., Fundamental Equations of State for Parahydrogen, Normal Hydrogen, and Orthohydrogen, J. Phys. Chem. Ref. Data, 2009, 38, 721
- [2009LEM] Lemmon, E.W., McLinden, M.O., and Wagner, W., Thermodynamic Properties of Propane. III. A Reference Equation of State for Temperatures from the Melting Line to 650 K and Pressures up to 1000 MPa, J. Chem. Eng. Data, 2009, 54, 3141
- [2010CEN] Censky, M.; Vrbka, P; Ruzicka, K.; Fulem, M.; Vapor pressure of selected aliphatic alcohols by ebulliometry. Part 2. Fluid Phase Equil., 2010, 298, 199
- [2010CEP] Cepeda, E., A.; Isobaric Vapor-Liquid Equilibrium for Binary Mixtures of 3-Methyl-1-butanol + 3-Methyl-1-butyl Ethanoate and 1-Pentanol + Pentyl Ethanoate at 101.3 kPa, J. Chem. Eng. Data, 2010, 55, 2349–2354
- [2010GAO] Gao, D.; Zhu, D.; Zhang, L.; Guan, H.; Sun, H.; Cheng, H.; Si, J.; Isobaric Vapor-Liquid Equilibria for Binary and Ternary Mixtures of Propanal, Propanol, and Propanoic Acid. J. Chem. Eng. Data, 2010, 55, 5887
- [2010HAY] Haynes, W.M., editor, CRC Handbook of Chemistry and Physics. 91st ed. Boca Raton, FL: CRC Press Inc., 2010
- [2010MAR] Marrufo, B.; Loras, S.; Sanchotello, M.; Isobaric Vapor-Liquid Equilibria for Binary and Ternary Mixtures with Cyclohexane, Cyclohexene, and 2-Methoxyethanol at 100 kPa. J. Chem. Eng. Data, 2010, 55, 62
- [2010SYN] Synquest Laboratories, Refrigerant Reference Table (July 3, 2010), http://www.synquestlabs.com/pdf/library_refrigerant_ref_table.pdf (accessed July 22, 2021)
- [2010YU] Yu, G.-B.; Guo, K.; Li, T.-X.; High-precision measurement and correlation of saturated vapor pressure for tetrafluoromethane R14 Huaxue Gongcheng, 2010, 38, 51
- [2010ZEN] Zeng, Z.-X.; Li, X.-N.; Xue, W.-L.; Zhang, C.-S.; Bian, S.-C.; Heat Capacity, Enthalpy of Formation, and Entropy of Methyl Carbamate. Ind. Eng. Chem. Res., 2010, 49, 5543
- [2011BAS] Baslioglu, B.; Cehreli, S.; Quaternary phase equilibrium of water-carboxylic acid mixture (formic-propionic acid or acetic-propionic acid)-solvent liquid systems at 298.15 K. Fluid Phase Equilib., 2011, 312, 85
- [2011DON] Dong, X.; Gong, M.; Liu, J.; Wu, J. Experimental measurement of vapor pressures and (vapor + liquid) equilibrium for {1,1,1,2-tetrafluoroethane (R134a) + propane (R290)} by a recirculation apparatus with view windows J. Chem. Thermodyn., 2011, 43, 505
- [2011GAO] Gao, D. M.; Zhang, L.; Guan, H.; Sun, H.; Chen, H.; Zhu, D.; Thermodynamic Properties for Vapor-Liquid Equilibrium of Diethyl Ether- Methanol-n-Butanol Multicomponent System. Gaoxiao Huaxue Gongcheng Xuebao, 2011, 25, 1
- [2011LLA] Lladosa, E.; Monton, J. B. ; Torre, J. D. L.; Martinez, N. F.; Liquid-Liquid and Vapor-Liquid-Liquid Equilibrium of the 2-Butanone + 2-Butanol + Water System. J. Chem. Eng. Data, 2011, 56, 1755
- [2011MAT] Mathuni, T.; J.-I. Kim, and S.-J. Park, J. Chem. Eng. Data, 2011, 56, 89

- [2011RAI] Rai, R. N.; Mudunuri, S. R.; Reddi, R. S. B.; Kumar Satuluri, V. S. A.; Ganeshmoorthy, S.; Gupta, P. K.; . J. Cryst. Growth, 2011, 321, 72
- [2011WU] Wu, J., Zhou, Y., and Lemmon, E.W., An Equation of State for the Thermodynamic Properties of Dimethyl Ether, J. Phys. Chem. Ref. Data, 2011, 40, 023104
- [2012AGA] Agafonova, L. E.; Varushchenko, R. M.; Druzhinina, A. I.; Polyakova, O. V.; Phase equilibria and the thermodynamic properties of methyl and ethyl esters of carboxylic acids. 1. Methyl n-butanoate and ethyl propanoate. J. Chem. Thermodyn., 2012, 47, 120
- [2012BEJ] Bejarano, A.; Poveda, L. J.; de la Fuente, J. C.; Supplementary vapor pressure data of the glycol ethers, 1-methoxy-2-propanol, and 2-methoxyethanol at a pressure range of (15 to 177) kPa. J. Chem. Thermodyn., 2012, 53, 114
- [2012GUN] Gundogdu, T.; Cehreli, S.; Ternary liquid-liquid phase equilibria of (water-carboxylic acid-1-undecanol) systems at 298.15 K. Fluid Phase Equilib., 2012, 331, 26
- [2012KAM] Kamihama, N.; Matsuda, H.; Kurihara, K.; Tochigi, K.; Oba, S. Isobaric Vapor-Liquid Equilibria for Ethanol + Water + Ethylene Glycol and Its Constituent Three Binary Systems J. Chem. Eng. Data, 2012, 57, 339
- [2012KRO] Kroenlein, K., Muzny, C., Kazakov, A., Diky, V., Chirico, R., Magee, J., Abdulagatov, I. and Frenkel, M. (2012), NIST/TRC Web Thermo Tables (WTT) NIST Standard Reference Subscription Database 3 - Professional Edition Version 2, NIST/TRC Web Thermo Tables (WTT) - Professional Edition, [online], <http://wtt-pro.nist.gov/> (Accessed August 10, 2021)
- [2012MAR] Marrufo, B.; Rigby, B.; Pla-Franco, J.; Loras, S.; Solvent Effects on Vapor Liquid Equilibria of the Binary System 1-Hexene + n-Hexane. J. Chem. Eng. Data, 2012, 57, 3721
- [2012MEJ] Mejia, A.; Segura, H.; Cartes, M.; Perez-Correa, J. R.; Experimental determination and theoretical modeling of the vapor-liquid equilibrium and surface tensions of hexane + tetrahydro-2H-pyran. Fluid Phase Equilib., 2012, 316, 55
- [2012RED] Reddy, K. R.; Kumar, D. B. K.; Rao, G. S.; Sri, P. B. S.; Rambabu, C.; Vapor-liquid equilibria and excess molar volumes of N-methyl-2-pyrrolidone with 2-alkoxyethanols. Fluid Phase Equilib., 2012, 336, 52
- [2012RIO] Rios, R.; Ortega, J.; Fernandez, L.; Measurements and Correlations of the Isobaric Vapor Liquid Equilibria of Binary Mixtures and Excess Properties for Mixtures Containing an Alkyl (Methyl, Ethyl) Butanoate with an Alkane (Heptane, Nonane) at 101.3 kPa. J. Chem. Eng. Data, 2012, 57, 3210
- [2012THO] Thol, M.; Lemmon, E. W.; Span, R.; Equation of state for benzene for temperatures from the melting line up to 725 K with pressures up to 500 MPa, High Temp. High Press., 2012, 41, 81

- [2012ZHO] Zhou, Y., Lemmon, E.W., and Wu, J., Thermodynamic Properties of o-Xylene, m-Xylene, p-Xylene, and Ethylbenzene, *J. Phys. Chem. Ref. Data*, 2012, 41, 023103
- [2013FER] Fernandez, L.; Ortega, J.; Perez, E.; Toledo, F.; Canosa, J.; Multiproperty Correlation of Experimental Data of the Binaries Propyl Ethanoate + Alkanes (Pentane to Decane). New Experimental Information for Vapor Liquid Equilibrium and Mixing Properties. *J. Chem. Eng. Data*, 2013, 58, 686
- [2013HAY] Haynes, W.M. (ed.). *CRC Handbook of Chemistry and Physics*. 94th Edition. CRC Press LLC, Boca Raton: FL 2013
- [2013LOM] Lomba, L.; Giner, B.; Lafuente, C.; Martin, S.; Artigas, H.; Thermophysical Properties of Three Compounds from the Acrylate Family. *J. Chem. Eng. Data*, 2013, 58, 1193
- [2013NIK] Nikitin, E. D.; Popov, A. P.; Bogatishcheva, N. S.; Critical properties of some alkyl naphthalenes. *Fluid Phase Equilib.*, 2013, 358, 296
- [2013ONE] O'Neil, M.J. (ed.). *The Merck Index - An Encyclopedia of Chemicals, Drugs, and Biologicals*. Cambridge, UK: Royal Society of Chemistry, 2013
- [2013PLA] Pla-Franco, J.; Lladosa, E.; Loras, S.; Monton, J. B.; Phase equilibria for the ternary systems ethanol, water + ethylene glycol or + glycerol at 101.3 kPa, *Fluid Phase Equilib.*, 2013, 341, 54
- [2013QIA] Qian, G.-F.; Liu, W.; Wang, L.-T.; Wang, D.-C.; Song, H.; (Vapour + liquid) equilibria in the ternary system (acetonitrile + n-propanol + ethylene glycol) and corresponding binary systems at 101.3 kPa, *J. Chem. Thermodyn.*, 2013, 67, 241
- [2013ZHA] Zhang, L.; Wu, Y. Sun, L. Li, B. Jiang, X. Li, N. Yang, and H. Ding, *J. Chem. Eng. Data*, 2013, 58, 1308
- [2014HAY] Haynes, W.M. (ed.). *CRC Handbook of Chemistry and Physics*. 95th Edition. CRC Press LLC, Boca Raton: FL 2014
- [2014NIC] Nicolae, M.; F. Oprea, *Fluid Phase Equilib.*, 2014, 370, 34
- [2014RIO] Rios, R.; Ortega, J.; Fernandez, L.; de Nuez, I.; Wisniak, J.; Improvements in the Experimentation and the Representation of Thermodynamic Properties (iso-p VLE and yE) of Alkyl Propanoate + Alkane Binaries. *J. Chem. Eng. Data*, 2014, 59, 125
- [2014SCH] Schroeder, J.A., Penoncello, S.G., and Schroeder, J.S., A New Fundamental Equation for Ethanol, *J. Phys. Chem. Ref. Data*, 2014, 43, 043102
- [2014SII] Siitsman, C.; Kamenev, I.; Oja, V.; Vapor pressure data of nicotine, anabasine and cotinine using differential scanning calorimetry. *Thermochim. Acta*, 2014, 595, 35
- [2014THO] Thol, M.; Piazza, L.; Span, R.; A New Functional Form for Equations of State for Some Polar and Weakly Associating Fluids, *Int. J. Thermophys.*, 2014, 35, 783
- [2014ZHO] Zhou, Y., Liu, J., Penoncello, S.G., and Lemmon, E.W., An Equation of State for the Thermodynamic Properties of Cyclohexane, *J. Phys. Chem. Ref. Data*, 2014, 43, 043105

- [2015GED] Gedanitz, H., Dávila, M.J., and Lemmon, E.W., Speed of Sound Measurements and a Fundamental Equation of State for Cyclopentane, *J. Chem. Eng. Data*, 2015, 60, 1311
- [2015SUS] Susial, P.; Rodriguez-Henriquez, J. J.; Castillo, V. D.; Estupinan, E. J.; Apolinario, J. C.; Susial, R.; Isobaric (vapor+liquid) equilibrium for n-propyl acetate with 1-butanol or 2-butanol. Binary mixtures at 0.15 and 0.6 MPa. *Fluid Phase Equilib.*, 2015, 385, 196
- [2015THO] Thol, M., Rutkai, G., Koester, A., Kortmann, M., Span, R., and Vrabec, J.; "Fundamental Equation of State for Ethylene Oxide Based On a Hybrid Dataset," *Chem. Eng. Sci.*, 2015, 121, 97
- [2016ANI] Aniya, V.; De, D.; Singh, A.; Bankupalli, S.; "Isobaric Phase Equilibrium of tert-Butyl Alcohol + Glycerol at Local and Subatmospheric Pressures, Volumetric Properties, and Molar Refractivity from 303.15 to 333.15 K of tert-Butyl Alcohol + Glycerol, tert-Butyl Alcohol + Water, and Water + Glycerol Binary Systems," *J. Chem. Eng. Data*, 2016, 61, 1850-1863
- [2016CHI] Chirico, R. D.; Steele, W. V.; Kazakov, A. F.; Thermodynamic properties of indan: Experimental and computational results. *J. Chem. Thermodyn.*, 2016, 96, 41
- [2016GAO] Gao, K., Wu, J., Zhang, P., and Lemmon, E.W., A Helmholtz Energy Equation of State for Sulfur Dioxide, *J. Chem. Eng. Data*, 2016, 61, 2859
- [2016HUR] Hurley, M.J., Gottuk, D.T., Hall Jr., J.R., Harada, K., Kuligowski, E.D., Puchovsky, M., Torero, J.L., Watts Jr., J.M., Wieczorek, C.J., editors, *SFPE Handbook of Fire Protection Engineering: 5th Edition*, Springer-Verlag New York, 2016
- [2016MAT] Matsuda, H.; Inaba, K.; Sumida, H.; Kurihara, K.; Tochigi, K.; Ochi, K.; Vapor-liquid equilibria of binary and ternary mixtures containing ethyl lactate and effect of ethyl lactate as entrainer. *Fluid Phase Equilib.*, 2016, 420, 50
- [2016QI] Qi, H., Fang, D., Gao, K., Meng, X., and Wu, J., "Compressed Liquid Densities and Helmholtz Energy Equation of State for Fluoroethane (R161)," *Int. J. Thermophys.*, 2016, 37, 55
- [2016THO] Thol, M., Rutkai, G., Köster, A., Dubberke, F.H., Windmann, T., Span, R., and Vrabec, J.; "Thermodynamic Properties for Octamethylcyclotetrasiloxane," *J. Chem. Eng. Data*, 2016, 61, 2580
- [2018GAO] K. Gao, J. Wu, I. H. Bell, A. H. Harvey, and E. W. Lemmon, "A reference equation of state with an associating term for the thermodynamic properties of ammonia," *J. Phys. Chem. Ref. Data* 52, 013102 (2023)
- [2018HE] He, N.; Jiang, T.; Experimental Research on the Vapor Pressure and Critical Parameters of Ethyl Fluoride (HFC-161) *J. Coastal Res.*, 2018, 83, 129
- [2018LEM] Lemmon, E. W.; Bell, I. H.; Huber, M. L.; McLinden, M. O.; Reference Fluid Thermodynamic and Transport Properties (REFPROP), NIST Standard Reference Database 23, Vers. 10.0, 2018

- [2018SHA] Shang, Wenyan; Cui, Xianbao; Yu, Xufeng; Zhang, Ying; Feng, Tianyang; He, Jie; Zhang, Xuemei; Wang, Jixiao; Isobaric vapor-liquid equilibrium for methanol + methyl acetate with ionic liquids [OMMIM][Tf2N] and [OMIM][Tf2N] as entrainers at 101.3 kPa. *Fluid Phase Equilib.*, 2018, 473, 90
- [2018THO] Thol, M., Dubberke, F.H., Baumhögger, E., Span, R., and Vrabec, J., Speed of Sound Measurements and a Fundamental Equation of State for Hydrogen Chloride, *J. Chem. Eng. Data* 63, 2533 (2018)
- [2019DIK] Diky, V.; Chirico, R. D.; Frenkel, M.; Bazyleva, A.; Magee, J. W.; Paulechka, E.; Kazakov, A.; Lemmon, E. W.; Muzny, C. D.; Smolyanitsky, A. Y.; Townsend, S.; Kroenlein, K.; ThermoData Engine (TDE), Vers. 10.4.2, NIST Standard Reference Database 103b, (2019)
- [2019JIA] Jiang, Shiliang; Gao, Jun; Li, Rui; Xu, Dongmei; Zhang, Lianzheng; Wang, Yinglong; Isobaric Vapor-Liquid Equilibrium Measurements for Separation of Azeotrope (Methanol + Methyl Acetate). *J. Chem. Eng. Data*, 2019, 64, 296
- [2020FIS] <https://www.fishersci.com/> (accessed February 11, 2020)
- [2020POD] Podgorbunskikh, E.M.; Bychkov, A.L.; Ryabchikova, E.I.; Lomovsky, O.I. The Effect of Thermomechanical Pretreatment on the Structure and Properties of Lignin-Rich Plant Biomass. *Molecules*, 2020, 25, 995
- [2021CHE] ChemSpider <http://www.chemspider.com/> (accessed July 22, 2021)
- [2021EIS] Eisenbach, T.; Scholz, C.; Span, R.; Cristancho, D.; Lemmon, E. W.; Thoi, M.; Speed-of-Sound Measurements and a Fundamental Equation of State for Propylene Glycol, *J. Phys. Chem. Ref. Data*, 2021, 50, 023105
- [2021EPA] EPA CompTox Chemicals Database, <https://comptox.epa.gov/dashboard/> (accessed July 22, 2021)
- [2021FIS] Fisher-Scientific online, <https://www.fishersci.com/> (Accessed August 10, 2021)
- [2021LIN] Linstrom P. J.; W.G. Mallard, Eds., NIST Chemistry WebBook, NIST Standard Reference Database Number 69, National Institute of Standards and Technology, Gaithersburg MD, 20899, <https://doi.org/10.18434/T4D303>, (accessed July 22, 2021)
- [2021TSU] Tsuji, T.; Sato, T.; Hoshina, T.; Oba, S.; Boiling Point of Five New Sulfur-free Odorants for LPG, 1-Pentyne, Cyclopentene, 1-Hexyne, 2-Hexyne and 1,5-Cyclooctadiene, and Bubble Point Pressures of Binaries with Propane. *J. Jpn. Petro. Inst.*, 2021, 64, 92
- [2022BEC] Beckmuller, R.; Span, R.; Lemmon, Thol, M.; A Fundamental Equation of State for the Calculation of Thermodynamic Properties of n-Octane, *J. Phys. Chem. Ref. Data*, in press (2022).
- [2022BLA] Blackham, T.M., Lemmon, A.K., and Lemmon, E.W., to be submitted to *Int. J. Thermophys.*, 2022
- [2022GAO] Gao, K., Wu, J., Bell, I.H., Harvey, A.H., and Lemmon, E.W., A Reference Equation of State with an Associating Term for Thermodynamic Properties of Ammonia, to be submitted to *J. Phys. Chem. Ref. Data*, 2022

- [2022TEN] Tenji, D., Thol, M., Lemmon, E.W. and Span, R., Fundamental Equation of State for n-Heptane, to be submitted to Int. J. Thermophys., 2022
- [2022THO] Thol, M., Uhde, T., Lemmon, E.W., and Span, R., Fundamental Equations of State for Hydrocarbons. Part I. n-Pentane, to be submitted, 2022
- [2022THO2] Thol, M., Wang, Y., Lemmon, E.W., and Span, R., Fundamental Equations of State for Hydrocarbons. Part II. n-Hexane, to be submitted, 2022
- [2023LEM] Lemmon, E.W., McLinden, M.O., Overhoff, U., and Wagner, W.; A Reference Equation of State for Propylene for Temperatures from the Melting Line to 575 K and Pressures up to 1000 MPa, to be submitted to J. Phys. Chem. Ref. Data, 2023

3. Latent Enthalpies of Vaporization $\Delta_v H(T_b)$ and Enthalpies of Fusion $\Delta_{fus} H(T_m)$

3.1. Overview

In this work we compiled, evaluated, and recommended enthalpies of vaporization $\Delta_v H$ at normal boiling points T_b ($P = 1$ atm, 101.325 kPa) and enthalpies of fusion $\Delta_{fus} H$ for hydrocarbons, substituted hydrocarbons, and a few other compounds of chemical and engineering interest. We provide about 850 values for enthalpies of vaporization and 110 enthalpies of fusion for about 330 compounds (see Table 3-3) from over 350 source references, with selected recommended values provided in Table 3-3. The substituted hydrocarbons include different classes of oxygen-, nitrogen-, sulfur-, and halogen-substituted hydrocarbons. The enthalpies of vaporization provided here were derived from direct calorimetric determinations, change in vapor pressures of pure components and in mixtures (liquid-vapor equilibrium), and Equation-of-State (EOS) formulations. The values were taken from the literature, from the NIST Chemistry WebBook,[2021LIN] the NIST TRC Thermodata Engine (TDE)[2019DIK], and from NIST REFPROP.[2018LEM] (the latter are usually the most accurate, being based on EOS formulations). The recommended values of latent enthalpies of vaporization and fusion provided here were selected from reliable original sources (they are not weighted averages). Although this is an extensive compilation, it is not comprehensive, but we believe that there are sufficient data to make recommendations.

We provide uncertainties from the original sources where available. About 40 % of the latent enthalpies of vaporization have uncertainties of less than 1 %, about 30 % have uncertainties of (1 to 2) %, about 20 % have uncertainties of (2 to 5) %, and about 10 % have uncertainties of greater than 5 %. For compounds with large uncertainties in the enthalpies of vaporization (> 5 %), we provide several values from the literature to provide a range of values. The uncertainties “unc” in Table 3-2 are as reported. In most cases, it is not stated whether it is a standard uncertainty u (1σ) or an expanded uncertainty U (2σ). For those derived from equation-of-state (EOS) formulations (such as those in NIST REFPROP[2018LEM] or those provided in NIST TDE[2019DIK], they are expanded uncertainties U (2σ). All our recommended uncertainties are expanded uncertainties U (2σ).

In addition, we derive latent enthalpies of vaporization for many compounds where vapor pressure curves were readily available from Antoine equation parameters (A , B , C).[1888ANT] [1973WAG]. Here we compute enthalpies of vaporization at the normal boiling point ($\Delta_v H$) assuming the ideal gas law ($PV=nRT$) using a Clausius-Clapeyron relationship.

$$(3.1) \quad \ln(P_1/P_0) = A - (B / (T + C))$$

$$(3.2) \quad \ln(P_1/P_0) = -\Delta_v H/R * (1/T_1 - 1/T_0)$$

where $P_0 = 1$ atm (101.325 kPa) and T_0 and T_1 are two different temperatures near the normal boiling point.

More accurate latent enthalpies of vaporization can be derived considering the compressibility of a real gas – how much the gas deviates from an ideal gas where a real gas includes a compressibility factor Z . Including a correction for compressibility reduces the derived enthalpies of vaporization by about 4 %. Our analysis of the data indicates that this is more than twice than the median (1.1 %) and average (1.7 %) uncertainties, but on the order of the expanded uncertainties for the data set – which is about (3 to 5) %.

$$(3.3) \quad PV = Z * nRT$$

$$(3.4) \quad \Delta_v H = -Z * RT * \delta \ln(P) / \delta(1/T)$$

The compressibility factor needs to be computed for each compound. This can be done directly using experimental densities, using the principle of corresponding states employing critical temperatures (T_c) and critical pressures (P_c), or other methods. Here, instead, we determined an average compressibility factor for the compounds. We utilized NIST REFPROP [2018LEM] to compute the vapor-phase compressibility factors of about 40 compounds at their normal boiling points: (11) C_1 and C_2 hydrocarbons, (5) aromatic hydrocarbons, (7) C_1 to C_4 oxidized hydrocarbons, and (18) fluorinated and chlorinated C_1 and C_2 hydrocarbons. We found an average compressibility factor of 0.961 ± 0.0055 (1σ), or an uncertainty on this estimated compressibility of only about 0.6 %; this is less than the uncertainties in experimental determinations of latent enthalpies of vaporization – which are on the order of (1 to 2) % for standard uncertainty (1σ) and (3 to 5) % for expanded uncertainty (2σ). In the tables below, we utilize this average compressibility factor to derive corrected enthalpies of vaporization $\Delta_v H$ from vapor pressure curves $\ln(P)$ vs. $(1/T)$ assuming ideal gases. We provide latent enthalpies of vaporization both assuming ideal gas $\Delta_v H(IG)$, and corrected $\Delta_v H$ for the estimated compressibility factors.

Note: To estimate the latent enthalpy of vaporization of hydrogen peroxide (H_2O_2) from vapor pressure curves, we utilized the compressibility factor of 0.985 for water (H_2O) from NIST REFPROP[2018LEM] and estimated the uncertainty in the latent enthalpy of vaporization as 1 %.

We also provide enthalpies of fusion $\Delta_{fus}H$ and sublimation $\Delta_{sub}H$ for compounds with melting points above about 273 K (0 °C). These values are important for determining enthalpies of combustion which require enthalpies of melting or sublimation. Note the column heading in the table T_{trs} is the normal boiling point T_b ($P = 1$ atm), the melting point T_{fus} , or the sublimation temperature T_{sub} . Correspondingly, the column heading $\Delta_{trs}H$ is the enthalpy of vaporization $\Delta_v H$, the enthalpy of fusion $\Delta_{fus}H$, or the enthalpy of sublimation $\Delta_{sub}H$.

In Table 3-1, we provide definitions of the Table columns in Tables 3-3 and 3-4. In Table 3-2, we provide the methods, as best we could, that were used to determine the enthalpies of vaporization (and fusion or sublimation) from the publications. In some cases, they were not recorded during the compilation. Selected values are indicated by “SEL” for the compound in the standard phase, while “SEL1” and “SEL2” are values in other phases. The notation “u” in the *Selected* column indicates a very uncertain value, usually because it was estimated. Table 3-3 provides all values from the literature that we compiled, while Table 3-4 just provides the recommended values. See Table 3-2 for notation used for methods.

Table 3-1. Notation used for Tables 3-3 and 3-4 transition temperatures and enthalpies of transition

Term	Definition
Phase	phase at standard state 298.15 K and 1 bar
trs	transition
BP	boiling point
MP	melting point
SP	sublimation point
TP	triple point
SEL	selected value in the phase at standard state
SEL1	selected value in another phase 1 (not standard state)
SEL2	selected value in another phase 2 (not standard state)
$\Delta_v H(\text{IG})$	enthalpy of vaporization (ideal gas), derived from vaporization curves
$\Delta_v H$	enthalpy of vaporization (real gas), estimated using compressibility factor $Z = 0.961$, see text
$\Delta_{\text{trs}} H$	transition enthalpy
unc	uncertainty as reported
U	expanded uncertainty (2σ , 95 % confidence level)
Method	method used to determine $\Delta_{\text{trs}} H$, see Table 3-1(2)
Reference	reference provided at the end of this section

Table 3-2. Notation for methods used for transition temperatures and enthalpies of transition

Method	Description
BP	boiling point method (unspec.)
cal	calorimetry
DfusH	enthalpy of fusion
DsubH	enthalpy of sublimation
DvH	enthalpy of vaporization
EOS	Equation-of-State determination
est	estimated
eval	evaluated
ext	extrapolated
MP	melting point method (unspec.)
rev	review
SP	sublimation point method (unspec.)
TP	triple point method (unspec.)
VLE	Vapor-Liquid Equilibrium method (multicomponent)
VP	vapor pressure method (unspec.)

3.2. Transition Temperatures T_{trs} and Enthalpies of Transition $\Delta_{\text{trs}}H$ (compilation)

This Table 3-3 contains all compiled transition temperatures and enthalpies of transition. The recommended values are provided in Table 3-4. See Table 3-2 for notation used for methods.

Table 3-3. Transition temperatures T_{trs} and enthalpies of transition $\Delta_{\text{trs}}H$

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_{\text{v}}H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}}H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference		
SMALL MOLECULES												
Hydrogen/Oxygen												
H ₂	hydrogen	gas	BP	20.37	SEL	1.07	0.91	0.01	EOS	2009LEA		
			BP	20.37					VP eval	2019DIK		
			BP	20.37					0.91	0.01	EOS-rev	2018LEM
O ₂	oxygen	gas	BP	90.19	SEL		6.82	0.01	EOS	1991STE		
			BP	90.19					6.81	0.07	DvH eval	2019DIK
			BP	90.19					6.82	0.01	EOS-rev	2018LEM
H ₂ O	water	liq	BP	373.124	SEL		40.65	0.02	EOS	2002WAG		
		sol	MP	273.15	SEL1				6.007	0.004	EOS	2006FEI
		sol	SP	273.15	SEL2				51.08	0.02	EOS-rev	2006FEI
		liq	BP	373.124					40.65	0.02	EOS-rev	2018LEM
H ₂ O ₂	hydrogen peroxide	liq	BP	426.4	SEL	46.43	45.73	0.46	VLE	2004MAN		
			MP	272.689	SEL1		12.5	0.1	DfusH	1954GIG		
			MP	272.74	SEL2		52.8	5.3	DsubH est	2019DIK		
			BP	426.4		46.03	45.34		VP eval	2019DIK		
			BP	426.4		48.53	47.81		VP	1924MAA		
Nitrogen/Hydrogen												
	nitrogen	gas	BP	77.36	SEL		5.58	0.00	EOS	2000SPA		
			BP	77.36					5.58	0.07	DvH eval	2019DIK
			BP	77.36					5.58	0.00	EOS-rev	2018LEM
	ammonia	gas	BP	239.82	SEL		23.33	0.01	EOS-rev	2018LEM		
			BP	239.82					23.32	0.12	DvH eval	2019DIK
Nitrogen/Oxygen												
N ₂ O	nitrous oxide	gas	BP	184.68	SEL		16.47	0.03	EOS	2006LEM		
			BP	184.68					16.47	0.03	EOS-rev	2018LEM

Sulfur/Hydrogen										
H ₂ S	hydrogen sulfide	gas	BP	212.85	SEL	19.10	18.35	0.25	VP	1996LOP
			BP	212.85		18.68	18.0	0.6	VP	1951CLA
			BP	212.85			18.67	0.11	DvH	1936GIA
			BP	212.85			18.1	0.5	DvH eval	2019DIK
			BP	212.85		19.23	18.48		VP	1999DYK
Boron/Hydrogen										
B ₂ H ₆	diborane	gas	BP	180.59	SEL	14.59		0.06	VP	1961DIT
			BP	180.59		14.39		0.06	VP eval	2019DIK
			BP	180.59		14.20		0.06	VP	1959PAR
Hydrogen Halides										
ClH	hydrogen chloride	gas	BP	188.173	SEL		16.18	0.08		2018LEM

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
HYDROCARBONS										
Carbon										
C	graphite	sol	SB	2603	SEL		712.9	0.8	VP	1948BRE
Alkanes										
CH ₄	methane	gas	BP	111.67	SEL		8.20	0.06	EOS	1991SET
			BP	111.67			8.15	0.06	DvH eval	2019DIK
			BP	111.67			8.20	0.06	EOS-rev	2018LEM
C ₂ H ₆	ethane	gas	BP	184.57	SEL		14.72	0.10	EOS	2006BUC
			BP	184.57			14.67	0.10	DvH eval	2019DIK
			BP	184.57			14.72	0.10	EOS-rev	2018LEM
			BP	184.48		15.27	14.68		VP	1926LOO
C ₃ H ₈	propane	gas	BP	231.04	SEL		18.77	0.19	EOS	2009LEM
			BP	231.04			18.74	0.19	DvH eval	2019DIK
			BP	231.04			18.77	0.19	EOS-rev	2018LEM
			BP	231.04		19.61	18.84		VP	2004GLO
			BP	231.04		19.58	18.82		VP	1959ZIE
C ₄ H ₁₀	isobutane	gas	BP	261.4	SEL		21.22	0.20	EOS	2006BUC
			BP	261.4			21.17	0.20	DvH eval	2019DIK
			BP	261.4			21.22	0.20	EOS-rev	2018LEM
			BP	261.41		22.25	21.38		VP	1940AST
	n-butane	gas	BP	272.66	SEL		22.42	0.24	EOS	2006BUC2
			BP	272.66			22.39	0.24	DvH eval	2019DIK
			BP	272.66			22.42	0.24	EOS-rev	2018LEM
			BP	272.77		23.14	22.24		VP	1940AST2
C ₅ H ₁₂	isopentane	liq	BP	300.98	SEL		24.77	0.33	EOS	2006LEM
			BP	300.98			24.7	0.3	DvH eval	2019DIK
			BP	300.98			24.77	0.33	EOS-rev	2018LEM
			BP	301.0		25.95	24.94		VP	1947STU
			BP	301.0		25.98	24.97		VP	1945WIL
	neopentane	gas	BP	282.65	SEL		22.78	0.02		2006LEM
			BP	282.64			22.7	0.3	DvH eval	2019DIK
			BP	282.64		23.66	22.74	0.3	VP	1975HOP
			BP	282.65		23.95	23.02		VP	1974OSB
			BP	282.65			22.78	0.02		2018LEM
	n-pentane	liq	BP	309.21	SEL		25.81	0.11	EOS-rev	2018LEM

C ₆ H ₁₄	2,3-dimethylbutane	liq	BP	309.21			25.73	0.11	DvH eval	2019DIK
			BP	309.22		27.13	26.07		VP	1974OSB
			BP	331.177	SEL		27.55	0.06		2018LEM
			BP	331.09			27.02	0.27	DvH eval	2019DIK
			BP	331.09		28.74	27.62		VP	1945WIL
			BP	336.43	SEL	29.49	28.34	0.30	VLE	1999LOR
	3-methylpentane	liq	BP	336.43			28.04	0.21	DvH eval	2019DIK
			BP	336.43			28.10	0.10	DvH	1949WAD2
			BP	336.43		29.61	28.45		VP	1945WIL
			BP	336.43						
	isohexane	liq	BP	333.42	SEL		27.84	0.31	EOS	2006LEM
			BP	333.42			27.7	0.3	DvH eval	2019DIK
			BP	333.42		29.95	28.78		VLE	2010SAP
			BP	333.42			27.79		DvH rev	1985MAJ
			BP	333.42		29.37	28.22		VP	1945WIL
			BP	333.36			27.84	0.06		2018LEM
	neohexane	liq	BP	322.846	SEL		26.41	0.11		2018LEM
			BP	322.89			25.77	0.26	DvH eval	2019DIK
			BP	322.89		28.05	26.96		VLE	2013HYE
			BP	322.89			26.31		DvH rev	1985MAJ
			BP	322.89			26.30	0.30	DvH	1947WAD
			BP	322.89		27.71	26.62		VP	1945WIL
	n-hexane	liq	BP	341.87	SEL		28.88	0.13	EOS-rev	2018LEM
			BP	341.87			28.78	0.13	DvH eval	2019DIK
			BP	341.87		30.65	29.46		VLE	2011MEJ
			BP	341.87		31.53	30.30		VLE	2010FER
			BP	341.87		30.71	29.51		VLE	1984MIC
			BP	341.89		30.51	29.32		VP	1945WIL
C ₇ H ₁₆	2,2,3-trimethylbutane	liq	BP	354.03	SEL		28.9	0.9	DvH eval	2019DIK
			BP	354.03		30.58	29.39		VP	1949FOR
	2,3-dimethylpentane	liq	BP	362.93	SEL		30.4	0.7	DvH eval	2019DIK
			BP	362.93		32.16	30.91		VP	1949FOR
	3,3-dimethylpentane	liq	BP	359.21	SEL		29.5	0.4	DvH eval	2019DIK
			BP	359.21		31.31	30.09		VP	1949FOR
	3-methylhexane	liq	BP	365.0	SEL		30.8	0.8	DvH eval	2019DIK
			BP	365.0		32.70	31.42		VP	1949FOR
	isoheptane	liq	BP	363.2	SEL		30.7	0.7	DvH eval	2019DIK
			BP	363.2		32.58	31.31		VP	1949FOR

	n-heptane	liq	BP	370.4	SEL	33.71	32.01	0.35	DvH	1926MAT				
			BP	371.55			31.71	0.27	DvH eval	2019DIK				
			BP	371.58			32.40		VP	1945WIL				
C ₈ H ₁₈	2,2,4-trimethylpentane	liq	BP	372.36	SEL	32.19	31.00	0.40	DvH	1982SVO				
			BP	372.36			30.5	0.6	DvH eval	2019DIK				
			BP	372.36			30.79		DvH	1985MAJ				
			BP	372.36			31.00		DvH	1940PIT				
			BP	372.36			30.93		VLE	2019BEN				
			BP	372.39			31.34		VP	1945WIL				
			2,2-dimethylhexane	liq			BP	379.77	SEL	33.57	32.26	0.50	VP	1965MCM
							BP	379.77			31.8	1.1	DvH eval	2019DIK
BP	379.77	34.39			33.05	VP	1947STU							
	n-octane	liq	BP	398.79	SEL	36.73	34.34	0.15	DvH eval	2019DIK				
			BP	398.79			35.63		VLE	2013RED				
			BP	398.82			35.29		VP	1945WIL				
C ₉ H ₂₀	n-nonane	liq	BP	423.91	SEL	40.72	37.0	1.7	DvH eval	2019DIK				
			BP	423.91			39.13		VLE	2018FER				
			BP	423.95			38.04		VP	1949FOR				
C ₁₀ H ₂₂	n-decane	liq	BP	447.27	SEL	42.74	39.2	0.6	DvH eval	2019DIK				
			BP	447.27			41.07		VLE	2018FER				
			BP	447.27			40.67		VP	1945WIL				
C ₁₂ H ₂₆	n-dodecane	liq	BP	489.44	SEL	36.84	43.63	0.33		2004LEM				
			MP	263.49			36.84		MP	1954FIN				
			BP	489.50			43.70	0.80	DvH	2019DIK				
			BP	489.44			43.63	0.33		2018LEM				
			MP	263.49			35.70		MP	2004MON				
			MP	263.49			36.82		MP rev	1996DOM				
C ₁₆ H ₃₄	n-hexadecane	liq	BP	559.903	SEL	56.48	51.75	0.33		2018LEM				
			MP	291.27			53.36		MP	1954FIN				
			BP	559.94			50.7	2.3	DvH est	2019DIK				
			BP	559.94			54.28		VP	1954CAM				
			MP	291.27			53.00		MP	2004MON				
			MP	291.27			51.46		MP rev	1996DOM				
Alkenes														
C ₂ H ₄	ethene	gas	BP	169.38	SEL		13.53	0.04	EOS	2000SMU				
			BP	169.38			13.51	0.04	DvH eval	2019DIK				

			BP	169.38			13.53	0.04	EOS-rev	2018LEM
			BP	169.37		14.06	13.51		VP	1950MIC
C ₃ H ₆	propene	gas	BP	225.53	SEL		18.47	0.26	EOS-rev	2018LEM
			BP	225.53			18.38	0.26	DvH eval	2019DIK
			BP	225.38		19.25	18.50		VP	1939POW
C ₄ H ₈	1-butene	gas	BP	266.84	SEL		22.01	0.06		2005LEM2
			BP	266.83			21.9	0.4	DvH eval	2019DIK
			BP	267.43		22.40	21.53		VP	1928COF
			BP	266.84			22.01	0.06		2018LEM
	2-butene, (E)-	gas	BP	274.03	SEL		22.73	0.07		2005LEM2
			BP	274.01			22.73	0.20	DvH eval	2019DIK
			BP	274.01		23.86	22.93		VP	1945GUT
			BP	274.03			22.73	0.07		2018LEM
	2-butene, (Z)-	gas	BP	276.87	SEL		23.24	0.05		2005LEM2
			BP	276.87			23.21	0.13	DvH eval	2019DIK
			BP	276.87		24.31	23.36		VP	1944SCO
			BP	276.87			23.24	0.05		2018LEM
	isobutene	gas	BP	266.15	SEL		21.97	0.11		2005LEM2
			BP	266.05			21.97	0.23	DvH eval	2019DIK
			BP	266.05		22.49	21.61		VP	1940LAM
			BP	266.15			21.97	0.11		2018LEM
C ₅ H ₁₀	1-pentene	liq	BP	303.101	SEL		25.215	0.010		2018LEM
			BP	303.12			25.1	0.5	DvH eval	2019DIK
			BP	303.12		26.44	25.41		VP	1950FOR
	2-methyl-1-butene	liq	BP	304.3	SEL		25.50	0.27	DvH eval	2019DIK
			BP	304.3		26.81	25.76		VP	1949SCO
	2-methyl-2-butene	liq	BP	311.72	SEL		26.17	0.28	DvH eval	2019DIK
			BP	311.72		27.57	26.49		VP eval	2019DIK
			BP	311.72		27.66	26.58		VP	1949SCO
	2-pentene, (E)-	liq	BP	309.51	SEL		26.00	0.22	DvH eval	2019DIK
			BP	309.51		27.41	26.34		VP	1950SCO2
	2-pentene, (Z)-	liq	BP	310.09	SEL		26.10	0.22	DvH eval	2019DIK
			BP	310.09		27.47	26.39		VP	1950SCO
	3-methyl-1-butene	gas	BP	293.21	SEL		23.97	0.26	DvH eval	2019DIK
			BP	293.21	SEL	25.28	24.29	0.25	VP	1950SCO
C ₆ H ₁₂	1-hexene	liq	BP	336.64	SEL	29.44	28.29	0.30	VLE	2009MAR
			BP	336.64			28.22	0.25	DvH eval	2019DIK

			BP	336.64		30.12	28.95		VLE	2001SEG
			BP	336.64		29.88	28.71		VP	1950FOR
C ₇ H ₁₄	1-heptene	liq	BP	366.71	SEL		31.20	0.26	DvH eval	2019DIK
			BP	366.71		33.64	32.33		VLE	1997SCH
			BP	366.71		33.09	31.80		VP	1950HAL
			BP	366.71		32.61	31.34		VP	1950FOR
C ₈ H ₁₆	1-octene	liq	BP	394.43	SEL		34.5	0.4	DvH eval	2019DIK
			BP	394.43		36.60	35.17		VLE	1991KUU
			BP	394.43		36.13	34.72		VP	1950FOR
C ₉ H ₁₈	1-nonene	liq	BP	420.02	SEL	39.03	37.51		VP	1950FOR
C ₁₀ H ₂₀	1-decene	liq	BP	443.72	SEL		38.7	0.4	DvH eval	2019DIK
			BP	443.72		41.74	40.11		VP eval	2019DIK
			BP	443.72		41.77	40.14		VP	1950FOR
Alkadienes										
C ₃ H ₄	allene	gas	BP	240.874	SEL		19.98	0.30		2018LEM
			BP	238.65			20.7	0.4	DvH eval	2019DIK
			BP	238.65		21.48	20.64		VP eval	2019DIK
			BP	238.32		20.70	19.90		VP	1947STU
			BP	238.65		20.26	19.47		VP	1930LIV
C ₄ H ₆	1,2-butadiene	gas	BP	284.06	SEL		24.0	0.5	DvH est	2019DIK
			BP	291.94		24.70	23.74		VP eval	2019DIK
			BP	284.06			24.02		DvH rev	1985MAJ
			BP	284.06		25.30	24.31		VP ext	1947AST
			BP	291.94		25.65	24.65		VP	1947STU
	1,3-butadiene	gas	BP	268.44	SEL		22.29	0.25	DvH eval	2019DIK
			BP	268.62		23.38	22.47		VP	1933HEI2
C ₅ H ₈	1,2-pentadiene	liq	BP	318.8	SEL		27.24	0.20	DvH eval	2019DIK
			BP	318.8		28.64	27.52		VP	1950FOR
			BP	318.8		28.43	27.32		VP	1969OSB
	1,3-pentadiene	liq	BP	314.9	SEL		26.8	1.1	DvH eval	2019DIK
			BP	315.4		28.14	27.05		VP	1947STU
	1,4-pentadiene	liq	BP	299.12	SEL		25.07	0.21	DvH eval	2019DIK
			BP	298.93		26.59	25.56		VP	1969OSB
			BP	298.9		26.63	25.59		VP	1940LAM
	2-methyl-1,3-butadiene	liq	BP	307.19	SEL		25.71	0.24	DvH eval	2019DIK
			BP	307.26		26.35	25.32		VP	1964GUB
Alkynes										

C ₂ H ₂	acetylene	gas	BP	188.41	SEL	17.2	0.4	VP eval	2019DIK
			TP	192.0	SEL1	3.8	0.4	TP eval	2019DIK
			BP	188.41		17.06	16.39	VP ext	1956AMB
C ₃ H ₄	propyne	gas	BP	249.8	SEL	23.5	0.5	DvH est	2019DIK
			BP	249.8		23.43	22.52	VP eval	2019DIK
			BP	249.8			21.96	DvH rev	1987STE
			BP	249.8		23.18	22.28	VP	1933HEI
			BP	249.8		23.84	22.91	VP	1967VAN
			BP	249.8					
C ₄ H ₆	1-butyne	gas	BP	281.23	SEL	24.70	0.05		2018LEM
			BP	281.18		24.95	0.17	DvH eval	2019DIK
			BP	281.18		25.92	24.91	VP	1950AST
C ₅ H ₈	1-pentyne	liq	BP	313.51	SEL	25.4	1.1	DvH eval	2019DIK
	2-pentyne	liq	BP	329.27	SEL	25.0	1.1	DvH eval	2019DIK
C ₆ H ₁₀	1-hexyne	liq	BP	344.53	SEL	30.0	0.4	DvH eval	2019DIK
			BP	344.53		32.02	30.77	VLE ext	2006BOU
			BP	344.53		31.73	30.49	VLE	1997BEL
	2-hexyne	liq	BP	357.59	SEL	33.52	32.21	VP eval	2019DIK
			BP	357.59		33.02	31.73	VP	1997NEG
	3-hexyne	liq	BP	354.58	SEL	31.6	0.4	DvH eval	2019DIK
			BP	354.58		32.53	31.26	VLE	1997NEG
			BP	356.34		33.49	32.18	VP	1965RON
Alkenynes									
C ₄ H ₄	butenyne	gas	BP	279.18	SEL	23.9	0.7	DvH eval	2019DIK
			BP	278.5		25.31	24.33	VP	1947STU
C ₅ H ₆	3-penten-1-yne	liq	BP	317.05	SEL	26.6	1.4	DvH eval	2019DIK
Alkadiynes									
C ₄ H ₂	butadiyne	gas	BP	283.65	SEL	24.0	1.0	DvH eval	2019DIK
			BP	282.48		24.87	23.90	VP	1947STU
Cycloalkanes									
C ₃ H ₆	cyclopropane	gas	BP	240.27	SEL	20.0	0.6	DvH eval	2019DIK
			BP	240.27		20.77	19.96	VP	1946RUE
C ₄ H ₈	cyclobutane	gas	BP	285.64	SEL	23.7	0.4	DvH eval	2019DIK
			BP	285.64		24.87	23.90	VP	1953RAT
C ₅ H ₁₀	cyclopentane	liq	BP	322.4	SEL	27.30	0.20	DvH	1959MCC2
			BP	322.4		27.1	0.4	DvH eval	2019DIK
			BP	322.4		27.49	0.17	DvH	1946SPI
			BP	322.41		28.41	27.30	VP	1945WIL

	methylcyclobutane	liq	BP	310.0	SEL	25.3	1.2	DvH eval	2019DIK		
C ₆ H ₁₂	cyclohexane	liq	BP	353.8649	SEL	29.99	0.15		2014ZHO		
			BP	353.87		29.93	0.15	DvH eval	2019DIK		
			BP	353.87	31.75	30.51		VLE	2013RED		
			BP	353.87		30.00		DvH	1988DON		
			BP	353.87		29.97		DvH	1985MAJ		
			BP	353.87		30.10	0.15	DvH	1973SVO		
			BP	353.87		33.33		DvH	1943AST		
			BP	354.04	31.15	29.94		VP	1974KER		
			BP	353.9	30.78	29.58		VP	1965MAR		
			BP	353.93		30.06	0.13	DvH	1946SPI		
			BP	353.89	31.30	30.08		VP	1945WIL		
			BP	353.2		30.15		DvH	1926MAT		
			BP	353.8649		29.99	0.15		2018LEM		
			methylcyclopentane	liq	BP	344.96	SEL	29.07	0.10	DvH	1959MCC2
					BP	344.96		29.02	0.25	DvH eval	2019DIK
					BP	344.96	31.05	29.84		VLE	2010SAP
					BP	344.96		29.08		DvH rev	1985MAJ
					BP	344.96	30.46	29.27		VP	1945WIL
C ₇ H ₁₄	1,1-dimethylcyclopentane	liq	BP	360.99	SEL	31.72	30.48	VP	1949FOR		
	ethylcyclopentane	liq	BP	376.61	SEL	33.58	32.27	VP	1949FOR		
	methylcyclohexane	liq	BP	374	SEL	31.23	na		2018LEM		
			BP	374.12		31.2	0.4	DvH eval	2019DIK		
			BP	374.12	33.41	32.11		VLE	2010SAP		
			BP	374.12		31.27		DvH rev	1985MAJ		
			BP	374.12		31.80	0.21	DvH	1946SPI		
			BP	373.1		31.57		DvH	1926MAT		
C ₈ H ₁₆	isopropylcyclopentane	liq	BP	399.57	SEL	35.56	34.17	VP	1949FOR		
	n-propylcyclopentane	liq	BP	404.1	SEL	36.57	35.15	VP	1949FOR		
C ₉ H ₁₈	isopropylcyclohexane	liq	BP	427.71	SEL	38.41	36.91	VP	1949FOR		
	propylcyclohexane	liq	BP	429.87	SEL	38.87	37.36	VP	1949FOR		
C ₁₀ H ₂₀	1,3-diethylcyclohexane	liq	BP	445.7	SEL		37.5	1.2	DvH eval	2019DIK	
	isobutylcyclohexane	liq	BP	444.47	SEL	40.27	38.69		VP	1949FOR	
	tert-butylcyclohexane	liq	BP	444.74	SEL	39.86	38.30		VP	1949FOR	
Cycloalkenes											
C ₃ H ₄	cyclopropene	gas	BP	237.7	SEL		20.50	0.60		2019DIK	
C ₄ H ₆	cyclobutene	gas	BP	275.73	SEL		23.19	0.46		2018LEM	

			BP	275.74			24.0	1.0	DvH eval	2019DIK
			BP	275.74		23.97	23.03		VP	1941HEI
C ₅ H ₈	cyclopentene	liq	BP	317.39	SEL		26.9	0.4	DvH eval	2019DIK
	methylenecyclobutane	liq	BP	315.0	SEL		25.3	1.2	DvH eval	2019DIK
C ₆ H ₁₀	cyclohexene	liq	BP	356.1	SEL		30.66	0.25	DvH	1973SVO
			BP	356.1			30.3	0.4	DvH eval	2019DIK
			BP	356.1			30.46		DvH	1985MAJ
			BP	356.1		31.65	30.42		VP	1973MEY
			BP	354.8			30.48		DvH	1926MAT
Cycloalkadienes										
C ₅ H ₆	1,3-cyclopentadiene	liq	BP	314.7	SEL	25.00	24.02		VLE	1967LES
C ₆ H ₈	1,3-cyclohexadiene	liq	BP	353.49	SEL	31.49	30.26		VP	1973MEY
Bicycloalkanes										
C ₁₀ H ₁₈	cis-decalin	liq	BP	468.93	SEL		39.9	0.4	DvH eval	2019DIK
			BP	468.93		42.12	40.48		VP	1955CAM
	trans-decalin	liq	BP	460.42	SEL		38.9	0.4	DvH eval	2019DIK
			BP	460.42		41.02	39.42		VP	1955CAM
C ₁₂ H ₂₂	cyclohexylcyclohexane	liq	BP	511.1	SEL		43.86	0.28	DvH eval	2019DIK
			MP	276.80	SEL1		6.86		MP	2002CHI
			MP	277.2			6.78		MP rev	1996DOM

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
AROMATIC HYDROCARBONS										
Alkylbenzenes										
C ₆ H ₆	benzene	liq	BP	353.22	SEL		30.75	0.06	EOS	2012THO
			MP	278.67	SEL1		9.87	0.01	MP	1948OLI
			BP	353.22			30.73	0.11	DvH eval	2019DIK
			BP	353.22			30.63		DvH	1990AMB
			BP	353.22			30.72		DvH rev	1985MAJ
			BP	353.22			30.75	0.06	EOS-rev	2018LEM
			BP	353.29		32.06	30.81		VP	1971EON
			BP	353.25		31.95	30.70		VP	1945WIL
			BP	353.4			30.83		DvH	1926MAT
			BP	353.22		32.14	30.89		VLE	2014NIC
			MP	278.67			9.87		MP rev	1996DOM
			MP	278.67			9.30		MP	1979SMI
			MP	278.67			9.92		MP	1942ZIE
C ₇ H ₈	toluene	liq	BP	383.75	SEL		33.23	0.16	EOS	2006LEM
			MP	178.15	SEL1		6.64	0.01	MP	1962SCO
			BP	383.75			33.16	0.16	DvH eval	2019DIK
			BP	383.75		35.33	33.95		VLE	2020MOO
			BP	383.75			33.234	0.163	EOS-rev	2018LEM
			BP	383.77		34.80	33.44		VP	1945WIL
			BP	382.8			33.35		DvH	1926MAT
			MP	178.15			6.61		MP rev	1996DOM
			MP	178.15			6.85		MP rev	1991ACR
			MP	178.15			6.55		MP	1942ZIE
			MP	178.15			6.61		MP	1930SOU
			MP	178.15			6.62		MP	1929KEL
C ₈ H ₁₀	1,2-dimethylbenzene	liq	BP	417.521	SEL		36.38	0.18		2012ZHO
			BP	417.56			36.5	0.3	DvH eval	2019DIK
			BP	417.56		39.33	37.79		VLE	2005ROD
			BP	417.56			36.38	0.18	EOS-rev	2018LEM
			BP	417.56			36.38	0.32	EOS	2012ZHO
			BP	417.56		38.39	36.90		VP	1945WIL
			BP	414.6			36.82		DvH	1926MAT
	1,3-dimethylbenzene	liq	BP	412.214	SEL		36.11	0.07		2012ZHO

			BP	412.25			36.0	0.4	DvH eval	2019DIK
			BP	412.214			36.11	0.07	EOS-rev	2018LEM
			BP	412.25	38.07		36.59		VLE	2005ROD
			BP	412.25			36.11	0.40	EOS	2012ZHO
			BP	411.11	36.69		35.25		VP	1967AMB
			BP	412.25	37.93		36.45		VP	1945WIL
			BP	411.5			36.36		DvH	1926MAT
	1,4-dimethylbenzene	liq	BP	411.47	SEL		35.71	0.07		2012ZHO
			BP	411.47			35.87	0.18	DvH eval	2019DIK
			BP	411.47	38.68		37.17		VLE	2005ROD
			BP	411.47			35.71	0.20	EOS	2012ZHO
			BP	411.53	37.69		36.22		VP	1974OSB
			BP	411.11	36.69		35.25		VP	1967AMB
			BP	411.5	37.67		36.20		VP	1945WIL
			BP	410.3			35.99	0.20	DvH	1926MAT
			BP	411.47			35.71	0.07	EOS-rev	2018LEM
	ethylbenzene	liq	BP	409.314	SEL		35.60	0.11		2012ZHO
			BP	408.96			35.47	0.23	DvH eval	2019DIK
			BP	408.96	37.46		36.00		VLE	2010MAT
			BP	408.96	36.56		35.14		VP	1967AMB
			BP	409.34	37.42		35.96		VP	1945WIL
			BP	408.3			36.02	0.25	DvH	1926MAT
			BP	409.314			35.60	0.11		2018LEM
C ₉ H ₁₂	1,2,3-trimethylbenzene	liq	BP	449.23	SEL	41.95	40.31		VP	1949FOR
	1,2,4-trimethylbenzene	liq	BP	442.5	SEL	41.29	39.68		VP	1949FOR
	1,3,5-trimethylbenzene	liq	BP	437.87	SEL	41.12	39.52		VP	1949FOR
	2-ethyltoluene	liq	BP	438.25	SEL	40.70	39.11		VP	1949FOR
	3-ethyltoluene	liq	BP	434.46	SEL	40.40	38.83		VP	1949FOR
	4-ethyltoluene	liq	BP	435.14	SEL	40.24	38.67		VP	1949FOR
	isopropylbenzene	liq	BP	425.54	SEL		37.0	0.4	DvH eval	2019DIK
			BP	425.54		39.03	37.50		VLE	1989CEP
			BP	425.54		39.03	37.51		VP	1945WIL
	n-propylbenzene	liq	BP	432.32	SEL	39.75	38.20		VP	1949FOR
C ₁₀ H ₁₄	1,2-diethylbenzene	liq	BP	456.57	SEL	42.71	41.04		VP	1949FOR
	1,3-diethylbenzene	liq	BP	454.25	SEL	42.65	40.98		VP	1949FOR
	1,4-diethylbenzene	liq	BP	456.9	SEL	42.68	41.01		VP	1949FOR
	2-isopropyltoluene	liq	BP	451.5	SEL		39.3	0.7	DvH eval	2019DIK

			BP	451.45		42.69	41.02		VP		1959MCD
3-isopropyltoluene	liq		BP	448.35	SEL		39.3	0.7	DvH eval		2019DIK
			BP	448.06		42.42	40.76		VP		1959MCD
4-isopropyltoluene	liq		BP	450.27	SEL		39.3	0.4	DvH eval		2019DIK
			BP	450.27		41.93	40.29		VP		1959MCD
n-butylbenzene	liq		BP	456.42	SEL		42.53		VP		1949FOR
tert-butylbenzene	liq		BP	442.27	SEL		40.61		VP		1949FOR
Alkenylbenzenes											
C ₈ H ₈	ethenylbenzene	liq	BP	418.45	SEL		36.8	0.4	DvH eval		2019DIK
			BP	418.45		38.83	37.32		VP		1949BUC
			BP	418.3		39.00	37.48		VP		1955DRE
Benzocycloalkanes											
C ₁₀ H ₁₂	benzocyclohexane	liq	BP	480.56	SEL		42.6	0.5	DvH eval		2019DIK
			BP	480.55		44.40	42.67		VP		1922HER
Benzocycloalkenes											
C ₉ H ₈	benzocyclopentene	liq	BP	454.72	SEL		40.36	0.4	DvH eval		2019DIK
			BP	454.72		44.77	43.02		VP		1947STU
Alkyl naphthalenes											
C ₁₀ H ₈	naphthalene	sol	MP	353.40	SEL		18.993	0.019	MP		2002CHI
			BP	491.11	SEL1		43.3	0.8	DvH eval		2019DIK
			SP	298.15	SEL2		72.44	0.3	VP		2005RUZ
			BP	490.73		45.79	44.01		VP		1968FOW
			BP	491.11		45.18	43.42		VP		1955CAM
			MP	353.40			19.10		MP		2008SHA
			MP	353.40			19.55		MP		2006KHI
			MP	353.40			19.10		MP rev		1991ACR
			MP	353.40			18.98	0.02	MP		1957MCC
			MP	353.40			19.00		MP		1926AND
			SP	353.37			70.38	0.3	VP		2005RUZ
			SP	298.15			73.00	0.25	VP		1972IRV
C ₁₁ H ₁₀	1-methylnaphthalene	liq	BP	517.84	SEL		46.3	0.3	DvH eval		2019DIK
			MP	242.65	SEL1		6.95		MP		1957MCC
			BP	517.84		48.51	46.62		VP		1955CAM
			MP	242.65			6.95		MP rev		1996DOM
2-methylnaphthalene	sol		MP	307.70	SEL		12.13		MP rev		1996DOM
			BP	514.2	SEL1	48.23	46.35		VP		1955CAM
			MP	307.20			11.97		MP		1931HUF

Polycyclic Aromatics										
C ₁₂ H ₈	acenaphthylene	sol	MP	366.40	SEL		10.96	MP rev	1996DOM	
			MP	550.60	SEL1		46.4	2.4	DvH est	2019DIK
	biphenylene	sol	MP	384.40	SEL		82.7	DsubH est	1995NAS	
			BP	548.50	SEL1		46.3	1.7	DvH est	2019DIK
			MP	384.40			82.4	1.0	DsubH est	2019DIK
			MP	364.02	SEL		21.46		MP	1977FIN
C ₁₂ H ₁₀	acenaphthene	sol	BP	550.39	SEL1	53.88	51.78	VP	1923MOR	
			MP	364.02			21.46		MP rev	1996DOM
			MP	387.90	SEL		19.10		MP	2000LIS
C ₁₃ H ₁₀	fluorene	sol	BP	570.19	SEL1	56.87	54.65	VP	1923MOR	
			MP	387.90			19.58		MP rev	1996DOM
			MP	489.50	SEL		29.37		MP	1970GOU
C ₁₄ H ₁₀	anthracene	sol	BP	614.87	SEL1	58.54	56.25	VP	1923MOR	
			MP	489.50			29.80		MP	2003ROJ
			MP	489.50			29.37		MP rev	1996DOM
			MP	372.74	SEL		16.60		MP	2003ROJ
	phenanthrene	sol	BP	609.0	SEL1	59.56	57.24	VP	1975OSB	
			BP	611.47		58.74	56.45		VP	1923MOR
			MP	372.74			16.20		MP	2000LIS
			MP	372.74			16.46		MP rev	1996DOM
			MP	372.74			15.72		MP	1992SAB
			MP	372.74			16.47		MP rev	1991ACR
			MP	381.00	SEL		18.73		MP	1971WON
			BP	659.82	SEL1	51.46	49.45		VP	1955TSY
C ₁₆ H ₁₀	fluoranthene	sol	MP	381.00			18.74	MP rev	1996DOM	
			MP	424.30	SEL		16.70		MP	2003ROJ
			BP	668.17	SEL1	56.54	54.34		VP	1955TSY
	pyrene	sol	MP	424.30			17.36		MP rev	1996DOM
			MP	424.30			17.11		MP rev	1991ACR
			MP	550.90	SEL		32.58		MP	1992SAB
			MP	550.90			31.88		MP rev	1996DOM
C ₂₀ H ₁₂	perylene	sol	MP	550.90			31.75		MP rev	1991ACR
			MP	710.50	SEL		21.20		MP	2009TOR
C ₂₄ H ₁₂	coronene	sol	MP	710.50			19.20		MP rev	1991ACR

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
OXIDIZED HYDROCARBONS										
Carbon Oxides										
CO ₂	carbon dioxide	gas	SP	194.67	SEL		25.230	0.002	DsubH	1937GIA
			TP	216.57	SEL1		25.64	0.25	TP eval	2019DIK
CO	carbon monoxide	gas	BP	81.64	SEL		6.01	0.02	EOS	2006LEM
			BP	81.64			6.03	0.17	DvH eval	2019DIK
			BP	81.64			6.01	0.02	EOS-rev	2018LEM
Alkanols										
CH ₄ O	methanol	liq	BP	337.63	SEL		35.28	0.21	EOS	1993REU
			BP	337.63			35.36	0.21	DvH eval	2019DIK
			BP	337.63		38.37	36.88		VLE	2018STU
			BP	337.63			35.28	0.21	EOS-rev	2018LEM
			BP	337.7		37.31	35.85		VP	1970AMB
			BP	337.0			35.30		DvH	1926MAT
C ₂ H ₆ O	ethanol	liq	BP	351.57	SEL		39.14	0.47	EOS	2014SCH
			BP	351.57			39.1	0.5	DvH eval	2019DIK
			BP	351.57			39.14	0.47	EOS-rev	2018LEM
			BP	351.45		40.63	39.04		VP	1970AMB
			BP	351.45		40.77	39.18		VLE	1949KRE
			BP	350.6			38.91		DvH	1926MAT
C ₃ H ₈ O	1-propanol	liq	BP	370.26	SEL		41.7	0.4	DvH eval	2019DIK
			BP	370.3		43.19	41.51		VP	1970AMB
			BP	370.35		43.75	42.05		VP	1969KEM
	isopropanol	liq	BP	354.4	SEL		40.04	0.25	DvH	1926MAT
			BP	355.39			39.92	0.24	DvH eval	2019DIK
			BP	355.39		41.99	40.35		VP	1963BID
C ₄ H ₁₀ O	1-butanol	liq	BP	390.77	SEL		42.9	0.5	DvH eval	2019DIK
			BP	391.02		44.64	42.90		VP	1969KEM
			BP	391.36		44.93	43.18		VP	1965HES
			BP	390.88		45.04	43.29		VP	1963BID
			BP	389.9			43.82		DvH	1926MAT
	2-butanol	liq	BP	372.66	SEL	42.90	41.22		VP	1963BID
	isobutanol	liq	BP	381.04	SEL	44.01	42.29	0.40	VP	1963BID
			BP	380.9			42.0	0.3	DvH eval	2019DIK
			BP	380.0			42.87		DvH	1926MAT

	tert-butanol	sol	MP	298.87	SEL		6.78		MP	1926PAR
			BP	355.5	SEL1		39.15	0.22	DvH eval	2019DIK
			SP	298.15	SEL2		49	5	est	2022BUR
			BP	355.47		41.54	39.92		VP	1969BRO
			BP	355.49		41.54	39.92		VP	1963BEY
			BP	355.5		41.36	39.75		VP	1963BID
			MP	298.87			6.70		MP rev	1996DOM
			MP	298.87			6.79		MP rev	1991ACR
C ₅ H ₁₂ O	1-pentanol	liq	BP	410.9	SEL		44.2	0.6	DvH eval	2019DIK
	2-methyl-1-butanol	liq	BP	402.14	SEL		43.5	0.5	DvH eval	2019DIK
			BP	403.8		45.99	44.20		VP	1935BUT
	2-pentanol	liq	BP	392.29	SEL	44.53	42.79		VP	1935BUT
	isopentanol	liq	BP	404.73	SEL		43.6	0.3	DvH eval	2019DIK
Glycols										
C ₂ H ₆ O ₂	1,2-ethanediol	liq	BP	470.313	SEL		54.56	na		2018LEM
			BP	470.73			52.7	1.0	DvH eval	2019DIK
			BP	470.73		54.59	52.47		VP	1952JON
C ₃ H ₈ O ₂	1,2-propanediol	liq	BP	461.22	SEL	53.58	51.49		VP	2002STE
			BP	461.22			51.54	0.7	DvH eval	2019DIK
			BP	461.22		53.26	51.18		VP eval	2019DIK
	2-methoxyethanol	liq	BP	397.54	SEL		39.59	0.26	DvH eval	2019DIK
			BP	397.29		41.71	40.08		VP	1956PIC
Sugars										
C ₃ H ₈ O ₃	glycerin	liq	BP	563.0	SEL		63.3	1.5	DvH eval	2019DIK
			MP	291.05	SEL1		18.29		MP	1923GIB
		sol	MP	291.05			18.28		MP rev	1991ACR
C ₆ H ₁₂ O ₆	alpha-D-glucose	sol	MP	414	SEL		31.4		MP rev	2019DIK
		liq	BP	706	SEL1(u)		84.4	4.2	DvH est	2019DIK
		sol	SP	425	SEL2		194	5	VP	1999OJA
C ₁₂ H ₂₂ O ₁₁	sucrose	sol	MP	424.4	SEL		32		cal	2014MAG
			SP	298.15	SEL2		94		est	2022BUR
Phenols										
C ₇ H ₈ O	2-methylphenol	sol	MP	304.20	SEL		15.82		MP	1967AND
			BP	464.14	SEL1	46.53	44.71		VP	1949DRE
			MP	304.14			15.90		MP	1998JAM
			MP	304.20			15.82		MP rev	1996DOM
			MP	304.14			14.80		MP	1990MEV

4-methylphenol		sol	MP	307.84	SEL		12.71	MP	1967AND	
			BP	474.99	SEL1	49.29	47.36	VP	1949DRE	
			MP	307.84			8.58	MP	1998JAM	
			MP	307.84			12.72	MP rev	1996DOM	
			MP	307.84			11.80	MP	1990MEV	
Alkyl Ethers										
C2H6O	dimethyl ether	gas	BP	248.37	SEL		21.43	0.90	EOS	2011WU
			BP	248.37			21.5	0.9	DvH eval	2019DIK
			BP	248.37			21.43	0.90	EOS-rev	2018LEM
			BP	248.34		22.28	21.41		VP	1941KEN
C4H10O	diethyl ether	liq	BP	307.56	SEL		26.55	0.27	EOS	2014THO
			BP	307.56			26.33	0.27	DvH eval	2019DIK
			BP	307.56			26.55	0.27	EOS-rev	2018LEM
			BP	307.1			26.69		DvH	1926MAT
			BP	307.63		27.99	26.90		VP	1922TAY
C6H14O	diisopropyl ether	liq	BP	341.49	SEL		28.99	0.17	DvH eval	2019DIK
			BP	341.49		31.03	29.82		VP	2007EFI
			BP	341.49		30.60	29.41		VLE	1999MON
			BP	341.49		30.86	29.65		VP	1969CID
di-n-propyl ether		liq	BP	363.25	SEL		33.16		VP	1973MEY
			BP	359.5			33.29		VP	1969CID
Aldehydes										
CH2O	formaldehyde	gas	BP	254.13	SEL		22.0	0.4	DvH est	2019DIK
			BP	254.13		23.30	22.39		VP ext	1935SPE
C2H4O	acetaldehyde	liq	BP	299.12	SEL		25.07	0.21	DvH eval	2019DIK
			BP	293.47	u	27.14	26.08		VLE	1963BUL
			BP	293.56		27.07	26.01		VLE	1950COL
C3H6O	propanal	liq	BP	321.19	SEL		28.3	0.4	DvH eval	2019DIK
			BP	321.19	u	32.38	31.12		VP	1951SMI
			BP	307.56			28.31		DvH rev	1985MAJ
C4H8O	butanal	liq	BP	347.95	SEL(u)		29.32	0.28	DvH eval	2019DIK
			BP	347.95	u	32.96	31.68		VLE	1963WOJ
			BP	348.33	u	31.34	30.11		VP	1959SEP
isobutanal		liq	BP	337.26	SEL		30.73		VP	1959SEP
Alkenals										
C3H4O	2-propenal	liq	BP	325.4	SEL		29.6	0.3	DvH eval	2019DIK
			BP	325.4		30.86	29.66		VP	1979MAR

			BP	325.71		29.38	28.23	VP	1947STU	
Ketones										
C ₃ H ₆ O	acetone	liq	BP	329.22	SEL	31.27	29.12	0.27	EOS	2006LEM
			BP	329.22			29.35	0.27	DvH eval	2019DIK
			BP	329.22			30.05		VLE ext	2008SON
			BP	329.22			29.12	0.27	EOS-rev	2018LEM
			BP	329.22			30.86		DvH rev	1987STE
			BP	329.42			29.70		VP	1974AMB
			BP	329.22			29.09		DvH	1957PEN
			BP	329.2			29.67		DvH	1926MAT
C ₄ H ₈ O	butanone	liq	BP	352.74	SEL	33.44	31.59	0.16	DvH eval	2019DIK
			BP	352.74			32.14		VLE	2008MAR
			BP	352.74			31.30		DvH rev	1985MAJ
			BP	352.74			31.30	0.10	DvH	1961NIC
			BP	352.63			31.56		VP	1961NIC
			BP	351.4			31.97		DvH	1926MAT
C ₅ H ₁₀ O	3-pentanone	liq	BP	375.11	SEL	35.31	33.93		VP	1965COL
			BP	376.41			29.76		VP	1949DRE
Carboxylic Acids										
CH ₂ O ₂	formic acid	liq	BP	374.05	SEL	SEL1	22.7	8.4	DvH est	2019DIK
			MP	281.47			12.68		MP	1941STO
			BP	374.05			21.0		DvH	1930COO
C ₂ H ₄ O ₂	acetic acid	liq	BP	391.2	SEL	38.67	37.16		VLE	2019LI
			MP	289.49			11.72		MP	1982MAR
			BP	391.01			37.45		VP	1959MCD
			MP	289.49			11.72		MP rev	1996DOM
C ₃ H ₆ O	propanoic acid	liq	BP	413.99	SEL		29.91	0.7	DvH	1989JAS
			BP	413.99			29.7	1.2	DvH eval	2019DIK
			BP	413.99			30.63		DvH	1926MAT
C ₄ H ₈ O	n-butanoic acid	liq	BP	436.42	SEL		38.4	2.4	DvH eval	2019DIK
			BP	436.42			35.9	1.5	DvH	1989JAS
Alkyl Esters										
C ₂ H ₄ O ₂	methyl formate	liq	BP	305.14	SEL		27.90	0.25	DvH	1976CIH
			BP	305.14			28.0	0.4	DvH eval	2019DIK
			BP	305.14			27.92		DvH rev	1985MAJ
			BP	304.5			28.24		DvH	1926MAT
C ₃ H ₆ O ₂	ethyl formate	liq	BP	327.29	SEL		29.80	0.20	DvH	1976CIH

			BP	327.29			29.7	0.3	DvH eval	2019DIK
			BP	327.29			29.91		DvH rev	1985MAJ
			BP	326.5			30.13		DvH	1926MAT
C ₄ H ₈ O ₂	ethyl acetate	liq	BP	349.2	SEL		32.30	0.30	DvH	1926MAT
			BP	350.21			32.05	0.27	DvH eval	2019DIK
			BP	350.21		33.60	32.29		VP	1965POL
Alkenyl Esters										
C ₄ H ₆ O ₂	vinyl acetate	liq	BP	345.33	SEL		31.7	0.3	DvH eval	2019DIK
			BP	345.33		34.45	33.11		VLE	1965SWA
			BP	345.88		33.12	31.83		VP	1963CAP
C ₅ H ₈ O ₂	ethyl propenoate	liq	BP	372.59	SEL		34.2	0.4	DvH eval	2019DIK
			BP	372.69		35.47	34.09		VP	1947STU
	methyl 2-methylpropenoate	liq	BP	369.73	SEL		34.14	0.28	DvH eval	2019DIK
			BP	369.73		38.82	37.30		VP	1956BRO
Oxacycloalkanes										
C ₂ H ₄ O	oxacyclopropane	gas	BP	283.59	SEL		25.47	0.19	EOS	2015THO
			BP	283.59			25.67	0.19	DvH eval	2019DIK
			BP	283.59			25.47	0.19	EOS-rev	2018LEM
			BP	283.59		26.49	25.46		VP	1959MCD
			BP	283.65		25.98	24.97		VLE	1950COL
C ₃ H ₆ O	methyloxacyclopropane	liq	BP	307.268			27.59	0.14	EOS-rev	2018LEM
Oxacycloalkadienes										
C ₄ H ₄ O	oxacyclopentadiene	liq	BP	304.51	SEL		27.03	0.18	DvH eval	2019DIK
			BP	304.51		28.05	26.96		VP	1952GUT
C ₅ H ₆ O	2-methyloxacyclopentadiene	liq	BP	338.39	SEL	30.86	29.66		VP	1971EON
Oxacycloketones										
C ₄ H ₆ O	cyclobutanone	liq	BP	367.79	SEL	37.15	35.70		VP	1942BEN
C ₆ H ₁₀ O	cyclohexanone	liq	BP	428.57	SEL	39.48	37.94		VP	1973MEY
Dioxanes and Trioxanes										
C ₃ H ₆ O ₃	1,3,5-trioxane	sol	MP	333.44	SEL		15.105	0.010	MP	1988VAN
			BP	387.2	SEL1		37.3	2.1	DvH est	2019DIK
			SP	298.15	SEL2		56.6	0.2	C	1969MAN
			MP	333.65			15.10		MP rev	1996DOM
			SP	298.15			57.06	0.37	DvH est	2019DIK
C ₄ H ₈ O ₂	1,3-dioxane	liq	BP	378.13	SEL		34.3	2.4	DvH est	2019DIK
	1,4-dioxane	liq	BP	374.25	SEL		35.58	0.3	DvH	1963VIN
			BP	374.25			35.3	0.3	DvH eval	2019DIK

			BP	374.25			34.16		DvH rev	1985MAJ
			BP	374.44		35.79	34.39		VP	1938CRE
Furandione and D-Camphor										
C ₄ H ₂ O ₃	2,5-furandione	sol	MP	325.70	SEL		12.26		MP	1978MAR
			BP	478.33	SEL1		44.7	2.3	DvH est	2019DIK
			SP	298.15	SEL2		68.85	0.42	cal	2017SOU
			BP	478.33	u	44.05	42.33		VP	1947STU
			MP	325.70			12.26		MP rev	1996DOM
			MP	325.72			13.60		MP	1983DEW
			SP	319.0			70.72	0.42	cal	2017SOU
C ₁₀ H ₁₆ O	D-camphor	sol	MP	463.38	SEL(u)		15.7		DfusH rev	1990DON
			BP	493.0	SEL1(u)		44.0	1.8	DvH est	2019DIK
			SP	298.15	SEL2		51.9	0.8	DsubHest	1977STE
			MP	451.80			5.30		MP rev	1979MJO
Aromatic Alcohols										
C ₆ H ₆ O	phenol	sol	MP	314.05	SEL		11.54		MP	1963AND
			BP	454.91	SEL1		45.47	0.23	DvH eval	2019DIK
			BP	454.91	u		48.10		DvH rev	1987STE
			BP	454.91		47.20	45.36		VP	1949DRE
			MP	314.05			11.51		MP rev	1996DOM
			MP	314.05			11.29		MP rev	1991ACR
			MP	314.05			11.51		MP	1972INO
			MP	314.05			12.13		MP	1957MAS
C ₇ H ₈ O	3-methylphenol	liq	BP	475.28	SEL		47.2	0.5	DvH eval	2019DIK
			MP	285.37	SEL1		10.67		MP	1998JAM
			BP	475.28		49.83	47.88		VP	1947GOL
			MP	285.37			10.71		MP rev	1996DOM
			MP	285.37			9.10		MP	1990MEV
			MP	285.37			10.71		MP	1967AND
	benzyl alcohol	liq	BP	477.4	SEL		50.49	1.00	DvH	1926MAT
			BP	478.5			47.3	0.8	DvH eval	2019DIK
			BP	478.45		50.39	48.42		VP	1949DRE
Aromatic Aldehydes and Acids										
C ₇ H ₆ O	benzaldehyde	liq	BP	451.15	SEL		40.9	0.3	DvH eval	2019DIK
			BP	451.68		42.38	40.73		VP	1947STU
C ₇ H ₆ O ₂	benzoic acid	sol	MP	395.50	SEL		17.10		MP	2002ROY
			BP	523.53	SEL1		56.5	1.0	DvH eval	2019DIK

			SP	298.15	SEL2		89.25	0.42	VP	1990DAS
			BP	523.53		59.25	56.94		VP	1947STU
			MP	395.50			16.99		MP	2009BRI
			MP	395.50			17.30		MP	2003SHA
			MP	395.50			17.99		MP	1984PIT
			MP	395.50			18.00		MP	1953GIN
Formula	Compound	Phase	trs	T _{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
NITROGEN-SUBSTITUTED HYDROCARBONS										
Hydrogen Cyanides										
CHN	hydrogen cyanide	liq	BP	298.79	SEL		25.22	0.25	DvH eval	2019DIK
			BP	298.79	u	27.91			VP rev	1987STE
			BP	298.79	u	27.70			VP	1939GIA
			BP	298.79	u	27.78			VP	1926SIN
Alkylamines										
CH ₅ N	methylamine	gas	BP	266.84	SEL		25.94	0.23	DvH eval	2019DIK
			BP	266.84		26.84	25.79		VP	1937AST
C ₂ H ₇ N	ethylamine	gas	BP	289.79	SEL(u)		27.4	0.4	DvH eval	2019DIK
			BP	289.79	u	27.34	26.27		VP	1947STU
C ₃ H ₉ N	n-propylamine	liq	BP	320.38	SEL		29.58	0.20	DvH eval	2019DIK
			BP	320.38			29.55		DvH rev	1985MAJ
			BP	320.38		31.04	29.83		VLE	1983WOL
			BP	320.38		30.80	29.60		VP	1968OSB
			BP	321.71		30.37	29.19		VP	1947STU
C ₄ H ₁₁ N	isopropylamine	liq	BP	304.92	SEL	29.03	27.90		VP	1968OSB
	n-butylamine	liq	BP	350.12	SEL		32.03	0.27	DvH eval	2019DIK
	tert-butylamine	liq	BP	317.19	SEL	29.53	28.38		VP	1968OSB
C ₂ H ₇ N	dimethylamine	gas	BP	280.02	SEL		26.48	0.06	DvH	1939AST
			BP	280.02			26.51	0.27	DvH eval	2019DIK
			BP	280.02			26.40		DvH rev	1985MAJ
			BP	280.02		27.96	26.87		VP	1970WOL
			BP	280.02		27.11	26.05		VP	1935SIM
C ₃ H ₉ N	methylethylamine	liq	BP	305.6	SEL		28.4	0.5	DvH eval	2019DIK
	trimethylamine	gas	BP	276.0	SEL		23.00	0.22	DvH eval	2019DIK
			BP	276.0		24.17	23.23		VP	1944AST
C ₆ H ₁₅ N	triethylamine	liq	BP	361.93	SEL		31.3	0.4	DvH eval	2019DIK
			BP	361.93			31.01		DvH rev	1985MAJ

C2H8N2	1,1-dimethylhydrazine	liq	BP	361.93		31.96	30.71		VP	1962BIT
			BP	335.57	SEL		32.1	0.9	DvH eval	2019DIK
			BP	335.57		34.41	33.06		VP	1953AST
Cycloalkylamines										
C6H13N	cyclohexylamine	liq	BP	406.98	SEL		35.8	0.6	DvH eval	2019DIK
			BP	406.98	u	38.90	37.38		VLE	2014RED
			BP	406.98	u	37.93	36.45		VP	1960NOV
Azacycloalkanes										
C4H9N	azacyclopentane	liq	BP	359.71	SEL	34.33	32.99		VP	1959MCC
C5H11N	azacyclohexane	liq	BP	379.37	SEL	35.14	33.77		VP	1968OSB
Methenamines										
C6H12N4	methenamine	sol	SP	298.15	SEL		79.64	0.18	VP	2002VER
Azacycloalkadienes										
C4H5N	azacyclopentadiene	liq	BP	402.93	SEL	40.33	38.76		VP	1968OSB
			BP	399.47		41.60	39.98		VP	1971EON
C5H7N	1-methylazacyclopentadiene	liq	BP	385.9	SEL	36.82	35.38		VP	1968OSB
			BP	384.66		37.84	36.36		VP	1971EON
Alkyl nitriles										
C2H3N	ethanenitrile	liq	BP	355.05	SEL		30.7	0.3	DvH eval	2019DIK
			BP	355.05			29.75		DvH rev	1985MAJ
			BP	355.05		32.51	31.24		VP	1974DOJ
C3H5N	propanenitrile	liq	BP	370.73	SEL		33.0	0.4	DvH eval	2019DIK
			BP	370.73		34.55	33.20		VLE	2013LAA
			BP	370.73		33.11	31.82		VP	1949DRE
C4H7N	butanenitrile	liq	BP	390.77	SEL		34.70	0.24	DvH eval	2019DIK
C5H9N	pentanenitrile	liq	BP	414.42	SEL	38.54	37.04		VP	1949DRE
C6H11N	hexanenitrile	liq	BP	436.61	SEL	41.24	39.63		VP	1973MEY
			BP	436.88		40.92	39.33		VP	1949DRE
C3H3N	propenenitrile	liq	BP	351.13	SEL(u)		30.7	0.7	DvH eval	2019DIK
			BP	351.13	u	34.40	33.06		VLE	2010HAI
			BP	351.13	u	34.18	32.85		VP rev	1990DAU
			BP	351.13	u	30.99	29.78		VLE	1964SEV
C2N2	cyanogen	gas	BP	252.0	SEL(u)		24.7	0.6	DvH eval	2019DIK
			BP	252.0	u		23.30	0.20	DvH	1939RUE
			BP	252.0	u	24.20	23.26		VP	1925PER
			BP	252.18	u	23.78	22.86		VP	1947STU
Melamines										

C ₃ H ₆ N ₆	melamine	liq	SP	497	SEL		120	4	VP	1960HIR
			BP	639	SEL1		79	4	DvH est	2019DIK
Anilines										
C ₆ H ₇ N	aniline	liq	BP	457.3	SEL		43.8	0.5	DvH eval	2019DIK
C ₈ H ₁₁ N	N,N-dimethylaniline	liq	BP	466.47	SEL		42.7	0.8	DvH eval	2019DIK
			MP	275.65	SEL1		11.56		MP	1972AHM
			BP	466.47	u	45.79	44.01		VP	1976RAD
			BP	466.58	u	45.81	44.02		VP	1925NEL
Pyridines										
C ₅ H ₅ N	pyridine	liq	BP	388.36	SEL	37.85	36.37		VP	1996CHI
C ₆ H ₇ N	2-methylpyridine	liq	BP	402.52	SEL	37.75	36.28		VP	1963SCO
Oxidized Amines										
CH ₄ N ₂ O	urea	sol	MP	406.5	SEL		14.79		MP	1987DEL
			SP	298.15	SEL1		95.5	0.3	VP	2006EME
			MP	406.17			14.60		MP	1999RAI
			MP	406.17			15.03		MP	1995FER
			MP	406.17			14.50		MP	1988GAM
			SP	298.15			94.6	2.0	VP	2003ZAI
C ₆ H ₁₅ NO ₃	triethanolamine	liq	BP	623.15	SEL		82.0	2.0	VP	2004WRI
			BP	623.15			76.1	4.2	DvH est	2019DIK
Alkyl Nitrates and Nitroalkanes										
CH ₃ NO ₃	methyl nitrate	liq	BP	338.0	SEL(u)		26.4	1.4	DvH est	2019DIK
C ₃ H ₅ N ₃ O ₉	nitroglycerin	sol	MP	286.65			21.87		MP rev	1991ACR
CN ₄ O ₈	tetranitromethane	liq	BP	397.95	SEL		40.74		DvH rev	1985MAJ
			SP	287.05	SEL1	47.40			DsubH	1941SEK
			BP	397.95	u	39.1		2.0	VP	1919MEN
			BP	397.95	u	39.56			VP	1952EDW
CH ₃ NO ₂	nitromethane	liq	BP	373.1	SEL		34.46	0.30	DvH	1926MAT
			BP	374.34	u	37.64	36.18		VLE	1979MAR2
			BP	374.34	u	35.91	34.51		VP	1954MCC
			BP	374.34			34.73	0.24	DvH eval	2019DIK
Isocyanates										
C ₉ H ₆ N ₂ O ₂	2,4-diisocyanatotoluene	liq	BP	524.83	SEL(u)		41.9	2.1	DvH est	2019DIK
			BP	524.83	u	53.84	51.74		VP	1975FRE
Nitro Aromatics										
C ₆ H ₅ NO ₂	nitrobenzene	liq	BP	483.81	SEL		43.9	0.5	DvH eval	2019DIK
			MP	278.89	SEL1		12.12		MP	1936PAR

C ₇ H ₅ N ₃ O ₆ trinitrotoluene sol	BP	483.81	45.97	44.18	VLE	1952BRO
	MP	278.89		12.12	MP rev	1996DOM
	MP	355.1	SEL	23.43	DfusH	1993ACR
	BP	620.5	SEL1	73.69	VP	1948ROB
	SP	298.15	SEL2	113.2	1.5 VP	1979KUD
	MP	355.1		22.41	DfusH	1990HU
	MP	355.10		21.23	MP rev	1991ACR

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
SULFUR-SUBSTITUTED HYDROCARBONS										
Alkanethiols										
CH ₄ S	methanethiol	gas	BP	279.05	SEL		24.4	0.9	DvH eval	2019DIK
			BP	279.1		25.34	24.35		VP	1942RUS
C ₂ H ₆ S	ethanethiol	liq	BP	308.19	SEL		26.6	0.6	DvH eval	2019DIK
			BP	308.15		27.78	26.70		VP	1966OSB
C ₃ H ₈ S	1-propanethiol	liq	BP	340.87	SEL	30.80	29.60		VP	1956PEN
	isopropanethiol	liq	BP	325.71	SEL	29.12	27.99		VP	1966OSB
C ₄ H ₁₀ S	1-butanethiol	liq	BP	371.56		33.79	32.47		VP	1966OSB
	1-isobutanethiol	liq	BP	361.64	SEL	32.48	31.22		VP	1966OSB
	2-butanethiol	liq	BP	358.13	SEL	32.04	30.79		VP	1966OSB
C ₅ H ₁₂ S	1,1-dimethyl-1-propanethiol	liq	BP	372.28	SEL	32.95	31.66		VP	1966OSB
	1-pentanethiol	liq	BP	399.79	SEL	36.69	35.26		VP	1966OSB
C ₆ H ₁₄ S	1-hexanethiol	liq	BP	425.81	SEL	39.46	37.92		VP	1966OSB
C ₇ H ₁₆ S	1-heptanethiol	liq	BP	450.07	SEL	42.15	40.50		VP	1966OSB
Alkyl Sulfides										
CS ₂	carbon disulfide	liq	BP	318.4	SEL		26.78	0.15	DvH	1926MAT
			BP	318.35			26.78	0.15	DvH eval	2019DIK
			BP	319.37		27.56	26.48		VP	1962WAD
C ₂ H ₆ S ₂	dimethyl disulfide	liq	BP	382.89	SEL	34.98	33.61		VP	1950SCO
C ₂ H ₆ S	dimethyl sulfide	liq	BP	310.42	SEL		27.0	0.3	DvH eval	2019DIK
			BP	310.45		28.16	27.06		VP	1942OSB
C ₃ H ₈ S	ethyl methyl sulfide	liq	BP	339.8	SEL	30.78	29.58		VP	1952WHI
C ₄ H ₁₀ S	diethyl sulfide	liq	BP	365.25	SEL	33.26	31.97		VP	1966OSB
Thiacycloalkanes										
C ₄ H ₈ S	thiacyclopentane	liq	BP	394.27	SEL	35.94	34.53		VP	1966OSB
			BP	393.42		37.46	36.00		VP	1971EON
Thiacycloalkenes										
C ₄ H ₄ S	thiacyclopentadiene	liq	BP	357.31	SEL	32.63	31.35		VP	1949WAD
			BP	356.52		34.82	33.46		VP	1971EON
Aromatic Thiols										
C ₆ H ₆ S	benzenethiol	liq	BP	442.29	SEL	41.35	39.74		VP	1966OSB
Alkyl Sulfoxides										
C ₂ H ₆ OS	dimethyl sulfoxide	liq	BP	463.86	SEL		43.5	0.4	DvH eval	2019DIK
			MP	291.57	SEL1		14.37		MP	1970CLE
			BP	463.86		45.86	44.07		VP	1972JAK

	MP	291.57	14.37	MP rev	1996DOM
--	----	--------	-------	--------	---------

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
HALOGENATED HYDROCARBONS										
Fluoroalkanes										
CH ₃ F	fluoromethane	gas	BP	194.65	SEL		16.60	0.46	EOS	2006LEM
			BP	194.65			17.0	0.5	DvH eval	2019DIK
			BP	194.65			17.30	0.13	VLE	1997HOL
			BP	194.65			17.38		VP	1983OI
			BP	194.65			17.57		VP	1961FAR
			BP	194.65			17.40		VP	1961LI
			BP	194.65			17.36		VP	1948MIC
			BP	194.65			17.72		VP	1940GRO
			BP	194.65				0.46	EOS-rev	2018LEM
			BP	194.13			16.91		VLE	1999TAK
CH ₂ F ₂	difluoromethane	gas	BP	221.5	SEL		19.87	0.20	EOS	1997TIL
			BP	221.5			20.24	0.20	DvH eval	2019DIK
			BP	221.5			20.70		VP	1993WEB
			BP	221.5			20.76		VP	1987KAN
			BP	221.5				0.20	EOS-rev	2018LEM
			BP	221.46			20.72		VLE	1999TAK
CHF ₃	trifluoromethane	gas	BP	188.75	SEL		16.76	0.15	EOS	2003PEN
			BP	188.75			16.93	0.15	DvH eval	2019DIK
			BP	188.75			16.76	0.15	EOS-rev	2018LEM
			BP	190.98			17.56		VP	1962VAL
CF ₄	tetrafluoromethane	gas	BP	145.56	SEL		11.83	0.27	EOS	1990PLA
			BP	145.56			11.84	0.27	DvH eval	2019DIK
			BP	145.56			11.83	0.27	EOS-rev	2018LEM
			BP	145.56			11.83		VP	1972REG
			BP	145.15			12.56		VP	1933MEN
C ₂ H ₅ F	fluoroethane	gas	BP	236.05	SEL		20.26	0.04	EOS	2016QI
			BP	236.05			20.32	0.19	DvH eval	2019DIK
			BP	236.05			20.80		VP	1961LI
			BP	236.05			20.26	0.04	EOS-rev	2018LEM
			BP	235.69			21.03		VP	1975VID
C ₂ H ₄ F ₂	1,1-difluoroethane	gas	BP	249.13	SEL		21.79	0.72	EOS	1996OUT
			BP	249.13			21.8	0.7	DvH eval	2019DIK
			BP	249.13			21.791	0.715	EOS-rev	2018LEM

C ₂ H ₃ F ₃	1,1,1-trifluoroethane	gas	BP	246.66		23.24	22.33	VP	1947STU
			BP	225.91	SEL		19.05	0.56 EOS	2000LEM
			BP	225.91			19.1	0.6 DvH eval	2019DIK
			BP	225.91		21.60	20.76	VLE	2001KUL
			BP	225.91			18.61	DvH	1996WEB
			BP	225.91			19.34	DvH	1955MEA
			BP	225.91			19.20	0.10 DvH	1944RUS
			BP	225.91			19.05	0.56 EOS-rev	2018LEM
			BP	225.85		20.05	19.26	VP	1944RUS
	1,1,2-trifluoroethane	gas	BP	276.81	SEL(u)		25.7	0.4 DvH eval	2019DIK
C ₂ H ₂ F ₄	1,1,1,2-tetrafluoroethane	gas	BP	247.08	SEL		22.138	0.004 EOS	1994TIL
			BP	247.08			22.11	0.12 DvH eval	2019DIK
			BP	247.08			22.138	0.004 EOS-rev	2018LEM
	1,1,2,2-tetrafluoroethane	gas	BP	253.23	SEL		22.7	1.0 DvH eval	2019DIK
C ₂ HF ₅	pentafluoroethane	gas	BP	225.06	SEL		19.70	0.22 EOS	2005LEM
			BP	225.06			19.58	0.22 DvH eval	2019DIK
			BP	225.06			19.70	0.22 EOS-rev	2018LEM
C ₂ F ₆	hexafluoroethane	gas	BP	194.87	SEL		16.14	0.05 EOS	2006LEM
			BP	194.87			16.14	0.07 DvH eval	2019DIK
			BP	194.87			16.14	0.05 EOS-rev	2018LEM
			BP	194.87		16.96	16.30	VLE	1999TAK
Fluoroalkenes									
C ₂ H ₃ F	fluoroethene	gas	BP	201.65	SEL(u)		16.2	1.0 DvH est	2019DIK
			BP	201.65		15.69	15.08	VP	1947STU
C ₂ H ₂ F ₂	1,1-difluoroethene	gas	BP	190.15	SEL		14.9	0.4 DvH eval	2019DIK
	1,2-difluoroethene, (E)-	gas	BP	231.15	SEL(u)		20.6	1.1 DvH est	2019DIK
	1,2-difluoroethene, (Z)-	gas	BP	253.15	SEL(u)		16.4	0.9 DvH est	2019DIK
C ₂ HF ₃	trifluoroethene	gas	BP	211.7	SEL(u)		20.35	0.90 EOS	2016AKA
			BP	211.7			18.1	0.9 DvH est	2019DIK
			BP	211.7			20.35	0.90 EOS-rev	2018LEM
C ₂ F ₄	tetrafluoroethene	gas	BP	197.51	SEL		16.83	0.15 DvH eval	2019DIK
			BP	197.51		17.61	16.93	VP	1953FUR
Chloroalkanes									
CH ₃ Cl	chloromethane	gas	BP	249.0	SEL		21.61	0.41 EOS	2014THO
			BP	249.0		22.20	21.33	VP	1961LI
			BP	249.0			21.54	DvH	1940MES
			BP	249.0			21.9	0.4 DvH eval	2019DIK

CH ₂ Cl ₂	dichloromethane	liq	BP	249.0		21.613	0.409	EOS-rev	2018LEM
			BP	248.99	22.20	21.34		VP	1948GAN
			BP	249.31	22.35	21.48		VP	1947BEE
			BP	312.92	SEL	27.95	0.20	DvH	1976DEP
			BP	312.92		28.11	0.13	DvH eval	2019DIK
			BP	312.92	30.12	28.95		VLE	2010GAN
			BP	312.92	28.53	27.42		VLE	1994KRI
			BP	312.92	29.13	27.99	0.15	VLE	1960MUE
			BP	313.34	29.08	27.95		VP	1948GAN
CHCl ₃	trichloromethane	liq	BP	313.6		27.98	0.20	DvH	1926MAT
			BP	333.3	SEL	29.37	0.25	DvH	1926MAT
			BP	334.42		29.2	0.4	DvH eval	2019DIK
			BP	334.42		29.24		DvH rev	1985MAJ
			BP	334.42	30.86	29.66		VLE	1999TEO
			BP	334.42	30.68	29.48		VLE	1995CHE
CCl ₄	tetrachloromethane	liq	BP	334.42	30.66	29.46		VP	1947STU
			BP	348.6	SEL	29.96	0.40	DvH	1926MAT
			BP	349.65		29.6	0.5	DvH eval	2019DIK
			BP	349.65		29.82		DvH rev	1985MAJ
			BP	349.88	30.95	29.75		VP	1959HIL2
C ₂ H ₅ Cl	chloroethane	gas	BP	285.4	SEL	24.42	0.14	DvH eval	2019DIK
			BP	285.4	25.50	24.51		VP	1961LI
			BP	285.4	25.58	24.58		VP	1948GOR
C ₂ H ₄ Cl ₂	1,1-dichloroethane	liq	BP	330.42	SEL	29.02	0.16	DvH eval	2019DIK
			BP	330.42	30.45	29.26		VLE	1999IGO
			BP	330.42		28.85		DvH rev	1985MAJ
			BP	330.42	30.17	28.99		VP	1956LI
	1,2-dichloroethane	liq	BP	356.7	SEL	32.090	0.032	EOS	2017THO
			BP	356.7		31.9	0.4	DvH eval	2019DIK
			BP	356.7		31.98		DvH rev	1985MAJ
			BP	356.7	32.38	31.12		VLE	1973JAG
			BP	356.7		32.090	0.032	EOS-rev	2018LEM
			BP	356.98	33.62	32.31		VP	1929PEA
C ₂ H ₃ Cl ₃	1,1,1-trichloroethane	liq	BP	347.22	SEL(u)	29.69	0.20	DvH eval	2019DIK
			BP	347.22		29.86		DvH rev	1985MAJ
			BP	347.22	32.40	31.14		VLE	1977RAO
			BP	340.96	34.82	33.46		VP	1944RUB

	1,1,2-trichloroethane	liq	BP	386.78	SEL		34.94	0.23	DvH eval	2019DIK
			BP	386.78		38.11	36.63		VLE	1946TRE
			BP	386.82		36.11	34.70		VP	1949DRE
C ₂ H ₂ Cl ₄	1,1,1,2-tetrachloroethane	liq	BP	403.33	SEL		35.07	0.27	DvH eval	2019DIK
			BP	403.33		37.33	35.87		VP	1949DRE
	1,1,2,2-tetrachloroethane	liq	BP	419.25	SEL(u)		38.0	0.8	DvH eval	2019DIK
			BP	419.25		38.17	36.69		DvH	1978SUN
			BP	422.32		36.66	35.23		VP	1950MAT
			BP	418.2			38.64		DvH	1926MAT
C ₂ HCl ₅	pentachloroethane	liq	BP	433.45	SEL(u)		37.4	1.8	DvH est	2019DIK
			BP	433.45		37.40	35.94		VP rev	1943MCG
			BP	433.45		37.92	36.44		VP	1930NEL
C ₂ Cl ₆	hexachloroethane	sol	MP	458.15	SEL		9.75		MP rev	1996DOM
			BP	458.2	SEL1(u)		37.5	1.5	DvH est	2019DIK
			SP	439	SEL2		59.0	0.8	VP	1947IVI
			BP	462.96		44.60	42.87		VP	1947STU
Chloroalkenes										
C ₂ H ₃ Cl	chloroethene	gas	BP	259.443	SEL		22.31	0.07		2018LEM
			BP	259.35		22.97	22.07		VP	1949DRE
			BP	259.35			20.4	1.0	DvH eval	2019DIK
			BP	259.35		22.85	21.96		VP	1959MCD
C ₂ H ₂ Cl ₂	1,1-dichloroethene	liq	BP	304.71	SEL		26.46	0.26	DvH eval	2019DIK
			BP	304.71			26.14		DvH rev	1985MAJ
			BP	304.71		26.84	25.79		VP	1959HIL
			BP	304.71		27.07	26.02		VP	1959MCD
	1,2-dichloroethene, (E)-	liq	BP	320.73	SEL		27.9	0.4	DvH eval	2019DIK
			BP	320.73		29.07	27.94		VLE	1985MAC
			BP	320.73		28.67	27.55		VP	1947KET
			BP	320.73		29.89	28.7	0.5	VP	1943MCG
	1,2-dichloroethene, (Z)-	liq	BP	333.58	SEL		29.62	0.29	DvH eval	2019DIK
			BP	333.58		30.22	29.04		VP	1947KET
C ₂ HCl ₃	trichloroethene	liq	BP	358.8	SEL		31.47	0.25	DvH	1926MAT
			BP	360.5			31.23	0.28	DvH eval	2019DIK
			BP	360.5		32.13	30.88		VP	1944MCD
C ₂ Cl ₄	tetrachloroethene	liq	BP	394.29	SEL		34.41	0.24	DvH eval	2019DIK
			BP	394.29			34.68		DvH	1985MAJ
			BP	394.29		34.32	32.98		VLE	1996DEJ

			BP	394.29		36.23	34.82	VP	1972BOU				
			BP	394.29		36.00	34.60	VLE	1970POL				
Chloroaromatics													
C6H5Cl	chlorobenzene	liq	BP	405.21	SEL		35.53	0.44	EOS-rev	2018LEM			
			BP	405.21			37.49		36.03	VLE	1989RAM		
			BP	405.21			36.92		35.48	VLE	1973MOR		
			BP	405.21			36.89		35.45	VLE	1952BRO		
Bromoalkanes													
CH3Br	bromomethane	gas	BP	276.7	SEL		23.5	0.5	DvH eval	2019DIK			
			BP	276.7					23.9	0.3	DvH rev	1985MAJ	
			BP	276.7			24.70		23.74		Vp	1961LI	
			BP	276.7			24.90		23.93		VP	1947BEE	
			BP	276.7			24.81		23.84		VP	1938EGA	
			BP	276.7			24.96		23.98		VP	1931HIS	
CH2Br2	dibromomethane	liq	BP	370.1	SEL		33.4	0.4	DvH eval	2019DIK			
			BP	370.1			35.83		34.43		VLE ext	1973PHI	
			BP	371.67			34.82		33.46		VP	1947STU	
CHBr3	tribromomethane	liq	BP	422.36	SEL		37.5	0.3	DvH eval	2019DIK			
			BP	422.36		u			39.66		DvH rev	1985MAJ	
			BP	422.36					39.53	37.98		VP ext	1972BOU
			BP	421.83					38.20	36.71		VLE	1941SIM
CBr4	tetrabromomethane	sol	MP	367.60	SEL		3.95		MP rev	1996DOM			
			BP	462.55	SEL1	40.95	39.35		VP	1947STU			
			SP	298.15	SEL2		54.5	0.7	cal	1984BIC			
C2H5Br	bromoethane	liq	BP	311.54	SEL		26.9	0.3	DvH eval	2019DIK			
			BP	311.54			27.80		26.72		VP	1961LI	
			BP	311.54			27.94		26.85		VP	1930ZMA	
Bromoalkenes													
C2H3Br	bromoethene	liq	BP	311.54	SEL(u)		27.3	1.9	DvH est	2019DIK			
			BP	288.6			24.89		23.92		VP	1937GUY	
Chlorofluoromethanes													
CH2ClF	chlorofluoromethane	gas	BP	264.0	SEL(u)		19.7	1.1	DvH est	2019DIK			
CHClF2	chlorodifluoromethane	gas	BP	232.35	SEL		20.21	0.19	EOS	1995KAM			
			BP	232.35			20.21		0.19	DvH eval	2019DIK		
			BP	232.35			20.21		0.19	EOS-rev	2018LEM		
			BP	232.28			20.95		20.14		VP	1964KLE	
CHCl2F	dichlorofluoromethane	gas	BP	282.05	SEL		24.64	0.20	EOS	1990PLA			

			BP	276.7			25.21		EOS	1969ALT
			BP	282.27		25.52	24.52		VP	1947STU
			BP	282.05			25.3	0.5	DvH eval	2019DIK
			BP	282.05			24.64	0.20	EOS-rev	2018LEM
			BP	276.7		25.56	24.56		VP	1940BEN
CClF ₃	chlorotrifluoromethane	gas	BP	191.75	SEL		15.601	0.016	EOS	2000MAG
			BP	191.75		16.44	15.80		VP rev	1987STE
			BP	191.75		16.24	15.61	0.23	VP	1985FER
			BP	191.75		16.34	15.70		VP	1952ALB
			BP	191.75		15.99	15.36		VP	1933THO
			BP	191.75			15.61	0.02	DvH eval	2019DIK
			BP	191.75			15.601	0.200	EOS-rev	2018LEM
			BP	191.91		16.26	15.63		VP	1947STU
CCl ₂ F ₂	dichlorodifluoromethane	gas	BP	243.35	SEL		20.09	0.04	EOS	1992MAR
			BP	243.35			20.08	0.22	DvH eval	2019DIK
			BP	243.35			20.09	0.04	EOS-rev	2018LEM
			BP	243.35		21.05	20.23		VP	1992HAE
			BP	243.29		20.98	20.16		VP	1947STU
CCl ₃ F	trichlorofluoromethane	gas	BP	296.858	SEL		24.91	0.05	EOS	1992JAC
			BP	296.95			24.85	0.16	DvH eval	2019DIK
			BP	296.858			24.91	0.05	EOS-rev	2018LEM
			BP	296.78		25.94	24.93		VP	1941OSB
			BP	296.86		25.99	24.97		VP	1940BEN
C ₂ ClF ₃	chlorotrifluoroethene	gas	BP	244.8	SEL		20.71	0.25	DvH eval	2019DIK
			BP	244.8		21.77	20.93		VP	1951OLI
Bromofluoroalkanes										
CH ₂ BrF	bromofluoromethane	liq	BP	310.0	SEL(u)		30.6	1.6	DvH est	2019DIK
CHBrF ₂	bromodifluoromethane	gas	BP	258.65	SEL		22.5	1.0	DvH eval	2019DIK
			BP	258.65		23.95	23.01		VP	1992SAL
CBrF ₃	bromotrifluoromethane	gas	BP	215.35	SEL		17.48	0.17	DvH eval	2019DIK
			BP	215.26		18.27	17.56		VP	1959MCD
CBr ₂ F ₂	dibromodifluoromethane	gas	BP	297.65	SEL		24.45	0.19	DvH eval	2019DIK
			BP	295.94		25.53	24.53		VP	1959MCD
CBr ₃ F	tribromofluoromethane	liq	BP	379.65	SEL(u)		42.6	2.1	DvH est	2019DIK
Bromochloroalkanes										
CH ₂ BrCl	bromochloromethane	liq	BP	341.21	SEL		30.55	0.18	DvH eval	2019DIK
			BP	341.21		31.84	30.60		VP	1959MCD

CHBrCl ₂	bromodichloromethane	liq	BP	362.65	SEL(u)	31.8	1.7	DvH est	2019DIK
CHBr ₂ Cl	dibromochloromethane	liq	BP	393.0	SEL(u)	38.4	1.9	DvH est	2019DIK
CBrCl ₃	bromotrichloromethane	liq	BP	376.65	SEL(u)	32.1	1.4	DvH est	2019DIK
CBr ₂ Cl ₂	dibromodichloromethane	liq	BP	423.35	SEL(u)	34.1	1.8	DvH est	2019DIK
Bromochlorofluoroalkanes									
CHBrClF	bromochlorofluoromethane	liq	BP	312.6	SEL(u)	26.9	1.4	DvH est	2019DIK
CBrClF ₂	bromochlorodifluoromethane	gas	BP	270.65	SEL	22.15	0.10	DvH eval	2019DIK
			BP	269.19		23.21	22.31	VP	1960GLE
CBrCl ₂ F	bromodichlorofluoromethane	liq	BP	325.15	SEL(u)	29.5	1.6	DvH est	2019DIK
CBr ₂ ClF	dibromochlorofluoromethane	liq	BP	352.65	SEL(u)	32.9	1.7	DvH est	2019DIK
Carbonyl Halides									
CF ₂ O	carbonyl difluoride	gas	BP	188.58	SEL	18.92		VP	1968PAC
CCl ₂ O	phosgene	gas	BP	280.71	SEL	24.40	0.05	DvH	1948GIA
			BP	280.71		24.51		DvH est	1987STE
			BP	280.71		24.4	0.5	DvH eval	2019DIK
Chlorophenols									
C ₆ HCl ₅ O	pentachlorophenol	sol	MP	462.80	SEL	17.15		MP rev	1991ACR
			BP	582.5	SEL1(u)	59.1	2.8	DvH est	2019DIK
			SP	298.15	SEL2	91.64	0.22	VP	2007VER
			SP	403.2		88.4	0.3	VP	2007VER
			BP	582.5		60.73	58.36	VP eval	2019DIK
			BP	582.5		72.81		DvH rev	1947STU

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	<i>unc</i> (kJ mol ⁻¹)	Method	Reference
Siloxanes										
C ₆ H ₁₈ OSi ₂	hexamethyldisiloxane	liq	BP	373.66	SEL(u)		31.32		DvH	2011ABB
			BP	373.66		33.89			VP	2011ABB
			BP	373.66		33.70			VP	1961SCO
C ₈ H ₂₄ O ₄ Si ₄	octamethylcyclotetrasiloxane	liq	BP	448.02	SEL(u)		38.4	1.2	DvH eval	2019DIK
			MP	290.65	SEL1		23.77		MP rev	1996DOM
			BP	448.02		42.54			VP eval	2019DIK
			BP	448.02		42.91			VP	2011ABB
			BP	448.02		43.46			VP ext	1954OST

3.3. Recommended Transition Temperatures T_{trs} and Enthalpies of Transition $\Delta_{\text{trs}}H$

3.3.1. Data Tables

These are the recommended transition temperatures and enthalpies of transition. All compiled values are provided in Table 3-2. A discussion of uncertainties in enthalpies of vaporization follows in Section 3.3.2. See Table 3-1(2) for notation used for methods.

Table 3-4. Recommended transition temperatures T_{trs} and enthalpies of transition $\Delta_{\text{trs}}H$

This table is divided into sub-tables by compound class.

Table 3-4(1). Small Molecules

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	<i>unc</i> (kJ mol ⁻¹)	Method	Reference
Hydrogen/Oxygen										
H ₂	hydrogen	gas	BP	20.37	SEL		0.91	0.01	EOS	2009LEA
O ₂	oxygen	gas	BP	90.19	SEL		6.82	0.01	EOS	1991STE
H ₂ O	water	liq	BP	373.124	SEL		40.65	0.02	EOS	2002WAG
		sol	MP	273.15	SEL1		6.007	0.004	EOS	2006FEI
		sol	SP	273.15	SEL2		51.08	0.02	EOS	2006FEI
		liq	BP	426.4	SEL	46.43	45.73	0.46	VLE	2004MAN
H ₂ O ₂	hydrogen peroxide	sol	MP	272.699	SEL1		12.5	0.1	DfusH	1954GIG
		sol	MP	272.74	SEL2		52.8	5.3	DsubH est	2019DIK
Nitrogen/Hydrogen										
N ₂	nitrogen	gas	BP	77.36	SEL		5.58	0.00	EOS	2000SPA
H ₃ N	ammonia	gas	BP	239.82	SEL		23.33	0.01	EOS-rev	2018LEM
Nitrogen/Oxygen										
N ₂ O	nitrous oxide	gas	BP	184.68	SEL		184.75	0.07	cal	1974ATA
Sulfur/Hydrogen										
H ₂ S	hydrogen sulfide	gas	BP	212.85	SEL	19.10	18.35	0.25	VP	1996LOP
Boron/Hydrogen										
B ₂ H ₆	diborane	gas	BP	180.59	SEL	14.59		0.06	VP	1961DIT
Hydrogen Halides										
ClH	hydrogen chloride	gas	BP	188.173	SEL		16.18	0.08	EOS	2018LEM

Table 3-4(2). Hydrocarbons

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
Carbon										
C	graphite	sol	SB	2603	SEL		712.9	0.8	VP	1948BRE
Alkanes										
CH ₄	methane	gas	BP	111.67	SEL		8.20	0.06	EOS	1991SET
C ₂ H ₆	ethane	gas	BP	184.57	SEL		14.72	0.10	EOS	2006BUC
C ₃ H ₈	propane	gas	BP	231.04	SEL		18.77	0.19	EOS	2009LEM
C ₄ H ₁₀	isobutane	gas	BP	261.4	SEL		21.22	0.20	EOS	2006BUC
	n-butane	gas	BP	272.66	SEL		22.42	0.24	EOS	2006BUC2
C ₅ H ₁₂	isopentane	liq	BP	300.98	SEL		24.77	0.33	EOS	2006LEM
	neopentane	gas	BP	282.65	SEL		22.78	0.02	EOS	2006LEM
	n-pentane	liq	BP	309.21	SEL		25.81	0.11	EOS-rev	2018LEM
C ₆ H ₁₄	2,3-dimethylbutane	liq	BP	331.177	SEL		27.55	0.06	EOS	2018LEM
	3-methylpentane	liq	BP	336.43	SEL	29.49	28.34	0.30	VLE	1999LOR
	isohexane	liq	BP	333.42	SEL		27.84	0.31	EOS	2006LEM
	neohexane	liq	BP	322.846	SEL		26.41	0.11	EOS	2018LEM
	n-hexane	liq	BP	341.87	SEL		28.88	0.13	EOS-rev	2018LEM
	2,2,3-trimethylbutane	liq	BP	354.03	SEL		28.9	0.9	DvH eval	2019DIK
C ₇ H ₁₆	2,3-dimethylpentane	liq	BP	362.93	SEL		30.4	0.7	DvH eval	2019DIK
	3,3-dimethylpentane	liq	BP	359.21	SEL		29.5	0.4	DvH eval	2019DIK
	3-methylhexane	liq	BP	365.0	SEL		30.8	0.8	DvH eval	2019DIK
	isoheptane	liq	BP	363.2	SEL		30.7	0.7	DvH eval	2019DIK
	n-heptane	liq	BP	370.4	SEL		32.01	0.35	DvH	1926MAT
	2,2,4-trimethylpentane	liq	BP	372.36	SEL		31.00	0.40	DvH	1982SVO
C ₈ H ₁₈	2,2-dimethylhexane	liq	BP	379.77	SEL	33.57	32.26	0.50	VP	1965MCM
	n-octane	liq	BP	398.79	SEL		34.34	0.15	DvH eval	2019DIK
C ₉ H ₂₀	n-nonane	liq	BP	423.91	SEL		37.0	1.7	DvH eval	2019DIK
C ₁₀ H ₂₂	n-decane	liq	BP	447.27	SEL		39.2	0.6	DvH eval	2019DIK
C ₁₂ H ₂₆	n-dodecane	liq	BP	489.44	SEL		43.63	0.33		2004LEM
		sol	MP	263.49	SEL1		36.84		MP	1954FIN
		liq	BP	559.903	SEL		51.75	0.33		2018LEM
C ₁₆ H ₃₄	n-hexadecane	sol	MP	291.27	SEL1		53.36		MP	1954FIN
Alkenes										
C ₂ H ₄	ethene	gas	BP	169.38	SEL		13.53	0.04	EOS	2000SMU

C ₃ H ₆	propene	gas	BP	225.53	SEL	18.47	0.26	EOS-rev	2018LEM
C ₄ H ₈	1-butene	gas	BP	266.84	SEL	22.01	0.06		2005LEM2
	2-butene, (E)-	gas	BP	274.03	SEL	22.73	0.07		2005LEM2
	2-butene, (Z)-	gas	BP	276.87	SEL	23.24	0.05		2005LEM2
	isobutene	gas	BP	266.15	SEL	21.97	0.11		2005LEM2
C ₅ H ₁₀	1-pentene	liq	BP	303.101	SEL	25.215	0.010		2018LEM
	2-methyl-1-butene	liq	BP	304.3	SEL	25.50	0.27	DvH eval	2019DIK
	2-methyl-2-butene	liq	BP	311.72	SEL	26.17	0.28	DvH eval	2019DIK
	2-pentene, (E)-	liq	BP	309.51	SEL	26.00	0.22	DvH eval	2019DIK
	2-pentene, (Z)-	liq	BP	310.09	SEL	26.10	0.22	DvH eval	2019DIK
	3-methyl-1-butene	gas	BP	293.21	SEL	25.28	24.29	0.25 VP	1950SCO
C ₆ H ₁₂	1-hexene	liq	BP	336.64	SEL	29.44	28.29	0.30 VLE	2009MAR
C ₇ H ₁₄	1-heptene	liq	BP	366.71	SEL		31.20	0.26 DvH eval	2019DIK
C ₈ H ₁₆	1-octene	liq	BP	394.43	SEL		34.5	0.4 DvH eval	2019DIK
C ₉ H ₁₈	1-nonene	liq	BP	420.02	SEL	39.03	37.51	VP	1950FOR
C ₁₀ H ₂₀	1-decene	liq	BP	443.72	SEL		38.7	0.4 DvH eval	2019DIK
Alkadienes									
C ₃ H ₄	allene	gas	BP	240.874	SEL		19.98	0.30	2018LEM
C ₄ H ₆	1,2-butadiene	gas	BP	284.06	SEL		24.0	0.5 DvH est	2019DIK
	1,3-butadiene	gas	BP	268.44	SEL		22.29	0.25 DvH eval	2019DIK
C ₅ H ₈	1,2-pentadiene	liq	BP	318.8	SEL		27.24	0.20 DvH eval	2019DIK
	1,3-pentadiene	liq	BP	314.9	SEL		26.8	1.1 DvH eval	2019DIK
	1,4-pentadiene	liq	BP	299.12	SEL		25.07	0.21 DvH eval	2019DIK
	2-methyl-1,3-butadiene	liq	BP	307.19	SEL		25.71	0.24 DvH eval	2019DIK
Alkynes									
C ₂ H ₂	acetylene	gas	BP	188.41	SEL		17.2	0.4 VP eval	2019DIK
		sol	TP	192.0	SEL1		3.8	0.4 TP eval	2019DIK
C ₃ H ₄	propyne	gas	BP	249.8	SEL		23.5	0.5 DvH est	2019DIK
C ₄ H ₆	1-butyne	gas	BP	281.23	SEL		24.70	0.05	2018LEM
C ₅ H ₈	1-pentyne	liq	BP	313.51	SEL		25.4	1.1 DvH eval	2019DIK
	2-pentyne	liq	BP	329.27	SEL		25.0	1.1 DvH eval	2019DIK
C ₆ H ₁₀	1-hexyne	liq	BP	344.53	SEL		30.0	0.4 DvH eval	2019DIK
	2-hexyne	liq	BP	357.59	SEL	33.52	32.21	VP eval	2019DIK
	3-hexyne	liq	BP	354.58	SEL		31.6	0.4 DvH eval	2019DIK
Alkenynes									
C ₄ H ₄	butenyne	gas	BP	279.18	SEL		23.9	0.7 DvH eval	2019DIK
C ₅ H ₆	3-penten-1-yne	liq	BP	317.05	SEL		26.6	1.4 DvH eval	2019DIK

Alkadiynes									
C ₄ H ₂	butadiyne	gas	BP	283.65	SEL	24.0	1.0	DvH eval	2019DIK
Cycloalkanes									
C ₃ H ₆	cyclopropane	gas	BP	240.27	SEL	20.0	0.6	DvH eval	2019DIK
C ₄ H ₈	cyclobutane	gas	BP	285.64	SEL	23.7	0.4	DvH eval	2019DIK
C ₅ H ₁₀	cyclopentane	liq	BP	322.4	SEL	27.30	0.20	DvH	1959MCC2
	methylcyclobutane	liq	BP	310.0	SEL	25.3	1.2	DvH eval	2019DIK
C ₆ H ₁₂	cyclohexane	liq	BP	353.8649	SEL	29.99	0.15		2014ZHO
	methylcyclopentane	liq	BP	344.96	SEL	29.07	0.10	DvH	1959MCC2
C ₇ H ₁₄	1,1-dimethylcyclopentane	liq	BP	360.99	SEL	31.72	30.48	VP	1949FOR
	ethylcyclopentane	liq	BP	376.61	SEL	33.58	32.27	VP	1949FOR
	methylcyclohexane	liq	BP	374	SEL	31.23	na		2018LEM
C ₈ H ₁₆	isopropylcyclopentane	liq	BP	399.57	SEL	35.56	34.17	VP	1949FOR
	n-propylcyclopentane	liq	BP	404.1	SEL	36.57	35.15	VP	1949FOR
C ₉ H ₁₈	isopropylcyclohexane	liq	BP	427.71	SEL	38.41	36.91	VP	1949FOR
	propylcyclohexane	liq	BP	429.87	SEL	38.87	37.36	VP	1949FOR
C ₁₀ H ₂₀	1,3-diethylcyclohexane	liq	BP	445.7	SEL	37.5	1.2	DvH eval	2019DIK
	isobutylcyclohexane	liq	BP	444.47	SEL	40.27	38.69	VP	1949FOR
	tert-butylcyclohexane	liq	BP	444.74	SEL	39.86	38.30	VP	1949FOR
Cycloalkenes									
C ₃ H ₄	cyclopropene	gas	BP	237.7	SEL	20.50	0.60		2019DIK
C ₄ H ₆	cyclobutene	gas	BP	275.73	SEL	23.19	0.46		2018LEM
C ₅ H ₈	cyclopentene	liq	BP	317.39	SEL	26.9	0.4	DvH eval	2019DIK
	methylenecyclobutane	liq	BP	315.0	SEL	25.3	1.2	DvH eval	2019DIK
C ₆ H ₁₀	cyclohexene	liq	BP	356.1	SEL	30.66	0.25	DvH	1973SVO
Cycloalkadienes									
C ₅ H ₆	1,3-cyclopentadiene	liq	BP	314.7	SEL	25.00	24.02	VLE	1967LES
C ₆ H ₈	1,3-cyclohexadiene	liq	BP	353.49	SEL	31.49	30.26	VP	1973MEY
Bicycloalkanes									
C ₁₀ H ₁₈	cis-decalin	liq	BP	468.93	SEL	39.9	0.4	DvH eval	2019DIK
	trans-decalin	liq	BP	460.42	SEL	38.9	0.4	DvH eval	2019DIK
C ₁₂ H ₂₂	cyclohexylcyclohexane	liq	BP	511.1	SEL	43.86	0.28	DvH eval	2019DIK
		sol	MP	276.80	SEL1	6.86		MP	2002CHI

Table 3-4(3). Aromatic Hydrocarbons

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
Alkylbenzenes										
C ₆ H ₆	benzene	liq	BP	353.22	SEL		30.75	0.06	EOS	2012THO
		sol	MP	278.67	SEL1		9.87	0.01	MP	1948OLI
C ₇ H ₈	toluene	liq	BP	383.75	SEL		33.23	0.16	EOS	2006LEM
		sol	MP	178.15	SEL1		6.64	0.01	MP	1962SCO
C ₈ H ₁₀	1,2-dimethylbenzene	liq	BP	417.521	SEL		36.38	0.18		2012ZHO
	1,3-dimethylbenzene	liq	BP	412.214	SEL		36.11	0.07		2012ZHO
	1,4-dimethylbenzene	liq	BP	411.47	SEL		35.71	0.07		2012ZHO
	ethylbenzene	liq	BP	409.314	SEL		35.60	0.11		2012ZHO
C ₉ H ₁₂	1,2,3-trimethylbenzene	liq	BP	449.23	SEL	41.95	40.31		VP	1949FOR
	1,2,4-trimethylbenzene	liq	BP	442.5	SEL	41.29	39.68		VP	1949FOR
	1,3,5-trimethylbenzene	liq	BP	437.87	SEL	41.12	39.52		VP	1949FOR
	2-ethyltoluene	liq	BP	438.25	SEL	40.70	39.11		VP	1949FOR
	3-ethyltoluene	liq	BP	434.46	SEL	40.40	38.83		VP	1949FOR
	4-ethyltoluene	liq	BP	435.14	SEL	40.24	38.67		VP	1949FOR
	isopropylbenzene	liq	BP	425.54	SEL		37.0	0.4	DvH eval	2019DIK
	n-propylbenzene	liq	BP	432.32	SEL	39.75	38.20		VP	1949FOR
C ₁₀ H ₁₄	1,2-diethylbenzene	Liq	BP	456.57	SEL	42.71	41.04		VP	1949FOR
	1,3-diethylbenzene	Liq	BP	454.25	SEL	42.65	40.98		VP	1949FOR
	1,4-diethylbenzene	Liq	BP	456.9	SEL	42.68	41.01		VP	1949FOR
	2-isopropyltoluene	liq	BP	451.5	SEL		39.3	0.7	DvH eval	2019DIK
	3-isopropyltoluene	liq	BP	448.35	SEL		39.3	0.7	DvH eval	2019DIK
	4-isopropyltoluene	liq	BP	450.27	SEL		39.3	0.4	DvH eval	2019DIK
	n-butylbenzene	liq	BP	456.42	SEL	42.53	40.87		VP	1949FOR
	tert-butylbenzene	liq	BP	442.27	SEL	40.61	39.03		VP	1949FOR
Alkenylbenzenes										
C ₈ H ₈	ethenylbenzene	liq	BP	418.45	SEL		36.8	0.4	DvH eval	2019DIK
Benzocycloalkanes										
C ₁₀ H ₁₂	benzocyclohexane	liq	BP	480.56	SEL		42.6	0.5	DvH eval	2019DIK
Benzocycloalkenes										
C ₉ H ₈	benzocyclopentene	liq	BP	454.72	SEL		40.36	0.4	DvH eval	2019DIK

Alkyl naphthalenes									
C ₁₀ H ₈	naphthalene	sol	MP	353.40	SEL	18.993	0.019	MP	2002CHI
			BP	491.11	SEL1	43.3	0.8	DvH eval	2019DIK
			TP	353.37	SEL2	70.38	0.3	VP	2005RUZ
C ₁₁ H ₁₀	1-methylnaphthalene	liq	BP	517.84	SEL	46.3	0.3	DvH eval	2019DIK
		sol	MP	242.65	SEL1	6.95		MP	1957MCC
	2-methylnaphthalene	sol	MP	307.70	SEL	12.13		MP rev	1996DOM
			BP	514.2	SEL1	48.23	46.35	VP	1955CAM
Polycyclic Aromatics									
C ₁₂ H ₈	acenaphthylene	sol	MP	366.40	SEL	10.96		MP rev	1996DOM
		sol	MP	550.60	SEL1	46.4	2.4	DvH est	2019DIK
	biphenylene	sol	MP	384.40	SEL	82.7		DsubH est	1995NAS
			BP	548.50	SEL1	46.3	1.7	DvH est	2019DIK
C ₁₂ H ₁₀	acenaphthene	sol	MP	364.02	SEL	21.46		MP	1977FIN
			BP	550.39	SEL1	53.88	51.78	VP	1923MOR
C ₁₃ H ₁₀	fluorene	sol	MP	387.90	SEL	19.10		MP	2000LIS
			BP	570.19	SEL1	56.87	54.65	VP	1923MOR
C ₁₄ H ₁₀	anthracene	sol	MP	489.50	SEL	29.37		MP	1970GOU
			BP	614.87	SEL1	58.54	56.25	VP	1923MOR
	phenanthrene	sol	MP	372.74	SEL	16.60		MP	2003ROJ
			BP	609.0	SEL1	59.56	57.24	VP	1975OSB
C ₁₆ H ₁₀	fluoranthene	sol	MP	381.00	SEL	18.73		MP	1971WON
			BP	659.82	SEL1	51.46	49.45	VP	1955TSY
	pyrene	sol	MP	424.30	SEL	16.70		MP	2003ROJ
			BP	668.17	SEL1	56.54	54.34	VP	1955TSY
C ₂₀ H ₁₂	perylene	sol	MP	550.90	SEL	32.58		MP	1992SAB
C ₂₄ H ₁₂	coronene	sol	MP	710.50	SEL	21.20		MP	2009TOR

Table 3-4(4). Oxidized Hydrocarbons

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
Carbon Oxides										
CO ₂	carbon dioxide	sol	SP	194.67	SEL		25.230	0.002	DsubH	1937GIA
		sol	TP	216.57	SEL1		25.64	0.25	TP eval	2019DIK
CO	carbon monoxide	gas	BP	81.64	SEL		6.01	0.02	EOS	2006LEM
Alkanols										
CH ₄ O	methanol	liq	BP	337.63	SEL		35.28	0.21	EOS	1993REU
C ₂ H ₆ O	ethanol	liq	BP	351.57	SEL		39.14	0.47	EOS	2014SCH
C ₃ H ₈ O	1-propanol	liq	BP	370.26	SEL		41.7	0.4	DvH eval	2019DIK
	isopropanol	liq	BP	354.4	SEL		40.04	0.25	DvH	1926MAT
C ₄ H ₁₀ O	1-butanol	liq	BP	390.77	SEL		42.9	0.5	DvH eval	2019DIK
	2-butanol	liq	BP	372.66	SEL	42.90	41.22		VP	1963BID
	isobutanol	liq	BP	381.04	SEL	44.01	42.29	0.40	VP	1963BID
	tert-butanol	sol	MP	298.87	SEL		6.78		MP	1926PAR
			BP	355.5	SEL1		39.15	0.22	DvH eval	2019DIK
			SP	298.15	SEL2		49	5	est	2022BUR
C ₅ H ₁₂ O	1-pentanol	liq	BP	410.9	SEL		44.2	0.6	DvH eval	2019DIK
	2-methyl-1-butanol	liq	BP	402.14	SEL		43.5	0.5	DvH eval	2019DIK
	2-pentanol	liq	BP	392.29	SEL	44.53	42.79		VP	1935BUT
	isopentanol	liq	BP	404.73	SEL		43.6	0.3	DvH eval	2019DIK
Glycols										
C ₂ H ₆ O ₂	1,2-ethanediol	liq	BP	470.313	SEL		54.56	na		2018LEM
C ₃ H ₈ O ₂	1,2-propanediol	liq	BP	461.22	SEL	53.58	51.49		VP	2002STE
	2-methoxyethanol	liq	BP	397.54	SEL		39.59	0.26	DvH eval	2019DIK
Sugars										
C ₃ H ₈ O ₃	glycerin	liq	BP	563.0	SEL		63.3	1.5	DvH eval	2019DIK
		sol	MP	291.05	SEL1		18.29		MP	1923GIB
C ₆ H ₁₂ O ₆	alpha-D-glucose	sol	MP	414	SEL		31.4		MP rev	2019DIK
		liq	BP	706	SEL1(u)		84.4	4.2	DvH est	2019DIK
		sol	SP	425	SEL2		194	5	VP	1999OJA
C ₁₂ H ₂₂ O ₁₁	sucrose	sol	MP	424.4	SEL		32		cal	2014MAG
			SP	298.15	SEL2		94		est	2022BUR
			MP	456.40			46.20		MP	1988SOP

Phenols										
C ₇ H ₈ O	2-methylphenol	sol	MP	304.20	SEL		15.82		MP	1967AND
			BP	464.14	SEL1	46.53	44.71		VP	1949DRE
	4-methylphenol	sol	MP	307.84	SEL		12.71		MP	1967AND
			BP	474.99	SEL1	49.29	47.36		VP	1949DRE
Alkyl Ethers										
C ₂ H ₆ O	dimethyl ether	gas	BP	248.37	SEL		21.43	0.90	EOS	2011WU
C ₄ H ₁₀ O	diethyl ether	liq	BP	307.56	SEL		26.55	0.27	EOS	2014THO
C ₆ H ₁₄ O	diisopropyl ether	liq	BP	341.49	SEL		28.99	0.17	DvH eval	2019DIK
	di-n-propyl ether	liq	BP	363.25	SEL	33.16	31.86		VP	1973MEY
Aldehydes										
CH ₂ O	formaldehyde	gas	BP	254.13	SEL		22.0	0.4	DvH est	2019DIK
C ₂ H ₄ O	acetaldehyde	liq	BP	299.12	SEL		25.07	0.21	DvH eval	2019DIK
C ₃ H ₆ O	propanal	liq	BP	321.19	SEL		28.3	0.4	DvH eval	2019DIK
C ₄ H ₈ O	butanal	liq	BP	347.95	SEL(u)		29.32	0.28	DvH eval	2019DIK
	isobutanal	liq	BP	337.26	SEL	30.73	29.54		VP	1959SEP
Alkenals										
C ₃ H ₄ O	2-propenal	liq	BP	325.4	SEL		29.6	0.3	DvH eval	2019DIK
Ketones										
C ₃ H ₆ O	acetone	liq	BP	329.22	SEL		29.12	0.27	EOS	2006LEM
C ₄ H ₈ O	butanone	liq	BP	352.74	SEL		31.59	0.16	DvH eval	2019DIK
C ₅ H ₁₀ O	3-pentanone	liq	BP	375.11	SEL	35.31	33.93		VP	1965COL
Carboxylic Acids										
CH ₂ O ₂	formic acid	liq	BP	374.05	SEL		22.7	8.4	DvH est	2019DIK
		sol	MP	281.47	SEL1		12.68		MP	1941STO
C ₂ H ₄ O ₂	acetic acid	liq	BP	391.2	SEL	38.67	37.16		VLE	2019LI
		sol	MP	289.49	SEL1		11.72		MP	1982MAR
C ₃ H ₆ O	propanoic acid	liq	BP	413.99	SEL		29.91	0.7	DvH	1989JAS
C ₄ H ₈ O	n-butanoic acid	liq	BP	436.42	SEL		38.4	2.4	DvH eval	2019DIK
Alkyl Esters										
C ₂ H ₄ O ₂	methyl formate	liq	BP	305.14	SEL		27.90	0.25	DvH	1976CIH
C ₃ H ₆ O ₂	ethyl formate	liq	BP	327.29	SEL		29.80	0.20	DvH	1976CIH
C ₄ H ₈ O ₂	ethyl acetate	liq	BP	349.2	SEL		32.30	0.30	DvH	1926MAT
Alkenyl Esters										
C ₄ H ₆ O ₂	vinyl acetate	liq	BP	345.33	SEL		31.7	0.3	DvH eval	2019DIK
C ₅ H ₈ O ₂	ethyl propenoate	liq	BP	372.59	SEL		34.2	0.4	DvH eval	2019DIK
	methyl 2-methylpropenoate	liq	BP	369.73	SEL		34.14	0.28	DvH eval	2019DIK

Oxacycloalkanes									
C ₂ H ₄ O	oxacyclopropane	gas	BP	283.59	SEL	25.47	0.19	EOS	2015THO
C ₃ H ₆ O	methyloxacyclopropane	liq	BP	307.268	SEL	27.59	0.14	EOS-rev	2018LEM
Oxacycloalkadienes									
C ₄ H ₄ O	oxacyclopentadiene	liq	BP	304.51	SEL	27.03	0.18	DvH eval	2019DIK
C ₅ H ₆ O	2-methyloxacyclopentadiene	liq	BP	338.39	SEL	30.86	29.66	VP	1971EON
Oxacycloketones									
C ₄ H ₆ O	cyclobutanone	liq	BP	367.79	SEL	37.15	35.70	VP	1942BEN
C ₆ H ₁₀ O	cyclohexanone	liq	BP	428.57	SEL	39.48	37.94	VP	1973MEY
Dioxanes and trioxanes									
C ₃ H ₆ O ₃	1,3,5-trioxane	sol	MP	333.44	SEL	15.105	0.010	MP	1988VAN
			BP	387.2	SEL1	37.3	2.1	DvH est	2019DIK
			SP	298.15	SEL2	56.6	0.2	C	1969MAN
C ₃ H ₆ O ₃	1,3,5-trioxane	sol	MP	333.65	SEL	15.105		MP	1988VAN
			BP	387.2	SEL1	37.3	2.1	DvH est	2019DIK
C ₄ H ₈ O ₂	1,3-dioxane	liq	BP	378.13	SEL	34.3	2.4	DvH est	2019DIK
	1,4-dioxane	liq	BP	374.25	SEL	35.58	0.3	DvH	1963VIN
Furandione and D-Camphor									
C ₄ H ₂ O ₃	2,5-furandione	sol	MP	325.70	SEL	12.26		MP	1978MAR
			BP	478.33	SEL1	44.7	2.3	DvH est	2019DIK
			SP	319.0	SEL2	70.72	0.42	cal	2017SOU
C ₁₀ H ₁₆ O	D-Camphor	sol	MP	463.38	SEL(u)	15.7		DfusH rev	1990DON
			BP	493.0	SEL1(u)	44.0	1.8	DvH est	2019DIK
			SP	298.15	SEL2	51.9	0.8	VP	1977STE
Aromatic Alcohols									
C ₆ H ₆ O	phenol	sol	MP	314.05	SEL	11.54		MP	1963AND
			BP	454.91	SEL1	45.47	0.23	DvH eval	2019DIK
C ₇ H ₈ O	3-methylphenol	liq	BP	475.28	SEL	47.2	0.5	DvH eval	2019DIK
		sol	MP	285.37	SEL1	10.67		MP	1998JAM
		benzyl alcohol	liq	BP	477.4	SEL	50.49	1.00	DvH
Aromatic Aldehydes and Acids									
C ₇ H ₆ O	benzaldehyde	liq	BP	451.15	SEL	40.9	0.3	DvH eval	2019DIK
C ₇ H ₆ O ₂	benzoic acid	sol	MP	395.50	SEL	17.10		MP	2002ROY
			BP	523.53	SEL1	56.5	1.0	DvH eval	2019DIK
			SP	298.15	SEL2	89.25	0.42	VP	1990DAS

Table 3-4(5). Nitrogen-Substituted Hydrocarbons

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
Hydrogen Cyanides										
CHN	hydrogen cyanide	liq	BP	298.79	SEL		25.22	0.25	DvH eval	2019DIK
Alkyl Amines										
CH ₅ N	methylamine	gas	BP	266.84	SEL		25.94	0.23	DvH eval	2019DIK
C ₃ H ₉ N	n-propylamine	liq	BP	320.38	SEL		29.58	0.20	DvH eval	2019DIK
	isopropylamine	liq	BP	304.92	SEL	29.03	27.90		VP	1968OSB
C ₄ H ₁₁ N	n-butylamine	liq	BP	350.12	SEL		32.03	0.27	DvH eval	2019DIK
	tert-butylamine	liq	BP	317.19	SEL	29.53	28.38		VP	1968OSB
C ₂ H ₇ N	dimethylamine	gas	BP	280.02	SEL		26.48	0.06	DvH	1939AST
C ₃ H ₉ N	methylethylamine	liq	BP	305.6	SEL		28.4	0.5	DvH eval	2019DIK
	trimethylamine	gas	BP	276.0	SEL		23.00	0.22	DvH eval	2019DIK
C ₆ H ₁₅ N	triethylamine	liq	BP	361.93	SEL		31.3	0.4	DvH eval	2019DIK
C ₂ H ₈ N ₂	1,1-dimethylhydrazine	liq	BP	335.57	SEL		32.1	0.9	DvH eval	2019DIK
Cycloalkylamines										
C ₆ H ₁₃ N	cyclohexylamine	liq	BP	406.98	SEL		35.8	0.6	DvH eval	2019DIK
Azacycloalkanes										
C ₄ H ₉ N	azacyclopentane	liq	BP	359.71	SEL	34.33	32.99		VP	1959MCC
C ₅ H ₁₁ N	azacyclohexane	liq	BP	379.37	SEL	35.14	33.77		VP	1968OSB
Methenamines										
C ₆ H ₁₂ N ₄	methenamine	sol	SP	298.15	SEL		79.64	0.18	VP	2002VER
Azacycloalkadienes										
C ₄ H ₅ N	azacyclopentadiene	liq	BP	402.93	SEL	40.33	38.76		VP	1968OSB
C ₅ H ₇ N	1-methylazacyclopentadiene	liq	BP	385.9	SEL	36.82	35.38		VP	1968OSB
Alkyl nitriles										
C ₂ H ₃ N	ethanenitrile	liq	BP	355.05	SEL		30.7	0.3	DvH eval	2019DIK
C ₃ H ₅ N	propanenitrile	liq	BP	370.73	SEL		33.0	0.4	DvH eval	2019DIK
C ₄ H ₇ N	butanenitrile	liq	BP	390.77	SEL		34.70	0.24	DvH eval	2019DIK
C ₅ H ₉ N	pentanenitrile	liq	BP	414.42	SEL	38.54	37.04		VP	1949DRE
C ₆ H ₁₁ N	hexanenitrile	liq	BP	436.61	SEL	41.24	39.63		VP	1973MEY
C ₃ H ₃ N	propenenitrile	liq	BP	351.13	SEL(u)		30.7	0.7	DvH eval	2019DIK
C ₂ N ₂	cyanogen	gas	BP	252.0	SEL(u)		24.7	0.6	DvH eval	2019DIK
C ₂ H ₈ N ₂	1,1-dimethylhydrazine	liq	BP	335.57	SEL		32.1	0.9	DvH eval	2019DIK
Melamines										

C ₃ H ₆ N ₆	melamine	liq	SP	497	SEL	120	4	VP	1960HIR
			BP	639	SEL1	79	4	DvH est	2019DIK
Anilines									
C ₆ H ₇ N	aniline	liq	BP	457.3	SEL	43.8	0.5	DvH eval	2019DIK
			BP	456.99		45.31		VP	1962HAT
C ₈ H ₁₁ N	N,N-dimethylaniline	liq	BP	466.47	SEL	42.7	0.8	DvH eval	2019DIK
		sol	MP	275.65	SEL1	11.56		MP	1972AHM
Pyridines									
C ₅ H ₅ N	pyridine	liq	BP	388.36	SEL	37.85		VP	1996CHI
C ₆ H ₇ N	2-methylpyridine	liq	BP	402.52	SEL	37.75		VP	1963SCO
Oxidized Amines									
CH ₄ N ₂ O	urea	sol	MP	406.5	SEL	14.79		MP	1987DEL
			SP	298.15	SEL1	94.6	2.0	VP	2003ZAI
C ₆ H ₁₅ NO ₃	triethanolamine	liq	BP	623.15	SEL	82.0	2.0	VP	2004WRI
Alkyl Nitrates and Nitroalkanes									
CH ₃ NO ₃	methyl nitrate	liq	BP	338.0	SEL(u)	26.4	1.4	DvH est	2019DIK
C ₃ H ₅ N ₃ O ₉	nitroglycerin	sol	MP	286.65	SEL	21.87		MP rev	1991ACR
CN ₄ O ₈	tetranitromethane	liq	BP	397.95	SEL	40.74		DvH rev	1985MAJ
		sol	SP	287.05	SEL1	47.40		DsubH	1941SEK
CH ₃ NO ₂	nitromethane	liq	BP	373.1	SEL	34.46	0.30	DvH	1926MAT
Isocyanates									
C ₉ H ₆ N ₂ O ₂	2,4-diisocyanatotoluene	liq	BP	524.83	SEL(u)	41.9	2.1	DvH est	2019DIK
Nitro Aromatics									
C ₆ H ₅ NO ₂	nitrobenzene	liq	BP	483.81	SEL	43.9	0.5	DvH eval	2019DIK
		sol	MP	278.89	SEL1	12.12		MP	1936PAR
C ₇ H ₅ N ₃ O ₆	trinitrotoluene	sol	MP	355.1	SEL	23.43		DfusH	1993ACR
			BP	620.5	SEL1	73.69		VP	1948ROB
			SP	298.15	SEL2	113.2	1.5	VP	1979KUD

Table 3-4(6). Sulfur-Substituted Hydrocarbons

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(IG)$ (kJ mol ⁻¹)	$\Delta_{trs} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
Alkanethiols										
CH ₄ S	methanethiol	gas	BP	279.05	SEL		24.4	0.9	DvH eval	2019DIK
C ₂ H ₆ S	ethanethiol	liq	BP	308.19	SEL		26.6	0.6	DvH eval	2019DIK
C ₃ H ₈ S	1-propanethiol	liq	BP	340.87	SEL	30.80	29.60		VP	1956PEN
	isopropanethiol	liq	BP	325.71	SEL	29.12	27.99		VP	1966OSB
C ₄ H ₁₀ S	1-butanethiol	liq	BP	371.56	SEL	33.79	32.47		VP	1966OSB
	1-isobutanethiol	liq	BP	361.64	SEL	32.48	31.22		VP	1966OSB
	2-butanethiol	liq	BP	358.13	SEL	32.04	30.79		VP	1966OSB
C ₅ H ₁₂ S	1,1-dimethyl-1-propanethiol	liq	BP	372.28	SEL	32.95	31.66		VP	1966OSB
	1-pentanethiol	liq	BP	399.79	SEL	36.69	35.26		VP	1966OSB
C ₆ H ₁₄ S	1-hexanethiol	liq	BP	425.81	SEL	39.46	37.92		VP	1966OSB
C ₇ H ₁₆ S	1-heptanethiol	liq	BP	450.07	SEL	42.15	40.50		VP	1966OSB
Alkyl Sulfides										
CS ₂	carbon disulfide	liq	BP	318.4	SEL		26.78	0.15	DvH	1926MAT
C ₂ H ₆ S ₂	dimethyl disulfide	liq	BP	382.89	SEL	34.98	33.61		VP	1950SCO
C ₂ H ₆ S	dimethyl sulfide	liq	BP	310.42	SEL		27.0	0.3	DvH eval	2019DIK
C ₃ H ₈ S	ethyl methyl sulfide	liq	BP	339.8	SEL	30.78	29.58		VP	1952WHI
C ₄ H ₁₀ S	diethyl sulfide	liq	BP	365.25	SEL	33.26	31.97		VP	1966OSB
Thiacycloalkanes										
C ₄ H ₈ S	thiacyclopentane	liq	BP	394.27	SEL	35.94	34.53		VP	1966OSB
Thiacycloalkenes										
C ₄ H ₄ S	thiacyclopentadiene	liq	BP	357.31	SEL	32.63	31.35		VP	1949WAD
Aromatic Thiols										
C ₆ H ₆ S	benzenethiol	liq	BP	442.29	SEL	41.35	39.74		VP	1966OSB
Alkyl Sulfoxides										
C ₂ H ₆ OS	dimethyl sulfoxide	liq	BP	463.86	SEL		43.5	0.4	DvH eval	2019DIK
		sol	MP	291.57	SEL1		14.37		MP	1970CLE

Table 3-4(7). Halogenated Hydrocarbons

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
Fluoroalkanes										
CH ₃ F	fluoromethane	gas	BP	194.65	SEL		16.60	0.46	EOS	2006LEM
CH ₂ F ₂	difluoromethane	gas	BP	221.5	SEL		19.87	0.20	EOS	1997TIL
CHF ₃	trifluoromethane	gas	BP	188.75	SEL		16.76	0.15	EOS	2003PEN
CF ₄	tetrafluoromethane	gas	BP	145.56	SEL		11.83	0.27	EOS	1990PLA
C ₂ H ₅ F	fluoroethane	gas	BP	236.05	SEL		20.26	0.04	EOS	2016QI
C ₂ H ₄ F ₂	1,1-difluoroethane	gas	BP	249.13	SEL		21.79	0.72	EOS	1996OUT
C ₂ H ₃ F ₃	1,1,1-trifluoroethane	gas	BP	225.91	SEL		19.05	0.56	EOS	2000LEM
	1,1,2-trifluoroethane	gas	BP	276.81	SEL(u)		25.7	0.4	DvH est	2019DIK
C ₂ H ₂ F ₄	1,1,1,2-tetrafluoroethane	gas	BP	247.08	SEL		22.138	0.004	EOS	1994TIL
	1,1,2,2-tetrafluoroethane	gas	BP	253.23	SEL		22.7	1.0	DvH eval	2019DIK
C ₂ HF ₅	pentafluoroethane	gas	BP	225.06	SEL		19.70	0.22	EOS	2005LEM
C ₂ F ₆	hexafluoroethane	gas	BP	194.87	SEL		16.14	0.05	EOS	2006LEM
Fluoroethenes										
C ₂ H ₃ F	fluoroethene	gas	BP	201.65	SEL(u)		16.2	1.0	DvH est	2019DIK
C ₂ H ₂ F ₂	1,1-difluoroethene	gas	BP	190.15	SEL		14.9	0.4	DvH eval	2019DIK
	1,2-difluoroethene, (E)-	gas	BP	231.15	SEL(u)		20.6	1.1	DvH est	2019DIK
	1,2-difluoroethene, (Z)-	gas	BP	253.15	SEL(u)		16.4	0.9	DvH est	2019DIK
C ₂ HF ₃	trifluoroethene	gas	BP	211.7	SEL(u)		20.35	0.90	EOS	2016AKA
C ₂ F ₄	tetrafluoroethene	gas	BP	197.51	SEL		16.83	0.15	DvH eval	2019DIK
Chloromethanes										
CH ₃ Cl	chloromethane	gas	BP	249.0	SEL		21.61	0.41	EOS	2014THO
CH ₂ Cl ₂	dichloromethane	liq	BP	312.92	SEL		27.95	0.20	DvH	1976DEP
CHCl ₃	trichloromethane	liq	BP	333.3	SEL		29.37	0.25	DvH	1926MAT
CCl ₄	tetrachloromethane	liq	BP	348.6	SEL		29.96	0.40	DvH	1926MAT
Chloroethanes										
C ₂ H ₅ Cl	chloroethane	gas	BP	285.4	SEL		24.42	0.14	DvH eval	2019DIK
C ₂ H ₄ Cl ₂	1,1-dichloroethane	liq	BP	330.42	SEL		29.02	0.16	DvH eval	2019DIK
	1,2-dichloroethane	liq	BP	356.7	SEL		32.090	0.032	EOS	2017THO
C ₂ H ₃ Cl ₃	1,1,1-trichloroethane	liq	BP	347.22	SEL(u)		29.69	0.20	DvH eval	2019DIK
	1,1,2-trichloroethane	liq	BP	386.78	SEL		34.94	0.23	DvH eval	2019DIK

C ₂ H ₂ Cl ₄	1,1,1,2-tetrachloroethane	liq	BP	403.33	SEL	35.07	0.27	DvH eval	2019DIK
	1,1,2,2-tetrachloroethane	liq	BP	419.25	SEL(u)	38.0	0.8	DvH eval	2019DIK
C ₂ HCl ₅	pentachloroethane	liq	BP	433.45	SEL(u)	37.4	1.8	DvH est	2019DIK
C ₂ Cl ₆	hexachloroethane	sol	MP	458.15	SEL	9.75		MP rev	1996DOM
			BP	458.2	SEL1(u)	37.5	1.5	DvH est	2019DIK
			SP	439	SEL2	59.0	0.8	VP	1947IVI
Chloroethenes									
C ₂ H ₃ Cl	chloroethene	gas	BP	259.443	SEL	22.31	0.07		2018LEM
C ₂ H ₂ Cl ₂	1,1-dichloroethene	liq	BP	304.71	SEL	26.46	0.26	DvH eval	2019DIK
	1,2-dichloroethene, (E)-	liq	BP	320.73	SEL	27.9	0.4	DvH eval	2019DIK
	1,2-dichloroethene, (Z)-	liq	BP	333.58	SEL	29.62	0.29	DvH eval	2019DIK
C ₂ HCl ₃	trichloroethene	liq	BP	358.8	SEL	31.47	0.25	DvH	1926MAT
C ₂ Cl ₄	tetrachloroethene	liq	BP	394.29	SEL	34.41	0.24	DvH eval	2019DIK
Chloroaromatics									
C ₆ H ₅ Cl	chlorobenzene	liq	BP	405.21	SEL	35.53	0.44	EOS-rev	2018LEM
Bromoalkanes									
CH ₃ Br	bromomethane	gas	BP	276.7	SEL	23.5	0.5	DvH eval	2019DIK
CH ₂ Br ₂	dibromomethane	liq	BP	370.1	SEL	33.4	0.4	DvH eval	2019DIK
CHBr ₃	tribromomethane	liq	BP	422.36	SEL	37.5	0.3	DvH eval	2019DIK
CBr ₄	tetrabromomethane	sol	MP	367.60	SEL	3.95		MP rev	1996DOM
			BP	462.55	SEL1	40.95	39.35	VP	1947STU
			SP	298.15	SEL2	54.5	0.7	cal	1984BIC
C ₂ H ₅ Br	bromoethane	liq	BP	311.54	SEL	26.9	0.3	DvH eval	2019DIK
Bromoethenes									
C ₂ H ₃ Br	bromoethene	liq	BP	311.54	SEL(u)	27.3	1.9	DvH est	2019DIK
Chlorofluoromethanes									
CH ₂ ClF	chlorofluoromethane	gas	BP	264.0	SEL(u)	19.7	1.1	DvH est	2019DIK
CHClF ₂	chlorodifluoromethane	gas	BP	232.35	SEL	20.21	0.19	EOS	1995KAM
CHCl ₂ F	dichlorofluoromethane	gas	BP	282.05	SEL	24.64	0.20	EOS	1990PLA
CClF ₃	chlorotrifluoromethane	gas	BP	191.75	SEL	15.601	0.016	EOS	2000MAG
CCl ₂ F ₂	dichlorodifluoromethane	gas	BP	243.35	SEL	20.09	0.04	EOS	1992MAR
CCl ₃ F	trichlorofluoromethane	gas	BP	296.858	SEL	24.91	0.05	EOS	1992JAC
C ₂ ClF ₃	chlorotrifluoroethene	gas	BP	244.8	SEL	20.71	0.25	DvH eval	2019DIK
Bromofluoromethanes									
CH ₂ BrF	bromofluoromethane	liq	BP	310.0	SEL(u)	30.6	1.6	DvH est	2019DIK
CHBrF ₂	bromodifluoromethane	gas	BP	258.65	SEL	22.5	1.0	DvH eval	2019DIK
CBrF ₃	bromotrifluoromethane	gas	BP	215.35	SEL	17.48	0.17	DvH eval	2019DIK

CBr ₂ F ₂	dibromodifluoromethane	gas	BP	297.65	SEL	24.45	0.19	DvH eval	2019DIK
CBr ₃ F	tribromofluoromethane	liq	BP	379.65	SEL(u)	42.6	2.1	DvH est	2019DIK
Bromochloromethanes									
CH ₂ BrCl	bromochloromethane	liq	BP	341.21	SEL	30.55	0.18	DvH eval	2019DIK
CHBrCl ₂	bromodichloromethane	liq	BP	362.65	SEL(u)	31.8	1.7	DvH est	2019DIK
CHBr ₂ Cl	dibromochloromethane	liq	BP	393.0	SEL(u)	38.4	1.9	DvH est	2019DIK
CBrCl ₃	bromotrichloromethane	liq	BP	376.65	SEL(u)	32.1	1.4	DvH est	2019DIK
CBr ₂ Cl ₂	dibromodichloromethane	liq	BP	423.35	SEL(u)	34.1	1.8	DvH est	2019DIK
Bromochlorofluoromethanes									
CHBrClF	bromochlorofluoromethane	liq	BP	312.6	SEL(u)	26.9	1.4	DvH est	2019DIK
CBrClF ₂	bromochlorodifluoromethane	gas	BP	270.65	SEL	22.15	0.10	DvH eval	2019DIK
CBrCl ₂ F	bromodichlorofluoromethane	liq	BP	325.15	SEL(u)	29.5	1.6	DvH est	2019DIK
CBr ₂ ClF	dibromochlorofluoromethane	liq	BP	352.65	SEL(u)	32.9	1.7	DvH est	2019DIK
Carbonyl Halides									
CF ₂ O	carbonyl difluoride	gas	BP	188.58	SEL	18.92		VP	1968PAC
CCl ₂ O	phosgene	gas	BP	280.71	SEL	24.40	0.05	DvH	1948GIA
Chlorophenols									
C ₆ HCl ₅ O	pentachlorophenol	sol	MP	462.80	SEL	17.15		MP rev	1991ACR
			BP	582.5	SEL1(u)	59.1	2.8	DvH est	2019DIK
			SP	298.15	SEL2	91.64	0.22	VP	2007VER

Table 3-4(8). Siloxanes

Formula	Compound	Phase	trs	T_{trs} (K)	SEL	$\Delta_v H(\text{IG})$ (kJ mol ⁻¹)	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc (kJ mol ⁻¹)	Method	Reference
C ₆ H ₁₈ OSi ₂	hexamethyldisiloxane	liq	BP	373.66	SEL(u)		31.32		DvH	2011ABB
C ₈ H ₂₄ O ₄ Si ₄	octamethylcyclotetrasiloxane	liq	BP	448.02	SEL(u)		38.4	1.2	DvH eval	2019DIK
		sol	MP	290.65	SEL1		23.77		MP rev	1996DOM

3.3.2. Discussion of Uncertainties in Heats of Vaporization

As discussed earlier, in Section 3.1, we derived an average compressibility factor of 0.961 ± 0.0055 (1σ), or an uncertainty of only about 0.6 %; this is less than the uncertainties in experimental determinations of latent enthalpies of vaporization, which are on the order of (1 to 2) %. We used this compressibility factor to estimated corrected real gas enthalpies of vaporization $\Delta_v H$ from ideal gas enthalpies of vaporization $\Delta_v H(\text{IG})$, derived using vapor pressure curves $\ln(P)$ vs. $(1/T)$.

A good source of normal boiling points and latent enthalpies of vaporizations for reference fluids can be found in REFPROP[2018LEM]. It does not, however, explicitly provide uncertainties in normal boiling points or enthalpies of vaporization, but can be derived from other information contained in REFPROP. Table 3-3(9) below provides normal boiling points and enthalpies of vaporization along with uncertainties for 75 compounds in NIST REFPROP (plus formic acid and 1,2-propanediol from NIST TRC TDE).[2019DIK] The change in pressure with temperature $\delta P/\delta T$ at the normal boiling point is also given. The uncertainties in normal boiling points were calculated using the uncertainties in pressure and $\delta P/\delta T$ as provided in NIST REFPROP (or TDE). Note here that we use the infinitesimal change notations $\delta P/\delta T$, $\delta \ln(P/P_0)$, and $\delta(1/T)$ since the values were computed numerically rather than the analytical derivative notations dP/dT , $d\ln(P/P_0)$, and $d(1/T)$.

Considering the equation for enthalpies (heats) of vaporization:

$$(3.5) \quad \Delta_v H = -R \delta \ln(P/P_0)/\delta(1/T)$$

$$(3.6) \quad U(\Delta_v H) \approx c R U(T)$$

where c is a proportionality constant, and R is the gas constant.

We considered all of the 77 compounds in Table 3-4(9) and calculated the uncertainties in the enthalpies of vaporization $U(\Delta_v H)$ using:

$$(3.7) \quad \Delta_v H = -R \delta \ln(P/P_0)/\delta(1/T)$$

$$(3.8) \quad \Delta_v H = -R \Delta \ln(P) / \Delta(1/T)$$

$$(3.9) \quad \Delta_v H = -R \{ \ln(P_1) - \ln(P_0) \} / (1/T_1 - 1/T_0)$$

where T_1 and T_0 were derived from the boiling points T_b for each compound using $P_1 = 2$ atm and $P_0 = 1/2$ atm, respectively (or 2σ centered about $P = 1$ atm). $U(\Delta_v H)$ was then determined by the difference using the expanded uncertainty in pressure $U(P) = \Delta P$:

$$(3.10) \quad U(\Delta_v H) = -R [\ln(2+\Delta P) - \ln(0.5-\Delta P)] / \Delta(1/T) - \Delta_v H$$

That is, at T_1 the pressure is higher by the uncertainty in pressure $U(P) = \Delta P$ and at T_0 the pressure is lower by $U(P) = \Delta P$. Consequently, the change in slope (determined numerically) can be used to derive the uncertainty in the enthalpy of vaporization $U(\Delta_v H)$.

We found a good correlation (see Figure 3-1) for the uncertainty in the enthalpies of vaporization with:

$$(3.11) \quad U(\Delta_v H) [\text{kJ mol}^{-1}] = 1.684 [\text{J mol}^{-1} \text{K}^{-1}] U(T_b), \text{ or}$$

$$(3.12) \quad U(\Delta_v H) [\text{kJ mol}^{-1}] = 0.2025 R U(T_b), \text{ or}$$

$$(3.13) \quad U(\Delta_v H) [\text{kJ mol}^{-1}] = 0.2025 R U(P)/(\delta P/\delta T)$$

having an expanded uncertainty (2σ) of 0.039 % (see Figure 3-2). This expanded uncertainty corresponds to about $U(\Delta_v H) = 0.010 \text{ kJ mol}^{-1}$ or very nearly zero. See Table 3-4(10) for the compounds with the largest deviations. In this analysis, we excluded 4 compounds that had deviations greater than 4σ – formic acid, 1,2-ethanediol, 1,2-propanediol, and *n*-hexadecane. This expanded uncertainty is much less than the significance in reported uncertainties in pressure $U(P)$ which is on the order of a factor of (2 to 2.5), since $U(P)$ provided with equation-of-state (EOS) formulations is usually reported in round numbers such as (0.02, 0.05, 0.1, 0.2, 0.5) %. This analysis is purely an enthalpic correlation and does not include an entropic correction. When we include an entropic correction (derived below), this uncertainty can be reduced about a factor of five to 0.008 % (see Figure 3-3). This uncertainty corresponds to about $U(\Delta_v H) = 0.002 \text{ kJ mol}^{-1}$, or essentially zero. See Table 3-4(10) for the compounds with the largest deviations. Here, we excluded 5 compounds (hydrogen, formic acid, *n*-undecane, *n*-dodecane, and *n*-hexadecane) that had deviations greater than 4σ . Each of these compounds is a special case. Hydrogen has very low boiling temperature. Formic acid forms dimers in the gas phase, and thus, it is an extremely non-ideal gas with a compressibility factor Z of about 0.6 at the normal boiling point.[1999MIY] For *n*-undecane, *n*-dodecane and *n*-hexadecane, the uncertainties in the normal boiling temperatures are likely much higher than those computed from the estimated uncertainty in pressure (provided in NIST REFPROP)[2018LEM] of (0.2 %) and $U(P)/(\delta P/\delta T)$ – or $U(T_b)$ of (0.5 to 1.5) K versus 0.1 K; correspondingly, the uncertainties in enthalpies of vaporization would also be much higher than computed in the same fashion – or $U(\Delta_v H)$ of (1 to 2) kJ mol^{-1} versus 0.2 kJ mol^{-1} : a factor of (5 to 10) higher.

We now derive the entropic correction. Trouton's Rule says that the molar entropy of vaporization for all liquids at their boiling points is approximately the same and can be used to estimate enthalpies of vaporization from normal boiling points: $S_b = \Delta_v H/T_b \approx (85 \text{ to } 88) \text{ J mol}^{-1} \text{ K}^{-1}$. [1884TRO] The entropy of vaporization ΔS_b is the difference between the entropy in the gas phase and that in the liquid phase $S_b = S_{\text{gas}} - S_{\text{liq}}$. However, $S_{\text{gas}} \gg S_{\text{liq}}$, and, consequently, the entropy of vaporization is a measure of the cohesive energy holding the molecules together in the liquid phase. For small molecules, such as for hydrogen, methane, oxygen, and nitrogen, this energy is less and the entropy of vaporization is less – about (44.4, 73.4, and 75.6, and 82.1) $\text{J mol}^{-1} \text{ K}^{-1}$, respectively, see Table 3-4(9). Increased order in the liquid phase and increased order in the gas phase increase and decrease the entropy of vaporization, respectively. For polar molecules, where there is hydrogen bonding (ordered structure) in the liquid phase, such as for ethanol, water, and ammonia, this energy is more and the entropy of vaporization is more – about (111.3, 108.9, and 97.3) $\text{J mol}^{-1} \text{ K}^{-1}$, respectively. In contrast, for some molecules, there is hydrogen bonding (ordered structure) in the gas phase, such as for dimers; thus, the entropy of vaporization is less – for example, about 60.6 $\text{J mol}^{-1} \text{ K}^{-1}$ for formic acid. Water is a special case, where there is both hydrogen bonding in the liquid and gas phases – consequently, the entropy of vaporization of water of 108.9 $\text{J mol}^{-1} \text{ K}^{-1}$ is less than that for ethanol, because of the offset due to hydrogen bonding of water in the gas phase.

From our analysis, we find corrections to the enthalpies of vaporization due to entropic contributions, where the corrected latent enthalpy $U(\Delta_v H)'$ is given by:

$$(3.14) \quad U'(\Delta_v H) = U(\Delta_v H) (\Delta_v H / T_b)^2 / S_{b0}^2$$

$$(3.15) \quad U'(\Delta_v H) = U(\Delta_v H) S_b^2 / S_{b0}^2$$

Here, $S_{b0} = 86.1 \text{ J mol}^{-1} \text{ K}^{-1}$ is a reference entropy of vaporization at the boiling point (statistically identical to Trouton's Rule from 140 years ago). Thus, the overall uncertainty in a latent enthalpy of vaporization $\Delta_v H$ can be calculated using:

$$(3.16) \quad U(\Delta_v H) [\text{kJ mol}^{-1}] = 0.2025 R U(P) S_b^2 / S_{b0}^2 / (\delta P / \delta T)$$

where, R is the gas constant, $U(P)$ is the uncertainty in pressure, T_b is the normal boiling point, $(\delta P / \delta T)$ is the change in pressure P with temperature, $S_b = \Delta_v H / T_b$ is the latent entropy of vaporization, and $S_{b0} = (86.1 \pm 1.3) \text{ J mol}^{-1} \text{ K}^{-1}$ is the reference entropy of vaporization at the boiling point as determined from a least squares fit to this expression. This uncertainty in S_{b0} of about 1.5 % is entirely due to the reported (NIST REFPROP) estimated and rounded uncertainties in pressure (*e.g.*, 0.001, 0.0015, 0.002, 0.0025, 0.003, 0.004, 0.005, 0.007).

Table 3-4(9). Derived uncertainties in EOS boiling points

This table provides boiling points (T_b), change in pressure with temperature ($\delta P/\delta T$), latent enthalpy of vaporization ($\Delta_v H$), Trouton's Rule entropy of vaporization ($S_b = \Delta_v H/T_b$). The quantities T_b , $U(P)$, $\delta P/\delta T$, and $\Delta_v H$ were taken from NIST REFPROP.[2018LEM] $U(T_b)$ was calculated from $U(P)$ and $\delta P/\delta T$. $U(\Delta_v H)$ was computed (see text). $U(P)$ values that are underlined were not taken from NIST REFPROP estimates but taken directly from uncertainties provided in the literature. $U(P)$ for 1,2-dimethylbenzene was assigned the same as for the other two dimethylbenzenes. Note: a legend for the column headings is given in Table 3-3(11). Those molecules whose Trouton's rule S_b entropies of vaporization differ substantially are provided in **bold** in this table. The low values are for the small, largely non-polar molecules reflects that lack of cohesive energy in the liquid state, while the high values are for largely polar molecules, where there is hydrogen bonding in the liquid phase.

Compound	T_b (K)	$U(T_b)$ (K)	$U(P)$ (atm)	$\delta P/\delta T$ (atm K ⁻¹)	$\Delta_v H$ (kJ mol ⁻¹)	$U(\Delta_v H)$ (kJ mol ⁻¹)	S_b (J mol ⁻¹ K ⁻¹)
hydrogen	20.369	0.007	0.002	0.295	0.905	0.003	44.4
nitrogen	77.355	0.0017	0.0002	0.118	5.58	0.002	72.1
carbon monoxide	81.640	0.0018	0.0002	0.114	6.01	0.002	73.7
oxygen	90.1878	0.0143	0.0015	0.105	6.82	0.018	75.6
methane	111.667	0.004	0.0003	0.082	8.20	0.004	73.4
tetrafluoromethane	145.10	0.07	<u>0.005</u>	0.070	11.83	0.11	81.5
ethene	169.379	0.008	0.0005	0.059	13.53	0.012	79.9
ethane	184.566	0.002	0.0001	0.054	14.72	0.003	79.7
nitrous oxide	184.68	0.03	0.002	0.060	16.47	0.06	89.2
hydrogen chloride	188.173	0.09	0.005	0.057	16.18	0.15	86.0
trifluoromethane	191.13	0.03	0.002	0.058	16.76	0.06	87.7
chlorotrifluoromethane	191.67	0.02	0.001	0.054	15.60	0.03	81.4
fluoromethane	194.84	0.05	0.0025	0.055	16.60	0.07	85.2
hexafluoroethane	195.06	0.06	0.003	0.054	16.14	0.09	82.7
hydrogen sulfide	212.85	0.05	0.0025	0.051	18.62	0.08	87.5
trifluoroethene	214.060	0.018	0.001	0.056	20.35	0.04	95.0
difluoromethane	221.499	0.004	0.0002	0.051	19.87	0.007	89.7
pentafluoroethane	225.06	0.02	0.001	0.049	19.70	0.04	87.5
propene	225.531	0.004	0.0002	0.045	18.47	0.007	81.9
1,1,1-trifluoroethane	225.91	0.02	0.0010	0.047	19.05	0.03	84.3
propane	231.039	0.045	0.002	0.044	18.77	0.07	81.2
chlorodifluoromethane	232.35	0.04	0.002	0.047	20.21	0.07	87.0
fluoroethane	235.61	0.04	0.002	0.046	20.26	0.07	86.0
ammonia	239.832	0.010	0.0005	0.050	23.33	0.02	97.3
allene	240.87	0.12	<u>0.0053</u>	0.043	19.98	0.19	82.9

dichlorodifluoromethane	243.40	0.05	0.002	0.043	20.09	0.07	82.5
1,1,1,2-tetrafluoroethane	247.076	0.004	0.0002	0.046	22.14	0.008	89.6
dimethyl ether	248.37	0.05	0.002	0.044	21.43	0.08	86.3
1,1-difluoroethane	249.130	0.004	0.0002	0.046	22.14	0.008	88.9
chloromethane	249.17	0.05	0.002	0.044	21.61	0.08	86.7
propyne	250.0	0.1	<u>0.004</u>	0.045	22.60	0.17	90.4
chloroethene	259.44	0.07	0.003	0.041	22.31	0.12	86.0
isobutane	261.40	0.05	<u>0.002</u>	0.039	21.22	0.08	81.2
sulfur dioxide	263.137	0.04	0.002	0.045	24.96	0.09	94.8
isobutene	266.15	0.08	<u>0.003</u>	0.039	21.97	0.12	82.6
1-butene	266.84	0.06	0.0025	0.039	22.01	0.10	82.5
<i>n</i> -butane	272.660	0.005	0.0002	0.038	22.42	0.008	82.2
2-butene, (<i>E</i>)-	274.03	0.08	0.003	0.038	22.73	0.12	83.0
cyclobutene	275.73	0.18	<u>0.007</u>	0.038	23.19	0.29	84.1
2-butene, (<i>Z</i>)-	276.87	0.05	0.002	0.038	23.24	0.08	83.9
1-butyne	281.23	0.05	0.002	0.039	24.70	0.09	87.8
dichlorofluoromethane	282.01	0.13	0.005	0.039	24.64	0.22	87.4
neopentane	282.65	0.03	0.001	0.036	22.78	0.04	80.6
oxacyclopropane	283.66	0.04	<u>0.002</u>	0.039	25.47	0.08	89.8
trichlorofluoromethane	296.858	0.06	0.002	0.035	24.91	0.09	83.9
isopentane	300.98	0.02	<u>0.0007</u>	0.035	24.77	0.03	82.3
1-pentene	303.101	0.012	0.0004	0.035	25.21	0.02	83.2
methyloxacyclopropane	307.268	0.027	<u>0.001</u>	0.036	27.59	0.05	89.8
diethyl ether	307.56	0.11	0.004	0.036	26.55	0.19	86.3
<i>n</i> -pentane	309.21	0.06	0.002	0.034	25.81	0.09	83.5
cyclopentane	322.4	0.03	0.001	0.033	27.30	0.05	84.7
neohexane	322.85	0.05	<u>0.002</u>	0.032	26.41	0.08	81.8
acetone	329.22	0.07	0.0025	0.034	29.12	0.13	88.5
2,3-dimethylbutane	331.18	0.06	0.002	0.032	27.55	0.10	83.2
isohexane	333.36	0.06	0.002	0.032	27.84	0.10	83.5
methanol	337.63	0.01	<u>0.0005</u>	0.039	35.28	0.03	104.5
<i>n</i> -hexane	341.87	0.06	0.002	0.031	28.88	0.10	84.5
ethanol	351.57	0.01	<u>0.0005</u>	0.039	39.14	0.04	111.3
benzene	353.219	0.006	0.0002	0.031	30.75	0.011	87.1
cyclohexane	353.87	0.02	0.0005	0.030	29.99	0.03	84.8
1,2-dichloroethane	356.65	0.03	0.001	0.031	32.09	0.06	90.0
<i>n</i> -heptane	371.55	0.07	0.002	0.029	31.73	0.11	85.4

water	373.124	0.007	0.00025	0.036	40.65	0.02	108.9
methylcyclohexane	374.0	0.1	<u>0.0026</u>	0.028	31.23	0.15	83.5
formic acid	374.345	0.235	<u>0.007</u>	0.024	22.70	0.28	60.6
toluene	383.750	0.018	0.0005	0.028	33.23	0.03	86.6
<i>n</i> -octane	398.794	0.07	0.002	0.028	34.44	0.12	86.4
chlorobenzene	405.21	0.11	<u>0.003</u>	0.027	35.53	0.19	87.7
ethylbenzene	409.31	0.11	0.003	0.027	35.60	0.19	87.0
1,4-dimethylbenzene	411.47	0.07	0.002	0.027	35.71	0.13	86.8
1,3-dimethylbenzene	412.214	0.075	0.002	0.027	36.11	0.13	87.6
1,2-dimethylbenzene	417.52	0.08	<u>0.002</u>	0.026	36.38	0.13	87.1
<i>n</i> -nonane	423.91	0.08	0.002	0.026	37.01	0.13	87.3
<i>n</i> -decane	447.27	0.08	0.002	0.025	39.33	0.14	87.9
1,2-propanediol	461.224	0.033	0.001	0.030	47.91	0.09	103.9
<i>n</i> -undecane	468.93	0.08	0.002	0.025	41.43	0.15	88.3
1,2-ethanediol	470.313	0.033	0.001	0.030	54.56	0.10	116.0
<i>n</i> -dodecane	489.44	0.08	0.002	0.024	43.63	0.16	89.1
<i>n</i> -hexadecane	559.90	0.09	0.002	0.022	51.75	0.19	92.4

Table 3-4(10). Analytical expression for derived uncertainties in heats of vaporization

This table demonstrates that the analytic expression of equation (3-16) accurately derives uncertainties in the latent heats of vaporization $U(\Delta_v H)$ $U(T_b)$ to within $0.006 \text{ kJ mol}^{-1}$ or about 0.02 %. This table provides correlations for compounds with the largest deviations between uncertainties in latent enthalpies of vaporization with uncertainties in temperature/pressure and latent entropies of vaporizations. Note: a legend for the column headings is given in Table 3-4(11). The quantity $f = U_0(\Delta_v H)/U(\Delta_v H)$ is a ratio of the correlated uncertainty versus derived uncertainty. Note that the uncertainties in computed uncertainties in the latent heats of vaporization using the uncertainties in pressure $U(P)$ and computed entropies of vaporization $S_b = \Delta_v H/T_b$ using equation (3-16) are less than about $0.006 \text{ kJ mol}^{-1}$ or about 0.02 % – negligible amounts. The exception here is the special case of formic acid with significant hydrogen bonding, existing in the gas phase as a dimer, with a derived uncertainty in the enthalpy of vaporization of $22.7 \pm 0.1 \text{ kJ mol}^{-1}$ or about 0.5 % – still a small number. We note that the uncertainty in the boiling point of formic acid $T_b = 374.35 \pm 0.24 \text{ K}$ (about 6 %) is large compared to the other molecules which have a median of about $U(T_b) = 0.05 \text{ K}$ and generally less than (0.08 to 0.12) K, which translates in a median uncertainty of about 0.02 % and generally less than (0.03 to 0.05) %.

Compound	T_b (K)	$\Delta_v H$ (kJ mol ⁻¹)	S_b (J mol ⁻¹ K ⁻¹)	S_b/S_0	$U(\Delta_v H)$ (kJ mol ⁻¹)	$U_0(\Delta_v H)$ (kJ mol ⁻¹)	$\Delta U(H)$ (kJ mol ⁻¹)	$\Delta U(H)/L$	f	$\Delta U(H,S)$ (kJ mol ⁻¹)	$\Delta U(H,S)/L$
hydrogen	20.37	0.905	44.41	0.52	0.003	0.011	-0.008	-0.90%	3.49	-0.0002	-0.025%
nitrogen	77.36	5.58	72.1	0.84	0.002	0.003	-0.0008	-0.02%	1.42	-0.00001	0.000%
carbon monoxide	81.64	6.01	73.7	0.86	0.002	0.003	-0.0008	-0.013%	1.36	0.00000	0.000%
oxygen	90.19	6.82	75.6	0.88	0.018	0.024	-0.006	-0.08%	1.31	0.0002	0.002%
methane	111.67	8.20	73.4	0.85	0.004	0.006	-0.002	-0.02%	1.38	0.00002	0.000%
tetrafluoromethane	145.10	11.83	81.5	0.95	0.107	0.120	-0.013	-0.11%	1.12	0.0003	0.003%
ammonia	239.83	23.33	97.3	1.13	0.021	0.017	0.004	0.02%	0.80	0.0004	0.002%
sulfur dioxide	263.14	24.96	94.8	1.10	0.090	0.075	0.015	0.06%	0.84	0.0014	0.005%
methanol	337.63	35.28	104.5	1.21	0.032	0.021	0.010	0.03%	0.67	-0.0003	-0.001%
ethanol	351.57	39.14	111.3	1.29	0.035	0.021	0.014	0.04%	0.60	0.0004	0.001%
water	373.12	40.65	108.9	1.27	0.018	0.012	0.007	0.02%	0.64	0.0006	0.001%
formic acid	374.345	22.7	60.6	0.70	0.284	0.396	-0.111	-0.49%	1.39	0.11	0.49%
<i>n</i> -octane	398.79	34.44	86.4	1.00	0.124	0.121	0.003	0.009%	0.98	-0.0023	-0.007%
<i>n</i> -nonane	423.91	37.01	87.3	1.01	0.134	0.127	0.006	0.02%	0.95	-0.0028	-0.008%
<i>n</i> -decane	447.27	39.33	87.9	1.02	0.142	0.132	0.010	0.02%	0.93	-0.0038	-0.010%
1,2-propanediol	461.22	47.91	103.9	1.21	0.085	0.055	0.030	0.06%	0.65	-0.0049	-0.010%
<i>n</i> -undecane	468.93	41.43	88.3	1.03	0.150	0.137	0.012	0.03%	0.92	-0.0049	-0.012%
1,2-ethanediol	470.31	54.56	116.0	1.35	0.097	0.055	0.042	0.08%	0.57	0.0032	0.006%
<i>n</i> -dodecane	489.44	43.63	89.1	1.04	0.158	0.142	0.016	0.04%	0.90	-0.0058	-0.013%
<i>n</i> -hexadecane	559.90	51.75	92.4	1.07	0.187	0.155	0.032	0.06%	0.83	-0.0082	-0.016%

Table 3-4(11). Legend for column headings in Tables 3-4(10) and 3-4(11)

Symbol	Definition
T_b	Normal boiling point
$\Delta_v H$	Latent enthalpy of vaporization
S_b	Latent entropy of vaporization $\Delta_v H/T_b$
S_0	Reference entropy of vaporization ($86.1 \text{ J mol}^{-1} \text{ K}^{-1}$)
$\delta P/\delta T$	Change in pressure with temperature
$U(T)$	Expanded uncertainty in T
$U(P)$	Expanded uncertainty in P
$U(\Delta_v H)$	Derived expanded uncertainty in $\Delta_v H$
$U_0(\Delta_v H)$	Correlated expanded uncertainty $U(\Delta_v H) = 1.684 U(T_b)$, see text
$\Delta U(H)$	Enthalpic (H) deviation of expanded uncertainty (with $U(T_b)$ correlation)
$\Delta U(H)/L$	Deviation of $\Delta U(H)$ relative to latent enthalpy of vaporization L (%)
$\Delta U(H,S)$	Enthalpic (H) deviation of expanded uncertainty (with both $U(T_b)$ and S_b correlations)
$\Delta U(H,S)/L$	Deviation of $\Delta U(H,S)$ relative to latent enthalpy of vaporization L (%)
f	$U_0(\Delta_v H)/U(\Delta_v H)$ correlated versus derived uncertainty

Figure 3-1. Correlation between uncertainty in latent enthalpy of vaporization $U(\Delta_v H)$ and uncertainty in boiling point $U(T_b)$

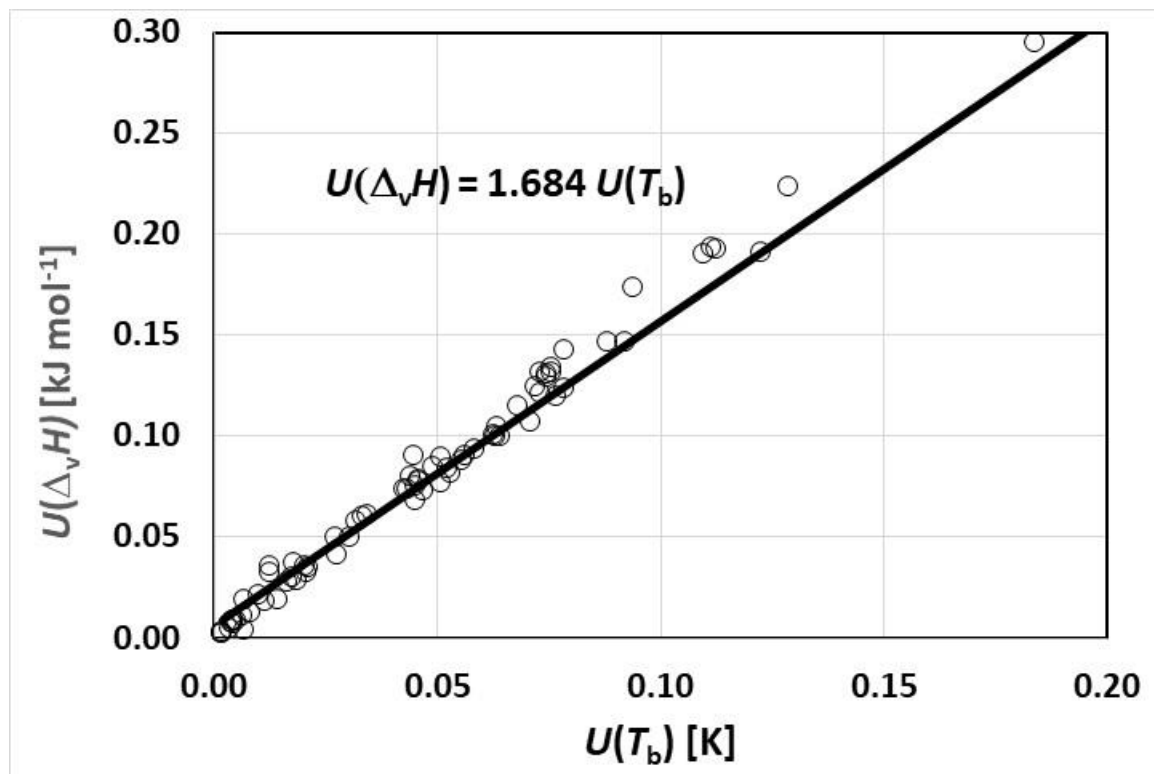


Figure 3-2. Relative deviation in uncertainty $U_{\Delta T}(\Delta_v H)/\Delta_v H$ in enthalpy of vaporization from correlated value

Relative deviation in uncertainty $U_{\Delta T}(\Delta_v H)/\Delta_v H$ in enthalpy of vaporization from $U(\Delta_v H) = 1.684 \text{ (kJ mol}^{-1}\text{)}$ $U(T_b)$ correlation.
Expanded deviation (2σ) is 0.039 % (about 0.01 kJ mol^{-1}) – very nearly zero.

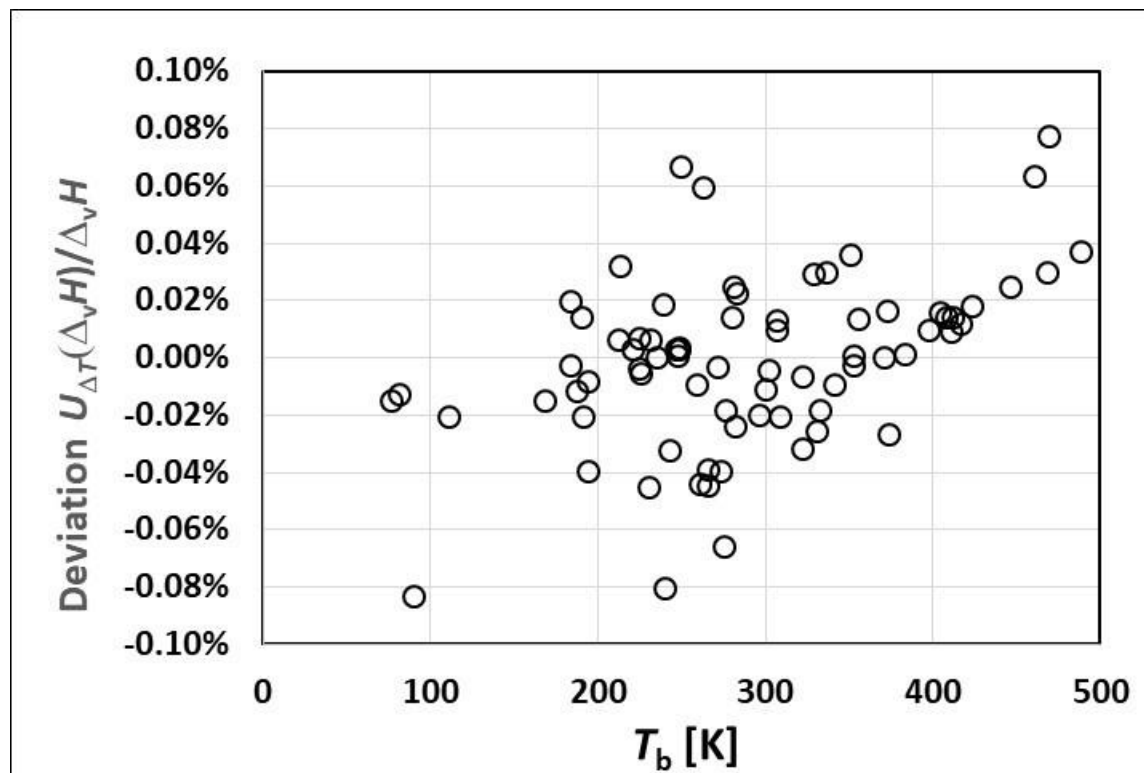
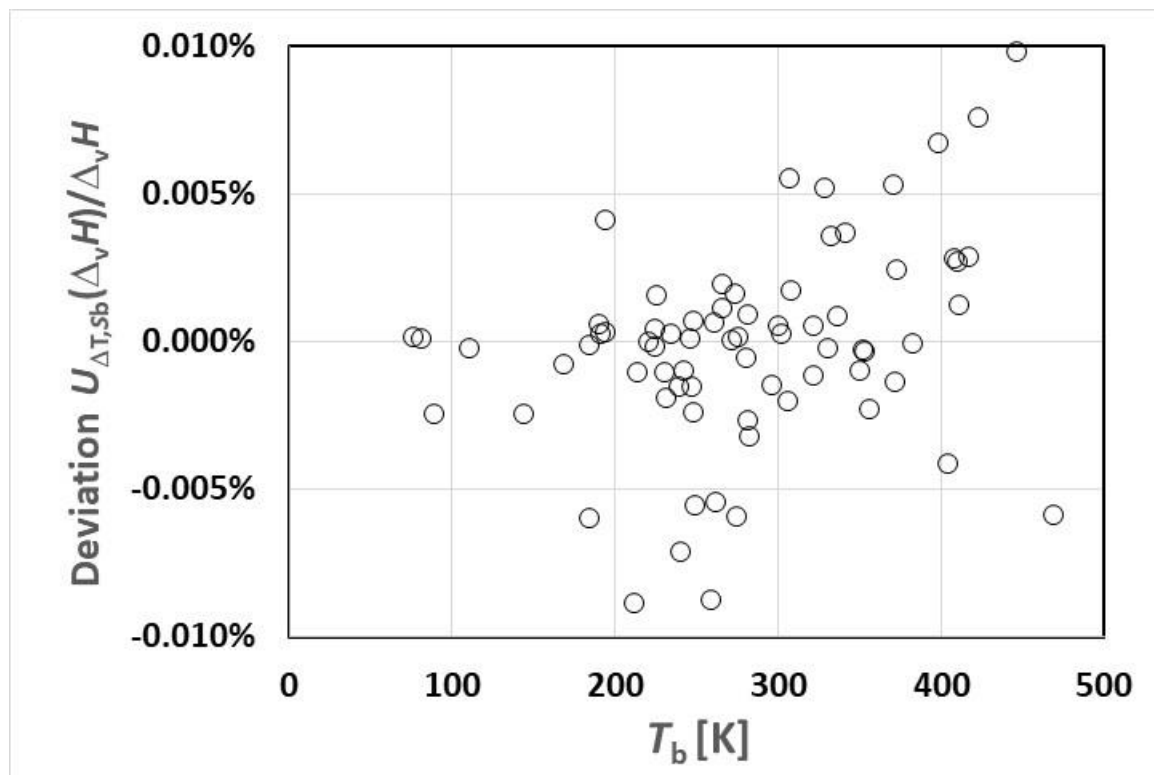


Figure 3-3. Relative deviation in uncertainty $U_{\Delta T}(\Delta_v H)/\Delta_v H$ in enthalpy of vaporization from correlated value (considering both enthalpic and entropic contributions)



3.3.3. References for Heats of Vaporization and Fusion

Reference	Source
[1884TRO]	Trouton, F.; IV. On molecular latent heat, Philosophical Magazine. Series 5., 1884, 18, 54–57
[1888ANT]	Antoine, C. Tensions des vapeurs; nouvelle relation entre les tensions et les températures, Comptes Rendus 107, 681 (1888)
[1919MEN]	Menzies, A. W. C.; The vapor pressure of tetranitromethane. J. Am. Chem. Soc., 1919, 41, 1336-1337
[1922HER]	Herz, W.; Schuftan, P., Physikalisch-chemische Untersuchungen an Tetralin und Dekalin, Z. Phys. Chem. (Frankfurt/Main), 1922, 101, 269-285
[1922TAY]	Taylor, Robert S.; Smith, Leighton B., The vapor pressures, densities and some derived quantities for ether at low temperatures, J. Am. Chem. Soc., 1922, 44, 2450-2463
[1923GIB]	Gibson, G.E.; Giaque, W.F., The third law of thermodynamics. Evidence from the specific heats of glycerol that the entropy of a glass exceeds that of a crystal at the absolute zero, J. Am. Chem. Soc., 1923, 45, 93-104
[1923MOR]	Mortimer, F. Spencer.; Murphy, Ray V., The Vapor Pressures of Some Substances Found in Coal Tar., Ind. Eng. Chem., 1923, 15, 1140-1142
[1924MAA]	Maass, O.; Hiebert, P.G., The properties of pure hydrogen peroxide. V. Vapor pressure, J. Am. Chem. Soc., 1924, 46, 2693-2700
[1925NEL]	Nelson, O.A.; Wales, H., Vapor Pressures and Boiling Points of Mono- and Dimethylanilines and Mono- and Diethylanilines, J. Am. Chem. Soc., 1925, 47, 867-872
[1925PER]	Perry, John H.; Bardwell, D.C., The vapor pressures of solid and liquid cyanogen 1, J. Am. Chem. Soc., 1925, 47, 2629-2632
[1926AND]	Andrews, D.H.; Lynn, G.; Johnston, J., The heat capacities and heat of crystallization of some isomeric aromatic compounds, J. Am. Chem. Soc., 1926, 48, 1274-1287
[1926LOO]	Loomis, A.G.; Walters, J.E., The vapor pressure of ethane near the normal boiling point 1, J. Am. Chem. Soc., 1926, 48, 2051-2055
[1926MAT]	Mathews, J. H.; The Accurate Measurement of Heats of Vaporization of Liquids. J. Am. Chem. Soc., 1926, 48, 562-576
[1926PAR]	Parks, G.S.; Anderson, C.T., Thermal data on organic compounds. III. The heat capacities, entropies and free energies of tertiary butyl alcohol, mannitol, erythritol and normal butyric acid, J. Am. Chem. Soc., 1926, 48, 1506-1512

- [1926SIN] Sinozaki, Heima; Hara, Ryoaburo; Mitsukuri, Shinroku, The vapour pressures of hydrogen cyanide, Bull. Chem. Soc. Jpn., 1926, 1, 59-61
- [1928COF] Coffin, C.C.; Maass, O., The Preparation and Physical Properties of α , β - and γ -Butylene and Normal and Isobutane, J. Am. Chem. Soc., 1928, 50, 1427-1437
- [1928NEL] Nelson, O.A., Vapor Pressures of Fumigants: II---Methyl, Ethyl, n-Propyl, Isopropyl, n-Butyl, Secondary Butyl and Isobutyl Formates 1, Ind. Eng. Chem., 1928, 20, 1382-1384
- [1929KEL] Kelley, K.K., The heat capacity of toluene from 14 K to 298 K. The entropy and the free energy of formation, J. Am. Chem. Soc., 1929, 51, 2738-2741
- [1929PEA] Pearce, J.N.; Peters, P.E., The Vapor Pressure of Ethylene Chloride Between -30° and 100°, J. Phys. Chem., 1929, 33, 873-878
- [1930COO] Coolidge, A. S.; The vapor pressure and heats of fusion and vaporization of formic acid. J. Am. Chem. Soc., 1930, 52, 1874-1887
- [1930LIV] Livingston, R.; Heisig, G. B.; The Vapor Pressure of Allene as A Function of Temperature. J. Am. Chem. Soc., 1930, 52, 2409-2410
- [1930NEL] Nelson, O.A., Vapor Pressures of Fumigants, Ind. Eng. Chem., 1930, 22, 971-972
- [1930SOU] Southard, John C.; Andrews, Donald H., The heat capacities of organic compounds at low temperatures. iii. an adiabatic calorimeter for heat capacities at low temperatures, Journal of the Franklin Institute, 1930, 209, 349-360
- [1930ZMA] Zmaczynski, M.A., Recherches Ebullioscopiques et Tonometriques Comparatives de 8 Substances Organiques Etalons, J. Chim. Phys. Phys. Chim. Biol., 1930, 27, 503-517
- [1931HIS] Hsia, A. W.; The Thermal Properties of High Molecular Weight Substances with Special Considerations of their Use in Refrigerating Machines with Turbo Compressors. Z. Gesamte Kalte-Ind., 1931, 1-40
- [1931HUF] Huffman, H.M.; Parks, G.S.; Barmore, M., Thermal data on organic compounds. X. Further studies on the heat capacities, entropies and free energies of hydrocarbons, J. Am. Chem. Soc., 1931, 53, 3876-3888
- [1933HEI] Heisig, G. B.; Hurd, C. D.; Vapor Pressure and Boiling Point of Pure Methylacetylene. J. Am. Chem. Soc., 1933, 55, 3485-3487
- [1933HEI2] Heisig, G.B., Action of Radon on Some Unsaturated Hydrocarbons. III. Vinylacetylene and Butadiene, J. Am. Chem. Soc., 1933, 55, 2304-2311
- [1933MEN] Menzel, Walter; Mohry, Friedrich, Die Dampfdrucke des CF₄ und NF₃ und der Tripelpunkt des CF₄, Z. Anorg. Allg. Chem., 1933, 210, 257-263
- [1933THO] Thornton, N. V.; Burg, A. B.; Schlesinger, H. I.; The Behaviour of Dichlorodifluoromethane and of Chlorotrifluoromethane in the Electric Discharge. J. Am. Chem. Soc., 1933, 55, 3177-3182

- [1935BUT] Butler, J.A.V.; Ramchandani, C.N.; Thomson, D.W., The Solubility of Non-Electrolytes. Part 1. The Free Energy of Hydration of Some Aliphatic Alcohols, *J. Chem. Soc.*, 1935, 280-285
- [1935SIM] Simon, A.; Huter, J.; Vapor Pressure Curves, Melting Point and Chemical Constants of Dimethyl, Trimethyl- and Isobutylamines. *Z. Elektrochem.*, 1935, 41, 28
- [1935SPE] Spence, R.; Wild, W.; The Vapor Pressure Curve of Formaldehyde and Some Related Data. *J. Chem. Soc.*, 1935, 506-9
- [1936GIA] Giauque, W. F.; Blue, R. W.; Hydrogen Sulfide. The Heat Capacity and Vapor Pressure of Solid and Liquid. The Heat of Vaporization. A Comparison of Thermodynamic and Spectroscopic Values of the Entropy. *J. Am. Chem. Soc.*, 1936, 58, 831-837
- [1936PAR] Parks, G.S.; Todd, S.S.; Moore, W.A., Thermal data on organic compounds. XVI. Some heat capacity, entropy and free energy data for typical benzene derivatives and heterocyclic compounds, *J. Am. Chem. Soc.*, 1936, 58, 398-401
- [1937AST] Aston, J.G.; Siller, C.W.; Messerly, G.H., Heat capacities and entropies of organic compounds. III. Methylamine from 12 K to the boiling point. Heat of vaporization and vapor pressure. The entropy from molecular data, *J. Am. Chem. Soc.*, 1937, 59, 1743-1751
- [1937GIA] Giauque, W.F.; Egan, C.J., Carbon Dioxide. The Heat Capacity and Vapor Pressure of the Solid. The Heat of Sublimation. Thermodynamic and Spectroscopic Values of the Entropy, *J. Chem. Phys.*, 1937, 5, 45
- [1937GUY] Guyer, A.; Schutze, H.; Weidenmann, M., Über einige thermische Eigenschaften des Vinylbromids, *Helv. Chim. Acta*, 1937, 20, 1, 936-949
- [1938CRE] Crenshaw, J.L.; Cope, A.C.; Finkelstein, N.; Rogan, R., The Dioxanates of the Mercuric Halides, *J. Am. Chem. Soc.*, 1938, 60, 2308-2311
- [1938EGA] Egan, C.J.; Kemp, J.D., Methyl bromide. The heat capacity, vapor pressure, heats of transition, fusion and vaporization. Entropy and density of the gas, *J. Am. Chem. Soc.*, 1938, 60, 2097-2101
- [1939AST] Aston, J.G.; Eidinoff, M.L.; Forster, W.S., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of dimethylamine, *J. Am. Chem. Soc.*, 1939, 61, 1539-1543
- [1939GIA] Giauque, W.F.; Ruehrwein, R.A., The entropy of hydrogen cyanide. Heat capacity, heat of vaporization and vapor pressure. Hydrogen bond polymerization of the gas in chains of indefinite length, *J. Am. Chem. Soc.*, 1939, 61, 2626-2633
- [1939POW] Powell, T.M.; Giauque, W.F., Propylene. The heat capacity, vapor pressure, heats of fusion and vaporization. The third law of thermodynamics and orientation equilibrium in the solid, *J. Am. Chem. Soc.*, 1939, 61, 2366-2370
- [1939RUE] Ruehrwein, R. A.; Giauque, W. F.; The entropy of cyanogen. Heat capacity and vapor pressure of solid and liquid. Heats of fusion and vaporization. *J. Am. Chem. Soc.*, 1939, 61, 2940-2944

- [1940AST] Aston, J.G.; Kennedy, R.M.; Schumann, S.C., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of isobutane, J. Am. Chem. Soc., 1940, 62, 2059-2063
- [1940AST2] Aston, J.G.; Messerly, G.H., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of n-butane, J. Am. Chem. Soc., 1940, 62, 1917-1923
- [1940BEN] Benning, A.F.; McHarness, R.C., Thermodynamic Properties of Fluorochloromethanes and -Ethan's Vapor Pressure of Three Fluorochloromethanes and Trifluorotichloroethane, Ind. Eng. Chem., 1940, 32, 497-499
- [1940GRO] Grosse, A. V.; Wackher, R. C.; Linn, C. B.; Physical Properties of the Alkyl Fluorides and A Comparison of the Alkyl Fluorides with the Other Alkyl Halides and with the Alkyls of the Elements of Period II. J. Phys. Chem., 1940, 44, 275-296
- [1940LAM] Lamb, Arthur B.; Roper, Edwin E., The Vapor Pressures of Certain Unsaturated Hydrocarbons, J. Am. Chem. Soc., 1940, 62, 806-814
- [1940MES] Messerly, G.H.; Aston, J.G., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of methyl chloride, J. Am. Chem. Soc., 1940, 62, 886-890
- [1940PIT] Pitzer K.S., The thermodynamics of n-heptane and 2,2,4-trimethylpentane, including heat capacities, heats of fusion and vaporization and entropies, J. Am. Chem. Soc., 1940, 62, 1224-1227
- [1941HEI] Heisig, G.B., The Preparation and the Vapor Pressures of Cyclobutene and Cyclobutane, J. Am. Chem. Soc., 1941, 63, 1698-1699
- [1941KEN] Kennedy, R.M.; Sagenkahn, M.; Aston, J.G., The heat capacity and entropy, heats of fusion and vaporization, and the vapor pressure of dimethyl ether. The density of gaseous dimethyl ether, J. Am. Chem. Soc., 1941, 63, 2267-2272
- [1941OSB] Osborne, D.W.; Garner, C.S.; Doescher, R.N.; Yost, D.M., The heat capacity, entropy, heats of fusion and vaporization and vapor pressure of fluorotrichloromethane, J. Am. Chem. Soc., 1941, 63, 3496-3499
- [1941SEK] Seki, S.; Nitta, I., Vapor pressure of molecular crystals. III. Tetranitromethane, J. Chem. Soc. Jpn., 1941, 62, 907
- [1941SIM] Simnikov, I.P., Increase of Enthalpy During Formation of Azeotropic Mixtures, Zh. Obshch. Khim., 1941, 14, 483-492
- [1941STO] Stout, J.W.; Fisher, L.H., The entropy of formic acid. The heat capacity from 15 to 300 K. Heats of fusion and vaporization, J. Chem. Phys., 1941, 9, 163-168
- [1942BEN] Benson, S.W.; Kistiakowsky, G.B., The Photochemical Decomposition of Cyclic Ketones, J. Am. Chem. Soc., 1942, 64, 80-86
- [1942OSB] Osborne, D.W.; Doescher, R.N.; Yost, D.M., The heat capacity, heats of fusion and vaporization, vapor pressure and entropy of dimethyl sulfide, J. Am. Chem. Soc., 1942, 64, 169-172
- [1942RUS] Russell, H., Jr.; Osborne, D.W.; Yost, D.M., The heat capacity, entropy, heats of fusion, transition, and vaporization and vapor pressures of methyl mercaptan, J. Am. Chem. Soc., 1942, 64, 165-169

- [1942ZIE] Ziegler, W.T.; Andrews, D.H., The heat capacity of benzene-d₆, J. Am. Chem. Soc., 1942, 64, 2482-2485
- [1943AST] Aston, J.G.; Szasa, G.J.; Fink, H.L., The heat capacity and entropy, heats of transition, fusion and vaporization and the vapor pressures of cyclohexane. The vibrational frequencies of alicyclic ring systems, J. Am. Chem. Soc., 1943, 65, 1135-1139
- [1943MCG] McGovern, E. W.; Chlorohydrocarbon Solvents. Ind. Eng. Chem., 1943, 35, 1230-9
- [1944AST] Aston, J.G.; Sagenkahn, M.L.; Szasa, G.J.; Moessen, G.W.; Zuhr, H.F., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of trimethylamine. The entropy from spectroscopic and molecular data, J. Am. Chem. Soc., 1944, 66, 1171-1177
- [1944MCD] McDonald, H.J., The vapor pressure and heat of vaporization of trichloroethylene, J. Phys. Chem., 1944, 48, 47-50
- [1944RUB] Rubin, T.R.; Levedahl, B.H.; Yost, D.M., The heat capacity, heat of transition, vaporization, vapor pressure and entropy of 1,1,1-trichloroethane, J. Am. Chem. Soc., 1944, 66, 279-282
- [1944RUS] Russell, H., Jr.; Golding, D.R.V.; Yost, D.M., The heat capacity, heats of transition, fusion and vaporization, vapor pressure and entropy of 1,1,1-trifluoroethane, J. Am. Chem. Soc., 1944, 66, 16-20
- [1944SCO] Scott, R.B.; Ferguson, W.J.; Brickwedde, F.G., Thermodynamic properties of cis-2-butene from 15° to 1,500 K, J. Res. Natl. Bur. Stand., 1944, 33, 1-20
- [1945GUT] Guttman, L.; Pitzer, K.S., trans-2-Butene. The heat capacity, heats of fusion and vaporization, and vapor pressure. The entropy and barrier to internal rotation, J. Am. Chem. Soc., 1945, 67, 324-327
- [1945WIL] Williamham, C.B.; Taylor, W.J.; Pignocco, J.M.; Rossini, F.D., Vapor Pressures and Boiling Points of Some Paraffin, Alkylcyclopentane, Alkylcyclohexane, and Alkylbenzene Hydrocarbons, J. Res. Natl. Bur. Stand., 1945, 35, 219-244
- [1946RUE] Ruehrwein, R.A.; Powell, T.M., The Heat Capacity, Vapor Pressure, Heats of Fusion and Vaporization of Cyclopropane. Entropy and Density of the Gas, J. Am. Chem. Soc., 1946, 68, 1063-1066
- [1946SPI] Spitzer, Ralph; Pitzer, Kenneth S., The Heat Capacity of Gaseous Cyclopentane, Cyclohexane and Methylcyclohexane, J. Am. Chem. Soc., 1946, 68, 2537-2538
- [1946TRE] Treybal, R. E.; Weber, L. D.; Daley, J. F.; The system acetone-water-1,1,2-trichloroethane - ternary liquid and binary vapour equilibria.. Ind. Eng. Chem., 1946, 38, 817
- [1947AST] Aston, J. G.; Szasz, G. J.; The Thermodynamics of Butadiene-1,2 from Calorimetric and Spectroscopic Data. J. Am. Chem. Soc., 1947, 69, 3108-3114
- [1947BEE] Beersmans, J.; Jungers, J.C., Synthese et Etude des Chlorure, Bromure et Iodure de Deuteromethyle, Bull. Soc. Chim. Belg., 1947, 56, 238-250
- [1947GOL] Goldblum, K.B.; Martin, R.W.; Young, R.B., Vapor Pressure Data for Phenols, Ind. Eng. Chem., 1947, 39, 1474-1476

- [1947IVI] Ivin, K.J.; Dainton, F.S., The vapour pressures, latent heats of sublimation and transition points of solid hexachloroethane, *Trans. Faraday Soc.*, 1947, 43, 32
- [1947KET] Ketelaar, J.A.A.; van Velden, P.F.; Zalm, P., Vapour Pressures and Heats of Vaporization of cis- and trans-1,2-Dichloroethene, *Recl. Trav. Chim. Pays-Bas*, 1947, 66, 721-732
- [1947STU] Stull, Daniel R., Vapor Pressure of Pure Substances. Organic and Inorganic Compounds, *Ind. Eng. Chem.*, 1947, 39, 517-540
- [1947WAD] Waddington, Guy; Douslin, Donald R., Experimental Vapor Heat Capacities and Heats of Vaporization of n-Hexane and 2,2-Dimethylbutane, *J. Am. Chem. Soc.*, 1947, 69, 10, 2275-2279
- [1948BRE] Brewer, L.; Gilles, P. W.; Jenkins, F. A.; The vapor pressure and heat of sublimation of graphite, *J. Chem. Phys.* 16, 797 (1948).
- [1948GAN] Ganef, J.M.; Jungers, J.C., Tensions de Vapeur du Systeme CH₃Cl - CH₂Cl₂, *Bull. Soc. Chim. Belg.*, 1948, 57, 82-87
- [1948GIA] Giauque, W.F.; Jones, W.M., Carbonyl chloride. Entropy. Heat capacity. Vapor pressure. Heats of fusion and vaporization. Comments on solid sulfur dioxide structure, *J. Am. Chem. Soc.*, 1948, 70, 120
- [1948GOR] Gordon, J.; Giauque, W.F., The entropy of ethyl chloride. Heat capacity from 18 to 287K. Vapor pressure. Heats of fusion and vaporization, *J. Am. Chem. Soc.*, 1948, 70, 1506-1510
- [1948MIC] Michels, A.; Wassenaar, T.; Vapour pressure of methylfluoride. *Physica (Amsterdam)*, 1948, 14, 104
- [1948OLI] Oliver, G.D.; Eaton, M.; Huffman, H.M., The heat capacity, heat of fusion and entropy of benzene, *J. Am. Chem. Soc.*, 1948, 70, 1502-1505
- [1948ROB] Robertson, A. J. B.; The decomposition, boiling, and explosion of trinitrotoluene at high temperatures. *Trans. Faraday Soc.*, 1948, 44, 977-983
- [1949BUC] Buck, F. R.; Coles, K. F.; Kennedy, G. T.; Morton, F.; Some Nuclear-methylated Styrenes and Related Compounds. *J. Chem. Soc.*, 1949, 2377-83
- [1949DRE] Dreisbach, R.R.; Shrader, S.A., Vapor Pressure--Temperature Data on Some Organic Compounds, *Ind. Eng. Chem.*, 1949, 41, 2879-2880
- [1949FOR] Forziati, A.F.; Norris, W.R.; Rossini, F.D., Vapor Pressures and Boiling Points of Sixty API-NBS Hydrocarbons, *J. Res. Natl. Bur. Stand.*, 1949, 43, 555-563
- [1949KRE] Kretschmer, Carl B.; Wiebe, Richard., Liquid-Vapor Equilibrium of Ethanol--Toluene Solutions, *J. Am. Chem. Soc.*, 1949, 71, 1793-1797
- [1949SCO] Scott, D.W.; Waddington, G.; Smith, J.C.; Huffman, H.M., Thermodynamic properties of three isomeric pentenes, *J. Am. Chem. Soc.*, 1949, 71, 2767-2773

- [1949WAD] Waddington, G.; Knowlton, J.W.; Scott, D.W.; Oliver, G.D.; Todd, S.S.; Hubbard, W.N.; Smith, J.C.; Huffman, H.M., Thermodynamic properties of thiophene, J. Am. Chem. Soc., 1949, 71, 797-808
- [1949WAD2] Waddington, G.; Smith, J.C.; Scott, D.W.; Huffman, H.M., Experimental vapor heat capacities and heats of vaporization of 2-methylpentane, 3-methylpentane and 2,3-dimethylbutane, J. Am. Chem. Soc., 1949, 71, 3902-3906
- [1950AST] Aston, J.G.; Mastrangelo, S.V.R.; Moessen, G.W., The thermodynamics of 1-butyne from calorimetric and spectroscopic data, J. Am. Chem. Soc., 1950, 72, 5287-5291
- [1950COL] Coles, K.F.; Popper, Felix, Vapor-Liquid Equilibria. Ethylene Oxide - Acetaldehyde and Ethylene Oxide - Water Systems, Ind. Eng. Chem., 1950, 42, 1434-1438
- [1950FOR] Forziati, A.F.; Camin, D.L.; Rossini, F.D., Density, refractive index, boiling point, and vapor pressure of eight monoolefin (1-alkene), six pentadiene, and two cyclomonoolefin hydrocarbons, J. Res. NBS, 1950, 45, 406-410
- [1950HAL] Halford, J. O.; Energy Levels and Thermodynamic Properties of the Internal Rotator, J. Chem. Phys., 1950, 18, 444
- [1950MAT] Matthews, J.B.; Sumner, J.F.; Moelwyn-Hughes, E.A., The Vapour Pressures of Certain Liquids, Trans. Faraday Soc., 1950, 46, 797-803
- [1950MIC] Michels, A.; Wassenaar, T., The vapour pressure of ethylene, Physica, 1950, 16, 221-224
- [1950SCO] Scott, D.W.; Finke, H.L.; Gross, M.E.; Guthrie, G.B.; Huffman, H.M., 2,3-Dithiabutane: low temperature heat capacity, heat of fusion, heat of vaporization, vapor pressure, entropy and thermodynamic functions, J. Am. Chem. Soc., 1950, 72, 2424-2430
- [1950SCO2] Scott, Donald W.; Waddington, Guy, Vapor Pressure of cis-2-Pentene, trans-2-Pentene and 3-Methyl-1-butene, J. Am. Chem. Soc., 1950, 72, 4310-4311
- [1951CLA] Clark, A.M.; Crockett, A. H.; Eisner, H. S.; The vapour pressure of hydrogen sulphide. Proc. R. Soc. London, Ser. A, 1951, 209, 408-415
- [1951OLI] Oliver, G. D.; Grisard, J. W.; Cunningham, C. W.; Thermodynamic Properties and P-V-T Relations of Chlorotrifluoroethylene.. J. Am. Chem. Soc., 1951, 73, 5719-5722
- [1951SMI] Smith, Thomas E.; Bonner, Robert F., Acetaldehyde, Propionaldehyde, and n-Butyraldehyde, Ind. Eng. Chem., 1951, 43, 1169-1173
- [1952ALB] Albright, L. F.; Martin, J. J.; Thermodynamic Properties of Chlorotrifluoromethane. Ind. Eng. Chem., 1952, 44, 188-198
- [1952BRO] Brown, I., Liquid-Vapour Equilibria. III. The Systems Benzene-n-Heptane, n-Hexane-Chlorobenzene, and cycloHexane-Nitrobenzene, Aust. J. Sci. Res. Ser. A, 1952, 5, 530-540
- [1952EDW] Edwards, G., The Vapour Pressure of Tetranitromethane, Trans. Faraday Soc., 1952, 48, 513-515

- [1952GUT] Guthrie, G.B., Jr.; Scott, D.W.; Hubbard, W.N.; Katz, C.; McCullough, J.P.; Gross, M.E.; Williamson, K.D.; Waddington, G., Thermodynamic properties of furan, *J. Am. Chem. Soc.*, 1952, 74, 4662-4669
- [1952JON] Jones, W.S.; Tamplin, W.S., Chapter 3. Physical Properties of Ethylene Glycol in Glycols, George O. Curme, Jr., ed(s), Reinhold Publishing Corporation, New York, 1952
- [1952WHI] White, P.T.; Barnard-Smith, D.G.; Fidler, F.A., Vapor Pressure--Temperature Relationships of Sulfur Compounds Related to Petroleum, *Ind. Eng. Chem.*, 1952, 44, 1430-1438
- [1953AST] Aston, J.G.; Wood, J.L.; Zolki, T.P., The thermodynamic properties and configuration of unsymmetrical dimethylhydrazine, *J. Am. Chem. Soc.*, 1953, 75, 6202-6204
- [1953FUR] Furukawa, G.T.; McCoskey, R.E.; Reilly, M.L., Heat capacity, heats of fusion and vaporization, and vapor pressure of tetrafluoroethylene, *J. Res. Natl. Bur. Stand.*, 1953, 51, 69-72
- [1953GIN] Ginnings, D.C.; Furukawa, G.T., Heat capacity standards for the range 14 to 1200°K, *J. Am. Chem. Soc.*, 1953, 75, 522-527
- [1953RAT] Rathjens, G.W., Jr.; Gwinn, W.D., Heat capacities and entropy of cyclobutane, *J. Am. Chem. Soc.*, 1953, 75, 5629-56
- [1954CAM] Camin, David L.; Forziati, Alphonse F.; Rossini, Frederick D., Physical Properties of n-Hexadecane, n-Decylcyclopentane, n-Decylcyclohexane, 1-Hexadecene and n-Decylbenzene, *J. Phys. Chem.*, 1954, 58, 440-442
- [1954FIN] Finke, H.L.; Gross, M.E.; Waddington, G.; Huffman, H.M., Low-temperature thermal data for the nine normal paraffin hydrocarbons from octane to hexadecane, *J. Am. Chem. Soc.*, 1954, 76, 333-341
- [1954GIG] Giguere, P. A.; Liu, I. D.; Dugdale, J. S.; Morrison, J. A.; Hydrogen peroxide: the low temperature heat capacity of the solid and the third law entropy, *Can. J. Chem.*, 1954, 32, 117-128
- [1954MCC] McCullough, J.P.; Scott, D.W.; Pennington, R.E.; Hossenlopp, I.A.; Waddington, Guy, Nitromethane: The Vapor Heat Capacity, Heat of Vaporization, Vapor Pressure and Gas Imperfection; the Chemical Thermodynamic Properties from 0 to 1500°K., *J. Am. Chem. Soc.*, 1954, 76, 4791-4796
- [1954OST] Osthoff, Robert C.; Grubb, W.T., Physical Properties of Organosilicon Compounds. III. Thermodynamic Properties of Octamethylcyclotetrasiloxane, *J. Am. Chem. Soc.*, 1954, 76, 399-401
- [1955CAM] Camin, David L.; Rossini, Frederick D., Physical Properties of Fourteen API Research Hydrocarbons, C9 to C15, *J. Phys. Chem.*, 1955, 59, 1173-1179
- [1955DRE] Dreyer, R.; Martin, W.; von Weber, U., Die Sättigungsdampfdrucke von Benzol, Toluol, ethylbenzol, Styrol, Cumol und Brombenzol zwischen 10 und 760 Torr, *J. Prakt. Chem.*, 1955, 1, 324-328
- [1955MEA] Mears, Whitney H.; Stahl, Richard F.; Orfeo, S. Robert; Shair, Robert C.; Kells, Lyman F.; Thompson, Walter; McCann, Harold, Thermodynamic Properties of Halogenated Ethanes and Ethylenes, *Ind. Eng. Chem.*, 1955, 47, 1449-1454

- [1955TSY] Tsypkina, O.Y., Study of Vacuum Pressure Influence on Efficiency of Separation of Some Polynuclear Compounds of Coal Tar Rectifications, *Zh. Prikl. Khim.* (Moscow), 1955, 28, 185-192
- [1956AMB] Ambrose, D., The Vapour Pressures and Critical Temperatures of Acetylene and Carbon Dioxide, *Trans. Faraday Soc.*, 1956, 52, 772-781
- [1956BRO] Brockhaus, Von A.; Jenckel, E., Über die kinetik des thermischen abbaues von polymethacrylsäuremethylester, *Makromol. Chem.*, 1956, 18, 262-293
- [1956LI] Li, J.C.M.; Pitzer, K.S., The thermodynamic properties of 1,1-dichloroethane: heat capacities from 14 to 294 K, heats of fusion and vaporization, vapor pressure and entropy of the ideal gas. The barrier to internal rotation, *J. Am. Chem. Soc.*, 1956, 78, 1077-1080
- [1956PEN] Pennington, R.E.; Scott, D.W.; Finke, H.L.; McCullough, J.P.; Messerly, J.F.; Hossenlopp, I.A.; Waddington, G., The chemical thermodynamic properties and rotational tautomerism of 1-propanethiol, *J. Am. Chem. Soc.*, 1956, 78, 3266-3272
- [1956PIC] Pick, J.; Fried, V.; Hala, E.; Vilim, O., Der Dampfdruck des Athylenglykolmonomethylathers und des Athylenglykolmonoathylathers, *Collect. Czech. Chem. Commun.*, 1956, 21, 260-261
- [1957MAS] Mastrangelo, S.V.R., Adiabatic calorimeter for determination of cryoscopic data, *Anal. Chem.*, 1957, 29, 841-845
- [1957MCC] McCullough, J.P.; Finke, H.L.; Messerly, J.F.; Kincheloe, T.C.; Waddington, G., The low temperature thermodynamic properties of naphthalene, 1-methylnaphthalene, 2-methylnaphthalene, 1,2,3,4-tetrahydronaphthalene, trans-decahydronaphthalene and cis-decahydronaphthalene, *J. Phys. Chem.*, 1957, 61, 1105-1116
- [1957PEN] Pennington, R. E.; Kobe, K. A.; The Thermodynamic Properties of Acetone. *J. Am. Chem. Soc.*, 1957, 79, 300-305
- [1959HIL] Hildenbrand, D. L.; McDonald, R. A.; Kramer, W. R.; Stull, D. R.; Thermodynamic and spectroscopic study of vinylidene chloride. I. Thermodynamic properties of the solid, liquid, and ideal gas. *J. Chem. Phys.*, 1959, 30, 930-4
- [1959HIL2] Hildenbrand, D.L.; McDonald, R.A., The Heat of Vaporization and Vapor Pressure of Carbon Tetrachloride; the Entropy from Calorimetric Data, *J. Phys. Chem.*, 1959, 63, 1521-1523
- [1959MCC] McCullough, J.P.; Douslin, D.R.; Hubbard, W.N.; Todd, S.S.; Messerly, J.F.; Hossenlopp, I.A.; Frow, F.R.; Dawson, J.P.; Waddington, G., Pyrrolidine: Chemical thermodynamic properties between 0 and 1500° K; effect of pseudorotation; and an unusual thermal anomaly in the liquid state, *J. Am. Chem. Soc.*, 1959, 81, 5884-5890
- [1959MCC2] McCullough, J.P.; Pennington, R.E.; Smith, J.C.; Hossenlopp, I.A.; Waddington, G., Thermodynamics of cyclopentane, methylcyclopentane and 1,cis-3-dimethylcyclopentane: Verification of the concept of pseudorotation, *J. Am. Chem. Soc.*, 1959, 81, 5880-5883
- [1959MCD] McDonald, R.A.; Shrader, S.A.; Stull, D.R., Vapor Pressures and Freezing Points of Thirty Pure Organic Compounds., *J. Chem. Eng. Data*, 1959, 4, 311-313

- [1959PAR] Paridon, Leo J.; MacWood, George E.; Hu, Jih-Heng, The Heat of Vaporization of Diborane, J. Phys. Chem., 1959, 63, 1998-1999
- [1959SEP] Seprakova, M.; Paulech, J.; Dykyj, J., Dampfdruck der Butyraldehyde, Chem. Zvesti, 1959, 13, 313-316
- [1959ZIE] Ziegler, W. T.; The vapor pressures of some hydrocarbons in the liquid and solid state at low temperatures. Natl. Bur. Stand., 1959, NBS Technical Note 4 (NBS Report 6038), Washington, D.C.
- [1960GLE] Glew, D.N., THE GAS HYDRATE OF BROMOCHLORODIFLUOROMETHANE, Can. J. Chem., 1960, 38, 208-221
- [1960HIR] Hirt, R.C.; Steger, J.E.; Simard, G.L., Vapor pressure of 2,4,6-triamino-s-triazine (melamine), J. Polym. Sci., 1960, 43, 319-323
- [1960MUE] Mueller, C.R.; Ignatowski, A.J., Equilibrium and Transport Properties of the Carbon Tetrachloride-Methylene Chloride System, J. Chem. Phys., 1960, 32, 1430-1434
- [1960NOV] Novák, J.; Matous, J.; Pick, J., Gleichgewicht flüssigkeit-dampf XXIV. Gleichgewicht flüssigkeit-dampf im system cyclohexylamin-cyclohexanol-anilin, Collect. Czech. Chem. Commun., 1960, 25, 2405-2413
- [1961DIT] Ditter, J. F.; Perrine, J. C.; Shapiro, I.; Vapor Pressure of Deuterodiborane. J. Chem. Eng. Data, 1961, 6, 271-271
- [1961FAR] Farhat-Aziz; Moelwyn-Hughes, E. A.; The kinetics of the reaction between methyl fluoride and lithium iodide in Acetone Solution.. J. Chem. Soc., 1961, 1523
- [1961LI] Li, J.C.M.; Rossini, F.D., Vapor Pressures and Boiling Points of the 1-Fluoroalkanes, 1-Chloroalkanes, 1-Bromoalkanes, and 1-Iodoalkanes, C1 to C20 ., J. Chem. Eng. Data, 1961, 6, 268-270
- [1961NIC] Nickerson, J.K.; Kobe, K.A.; McKetta, John J., The thermodynamic properties of the methyl ketone series, J. Phys. Chem., 1961, 65, 1037-1043
- [1961SCO] Scott, D.W.; Messerly, J.F.; Todd, S.S.; Guthrie, G.B.; Hossenlopp, I.A.; Moore, R.T.; Osborn, A.; Berg, W.T.; McCullough, J.P., Hexamethyldisiloxane: chemical thermodynamic properties and internal rotation about the siloxane linkage, J. Phys. Chem., 1961, 65, 1320-1326
- [1962BIT] Bittrich, H.J.; Kauer, E., Zur Thermodynamik des Systems Diathylamin-Triathylamin. I. Das Flüssigkeits-Dampf-Gleichgewicht, Z. Phys. Chem. (Frankfurt/Main), 1962, 219, 224-238
- [1962HAT] Hatton, W.E.; Hildenbrand, D.L.; Sinke, G.C.; Stull, D.R., Chemical thermodynamic properties of aniline, J. Chem. Eng. Data, 1962, 7, 229-231
- [1962SCO] Scott, D.W.; Guthrie, G.B.; Messerly, J.F.; Todd, S.S.; Berg, W.T.; Hossenlopp, I.A.; McCullough, J.P., Toluene: thermodynamic properties, molecular vibrations, and internal rotation, J. Phys. Chem., 1962, 66, 911-914
- [1962VAL] Valentine, R.H.; Brodale, G.E.; Giauque, W.F., Trifluoromethane: entropy, low temperature heat capacity, heats of fusion and vaporization, and vapor pressure, J. Phys. Chem., 1962, 66, 392-395

- [1962WAD] Waddington, Guy; Smith, J.C.; Williamson, K.D.; Scott, D.W., Carbon disulfide as a reference substance for vapor-flow calorimetry; the chemical thermodynamic properties, *J. Phys. Chem.*, 1962, 66, 1074-1077
- [1963AND] Andon, R.J.L.; Counsell, J.F.; Herington, E.F.G.; Martin, J.F., Thermodynamic properties of organic oxygen compounds, *Trans. Faraday Soc.*, 1963, 59, 830-835
- [1963BEY] Beynon, Eugene T.; McKetta, John J., The thermodynamic properties of 2-methyl-2-propanol, *J. Phys. Chem.*, 1963, 67, 2761-2765
- [1963BID] Biddiscombe, D.P.; Collerson, R.R.; Handley, R.; Herington, E.F.G.; Martin, J.F.; Sprake, C.H.S., Thermodynamic Properties of Organic Oxygen Compounds. Part 8. Purification and Vapor Pressures of the Propyl and Butyl Alcohols, *J. Chem. Soc.*, 1963, 1954-1957
- [1963BUL] Bull, S.Sh.; Seregennaja, I.I.; Tsherbakora, P.R., Isothermic Equilibrium of Liquid-Steam in System Water-Acetoaldehyde, *Khim. Prom. (Moscow)*, 1963, 7, 507-509
- [1963CAP] Capková, A.; Fried, V., Gleichgewicht flüssigkeit-dampf XXX. System vinylacetat-essigsäure, *Collect. Czech. Chem. Commun.*, 1963, 28, 2235-2239
- [1963SCO] Scott, D.W.; Hubbard, W.N.; Messerly, J.F.; Todd, S.S.; Hossenlopp, I.A.; Good, W.D.; Douslin, D.R.; McCullough, J.P., Chemical Thermodynamic Properties and Internal Rotation of Methylpyridines I. 2-Methylpyridine, *J. Phys. Chem.*, 1963, 67, 680
- [1963VIN] Vinson, C. G.; Martin, J. J.; Heat of Vaporization and Vapor Pressure of 1,4-Dioxane. *J. Chem. Eng. Data*, 1963, 8, 74-75
- [1963WOJ] Wojtasinski, J. G.; Measurement of Total Pressures for Determining Liquid-Vapor Equilibrium Relations of the Binary System Isobutyraldehyde-n-Butyraldehyde. *J. Chem. Eng. Data*, 1963, 8, 381-385
- [1964GUB] Gubkov, A.N.; Fermor, N.A.; Smirnov, N.I., Vapor Pressure of Mono-Poly Systems, *Zh. Prikl. Khim.*, 1964, 37, 2204-2210
- [1964KLE] Kletskii, A.V., Dependence of Vapor Pressure Freon - 22, *Inzh. Fiz. Zh.*, 1964, 7, 40-43
- [1964SEV] Sevrugova, N.N.; Sokorskii, V.A.; Zhavoronkov, N.M., Phase of Equilibrium for Mixtures Acrylonitrile-Acetonitrile, *Zh. Prikl. Khim. (Leningrad)*, 1964, 37, 1989-1993
- [1965COL] Collerson, R.R.; Counsell, J.F.; Handley, R.; Martin, J.F.; Sprake, C.H.S., Thermodynamic Properties of Organic Oxygen Compounds. Part XV. Purification and Vapour Pressures of Some Ketones and Ethers, *J. Chem. Soc.*, 1965, 3697-3700
- [1965HES] Hessel, D.; Geiseler, G., Über die Druckabhängigkeit des heteroazeotropen Systems n-Butanol/Wasser, *Z. Phys. Chem. (Leipzig)*, 1965, 229, 199-209

- [1965MAR] Marinichev, A.N.; Susarev, M.P., Study of a Liquid-Vapor Equilibrium in the Systems of Acetone/Methanol and Acetone/Cyclohexane at Temperatures 35, 45, 55 °C and Pressure 760 Torr, Zh. Prikl. Khim. (Moscow), 1965, 38, 378-383
- [1965MCM] McMicking, J. H.; Kay, W. B.; Vapor Pressures and Saturated Liquid and Vapor Densities of The Isomeric Heptanes and Isomeric Octanes. Proc. Am. Pet. Inst., Sect. 3, 1965, 45, 75-90
- [1965POL] Polák, J.; Mertl, I., Saturated vapour pressure of methyl acetate, ethyl acetate, n-propyl acetate, methyl propionate, and ethyl propionate, Collect. Czech. Chem. Commun., 1965, 30, 3526-3528
- [1965RON] Rondeau, R.E.; Harrah, L.A., Melting Point and Vapor Pressures of 3-Hexyne., J. Chem. Eng. Data, 1965, 10, 84-85
- [1965SWA] Swamy, P.A.; Van Winkle, Matthew, Vapor-Liquid Equilibria at 760 mm Mercury for the System Vinyl Acetate-2,4-Dimethyl Pentane., J. Chem. Eng. Data, 1965, 10, 214-215
- [1966OSB] Osborn, A.G.; Douslin, D.R., Vapor Pressure Relations of 36 Sulfur Compounds Present in Petroleum., J. Chem. Eng. Data, 1966, 11, 502-509
- [1967AMB] Ambrose, D.; Broderick, B.E.; Townsend, R., The Vapour Pressures Above the Normal Boiling Point and the Critical Pressures of Some Aromatic Hydrocarbons, J. Chem. Soc. A, 1967, 633-641
- [1967AND] Andon, R.J.L.; Counsell, J.F.; Lees, E.B.; Martin, J.F.; Mash, C.J., Thermodynamic properties of organic oxygen compounds. Part 17. Low-temperature heat capacity and entropy of the cresols, Trans. Faraday Soc., 1967, 63, 1115-1121
- [1967LES] Lesteva, T.M.; Ogoradnikov, S.K.; Morozova, A., Kinetics of Dimerisation of Cyclopentadiene and Equilibrium of Liquid-Steam in System of Isoprene-Cyclopentadiene, Zh. Prikl. Khim. (Leningrad), 1967, 40, 891-894
- [1967VAN] van Hook, W.A., Vapor Pressures of the Methylacetylenes, H₃CCCH, H₃CCCD, D₃CCCH, and D₃CCCD, J. Chem. Phys., 1967, 46, 1907-1918
- [1968FOW] Fowler, Lewis; Trump, Walter N.; Vogler, Carl E., Vapor pressure of naphthalene. Measurements between 40.deg. and 180.deg., J. Chem. Eng. Data, 1968, 13, 209-210
- [1968OSB] Osborn, Ann G.; Douslin, Donald R., Vapor pressure relations of 13 nitrogen compounds related to petroleum, J. Chem. Eng. Data, 1968, 13, 534-537
- [1968PAC] Pace, E.L.; Reno, M.A., Thermodynamic Properties of Carbonyl Fluoride from 12 K to its Boiling Point. The Entropy from Molecular and Spectroscopic Data, J. Chem. Phys., 1968, 48, 1231-1235
- [1969ALT] Altunin, V. V.; Spiridonov, G. A.; Kaekin, V. S.; Equation of state and thermodynamic properties of freon-21. Teploenergetika, 1969, 16, 79-83
- [1969BRO] Brown, I.; Fock, W.; Smith, F., The thermodynamic properties of solutions of normal and branched alcohols in benzene and n-hexane, J. Chem. Thermodyn., 1969, 1, 273-291

- [1969CID] Cidlinský, J.; Polák, J., Saturated vapour pressures of some ethers, Collect. Czech. Chem. Commun., 1969, 34, 1317-1321
- [1969KEM] Kemme, Herbert R.; Kreps, Saul I., Vapor pressure of primary n-alkyl chlorides and alcohols, J. Chem. Eng. Data, 1969, 14, 98-102
- [1969MAN] Mansson, M.; Morawetz, E.; Nakase, Y.; Sunner, S., The enthalpies of formation of trioxane and tetroxane, Acta Chem. Scand., 1969, 23, 56-60.
- [1969OSB] Osborn, Ann G.; Douslin, Donald R., Vapor pressure relations for the seven pentadienes, J. Chem. Eng. Data, 1969, 14, 208-209
- [1970AMB] Ambrose, D.; Sprake, C.H.S., Thermodynamic properties of organic oxygen compounds XXV. Vapour pressures and normal boiling temperatures of aliphatic alcohols, J. Chem. Thermodyn., 1970, 2, 631-645
- [1970CLE] Clever, H.L.; Westrum, E.F., Jr., Dimethylsulfoxide and dimethylsulfone. Heat capacities, enthalpies of fusion, and thermodynamic properties, J. Phys. Chem., 1970, 74, 1309-1317
- [1970GOU] Goursot, P.; Girdhar, H.L.; Westrum, E.F., Jr., Thermodynamics of polynuclear aromatic molecules. III. Heat capacities and enthalpies of fusion of anthracene, J. Phys. Chem., 1970, 74, 2538-2541
- [1970POL] Polak, Jiri; Murakami, Sachio; Lam, V.T.; Benson, George C., Excess enthalpy, volume, and Gibbs free energy of cyclopentane-tetrachloroethylene mixtures at 25.deg., J. Chem. Eng. Data, 1970, 15, 323-328
- [1970WOL] Wolff, H.; Wuertz, R.; Hydrogen bonding and vapor pressure isotope effect of dimethylamine.. J. Phys. Chem., 1970, 74, 1600
- [1971EON] Eon, C.; Pommier, C.; Guiochon, G., Vapor pressures and second virial coefficients of some five-membered heterocyclic derivatives, J. Chem. Eng. Data, 1971, 16, 408-410
- [1971WON] Wong, W-K.; Westrum, E.F., Jr., Thermodynamics of polynuclear aromatic molecules. I. Heat capacities and enthalpies of fusion of pyrene, flouranthene, and triphenylene, J. Chem. Thermodynam., 1971, 3, 105-124
- [1972AHM] Ahmed, A.M.I.; Eades, R.G., Proton relaxation in solid aniline and some methyl derivatives, J. Chem. Soc., Faraday Trans. 2, 1972, 68, 2017
- [1972BOU] Boublik, T.; Aim, K.; Heats of Vaporization of Simple Non-spherical Molecule Compounds. Collect. Czech. Chem. Commun., 1972, 37, 3513
- [1972INO] Inozemtsev, P.P.; Liakumovich, A.G.; Gracheva, Z.D., Russ. J. Phys. Chem., 1972, 46, 914
- [1972IRV] Irving, R.J., The standard enthalpy of sublimation of naphthalene, J. Chem. Thermodyn., 1972, 4, 793-794
- [1972JAK] Jakli, G.; van Hook, W.A., The Vapor Pressures of Dimethyl Sulfoxide and Hexadeuterodimethyl Sulfoxide from about 313 to 453 K, J. Chem. Thermodyn., 1972, 4, 857-864
- [1972REG] Regnier, J., Tension de Vapeur de L'Ethane Entre 80 et 135 K, J. Chim. Phys., 1972, 69, 942-944

- [1973JAG] JagannadhaRao, Y.; Viswanath, D. S.; Integral isobaric heat of vaporization of benzene + 1,2-dichloroethane system. J. Chem. Eng. Data, 1973, 18, 49-50
- [1973MEY] Meyer, Edwin F.; Hotz, Roger D., High-precision vapor-pressure data for eight organic compounds, J. Chem. Eng. Data, 1973, 18, 359-362
- [1973MOR] Morke, K.; Kauer, E.; Kornig, J.; Bittrich, H.-J.; Excess free energy of binary systems of cyclohexanol, chlorocyclohexane, and cyclohexylamine.. Z. Phys. Chem. (Leipzig), 1973, 252, 323-36
- [1973PHI] Philippe, R.; Jambon, C.; Clechet, P.; Thermodynamic properties of dimethylsulfoxide + halomethane mixtures: II vapour pressures and excess thermodynamic functions. J. Chem. Thermodyn., 1973, 5, 431-444
- [1973SVO] Svoboda, V.; Vesely, F.; Holub, R.; Pick, J.; Enthalpy data of liquids: II the dependence of heats of vaporization of methanol, propanol, butanol, cyclohexane, cyclohexene and benzene on temperature. Collect. Czech. Chem. Commun., 1973, 38, 3539-43
- [1973WAG] Wagner, W.; "New vapour pressure measurements for argon and nitrogen and a new method for establishing rational vapour pressure equations," Cryogenics, 1973, 13, 470-82
- [1974AMB] Ambrose, D.; Sprake, C.H.S.; Townsend, R., Thermodynamic properties of organic oxygen compounds XXXIII. The vapour pressure of acetone, J. Chem. Thermodyn., 1974, 6, 693-700
- [1974DOJ] Dojcansky, J.; Heinrich, J., Saturated Vapour Pressure of Acetonitrile, Chem. Zvesti, 1974, 28, 157-159
- [1974KER] Kerns, W.J.; Anthony, R.G.; Eubank, P.T., Volumetric Properties of Cyclohexane Vapor, AIChE Symp. Ser., 1974, 70, 14-21
- [1974OSB] Osborn, Ann G.; Douslin, Donald R., Vapor-pressure relations for 15 hydrocarbons, J. Chem. Eng. Data, 1974, 19, 114-117
- [1975FRE] Frensdorff, H.K.; Adams, R.K., Vapor pressure of 2,4-tolylene diisocyanate, J. Chem. Eng. Data, 1975, 20, 13-15
- [1975HOP] Hopfner, A.; Parekh, N.; Horner, Ch.; Abdel-Hamid, A., Der Dampfdruck-Isotopie-Effekt von deuterierten Neopentanen, Ber. Bunsen-Ges. Phys. Chem., 1975, 79, 216-222
- [1975OSB] Osborn, Ann G.; Douslin, Donald R., Vapor pressures and derived enthalpies of vaporization for some condensed-ring hydrocarbons, J. Chem. Eng. Data, 1975, 20, 229-231
- [1975VID] Vidaurri, F.C., Ethyl Fluoride Vapor Pressure, J. Chem. Eng. Data, 1975, 20, 349-350
- [1976CIH] Cihlár, J.; Hynek, V.; Svoboda, V.; Holub, R., Heats of vaporization of alkyl esters of formic acid, Collect. Czech. Chem. Commun., 1976, 41, 1-6
- [1976DEP] Depaolo, P. H.; Venturello, V. S.; A Modified Method for the Determination of Heat of Vaporization or Heat Capacity of Selective Liquids Using an Empirical Differential Scanning Calorimetry Scan Technique, Thermochim. Acta, 1976, 16, 17-25

- [1976RAD] Radwan, M. H.; Hanna, A. A.; Binary Azeotropes Containing Butyric Acid. J. Chem. Eng. Data, 1976, 21, 285-289
- [1977FIN] Finke, H.L.; Messerly, J.F.; Lee, S.H.; Osborn, A.G.; Douslin, D.R., Comprehensive thermodynamic studies of seven aromatic hydrocarbons, J. Chem. Thermodyn., 1977, 9, 937-956
- [1977RAO] Rao, Yaddanapudi J.; Viswanath, Dabir S., Integral isobaric heats of vaporization of benzene-chloroethane systems, J. Chem. Eng. Data, 1977, 22, 36-38
- [1977STE] Steele, W.V., The standard enthalpies of formation of bicyclic compounds. III. 1,7,7-trimethylbicyclo[2.2.1]heptan-2-one, J. Chem. Thermodyn., 1977, 9, 311-314
- [1978MAR] Marchidan, D.I.; Ciopec, M., Thermodynamic properties of maleic, trimellitic and pyromellitic anhydrides, Rev. Roum. Chim., 1978, 23, 19-29
- [1978SUN] Sundaram, S.; Viswanath, D. S.; Thermodynamic properties of toluene-1,1,2,2,-tetrachloroethane. J. Chem. Eng. Data, 1978, 23, 62-64
- [1979KUD] Kudchadker, Shanti A.; Kudchadker, Arvind P.; Zwolinski, Bruno J., Chemical thermodynamic properties of anthracene and phenanthrene, J. Chem. Thermodyn., 1979, 11, 1051-1059
- [1979MAR] Markornik, V. S.; Sachek, A. I.; Peshchenko, A. D.; Sharro, O. V.; Andreevskii, D. N.; Olizarevich, N. M.; Temperature dependence of the vapor pressures of some aldehydes.. Termodin. Org. Soedin., 1979, 107
- [1979MAR2] Marsh, K. N.; French, H. T.; Rogers, H. P. D.; Excess Gibbs free energy for cyclohexane + nitromethane, + nitroethane, + 1-nitropropane, and + 2-nitropropane at 318.15 K.. J. Chem. Thermodyn., 1979, 11, 897-903
- [1979MJO] Mjojo, Chodziwadziwa C., Order-disorder phenomena. Part 2. Order-disorder phase equilibria in D- and L-systems of camphor and related compounds, J. Chem. Soc., Faraday Trans. 2, 1979, 75, 692
- [1979SMI] Smith, G.W., Phase behavior of some linear polyphenyls, Mol. Cryst. Liq. Cryst., 1979, 49, 207-209
- [1982MAR] Martin, J.F.; Andon, R.J.L., Thermodynamic properties of organic oxygen compounds. Part LII. Molar heat capacity of ethanoic, propanoic, and butanoic acids, J. Chem. Thermodynam., 1982, 14, 679-688
- [1982SVO] Svoboda, Václav; Charvátová, Vladimíra; Majer, Vladimír; Hynek, Vladimír, Determination of heats of vaporization and some other thermodynamic properties for four substituted hydrocarbons, Collect. Czech. Chem. Commun., 1982, 47, 543-549
- [1983DEW] DeWit, H.G.M.; Offringa, J.C.A.; De Kruif, C.G.; Van Miltenburg, J.C., Thermodynamic properties of molecular organic crystals containing nitrogen, oxygen and sulfur. III. Molar heat capacities measured by differential scanning calorimetry, Thermochim. Acta, 1983, 65, 43-51
- [1983OI] Oi, T.; Shulman, J.; Popowicz, A.; Ishida, Y.; Vapor pressure isotope effects in liquid methyl flouride. J. Phys. Chem., 1983, 87, 3153

- [1983WOL] Wolff, H.; Shadiakhy, A.; The vapour-pressure behaviour and the association of isomeric propylamines and N-deuteriopropylamines in mixtures with n-hexane. *Fluid Phase Equilib.*, 1983, 11, 267-287
- [1984BIC] Bickerton, J.; Minas Da Piedade, M.E.; Pilcher, G., Enthalpy of formation of tetrabromomethane by rotating-bomb calorimetry, *J. Chem. Thermodyn.*, 1984, 16, 661-668.
- [1984MIC] Michou-Saucet, Marie-Annie; Jose, Jacques; Michou-Saucet, Christian; Merlin, J.C., Pressions de vapeur et enthalpies libres d'excès de systèmes binaires: Hexaméthylphosphorotriamide (HMPT) + n-hexane; n-heptane; n-octane: 298,15 K; 303,15 K; 313,15 K; 323,15 K; 333,15 K, *Thermochimica Acta*, 1984, 75, 85-106
- [1984PIT] Pitzer, Kenneth S.; Peiper, J. Christopher; Busey, R.H., Thermodynamic Properties of Aqueous Sodium Chloride Solutions, *J. Phys. Chem. Ref. Data*, 1984, 13, 1
- [1985FER] Fernandez-Fassnacht, E.; Del Rio, F.; Vapour pressures of CCl₃F, CCl₂F₂ and CClF₃. *Cryogenics*, 1985, 25, 204
- [1985MAC] Machat, V.; Boublik, T.; Vapour-liquid equilibrium and excess molar volume of (trans-1,2-dichloroethene + cyclopentane) at 293.15 K. *J. Chem. Thermodyn.*, 1985, 17, 887-891
- [1985MAJ] Majer, V.; Svoboda, V., Enthalpies of Vaporization of Organic Compounds: A Critical Review and Data Compilation, Blackwell Scientific Publications, Oxford, 1985, 300
- [1987AMB] Ambrose, D.; Giassee, N. B.; Vapor Pressures and Critical Temperatures and Critical Pressures of Some Alkanoic Acids: C₁ to C₁₀. *J. Chem. Thermodyn.*, 1987, 19, 505-519
- [1987DEL] Della Gatta, G.; Ferro, D., Enthalpies of fusion and solid-to-solid transition, entropies of fusion for urea and twelve alkylureas, *Thermochim. Acta*, 1987, 122, 143-152
- [1987KAN] Kanungo, Arundhati.; Oi, Takao.; Popowicz, Anthony.; Ishida, Takanobu., Vapor pressure isotope effects in liquid methylene difluoride, *J. Phys. Chem.*, 1987, 91, 4198-4203
- [1987STE] Stephenson, Richard M.; Malanowski, Stanislaw, Handbook of the Thermodynamics of Organic Compounds, Springer, Dordrecht, 1987
- [1988DON] Dong, Jin-Quan; Lin, Rui-Sen; Yen, Wen-Hsing, Heats of vaporization and gaseous molar heat capacities of ethanol and the binary mixture of ethanol and benzene, *Can. J. Chem.*, 1988, 66, 783-790
- [1988GAM] Gambino, M.; Bros, J.P., Capacité calorifique de l'urée et de quelques mélanges eutectiques à base d'urée entre 30 et 140°C, *Thermochim. Acta*, 1988, 127, 223-236
- [1988SOP] Sopade, P.A.; Kearsley, M.W.; Le Grys, G.A., Heat of melting of sucrose, *Int. Sugar J.*, 1988, 90, 38
- [1988VAN] van Bommel, M.J.; van Miltenburg, J.C.; Schuijff, A., Heat-capacity measurements and thermodynamic functions of 1,3,5-triazine and 1,3,5-trioxane, *J. Chem. Thermodynam.*, 1988, 20, 397-403
- [1989CEP] Cepeda, E.; Gonzalez, C.; Resa, J. M.; Isobaric vapor-Liquid Equilibrium for the Cumene-Phenol System. *J. Chem. Eng. Data*, 1989, 34, 270-273

- [1989JAS] Jasperson, L. V.; Wilson, L. C.; Brady, C. J.; Wilding, W. V.; Wilson, G. M.; Vapor Association of Monocarboxylic Acids From Heat of Vaporization and PVT Measurements. AIChE Symp. Ser., 1989, 85, 102-39
- [1989RAM] Ramanjaneyulu, K.; Reddy, D. V. B.; Krishnaiah, A.; Viscosities of binary liquid mixtures of tetrachloroethylene with some aliphatic, alicyclic and substituted aromatic hydrocarbons. Phys. Chem. Liq., 1989, 20, 195
- [1990AMB] Ambrose, D.; Ewing, M.B.; Ghiassee, N.B.; Sanchez Ochoa, J.C., The ebulliometric method of vapour-pressure measurement: vapour pressures of benzene, hexafluorobenzene, and naphthalene, J. Chem. Thermodyn., 1990, 22, 589-605
- [1990DAS] Da Silva, Manuel A.V. Ribeiro; Monte, Manuel J.S., The construction, testing and use of a new knudsen effusion apparatus, Thermochimica Acta, 1990, 171, 169-183,
- [1990DAU] Daubert, T. E.; Hutchison, G.; Vapor pressure of 18 pure industrial chemicals. AIChE Symp. Ser., 1990, 86, 93-114
- [1990DON] Donnelly, J.R.; Drewes, L.A.; Johnson, R.L.; Munslow, W.D.; Knapp, K.K.; Sovocool, G.W., Purity and heat of fusion data for environmental standards as determined by differential scanning calorimetry, Thermochim. Acta, 1990, 167, 155
- [1990HU] Hu, R.; Li, J.; Liang, Y.; Wu, S.; Sun, L.; Wang, Y., Thermal behavior of 2,2,2-trinitroethyl-4,4,4-trinitrobutyrate, J. Therm. Anal., 1990, 36, 1155-60
- [1990MEV] Meva'a, L.M.; Lichanot, A., Proprietes thermodynamiques en phase condensee des ortho, meta et para fluorotoluene, cresol et toluidine, Thermochim. Acta, 1990, 158, 335-345
- [1990PLA] Platzer, B., Polt, A., and Maurer, G.; Thermophysical Properties of Refrigerants, Berlin, Springer-Verlag, 1990.
- [1991ACR] Acree, William E., Thermodynamic properties of organic compounds: enthalpy of fusion and melting point temperature compilation, Thermochimica Acta, 1991, 189, 37-56
- [1991KUU] Kuus, M.; Kirss, H.; Siimer, E.; Kudryavtseva, L. S.; Prediction of vapor-liquid equilibria and enthalpy of mixing by means of UNIFAC models. Thermochim. Acta, 1991, 182, 29-37
- [1991SET] Setzmann, U. and Wagner, W.; A New Equation of State and Tables of Thermodynamic Properties for Methane Covering the Range from the Melting Line to 625 K at Pressures up to 1000 MPa, J. Phys. Chem. Ref. Data, 20, 1061-1151, 1991
- [1991STE] Stewart, R.B., Jacobsen, R.T, and Wagner, W.; Thermodynamic Properties of Oxygen from the Triple Point to 300 K with Pressures to 80 MPa, J. Phys. Chem. Ref. Data, 20, 917-1021, 1991.
- [1992HAE] Haendel, G.; Kleinrahm, R.; Wagner, W.; Measurements of the (pressure, density, temperature) relations of dichlorodifluoromethane (R12) and of chlorodifluoromethane (R22) in parts of the homogeneous gas and liquid regions and on the coexistence curve. J. Chem. Thermodyn., 1992, 24, 697-713

- [1992JAC] Jacobsen, R.T, Penoncello, S.G., and Lemmon, E.W., "A Fundamental Equation for Trichlorofluoromethane (R-11)," Fluid Phase Equilib., 80, 45-56, 1992.
- [1992MAR] Marx, V., Pruss, A., and Wagner, W.; Neue Zustandsgleichungen fuer R 12, R 22, R 11 und R 113. Beschreibung des thermodynamischen Zustandsverhaltens bei Temperaturen bis 525 K und Druecken bis 200 MPa, Duesseldorf, VDI Verlag, Series 19 (Waermetechnik/Kaeltechnik), No. 57, 1992.
- [1992SAB] Sabbah, R.; El Watik, L., New reference materials for the calibration (temperature and energy) of differential thermal analysers and scanning calorimeters, J. Therm. Anal., 1992, 38, 855-863
- [1992SAL] Salvi-Narkhede, M.; Wang, B. -H.; Adcock, J. I.; Van Hook, W. A.; Vapor pressures, liquid molar volumes, vapor non-ideality, and critical properties of some partially fluorinated ethers (CF₃OCF₂CF₂H, CF₃OCF₂H, and CF₃OCH₃), some perfluoroethers (CF₃OCF₂OCF₃, c-CF₂OCF₂OCF₂, and c-CF₂CF₂CF₂O), and of CHF₂Br and CF₃CFHCF₃. J. Chem. Thermodyn., 1992, 24, 1065-1075
- [1993ACR] Acree, William E., Thermodynamic properties of organic compounds, Thermochimica Acta, 1993, 219, 97-104
- [1993REU] de Reuck, K.M. and Craven, R.J.B.; Methanol, International Thermodynamic Tables of the Fluid State-12, IUPAC, Blackwell Scientific Publications, London, 1993.
- [1993WEB] Weber, L. A.; Goodwin, A. R. H.; Ebulliometric measurement of the vapor pressure of difluoromethane. J. Chem. Eng. Data, 1993, 38, 254-256
- [1994KRI] Krishnaiah, A.; Viswanath, D. S.; Toghiani, H.; Isothermal Vapor-Liquid Equilibria of Dichloromethane + Isopropyl Ether dichloromethane. Experimental Results for DIPPR 1990-91 Projects on Phase Equilibria and Pure Component Properties, 1994, DIPPR Data Series No. 2
- [1994TIL] Tillner-Roth, R. and Baehr, H.D., "An International Standard Formulation of the Thermodynamic Properties of 1,1,1,2-Tetrafluoroethane (HFC-134a) for Temperatures from 170 K to 455 K at Pressures up to 70 MPa," J. Phys. Chem. Ref. Data, 23, 657-729, 1994.
- [1995CHE] Chen, G.; Wang, Q.; Ma, Z.-M.; Yan, X.-H.; Han, S.-J.; Phase equilibria at superatmospheric pressures for systems containing halohydrocarbon, aromatic hydrocarbon, and alcohol. J. Chem. Eng. Data, 1995, 40, 361-366
- [1995FER] Ferloni, Paolo; Gatta, Giuseppe Della, Heat capacities of urea, N-methylurea, N-ethylurea, N-(n)propylurea, and N-(n)butylurea in the range 200 to 360 K, Thermochimica Acta, 1995, 266, 203-212
- [1995KAM] Kamei, A., Beyerlein, S.W., and Jacobsen, R.T.; Application of Nonlinear Regression in the Development of a Wide Range Formulation for HCFC-22, Int. J. Thermophys., 16, 1155-1164, 1995.
- [1995NAS] Nass, Karen; Lenoir, Dieter; Kettrup, Antonius, Calculation of the Thermodynamic Properties of Polycyclic Aromatic Hydrocarbons by an Incremental Procedure, Angew. Chem. Int. Ed. Engl., 1995, 34, 1735-1736

- [1996CHI] Chirico, R. D.; Steele, W. V.; Nguyen, A.; Klots, T. D.; Knipmeyer, S. E.; "Thermodynamic properties of pyridine. I. Vapor pressures , high-temperature heat capacities, densities, critical properties, derived thermodynamic functions, vibrational assignment, and derivation of recommended values," J. Chem. Thermodyn. 28, 797-818 (1996).
- [1996DEJ] Dejoz, A.; Gonzalez-Alfaro, V.; Miguel, P. J.; Vazquez, M. I.; Isobaric Vapor-Liquid Equilibria of Tetrachloroethylene + 1-Propanol and +2-Propanol at 20 and 100 kPa. J. Chem. Eng. Data, 1996, 41, 1361-1365
- [1996DOM] Domalski, Eugene S.; Hearing, Elizabeth D., Heat Capacities and Entropies of Organic Compounds in the Condensed Phase. Volume III, J. Phys. Chem. Ref. Data, 1996, 25, 1
- [1996LOP] Lopes, J. N. C.; Calado, J. C. G.; Vapour pressure of liquid deuterium sulphide. J. Chem. Thermodyn., 1996, 28, 1395-1401
- [1996OUT] Outcalt, S.L. and McLinden, M.O.; A Modified Benedict-Webb-Rubin Equation of State for the Thermodynamic Properties of R152a (1,1-Difluoroethane), J. Phys. Chem. Ref. Data, 25, 605-636, 1996.
- [1996WEB] Weber, L.A.; Defibaugh, D.R., Vapor Pressures and PVT Properties of the Gas Phase of 1,1,1-Trifluoroethane, J. Chem. Eng. Data, 1996, 41, 1477-1480
- [1997BEL] Belaribi-Boukais, G.; Ait-Kaci, A.; Mokbel, I.; Jose, J.; Total vapor pressures between 253 K and 363 K of binary mixtures of hex-1-yne +methanol, +ethanol, or+ butan-1-ol and excess enthalpies at 298.15 K of binary mixtures of hex-1-yne + methanol or + butan-1-ol. ELDATA: The International Electronic Journal of Physico-Chemical Data, 1997, 3, 173-182
- [1997HOL] Holcomb, C.D.; Magee, J. W.; Scott, J. L.; Outcalt, S. L.; Haynes, W. M.; Selected Thermodynamic Properties for Mixtures of R-32 R-125 R-134A R-143A R-41 R-290 and R-744, NIST Technical Note 1397 (1997)
- [1997NEG] Negadi, L.; Guetachew, T.; Ait-Kaci, A.; Jose, J.; Static measurements of the total vapor pressure of binary mixtures of dimethyl sulfoxide + hex-1-yne, hex-2-yne or hex-3-yne at temperatures between 283 K and 363 K. ELDATA: The International Electronic Journal of Physico-Chemical Data, 1997, 3, 91-100
- [1997SCH] Schaar, E.; Int. DATA Ser., Sel. Data Mixtures, Ser. A, 1997, 25, 40-63
- [1997TIL] Tillner-Roth, R. and Yokozeki, A.; An International Standard Equation of State for Difluoromethane (R-32) for Temperatures from the Triple Point at 136.34 K to 435 K and Pressures up to 70 MPa, J. Phys. Chem. Ref. Data, 26, 1273-1328, 1997.
- [1998CHI] Chirico, R.D.; Cowell, A.B.; Good, W.D.; Klots, T.D.; Knipmeyer, S.E.; Nguyen, A.; Rau, A.P.; Reynolds, J.W.; Smith, N.K.; Steele, W.V., Heat capacities, enthalpy increments, phase transitions, and derived thermodynamic functions for the condensed phases of bicyclohexyl between the temperatures 6 K and 440 K, J. Chem. Thermodyn., 1998, 30, 1423-1439

- [1998JAM] Jamróz, Malgorzata E.; Palczewska-Tulinska, Marcela; Wyrzykowska-Stankiewicz, Danuta; Szafranski, Andrzej M.; Polaczek, Jerzy; Dobrowolski, Jan Cz.; Jamróz, Michal H.; Mazurek, Aleksander P., The urea--phenol(s) systems, *Fluid Phase Equilibria*, 1998, 152, 307-326
- [1999DYK] Dykyj, J.; Svoboda, J.; Wilhoit, R.C.; Frenkel, M.L.; Hall, K.R., Vapor Pressure of Chemicals: Part A. Vapor Pressure and Antoine Constants for Hydrocarbons and Sulfur, Selenium, Tellurium and Hydrogen Containing Organic Compounds, Springer, Berlin, 1999
- [1999IGO] Igoudjilene, O.; Ait-Kaci, A.; Jose, J.; Static measurements of the total vapor pressure of binary mixtures of butan-2-one with 1,1,1-trichloroethane or 1,1-dichloroethane in the temperature range from 263 K to 343 K. *ELDATA: The International Electronic Journal of Physico-Chemical Data*, 1999, 5, 165-171
- [1999LOR] Loras, Sonia; Aucejo, Antonio; Muñoz, Rosa, Vapor-liquid equilibria in the systems 3-methylpentane+methyl 1,1-dimethylethyl ether and 3-methylpentane+methyl 1,1-dimethylpropyl ether at 101.3 kPa, *Fluid Phase Equilibria*, 1999, 156, 185-195
- [1999MIY] Miyamoto, S.; Nakamura, S.; Iwai, Y.; Arai, Y.; Measurement of vapor-phase compressibility factors of monocarboxylic acids using a flow-type apparatus and their association constants, *J. Chem. Eng. Data*, 1999, 44, 48-51
- [1999MON] Montón, Juan B.; de la Torre, Javier; Burguet, M.C.; Muñoz, Rosa; Loras, Sonia, Isobaric Vapor-Liquid Equilibrium in the Systems 2,3-Dimethylpentane + Methyl 1,1-Dimethylethyl Ether, + Diisopropyl Ether and + Methyl 1,1-Dimethylpropyl Ether, *J. Chem. Eng. Data*, 1999, 44, 1158-1162
- [1999OJA] Oja, V.; Suuberg, E. M.; Vapor Pressures and Enthalpies of Sublimation of d-Glucose, d-Xylose, Cellobiose, and Levoglucosan, *J. Chem. Eng. Data* 44, 26 (1999).
- [1999RAI] Rai, U.S.; Rai, R.N., Some Physicochemical Studies on Organic Eutectics and Molecular Complex: Urea -- p-nitrophenol System, *J. Mater. Res.*, 1999, 14, 1299-1305
- [1999TAK] Takagi, T.; T. Sakura, T. Tsuji, M. Hongo, Bubble point pressure for binary mixtures of difluoromethane with pentafluoroethane and 1,1,1,2-tetrafluoroethane. *Fluid Phase Equilib.* 162 (1999) 171–179
- [1999TEO] Teodorescu, M.; Aim, K.; Wichterle, I.; Isothermal vapour-liquid equilibria for pentan-3-one + 1,4 dichlorobutane, + trichloromethane, + 1,1,1 trichloroethane, + 1,1,2,2-tetrachloroethane binary mixtures. *Fluid Phase Equilib.*, 1998, 147, 215-228
- [2000LEM] Lemmon, E.W. and Jacobsen, R.T.; An International Standard Formulation for the Thermodynamic Properties of 1,1,1-Trifluoroethane (HFC-143a) for Temperatures from 161 to 450 K and Pressures to 50 MPa, *J. Phys. Chem. Ref. Data*, 29, 521-552, 2000.

- [2000LIS] Lisicki, Zygmunt; Jamróz, Malgorzata E., (Solid + liquid) equilibria in (polynuclear aromatic+ tertiary amide) systems, *J. Chem. Thermodyn.*, 2000, 32, 1335-1353
- [2000MAG] Magee, J.W., Outcalt, S.L., and Ely, J.F.; Molar Heat Capacity $C(v)$, Vapor Pressure, and (p, ρ, T) Measurements from 92 to 350 K at Pressures to 35 MPa and a New Equation of State for Chlorotrifluoromethane (R13), *Int. J. Thermophys.*, 21, 1097-1121, 2000.
- [2000SMU] Smukala, J., Span, R., and Wagner, W.; A New Equation of State for Ethylene Covering the Fluid Region for Temperatures from the Melting Line to 450 K at Pressures up to 300 MPa, *J. Phys. Chem. Ref. Data*, 29, 1053-1122, 2000.
- [2000SPA] Span, R., Lemmon, E.W., Jacobsen, R.T, Wagner, W., and Yokozeki, A.; A Reference Equation of State for the Thermodynamic Properties of Nitrogen for Temperatures from 63.151 to 1000 K and Pressures to 2200 MPa, *J. Phys. Chem. Ref. Data*, 29, 1361-1433, 2000.
- [2001KUL] Kul, I.; DesMarteau, D. D.; Beyerlein, A. L.; Vapor-liquid equilibria for CF₃OCF₂H/fluorinated ethane and CF₃SF₅/fluorinated ethane mixtures as potential R22 alternatives. *Fluid Phase Equilib.*, 2001, 185, 241-253
- [2001SEG] Segura, Hugo; Lam, Elizabeth; Reich, Ricardo; Wisniak, Jaime, Isobaric Phase Equilibria in the Binary Systems Ethyl 1,1-Dimethylethyl Ether + 1-hexene and + Cyclohexene at 94.00 kPa, *Phys. Chem. Liq.*, 2001, 39, 43-54
- [2002CHI] Chirico, R.D.; Knipmeyer, S.E.; Steele, W.V., Heat capacities, enthalpy increments, and derived thermodynamic functions for naphthalene between the temperatures 5 K and 440 K, *J. Chem. Thermodyn.*, 2002, 34, 1873-1884
- [2002ROY] Roy, S.; Riga, A.T.; Alexander, K.S., Experimental design aids the development of a differential scanning calorimetry standard test procedure for pharmaceuticals, *Thermochimica Acta*, 2002, 392, 399-404
- [2002STE] W. V. Steele, R. D. Chirico, S. E. Knipmeyer, and A. Nguyen, *J. Chem. Eng. Data* 47, 689 (2002)
- [2002VER] Verevkin, Sergey P., Relationships among strain energies of mono- and poly-cyclic cyclohexanoid molecules and strain of their component rings, *J. Chem. Thermodyn.*, 2002, 34, 263-275
- [2002WAG] Wagner, W. and Pruss, A., "The IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use," *J. Phys. Chem. Ref. Data*, 31, 387-535, 2002
- [2003PEN] Penoncello, S.G., Lemmon, E.W., Jacobsen, R.T, and Shan, Z.; A Fundamental Equation for Trifluoromethane (R-23), *J. Phys. Chem. Ref. Data*, 32, 1473-1499, 2003
- [2003ROJ] Rojas, Aarón; Orozco, Eulogio, Measurement of the enthalpies of vaporization and sublimation of solids aromatic hydrocarbons by differential scanning calorimetry, *Thermochimica Acta*, 2003, 405, 93-107
- [2003SHA] Sharma, B.L.; Kant, Rajesh; Sharma, Ritu; Tandon, Sonika, Deviations of binary organic eutectic melt systems, *Materials Chemistry and Physics*, 2003, 82, 216-224

- [2003ZAI] Zaitsau, Dz; Kabo, G.J.; Kozyro, A.A.; Sevruck, V.M., The effect of the failure of isotropy of a gas in an effusion cell on the vapor pressure and enthalpy of sublimation for alkyl derivatives of carbamide, *Thermochimica Acta* 406, 17 (2003).
- [2004GLO] Glos, S.; Kleinrahm, R.; Wagner, W.; Measurement of the (p, ρ , T) relation of propane, propylene, n-butane, and isobutane in the temperature range from (95 to 340) K at pressures up to 12 MPa using an accurate two-sinker densimeter. *J. Chem. Thermodyn.*, 2004, 36, 1037-1059
- [2004LEM] Lemmon, E.W. and Huber, M.L., "Thermodynamic Properties of n-Dodecane," *Energy & Fuels*, 18, 960-967, 2004.
- [2004MAN] Manatt, S. L.; Manatt, M. R. R.; On the Analyses of Mixture Vapor Pressure Data: The Hydrogen Peroxide/ Water System and Its Excess Thermodynamic Functions. *Chem. Eur. J.*, 2004, 6540
- [2004MON] Mondieig, D.; Rajabalee, F.; Metivaud, V.; Oonk, H.A.J.; Cuevas-Diarte, M.A., n -Alkane Binary Molecular Alloys, *Chem. Mater.*, 2004, 16, 786-798
- [2004WRI] Wright, S. F.; Dollimore, D.; Dunn, J. G.; Alexander, K.; Determination of the vapor pressure curves of adipic acid and triethanolamine using thermogravimetric analysis, *Thermochim. Acta* 421, 25 (2004).
- [2005LEM] Lemmon, E.W. and Jacobsen, R.T.; A New Functional Form and New Fitting Techniques for Equations of State with Application to Pentafluoroethane (HFC-125), *J. Phys. Chem. Ref. Data*, 34, 69-108, 2005
- [2005LEM2] Lemmon, E.W. and Ihmels, E.C., Thermodynamic Properties of the Butenes. Part II. Short Fundamental Equations of State, *Fluid Phase Equilib.*, 228-229, 173-187, 2005
- [2005LID] CRC Handbook of Chemistry and Physics, CRC Press, Boca Raton, 2005
- [2005ROD] Rodrigues, W. L.; Mattedi, S.; Abreu, J. C. N.; Vapor-Liquid Equilibria Data for Binary Systems of Ethylbenzene + Xylene Isomers at 100.65 kPa. *J. Chem. Eng. Data*, 2005, 50, 1134-1138
- [2005RUZ] Ruzicka, K.; Fulem, M.; Ruzicka, V.; Recommended vapor pressure of solid naphthalene, *J. Chem. Eng. Data* 50, 1956 (2005)
- [2006BOU] Boukais-Belaribi, G.; Belaribi, B. F.; Ait-Kaci, A.; Jose, J.; Thermodynamics of isomeric hexynes +MTBE binary mixtures. *Fluid Phase Equilib.*, 2006, 248, 181-190
- [2006BUC] Bückner, D. and Wagner, W.; A Reference Equation of State for the Thermodynamic Properties of Ethane for Temperatures from the Melting Line to 675 K and Pressures up to 900 MPa, *J. Phys. Chem. Ref. Data*, 35, 205-266, 2006.
- [2006BUC2] Bückner, D. and Wagner, W.; Reference Equations of State for the Thermodynamic Properties of Fluid Phase n-Butane and Isobutane, *J. Phys. Chem. Ref. Data*, 35, 929-1019, 2006.

- [2006EME] Emel'yanenko, Vladimir N.; Kabo, Gennady J.; Verevkin, Sergey P., Measurement and Prediction of Thermochemical Properties: Improved Increments for the Estimation of Enthalpies of Sublimation and Standard Enthalpies of Formation of Alkyl Derivatives of Urea, *J. Chem. Eng. Data* 51, 79 (2006).
- [2006FEI] Feistel, R.; Wagner, W.; A New Equation of State for H₂O Ice Ih, *J. Phys. Chem. Ref. Data*, 2006, 35, 1021-1047
- [2006KHI] Khimeche, Kamel; Dahmani, Abdallah, Solid-Liquid Equilibria of Naphthalene + Alkanediamine Mixtures, *J. Chem. Eng. Data*, 2006, 51, 382-385
- [2006LEM] Lemmon, E.W. and Span, R.; Short Fundamental Equations of State for 20 Industrial Fluids, *J. Chem. Eng. Data*, 51, 785-850, 2006.
- [2007EFI] Efimova, A.A.; Pashchenko, L.L.; Varushchenko, R.M.; Krasnyh, E.; Levanova, S.V., The thermodynamics of vaporization of ethyl tert-butyl ether, isobutyl tert-butyl ether, and di-isopropyl ether, *J. Chem. Thermodyn.*, 2007, 39, 142-147
- [2007VER] Verevkin, Sergey P.; Emel'yanenko, Vladimir N.; Klamt, Andreas, Thermochemistry of Chlorobenzenes and Chlorophenols: Ambient Temperature Vapor Pressures and Enthalpies of Phase Transitions, *J. Chem. Eng. Data*, 2007, 52, 499-510
- [2008MAR] Martinez, N. F.; Lladosa, E.; Burguet, M.; Monton, J. B. ; Isobaric vapour liquid equilibria for binary systems of 2-butanone with ethanol, 1-propanol, and 2-propanol at 20 and 101.3 kPa. *Fluid Phase Equilib.*, 2008, 270, 62-68
- [2008SHA] Sharma, B.L.; Gupta, S.; Tandon, S.; Kant, R., Physico-mechanical properties of naphthalene--acenaphthene eutectic system by different modes of solidification, *Materials Chemistry and Physics*, 2008, 111, 423-430
- [2008SON] Soni, Minal; Ramjugernath, Deresh; Raal, J. David, Vapor--Liquid Equilibrium for Binary Systems of 2,3-Pentanedione with Diacetyl and Acetone, *J. Chem. Eng. Data*, 2008, 53, 745-749
- [2009BRI] Brittain, Harry G., Vibrational Spectroscopic Studies of Cocrystals and Salts. 2. The Benzylamine-Benzoic Acid System, *Crystal Growth & Design*, 2009, 9, 3497-3503
- [2009LEA] Leachman, J.W., Jacobsen, R.T, Penoncello, S.G., and Lemmon, E.W.; Fundamental Equations of State for Parahydrogen, Normal Hydrogen, and Orthohydrogen, *J. Phys. Chem. Ref. Data*, 38, 721-748, 2009.
- [2009LEM] Lemmon, E.W., McLinden, M.O., and Wagner, W.; Thermodynamic Properties of Propane. III. A Reference Equation of State for Temperatures from the Melting Line to 650 K and Pressures up to 1000 MPa, *J. Chem. Eng. Data*, 54, 3141-3180, 2009
- [2009MAR] Marrufo, Beatriz; Aucejo, Antonio; Sanchotello, Margarita; Loras, Sonia, Isobaric vapor--liquid equilibrium for binary mixtures of 1-hexene+n-hexane and cyclohexane+cyclohexene at 30, 60 and 101.3 kPa, *Fluid Phase Equilibria*, 2009, 279, 11-16
- [2009TOR] Torres, Luis A.; Campos, Myriam; Martínez, Melchor; Rojas, Aarón, The thermochemistry of coronene revisited, *J. Chem. Thermodyn.*, 2009, 41, 957-965

- [2010FER] Fernandez, L.; Perez, E.; Ortega, J.; Canosa, J.; Wisniak, J.; Measurements of the Excess Properties and Vapor-Liquid Equilibria at 101.32 kPa for Mixtures of Ethyl Ethanoate + Alkanes (from C5 to C10) . J. Chem. Eng. Data, 2010, 55, 5519-5533
- [2010GAN] Ganeff, Jean M.; Jungers, Joseph C., Tensions de vapeur du système CH₃Cl / CH₂Cl₂, Bull. Soc. Chim. Belges, 2010, 57, 82-87
- [2010HAI] Haimi, P.; Uusi-Kyyny, P.; Pokki, J. -P.; Alopaeus, V.; Phase equilibrium measurements for systems containing propanenitrile with tert-butyl ethyl ether and C₄-hydrocarbons. Fluid Phase Equilib., 2010, 299, 148-160
- [2010MAT] Matsuda, H.; Yokoyama, K.; Kyuzaki, H.; Kurihara, K.; Tochigi, K.; Isobaric Vapor Liquid Equilibria for Ethylbenzene + p-Xylene System and Separation by Distillation. J. Chem. Eng. Jpn., 2010, 43, 247-252
- [2010SAP] Sapei, Erlin; Uusi-Kyyny, Petri; Keskinen, Kari I.; Alopaeus, Ville, Phase equilibria of binary systems of 3-methylthiophene with four different hydrocarbons, Fluid Phase Equilibria, 2010, 288, 155-160
- [2011ABB] Abbas, R.; Schedemann, A.; Ihmels, C.; Enders, S.; Gmehling, J.; Measurement of Thermophysical Pure Component Properties for a Few Siloxanes Used as Working Fluids for Organic Rankine Cycles. Ind. Eng. Chem. Res., 2011, 50, 9748-9757
- [2011MEJ] Mejia, A.; Segura, H.; Cartes, M.; Measurement and theoretical prediction of the vapor-liquid equilibrium, densities and interfacial tensions of the system hexane + 2-methoxy-2-methylbutane. Fluid Phase Equilib., 2011, 308, 15-24
- [2011WU] Wu, J., Zhou, Y., and Lemmon, E.W.; An Equation of State for the Thermodynamic Properties of Dimethyl Ether, J. Phys. Chem. Ref. Data, 40, 023104, 2011.
- [2012THO] Thol, M., Lemmon, E.W., and Span, R., Equation of State for Benzene for Temperatures from the Melting Line up to 725 K with Pressures up to 500 MPa, High Temp.-High Press., 41, 81-97, 2012.
- [2012ZHO] Zhou, Y., Lemmon, E.W., and Wu, J.; Thermodynamic Properties of o-Xylene, m-Xylene, p-Xylene, and Ethylbenzene, J. Phys. Chem. Ref. Data, 41, 023103, 2012.
- [2013HYE] Hyeong, S.; Jang, S.; Lee, K.-s.; Kim, H.; Isothermal vapor-liquid equilibria for the binary systems of ethylene glycol monopropyl ether with 2,2-dimethylbutane and 2,3-dimethylbutane. Korean J. Chem. Eng., 2013, 30, 434-439
- [2013LAA] Laavi, H.; Pokki, J. -P.; Uusi-Kyyny, P.; Kim, Y.; Alopaeus, V.; Vapor Liquid Equilibrium, Excess Molar Enthalpies, and Excess Molar Volumes of Binary Mixtures Containing 2-Ethoxy-2-methylpropane or 2-Ethoxy-2-methylbutane and Acetonitrile or Propanenitrile. J. Chem. Eng. Data, 2013, 58, 943-950
- [2013RED] Reddy, P.; Raal, J. D.; Ramjugernath, D.; A novel dynamic recirculating apparatus for vapour-liquid equilibrium measurements at moderate pressures and temperatures. Fluid Phase Equilib., 2013, 358, 121-130
- [2014MAG] A. Magoń, A. Wurm, C. Schick, P. Pangloli, S. Zivanovic, M. Skotnicki, M. Pyda, Heat capacity and transition behavior of sucrose by standard, fast scanning and temperature-modulated calorimetry. Thermochim. Acta 589, 183–196 (2014)

- [2014NIC] Nicolae, M.; Oprea, F.; Vapor-liquid equilibrium for the binary mixtures of dipropylene glycol with aromatic hydrocarbons: Experimental and regression. *Fluid Phase Equilib.*, 2014, 370, 34-42
- [2014RED] Reddy, K. R.; Kumar, D. B. K.; Rao, G. S.; Sri, P. B. S.; Begum, Z.; Rambabu, C.; Ebulliometric determination of vapor liquid equilibria for binary mixtures of NMP with some cyclic compounds. *J. Mol. Liq.*, 2014, 193, 220-225
- [2014SCH] Schroeder, J.A., Penoncello, S.G., and Schroeder, J.S.; A New Fundamental Equation for Ethanol," *J. Phys. Chem. Ref. Data*, 43, 043102, 2014.
- [2014THO] Thol, M., Piazza, L., and Span, R.; A New Functional Form for Equations of State for Some Polar and Weakly Associating Fluids, *Int. J. Thermophys.*, 35, 783-811, 2014.
- [2014ZHO] Zhou, Y., Liu, J., Penoncello, S.G., and Lemmon, E.W., An Equation of State for the Thermodynamic Properties of Cyclohexane, *J. Phys. Chem. Ref. Data*, 43, 043105, 2014.
- [2015THO] Thol, M., Rutkai, G., Koester, A., Kortmann, M., Span, R., and Vrabec, J.; Fundamental Equation of State for Ethylene Oxide Based On a Hybrid Dataset, *J. Chem. Eng. Sci.*, 121, 87, 2015.
- [2016AKA] Akasaka, R., Fukushima, M., and Lemmon, E.W.; A Helmholtz Energy Equation of State for Trifluoroethylene (R-1123), International Refrigeration and Air Conditioning Conference at Purdue, July 11-14, 2016.
- [2016QI] Qi, H., Fang, D., Gao, K., Meng, X., and Wu, J.; Compressed Liquid Densities and Helmholtz Energy Equation of State for Fluoroethane (R161), *Int. J. Thermophys.*, 37, 55, 2016
- [2017SOU] C. Sousa, M. A. R. Matos and V. M. F. Morais, Experimental and Computational Thermochemical Study of Maleic Anhydride and Vinylene Carbonate, *J. Phys. Chem. A* 121, 9474 (2017).
- [2017THO] M. Thol, G. Rutkai, A. Köster, S. Miroshnichenko, W. Wagner, J. Vrabec and R. Span, Equation of state for 1,2-dichloroethane based on a hybrid data set. *Molecular Physics* 115, 1166 (2017).
- [2018FER] Ferreira, Machel; Schwarz, Cara E.; Low-Pressure VLE Measurements and Thermodynamic Modeling, with PSRK and NRTL, of Binary 1-Alcohol + n-Alkane Systems. *J. Chem. Eng. Data*, 2018, 63, 4614-4625
- [2018LEM] Lemmon, E. W.; Bell, I. H.; Huber, M. L.; McLinden, M. O.; Reference Fluid Thermodynamic and Transport Properties (REFPROP), NIST Standard Reference Database 23, Vers. 10.0, 2018
- [2018STU] Stuckenholtz, M.; Crespo, E. A.; Vega, L. F.; Carvalho, P. J.; Coutinho, J. A. P.; Schrer, W.; Kiefer, J.; Rathke, B.; Vapor Liquid Equilibria of Binary Mixtures of 1-Butyl-3-methylimidazolium Triflate and Molecular Solvents: n-Alkyl Alcohols and Water. *J. Phys. Chem. B*, 2018, 122, 6017-6032
- [2019BEN] Benecke, Travis Pio; Narasigadu, Caleb; Iwarere, Samuel A.; Nelson, Wayne M.; Ramjugernath, Deresh; Vapor-Liquid Equilibrium Measurements of Ether Alcohol Blends for Investigation on Reformulated Gas. *J. Chem. Eng. Data*, 2019, 64, 115-123

- [2019DIK] Diky, V.; Chirico, R. D.; Frenkel, M.; Bazyle, A.; Magee, J. W.; Paulechka, E.; Kazakov, A.; Lemmon, E. W.; Muzny, C. D.; Smolyanitsky, A. Y.; Townsend, S.; Kroenlein, K.; ThermoData Engine (TDE), Vers. 10.4.2, NIST Standard Reference Database 103b, (2019)
- [2019LI] Li, G.; Yin, X.; Jing, B.; Study of Vapor-Liquid Equilibria for Acetic Acid + n-Propyl Acetate + Isopropyl Acetate Systems. J. Chem. Eng. Data, 2019, 64, 2759-2766
- [2020MOO] Moodley, Kuveneshan; Excess properties and isothermal vapour-liquid equilibrium data for the ((toluene + pyridine)) system from 343 K to 383 K. J. Chem. Thermodyn., 2020, 141, 105946
- [2021LIN] P.J. Linstrom and W.G. Mallard, Eds., NIST Chemistry WebBook, NIST Standard Reference Database Number 69, National Institute of Standards and Technology, Gaithersburg MD, 20899, <https://doi.org/10.18434/T4D303>, (accessed July 22, 2021).
- [2022BUR] Estimated in this work

4. Heats of Combustion and Formation

4.1. Overview

In this section, we present compiled, evaluated, and recommended standard state ($T = 298.15\text{ K}$) heats (enthalpies) of combustion and formation for hydrocarbons, substituted hydrocarbons, and a few other compounds of physical, chemical, and engineering interest. We provide over 1500 values for enthalpies of combustion and formation in the gas (about 700), liquid (about 750), and solid (about 90) phases for about 270 compounds taken from about 550 different sources. The enthalpies of combustion and formation were determined using a variety of methods, mainly combustion calorimetry, heats of hydrogenation, and heats of hydrolysis. The values provided here were taken from the literature, from the NIST Chemistry WebBook,[2021LIN], and the NIST TRC Thermodata Engine (TDE).[2021DIK] The recommended values of enthalpies of combustion and formation provided here were selected from reliable original sources (not weighted averages).

All of the compiled values are contained in Table 4-8, while the selected recommended values are presented in Table 4-9. As a companion to the recommended combustion values in Table 4-9, we provide in Table 4-10 both “gross” and “net” heats of combustion, where “gross” are the heats of combustion where the products are at standard state (298.15 K and 1 bar) where water is in the liquid phase, while “net” are the heats of combustion where the products are all gaseous. For example, for combustion of chlorinated hydrocarbons the gross heat of combustion is where H_2O is liquid and HCl is in an aqueous solution, while the net heat of combustion is where both H_2O and HCl are in the gas phase. In the uncertainty column “unc”, we provide the uncertainty as provided in the original – in many cases, the workers did not identify whether the uncertainties were 1σ or 2σ . On the other hand, all of our recommended values are 2σ .

Most of the enthalpies of combustion/formation in these tables are derived from heats of combustion (in the liquid or gas phase), although there are a fair number derived from heats of hydrogenation, hydrolysis, or other chemical reactions. In cases where the original source data is not from combustion, the heat of combustion is derived from the heats of formation. The reverse is also true – heats of formation derived from heats of combustion. For many compounds, the “best” (selected) value of enthalpy of vaporization (or fusion) is used to compute enthalpies of formation in one phase from another phase, *i.e.*, $\Delta_f H(\text{gas}) = \Delta_f H(\text{liq}) + \Delta_v H$ or $\Delta_f H(\text{liq}) = \Delta_f H(\text{sol}) + \Delta_{\text{fus}} H$.

Tables 4-1, 4-2, 4-3, 4-4, and 4-5 provide descriptions of the notations used in the tables, while Tables 4-6 and 4-7 (and text in Sec. 4.2. Heats of Dilution) provide information about heats of dilution for aqueous solutions of HF , HCl , HBr , H_2SO_4 (sulfuric acid), and H_3BO_3 (boric acid).

In the tables in this section, as with other sections, the compounds are arranged first by primary and secondary chemical class and then by chemical formula.

Table 4-1. Notations for column headings

Symbol	Description
Formula	molecular formula
Compound	chemical name
T_{trs}	transition temperature: boiling (l), melting (s), or sublimation (s) note: normal boiling and melting points at 1 atm
Ph	phase below the transition temperature
SEL	selected (recommended) value
$\Delta_c H$	enthalpy of combustion at 298.15 K with H ₂ O (liq)
$\Delta_f H$	enthalpy of formation at 298.15 K with H ₂ O (liq)
$\Delta_{\text{trs}} H$	enthalpy of vaporization, fusion, or sublimation at 298.15 K
u	<u>reported</u> uncertainty in enthalpy of combustion, formation, or transition note: this is the reported uncertainty, not the standard uncertainty
M	method (see Table 4-4)
R	Denotes revised value (see notes below)
Ref	reference

Table 4-2. Notations for phase column

Notation	Phase
s	solid
l	liquid
g	gas
aq	aqueous solution
lg	denotes boiling point
sl	denotes melting point
sg	denotes sublimation point
s(d)	denotes decomposition temperature of a solid

Table 4-3. Notations for selected values column

SEL	Description
SEL	selected value at standard state (298 K, 1 atm)
SEL1	selected value in phase 1 (not standard state)
SEL2	selected value in another phase 2 (not standard state)
u	denotes a very uncertain value

Table 4-4. Notations for methods column

Method	Definition	Method	Definition
BDE	from bond dissociation energy	G3B3	calculated value using G3B3 method
Br2	from reaction involving Br ₂	H2	from enthalpy of hydrogenation
C	from enthalpy of combustion	H2O	from enthalpy of hydrolysis
C(l)	from enthalpy of combustion of liquid	HBr	from reaction using HBr
CaO2	from reaction involving CaO ₂ H ₂	HCl	from reaction using HCl
CBS	calculated value using complete basis set method	Hde	from enthalpy of dehydrogenation
CC	calculated value using coupled cluster method	I2	from heats of iodination
Cl2	from reaction using Cl ₂	ion	from heat of ion/molecule reaction
D	derived from enthalpy of decomposition	K	from enthalpy of combustion using K
def	by definition at standard state	LiAl	from enthalpy of reduction using Lithium aluminum hydride
DfusH	computed using reference value for enthalpy of fusion	LiB	from enthalpy of reduction using Lithium triethylborohydride
DsubH	computed using reference value for enthalpy of sublimation	Na	from enthalpy of combustion using Na
DvH	computed using reference value for enthalpy of vaporization	NF3	from reaction using NF ₃
E	derived using equilibrium reaction	QM	calculated value using quantum chemistry
est	estimated value	R	review value
f	reference value for enthalpy of fusion	s	reference value for enthalpy of sublimation
f(est)	derived using estimated enthalpy of fusion	TN	calculated using a thermochemical network
F2	from reaction using F ₂	v	reference value for enthalpy of vaporization

Table 4-5. Notations for revised values contained in column “R”

This table contains notation associated with the method to compute enthalpies of formation and combustion. In the first 4 cases (a through d), the values were recalculated by later work. In last 5 cases (e through k), the original aqueous dilutions that were measured or assumed for H₂SO₄ and HF are provided. The heats of formation of the aqueous dilutions are provided in Tables 4-8 and 4-9.

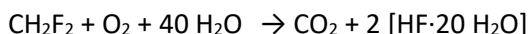
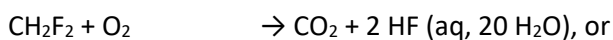
Note	Description
a	recalculated by 1965COX
b	recalculated by 1970COX
c	recalculated by 2002MAN
d	recalculated by 2019PAU
e	H ₂ SO ₄ (aq, 30 H ₂ O)
f	H ₂ SO ₄ (aq, 45 H ₂ O)
g	H ₂ SO ₄ (aq, 115 H ₂ O)
j	HF (aq, 22 H ₂ O)
k	HF (aq, 23 H ₂ O)

4.2. Heats of Dilution

Where the standard state for a product of combustion is an aqueous solution, we have computed heats of combustion and formation employing standard enthalpies of dilution for HF, HCl, HBr, sulfuric acid (H₂SO₄), and boric acid (H₃BO₃).

Aqueous Dilutions for HCl

The notation, for example, HF (aq, 20 H₂O) means a dilution of one part HF in 20 parts H₂O and the reaction for the combustion of difluoromethane (gas) can be written equivalently as



Various workers have used different heats of dilution for HF in aqueous solutions for combustion of fluorinated hydrocarbons depending on the calculated final concentrations of HF in water (or assumed concentrations). This impacts the enthalpies of formation derived from the enthalpies of combustion by on the order of 0.15 kJ mol⁻¹ with final dilutions ranging from HF (aq, 10 H₂O) to HF (aq, 50 H₂O). There are other factors that also impact the derived enthalpies of formation by as much as about 30 kJ mol⁻¹ as discussed by Cox et al [1965COX] and Paulechka and Kazakov [2019PAU]. The following table shows dilutions determined (or assumed) for a set of fluorinated hydrocarbons. We utilize the enthalpy of formation (-321.05±0.16) kJ mol⁻¹ from Ruscic and Bross [2021RUS] for HF (aq, 20 H₂O) to derive enthalpies of combustion and formation.

Table 4-6. Compounds and references for aqueous dilutions for HF

Compound	HF Dilution	Reference
CH ₂ F ₂	23 H ₂ O	2021RUS
CH ₂ F ₂	22 H ₂ O	1958NEU
CHF ₃	24 H ₂ O	1958NEU
CF ₄	20 H ₂ O	1965COX
CF ₄	20 H ₂ O	1967DOM
CF ₄	10 H ₂ O	1965DOM
CF ₄	10 H ₂ O	1956GOO
CF ₄	10 H ₂ O	1968GRE
CF ₄	10 H ₂ O	1955SCO
CH ₃ CHF ₂	20 H ₂ O	1968KOL
CH ₃ CF ₃	27 H ₂ O	1965KOL
CH ₂ CHF	40 H ₂ O	1970KOL
CH ₂ CF ₂	50 H ₂ O	1956NEU
CH ₂ CF ₂	30 H ₂ O	1962KOL
CHFCF ₂	30 H ₂ O	1962KOL

Aqueous Dilutions for HCl

Most workers have used a standard dilution for HCl of HCl (aq, 600 H₂O) to compute enthalpies for formation from enthalpies of combustion for the chlorinated hydrocarbons. This has been in standard since the work of Smith et al [1953SMI] based on the heats of dilution contained in NBS Circular 500.[1952ROS] The enthalpy of formation for HCl (aq, 600 H₂O) is (-166.452±0.024) kJ mol⁻¹. [2021RUS] We have used this value to compute enthalpies of combustion for enthalpies of formation. The following table lists a number of chlorinated hydrocarbons and references for the enthalpy of combustion measurements. Overall, for HCl, we consider the uncertainty due to dilution is about 0.3 kJ mol⁻¹ per HCl.

Table 4-7. Compounds and references for aqueous dilution for HCl

Compound	HCl Dilution	Reference
CH ₃ Cl	600 H ₂ O	1971FLE
CH ₂ Cl ₂	600 H ₂ O	1969HU
CHCl ₃	600 H ₂ O	1969HU
CCl ₄	600 H ₂ O	1969HU
CH ₃ CH ₂ Cl	600 H ₂ O	1971FLE
CH ₃ CH ₂ Cl	600 H ₂ O	1972HU
CH ₂ ClCH ₂ Cl	600 H ₂ O	1958SIN
CH ₃ CCl ₃	600 H ₂ O	1972HU
CH ₂ ClCCl ₃	600 H ₂ O	1978GUN
CH ₂ CHCl	600 H ₂ O	1964JOS
CH ₂ CCl ₂	600 H ₂ O	1971MAN
CH ₂ CCl ₂	600 H ₂ O	1958SIN
CHClCCl ₂	600 H ₂ O	1985PAP

Aqueous Dilutions for HBr, H₂SO₄, and H₃BO₃

HBr (aq, 600 H₂O) is considered the standard dilution for hydrogen bromide; we have utilized the enthalpy of formation of (-120.11±0.15) kJ mol⁻¹ taken from Ruscic et al [2021RUS]. H₂SO₄ (aq, 115 H₂O) is considered the standard dilution for sulfuric acid; we have utilized the enthalpy of formation of (-887.81±0.25) kJ mol⁻¹ taken from Wagman et al.[1982WAG]. H₃BO₃ (aq, 500 H₂O) is considered the standard dilution for boric acid; we have utilized the enthalpy of formation taken of (-1069.89±0.13) kJ mol⁻¹ from Smisko and Mason.[1950SMI]

4.3. Heats of Combustion, Formation, and Transition (compilation)

The following Table 4-8 contains all of the compiled heats of combustion, formation, and transition at standard state, while Table 4-9 contains only the selected/recommended values. The compounds are listed first by primary and secondary compound class, and then by molecular formula. The notations used for Ph (phase), SEL (selected), M (method), and R (revised) are given in Tables 4-2, 4-3, 4-4, and 4-5.

Table 4-8. Heats of Combustion, Formation, and Transition (compilation)

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_f H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
SMALL MOLECULES													
Hydrogen/Oxygen													
H ₂	hydrogen	20.37	g	SEL	-285.826	0.000	0.000	0.000			def		def
O ₂	oxygen	90.19	g	SEL	0.000	0.000	0.000	0.000			def		def
H ₂ O ₂	hydrogen peroxide	426.40	l	SEL	-98.48	0.79	-187.35	0.79	51.89	0.79	TN		2021RUS
		426.40	g	SEL1	-150.368	0.068	-135.458	0.063			TN		2021RUS
		272.74	s	SEL2	-86.26	0.79	-199.56	0.79	12.22	0.03	DfusH		1955GIG
		426.40	g		-150.6	2.7	-135.2	2.7			R		2019DIK
		426.40	g		-150.32	0.06	-135.51	0.05			BDE		1992LUO
		426.40	g		-149.1	0.4	-136.7	0.4			BDE		1988LIT
		426.40	g		-149.9	0.2	-135.9	0.2			D		1955GIG
		426.40	g		-98.49	0.17	-187.34	0.17			C		1951FOL
		426.40	l		-98.24	0.13	-187.59	0.13			C		1930ROT
		426.40	l		-98.198	0.084	-187.628	0.084			C		1929MAT
		426.40	lg						51.63				1955GIG
		272.74	sl	SEL					12.22	0.03	f		1951FOL
H ₂ O	water	373.12	l	SEL	0.000	0.037	-285.826	0.026	44.00	0.22	TN		2021RUS
		373.12	g	SEL1	-43.994	0.037	-241.832	0.026			TN		2021RUS
		273.153	s	SEL2	6.913	0.037	-292.739	0.026	6.91		TN		2021RUS
		373.12	g		-44.01	0.05	-241.82	0.04			R		2019DIK
		373.12	g		-44.00	0.05	-241.83	0.04			R		1984COX
		373.12	g		-43.99	0.22	-241.83	0.22			D		1968KIN
		373.12	g		-44.04	0.05	-241.79	0.04			D		1939ROS

		373.12	l		-0.06	0.19	-285.77	0.19	43.95	0.19	R	2019DIK
		373.12	l		0.00	0.05	-285.83	0.04			R	1984COX
		273.153	sg						50.88			2006REI
Nitrogen/Hydrogen												
N ₂	nitrogen	77.36	g	SEL	0.000	0.000	0.000	0.000			def	
H ₃ N	ammonia	239.82	g	SEL	-383.181	0.029	-45.558	0.029			TN	2021RUS
		239.82	l	SEL1	-363.32	0.11	-65.42	0.11			DvH	2021RUS
		239.82	lg	SEL					19.87	0.10	v	2019DIK
H ₄ N ₂	hydrazine	386.65	l	SEL	-622.34		50.69	0.18	-336.51	1.00		2021RUS
		386.65	g	SEL1	-669.22		97.6	1.0				2021RUS
		386.65	g		-667.60		95.95	0.63				2019DIK
		386.65	g		-664.7		93.1					1939HUG
		386.65	l		-621.80		50.15	0.32	45.80	0.54		2019DIK
		386.65	l		-622.3		50.6		42.47			1939HUG
Nitrogen/Oxygen												
ClH ₄ NO ₄	ammonium perchlorate	403.15	s(d)	SEL	-442.33	0.17	-295.77	0.17				1998CHA
		403.15	s(d)		-452.3		-285.8	1.3				2019DIK
Hydrogen Halides												
FH	hydrogen fluoride	292.7	g	SEL	na		-272.721	0.048			TN	2021RUS
		292.7	l	SEL1	na		-302.41	0.16			TN	2021RUS
		292.7	g		na		-273.3	0.7			R	1984COX
	hydrogen fluoride (aq)	293.15	aq		na		-334.50	0.16			TN	2021RUS
	hydrogen fluoride (aq, 10 H ₂ O)	293.15	aq		na		-320.88	0.16			TN	2021RUS
	hydrogen fluoride (aq, 20 H ₂ O)	293.15	aq	SEL	na		-321.05	0.16			TN	2021RUS
	hydrogen fluoride (aq, 22 H ₂ O)	293.15	aq		na		-321.06	0.16			TN	2021RUS
	hydrogen fluoride (aq, 23 H ₂ O)	293.15	aq		na		-321.07	0.16			TN	2021RUS
	hydrogen fluoride (aq, 24 H ₂ O)	293.15	aq		na		-321.08	0.16			TN	2021RUS
	hydrogen fluoride (aq, 27 H ₂ O)	293.15	aq		na		-321.10	0.16			TN	2021RUS
	hydrogen fluoride (aq, 30 H ₂ O)	293.15	aq		na		-321.11	0.16			TN	2021RUS
	hydrogen fluoride (aq, 40 H ₂ O)	293.15	aq		na		-321.14	0.16			TN	2021RUS
	hydrogen fluoride (aq, 50 H ₂ O)	293.15	aq		na		-321.16	0.16			TN	2021RUS
	hydrogen fluoride (aq, 60 H ₂ O)	293.15	aq		na		-321.18	0.16			TN	2021RUS
ClH	hydrogen chloride	188.1	g	SEL	na		-92.173	0.006			TN	2021RUS
		188.1	g		na		-92.31	0.10			R	1984COX
	hydrogen chloride (aq)	293.15	aq		na		-166.992	0.023			TN	2021RUS
	hydrogen chloride (aq, 600 H ₂ O)	293.15	aq	SEL	na		-166.452	0.024			TN	2021RUS
BrH	hydrogen bromide	206.77	g	SEL	na		-36.29	0.16			R	1984COX
		206.77	g		na		-35.45	0.14			TN	2021RUS

hydrogen bromide (aq)	293.15	aq		na		-120.59	0.15		TN	2021RUS
hydrogen bromide (aq, 600 H ₂ O)	293.15	aq	SEL	na		-120.11	0.15		TN	2021RUS
Sulfur/Hydrogen										
H ₂ S	212.85	g	SEL	-867.21	0.5	-20.6	0.5		R	1984COX
	212.85	l	SEL1	-853.55	0.64	-34.27	0.64		DvH	1984COX
	212.85	lg	SEL					13.67 0.40	v	2019DIK
Sulfur/Oxygen										
O ₂ S	263.14	l	SEL	na		-320.5	1.0	23.67	R	1982WAG
	263.14	g	SEL1	na		-296.83	0.20		R	1982WAG
	263.14	g		na		-296.81	0.20		R	2019DIK
	263.14	lg						22.93 1.03	R	2019DIK
H ₂ O ₄ S	283.46	l	SEL1	na		-814.0	1.0		R	1982WAG
	283.46	g	SEL2	na		-735.1	1.0		R	1998CHA
	283.46	aq	SEL	na		-887.81	0.25		R	1982WAG
	283.46	aq		na	35.98	-888.01	0.29		H2O	1963MAN
	283.46	aq		na	1.17	-887.72	0.25		H2O	1960GOO
	283.46	aq		na		-885.98	0.25		R	1982WAG
	283.46	aq		na		-886.62	0.25		R	1982WAG
Boron Compounds										
BH ₃ O ₃	444.05	s	SEL	na		-1091.94	0.13		D	1950SMI
	444.05	aq		na		-1069.26	0.13		H2O	1950SMI
	444.05	aq	SEL	na		-1069.89	0.13		H2O	1950SMI
B ₂ H ₆	180.59	g	SEL	-2175.76	1.17	36.0	1.2		D	1965GUN
	180.59	l	SEL1	-2174.75	1.17	35.0	1.2		DvH	1965GUN
	180.59	g		-2180.8	16.7	41	17		R	1998CHA
	180.59	g		-2175.34		35.56			R	1988RUS
	180.59	g		-2176.2		36.40			R	1982WAG
	180.59	g		-2175.1	2.3	35.3	2.3		D	1965GUN
	180.59	g		-2176.52	1.17	36.7	1.2		D	1965GUN
	180.59	g		-2177.11	0.63	37.33	0.63		H2O	1960GUN
	180.59	g		-2167.9	2.2	28.2	2.2		D	1958PRO
	180.59	l		-2166.9	2.2	27.2	2.2		D	1958PRO
	180.59	lg	SEL			36.8		1.01	v	1959PAR
B ₂ O ₃	723.15	s	SEL	na		-1279.9	2.1			1959JOH
	2130	g	SEL1	na		-836.0	4.2			1963HIL
	2130	l	SEL2	na		-1259.3	1.3			1961EVA
	2130	g		na		-834	13			2011JAC
	2130	g		na		-834.2	8.4			1978SCH

	2130	g	na	-848.5	8.7			DsubH	1959JOH
	2130	g	na	-829.2	3.9				1957SCH
	2130	l	na	-1253.4	2.5				1998CHA
	723.15	s	na	-1271.9	2.1				1998CHA
	723.15	s	na	-1273.5	1.4				1984COX
	723.15	s	na	-1277.5	1.4				1961EVA
	723.15	s	na	-1278.2	1.9				1960GUN
	723.15	s	na	-1280.9	3.1				1959PRO
	723.15	sg				431.4	8.4		1963HIL

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_f H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
HYDROCARBONS													
Carbon													
C	diamond	4300	s	SEL1	-395.347	0.070	1.871	0.070			C		1966HAW
	graphite	4800	s	SEL	-393.476	0.025	0.000	0.025			def		1966HAW
		5000	g	SEL1	-1110.359	0.047	716.883	0.045			TN		2021RUS
Alkanes													
CH ₄	methane	111.67	g	SEL	-890.602	0.050	-74.526	0.049			TN		2021RUS
		111.67	g		-890.52	0.36	-74.61	0.36			R		2019DIK
		111.67	g		-890.46	0.35	-74.67	0.35			C		2002YU
		111.67	g		-890.61	0.21	-74.52	0.21			C		2002DAL
		111.67	g		-890.53	0.30	-74.60	0.30			R		1989GUR
		111.67	g		-890.7	0.4	-74.4	0.4			C		1972PIT
		111.67	g		-890.35	0.30	-74.78	0.31			C		1945PRO
		111.67	g		-891.8	1.1	-73.3	1.1			C		1932ROT
		111.67	g		-890.16	0.30	-74.97	0.30			C		1931ROS
C ₂ H ₆	ethane	184.57	g	SEL	-1560.39	0.13	-84.04	0.13			TN		2021RUS
		184.57	l	SEL1	-1555.58	1.03	-88.85	1.03			DvH		2021RUS
		184.57	g	SEL	-1560.34	0.99	-84.09	0.99	4.81	0.29	v		2019DIK
		184.57	g		-1560.4	0.4	-84.0	0.4			R		1989GUR
		184.57	g		-1560.70	0.30	-83.73	0.30			C		1972PIT
		184.57	g		-1559.90	0.49	-84.53	0.49			C		1945PRO
		184.57	g		-1559.84	0.46	-84.59	0.46			C		1934ROS
		184.57	lg						9.76		R		1985MAJ
C ₃ H ₈	propane	231.04	g	SEL	-2218.66	0.16	-105.07	0.16			TN		2021RUS
		231.04	l	SEL1	-2203.31	0.26	-120.42	0.26			DvH		2021RUS
		231.04	g		-2219.28	1.10	-104.45	1.10			R		2019DIK
		231.04	g		-2219.2	0.5	-104.5	0.5			C		1972PIT
		231.04	g		-2219.93	0.54	-103.80	0.59			DvH		1945PRO
		231.04	g		-2219.9	0.5	-103.8	0.5			C		1934ROS
		231.04	g		-2206.6		-117.1				C		1918GUI
		231.04	l		-2204.49	1.12	-119.24	1.12	14.79	0.18	R		2019DIK
		231.04	lg						16.25		R		1985MAJ
		231.04	lg	SEL					15.4		v		1945PRO
C ₄ H ₁₀	isobutane	261.40	g	SEL	-2868.41	0.30	-134.62	0.30			TN		2021RUS
		261.40	l	SEL1	-2848.95	0.37	-154.08	0.37			DvH		2021RUS
		261.40	g		-2868.1	1.5	-134.9	1.5			R		2019DIK

		261.40	g		-2869.00	0.59	-134.03	0.63			C	1972PIT
		261.40	g		-2867.60	0.46	-135.43	0.54			C	1951PRO
		261.40	l	SEL	-2848.6	1.7	-154.4	1.7	19.46	0.21	v	2019DIK
		261.40	lg						19.99		R	1985MAJ
	n-butane	272.66	g	SEL	-2877.28	0.24	-125.75	0.24			TN	2021RUS
		272.66	l	SEL1	-2854.88	0.69	-148.15	0.69			DvH	2021RUS
		272.66	g		-2876.7	1.5	-126.3	1.5			R	2019DIK
		272.66	g		-2877.50	0.63	-125.53	0.67			C	1972PIT
		272.66	g		-2876.20	0.67	-126.83	0.67			C	1951PRO
		272.66	g		-2878.34	0.63	-124.69	0.63			C	1934ROS
		272.66	l		-2856.33	1.16	-146.70	1.16	20.41	0.21	R	2019DIK
		272.66	lg	SEL					22.40		v	1971WIL
C ₅ H ₁₂	isopentane	300.98	l	SEL	-3503.95	0.43	-178.39	0.43			TN	2021RUS
		300.98	g	SEL1	-3529.17	0.43	-153.17	0.43			TN	2021RUS
		300.98	g		-3528.69	0.92	-153.65	0.92			R	2019DIK
		300.98	g		-3528.66	0.59	-153.68	0.59			DvH	1970GOO
		300.98	g		-3528.40	0.92	-153.94	0.96			C	1967PIL
		300.98	g		-3528.60	0.63	-153.74	0.63			C	1939KNO
		300.98	g		-3527.6	3.5	-154.7	3.5			C	1936ROT
		300.98	l		-3503.89	0.97	-178.45	0.97	24.80	0.32	R	2019DIK
		300.98	l		-3504.40	0.84	-177.94	0.88	24.50		C	1979GOO
		300.98	l		-3503.30	0.75	-179.04	0.84	24.80		C	1945PRO
		300.98	lg	SEL					25.22		v	1985MAJ
		300.98	lg						24.80			1971WIL
		300.98	lg						24.84	0.13		1951SCO
		300.98	lg						25.00			1942SCH
	neopentane	282.65	l	SEL	-3492.56	0.39	-189.78	0.39	22.39	0.30	TN	2021RUS
		282.65	g	SEL1	-3514.95	0.39	-167.39	0.39			TN	2021RUS
		282.65	g		-3514.03	0.88	-168.30	0.88			R	2019DIK
		282.65	g		-3514.10	0.92	-168.24	0.96			C	1967PIL
		282.65	l		-3492.20	0.88	-190.14	0.88	21.84	0.06	R	2019DIK
		282.65	l		-3492.43	0.84	-189.91	0.59	22.38	0.59	C	1970GOO
		282.65	l		-3494.4	0.8	-187.9	1.0	22.20		C	1945PRO
		282.65	lg						22.39			1985MAJ
		282.65	lg						21.80			1981HOS
		282.65	lg						21.85			1971WIL
	n-pentane	309.21	l	SEL	-3509.35	0.43	-172.99	0.43	26.74	0.20	TN	2021RUS
		309.21	g	SEL1	-3536.09	0.31	-146.25	0.31			TN	2021RUS
		309.21	g		-3535.27	0.73	-147.06	0.73			R	2019DIK

		309.21	g		-3535.54	0.59	-146.80	0.59			DvH	1979GOO
		309.21	g		-3535.4	1.0	-146.9	1.0			C	1967PIL
		309.21	g		-3536.60	0.88	-145.74	0.88			C	1934ROS
		309.21	l		-3508.90	0.72	-173.44	0.72	26.38	0.12	R	2019DIK
		309.21	l		-3509.04	0.46	-173.30	0.47	26.73	0.42	C	1970GOO
		309.21	l		-3509.50	0.59	-172.84	0.67	26.70		C	1945PRO
		309.21	l		-3509.16	0.75	-173.17	0.75			C	1944PRO
		309.21	lg						26.20			1975HOR
		309.21	lg						26.40			1994RUZ
		309.21	lg						26.75		R	1985MAJ
		309.21	lg						26.60	0.10		1984LAW
		309.21	lg						26.60	0.10		1982FUC
		309.21	lg						26.40			1981HOS
		309.21	lg						26.40			1971WIL
		309.21	lg						26.42			1947OSB
		309.21	lg						27.40			1945WIL
		309.21	lg						26.20			1940MES
C ₆ H ₁₄	3-methylpentane	336.43	l	SEL	-4159.90	0.88	-201.74	0.96	30.30	0.10	C	1945PRO
		336.43	g	SEL1	-4190.2	1.0	-171.4	1.0			DvH	1945PRO
		336.43	g		-4189.5	1.9	-172.2	1.9			R	2019DIK
		336.43	l		-4159.2	1.9	-202.4	1.9	30.24	0.18	R	2019DIK
		336.43	lg						30.47		R	1985MAJ
		336.43	lg						30.30	0.10		1979MAJ
		336.43	lg						30.30			1971WIL
		336.43	lg						30.19	0.03		1966WAD
		336.43	lg	SEL					30.27	0.10	v	1947OSB
	isohexane	333.42	l	SEL	-4157.65	0.83	-203.99	0.83	30.11	0.20	TN	2021RUS
		333.42	g	SEL1	-4187.76	0.84	-173.88	0.84			TN	2021RUS
		333.42	g		-4186.8	2.0	-174.9	2.0			R	2019DIK
		333.42	l		-4157.0	1.9	-204.7	1.9	29.79	0.26	R	2019DIK
		333.42	l		-4157.3	1.0	-204.3	1.0	30.0		C	1945PRO
		333.42	lg						30.10		R	1985MAJ
		333.42	lg						29.90			1971WIL
		333.42	lg						29.89	0.10		1949WAD
		333.42	lg						24.86	0.10		1947OSB
	neohexane	322.89	l	SEL	-4148.50	0.88	-213.14	0.96	27.68	0.20	C	1945PRO
		322.89	g	SEL1	-4176.18	0.96	-185.46	0.96			DvH	1945PRO
		322.89	g		-4175.4	2.0	-186.3	2.0			R	2019DIK
		322.89	l		-4147.8	1.9	-213.8	1.9	27.56	0.34	R	2019DIK

	322.89	lg						27.93		R	1985MAJ	
	322.89	lg						27.70			1971WIL	
	322.89	lg	SEL					27.68		v	1947OSB	
n-hexane	341.87	l	SEL	-4163.04	0.34	-198.60	0.34	31.71	0.20	TN	2021RUS	
	341.87	g	SEL1	-4194.75	0.39	-166.89	0.39			TN	2021RUS	
	341.87	g		-4194.52	0.93	-167.12	0.93			R	2019DIK	
	341.87	l		-4163.02	0.91	-198.61	0.91	31.50	0.19	R	2019DIK	
	341.87	l		-4163.20	0.67	-198.44	0.67	31.6		C	1969GOO	
	341.87	l		-4163.10	0.54	-198.54	0.79	31.6		C	1945PRO	
	341.87	l		-4163.00	0.71	-198.64	0.71			C	1944ROS	
	341.87	l		-4163.7	1.4	-197.9	1.4			C	1937JES	
	341.87	lg						31.40	0.20		2007PAS	
	341.87	lg						31.50	0.10		1996VAR	
	341.87	lg						31.50			1994RUZ	
	341.87	lg						32.10			1992GRA	
	341.87	lg						31.73		R	1985MAJ	
	341.87	lg						31.60	0.10		1979MAJ	
	341.87	lg						31.55			1971WIL	
	341.87	lg						29.96			1957GLA	
	341.87	lg						31.54	0.10		1947OSB	
	341.87	lg						31.00	0.20		1943LEM	
C7H16	n-heptane	371.55	l	SEL	-4816.83	0.46	-224.11	0.46	36.65	0.20	TN	2021RUS
		371.55	g	SEL1	-4853.48	0.46	-187.46	0.46			TN	2021RUS
		371.55	g		-4853.4	1.3	-187.6	1.3			R	2019DIK
		371.55	l		-4816.8	1.3	-224.2	1.3	36.59	0.33	R	2019DIK
		371.55	l		-4825.0		-215.9				C	1973DEL
		371.55	l		-4818.7		-222.2				C	1957SKU
		371.55	l		-4816.91	0.63	-224.03	0.79	36.6		C	1945PRO
		371.55	l		-4817.16	0.88	-223.78	0.88	35.98		C	1944PRO
		371.55	l		-4810.3		-230.6				C	1941JON
		371.55	l		-4815.0	1.0	-225.9	1.3	36.6		C	1941DAV
		371.55	l		-4817.0	1.5	-223.9	1.5			C	1937JES
		371.55	lg					36.60				1994RUZ
		371.55	lg					36.66		R		1985MAJ
		371.55	lg					36.60	0.10			1979MAJ
		371.55	lg					36.90				1979JAK
		371.55	lg					36.55				1971WIL
		371.55	lg					36.54				1947OSB
		371.55	lg					32.01	0.17			1926MAT

C ₈ H ₁₈	n-octane	398.79	l	SEL	-5470.63	0.55	-249.61	0.55	41.53	0.20	TN	2021RUS
		398.79	g	SEL1	-5512.16	0.54	-208.08	0.54			TN	2021RUS
		398.79	G		-5511.61	1.09	-208.63	1.09			R	2019DIK
		398.79	L		-5470.07	1.05	-250.17	1.05	41.54	0.27	R	2019DIK
		398.79	L		-5470.3	1.6	-249.9	1.8	41.6		C	1972GOO
		398.79	L		-5470.10	0.67	-250.14	0.84	41.6		C	1945PRO
		398.79	L		-5466.6		-253.6				C	1937JES
		398.79	L		-5471.8	5.4	-248.4	5.4			C	1933BAN
		398.79	lg						41.60			1994RUZ
		398.79	lg						41.5		R	1985MAJ
		398.79	lg						41.50	0.10		1981HOS
		398.79	lg						41.50	0.10		1979MAJ
		398.79	lg						41.5			1971WIL
		398.79	lg						41.59	0.08		1966WAD
		398.79	lg						41.59	0.08		1960WAD
		398.79	lg						41.48	0.20		1947OSB
		398.79	lg						33.97			1920FAJ
C ₉ H ₂₀	n-nonane	423.91	l	SEL	-6124.85	0.50	-274.69	0.50	46.44	0.20	TN	2021RUS
		423.91	g	SEL1	-6171.29	0.51	-228.25	0.51			TN	2021RUS
		423.91	g		-6170.77	1.14	-228.77	1.14			R	2019DIK
		423.91	l		-6124.42	0.99	-275.13	0.99	46.36	0.55	R	2019DIK
		423.91	l		-6125.21	0.54	-274.70	0.54	46.4		C	1969GOO
		423.91	l		-6124.9	1.1	-274.6	1.1			C	1944PRO
		423.91	l		-6119.8		-279.7				C	1937JES
		423.91	lg						46.50	0.20		2007PAS
		423.91	lg						46.60	0.20		1996VAR
		423.91	lg						46.60			1994RUZ
		423.91	lg						46.43		R	1985MAJ
		423.91	lg						46.40			1971WIL
		423.91	lg						46.43			1947OSB
C ₁₀ H ₂₂	n-decane	447.27	l	SEL	-6778.29	0.80	-300.56	0.80	51.39	0.20	TN	2021RUS
		447.27	g	SEL1	-6829.68	0.91	-249.17	0.91			TN	2021RUS
		447.27	g		-6829.4	2.7	-249.5	2.7			R	2019DIK
		447.27	l		-6777.8	2.6	-301.0	2.6	51.55	0.39	R	2019DIK
		447.27	l		-6778.3	0.9	-300.5	1.1	51.3		C	1945PRO
		447.27	l		-6778.6	1.5	-300.2	1.5	50.9		C	1944PRO
		447.27	l		-6774.2		-304.6				C	1937JES
		447.27	lg						51.1	3.9		2000NIC
		447.27	lg						50.9			1995CHI

C ₁₆ H ₃₄	n-hexadecane	447.27	lg						51.40			1994RUZ
		447.27	lg						51.37	0.02		1989CHI
		447.27	lg						51.4		R	1985MAJ
		447.27	lg						51.40	0.10		1982FUR
		447.27	lg						51.38	0.12		1972IRV
		447.27	lg						51.40			1971WIL
		447.27	lg						51.36			1947OSB
		559.94	l	SEL	-10699.1	1.8	-455.6	2.0	81.40	0.40	C	1955FRA
		559.94	g	SEL1	-10780.5	1.8	-374.2	2.0			DvH	1955FRA
		559.94	g		-10777.9	2.9	-376.7	2.9			R	2019DIK
Alkenes		559.94	l		-10697.0	2.5	-457.6	2.5	80.90	1.40	R	2019DIK
		559.94	l		-10699.1	4.1	-455.6	4.1	81.4		C	1939RIC
		559.94	lg						81.80	1.30		2000NIC
		559.94	lg	SEL					81.4		v	1995CHI
		559.94	lg						81.40			1994RUZ
		559.94	lg						81.4		R	1985MAJ
		559.94	lg						81.40	0.40		1972MOR
		559.94	lg						81.1			1971WIL
		559.94	lg						80.60	1.50		1949PAR
		C ₂ H ₄	ethene	169.38	g	SEL	-1410.92	0.12	52.32	0.12		
C ₃ H ₆	propene	169.38	l	SEL1	-1402.48	0.17	43.88	0.17			DvH	2021RUS
		169.38	g		-1411.12	0.57	52.52	0.57			R	2019DIK
		169.38	g		-1411.0	0.5	52.4	0.5			R	1989GUR
		169.38	g		-1411.20	0.30	52.60	0.30			C	1937ROS
		169.38	l	SEL	-1402.7	2.7	44.1	2.7	8.44	0.13	v	2019DIK
		225.53	g	SEL	-2057.79	0.20	19.88	0.19			TN	2021RUS
		225.53	l	SEL1	-2043.57	0.33	5.66	0.33			DvH	2021RUS
		225.53	g		-2056.5	2.5	18.6	2.5			R	2019DIK
		225.53	g		-2058.3		20.4				E	1969FUR
		225.53	g		-2057.80	1.10	19.89	1.10			C	1968WIB
C ₄ H ₈	1-butene	225.53	g		-2058.3		20.4				HBr	1950LAC
		225.53	g		-2057.7	0.6	19.8	0.6			C	1937ROS
		225.53	l	SEL	-2042.3	2.5	4.4	2.5	14.22	0.27	v	2019DIK
		266.83	g	SEL	-2717.30	0.36	0.09	0.36			TN	2021RUS
		266.83	l	SEL1	-2697.20	0.45	-20.01	0.45			DvH	2021RUS
		266.83	g		-2716.65	1.03	-0.56	1.03			R	2019DIK
		266.83	g		-2716.80	0.75	-0.41	0.76			C	1951PRO
		266.83	l		-2696.3	2.0	-20.9	2.0	20.32	0.30	R	2019DIK

		266.83	lg					20.88		R	1985MAJ
		266.83	lg	SEL				20.1		v	1971WIL
	2-butene, (E)-	274.01	g	SEL	-2706.07	0.38	-11.14	0.38		TN	2021RUS
		274.01	l	SEL1	-2684.77	0.41	-32.44	0.41		DvH	2021RUS
		274.01	g		-2706.6	1.0	-10.6	1.0		C	2019PRO
		274.01	g		-2706.48	1.18	-10.73	1.18		R	2019DIK
		274.01	g		-2706.63	0.96	-10.58	0.96		C	1951PRO
		274.01	l		-2685.09	1.19	-32.12	1.19	21.39	R	2019DIK
		274.01	lg						21.97	R	1985MAJ
		274.01	lg	SEL					21.3	v	1971WIL
	2-butene, (Z)-	276.87	g	SEL	-2710.16	0.40	-7.05	0.40		TN	2021RUS
		276.87	l	SEL1	-2688.06	0.59	-29.15	0.59		DvH	2021RUS
		276.87	g		-2709.6	1.4	-7.6	1.4		R	2019DIK
		276.87	g		-2709.8	1.3	-7.4	1.3		C	1951PRO
		276.87	l		-2688.8	7.6	-28.4	7.6	20.80	R	2019DIK
		276.87	lg						22.70	R	1985MAJ
		276.87	lg	SEL					22.1	v	1971WIL
	isobutene	266.05	g	SEL	-2700.20	0.39	-17.01	0.39		TN	2021RUS
		266.05	l	SEL1	-2679.60	0.50	-37.61	0.49		DvH	2021RUS
		266.05	g		-2699.3	1.3	-17.9	1.3		R	2019DIK
		266.05	g		-2699.50	1.10	-17.71	1.10		C	1951PRO
		266.05	l		-2677.8	7.2	-39.4	7.2	21.50	R	2019DIK
		266.05	lg	SEL					20.60	v	1971WIL
C ₅ H ₁₀	1-pentene	303.12	l	SEL	-3349.72	0.64	-46.79	0.64		C	1979GOO
		303.12	g	SEL1	-3375.22	0.67	-21.29	0.67		DvH	1979GOO
		303.12	g		-3374.93	0.93	-21.58	0.93		R	2019DIK
		303.12	l		-3349.5	0.8	-47.0	0.8	25.42	R	2019DIK
		303.12	l		-3349.57	0.43	-46.94	0.42		H2O	1984WIB
		303.12	l		-3349.6	2.5	-46.9	2.5		H2	1984MOL
		303.12	l		-3348.0	1.9	-48.5	1.9		H2	1984MOL
		303.12	l		-3345.6	2.5	-50.9	2.5		H2	1974ROG
		303.12	l		-3342.0	1.2	-54.5	1.2		H2	1971ROG
		303.12	lg						29.82	R	1985MAJ
		303.12	lg	SEL					25.5	v	1971WIL
		303.12	lg						25.50		1949SCO
	2-methyl-1-butene	304.30	l	SEL	-3335.74	0.84	-60.77	0.84		C	1979GOO
		304.30	g	SEL1	-3361.64	0.86	-34.87	0.86		DvH	1979GOO
		304.30	g		-3361.38	0.96	-35.13	0.96		R	2019DIK
		304.30	g		-3361.41	0.84	-35.10	0.84		H2O	1991WIB

		304.30	l		-3335.53	0.93	-60.98	0.93	25.85	0.26	R	2019DIK
		304.30	lg						26.19		R	1985MAJ
		304.30	lg	SEL					25.9		v	1971WIL
		304.30	lg						25.90	0.10		1949SCO
2-pentene, (E)-		309.51	l	SEL	-3338.71	0.76	-57.80	0.76			C	1979GOO
		309.51	g	SEL1	-3365.4	0.8	-31.1	0.8			DvH	1979GOO
		309.51	g		-3365.2	1.4	-31.3	1.4			R	2019DIK
		309.51	g		-3364.87	0.45	-31.64	0.45			H2O	1984WIB
		309.51	g		-3363.4	1.3	-33.1	1.3			E	1966EGG
		309.51	l		-3338.5	1.3	-58.0	1.3	26.67	0.63	R	2019DIK
		309.51	l		-3338.27	0.42	-58.24	0.42			H2O	1984WIB
		309.51	l		-3336.8	1.0	-59.7	1.0			E	1966EGG
		309.51	l		-3335.9	0.4	-60.6	0.4			C	1946COO
		309.51	lg	SEL					26.7		v	1971WIL
2-pentene, (Z)-		310.09	l	SEL	-3343.21	0.62	-53.30	0.62			C	1979GOO
		310.09	g	SEL1	-3370.0	0.7	-26.5	0.7			DvH	1979GOO
		310.09	g		-3369.80	0.79	-26.71	0.79			R	2019DIK
		310.09	g		-3368.5	0.8	-28.0	0.8			E	1966EGG
		310.09	l		-3343.00	0.77	-53.51	0.77	26.80	0.19	R	2019DIK
		310.09	l		-3343.16	0.43	-53.35	0.42			H2O	1984WIB
		310.09	l		-3336.66	0.30	-59.85	0.29			C	1946COO
		310.09	lg	SEL					26.8		v	1971WIL
3-methyl-1-butene		293.21	g	SEL	-3369.00	0.68	-27.51	0.68			DvH	1979GOO
		293.21	l	SEL1	-3345.10	0.62	-51.41	0.62			C	1979GOO
		293.21	g		-3368.59	0.84	-27.92	0.84			R	2019DIK
		293.21	l		-3344.88	0.79	-51.63	0.79	23.71	0.29	R	2019DIK
		293.21	lg	SEL					23.9		v	1971WIL
C ₆ H ₁₂ 1-hexene		336.64	l	SEL	-4003.1	1.2	-72.7	1.2			H2	1984MOL
		336.64	g	SEL1	-4033.7	1.2	-42.1	1.2			DvH	1984MOL
		336.64	l		-4002.7	2.7	-73.2	2.7	-30.55	0.20	R	2019DIK
		336.64	l		-4003.5	1.2	-72.3	1.2			H2O	1981WIB
		336.64	l		-4003.8	0.8	-72.0	0.8			H2	1979ROG
		336.64	l		-4000.4	1.0	-75.4	1.0			H2	1975ROG
		336.64	l		-4003.7	1.2	-72.1	1.2			H2	1974ROG
		336.64	l		-4002.2	1.5	-73.6	1.5			H2	1970BRE
		336.64	l		-4003.7	1.0	-72.1	1.0			H2	1959SKI
		336.64	l		-4003.1	2.1	-72.7	2.1			H2	1957FLI
		336.64	lg	SEL					30.60		v	1971WIL
		336.64	lg						30.60			1956CAM

C ₇ H ₁₄	1-heptene	366.71	l	SEL	-4657.41	0.63	-97.70	0.63			H ₂ O	1984WIB
		366.71	g	SEL1	-4693.1	0.7	-62.0	0.7			DvH	1984WIB
		366.71	g		-4691.3	4.8	-63.8	4.8			R	2019DIK
		366.71	l		-4655.7	4.8	-99.4	4.8	35.59	0.21	R	2019DIK
		366.71	l		-4657.0	0.8	-98.1	1.2			C	1976GOO
		366.71	l		-4658.3	1.2	-96.8	1.2	36.07		C	b 1961ROC
		366.71	l		-4650.98	0.79	-104.13	0.79			C	1947COO
		366.71	lg	SEL					35.7		v	1971WIL
C ₈ H ₁₆	1-octene	394.43	l	SEL	-5312.9	1.2	-121.5	1.2			C	1961ROC
		394.43	g	SEL1	-5353.5	1.2	-80.9	1.2			DvH	1961ROC
		394.43	g		-5353.1	1.7	-81.4	1.7			R	2019DIK
		394.43	g		-5351.5		-82.9				H ₂	1970BRE
		394.43	l		-5312.6	1.6	-121.8	1.6	40.44	0.54	R	2019DIK
		394.43	lg	SEL					40.60		v	1995CHI
		394.43	lg						40.44		R	1985MAJ
		394.43	lg						40.27	0.20		1977MAN
		394.43	lg						40.27	0.20		1977MAN
		394.43	lg						38.0			1971WIL
Alkadienes												
C ₃ H ₄	allene	238.65	g	SEL	-1941.86	0.24	189.78	0.24			TN	2021RUS
		238.65	l	SEL1	-1925.30	0.69	173.22	0.69			DvH	2021RUS
		238.65	g		-1947.9	4.8	195.8	4.8			R	2019DIK
		238.65	l	SEL	-1931.3	4.8	179.3	4.8	16.56	0.65	v	2019DIK
C ₄ H ₆	1,2-butadiene	284.06	g	SEL	-2593.79	0.46	162.41	0.46	23.68	0.30	TN	2021RUS
		284.06	l	SEL1	-2570.11	0.66	138.73	0.66			TN	2021RUS
		284.06	g		-2593.64	0.89	162.26	0.89			R	2019DIK
		284.06	g		-2593.80	0.54	162.42	0.54			C	1951PRO
		284.06	g		-2597.0	1.2	165.6	1.2			C	1949PRO
		284.06	l		-2570.36	0.92	138.98	0.92	23.28	0.22	R	2019DIK
		284.06	lg						23.68		R	1985MAJ
		284.06	lg						23.90			1971WIL
	1,3-butadiene	268.44	g	SEL	-2542.28	0.33	110.90	0.33			TN	2021RUS
		268.44	l	SEL1	-2521.18	0.43	89.80	0.43			DvH	2021RUS
		268.44	l	u	-2523	7	92	7	20.74	0.28	R	2019DIK
		268.44	g		-2540.21	1.03	108.83	1.03			R	2019DIK
		268.44	g		-2540.40	0.96	109.02	0.96			C	1951PRO
		268.44	l		-2522.10	0.96	87.62	0.96	21.40		C	1945PRO
		268.44	lg						21.47		R	1985MAJ
		268.44	lg	SEL					21.1		v	1971WIL

C ₅ H ₈	1,2-pentadiene	318.80	l	SEL	-3222.90	0.69	112.22	0.69			DvH	1955FRA	
		318.80	g	SEL1	-3251.60	0.67	140.92	0.67			C	1955FRA	
		318.80	g		-3251.4	1.5	140.7	1.5			R	2019DIK	
		318.80	l		-3223.0	1.5	112.3	1.5	28.41	0.17	R	2019DIK	
		318.80	lg	SEL					28.7		v	1971WIL	
	1,3-pentadiene	314.90	lg						27.78	1.19	R	2019DIK	
	1,3-pentadiene, (E)-	315.32	l	SEL	-3158.90	0.78	48.22	0.78			DvH	1955FRA	
		315.32	g	SEL1	-3186.70	0.67	76.02	0.67			C	1955FRA	
		315.32	g		-3193.4	3.7	82.7	3.7			R	2012KRO	
		315.32	l		-3165.8	3.8	55.1	3.8	27.6		R	2012KRO	
		315.32	lg	SEL					27.8		v	1971WIL	
	1,3-pentadiene, (Z)-	317.18	l	SEL	-3165.3	1.4	54.6	1.4			DvH	1955FRA	
		317.18	g	SEL1	-3193.60	0.92	82.92	0.92			C	1955FRA	
		317.18	lg	SEL					28.30	0.40	v	1971WIL	
	1,4-pentadiene	299.12	g	SEL	-3217.2	1.3	106.5	1.3			C	1955FRA	
		299.12	l	SEL1	-3192.0	1.3	81.3	1.3	25.20	0.20	DvH	1955FRA	
		299.12	g		-3217.1	2.6	106.4	2.6			R	2019DIK	
		299.12	l		-3191.9	2.7	81.3	2.7	25.13	0.21	R	2019DIK	
	2-methyl-1,3-butadiene	299.12	lg	SEL					25.2		v	1971WIL	
		307.19	l	SEL	-3159.50	0.72	48.82	0.72			DvH	2021RUS	
		307.19	g	SEL1	-3186.28	0.67	75.60	0.67			TN	2021RUS	
		307.19	l		-3160.2	2.1	49.5	2.1	26.22	0.22	R	2019DIK	
		307.19	l		-3158.2	1.6	47.5	1.6			C	1938JES	
		307.19	g		-3186.4	2.1	75.8	2.1			R	2019DIK	
		307.19	g		-3186.6	1.0	75.9	1.0			C	1955FRA	
		307.19	lg						26.40			1971WIL	
		307.19	lg	SEL					26.78	0.25	v	1971ROG	
		Alkynes											
	C ₂ H ₂	acetylene	188.41	g	SEL	-1301.06	0.13	228.28	0.13			TN	2021RUS
			188.41	l	SEL1	-1293.7	4.4	220.9	4.4			DvH	2021RUS
			188.41	g		-1299.1	1.6	226.4	1.6			R	2019DIK
			188.41	g		-1300.2	0.8	227.4	0.8			R	1989GUR
188.41			g		-1299.48	0.79	226.70	0.79			E	1945WAG	
188.41			l	SEL	-1291.8	4.7	219.0	4.7	7.4	4.4	v	2019DIK	
C ₃ H ₄	propyne	249.80	g	SEL	-1937.72	0.22	185.64	0.22			TN	2021RUS	
		249.80	l	SEL1	-1918.14	0.56	166.06	0.56			DvH	2021RUS	
		249.80	g		-1937.5	1.7	185.4	1.7			R	2019DIK	
		249.80	g		-1937.48	0.88	185.40	0.88			E	1945WAG	

C ₄ H ₆	1-butyne	249.80	l	SEL	-1918.5	3.8	166.5	3.8	19.58	0.51	v	2019DIK
		281.18	g	SEL	-2598.06	0.67	166.68	0.67			TN	2021RUS
		281.18	l	SEL1	-2574.36	0.71	142.98	0.71			DvH	2021RUS
		281.18	g		-2596.61	1.06	165.23	1.06			R	2019DIK
		281.18	g		-2596.76	0.84	165.38	0.84			C	1951PRO
		281.18	g		-2597.5	2.1	166.1	2.1			E	1945WAG
		281.18	g		-2596.80	0.88	165.42	0.88			C	1945WAG
		281.18	l		-2572.60	1.09	141.22	1.09	24.00	0.23	R	2019DIK
		281.18	lg						23.79		R	1985MAJ
		281.18	lg	SEL					23.7		v	1971WIL
		281.18	lg						23.38			1950AST
	2-butyne	300.12	l	SEL	-2551.15	0.58	119.77	0.57			DvH	2021RUS
		300.12	g	SEL1	-2577.85	0.49	146.47	0.49			TN	2021RUS
		300.12	g		-2576.9	2.9	145.5	2.9			R	2019DIK
		300.12	g		-2576.7	1.0	145.3	1.0			C	1951PRO
		300.12	g		-2579.4	0.2	148.0	0.2			E	1945WAG
		300.12	l		-2550.3	2.9	118.9	2.9	26.62	0.48	R	2019DIK
		300.12	lg	SEL					26.7		v	1971WIL
C ₅ H ₈	1-pentyne	313.51	l	SEL	-3226.6	2.3	115.9	2.3			DvH	1945WAG
		313.51	g	SEL1	-3255.0	2.1	144.3	2.1			E	1945WAG
		313.51	g		-3255.0	4.2	144.4	4.2			R	2019DIK
		313.51	l		-3228.9	4.3	118.3	4.3	26.09	1.09	R	2019DIK
		313.51	lg	SEL					28.4		v	1971WIL
		329.27	l	SEL	-3208.8	2.2	98.1	2.2			DvH	1945WAG
	2-pentyne	329.27	g	SEL1	-3239.6	2.1	128.9	2.1			E	1945WAG
		329.27	g		-3239.6	4.2	128.9	4.2			R	2019DIK
		329.27	l		-3213.4	4.3	102.8	4.3	26.12	1.14	R	2019DIK
		329.27	lg	SEL					30.8		v	1971WIL
		344.53	l	SEL	-3879.3	1.6	89.4	1.6			DvH	1979ROG
		344.53	g	SEL1	-3912.3	1.2	122.3	1.2			H2	1979ROG
C ₆ H ₁₀	1-hexyne	344.53	g		-3914.3	2.1	124.3	2.1			R	2019DIK
		344.53	l	SEL	-3881.3	2.1	91.4	2.1	32.94	0.29	v	2019DIK
	2-hexyne	357.59	l	SEL	-3861.9	2.5	71.9	2.5			DvH	1979ROG
		357.59	g	SEL1	-3897.7	2.4	107.7	2.4			H2	1979ROG
		357.59	g		-3894.6	7.5	104.6	7.5			R	2012KRO
	3-hexyne	357.59	l		-3859.0	7.6	69.0	7.6	35.6		R	2012KRO
		357.59	lg	SEL					35.8		v	2007BOU
		354.58	l	SEL	-3860.2	1.9	70.2	1.9			DvH	1979ROG
		354.58	g	SEL1	-3895.4	1.9	105.4	1.9			H2	1979ROG

		354.58	g		-3900.1	2.6	110.1	2.6		R	2019DIK
		354.58	l	SEL	-3864.8	2.6	74.9	2.6	35.21 0.29	v	2019DIK
Alkenynes											
C ₄ H ₄	butenyne	279.18	g	SEL	-2434.97	0.58	289.41	0.58		TN	2021RUS
		279.18	l	SEL1	-2411.8	2.1	266.2	2.1	23.21 0.40	TN	2021RUS
		279.18	g		-2434.4	2.8	288.9	2.8		R	2019DIK
		279.18	g		-2440.6	5.0	295.0	5.0		QM	1991ROT
		279.18	l		-2411.5	2.9	266.0	2.9	22.86 0.76	R	2019DIK
		279.18	l		-2380.0		234.4			C	1948REP
C ₅ H ₆	3-penten-1-yne	317.05	l	SEL	-3050.7	3.6	225.9	3.6	27.58 1.50	R	2019DIK
		317.05	g	SEL1	-3078.3	3.3	253.4	3.3		R	2019DIK
	3-penten-1-yne, (E)-	323.65	l	SEL	-3050.7	3.2	225.9	3.2	28.35 1.54	R	2019DIK
323.65		g	SEL1	-3079.1	2.8	254.2	2.8		R	2019DIK	
323.65		g		-3083.9	5.0	259.0	5.0		QM	1991ROT	
	3-penten-1-yne, (Z)-	317.15	l	SEL	-3050.2	2.9	225.4	2.9	27.28 0.77	R	2019DIK
317.15		g	SEL1	-3077.5	2.8	252.6	2.8		R	2019DIK	
317.15		g		-3082.9	5.0	258.0	5.0		QM	1991ROT	
Alkadiynes											
C ₄ H ₂	butadiyne	283.65	g	SEL	-2319.73	0.58	460.00	0.58		TN	2021RUS
		283.65	l	SEL1	-2296.49	1.17	436.76	1.16		DvH	2021RUS
		283.65	g		-2320.4	2.9	460.7	2.9		R	2019DIK
		283.65	g		-2323.7		464.0			D	1992KIE
		283.65	l	SEL	-2297.1	3.1	437.4	3.1	23.24 1.01	v	2019DIK
Cycloalkanes											
C ₃ H ₆	cyclopropane	240.27	g	SEL	-2091.75	0.40	53.84	0.40		TN	2021RUS
		240.27	l	SEL1	-2073.63	0.53	35.72	0.53	18.12 0.20	TN	2021RUS
		240.27	g		-2074.33	0.62	36.43	0.62		R	2019DIK
		240.27	g		-2077.2		39.3			HBr	1950LAC
		240.27	g		-2091.37	0.54	53.47	0.55		C	1949KNO
		240.27	l		-2091.2	0.5	53.3	0.5	16.88 0.37	R	2019DIK
		240.27	lg						18.11	R	1985MAJ
		240.27	lg						17.02		1970LIN
		240.27	lg						17.02		1970LIN
C ₄ H ₈	cyclobutane	285.64	g	SEL	-2745.02	0.39	27.81	0.39		TN	2021RUS
		285.64	l	SEL1	-2721.14	0.40	3.93	0.40	23.88 0.30	TN	2021RUS
		285.64	g		-2743.9	1.4	26.7	1.4		R	2019DIK
		285.64	l		-2720.8	1.3	3.6	1.3	23.04 0.47	R	2019DIK
		285.64	l		-2720.4	0.5	3.2	0.5	23.20	C	1952KAA
		285.64	lg						23.92	R	1985MAJ

C ₅ H ₁₀	cyclopentane	285.64	lg					23.26	0.40			1950COO
		322.40	l	SEL	-3291.40	0.43	-105.11	0.42	28.69	0.20	TN	2021RUS
		322.40	g	SEL1	-3320.09	0.43	-76.42	0.42			TN	2021RUS
		322.40	g		-3319.8	1.2	-76.7	1.2			R	2019DIK
		322.40	g		-3320.11	0.80	-76.40	0.79			R	1959MCC
		322.40	l		-3291.37	1.19	-105.14	1.19	28.41	0.34	R	2019DIK
		322.40	l		-3291.2	1.3	-105.3	1.3			C	1947SPI
		322.40	l		-3290.90	0.71	-105.61	0.71	28.66		C	1946PRO
		322.40	lg						28.72		R	1985MAJ
		322.40	lg						28.50			1971WIL
		322.40	lg						28.58	0.10		1959MCC
		322.40	lg						28.90			1952KAA
		322.40	lg						29.29			1947SPI
		322.40	lg						29.20			1943AST
	methylcyclobutane	310.00	l	SEL	-3339.5	1.3	-57.1	1.3			DvH	1950HUM
		310.00	g	SEL1	-3352.0	1.3	-44.5	1.3			C	1950HUM
		310.00	g		-3391.3	2.6	-5.2	2.6			R	2019DIK
		310.00	l	SEL	-3365.4	2.9	-31.1	2.9	25.92	1.30	v	2019DIK
C ₆ H ₁₂	cyclohexane	353.87	l	SEL	-3919.73	0.30	-156.08	0.29	33.14	0.15	TN	2021RUS
		353.87	g	SEL1	-3952.87	0.31	-122.94	0.30			TN	2021RUS
		353.87	l		-3952.28	0.79	-123.53	0.78			R	2019DIK
		353.87	l		-3919.28	0.76	-156.53	0.76	33.00	0.21	R	2019DIK
		353.87	l		-3918.6	0.7	-157.2		33.50		C	1952KAA
		353.87	l		-3918.4	1.0	-157.4	1.0	33.47		C	1947SPI
		353.87	l		-3919.90	0.71	-155.91	0.71	33.1		C	1946PRO
		353.87	l		-3919.6	1.3	-156.2	1.3			C	1940MOO
		353.87	l		-3917.98	0.75	-157.83				C	1939MOO
		353.87	l		-3935.3		-140.5				C	1915RIC
		353.87	l		-3946.8		-129.0				C	1901ZUB
		353.87	lg						33.10			1995DIO
		353.87	lg						33.12		R	1985MAJ
		353.87	lg						32.79	0.14		1982JOC
		353.87	lg						33.00	0.10		1982FUR
		353.87	lg						33.00			1981SHI
		353.87	lg						33.00	0.10		1979MAJ
		353.87	lg						32.90	0.30		1972SAB
		353.87	lg						32.90	0.30		1972SAB
		353.87	lg						33.00			1971WIL
		353.87	lg						32.90			1971MOR

		353.87	lg					33.01	0.15		1960WAD
		353.87	lg					30.12			1957GLA
		353.87	lg					33.04			1956BRE
		353.87	lg					33.03			1947OSB
		353.87	lg					33.10			1946PRO
		353.87	lg					33.30	0.10		1943AST
		353.87	lg					33.50			1927NAG
		353.87	lg					30.17	0.29		1926MAT
methylcyclopentane	344.96	l	SEL	-3937.74	0.37	-138.07	0.36	31.78	0.15	TN	2021RUS
	344.96	g	SEL1	-3969.52	0.38	-106.29	0.37			TN	2021RUS
	344.96	g		-3968.9	1.5	-106.9	1.5			R	2019DIK
	344.96	l		-3937.4	1.4	-138.5	1.4	31.59	0.22	R	2019DIK
	344.96	l		-3938.32	0.59	-137.50	0.59			C	1969GOO
	344.96	l		-3937.70	0.75	-138.11	0.75	31.7		C	1946PRO
	344.96	l		-3936.2	1.7	-139.6	1.7			C	1940MOO
	344.96	l		-3956.4		-119.4				C	1901ZUB
	344.96	lg						31.78		R	1985MAJ
	344.96	lg						31.60			1971WIL
	344.96	lg						31.71			1960SCO
	344.96	lg						31.63	0.10		1947OSB
	C ₇ H ₁₄ methylcyclohexane	374.12	l	SEL	-4565.29	0.96	-189.82	1.05	35.40	0.20	C
374.12		g	SEL1	-4600.69	1.08	-154.42	1.08			DvH	1946PRO
374.12		g		-4600.2	1.7	-154.9	1.7			R	2019DIK
374.12		l		-4564.9	1.6	-190.2	1.6	35.33	0.27	R	2019DIK
374.12		l		-4564.3	1.9	-190.8	1.9			C	1940MOO
374.12		lg						35.10	0.40		1987AZA
374.12		lg						35.44		R	1985MAJ
374.12		lg						35.40	0.10		1979MAJ
374.12		lg						35.40	0.29		1978FUC
374.12		lg						35.30			1975KUS
374.12		lg						35.23	0.17		1966WAD
374.12		lg						35.35			1947OSB
374.12		lg						31.59	0.01		1926MAT
C ₁₀ H ₂₀ 1,3-diethylcyclohexane	445.70	l	SELu	-6515.9	6.2	-277.1	6.2	46.83	1.55	R	2019DIK
	445.70	g	SEL1u	-6562.7	6.0	-230.3	6.0			R	2019DIK
Cycloalkenes											
C ₄ H ₆ cyclobutene	275.74	g	SEL	-2591.98	0.71	160.60	0.71			TN	2021RUS
	275.74	l	SEL1	-2569.73	0.74	138.35	0.74			DvH	2021RUS
	275.74	g		-2588.1	1.6	156.7	1.6			R	2019DIK

C ₅ H ₈	cyclopentene	275.74	g		-2588.2	1.5	156.8	1.5		C	1968WIB
		275.74	l	SEL	-2565.8	1.7	134.4	1.7	22.26 0.21	v	2019DIK
		317.39	l	SEL	-3117.56	0.52	6.88	0.52	28.29 0.30	TN	2021RUS
		317.39	g	SEL1	-3145.85	0.47	35.17	0.47		TN	2021RUS
		317.39	g		-3142.63	0.99	31.94	0.99		R	2019DIK
		317.39	g		-3146.2		35.6			H2	1982ALL
		317.39	g		-3145.0		34.3			E	1970FUR
		317.39	l		-3114.77	0.93	4.09	0.93	27.86 0.34	R	2019DIK
		317.39	l		-3115.20	0.59	4.51	0.59	28.29	C	1961LAB
		317.39	l		-3115.54	0.67	4.85	0.67	28.35	E	1949EPS
	methylenecyclobutane	317.39	lg					28.37			1941LIS
		315.00	l	SEL	-3204.7	0.5	94.0	0.5	27.70 0.42	C	1974GOO
		315.00	g	SEL1	-3232.40	0.65	121.72	0.65		C	1974GOO
		315.00	g		-3230.8	1.6	120.1	1.6		R	2019DIK
		315.00	g		-3217.0	5.0	106.3	5.0		QM	1991ROT
C ₆ H ₁₀	cyclohexene	315.00	l		-3204.5	0.9	93.8	0.9	26.24 1.29	R	2019DIK
		356.10	l	SEL	-3752.00	0.32	-37.99	0.31	33.56 0.53	TN	2021RUS
		356.10	g	SEL1	-3785.56	0.33	-4.43	0.32		TN	2021RUS
		356.10	g		-3785.32	1.07	-4.67	1.07		R	2019DIK
		356.10	l		-3751.90	1.03	-38.09	1.03	33.42 0.29	R	2019DIK
		356.10	l		-3752.39	0.49	-37.60	0.49	33.50 0.53	C	1996STE
		356.10	l		-3752.0	0.5	-38.0	0.5		C	1969GOO
		356.10	l		-3751.5	0.5	-38.5	0.5		C	1961LAB
		356.10	l		-3749.39	0.79	-40.60	0.79		E	1949EPS
		356.10	lg								1996STE
		356.10	lg					33.57	R	1985MAJ	
		356.10	lg					30.50 0.25			1926MAT
Bicycloalkanes											
C ₁₀ H ₁₂	tetralin	480.56	l	SEL	-5621.5	1.0	-28.2	1.0		C	1976GOO
		480.56	g	SEL1	-5676.5	1.4	26.8	1.4		DvH	1976GOO
		480.56	g		-5675.7	2.0	26.0	2.0			1971BOY
		480.56	l		-5598.6		-51.1			C	1953KAR
		480.56	l		-5581.9		-67.8			C	1950HOC
		480.56	lg	SEL					55.00 1.00	v	1971BOY
		480.56	lg						58.60		1971BOY
C ₁₀ H ₁₈	cis-decalin	468.93	l	SEL	-6287.7	1.3	-219.5	1.3	50.2 2.1	C	1941DAV
		468.93	g	SEL1	-6337.9	2.5	-169.3	2.5		DvH	1941DAV
		468.93	g		-6338.2	1.9	-169.0	1.9		R	2019DIK
		468.93	g		-6338.0	2.3	-169.2	2.3		DvH	1960SPE

		468.93	l		-6287.7	1.6	-219.5	1.6	50.47	0.91	R	2019DIK
		468.93	l		-6288.2	0.9	-219.0	0.9			C	1960SPE
		468.93	l		-6275.6		-231.6				C	1926HUC
		468.93	l		-6280.9	4.1	-226.3				C	1925ROT
		468.93	lg	SEL					50.2	2.1	v	1960SPE
trans-decalin		460.42	l	SEL	-6278.9	1.3	-228.3	1.3			C	1941DAV
		460.42	g	SEL1	-6327.4		-179.8				DvH	1941DAV
		460.42	g		-6325.2	2.8	-182.0	2.8			R	2019DIK
		460.42	g		-6325.0	2.3	-182.2	2.3			DvH	1960SPE
		460.42	l		-6277.0	2.6	-230.2	2.6	48.15	0.99	R	2019DIK
		460.42	l		-6276.96	0.92	-230.23	0.92			C	1960SPE
		460.42	l		-6263.9	2.1	-243.3				C	1937HUC
		460.42	l		-6255.9		-251.3				C	1926HUC
		460.42	l		-6268.2	4.6	-239.0				C	1925ROT
		460.42	lg	SEL					48.5	2.1	v	1960SPE
		460.42	lg						42.84			1957GLA
C ₁₂ H ₂₂	1,1'-bicyclohexyl	511.10	l	SEL	-7593.0	1.2	-272.8	1.2			C	1976GOO
		511.10	g	SEL1	-7651.0	0.3	-214.8	1.2			DvH	1976GOO
		511.10	g		-7650.6	3.0	-215.2	3.0			R	2012KRO
		511.10	g		-7647.4	3.1	-218.4	3.1			DvH	1978MON
		511.10	l		-7592.0	2.3	-273.8	2.3	58.60	1.90	R	2012KRO
		511.10	l		-7589.9	2.6	-275.9	2.6	57.99	0.25	C	1978MON
		511.10	l		-7439.7		-426.1				C	1962GOL
		511.10	l		-7596.5		-269.3				C	1951WIS
		511.10	l		-7537.5	6.3	-328.3				C	1935BRU
		511.10	lg						58.90	1.70	R	2019DIK
		511.10	lg	SEL					57.98		v	1985MAJ
C ₁₀ H ₁₂	tetralin	480.56	l	SEL	-5621.5	0.9	-28.2	1.0			C	1976GOO
		480.56	g	SEL1	-5676.54	1.05	26.82	1.04			DvH	1976GOO
		480.56	g		-5674.2	4.6	24.5	4.6			R	2019DIK
		480.56	l		-5620.6	4.0	-29.1	4.0	53.6	2.2	R	2019DIK
		480.56	l		-5617.5	2.1	-32.2	2.2			C	1971BOY
		480.56	lg	SEL					55.00	0.30	v	1971BOY
AROMATIC HYDROCARBONS												
Alkylbenzenes												
C ₆ H ₆	benzene	353.22	l	SEL	-3267.51	0.23	49.18	0.22	33.94	0.15	TN	2021RUS
		353.22	g	SEL1	-3301.45	0.23	83.12	0.22			TN	2021RUS
		278.70	s	SEL2	-3256.83	0.92	38.50	0.92	10.50	0.20	DfusH	2008ROU
		353.22	g		-3301.20	0.76	82.86	0.76			R	2019DIK

		353.22	g		-3301.2	0.9	82.9	0.9			R	2008ROU
		353.22	l		-3267.34	0.75	49.01	0.75	33.85	0.14	R	2019DIK
		353.22	l		-3267.49	0.42	49.16	0.42			C	1969GOO
		353.22	l		-3276.1		57.7				C	1948REP
		353.22	l		-3266.95	0.38	48.62				C	1947COO
		353.22	l		-3267.62	0.42	49.29	0.42	33.94		C	1945PRO
		353.22	l		-3264.8		46.4				C	1929LAN
		353.22	l		-3277.7		59.4				C	1925BAR
		353.22	l		-3271.3		52.9				C	1920RIC
		353.22	l		-3246.8		28.5				C	1918GUI
		353.22	l		-3267.2		48.9				C	1915RIC
		353.22	lg						33.92	0.06	R	2008ROU
		353.22	lg						33.90	0.10	R	2002MAN
		353.22	lg						33.74	0.11		1987SAB
		353.22	lg						33.80	0.10		1973SVO
		353.22	lg						33.90	0.10		1973KON
		353.22	lg						33.90			1971WIL
		353.22	lg						33.85			1957AND
		353.22	lg						33.84			1947OSB
		353.22	lg						34.00			1927NAG
		353.22	lg						30.84	0.13		1926MAT
		278.70	sl	SEL					10.50	0.20	f	2008ROU
C ₇ H ₈	toluene	383.75	l	SEL	-3909.71	0.34	12.07	0.33	38.04	0.25	TN	2021RUS
		383.75	g	SEL1	-3947.7		50.1				TN	2021RUS
		178.20	s	SEL2	-3899.8	1.5	2.1	1.5	9.90	1.10	R	2008ROU
		383.75	g		-3947.75	0.92	50.11	0.92			R	2019DIK
		383.75	g		-3947.74	1.10	50.10	1.10			R	2008ROU
		383.75	l		-3909.66	0.89	12.02	0.89	38.09	0.25	R	2019DIK
		383.75	l		-3909.68	1.05	12.04	1.05			R	2008ROU
		383.75	l		-3909.9	0.5	12.3	0.5	38.04		C	1945PRO
		383.75	l		-3938.4		40.8				C	1925BAR
		383.75	l		-3913.0		15.4				C	1920RIC
		383.75	l		-3920.7		23.1				C	1915RIC
		383.75	l		-3927.0		29.4				C	1906SCH
		383.75	lg						38.06	0.04	R	2008ROU
		383.75	lg						38.10	0.10	R	2002MAN
		383.75	lg						37.79	0.13		1982JOC
		383.75	lg						38.00			1971WIL
		383.75	lg						38.14			1957AND

C ₈ H ₁₀	ethylbenzene	383.75	lg					37.99			1947OSB	
		383.75	lg					33.35	0.04		1926MAT	
		383.75	lg					33.35	0.04		1926MAT	
		408.96	l	SEL	-4564.41	0.55	-12.53	0.55	42.26	0.40	TN	2021RUS
		408.96	g	SEL1	-4606.67	0.55	29.73	0.55			TN	2021RUS
		408.96	g		-4605.8	3.3	28.8	3.3			R	2019DIK
		408.96	l		-4563.3	3.2	-13.7	3.2	42.49	0.43	R	2019DIK
		408.96	l		-4562.4	1.0	-14.5	1.0			C	1953COO
		408.96	l		-4564.87	0.71	-12.07	0.71	42.26		C	1945PRO
		408.96	l		-4566.4		-10.5				C	1922AUW
		408.96	l		-4557.6		-19.3				C	1922AUW
		408.96	l		-4565.9		-11.0				C	1915RIC
		408.96	lg						42.40	0.10		1981HOS
		408.96	lg						42.30			1971WIL
		408.96	lg						42.24	0.10		1947OSB
		408.96	lg						36.02	0.17		1926MAT
	m-xylene	412.25	l	SEL	-4551.86	0.63	-25.08	0.63	42.60	0.30	C	1946PRO
		412.25	g	SEL1	-4594.46	0.69	17.52	0.69			DvH	1946PRO
		412.25	lg		-4551.6	1.3	-25.4	1.3	42.77	0.28	v	2019DIK
		412.25	g		-4594.3	1.4	17.4	1.4			R	2019DIK
		412.25	l		-4549.56	0.54	-27.38	0.54			C	1946COO
		412.25	l		-4567.6		-9.3				C	1915RIC
		412.25	l		-4581.4		4.5				C	1910RIC
		412.25	lg						42.70			1971WIL
		412.25	lg						36.36			1957GLA
		412.25	lg						43.04			1957AND
		412.25	lg						42.65	0.10		1947OSB
		412.25	lg						42.60			1946PRO
		412.25	lg						36.36	0.38		1926MAT
o-xylene	417.56	l	SEL	-4552.9	1.0	-24.1	1.0	43.40	0.30	C	1945PRO	
	417.56	g	SEL1	-4596.26	1.04	19.32	1.04			DvH	1945PRO	
	417.56	g		-4595.9	1.8	19.0	1.8			R	2019DIK	
	417.56	l		-4552.4	1.8	-24.5	1.8	43.48	0.26	R	2019DIK	
	417.56	l		-4551.5	0.5	-25.5	0.5			C	1946COO	
	417.56	l		-4567.6		-9.3				C	1915RIC	
	417.56	l		-4581.4		4.5				C	1910RIC	
	417.56	lg						42.90			1995CHI	
	417.56	lg						43.40			1971WIL	
417.56	lg						36.90			1957GLA		

				417.56	lg					43.96			1957AND	
				417.56	lg					43.43	0.10		1947OSB	
				417.56	lg					43.40			1945PRO2	
				417.56	lg					36.82	0.38		1926MAT	
p-xylene				411.47	l	SEL	-4552.86	0.92	-24.08	0.92	42.30	0.30	C	1946PRO
				411.47	g	SEL1	-4595.16	0.97	18.22	0.97			C(l)	1946PRO
				411.47	g		-4594.6	2.2	17.7	2.2			R	2019DIK
				411.47	l		-4552.4	1.9	-24.5	1.9	42.20	1.08	R	2019DIK
				411.47	l		-4551.4	0.5	-25.5	0.5			C	1946COO
				411.47	l		-2254.7		-25.2				C	1915RIC
				411.47	l		-4565.9		-11.0				C	1910RIC
				411.47	lg						42.30			1995CHI
				411.47	lg						43.00	0.10		1990SMI
				411.47	lg						42.35	0.10		1988MES
				411.47	lg						42.30	0.10		1981HOS
				411.47	lg						42.60			1974AMB
				411.47	lg						42.40			1971WIL
				411.47	lg						35.94			1957GLA
				411.47	lg						42.72			1957AND
				411.47	lg						42.37	0.10		1947OSB
				411.47	lg						42.30			1946PRO
				411.47	lg						35.98	0.08		1926MAT
C ₉ H ₁₂	isopropylbenzene			425.54	l	SEL	-5215.44	0.96	-40.80	0.97			C	1945PRO
				425.54	g	SEL1	-5260.54	1.09	4.30	1.09			DvH	1945PRO
				425.54	g		-5260.2	1.7	4.0	1.7			R	2019DIK
				425.54	l		-5215.1	1.6	-41.1	1.6	45.10	0.39	DvH	2019DIK
				425.54	l		-5218.6		-37.6				C	1915RIC
				425.54	lg						45.10			1994RUI
				425.54	lg						45.10	0.10		1982FUC
				425.54	lg						44.00			1975KUS
				425.54	lg						45.10			1971WIL
				425.54	lg						37.03			1957GLA
				425.54	lg	SEL					45.14		v	1947OSB
				425.54	lg						45.10			1945PRO2
C ₁₀ H ₁₄	m-isopropyltoluene			448.35	l	SEL	-5857.3	1.0	-78.2	1.0			C	1973GOO
				448.35	g	SEL1	-5906.9	1.3	-28.7	1.3			DvH	1973GOO
				448.35	g		-5905.9	1.9	-29.7	1.9			R	2019DIK
				448.35	l	SEL	-5856.3	1.7	-79.2	1.7	49.55	0.88	v	2019DIK
				448.35	l		-5857.8	2.2	-77.7	2.2			C	1946PRO

o-isopropyltoluene	448.35	lg						50.00			1971WIL
	451.50	l	SEL	-5862.62	0.84	-72.92	0.84			C	1973GOO
	451.50	g	SEL1	-5912.26	1.19	-23.28	1.19			DvH	1973GOO
	451.50	g		-5911.3	1.8	-24.2	1.8			R	2019DIK
	451.50	l	SEL	-5861.7	1.6	-73.9	1.6	49.64	0.84	v	2019DIK
	451.50	l		-5859.9	2.2	-75.6	2.2			C	1946PRO
	451.50	lg						50.60			1971WIL
p-isopropyltoluene	450.27	l	SEL	-5857.9	1.0	-77.6	1.0			C	1973GOO
	450.27	g	SEL1	-5907.28	1.11	-28.26	1.10			DvH	1973GOO
	450.27	g		-5906.3	1.7	-29.3	1.7			R	2019DIK
	450.27	l	SEL	-5856.9	1.6	-78.7	1.6	49.39	0.47	v	2019DIK
	450.27	l		-5857.6	2.2	-77.9	2.2			C	1946PRO
	450.27	lg						49.20			2005HOS
	450.27	lg						48.90			1994RUI
	450.27	lg						50.30			1971WIL
Alkenylbenzenes											
C ₈ H ₈ styrene	418.45	l	SEL	-4395.42	0.48	104.31	0.48	44.03	0.40	TN	2021RUS
	418.45	g	SEL1	-4439.45	0.48	148.34	0.48			TN	2021RUS
	418.45	g		-4439.1	1.5	148.0	1.5			R	2019DIK
	418.45	l		-4394.9	1.4	103.8	1.4	44.16	0.60	R	2019DIK
	418.45	l		-4395.63	0.59	104.52	0.59	43.93	0.42	C	1947ROB
	418.45	l		-4394.87	0.84	103.76	0.84	43.50		C	1945PRO
	418.45	g		-4438.35	0.92	147.24	0.92			C	1945PRO
	418.45	l		-4379.8	6.3	88.7	6.3			C	1938LUS
	418.45	l		-4399.5		108.4				C	1929LAN
	418.45	l		-4374.4		83.3				C	1918GUI
	418.45	l		-4379.0		87.9				C	1914MOU
	418.45	l		-4431.3		140.2				C	1911LEM
	418.45	l		-4336.7		45.6				C	1910AUW
	418.45	lg						43.93	0.42		1946PIT
	418.45	lg						43.20			1939PAT
Alkyl naphthalenes											
C ₁₀ H ₈ naphthalene	353.65	s	SEL	-5154.7	3.4	78.0	1.5	16.90	0.70	R	2008ROU
	353.65	l	SEL1	-5173.3	2.1	95.2	2.1	55.40	1.40	DvH	2008ROU
	491.11	g	SEL2	-5228.7	1.5	150.6	1.5			R	2008ROU
	353.65	s		-5155.9	0.1	77.8	1.5			R	2019DIK
	353.65	s		-5150.1		72.0		-		C	1996BAL
	353.65	s		-5153.9	5.1	75.8	5.1	-		C	1983HOL
	353.65	s		-5156.4	0.6	78.3	0.6	-		C	1977AMM

		353.65	s		-5156.15	0.92	78.09	0.92	-	C	1966COL	
		353.65	s		-5156.9	1.6	78.8	1.6	-	C	1960SPE	
		353.65	s		-5156.8		78.8		-	C	1932MIL	
		353.65	s		-5151.2	0.2	73.1	0.2	-	C	1931KEF	
		353.65	s		-5152.7		74.7		-	C	1931BUR	
		491.11	g		-5228.4	1.5	150.3	1.5		R	2019DIK	
		491.11	g		-5229.7	1.9	151.7	1.9		DfusH	1960SPE	
		491.11	lg						54.64	0.95	2008ZHA	
		491.11	lg	SEL					55.40	1.40	2008ROU	
		491.11	lg						60.30	1.10	2006HAF	
		491.11	lg						53.40		1998CHI	
		491.11	lg						56.60		1995CHI	
		491.11	lg						54.80	0.40	1993CHI	
		491.11	lg						55.00	1.20	1981DEK	
		491.11	lg						55.10	1.10	1968FOW	
		491.11	lg						56.4	2.0	1955CAM	
		491.11	lg						55.50	1.80	1923MOR	
		353.65	sg						72.47	0.06	2019DIK	
		353.65	sg						72.60	0.30	2008ROU	
		353.65	sg						70.4		1998CHI	
		353.65	sg						72.30	0.40	1988TOR	
		353.65	sg						72.60	0.10	1983EKE	
		353.65	sg						72.40	0.70	1982MUR	
		353.65	sg						72.5		1974SIN	
		353.65	sg						72.10	0.25	1972MOR	
		353.65	sg						73.00	0.25	1972IRV	
		353.65	sg						72.9		1966COL	
		353.65	sg						72.70	0.30	1963MIL	
		353.65	sg						73.2	3.1	1960SPE	
		353.65	sl	SEL					16.90	0.70	2008ROU	
		353.65	sl						16.80	0.40	2002CHI	
		353.65	sl						16.81	0.70	2001RIJ	
		353.65	sl						16.80	0.70	1993CHI	
		353.65	sl						16.73	0.70	1981DEK	
		353.65	sl						16.90	0.70	1980AND	
		353.65	sl						16.80	0.70	1957MCC	
C ₁₁ H ₁₀	1-methylnaphthalene	517.84	l	SEL	-5814.0	1.7	56.7	1.7	60.10	0.80	C	1960SPE
		517.84	g	SEL1	-5874.1	1.9	116.8	1.9			DvH	1960SPE
		242.52	s	SEL2	-5807.1	1.7	49.7	1.7			DfusH	1960SPE

	517.84	l		-5813.6	2.3	56.3	2.3	59.32	1.44	R	2019DIK
	517.84	l		-5783.0	5.8	25.6				C	1996BAL
	517.84	l		-5627.5		-129.9				C	1954HIP
	517.84	g		-5873.0	2.7	115.6	2.7			R	2019DIK
	517.84	lg						65.10	1.10		2006HAF
	517.84	lg	SEL					60.10	0.80	v	2003VER
	517.84	lg						62.4			2002LEI
	517.84	lg						57.32	0.42		1974SAB
	517.84	lg						60.7	2.1		1960SPE
	517.84	lg						46.48			1957GLA
	242.52	sl	SEL					6.95	1.00	f	1996DOM
2-methylnaphthalene	307.69	s	SEL	-5802.7	1.5	45.3	1.5	12.13	1.00	C	1960SPE
	514.25	l	SEL1	-5814.8	2.2	57.5	2.4			DfusH	1960SPE
	514.25	g	SEL2	-5860.1	1.8	102.8	2.6			DvH	1960SPE
	307.69	s	u	-5796	11	39	11	70.6	5.0	R	2019DIK
	307.69	s		-5792.7	2.2	35.3	2.2			C	1939RIC
	514.25	l	u	-5815	8	58	8	58.69		R	2019DIK
	514.25	g	u	-5867	12	110	12			R	2019DIK
	514.25	lg	SEL					45.31	1.00	v	1957GLA
	307.69	sl	SEL					12.13	1.00	f	1996DOM

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_f H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
OXIDIZED HYDROCARBONS													
Carbon Oxides													
CO ₂	carbon dioxide	194.69	g	SEL	0.000	0.021	-393.476	0.015			TN		2021RUS
		194.69	g		0.03	0.13	-393.51	0.13			R		2019DIK
		194.69	g		0.03	0.13	-393.51	0.13			R		1984COX
CO	carbon monoxide	81.64	g	SEL	-282.952	0.030	-110.524	0.026			TN		2021RUS
		81.64	g		-282.93	0.28	-110.55	0.28			R		2019DIK
		81.64	g		-282.95	0.17	-110.53	0.17			R		1984COX
		81.64	g		-283.036	0.146	-110.440	0.146			C		1933FEN
Alkanols													
CH ₄ O	methanol	337.63	l	SEL	-726.52	0.16	-238.61	0.15	37.70	0.05	TN		2021RUS
		337.63	g	SEL1	-764.22	0.16	-200.91	0.15			TN		2021RUS
		337.63	g		-763.44	0.92	-201.69	0.92			R		2019DIK
		337.63	g		-764.3		-200.8				E		1976HIN
		337.63	g		-763.68	0.20	-201.45				C		1932ROS
		337.63	l		-725.89	0.88	-239.24	0.88	37.55	0.27	R		2019DIK
		337.63	l		-726.8		-238.4				C		1972BAR
		337.63	l		-725.71	0.13	-239.41	0.13			C		1965CHA
		337.63	l		-726.51	0.21	-238.62	0.21			C		1960GRE
		337.63	l		-726.34	0.20	-238.79				C		1931ROS
		337.63	l		-715.0		-250.1				C		1925PAR
		337.63	l		-713.8		-251.3				C		1920RIC
		337.63	lg						38.00				2004NAS
		337.63	lg						37.83		R		1985MAJ
		337.63	lg						37.80				1983SCH
		337.63	lg						37.40	0.10			1973SVO
		337.63	lg						37.43	0.02			1971POL
		337.63	lg						37.30	0.10			1966WAD
		337.63	lg						37.70	0.10			1963MCC
		337.63	lg						37.40	0.04			1960GRE
		337.63	lg						35.30	0.04			1926MAT
C ₂ H ₆ O	ethanol	351.57	l	SEL	-1366.92	0.23	-277.51	0.22	42.47		TN		2021RUS
		351.57	g	SEL1	-1409.39	0.23	-235.04	0.22			TN		2021RUS
		351.57	g		-1409.8	1.2	-234.6	1.2			R		2019DIK
		351.57	l		-1367.24	1.12	-277.19	1.12	42.57	0.66	R		2019DIK
		351.57	l		-1367.58	0.25	-276.85	0.25			C		1965CHA

		351.57	l		-1366.95	0.42	-277.48	0.42	42.34	0.08	C	1960GRE
		351.57	l		-1366.31	0.40	-278.12	0.40			C	1932ROS
		351.57	l		-1366.9		-277.6					1929KEL
		351.57	l		-1370.9		-273.5				C	1925PAR
		351.57	l		-1368.3		-276.1				C	1920RIC
		351.57	l		-1368.6		-275.8				C	1911EME
		351.57	lg						41.70			2004NAS
		351.57	lg						42.40			1995DIO
		351.57	lg						42.30			1983SCH
		351.57	lg						42.26	0.02		1971POL
		351.57	lg						42.30	0.08		1966WAD
		351.57	lg						42.20	0.10		1963MCC
		351.57	lg						42.59			1961GRE
		351.57	lg						38.91	0.01		1926MAT
C ₃ H ₈ O	1-propanol	370.26	l	SEL	-2021.03	0.25	-302.70	0.24	47.46		TN	2021RUS
		370.26	g	SEL1	-2068.49	0.26	-255.24	0.25			TN	2021RUS
		370.26	g		-2068.2	2.3	-255.6	2.3			R	2019DIK
		370.26	l		-2020.1	2.2	-303.6	2.2	48.04	0.69	R	2019DIK
		370.26	l		-2021.31	0.25	-302.42	0.26			C	1975MOS
		370.26	l		-2020.7	1.3	-303.0	1.3	47.41	0.08	Hde	1972CON
		370.26	l		-2021.17	0.25	-302.56	0.25			C	1970DEK
		370.26	l		-2020.41	0.63	-303.32	0.63			C	1967BUC
		370.26	l		-2019.37	0.29	-304.37	0.30			C	1965CHA
		370.26	l		-2021.37	0.75	-302.36	0.76	47.86	0.21	C	1961SNE
		370.26	l		-2017.7	1.0	-306.1	1.0	47.50		C	1960GRE
		370.26	l		-2032.6		-291.1				C	1920RIC
		370.26	lg						46.30			1983SCH
		370.26	lg						45.70			2004NAS
		370.26	lg						47.80			2004LUB
		370.26	lg						49.20			1995CHI
		370.26	lg						47.49	0.02		1971POL
		370.26	lg						47.32	0.08		1966WAD
		370.26	lg						46.60			1963MCC
	isopropanol	355.39	l	SEL	-2004.83	0.36	-318.90	0.36	45.44	0.10	TN	2021RUS
		355.39	g	SEL1	-2050.27	0.36	-273.46	0.36			TN	2021RUS
		355.39	g		-2051.9	1.8	-271.8	1.8			R	2019DIK
		355.39	g		-2051.0		-272.8				E	1965BUC
		355.39	l		-2006.3	1.7	-317.5	1.7	45.65	0.62	R	2019DIK

		355.39	l		-2006.90	0.21	-316.83	0.22	45.73		C	1965CHA
		355.39	l		-2005.77	0.42	-317.96	0.42	45.90	0.21	C	1961SNE
		355.39	l		-2005.1		-318.6		44.40		C	1950PAR
		355.39	lg						44.00			2004NAS
		355.39	lg						45.34	0.02		1971POL
		355.39	lg						45.23	0.13		1966WAD
		355.39	lg						44.00			1963MCC
		355.39	lg						45.51			1963HAL
		355.39	lg						40.04	0.04		1926MAT
C ₄ H ₁₀ O	1-butanol	390.77	l	SEL	-2676.18	0.24	-326.85	0.25			C	1975MOS
		390.77	g	SEL1	-2728.68	0.39	-274.35	0.38			DvH	1975MOS
		390.77	g		-2726.9	2.8	-276.1	2.8			R	2019DIK
		390.77	g		-2723.04	0.92	-279.99	0.92			C	1960GRE
		390.77	l		-2674.4	2.6	-328.7	2.6	52.57	1.07	R	2019DIK
		390.77	l		-2636	115	-367				C	1973DEL
		390.77	l		-2676.01	0.24	-327.02	0.25			C	1969MOS
		390.77	l		-2675.61	0.45	-327.42	0.45	52.30		C	1969GUN
		390.77	l		-2677.43	0.63	-325.61	0.63			C	1965CHA
		390.77	l		-2670.44	0.42	-332.60	0.42			C	1960TJE
		390.77	l		-2674.87	0.84	-328.16	0.84	51.88	0.42	C	1960SKI
		390.77	l		-2670.4		-332.6				C	1928VER
		390.77	l		-2676.0		-327.0				C	1925PAR
		390.77	l		-2670.8		-332.3				C	1920RIC
		390.77	lg						48.40			2004NAS
		390.77	lg	SEL					52.50		v	1995CHI
		390.77	lg						55.20			1992GRA
		390.77	lg						52.10			1984LAW
		390.77	lg						51.70			1983SCH
		390.77	lg						52.34	0.02		1971POL
		390.77	lg						52.30	0.08		1966WAD
		390.77	lg						51.00	0.10		1963MCC
		390.77	lg						52.80	0.84		1960GRE
		390.77	lg						43.81	0.04		1926MAT
	isobutanol	380.90	l	SEL	-2669.64	0.59	-333.39	0.59			C	1965CHA
		380.90	g	SEL1	-2720.48	0.59	-282.56	0.59			DvH	1965CHA
		380.90	g		-2720.5	1.2	-282.5	1.2			R	2019DIK
		380.90	l		-2669.18	1.13	-333.85	1.13	51.32	0.49	R	2019DIK
		380.90	l		-2668.4	0.9	-334.6	0.9	50.80		Hde	1975CON

		380.90	l		-2668.51	0.84	-334.52	0.84	50.70		C	1960SKI
		380.90	l		-2665.8		-337.2				C	1920RIC
		380.90	lg						48.80			1999ORT
		380.90	lg						50.76	0.03		1984MAJ
		380.90	lg						50.79	0.02		1971POL
		380.90	lg	SEL					50.84	0.08	v	1966WAD
		380.90	lg						49.80			1963MCC
		380.90	lg						42.89	0.01		1926MAT
	tert-butanol	355.50	l	SEL	-2643.95	0.79	-359.08	0.80	46.30		C	1960SKI
		355.50	g	SEL1	-2689.7	1.5	-313.4	1.5			DvH	1960SKI
		298.87	s	SEL2	-2636.7	5.1	-366.4	5.1	7.24		R	2019DIK
		355.50	g		-2690.72	1.08	-312.31	1.08			R	2019DIK
		355.50	l		-2643.91	0.97	-359.13	0.96	46.81	0.49	R	2019DIK
		355.50	l		-2643.80	0.84	-359.24	0.84	46.60		H2O	1991WIB
		355.50	l		-2647.4		-355.6		45.94			1955TAF
		355.50	lg						47.40			2007MAL
		355.50	lg						45.40			1995CHI
		355.50	lg						46.94	0.02		1971POL
		355.50	lg						46.61	0.08		1966WAD
		355.50	lg						44.90			1963MCC
C ₅ H ₁₂ O	1-pentanol	410.90	l	SEL	-3330.91	0.28	-351.43	0.29			C	1975MOS
		410.90	g	SEL1	-3387.60	0.35	-294.74	0.34			DvH	1975MOS
		410.90	g		-3387.0	1.8	-295.4	1.8			R	2019DIK
		410.90	g		-3381.7	2.0	-300.6	2.0			C	1960GRE
		410.90	l		-3330.3	1.3	-352.1	1.3	56.69	1.21	R	2019DIK
		410.90	l		-3329.92	0.67	-352.42	0.67	-1615.00		C	1971HAY
		410.90	l		-3329.96	0.64	-352.38	0.64	56.94	0.17	C	1969GUN
		410.90	l		-3324.65	0.38	-357.69	0.38			C	1965CHA
		410.90	lg						55.40			2006NAS
		410.90	lg						57.40			1995CHI
		410.90	lg						57.80			1995CHI
		410.90	lg						55.40			1983SCH
		410.90	lg						56.94	0.17		1966WAD
		410.90	lg	SEL					56.69		v	1965CHA
		410.90	lg						57.74	1.05		1960GRE
	2-methyl-1-butanol	402.14	l	SEL	-3325.95	0.54	-356.39	0.55	54.56	1.26	C	1965CHA
		402.14	g		-3381.2	1.3	-301.2	1.3			R	2019DIK
		402.14	l		-3325.75	0.97	-356.59	0.96	55.41	0.79	R	2019DIK

isopentanol				402.14	lg					54.10			1963MCC	
				404.73	l	SEL	-3326.2	0.5	-356.1	0.5	55.02	1.26	C	1965CHA
				404.73	g	SEL1	-3381.2	1.4	-301.1	1.4			DvH	1965CHA
				404.73	g		-3381.9	1.2	-300.5	1.2			R	2019DIK
				404.73	g		-3326.6	1.6	-300.4	1.6			C	1987MAR
				404.73	l		-3326.00	0.94	-356.33	0.94	55.88	0.75	R	2019DIK
				404.73	l		-3326.6	1.6	-355.7	1.6			C	1987MAR
				404.73	lg						55.30			1995CHI
				404.73	lg						54.30			1963MCC
				404.73	lg						44.18	0.01		1926MAT
Glycols														
C ₂ H ₆ O ₂	1,2-ethanediol	470.73	l	SEL	-1189.19	0.37	-455.24	0.37	65.87	0.20	TN	2021RUS		
		470.73	g	SEL1	-1255.06	0.40	-389.37	0.40			TN	2021RUS		
		470.73	g		-1256.6	5.0	-387.8	5.0			R	2019DIK		
		470.73	g		-1250.0	2.8	-394.4	2.8				1990KNA		
		470.73	l		-1189.9	2.1	-454.6	2.1	66.8	4.6	R	2019DIK		
		470.73	l		-1184.6	2.8	-459.8	2.8			C	1990KNA		
		470.73	l		-1188.84	0.67	-455.59	0.67	67.60		C	1972GAR		
		470.73	l		-1190.2		-454.2				C	1952PAR		
		470.73	l		-1188.67	0.84	-455.76	0.84	65.46		C	1947MCC		
		470.73	l		-1189.59	0.71	-454.84	0.71	65.54		C	1946PAR		
		470.73	l		-1202.61	1.21	-441.82	1.21			C	1942JUN		
		470.73	l		-1191.4	1.2	-453.0	1.2			C	1937MOU		
		470.73	lg						65.40	0.30		2005VAS		
		470.73	lg						66.00	0.20		2004VER		
		470.73	lg						66.20			2003DER		
		470.73	lg						63.40	0.10		1999MOK		
		470.73	lg						65.60	0.30		1988KNA		
		470.73	lg						68.20	0.80		1981JOO		
		470.73	lg						66.10	0.30		1981HAL		
		470.73	lg						63.70	0.10		1981AMB		
		470.73	lg									1972GAR		
		470.73	lg						63.80			1937GAL		
C ₃ H ₈ O ₂	2-methoxyethanol	397.54	l	SEL	-1901.7	7.6	-422.0	7.6	45.1	2.7	H2O	1977GUT		
		397.54	g	SEL1	-1947	8	-377	8			DvH	1977GUT		
		397.54	g		-1950.9	2.7	-372.9	2.7			R	2019DIK		
		397.54	l		-1905.1	2.8	-418.6	2.8	45.72	0.70	R	2019DIK		
		397.54	l		-1882.8		-440.9		45.03		C	1947SIM		

C ₃ H ₈ O ₃	glycerin	397.54	lg					45.20	0.20			1971KUS2
		563.00	l	SEL	-1654.3	0.4	-669.4	0.4	91.70	0.90	C	1988BAS
		563.00	g	SEL1	-1746.00	0.99	-577.73	0.98			DvH	1988BAS
		563.00	s	SEL2	-1636.02	0.99	-687.71	0.98			DfusH	1988BAS
		563.00	g		-1743.9	3.2	-579.9	3.2			R	2019DIK
		563.00	l		-1654.5	1.2	-669.3	1.2	89.4	2.9	R	2019DIK
		563.00	l		-1654.1		-669.6				C	1952PAR
		563.00	l		-1655.3	1.0	-668.4	1.0	91.52		C	1946PAR
		563.00	l		-1665.1		-658.6				C	1911EME
		563.00	lg	SEL					85.77		v	1962ROS
		291.05	sl	SEL				18.28		f	1991ACR	
Sugars												
C ₆ H ₁₂ O ₆	alpha-D-glucose	415.0	s	SEL	-2805.0	1.3	-1270.8	1.3			C	1960PON
		706	g	SEL1	-2944	49	-1132	49			DsubH	1960PON
		706	g		-2940	49	-1136	49			R	2019DIK
		415.0	s		-2801.51	1.10	-1274.30	1.10			R	2019DIK
		415.0	sg	SEL					139	49	s	2019DIK
C ₆ H ₁₀ O ₅	cellulose	553	s(d)		-2825.5	0.8	-964.5	0.8			C	1950JES
C ₆ H ₁₂ O ₆	fructose	415.0	s(d)	SEL	-2810.4	0.3	-1265.4	0.3			C	1939CLA
C ₁₂ H ₂₂ O ₁₁	sucrose	415.0	s(d)	SEL	-5643.4	1.8	-2222.4	1.8			C	1960PON
		415.0	s(d)		-5644.2		-2221.6				C	1939CLA
		415.0	s(d)		-5637.4	1.7	-2228.4	1.7			C	1935HUF
Alkyl Ethers												
C ₂ H ₆ O	dimethyl ether	248.37	g	SEL	-1460.41	0.40	-184.02	0.40			TN	2021RUS
		248.37	l	SEL1	-1441.9	1.1	-202.5	1.1			DvH	2021RUS
		248.37	l	u	-1443.6	7.2	-200.8	7.2	18.60	0.97	R	2019DIK
		248.37	g		-1460.4	1.0	-184.0	1.0			R	2019DIK
		248.37	g		-1460.38	0.46	-184.05	0.46			C	1964PIL
		248.37	l		-1460.38	0.61	-202.55	0.61			DvH	1964PIL
		248.37	lg	SEL					18.50		v	1976AMB
C ₃ H ₈ O	ethyl methyl ether	280.59	g	SEL	-2106.41	0.58	-217.32	0.58			DvH	2021RUS
		280.59	l	SEL1	-2082.5	1.2	-241.2	1.2			TN	2021RUS
		280.59	g		-2107.32	1.36	-216.41	1.36			R	2019DIK
		280.59	g		-2107.40	0.63	-216.33	0.63			C	1964PIL
		280.59	l	SEL	-2083.4	1.9	-240.3	1.9	23.90	1.10	v	2019DIK
C ₄ H ₁₀ O	diethyl ether	307.56	l	SEL	-2725.25	0.82	-277.78	0.82	27.37	0.25	TN	2021RUS
		307.56	g	SEL1	-2752.62	0.82	-250.41	0.82			TN	2021RUS
		307.56	g		-2751.8	2.7	-251.2	2.7			R	2019DIK

		307.56	g		-2726.3	1.8	-276.7	1.8			C	1968PIH
		307.56	g		-2751.06	0.75	-251.97	0.76			C	1963PIL
		307.56	l		-2724.9	2.7	-278.2	2.7	26.95	0.22	R	2019DIK
		307.56	l		-2732.1	1.9	-270.9	1.9			C	1957MUR
		307.56	lg						27.14	0.10		1980MAJ
		307.56	lg						27.20			1976AMB
		307.56	lg						26.61			1968PIH
		307.56	lg						27.15	0.25		1926MAT
C ₆ H ₁₄ O	diisopropyl ether	341.49	l	SEL	-4010.4	1.3	-351.2	1.3			C	1965COL
		341.49	g	SEL1	-4043.1	1.4	-318.5	1.4			DvH	1965COL
		341.49	g		-4041.7	4.2	-319.9	4.2			R	2019DIK
		341.49	l		-4010.2	1.7	-351.5	1.7	32.11	0.21	R	2019DIK
		341.49	lg	SEL					32.70	0.50	v	2007EFI
		341.49	lg						32.26		R	1985MAJ
		341.49	lg						32.10	0.10		1980MAJ
Aldehydes												
CH ₂ O	formaldehyde	254.13	g	SEL	-570.088	0.096	-109.214	0.096			TN	2021RUS
		254.13	l	SEL1	-549.65	0.55	-129.65	0.55	20.44	0.54	TN	2021RUS
		254.13	g		-570.61	0.41	-108.70	0.41			R	2019DIK
		254.13	g		-570.78	0.42	-108.52	0.42			C	1970FLE
		254.13	g		-561.1		-118.2				C	1925WAR
		254.13	l		-550.17	0.68	-129.14	0.68	20.44	0.54	R	2019DIK
C ₂ H ₄ O	acetaldehyde	293.56	g	SEL	-1193.06	0.25	-165.54	0.25			TN	2021RUS
		293.56	l	SEL1	-1166.84	0.47	-191.76	0.47	26.22	0.40	TN	2021RUS
		293.56	g		-1192.4	2.6	-166.2	2.6			R	2019DIK
		293.56	g		-1187.9	1.5	-170.7	1.5			LiB	1991WIB
		293.56	l		-1167.5	2.6	-191.1	2.6	24.86	0.38	R	2019DIK
		293.56	lg						25.70			1991WIB
		293.56	lg						26.12		R	1985MAJ
		293.56	lg						26.90			1963BUL
C ₃ H ₆ O	propanal	321.19	l	SEL	-1821.20	0.36	-216.71	0.35	29.94	0.25	TN	2021RUS
		321.19	g	SEL1	-1851.14	0.26	-186.77	0.25			TN	2021RUS
		321.19	g	u	-1846	8	-192	8			R	2019DIK
		321.19	l	u	-1816	8	-221	8	29.63	0.34	R	2019DIK
		321.19	l		-1819.58	0.63	-218.32	0.63			LiB	1991WIB
		321.19	l		-1822.2	1.5	-215.7	1.5			Hde	1972CON
		321.19	l		-1817.66	0.96	-220.25	0.96			Hde	1967BUC
		321.19	l		-1816.53	0.75	-221.38	0.75			C	1962TJE

		321.19	lg					29.60				1991WIB
		321.19	lg					29.64	0.07	Hde		1972CON
		321.19	lg					29.66	0.42	Hde		1967BUC
		321.19	lg					29.50				1962TJE
C ₄ H ₈ O	butanal	347.95	l	SEL	-2477.72	0.84	-239.49	0.84	33.68	0.42	TN	2021RUS
		347.95	g	SEL1	-2511.40	0.85	-205.81	0.85			TN	2021RUS
		347.95	g		-2511.8	1.9	-205.4	1.9			R	2019DIK
		347.95	l		-2478.1	1.9	-239.1	1.9	33.68	0.30	R	2019DIK
		347.95	l		-2471.77	0.84	-245.43	0.84			LiB	1991WIB
		347.95	l		-2479.1	1.5	-238.1	1.5			Hde	1967BUC
		347.95	l		-2478.69	0.71	-238.52	0.71			C	1960TJE
		347.95	l		-2477.1	1.4	-240.1	1.4			C	1960NIC
		347.95	lg						33.20			1995CHI
		347.95	lg						33.60			1991WIB
		347.95	lg						33.68	0.42	Hde	1967BUC
C ₃ H ₄ O	1-propenal	264.25	g	SEL	-1687.9		-64.2	1.1			TN	2021RUS
		264.25	g	u	-1647		-105				R	1998LIA
		264.25	g	u	-1659		-93				ion	1997AUB
		264.25	g	u	-1655		-97				ion	1996BOU
		264.25	g		-1689.1		-63.0	2.6			R	2019DIK
	2-propenal	325.40	l	SEL	-1656.4	1.4	-95.7	1.4	30.00	0.30	TN	2021RUS
		325.40	g	SEL1	-1686.4	1.2	-65.7	1.2			TN	2021RUS
		325.40	g	u	-1678.8	5.9	-73.3	5.9			R	2019DIK
		325.40	l	u	-1647.9	5.9	-104.2	5.9	30.91	0.29	R	2019DIK
		325.40	l		-1632.2		-119.9				C	1920MOU
Ketones												
C ₃ H ₆ O	acetone	329.22	l	SEL	-1789.79	0.35	-248.12	0.35	31.38	0.10	TN	2021RUS
		329.22	g	SEL1	-1821.17	0.35	-216.74	0.35			TN	2021RUS
		329.22	g		-1821.3	2.1	-216.6	2.1			R	2019DIK
		329.22	g		-1819.41	0.59	-218.50	0.59			LiB	1991WIB
		329.22	g		-1820.8	0.5	-217.1	0.5			R	1976CHA
		329.22	g		-1820.38	0.67	-217.53	0.67			E	1965BUC
		329.22	g		-1821.5		-216.4					1957PEN
		329.22	g		-1821.40	0.84	-216.51	0.84			C	1941MIL
		329.22	l		-1790.1	2.1	-247.8	2.1	31.18	0.39	R	2019DIK
		329.22	l		-1788.54	0.63	-249.37	0.63	30.84		LiB	1991WIB
		329.22	l		-1771.5		-266.4				C	1918GUI
		329.22	l		-1804.2		-233.7				C	1911EME

C ₄ H ₈ O	butanone	329.22	lg					31.30			1975AMB	
		329.22	lg					29.66	0.01		1926MAT	
		352.74	l	SEL	-2443.83	0.79	-273.38	0.79	34.92	0.10	TN	2021RUS
		352.74	g	SEL1	-2478.75	0.79	-238.46	0.79			TN	2021RUS
		352.74	g	u	-2477.8	6.7	-239.4	6.7			R	2019DIK
		352.74	l	u	-2442.9	6.7	-274.3	6.7	34.84	0.18	R	2019DIK
		352.74	g		-2478.64	0.84	-238.57	0.84			R	1976CHA
		352.74	g		-2478.51	0.96	-238.70	0.96			E	1965BUC
		352.74	l		-2444.2		-273.0		34.97		C	1964SIN
		352.74	l		-2438.4		-278.8				C	1950PAR
		352.74	l		-2436.3	1.5	-280.9				C	1942CRO
		352.74	lg						34.92		R	1985MAJ
		352.74	lg						34.80	0.10		1983UCH
		352.74	lg						35.51	0.08		1979SUN
		352.74	lg						31.97	0.10		1926MAT
Carboxylic Acids												
CH ₂ O ₂	formic acid	374.05	l	SEL	-254.59	0.20	-424.71	0.20	46.33		TN	2021RUS
		374.05	g	SEL1	-300.92	0.22	-378.38	0.22			TN	2021RUS
		281.47	s	SEL2	-241.9	1.0	-437.4	1.0	12.68	1.00	DfusH	2021RUS
		374.05	g	u	-275	9	-405	9			R	2019DIK
		374.05	l		-254.45	0.85	-424.85	0.85	20.1	9.1	R	2019DIK
		374.05	l		-254.2		-425.1		46.48		H2O	1974GUT
		374.05	l		-253.80	0.30	-425.50	0.30			C	1964LEB
		374.05	l		-254.60	0.30	-424.70	0.30	46.30	0.50	C	1959SIN
		374.05	lg						36.00		R	1987STE
		281.47	lg						46.30		R	1985MAJ
		374.05	lg						46.30	0.50		1970KON
		374.05	lg						19.90			1941STO
		281.47	sl	SEL					12.68	1.00	f	1941STO
C ₂ H ₄ O ₂	acetic acid	391.20	l	SEL	-874.95	0.18	-483.65	0.18	50.91	0.30	TN	2021RUS
		391.20	g	SEL1	-925.86	0.38	-432.74	0.38			TN	2021RUS
		289.70	s	SEL2	-863.23	0.35	-495.37	0.35			DfusH	2021RUS
		391.20	l		-874.72	0.83	-483.89	0.82			R	2019DIK
		391.20	l		-875.16	0.34	-483.44	0.34			C	1997STE
		391.20	l		-874.20	0.21	-484.40	0.21			C	1964LEB
		391.20	l		-874.46	0.42	-484.15	0.42			C	1959EVA
		391.20	lg						50.2			2000VER
		391.20	lg						50.3			2000VER

		391.20	lg					51.60	1.50			1970KON
		289.70	sl	SEL				11.72		f		1996DOM
C ₃ H ₆ O	propanoic acid	413.99	l	SEL	-1528.31	0.18	-509.60	0.18	55.36	0.20	TN	2021RUS
		413.99	g	SEL1	-1583.67	0.28	-454.24	0.28			TN	2021RUS
		413.99	g		-1558.23	0.88	-479.67	0.88			R	2019DIK
		413.99	l		-1527.17	0.59	-510.74	0.58	31.06	0.66	R	2019DIK
		413.99	l		-1527.29	0.13	-510.62	0.14			C	1964LEB
		413.99	l		-1528.0		-509.9				C	1935SCH
		413.99	lg						54.50			2000VER
		413.99	lg						54.40			2000VER
		413.99	lg						54.90			1995CHI
		413.99	lg						55.0	2.0		1970KON
		413.99	lg						30.63			1926MAT
C ₄ H ₈ O	n-butanoic acid	436.42	l	SEL	-2183.50	0.59	-533.70	0.59			C	1964LEB
		436.42	g	SEL1	-2241.73	0.43	-475.47	0.42			DvH	1964LEB
		436.42	g		-2224.2	2.7	-493.0	2.7			R	2019DIK
		436.42	l		-2183.3	1.3	-533.9	1.3	40.8	2.4	R	2019DIK
		436.42	lg	SEL					58.23	0.30	v	2000VER
		436.42	lg						58.10			2000VER
		436.42	lg						58.50			2000VER
		436.42	lg						60.70			1995CHI
		436.42	lg						58.0	4.0		1970KON
Alkyl Esters												
C ₃ H ₆ O ₂	ethyl formate	327.29	l	SEL	-1611.8	1.2	-426.1	1.2	32.10	0.15	TN	2021RUS
		327.29	g	SEL1	-1643.9	1.2	-394.0	1.2			TN	2021RUS
		327.29	g		-1639.3	8.4	-398.6	8.4			R	2019DIK
		327.29	l		-1607.4	8.4	-430.5	8.4	31.92	0.30	R	2019DIK
		327.29	l		-1643.7	0.8	-394.2	0.8			E	1991FRO
		327.29	l		-1607.4		-430.5				E	1974HIN
		327.29	lg						32.11		R	1985MAJ
		327.29	lg						32.17			1974HIN
		327.29	lg						32.50			1974HIN
		327.29	lg						30.12	0.10		1926MAT
C ₂ H ₄ O ₂	methyl formate	305.14	l	SEL	-970.10	0.86	-388.50	0.86	28.58	0.20	TN	2021RUS
		305.14	g	SEL1	-998.68	0.84	-359.92	0.84			TN	2021RUS
		305.14	g		-1000.9	1.9	-357.7	1.9			R	2019DIK
		305.14	g		-996.3		-362.3				E	1974HIN
		305.14	l		-972.5	1.8	-386.1	1.8	28.37	0.43	R	2019DIK

		305.14	l		-992.7		-365.9			E	1990GLA	
		305.14	l		-967.8		-390.8		28.37	E	1974HIN	
		305.14	l		-980.6		-378.0			H2O	1974GUT	
		305.14	l		-972.61	0.59	-385.99	0.59	30.59	C	1971HAL	
		305.14	lg						28.60	R	1985MAJ	
		305.14	lg						28.24	0.04		1926MAT
C4H8O2	ethyl acetate	350.21	l	SEL	-2237.35	0.46	-479.86	0.46	35.15	0.21	H2O	1991WIB2
		350.21	g	SEL1	-2272.49	0.51	-444.72	0.51			H2O	1991WIB2
		350.21	g		-2273.73	0.80	-443.48	0.79			R	2019DIK
		350.21	l		-2238.39	0.75	-478.82	0.75	35.34	0.25	R	2019DIK
		350.21	l		-2236.63	0.80	-480.57	0.79			LiB	1991WIB
		350.21	l		-2238.54	0.48	-478.67	0.48			C	1978FEN
		350.21	l		-2235.4	3.9	-481.8	3.9			C	1970BUT
		350.21	l		-1350.5		-461.6				C	1929ROT
		350.21	l		-2246.4		-470.8				C	1918GUI
		350.21	lg						35.00			1995CHI
		350.21	lg						35.60	0.10		1980SVO
		350.21	lg						35.15	0.21		1966WAD
		350.21	lg						32.30	0.42		1926MAT
		C4H6O2	vinyl acetate	345.33	l	SEL	-2080.6	2.0	-350.8	2.0	37.15	0.84
345.33	g			SEL1	-2117.7	2.2	-313.6	2.2			DvH	1980VIL
345.33	l			u	-2086	10	-346	10			C	1980VIL
345.33	g				-2116.6	2.8	-314.8	2.8			R	2019DIK
345.33	l				-2081.5	2.8	-349.9	2.8	35.12	0.35	R	2019DIK
C5H8O2	ethyl acrylate	372.59	l	SEL	-2740.1		-370.6				H2	1980VIL
		372.59	g	SEL1	-2779.3		-331.4				DvH	1980VIL
		372.59	g		-2773.2	2.8	-337.5	2.8			R	2019DIK
		372.59	l		-2734.1	2.8	-376.6	2.8			R	2019DIK
		372.59	l		-2715.0		-395.6		39.20		C	1975VIL
		372.59	l		-2743.3		-367.4				C	1972BAR
		372.59	lg	SEL					39.20		v	1975VIL
	methyl methacrylate	369.73	l	SEL	-2728.3	4.0	-382.4	4.0				1980VIL
		369.73	g	SEL1	-2768.3	4.0	-342.3	4.0				1980VIL
		369.73	g		-2775.11	0.78	-335.58	0.78			R	2019DIK
		369.73	l		-2735.60	0.74	-375.09	0.74	39.51	0.25	R	2019DIK
		369.73	l		-2724.6	4.0	-386.1	4.0			C	1980VIL
		369.73	lg						40.08			1975VIL
Oxacyclohydrocarbons												

C ₂ H ₄ O	oxirane	283.59	g	SEL	-1306.00	0.37	-52.60	0.37			TN	2021RUS
		283.59	l	SEL1	-1281.07	0.44	-77.53	0.44	24.93	0.24	TN	2021RUS
		283.59	g		-1306.8	2.0	-51.8	2.0			R	2019DIK
		283.59	g		-1306.04	0.59	-52.57	0.59			C	1965PEL
		283.59	g		-1307.71	0.84	-50.89	0.84			C	1942CRO
		283.59	l		-1281.8	2.0	-76.8	2.0	24.93	0.24	R	2019DIK
		283.59	l		-1262.9	1.3	-95.7	1.3			C	1937MOU
		283.59	lg						25.90		R	1987STE
		283.59	lg						25.51		R	1985MAJ
C ₄ H ₄ O	furan	304.51	l	SEL	-2083.5	0.6	-62.0	0.6	27.60	0.20	C	1952GUT
		304.51	g	SEL1	-2111.1	0.6	-34.4	0.6			DvH	1952GUT
		304.51	g		-2110.76	0.86	-34.80	0.86			R	2019DIK
		304.51	g		-2117.9		-27.7					1991ZAH
		304.51	l		-2083.34	0.85	-62.21	0.84	27.41	0.18	R	2019DIK
		304.51	l		-2088.2		-57.3				C	1929LAN
		304.51	lg						27.71		R	1985MAJ
		304.51	lg						28.20			1970MOI
C ₃ H ₆ O ₃	1,3,5-trioxane	335.65	s	SEL	-1515.70	0.30	-522.21	0.30			C	1969MAN
		387.20	l	SEL1	-1526.4	4.8	-511.5	4.8			DfusH	1969MAN
		387.20	g	SEL2	-1572.3	0.5	-465.6	0.5			DsubH	1969MAN
		387.20	g		-1571.78	0.62	-466.13	0.62			R	2019DIK
		387.20	l		-1525.4	5.6	-512.5	5.6	46.3	4.9	DvH	2019DIK
		335.65	s	SEL	-1514.7	0.5	-523.2	0.5	10.73	0.20	f	2019DIK
		335.65	s		-1513.0	0.4	-524.9	0.4			C	1971HAY
		335.65	s		-1492.0	1.0	-545.9	1.0			C	1943WAL
		335.65	s		-1518.0		-519.9				C	1942DEL
		335.65	sg						57.06	0.37	R	2019DIK
		387.20	sg						55.6			1983WIT
		387.20	sg	SEL					56.60	0.20	s	1969MAN
		387.20	sg						48.5	2.5		1943WAL
C ₄ H ₈ O ₂	1,3-dioxane	378.13	l	SEL	-2339.86	0.99	-377.35	0.99	39.09	0.05	C	1982BYS
		378.13	g	SEL1	-2378.95	0.99	-338.26	0.99			DvH	1982BYS
		378.13	g		-2372.5	5.6	-344.7	5.6			R	2019DIK
		378.13	l		-2333.8	5.4	-383.4	5.4	38.69	1.60	R	2019DIK
		378.13	l		-2339.86	0.99	-377.35	0.99			C	1982MOS
		378.13	l		-2332.8	1.8	-384.4	1.8			C	1968PIH
		378.13	l		-2331.8	0.8	-385.4	1.8			C	1968PIH
		378.13	l		-2331.91	0.84	-385.30	0.84	35.00	1.91	C	1967PIH

	378.13	l		-2340.82	0.59	-376.39	0.59	35.60		C	1961SNE
	378.13	l		-2322.1		-395.1		35.56	0.84	C	1959FLE
	378.13	l		-2331.32	0.84	-385.88	0.84			C	1957SKU
1,4-dioxane	374.25	l	SEL	-2362.23	0.99	-354.98	0.99	38.64	0.05	C	1982BYS
	374.25	g	SEL1	-2362.28	0.99	-354.93	0.99			DvH	1982BYS
	374.25	g		-2402.1	2.2	-315.1	2.2			R	2019DIK
	374.25	g		-2399.4	1.7	-317.8	1.7			DvH	1961SNE
	374.25	l		-2363.4	2.2	-353.8	2.2	38.73	0.39	R	2019DIK
	374.25	l		-2363.9	0.5	-353.3	0.5	35.50	1.60	C	1961SNE
	374.25	l		-2346.2		-371.0				C	1933ROT
	374.25	l		-2186.8		-530.4				C	1929HER
	374.25	lg						39.80			1982BYS
	374.25	lg						34.10			1957GLA
	374.25	lg						37.49			1938CRE
Furandione and D-Camphor											
C ₄ H ₂ O ₃ 2,5-furandione	325.65	s	SEL	-1389.50	0.67	-470.23	0.67			C	1964WIL
	478.33	g	SEL1	-1458.5	2.1	-401.2	2.1			DvH	1964WIL
	478.33	l	SEL2	-1401.0	2.1	-458.8	2.1			DfusH	1964WIL
	478.33	g		-1458.0	2.3	-401.8	2.3			R	2019DIK
	478.33	l	SEL	-1400	10	-459	10	57.5	9.5	v	2019DIK
	325.65	s	SEL	-1389.8	1.1	-470.0	1.1	10.68		f	2019DIK
	325.65	s		-1390.30	0.71	-469.43	0.71			C	1950PAR
	325.65	sg						68.2	2.0	R	2019DIK
	325.65	sg						70.0	2.0		1978VIL
C ₁₀ H ₁₆ O D-camphor	451.8	s	SEL	-5902.0	3.0	-319.4	3.0			C	1977STE
	493.00	g	SEL1	-5953.9	3.1	-267.5	3.1			DsubH	1977STE
	451.8	l	SEL2	-5908.8	3.2	-312.5	3.2			DvH	1977STE
	451.8	sg	u					59	23	R	2019DIK
	493.00	lg						55.30	na		2002ROO
	493.00	lg	SEL					6.82		v	1995CHI
	451.8	s		-5891.2		-330.2				C	1932SHC
	451.8	s		-5898.6	3.8	-322.8	3.8			C	1929SWI
	451.8	s		-5910.7		-310.7				C	1913ROT
	451.8	sg						54.40			1995CHI
	451.8	sg						54.50			1995CHI
	451.8	sg						55.20			1995CHI
	451.8	sg	SEL					51.88	0.84	s	1977STE
Aromatic Alcohols											

C ₆ H ₆ O	phenol	314.15	s	SEL	-3055.98	0.65	-162.35	0.65	69.01	0.30	TN	2021RUS
		454.91	g	SEL1	-3124.99	0.63	-93.34	0.63			TN	2021RUS
		454.91	l	SEL2	-3071.06	0.70	-147.28	0.70			DfusH	2021RUS
		454.91	g		-3122.5		-95.9				R	2019DIK
		454.91	g		-3121.98	0.59	-96.36	0.59			DsubH	1961COX
		454.91	g		-3121.89	0.63	-96.44	0.63			C	1960AND
		454.91	l		-3068.4	2.8	-149.9	2.8	58.34	1.04	R	2019DIK
		314.15	lg						58.8			1995CHI
		314.15	s		-3053.32	0.68	-165.01	0.67	15.07		DfusH	2019DIK
		314.15	s		-3053.6	0.3	-164.7	0.3			C	1961COX
		314.15	s		-3053.50	0.67	-164.83	0.67			C	1960AND
		314.15	s		-3055.8		-162.5				C	1954PAR
		314.15	s		-3054.7		-163.6				C	1941BAD
		314.15	sg						69.14	0.49	R	2019DIK
		314.15	sg						69.70	0.90		1971PAR
		314.15	sg						68.6			1961COX
		314.15	sg						68.66	0.50		1960AND
C ₇ H ₈ O	3-methylphenol	475.28	l	SEL	-3703.89	0.59	-193.75	0.59	61.71	1.05	C	1960AND
		475.28	g	SEL1	-3765.6	1.2	-132.0	1.2			DvH	1960AND
		285.37	s	SEL2	-3693.18	0.78	-204.46	0.78			DfusH	1960AND
		475.28	g		-3769.9	4.0	-127.7	4.0			R	2019DIK
		475.28	l		-3704.11	1.15	-193.53	1.15	65.8	3.8	R	2019DIK
		475.28	l		-3704.68	0.25	-192.96	0.26	59.70		C	1961COX
		475.28	l		-3721.2		-176.4				C	1954PUS
		475.28	l		-3704.4		-193.2				C	1941BAD
		475.28	l		-3694.5		-203.2				C	1925BAR
		475.28	lg						65.00	0.70		2007RIC
		475.28	lg						62.50			1995CHI
		475.28	lg						61.70		R	1987STE
		475.28	lg						61.70			1958BID
		475.28	lg						46.28			1957GLA
		285.37	sg						56.10		R	1987STE
		285.37	sl	SEL					10.71		f	1996DOM
	benzyl alcohol	478.50	l	SEL	-3736.9	1.3	-160.7	1.3			C	1954PAR
		478.50	g	SEL1	-3801.7	1.4	-95.9	1.4			DvH	1954PAR
		478.50	g	u	-3776.7	5.8	-120.9	5.8			R	2019DIK
		478.50	l	u	-3711.5	6.1	-186.1	6.1	65.17	1.94	R	2019DIK
		478.50	l	u	-3723.3		-174.3				C	1929LAN

		478.50	l	u	-3723.8		-173.9			C	1925BAR
		478.50	l		-3743.0	2.8	-154.6	2.8	60.30 0.30	C	1995PAP
		478.50	l		-3731.3		-166.3			C	1906SCH
		478.50	lg						65.50 0.40		2005VAS
		478.50	lg	SEL					64.80 0.60	v	1999VER
		478.50	lg						69.50		1995CHI
		478.50	lg						62.50 0.30		1990AMB
		478.50	lg						62.10 0.30		1949DRE
		478.50	lg						63.0 2.2		1937GAR
		478.50	lg						60.29 0.42		1926MAT
Aromatic Aldehydes and Acids											
C ₇ H ₆ O	benzaldehyde	451.15	l	SEL	-3525.01	1.10	-86.80	1.10	49.73 0.40	TN	2021RUS
		451.15	g	SEL1	-3574.74	0.91	-37.07	0.91		TN	2021RUS
		451.15	g		-3576.1	5.4	-35.8	5.4		R	2019DIK
		451.15	g		-3578.5		-33.3				1975KUD
		451.15	g		-3574.6	4.2	-37.2	4.2			1971SOL
		451.15	l		-3526.0	5.4	-85.8	5.4	50.01 0.40	R	2019DIK
		451.15	l		-3525.0	2.0	-86.8	2.0	50.30	C	1975AMB
		451.15	l		-3528.79	0.84	-83.02	0.84	49.79 0.84	C	1972LEB
		451.15	l		-3525.9		-86.0			C	1929LAN
		451.15	lg						49.00 0.70		2007EME
		451.15	lg						49.10		1995CHI
		451.15	lg						45.90		1972LEB
		451.15	lg						40.92		1957GLA
C ₇ H ₆ O ₂	benzoic acid	395.50	s	SEL	-3227.01	0.18	-384.80	0.17		TN	2021RUS
		523.53	g	SEL1	-3317.62	0.21	-294.19	0.20	90.61 0.50	TN	2021RUS
		523.53	l	SEL2	-3238.72	0.21	-373.09	0.20		DvH	2021RUS
		523.53	g		-3318.7	1.4	-293.1	1.4		R	2019DIK
		523.53	g		-3316.30	0.54	-295.51	0.54		DsubH	1969GUN
		523.53	lg	SEL					78.90	v	1995CHI
		395.50	s		-3227.7	1.4	-384.1	1.4		R	2019DIK
		395.50	s		-3228.06	0.44	-383.75	0.44	-	C	1984PIL
		395.50	s		-3226.87	0.18	-384.94	0.18	-	C	1969MOS
		395.50	s		-3227.0	0.2	-384.8	0.2	-	C	1969GUN
		395.50	s		-3228.0	0.5	-383.8	0.5	-	C	1960LI
		395.50	s		-3227.30	0.30	-384.51	0.30	-	C	1960COR
		395.50	s		-3227.40	0.30	-384.41	0.30	-	C	1958GUN
		395.50	s		-3227.2	0.5	-384.6	0.5	-	C	1956COO

	395.50	s	-3227.30	0.30	-384.51	0.30	-	C	1955CHA
	395.50	s	-3227.51	0.32	-384.30	0.32	-	C	1946JES
	395.50	s	-3227.60	0.30	-384.21	0.30	-	C	1944PRO
	395.50	s	-3226.39	0.32	-385.42	0.32	-	C	1942JES
	395.50	s	-3229.8		-382.0		-	C	1939MAT
	395.50	s	-3228.8		-383.0		-	C	1934HIR
	395.50	s	-3231.3		-380.5		-	C	1932MIL
	395.50	s	-3229.0		-382.8		-	C	1931BUR
	395.50	s	-3226.0		-385.8		-	C	1929LAN
	395.50	s	-3232.0		-379.8		-	C	1911WRE
	395.50	sg					91.04 0.14	R	2019DIK
	395.50	sg					93.30 1.20		2008FRE
	395.50	sg					90.40 0.50		2006RIB
	395.50	sg					88.30 0.50		2001KIY
	395.50	sg	SEL				89.30 0.50	s	1990SIL
	395.50	sg					89.23 0.50		1990SIL
	395.50	sg					89.01 0.33		1973MAL
	395.50	sg					86.6		1972WIE
	395.50	sg					89.50 0.20		1972MOR
	395.50	sg					89.70 0.60		1969CHA
	395.50	sg					91.40 0.40		1954DAV

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_f H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
NITROGEN-SUBSTITUTED HYDROCARBONS													
Hydrogen Cyanide													
CHN	hydrogen cyanide	298.79	l	SEL	-639.08	0.30	102.69	0.30			DvH		2021RUS
		298.79	g	SEL1	-665.669	0.092	129.280	0.089			TN		2021RUS
		298.79	g	SEL	-665.4	2.7	129.0	2.7	26.59	0.29	v		2019DIK
		298.79	l		-638.8	2.7	102.4	2.7			R		2019DIK
Alkylamines													
CH ₅ N	methylamine	266.84	g	SEL	-1086.81	0.24	-21.23	0.24			TN		2021RUS
		266.84	l	SEL1	-1062.79	0.45	-45.25	0.45	24.02	0.30	TN		2021RUS
		266.84	l	u	-1093.7		-14.3				C		1910MUL
		266.84	g		-1085.6	2.7	-22.4	2.7			R		2019DIK
		266.84	g		-1086.28	0.86	-21.76	0.86			E		1960ISS
		266.84	l		-1060.75	0.95	-47.29	0.94	23.76	0.28	R		2019DIK
		266.84	l		-1060.81	0.38	-47.23	0.38			C		1937AST
		266.84	lg						23.85		R		1985MAJ
C ₂ H ₇ N	ethylamine	289.79	g	SEL	-1737.31	0.54	-50.03	0.54			TN		2021RUS
		289.79	l	SEL1	-1713.25	0.55	-74.09	0.55	24.06		TN		2021RUS
		289.79	g		-1739.8	1.4	-47.5	1.4			R		2019DIK
		289.79	l		-1713.1	1.3	-74.3	1.3	26.76	0.44	R		2019DIK
		289.79	l		-1715.0		-72.3				C		1907LEM
		289.79	lg						27.00	0.50			1970COX
C ₃ H ₉ N	1-propylamine	320.38	l	SEL	-2365.13	0.41	-101.52	0.41	31.42	0.15	TN		2021RUS
		320.38	g	SEL1	-2396.55	0.42	-70.10	0.42			TN		2021RUS
		320.38	g		-2396.42	0.61	-70.23	0.61			R		2019DIK
		320.38	g		-2396.77	0.84	-69.87	0.84			R		1971SCO
		320.38	l		-2365.14	0.58	-101.51	0.57	31.28	0.21	R		2019DIK
		320.38	l		-2365.3	2.9	-101.4	2.9	31.34	0.21	C		1967SMI
		320.38	lg						31.30				1979MAJ2
		320.38	lg						29.20				1957GLA
	isopropylamine	304.92	l	SEL	-2354.37	0.52	-112.28	0.52	28.71	0.20	TN		2021RUS
		304.92	g	SEL1	-2383.08	0.53	-83.57	0.53			TN		2021RUS
		304.92	g		-2382.9	0.8	-83.7	0.8			R		1971SCO
		304.92	l		-2354.50	0.50	-112.15	0.71	28.50	0.20	C		1967SMI
C ₄ H ₁₁ N	n-butylamine	350.12	l	SEL	-3018.46	1.13	-127.48	1.13	32.64	1.26	C		1959EVA
		350.12	g	SEL1	-3054.17	1.13	-91.77	1.13			DvH		1959EVA
		350.12	g		-3054.22	1.32	-91.73	1.32			R		2019DIK

		350.12	l		-3018.31	1.30	-127.64	1.30	35.91	0.25	R	2019DIK
		350.12	lg						36.00			1995WOL
		350.12	lg						35.60			1995CHI
		350.12	lg						35.70	0.20		1985KUS
		350.12	lg						35.70	0.10		1979MAJ2
		350.12	lg	SEL					35.71	0.06	v	1969WAD
		350.12	lg						27.15			1957GLA
C ₂ H ₇ N	dimethylamine	280.02	g	SEL	-1769.79	0.42	-17.55	0.42			TN	2021RUS
		280.02	l	SEL1	-1743.91	0.37	-43.43	0.37	25.88	0.20	TN	2021RUS
		280.02	g		-1767.8	2.1	-19.5	2.1			R	2019DIK
		280.02	l		-1743.44	0.68	-43.90	0.68	24.65	0.36	R	2019DIK
		280.02	l		-1743.51	0.42	-43.83	0.42	25.40	0.04	C	1960ISS
		280.02	l		-1750.0		-37.3				C	1907LEM
		280.02	lg						26.00			1960ISS
		280.02	lg						26.48	0.13		1939AST
C ₃ H ₉ N	methylethylamine	305.60	l	SEL	-2391.4	2.3	-75.2	2.3			TN	2021RUS
		305.60	g	SEL1	-2420.7	1.1	-45.9	1.1			TN	2021RUS
		305.60	g		-2421.1	2.7	-45.6	2.7			R	2019DIK
		305.60	l		-2392.1	2.8	-74.5	2.8	28.99	0.51	R	2019DIK
	trimethylamine	276.00	g	SEL	-2439.91	0.76	-26.74	0.76			TN	2021RUS
		276.00	l	SEL1	-2417.80	0.87	-48.85	0.87	22.11	0.10	TN	2021RUS
		276.00	g		-2442.92	0.76	-23.72	0.75			R	2019DIK
		276.00	g		-2442.92	0.76	-23.72	0.75			DvH	1960ISS
		276.00	l		-2420.90	0.63	-45.74	0.62	21.68	0.27	R	2019DIK
		276.00	l		-2421.03	0.63	-45.62	0.63			C	1970COX
		276.00	l		-2420.92	0.71	-45.73	0.71	22.01	0.08	E	1960ISS
C ₆ H ₁₅ N	triethylamine	361.93	l	SEL	-4377.09	0.54	-127.46	0.55			C	1966LEB
		361.93	g	SEL1	-4412.49	0.59	-92.06	0.58			DvH	1966LEB
		361.93	g		-4411.0	4.3	-93.6	4.3			R	2019DIK
		361.93	l		-4376.3	4.3	-128.3	4.3	34.72	0.30	R	2019DIK
		361.93	l		-4335.1		-169.5				C	1972BAR
		361.93	lg	SEL					35.40	0.20	v	2009MOK
		361.93	lg						35.50		R	1987STE
		361.93	lg						34.80	0.20		1979MAJ2
		361.93	lg						35.10			1975CHU
		361.93	lg						34.94	0.06		1969WAD
C ₆ H ₁₃ N	cyclohexylamine	406.98	l	SEL	-4071.3	1.3	-147.4	1.3			C	1979STE
		406.98	g	SEL1	-4113.8	1.3	-104.9	1.3			DvH	1979STE

		406.98	g		-4115.7	3.0	-103.1	3.0			R	2019DIK
		406.98	l		-4071.2	2.8	-147.6	2.8	44.51	0.98	R	2019DIK
		406.98	l		-4077.5	1.6	-141.2	1.6			C	1982GUT
		406.98	lg	SEL					42.80	0.10	v	1975BER
Alkyl nitriles												
C ₂ H ₃ N	ethanenitrile	355.05	l	SEL	-1256.29	0.22	40.60	0.22	33.45	0.21	TN	2021RUS
		355.05	g	SEL1	-1289.74	0.25	74.05	0.25			TN	2021RUS
		355.05	l	u	-1247.2	7.2	31.5	7.2			C	1971HAL
		355.05	g		-1289.30	1.06	73.61	1.06			R	2019DIK
		355.05	l		-1256.28	0.94	40.59	0.94	33.02	0.49	R	2019DIK
		355.05	l		-1256.33	0.30	40.64	0.30	33.02	0.49	C	1983AN
		355.05	l		-1271.9		56.2				C	1909LEM
		355.05	lg						33.00			2004CHI
		355.05	lg						33.00			2004ANT
		355.05	lg						33.45	0.21		1983AN
		355.05	lg						32.94	0.06		1970HOW
C ₃ H ₅ N	propanenitrile	370.73	l	SEL	-1910.62	0.54	15.63	0.55	35.98		C	1971HAL
		370.73	g	SEL1	-1946.60	0.62	51.61	0.62			C	1971HAL
		370.73	g		-1947.0	1.8	52.0	1.8			R	2019DIK
		370.73	l		-1910.5	1.8	15.5	1.8	36.44	0.34	R	2019DIK
		370.73	l		-1918.4		23.4				C	1909LEM
		370.73	lg						37.10	0.30		2004CHI
		370.73	lg						37.10	0.30		2004ANT
		370.73	lg						36.03	0.02		1970HOW
		370.73	lg						36.70	0.30		1933HEI
C ₄ H ₇ N	butanenitrile	390.77	l	SEL	-2568.68	0.92	-5.61	0.92	36.99		C	1959EVA
		390.77	g	SEL1	-2605.67	0.92	31.37	0.95			DvH	1959EVA
		390.77	l		-2568.5	1.1	-5.8	1.1			R	2019DIK
		390.77	l		-2608.33	1.14	34.03	1.14			R	2019DIK
		390.77	l		-2578.6		4.3				C	1909LEM
		390.77	lg						39.20	0.10		1982FUC
		390.77	lg						39.33	0.08		1970HOW
		390.77	lg						40.50	0.20		1933HEI
C ₃ H ₃ N	propenenitrile	351.13	l	SEL	-1756.40	0.42	147.23	0.42	32.64		C	1971HAL
		351.13	g	SEL1	-1789.04	0.72	179.87	0.72			DvH	1971HAL
		351.13	g		-1790.4	4.2	181.2	4.2			R	2019DIK
		351.13	l		-1757.1	4.1	148.0	4.1	33.24	0.70	R	2019DIK
		351.13	l		-1749.3		140.2				C	1972BAR

		351.13	l		-1759.4	2.9	150.2	2.9	32.64		C	1945DAV
		351.13	lg						31.60		R	1987STE
Azahydrocarbons												
C ₂ N ₂	cyanogen	252.00	g	SEL	-1097.07	0.41	310.12	0.41			TN	2021RUS
		252.00	l	SEL1	-1076.13	0.42	289.18	0.42	20.94		TN	2021RUS
		252.00	g		-1094.1	2.5	307.1	2.5			R	2019DIK
		252.00	g		-1096.0	1.8	309.0	1.8			C	1951KNO
		252.00	g		-1093.70	0.84	306.75	0.84			C	1933WAR
		252.00	g		-1051.9	5.0	264.9	5.0			C	1933MCM
		252.00	l		-1072.3	2.8	285.4	2.8	21.73	1.24	R	2019DIK
		252.00	lg						20.8		R	1985MAJ
C ₂ H ₈ N ₂	1,1-dimethylhydrazine	335.57	l	SEL	-1978.7	3.6	48.4	3.6			C	1960DON
		335.57	g	SEL1	-2013.7	3.6	83.4	3.6			DvH	1960DON
		335.57	g		-2014.2	2.7	83.9	2.7			R	2019DIK
		335.57	l		-1979.2	2.5	48.9	2.5	35.02	0.76	R	2019DIK
		335.57	l		-1979.7	2.9	49.4				C	1952AST
		335.57	l	SEL					35.00	0.17	v	1952AST
C ₃ H ₆ N ₆	1,3,5-triazine-2,4,6-triamine	618	s	SEL	-1972.0	3.0	-65.9	3.0			C	1973SEL
		618	s		-1966.24	0.53	-71.67	0.53			R	2019DIK
		618	s		-1951.0		-86.9				C	1972BAR
		618	s		-1966.4	2.0	-71.5	2.0			C	1956HAV
		618	s		-1966.5		-71.4				C	1955SAL
C ₆ H ₇ N	aniline	457.30	l	SEL	-3393.06	0.84	31.81	0.84	55.77		C	1962HAT
		457.30	g	SEL1	-3448.27	0.88	87.03	0.88			DvH	1962HAT
		457.30	g		-3447.6	2.1	86.4	2.1			R	2019DIK
		457.30	l		-3392.6	1.7	31.3	1.7	55.04	1.20	R	2019DIK
		457.30	l		-3392.0		30.7		52.72		C	1952VRI
		457.30	l		-3392.3		31.1				C	1951COL
		457.30	l		-3391	13	30		50.21		C	1942AND
		457.30	lg						55.83	0.02		1971KUS
	pyridine	388.36	l	SEL	-2782.4	1.5	100.2	1.5			C	1954COX
		388.36	g	SEL1	-2822.6	1.5	140.4	1.5	40.16	0.06	DvH	1996CHI
C ₈ H ₁₁ N	N,N-dimethylaniline	466.47	l	SEL	-4767.8	3.2	47.9	3.2	52.83	0.03	C	1982FUR
		466.47	g	SEL1	-4820.6	3.2	100.8	3.2			DvH	1982FUR
		275.65	s	SEL2	-4756.2	3.2	36.4	3.2			DfusH	1982FUR
		466.47	g		-4819.5	3.0	99.7	3.0			R	2019DIK
		466.47	l		-4766.2	2.6	46.3	2.6	53.37	1.42	R	2019DIK
		466.47	l		-4754.3		34.4		49.79		C	1952VRI

C ₆ H ₁₅ NO ₃	triethanolamine	275.65	sl	SEL					11.56		f	1972AHN
		623.15	l	SEL	-3840.6	1.5	-664.0	1.5	105.9	2.2	C	1982MIN
		623.15	g	SEL1	-3946.3	2.7	-558.3	2.7			DvH	1982MIN
		623.15	g	u	-3941	26	-563	26			R	2019DIK
		623.15	l		-3840.4	2.9	-664.2	2.9	101	26	R	2019DIK
Alkyl Nitrates and Nitroalkanes												
CH ₃ NO ₃	methyl nitrate	338.00	g	SEL	-700.0	1.3	-122.2	1.3			E	1959RAY
		338.00	g		-700.0	2.5	-122.2	2.5			R	2019DIK
		338.00	l		-672.4	2.9	-149.8	2.9	27.63	1.46	R	2019DIK
		338.00	lg						34.10			1957GRA
CN ₄ O ₈	tetranitromethane	397.95	l	SEL	-431.8	1.7	38.3	1.7	43.93	0.42	C	1975LEB
		397.95	g	SEL1	-475.7	1.7	82.2	1.7			DvH	1975LEB
		287.05	s	SEL2	-395.4	4.3	1.9	4.3			DfusH	1975LEB
		397.95	l		-432.7	3.1	39.2	3.1			R	2019DIK
		397.95	l		-430.3	2.9	36.8	2.9			C	1963GAR
		397.95	l		-434.7	4.2	41.2	4.2			C	1944ROT
		287.05	sl	SEL					36.4	4.0	f(est)	2021CHE
CH ₃ NO ₂	nitromethane	374.34	l	SEL	-709.25	0.43	-112.97	0.43	38.26	0.10	TN	2021RUS
		374.34	g	SEL1	-747.51	0.46	-74.71	0.46			TN	2021RUS
		374.34	g		-749.9	3.5	-72.3	3.5			R	2019DIK
		374.34	l		-711.4	3.5	-110.8	3.5	38.46	0.30	R	2019DIK
		374.34	l		-712.51	0.55	-109.71	0.55			C	1973LEB
		374.34	l		-709.61	0.42	-112.61	0.42			C	1973LEB
		374.34	l		-702.91	1.26	-119.30	1.26			C	1971KNO
		374.34	l		-709.23	0.59	-112.99	0.59			C	1958CAS
		374.34	l		-733.25	0.75	-88.97	0.75	38.03	0.38	C	1949HOL
		374.34	l		-709.2		-113.0				C	1910SWI
		374.34	lg						37.20			1995CHI
		374.34	lg						38.37			1954MCC
		374.34	lg						38.30	0.10		1947JON
		374.34	lg						34.48	0.08		1926MAT
C ₉ H ₆ N ₂ O ₂	2,4-diisocyanatotoluene	524.83	l	SEL	-4235.0		-163.7		49.5		C	1962STR
		524.83	g	SEL1	-4284.5		-114.2				DvH	1962STR
		293.65	s	SEL2	-4168.0		-230.8				est	1985CHR
		524.83	lg	SEL					49.5	5.2	v	2019DIK
C ₆ H ₅ NO ₂	nitrobenzene	483.81	l	SEL	-3082.0	1.4	6.6	1.4	55.00	0.05	TN	2021RUS
		483.81	g	SEL1	-3137.0	1.4	61.6	1.4			TN	2021RUS
		483.81	g		-3142.4	3.2	67.0	3.2			R	2019DIK

	483.81	I		-3087.4	2.9	11.9	2.9	55.03	1.37	R	2019DIK
	483.81	I		-3088.08	0.42	12.66	0.42	56.03	0.40	C	1971LEB
	483.81	I		-3095.7		20.3				C	1921GAR
	483.81	lg						54.50	na		1995CHI
	483.81	lg						56.10	1.70		1971LEB
	483.81	lg						55.01	0.02		1971KUS
C ₇ H ₅ N ₃ O ₆ trinitrotoluene	355.10	s	SEL	-3406.0	3.0	-62.9	3.0			C	1976ROU
	621.00	g	SEL1	-3519.2	3.4	50.3	3.4			DsubH	1976ROU
	621.00	I	SEL2	-3493.0	3.6	24.1	3.6			DvH	1976ROU
	621.00	g		-3516.7	6.5	47.8	6.5			R	2019DIK
	621.00	g		-3493.0	3.5	24.1	3.5				1971LEN
	621.00	lg	SEL					87.00	1.90	v	1978CUN
	355.10	s		-3404.3	6.0	-64.6	6.0			R	2019DIK
	355.10	s		-3388.6	3.0	-80.3	3.0			C	1971LEN
	355.10	s		-3404.5	2.1	-64.4	2.1			C	1956YOU
	355.10	s		-3407.0		-61.9				C	1956HAN
	355.10	s		-3402.3	3.4	-66.6	3.4			C	1939BUR
	355.10	s		-3401.8	3.4	-67.1	3.4			C	1939BAD
	355.10	s		-3419.2		-49.7				C	1930RIN
	355.10	s		-3434.0		-34.9				C	1921GAR
	355.10	sg						112.4	2.6	R	2019DIK
	355.10	sg	SEL					113.20	1.50	s	1979KUD
	355.10	sg						104.60	1.70		1971LEN
	355.10	sg						118			1950EDW

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_f H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
SULFUR-SUBSTITUTED HYDROCARBONS													
Akanethiols													
CH ₄ S	methanethiol	279.05	g	SEL	-1544.57	0.51	-22.55	0.51			DvH		1961GOO
		279.05	l	SEL1	-1520.8	0.5	-46.35	0.5	23.81	0.08	R	a	1961GOO
		279.05	g		-1544.9	4.7	-22.2	4.7			R		2019DIK
		279.05	l		-1522.07	1.0	-45.04	1.01	23.51	0.94	R		2019DIK
		279.05	l		-1519.0	0.5	-46.7				C	e	1961GOO
		279.05	lg						23.80				1971WIL
C ₂ H ₆ S	ethanethiol	308.19	l	SEL	-2173.17	0.42	-73.68	0.42			C	a	1957MCC
		308.19	g	SEL1	-2200.3	0.43	-46.1	0.43			DvH		1957MCC
		308.19	g		-2199.7		-46.8	7.7			R		2019DIK
		308.19	l		-2172.5	7.7	-73.9	7.7	27.17	0.50	R		2019DIK
		308.19	l		-2172.00	0.42	-73.26	0.42	27.30	0.08	C	f	1957MCC
		308.19	l		-2164.0						C		1901BER
		308.19	lg						27.30				1971WIL
		308.19	lg						27.53				1957MCC
Sulfides													
CS ₂	carbon disulfide	318.35	l	SEL	-1687.2	0.5	89.71				C	a	1961GOO
		318.35	g	SEL1	-1714.5	0.5	117.1				DvH		1961GOO
		318.35	g		-1716.10	1.24	118.66	1.24			R		2019DIK
		318.35	g		-1112.1						C		1949GUE
		318.35	l		-1688.61	1.23	91.17	1.23	27.50	0.15	R		2019DIK
		318.35	l		-1684.6	0.5	89.4		27.66	0.08	C	f	1961GOO
		318.35	lg						27.52	0.13			1962WAD
		318.35	lg						26.78	0.01			1926MAT
C ₂ H ₆ S	dimethyl sulfide	310.42	l	SEL	-2181.45	0.33	-64.96				C	a	1957MCC
		310.42	g	SEL1	-2208.8	0.59	-37.57	0.59			DvH		1957MCC
		310.42	g		-2208.4	7.7	-38.1	7.7			R		2019DIK
		310.42	l		-2180.8	7.7	-65.7	7.7	27.59	0.25	R		2019DIK
		310.42	l		-2181.0	1.5	-65.4	1.5	27.90	0.60			1989VOR
		310.42	l		-2180.24	0.33	-65.44		27.82	0.04	C	f	1957MCC
		310.42	l		-2186.2		-60.2						1946DOU
		310.42	lg						27.50				1981SHI
		310.42	lg						27.70				1971WIL
C ₂ H ₆ OS	dimethyl sulfoxide	463.86	l	SEL	-2037.30	1.25	-203.40		52.90	0.83	C	g	1994MAS
		463.86	g	SEL1	-2095.9	1.5	-150.5	1.5			DvH		1994MAS

	463.86	g	-2095.8	1.23	-150.64	1.23			R	2019DIK
	463.86	l	-2043.1	1.03	-203.30	1.03	52.66	0.68	R	2019DIK
	463.86	lg					52.89	0.42		1948DOU

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_f H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
HALOGENATED HYDROCARBONS													
Fluoromethanes													
CH ₃ F	fluoromethane	194.65	g	SEL	-764.86	0.28	-235.49	0.23			TN	k	2021RUS
		194.65	l	SEL1	-756.33	1.15	-244.02	1.14			DvH		2021RUS
		194.65	g		-753.5		-246.9				R		1985LIA
		194.65	lg	SEL					8.53	1.12	v		2019DIK
CH ₂ F ₂	difluoromethane	221.50	g	SEL	-584.82	0.40	-450.76	0.33			TN	k	2021RUS
		221.50	l	SEL1	-566.84	0.43	-468.74	0.36	17.98	0.20	TN		2021RUS
		221.50	g	u	-582.5	2.3	-453.0	2.3			R		2019DIK
		221.50	l	u	-568.6	2.3	-467.0	2.3	13.95	0.39	R		2019DIK
		221.50	g		-583.3	2.6	-452.3	2.6			CC		2019PAU
		221.50	g		-583.08	0.92	-452.21	0.92			C	j	1958NEU
CHF ₃	trifluoromethane	188.75	g	SEL	-374.80		-696.00	0.41			TN	k	2021RUS
		188.75	l	SEL1	-366.34		-704.46	0.43	8.46	0.30	TN		2021RUS
		188.75	g	u	-390.5	2.7	-680.3	2.7			R		2019DIK
		188.75	l	u	-388.1	2.8	-682.7	2.7	2.42	0.32	R		2019DIK
		188.75	g		-372.5	2.7	-698.3	2.7			CC		2019PAU
		188.75	g		-374.2	2.8	-696.6	2.8			C	d	1958NEU
		188.75	g		-375.4	2.7	-695.4	2.7			C	b	1957NEU
CF ₄	tetrafluoromethane	145.56	g	SEL	-172.55		-933.47	0.25			TN	k	2021RUS
		145.56	l	SEL1	-160.74		-945.28	0.56			DvH		2021RUS
		145.56	g	u	-185.1	5.9	-920.9	5.9			D		1961BAI
		145.56	g	u	-189	10	-917	10			C		1960VOR
		145.56	g	u	-197.7	5.0	-908.3	5.0			C		1957NEU
		145.56	g	u	-197.7	5.0	-908.3	5.0			C		1956NEU
		145.56	g	u	-191	17	-915	17			F2		1955JES
		145.56	g	u	-184	17	-922	17			F2		1955JES
		145.56	g	u	-194	17	-912	17			K		1954KIR
		145.56	g	u	-163	17	-943	17			K		1954KIR
		145.56	g	u	-193.9		-912.1						1954DUU
		145.56	g		-171.0	2.9	-935.0	2.9			CC		2019PAU
		145.56	g		-172.6	2.0	-933.4	1.9			R		2019DIK
		145.56	g		-173.0		-933.0				R		1979PRI
		145.56	g		-172.8		-933.2				R		1972WAL
		145.56	g		-172.82		-933.20	0.75			F2		1968GRE
		145.56	g		-171.7	1.7	-934.3	1.7			F2		1967WOO

		145.56	g		-173.54		-932.49	0.17		F2	1967DOM
		145.56	g		-178.4		-927.6			D	1965DOM
		145.56	g		-162.0	2.7	-944.0	2.7		R	a 1965COX
		145.56	g		-173.13	1.34	-914.46	1.34		R	1965COX
		145.56	g		-183.0	5.9	-923.0	5.9			1962BAI
		145.56	g		-158	10	-949	10			1960VOR
		145.56	g		-168.0	5.9	-938.1	5.9			1956NEU
		145.56	g		-173.6	4.2	-913.4	4.2		H2O	1956GOO
		145.56	g		-163.0	5.0	-943.1	5.0		H2O	1956GOO
		145.56	g		-173.6	4.2	-913.4	4.2		H2O	1955SCO
		145.56	g		-181.95	0.00	-924.08	0.00			1955SCO
		145.56	lg	SEL					11.81	v	1969SMI
Fluoroethanes											
C ₂ H ₅ F	fluoroethane	236.05	g	SEL	-1407.52	0.40	-272.13	0.36		TN	2021RUS
		236.05	l	SEL1	-1391.21	0.47	-288.45	0.44		DvH	2021RUS
		236.05	lg	SEL					16.32 0.25	v	2019DIK
C ₂ H ₄ F ₂	1,1-difluoroethane	249.13	g	SEL	-1212.14	0.60	-502.74	0.55		DvH	2021RUS
		249.13	l	SEL1	-1193.66	0.88	-521.22	0.84		TN	2021RUS
		249.13	g	u	-1217	16	-498	16		R	2019DIK
		249.13	g	u	-1209	8	-505	8		G3B3	2016GOO
		249.13	g	u	-1219.6	8.4	-497.0	8.4		C	1968KOL
		249.13	l	u	-1199	16	-516	16	18.48 0.64	R	2019DIK
		249.13	g		-1210.9	2.1	-504.0	2.1		C	1968KOL
		249.13	l		-1210.3	2.6	-504.6	2.6		CC	2019PAU
	1,2-difluoroethane	293.00	g	SEL	-1267.04	0.84	-447.84	0.81		TN	2021RUS
		293.00	l	SEL1	-1245.91	0.98	-468.97	0.95	21.13 0.50	DvH	2021RUS
		293.00	g		-1267.3		-447.6			G3B3	2016GOO
		293.00	lg	SEL					21.13 0.50	v	2022BUR1
C ₂ H ₃ F ₃	1,1,1-trifluoroethane	225.91	g	SEL	-1008.3	1.7	-748.5	1.7		C	1965KOL
		225.91	l	SEL1u	-988.2	2.0	-761.9	1.9		DvH	1965KOL
		225.91	g	u	-994	8	-756	8		G3B3	2016GOO
		225.91	lg	SEL					13.43 0.97	v	2019DIK
		225.91	g		-996.2	2.7	-753.9	2.7		CC	2019PAU
		225.91	g		-1001.4	3.2	-748.7	3.2		E	1974WU
	1,1,2-trifluoroethane	276.81	g	SEL	-1059	10	-691	10		H2	1965KOL
		276.81	l	SEL1	-1035	10	-715	10		DvH	1965KOL
		276.81	g		-1080.8	2.7	-669.3	2.7		CC	2019PAU
		276.81	g		-1053.8	3.0	-696.3	3.0		E	1956LAC

C ₂ H ₂ F ₄	1,1,1,2-tetrafluoroethane	276.81	lg	SEL					24.28	0.45	v	2019DIK
		247.08	g	SEL	-883.7	2.0	-901.6	2.0			CBS	1988BER
		247.08	l	SEL1	-865.6	2.0	-919.7	2.0	18.10	0.16	DvH	1988BER
		247.08	lg	SEL					18.10	0.16	v	2019DIK
	1,1,2,2-tetrafluoroethane	253.23	g	SEL	-910.1	2.0	-875.2	2.0			CBS	1988BER
		253.23	l	SEL1	-890.3	2.3	-895.0	2.3			G3B3	2016GOO
		253.23	lg	SEL					19.79	1.13	v	2019DIK
C ₂ HF ₅	pentafluoroethane	225.06	g	SEL	-701	8	-1120	8			G3B3	2016GOO
		225.06	l	SEL1	-687	8	-1133	8			DvH	2016GOO
		225.06	g		-705.2	3.1	-1115.4	3.1			CC	2019PAU
		225.06	lg	SEL					13.44	0.16	v	2019DIK
C ₂ F ₆	hexafluoroethane	194.87	g	SEL	-513.33		-1342.44	0.97			TN	2021RUS
		194.87	l	SEL1	-506.33		-1349.44	1.09	7.00	0.50	DvH	2021RUS
		194.87	g	u	-588	8	-1268	8			R	2019DIK
		194.87	g	u	-524	8	-1331	8			E	1966SIN
		194.87	g		-510.0	3.3	-1345.8	3.3			CC	2019PAU
		194.87	g		-504.3		-1351.5				G3B3	2016GOO
		194.87	g		-511.9	5.0	-1343.9	5.0				1970WAL
		194.87	g		-525.3		-1330.5				E	1967COO
		194.87	g		-511.8	3.0	-1344.0	3.0			NF3	d 1966SIN
		194.87	g		-511.8	4.0	-1344.0	4.0			K	c 1954KIR
		194.87	lg	SEL					7.00	0.50	v	2022BUR1
Fluoroethenes												
C ₂ H ₃ F	fluoroethene	201.65	g	SEL	-1251.41	0.48	-142.42	0.45			TN	2021RUS
		201.65	l	SEL1	-1240.61	0.90	-153.22	0.88			DvH	2021RUS
C ₂ H ₃ F		201.65	g		-1250.2	2.6	-143.6	2.6			CC	2019PAU
C ₂ H ₃ F ₃		201.65	g		-1254.8	3.2	-139.0	3.2			R	2019DIK
C ₂ H ₃ F		201.65	g		-1256.0	1.7	-135.6	1.7			C	1970KOL
		201.65	g		-1255.1	1.9	-138.7	1.9			C	d 1970KOL
C ₂ H ₃ F ₃		201.65	l	SEL	-1244.0	3.3	-149.8	3.3	10.80	0.76	v	2019DIK
C ₂ H ₂ F ₂	1,1-difluoroethene	190.15	g	SEL	-1078.41	0.80	-350.64	0.77			TN	2021RUS
		190.15	l	SEL1	-1074.0	2.7	-355.1	2.7			DvH	2021RUS
		190.15	g	u	-1087	10	-344	10			C	1962KOL
		190.15	g		-1077.9	2.7	-351.2	2.7			CC	2019PAU
		190.15	g		-1094.9	3.4	-334.1	3.4			R	2019DIK
		190.15	g		-1104.0	3.4	-325.1	3.3			C	1957NEU
		190.15	g		-1097.0	3.4	-334.0	3.3			C	1956NEU
		190.15	lg	SEL	-1090.5	4.3	-338.6	4.3	4.4	2.6	v	2019DIK

1,2-difluoroethene, (E)-	231.15	g	SEL	-1123.0	0.9	-306.1	0.9	14.90	1.04	TN	2021RUS	
	231.15	l	SEL1	-1108.05	1.39	-321.00	1.38			DvH	2021RUS	
	231.15	lg	SEL							v	2019DIK	
1,2-difluoroethene, (Z)-		g	SEL	-1119.4	0.9	-309.7	0.9	13.79	0.94	TN	2021RUS	
	253.15	l	SEL1	-1105.6	1.3	-323.4	1.3			DvH	2021RUS	
	253.15	lg	SEL							v	2019DIK	
C ₂ HF ₃	trifluoroethene	211.70	g	SEL	-966.2	1.6	-498.1	1.6	11.37	0.92	TN	2021RUS
		211.70	l	SEL1	-954.8	1.9	-509.5	1.8			DvH	2021RUS
		211.70	g	u	-971	21	-493	21			R	2019DIK
		211.70	g	u	-977	8	-474	8			C	1962KOL
		211.70	g		-965.5	2.8	-498.8	2.8			CC	2019PAU
		211.70	g		-977.9	3.9	-486.4	3.9			C	d 1962KOL
		211.70	lg	SEL							v	2019DIK
C ₂ F ₄	tetrafluoroethene	197.51	g	SEL	-825.10	0.61	-674.40	0.52	6.10	1.30	TN	2021RUS
		197.51	l	SEL1	-819.00	1.44	-680.50	1.40			DvH	2021RUS
		197.51	g	u	-850	23	-650	23			R	2019DIK
		197.51	lg	SEL	-844	23	-656	23			v	2019DIK
		197.51	g		-826.3	3.0	-673.2	3.0			CC	2019PAU
		197.51	g		-839.4	5.5	-660.1	5.5			Na	d 1962KOL2
		197.51	g		-838.8	3.4	-660.7	3.3			Na	1962KOL
		197.51	g		-842.4	4.8	-657.1	4.8			H2	d 1956NEU
		197.51	g		-840.0	2.0	-659.5	2.0			D	d 1956NEU
		197.51	g		-838.8	3.4	-660.7	3.3			H2	1956NEU
		197.51	g		-787	10	-713	10			K	d 1955WAR
		197.51	g		-813.3		-686.2				K	1955WAR
		197.51	g		-823	7	-676	7			D	d 1955DUU
		197.51	g		-841.7	5.7	-657.8	5.7			H2	d 1955DUU
		197.51	g		-796	8	-704	8			K	d 1954KIR
		197.51	g		-821.7	4.2	-677.8	4.2			K	1954KIR
Chloromethanes												
CH ₃ Cl	chloromethane	249.00	g	SEL	-763.51	0.39	-82.24	0.39	19.90	0.30	TN	2021RUS
		249.00	l	SEL1	-743.61	0.25	-102.14	0.25			TN	2021RUS
		249.00	g		-763.81	0.63	-81.94	0.63			R	2019DIK
		249.00	g		-763.9	1.5	-81.9	1.5			R	2002MAN
		249.00	g		-763.8	0.5	-82.0	0.5			C	1971FLE
		249.00	g		-759.7	0.6	-86.1	0.6			H2	b 1956LAC
		249.00	g		-759.86	0.59	-85.90	0.59			H2	1956LAC
		249.00	g		-760.4	0.6	-85.4	0.6			H2	b 1965FOW

CH ₂ Cl ₂	dichloromethane	249.00	l		-743.4	1.5	-102.4	1.5	20.50	0.30	R	2002MAN
		312.92	g	SEL	-633.00	0.52	-93.38	0.52			TN	2021RUS
		312.92	l	SEL1	-603.99	0.54	-122.39	0.54	29.01	0.10	TN	2021RUS
		312.92	g		-631.0	0.4	-95.4	0.4			R	2019DIK
		312.92	g		-631.3	2.5	-95.1	2.5			R	2002MAN
		312.92	g		-602.5	0.8	-95.1	0.8			C	c 1969HU
		312.92	g		-630.4	1.3	-96.0	1.3			H2	1967LAC
		312.92	g		-630.69	1.34	-95.69	1.34			H2	b 1967LAC
		312.92	g		-634.6	8.4	-91.8	8.4			C	c 1938EFT
		312.92	l		-602.13	0.38	-124.25	0.38	28.88	0.13	R	2019DIK
		312.92	l		-602.3	2.5	-124.1	2.5	29.03	0.08	R	2002MAN
		312.92	l		-602.5		-124.3				C	1969HU
		312.92	l		-605.8	8.4	-120.6	8.4			C	b 1938EFT
		312.92	l		-604.6	8.4	-121.8	8.4			C	1938EFT
		312.92	lg						30.60	0.10		1989AN
		312.92	lg						28.80			1980HUG
		312.92	lg						28.45	0.42		1926MAT
CHCl ₃	trichloromethane	334.42	g	SEL	-504.94	0.50	-102.07	0.50			TN	2021RUS
		334.42	l	SEL1	-473.54	0.51	-133.47	0.51	31.40	0.10	TN	2021RUS
		334.42	g		-503.8	1.7	-103.2	1.7			R	2019DIK
		334.42	g		-504.1	2.5	-102.9	2.5			R	2002MAN
		334.42	g		-502.0	1.4	-105.0	1.4			Br2	c 1973MEN
		334.42	g		-473.2		-96.2				C	c 1969HU
		334.42	g		-473.2	0.8	-102.9	0.8			C	c 1956KIR
		334.42	g		-504.9	8.4	-102.1	8.4			C	c 1938EFT
		334.42	l		-472.7	1.7	-134.3	1.7	31.12	0.31	R	2019DIK
		334.42	l		-472.9	2.5	-134.1	2.5	31.32	0.08	R	2002MAN
		334.42	l		-473.2	0.8	-134.3	0.8			C	1969HU
		334.42	l		-474.0	8.4	-133.0	8.4			C	b 1938EFT
		334.42	l		-473.4	8.4	-133.7	8.4			C	1938EFT
		334.42	lg						31.10			1980HUG
		334.42	lg						30.54	0.42		1926MAT
CCl ₄	tetrachloromethane	349.65	g	SEL	-392.02	0.43	-95.61	0.43			TN	2021RUS
		349.65	l	SEL1	-359.52	0.43	-128.11	0.43	32.5		TN	2021RUS
		349.65	g		-391.69	1.31	-95.94	1.31			R	2019DIK
		349.65	g		-392.0	2.5	-95.6	2.5			R	2002MAN
		349.65	g		-393.9	1.7	-93.7	1.7			Br2	1973MEN
		349.65	g		-373.0	2.9	-114.6	2.9			D	1969LOR

		349.65	g		-362.9	4.6	-124.7	4.6		D	1969LOR
		349.65	g		-385	8	-103	8		D	1961BAI
		349.65	g		-394.1	1.4	-93.5	1.4		Br2	c 1937MEN
		349.65	l		-359.24	1.29	-128.39	1.29	32.45 0.22	R	2019DIK
		349.65	l		-359.5	2.5	-128.1	2.5	32.55 0.07	R	2002MAN
		349.65	l		-359.9		-128.4			C	1969HU
		349.65	l		-365.7	8.4	-121.9	8.4		C	b 1938EFT
		349.65	l		-365.2	8.4	-122.4	8.4		C	1938EFT
		349.65	lg						32.40		1980HUG
		349.65	lg						32.54 0.10		1973KON
		349.65	lg						32.43 0.01		1966WAD
		349.65	lg						32.41 0.01		1959HIL1
		349.65	lg						29.96 0.01		1926MAT
Chloroethanes											
C ₂ H ₅ Cl	chloroethane	285.40	g	SEL	-1413.57	0.26	-111.49	0.26		TN	2021RUS
		285.40	l	SEL1	-1389.14	0.40	-135.92	0.40	24.43 0.30	TN	2021RUS
		285.40	g	u	-1427	10	-100	10		C	1951CAS
		285.40	g		-1412.78	0.64	-112.28	0.64		R	2019DIK
		285.40	g		-1413.0	0.7	-112.1	0.7		R	2002MAN
		285.40	g		-1413.06	0.59	-112.26	0.59		C	1971FLE
		285.40	g		-1418.2	0.4	-106.9	0.4		H2	b 1956LAC
		285.40	g		-1417.36	0.59	-107.70	0.59		H2	1956LAC
		285.40	g		-1415.9	1.7	-109.2	1.7		HCl	b 1955HOW
		285.40	g		-1413.6	2.0	-111.5	2.0		HCl	1955HOW
		285.40	g		-1413.2	3.0	-111.9	3.0		HCl	1955HOW
		285.40	g		-1412.7	3.0	-112.4	3.0		HCl	1955HOW
		285.40	g		-1413.3		-111.7			HCl	1953LAN
		285.40	l		-1389.01	0.66	-136.05	0.66	23.77 0.15	R	2019DIK
		285.40	l		-1388.4	1.0	-136.7	1.0	24.60 0.30	R	2002MAN
C ₂ H ₄ Cl ₂	1,1-dichloroethane	330.42	g	SEL	-1272.61	0.44	-133.07	0.44		TN	2021RUS
		330.42	l	SEL1	-1241.81	0.48	-163.87	0.48	30.8	TN	2021RUS
		330.42	g		-1273.2	3.5	-132.5	3.5		R	2002MAN
		330.42	g		-1274.2	3.5	-131.5	3.5		HCl	c 1976LEV
		330.42	g		-1273.2	3.5	-132.5	3.5		HCl	c 1976LEV
		330.42	g		-1272.5	3.5	-133.2	3.5		HCl	c 1976LEV
		330.42	g		-1278.07	1.13	-127.61	1.13		H2	1967LAC
		330.42	l		-1242.4	3.5	-163.3	3.5	30.83 0.08	R	2002MAN
		330.42	l		-1246.8	8.4	-158.9	8.4		C	b 1938EFT

		330.42	l		-1244.6	8.4	-161.1	8.4			C	1938EFT
		330.42	lg						30.73	0.15	R	2019DIK
		330.42	lg						30.62	0.14		1972LAY
1,2-dichloroethane		356.70	l	SEL	-1238.5	3.5	-167.2	3.5	35.21	0.05	R	2002MAN
		356.70	g	SEL1	-1273.7	3.5	-132.0	3.5			R	2002MAN
		356.70	g		-1271.2	2.3	-134.5	2.3			R	2019DIK
		356.70	g		-1280.2	1.0	-125.4	1.0			H2	b 1967LAC
		356.70	g		-1279.4	1.0	-126.3	1.0			H2	1967LAC
		356.70	g		-1276.5	1.7	-129.2	1.7			C	c 1958SIN
		356.70	g		-1276.8	4.5	-128.9	4.5			Cl2	1956KIR
		356.70	g		-1275.3	0.8	-130.4	0.8			Cl2	c 1938CON
		356.70	l		-1236.0	2.2	-169.7	2.2	35.14	0.25	R	2019DIK
		356.70	l		-1236.4		-169.7				C	1969HU
		356.70	l		-1246.4	8.4	-159.3	8.4			C	b 1938EFT
		356.70	l		-1243.9	8.4	-161.8	8.4			C	1938EFT
		356.70	lg						35.20	0.40		2007VAR
		356.70	lg						34.40			1994CAR
		356.70	lg						35.10	0.10		1989AN
		356.70	lg						35.20	0.10		1980HUG
		356.70	lg						35.15	0.01		1968WAD
		356.70	lg						35.44	0.08		1926MAT
C ₂ H ₃ Cl ₃	1,1,1-trichloroethane	347.22	l	SEL	-1109.1	2.0	-177.2	2.0	32.59	0.07	R	2002MAN
		347.22	g	SEL1	-1141.7	2.0	-144.6	2.0			R	2002MAN
		347.22	g		-1143.9	5.3	-142.4	5.3			R	2019DIK
		347.22	g		-1141.7	1.6	-144.6	1.6			HCl	c 1972HU
		347.22	g		-1140.31	0.84	-146.00	0.84			HCl	1972HU
		347.22	g		-1148.1	1.0	-138.2	1.0			HCl	1969HU
		347.22	l		-1111.6	5.3	-174.7	5.3	32.31	0.22	R	2019DIK
		347.22	l		-1111.8		-174.5				HCl	1975LEV
		347.22	l		-1107.73	0.80	-178.57	0.79			C	1972HU
		347.22	l		-1111.51	1.28	-174.80	1.28	32.47	0.08	C	1971MAN
		347.22	lg						32.50	0.10		1980HUG
		347.22	lg						32.47	0.06		1972LAY
1,1,2-trichloroethane		386.78	l	SEL	-1098.0	4.0	-188.3	4.0	40.30	0.07	R	2002MAN
		386.78	g	SEL1	-1138.3	4.0	-148.0	4.0			R	2002MAN
		386.78	g		-1137.7	8.2	-148.6	8.2			R	2019DIK
		386.78	g		-1139.1	4.0	-147.2	4.0			HCl	c 1976LEV
		386.78	g		-1148.0		-138.3				C	c 1969STU

		386.78	g		-1135.4	4.5	-150.9	4.5			C	c	1956KIR
		386.78	l		-1097.5	8.2	-188.8	8.2	40.23	0.23	R		2019DIK
		386.78	l		-1097.8	4.4	-188.5	4.4			C		1987PAP
		386.78	lg						40.10	0.60			2007VAR
		386.78	lg						40.20	0.10			1980HUG
		386.78	lg						40.28	0.06			1972LAY
C ₂ H ₂ Cl ₄	1,1,1,2-tetrachloroethane	403.33	l	SEL	-973.5	2.3	-193.4	2.3	41.10	0.50	R		2002MAN
		403.33	g	SEL1	-1014.6	2.4	-152.3	2.4			R		2002MAN
		403.33	g		-1014.14	1.13	-152.79	1.13			R		2019DIK
		403.33	g		-1191.5		24.6						1984PLA
		403.33	g		-1017.83	0.40	-149.10	0.40			HCl	c	1976LBR
		403.33	l		-973.10	1.10	-193.83	1.10			R		2019DIK
		403.33	l		-973.9	1.3	-183.5	1.4			C		1978GUN
		403.33	lg						54.80				1994SPI
		403.33	lg						47.80				1984BOU
	1,1,2,2-tetrachloroethane	419.25	l	SEL	-964.5	3.5	-202.4	3.5	45.73	0.09	R		2002MAN
		419.25	g	SEL1	-1010.2	3.5	-156.7	3.5			R		2002MAN
		419.25	g	u	-1011	8	-156	8			Cl ₂		1956KIR
		419.25	g		-1008.2		-158.7				CaO ₂	c	1956KIR
		419.25	l		-972.8	8.4	-194.1	8.4			C	b	1938EFT
		419.25	l		-971.2	8.4	-195.7	8.4			C		1938EFT
		419.25	lg						45.58	1.14	R		2019DIK
		419.25	lg						45.78	0.16			1972LAY
		419.25	lg						45.19	1.26			1926MAT
C ₂ HCl ₅	pentachloroethane	433.45	l	SEL	-844.3	4.0	-203.3	4.0	47.40	1.50	R		2002MAN
		433.45	g	SEL1	-891.7	4.3	-155.9	4.3			R		2002MAN
		433.45	g		-892.6		-155.0				HCl	c	1979LEV
		433.45	g		-887.6		-160.0				HCl	c	1979LEV
		433.45	g		-902.4		-145.2				Cl ₂		1956KIR
		433.45	g		-891.8		-155.8				CaO ₂	b	1956KIR
		433.45	g		-891.7		-155.9				Cl ₂	b	1956KIR
		433.45	l		-862.3	8.4	-185.3	8.4			C	b	1938EFT
		433.45	l		-861.4	8.4	-186.2	8.4			C		1938EFT
		433.45	lg						43.9	2.4	R		2019DIK
C ₂ Cl ₆	hexachloroethane	462.96	l	SEL	-711.7	2.8	-216.5	2.8	67.1		TN		2021RUS
		462.96	g	SEL1	-778.8	1.3	-149.4	1.3			TN		2021RUS
		462.96	g	u	-784	28	-144	28			R		2019DIK
		462.96	g	u	-780.0	5.7	-148.2	5.7			R		2002MAN

		462.96	g	u	-785	9	-143	9			D	1963PUY
		462.96	g	u	-778.9		-149.3				D	c 1963PUY
		462.96	l	u	-739	29	-189	29	45		R	2019DIK
		462.96	l	u	-729.0	6.1	-199.2	6.1	51.0	2.3	R	2002MAN
		462.96	g		-783.7		-144.5				D	c 1996HUY
		462.96	g		-781.8		-146.4				Cl2	c 1996HUY
		462.96	g		-767.5		-160.7				D	c 1979BUS
		462.96	g		-764.9		-163.3				D	c 1979BUS
		462.96	g		-777.7		-150.5				D	c 1956KIR
		462.96	g		-779.1		-149.1				D	c 1950DAI
		462.96	l		-727.4	8.4	-200.7	8.4			C	1938EFT
		462.96	sg						69.0	2.1		1974CHA
Chloroethenes												
C ₂ H ₃ Cl	chloroethene	259.35	l	SEL	-1241.13	0.36	1.90	0.36	19.80	0.20	TN	2021RUS
		259.35	g	SEL1	-1260.93	0.30	21.70	0.30			TN	2021RUS
		259.35	g		-1261.2	3.0	22.0	3.0			R	2002MAN
		259.35	g		-1268.5	0.7	29.3	0.7			HCl	1976LEV
		259.35	g		-1261.9	0.7	22.7	0.7			HCl	c 1976LEV
		259.35	g		-1261.2	0.7	22.0	0.7			HCl	c 1976LEV
		259.35	g		-1260.2	0.7	21.0	0.7			HCl	c 1976LEV
		259.35	g		-1260.2		20.9				E	1973ALF
		259.35	g		-1274.5	1.4	35.3	1.4			HCl	1962LAC
		259.35	g		-1273.8	1.5	34.6	1.5			HCl	c 1962LAC
		259.35	g		-1262.0	2.2	22.8	2.2			C	c 1958SIN
		259.35	g		-1276.8	1.7	37.6	1.7			H2	c 1956LAC
		259.35	g		-1277.30	0.84	38.07	0.84			H2	1956LAC
		259.35	l		-1240.1	3.2	0.9	3.2	21.10	1.00	R	2002MAN
		259.35	lg						19.96	0.17	R	2019DIK
C ₂ H ₂ Cl ₂	1,1-dichloroethene	304.71	l	SEL	-1096.41	0.57	-23.45	0.57	26.75	0.15	TN	2021RUS
		304.71	g	SEL1	-1123.16	0.56	3.30	0.56			TN	2021RUS
		304.71	g		-1121.75	0.74	1.89	0.74			R	2019DIK
		304.71	g		-1122.3	2.0	2.4	2.0			R	2002MAN
		304.71	g		-1121.4	2.1	1.5	2.1			HCl	c 1975LEV
		304.71	g		-1117.2	2.1	-2.7	2.1			HCl	c 1975LEV
		304.71	g		-1122	8	2	8			HCl	c 1972HU
		304.71	g		-1122.5	1.4	2.6	1.4			C	c 1971MAN
		304.71	g		-1122.8	1.3	2.9	1.3			C	b 1958SIN
		304.71	l		-1095.39	0.73	-24.47	0.73	26.36	0.17	R	2019DIK

		304.71	l		-1095.6	2.0	-24.3	2.0	26.74	0.09	R	2002MAN
		304.71	l		-1096.0	1.4	-24.3	1.4	26.48	0.08	C	1971MAN
		304.71	l		-1095.9		-25.1				C	1958SIN
		304.71	lg						26.48	0.08		1959HIL
1,2-dichloroethene, (E)-		320.73	l	SEL	-1119.74	0.54	-0.12	0.54			DvH	2021RUS
		320.73	g	SEL1	-1149.24	1.14	29.38	1.14			TN	2021RUS
		320.73	g		-1119.4	2.0	-0.5	2.0			R	2002MAN
		320.73	g		-1121.5		1.7				E	1974ROZ
		320.73	g		-1119.6		-0.3				E	c 1974ROZ
		320.73	g		-1119.4		-0.5				E	c 1974ROZ
		320.73	l		-1089.9	2.2	-30.0	2.2	29.50	1.00	R	2002MAN
		320.73	l		-1095.8	8.4	-24.1	8.4			C	b 1938EFT
		320.73	l		-1094.7	8.4	-25.2	8.4			C	1938EFT
		320.73	lg						29.01	0.47	R	2019DIK
		320.73	lg	SEL					29.50	1.00	v	2002MAN
		320.73	lg						29.47			1947KET
1,2-dichloroethene, (Z)-		333.58	l	SEL	-1086.23	1.14	-33.63	1.14			TN	2021RUS
		333.58	g	SEL1	-1117.33	0.54	-2.53	0.54			TN	2021RUS
		333.58	g		-1116.9	2.0	-3.0	2.0			R	2002MAN
		333.58	g		-1121.6	2.1	1.7	2.1			HCl	c 1975LEV
		333.58	g		-1116.3	2.1	-3.6	2.1			HCl	c 1975LEV
		333.58	g		-1124.1		4.3				E	1974ROZ
		333.58	g		-1116.7		-3.2				E	c 1974ROZ
		333.58	g		-1117.5		-2.4				E	c 1956KIR
		333.58	l		-1085.8	2.2	-34.1	2.2	31.10	1.00	R	2002MAN
		333.58	l		-1091.4		-28.5				Cl2	1956KIR
		333.58	l		-1093.7	8.4	-26.2	8.4			C	b 1938EFT
		333.58	l		-1092.3	8.4	-27.6	8.4			C	1938EFT
		333.58	lg						31.36	0.28	R	2019DIK
		333.58	lg						30.9			1947KET
C ₂ HCl ₃ trichloroethene		360.50	l	SEL	-951.8	1.5	-48.7	1.5	34.60	0.15	TN	2021RUS
		360.50	g	SEL1	-986.4	1.5	-14.1	1.5			TN	2021RUS
		360.50	g		-981.6	4.0	-18.9	4.0			R	2019DIK
		360.50	g		-983.0	2.0	-17.5	2.0			R	2002MAN
		360.50	g		-994.6	2.9	-5.9	2.9			Cl2	1956KIR
		360.50	g		-982.0	2.9	-18.5	2.9			Cl2	c 1956KIR
		360.50	l		-947.0	4.0	-53.5	4.0	34.56	0.26	R	2019DIK
		360.50	l		-948.4	3.0	-52.1	3.0	34.57	0.09	R	2002MAN

		360.50	l		-947.7	2.9	-53.1	2.9	33.97	0.13	C	1985PAP
		360.50	l		-962.4	8.4	-1000.5	8.4			C	1938EFT
		360.50	l		-956.5	8.4	-1000.5	8.4			C	b 1938EFT
		360.50	lg									1985PAP
		360.50	lg						34.50	0.10		1980HUG
		360.50	lg						34.73	0.42		1926MAT
C ₂ Cl ₄	tetrachloroethene	394.29	l	SEL	-818.2	1.1	-62.9	1.1	39.70	0.10	TN	2021RUS
		394.29	g	SEL1	-857.9	1.1	-23.2	1.1			TN	2021RUS
		394.29	g		-857.1	4.0	-24.0	4.0			R	2002MAN
		394.29	g		-855.2		-25.9				Cl ₂	c 1996HUY
		394.29	g		-853.3		-27.8				D	c 1996HUY
		394.29	g		-856.0		-25.1				HCl	c 1979LEV
		394.29	g		-861.3		-19.8				HCl	c 1979LEV
		394.29	g		-872.4		-8.7				HCl	c 1979BUS
		394.29	g		-869.4		-11.7				HCl	c 1979BUS
		394.29	g		-859.5		-21.6				HCl	c 1963PUY
		394.29	g		-858.9		-22.2				Cl ₂	c 1956KIR
		394.29	g		-856.8		-24.3				CaO ₂	b 1956KIR
		394.29	g		-857.8		-23.3				D	c 1950DAI
		394.29	g		-831.0	8.4	-50.2	8.4			C	1938EFT
		394.29	l		-817.1	4.0	-64.0	4.0	39.72	0.05	R	2002MAN
		394.29	lg						39.75	0.24	R	2019DIK
		394.29	lg						39.70	0.10		1980HUG
		394.29	lg						39.75	0.84		1926MAT
Chloroethynes												
C ₂ HCl	chloroethyne	243.6	g	SEL	-1182.39	0.70	228.99	0.70			TN	2021RUS
		243.6	l	SEL1	-1162.4	1.2	209.0	1.2			DvH	2021RUS
		243.6	g			10	226	10				2002MAN
		243.6	l			10	207	10			DvH	2002MAN
		243.6	lg						32.6	1.7		2019DIK
		243.6	lg	SEL					20.0	1.0	v	2002MAN
C ₂ Cl ₂	dichloroethyne	306.00	g		-1067.47	0.94	233.44	0.94			TN	2021RUS
		306.00	g		-1061	14	227	14			R	2002MAN
		306.00	l		-1040.1	1.5	206.0	1.5			DvH	2021RUS
		306.00	l		-1033	14	199	14			DvH	2002MAN
		306.00	lg	SEL					27.4	1.2	v	2002MAN
Bromomethanes												
CH ₃ Br	bromomethane	276.70	g	SEL	-763.80	0.26	-35.61	0.26			TN	2021RUS

		276.70	l	SEL1	-740.47	0.27	-58.94	0.27		TN	2021RUS
		276.70	g		-765.1	0.8	-34.3	0.8		Br2	1973FER
		276.70	g		-761.8	1.3	-37.7	1.3		LiAl	1966ADA
		276.70	g		-761.9	1.5	-37.5	1.5		H2	1965FOW
		276.70	l		-738.8	1.3	-60.6	1.3	23.43	LiAl	1966ADA
		276.70	lg						23.09	R	2019DIK
		276.70	lg						23.91		1938EG
CH ₂ Br ₂	dibromomethane	370.10	l	SEL	-601.3	1.1	-32.4	1.1		TN	2021RUS
		370.10	g	SEL1	-638.3	1.1	4.6	1.1		TN	2021RUS
		370.10	lg						36.91	R	2019DIK
		370.10	lg						36.97		1972LAY
CHBr ₃	tribromomethane	422.36	l	SEL	-471.5	1.3	3.5	1.3	46.10	TN	2021RUS
		422.36	g	SEL1	-517.6	1.3	49.6	1.3		TN	2021RUS
		422.36	l		-477.4	3.3	9.4	3.3	44.42	C	1982PAP
		422.36	lg							R	2019DIK
		422.36	lg						46.05		1972LAY
CB ₄	tetrabromomethane	462.55	l	SEL	-351.6	1.4	49.3	1.4		TN	2021RUS
		462.55	g	SEL1	-406.1	1.3	103.8	1.3		TN	2021RUS
		462.55	g		-386.2	3.4	83.9	3.4		DsubH	1984BIC
		462.55	s		-331.6	1.1	29.4	1.1		R	2019DIK
		367.60	s		-331.7	3.3	29.4	3.4		C	1984BIC
		462.55	sg	SEL					54.50	s	1984BIC
		365.00	sg						54.40		1984BIC
Bromomethanes											
C ₂ H ₅ Br	bromoethane	311.54	l	SEL	-1387.66	0.26	-91.05	0.26		TN	2021RUS
		311.54	g	SEL1	-1415.65	0.25	-63.06	0.25		TN	2021RUS
		311.54	g	u	-1413.4	6.3	-65.3	6.3		LiAl	1965ASH
		311.54	g		-1415.1		-63.6			R	1979KUD
		311.54	g		-1416.8	1.0	-61.9	1.0		H2	1965FOW
		311.54	g		-1414.1	2.1	-64.6	2.1		HBr	1953LAN
		311.54	l		-1383.2	2.1	-95.5	2.1		LiAl	b 1965ASH
		311.54	l		-1382.5	6.3	-96.2	6.3		LiAl	1965ASH
		311.54	lg						27.50	R	2019DIK
		311.54	lg						27.61		1953LAN
Bromoethenes											
C ₂ H ₃ Br	bromoethene	288.60	g	SEL	-1266.79	0.57	73.90	0.57		TN	2021RUS
		288.60	l	SEL1	-1233.0	1.8	40.1	1.8		DvH	2021RUS

		288.60	g		-1272.1	1.9	79.2	1.9		H2	b	1957LAC
		288.60	g		-1271.1	1.9	78.2	1.9		H2		1957LAC
		288.60	lg	SEL					33.76 1.74	v		2019DIK
Chlorofluoromethanes												
CH ₂ ClF	chlorofluoromethane	264.00	g	SEL	-617.7	1.1	-263.3	1.1		TN		2021RUS
		264.00	l	SEL1	-599.1	1.5	-281.9	1.5		DvH		2021RUS
		264.00	g		-619.1	13.0	-261.9	13.0		R		1998CHA
		264.00	g		-657.0		-224.0			R		1989DAU
		264.00	lg	SEL					18.62 0.96	v		2019DIK
CHClF ₂	chlorodifluoromethane	232.35	g	SEL	-433.99	1.01	-482.21	0.98		TN		2021RUS
		232.35	l	SEL1	-418.29	1.05	-497.91	1.03		DvH		2021RUS
		232.35	g		-433.4	2.9	-482.8	2.9		D		1965EDW
		232.35	lg	SEL					15.70 0.24	v		2019DIK
CHCl ₂ F	dichlorofluoromethane	282.05	g	SEL	-478.01	0.92	-283.59	0.91		TN		2021RUS
		282.05	l	SEL1	-454.4	1.5	-307.2	1.5		DvH		2021RUS
		282.05	g		-478.3	13.0	-283.3	13.0		R		1998CHA
		282.05	lg	SEL					23.64 1.18	v		2019DIK
CClF ₃	chlorotrifluoromethane	191.75	g	SEL	-241.78	0.63	-709.65	0.57		TN		2021RUS
		191.75	l	SEL1	-225.28	0.70	-726.15	0.64		TN		2021RUS
		191.75	g	u	-242	28	-710	28		R		2019DIK
		191.75	g	u	-232.4	4.0	-719.0	4.0		H2		1963KOL
		191.75	g	u	-211.9	5.0	-739.5	5.0		H2	b	1963KOL
		191.75	g	u	-256	9	-695	9		D		1961BAI
		191.75	l	u	-238	28	-714	28	4.3 2.9	R		2019DIK
		191.75	g		-250.6	3.4	-700.8	3.4		I2		1967LOR
		191.75	g		-246.4	3.4	-705.0	3.4		I2	b	1967LOR
		191.75	g		-242.7	4.2	-708.8	4.2		Br2		1967COO
		191.75	g		-252.7		-698.7					1955WAR
		191.75	g		-236.0	4.2	-715.5	4.2		C		1954KIR
		191.75	lg						16.50			1987STE
CCl ₂ F ₂	dichlorodifluoromethane	243.35	g	SEL	-303.18	0.71	-493.65	0.67		TN		2021RUS
		243.35	l	SEL1	-286.38	0.73	-510.45	0.70		DvH		2021RUS
		243.35	g	u	-327.8	5.0	-469.0	5.0		H2		1963KOL
		243.35	g	u	-319.2	5.6	-477.6	5.6		H2	b	1963KOL
		243.35	g	u	-324	13	-473	13		C		1955WAR
		243.35	g	u	-320	13	-477	13		C	b	1955WAR
		243.35	g	u	-328.2	8.4	-468.6	8.4		K		1954KIR
		243.35	l	u	-324	13	-473	13		R		2019DIK

CCl ₃ F	trichlorofluoromethane	243.35	lg	SEL					16.80	0.23	v	2019DIK
		296.95	g	SEL	-353.80	0.80	-288.43	0.78			TN	2021RUS
		296.95	l	SEL1	-329.01	0.81	-313.22	0.80			TN	2021RUS
		296.95	g	u	-369	27	-273	27			R	2019DIK
		296.95	g	u	-364.4	8.8	-277.8	8.8			D	1961BAI
		296.95	g	u	-373.9	8.4	-268.3	8.4			C	b 1955WAR
		296.95	g	u	-362.2	8.4	-280.0	8.4			C	1955WAR
		296.95	g	u	-349	17	-293	17			C	1954KIR
		296.95	l	u	-345	27	-298	27	24.79	0.16	R	2019DIK
Chlorofluoroethenes												
C ₂ ClF ₃	chlorotrifluoroethene	244.80	g	SEL	-833.2	2.8	-511.7	4.7			DvH	1982ERA
		244.80	l	SEL1	-816.0	2.8	-528.9	2.8	17.2	3.8	C	1982ERA
		244.80	g	u	-814	45	-531	45			R	2019DIK
		244.80	g	u	-843	13	-502	13			C	1955WAR
		244.80	g	u	-835	13	-510	13			C	b 1955WAR
		244.80	g	u	-817.7	8.4	-527.2	8.4			C	1954KIR
		244.80	l	u	-812.1	5.9	-532.8	5.9			R	2019DIK
		244.80	g		-800.1	5.4	-544.8	5.4			H2	1963KOL
		244.80	g		-780.1	5.8	-564.8	5.8			H2	b 1963KOL
244.80	lg						17.13	0.50	R	2019DIK		
Bromofluoroalkanes												
CH ₂ BrF	bromofluoromethane	310.00	l	SEL	-591.3	5.2	-243.4	5.2			DvH	2021RUS
		310.00	g	SEL1	-622.7	4.9	-211.9	4.9			TN	2021RUS
CHBrF ₂	bromodifluoromethane	310.00	lg	SEL					31.47	1.72	v	2019DIK
		258.65	g	SEL	-445.5	1.2	-424.3	1.2			TN	2021RUS
		258.65	l	SEL1	-425.5	1.6	-444.4	1.5			DvH	2021RUS
		258.65	g		-444.6	0.9	-425.3	0.9			Br2	1974OKA
		258.65	lg	SEL					20.02	0.96	v	2019DIK
CHBr ₂ F	dibromofluoromethane	338.05	l	SEL	-456.9	5.2	-212.0	5.2			DvH	2021RUS
		338.05	g	SEL1	-489.6	4.9	-179.3	4.9			TN	2021RUS
		338.05	lg	SEL					32.73	1.81	v	2019DIK
CBrF ₃	bromotrifluoromethane	215.35	g	SEL	-254.08	0.52	-651.00	0.44			TN	2021RUS
		215.35	l	SEL1	-242.31	0.53	-662.78	0.46			TN	2021RUS
		215.35	g		-255.6	3.2	-649.4	3.2			I2	1967LOR
		215.35	g		-257.7	2.9	-647.3	2.9			Br2	1967COO
		215.35	lg						11.78	0.12	R	2019DIK
CBr ₂ F ₂	dibromodifluoromethane	297.65	g	SEL	-322.1	1.3	-382.0	1.3			TN	2021RUS

CBr ₃ F	tribromofluoromethane	297.65	l	SEL1	-297.7	1.3	-406.4	1.3	24.42	0.19	DvH	2021RUS
		297.65	lg	SEL							v	2019DIK
		379.65	l	SEL	-324.4	3.0	-178.9	3.0	49.4	2.6	DvH	2021RUS
		379.65	g	SEL1	-373.7	1.5	-129.5	1.5			TN	2021RUS
		379.65	lg	SEL							v	2019DIK
Bromochloromethanes												
CH ₂ BrCl	bromochloromethane	341.21	l	SEL	-605.1	1.3	-74.9	1.3	32.75	0.18	DvH	2021RUS
		341.21	g	SEL1	-637.8	1.3	-42.2	1.3			TN	2021RUS
		341.21	g	u	-660	7	-20	7	35.01	1.84	Br2	1996SKO
		341.21	lg	SEL							v	2019DIK
CHBrCl ₂	bromodichloromethane	362.65	l	SEL	-476.6	2.2	-84.1	2.2	44.0	2.3	DvH	2021RUS
		362.65	g	SEL1	-511.6	1.2	-49.1	1.2			TN	2021RUS
		362.65	lg	SEL							v	2019DIK
CHBr ₂ Cl	dibromochloromethane	393.00	l	SEL	-470.4	4.1	-44.0	4.1	35.64	1.49	DvH	2021RUS
		393.00	g	SEL1	-514.3	3.4	0.0	3.4			TN	2021RUS
		393.00	lg	SEL							v	2019DIK
CBrCl ₃	bromotrichloromethane	376.65	l	SEL	-363.8	1.6	-77.5	1.6	39.5	4.0	DvH	2021RUS
		376.65	g	SEL1	-399.40	0.59	-41.89	0.59			TN	2021RUS
		376.65	g		-399.3	1.0	-42.0	1.0			Br2	1973MEN
		376.65	g		-402.3		-39.0				Br2	1951SUL
		376.65	lg	SEL							v	2019DIK
CBr ₂ Cl ₂	dibromodichloromethane	423.35	l	SEL	-360.5	4.2	-34.4	4.2	27.6	1.52	DvH	2021RUS
		423.35	g	SEL1	-400.0	1.3	5.1	1.3			TN	2021RUS
		423.35	lg	SEL							v	2019DIK
CBr ₃ Cl	tribromochloromethane	433.00	l	SEL	-355.0	2.9	6.4	2.9	20.72	0.12	DvH	2021RUS
		433.00	g	SEL1	-403.0	1.5	54.4	1.5			TN	2021RUS
		433.00	lg	SEL							v	2019DIK
Bromochlorofluoromethanes												
CHBrClF	bromochlorofluoromethane	312.60	l	SEL	-456.7	5.1	-258.5	5.1	27.62	1.52	DvH	2021RUS
		312.60	g	SEL1	-484.4	4.9	-230.9	4.9			TN	2021RUS
		312.60	lg	SEL							v	2019DIK
CBrClF ₂	bromochlorodifluoromethane	270.65	g	SEL	-310.6	4.9	-439.9	4.9	30.78	1.70	TN	2021RUS
		270.65	l	SEL1	-289.9	4.9	-460.6	4.9			DvH	2021RUS
		270.65	lg	SEL							v	2019DIK
CBrCl ₂ F	bromodichlorofluoromethane	325.15	l	SEL	-326.8	5.2	-269.1	5.2	30.78	1.70	DvH	2021RUS
		325.15	g	SEL1	-357.6	4.9	-238.3	4.9			TN	2021RUS
		325.15	lg	SEL							v	2019DIK
CBr ₂ ClF	dibromochlorofluoromethane	352.65	l	SEL	-327.2	5.2	-222.3	5.2			DvH	2021RUS

	352.65	g	SEL1	-362.7	4.9	-186.8	4.9			TN	2021RUS
	352.65	l	SEL					35.51	1.93	v	2019DIK
Carbonyl Halides											
CCl ₂ O	phosgene	280.71	g	SEL	-221.41	0.27	-219.14	0.27		TN	2021RUS
		280.71	l	SEL1	-198.58	0.47	-241.98	0.47		DvH	2021RUS
		280.71	g		-220.43	0.03	-220.12	0.04		R	2019DIK
		280.71	g		-220.43	0.96	-220.12	0.96		H2O	1972DAV
		280.71	g		-221.44	0.59	-219.12	0.59		Cl2	1970LOR
		280.71	g		-231.06	1.17	-209.49	1.17		Cl2	1969LOR
		280.71	l	SEL	-197.60	0.39	-242.96	0.39	22.84	v	2019DIK
Chlorophenols											
C ₆ HCl ₅ O	pentachlorophenol	463.84	s	SEL	-2329.0	2.9	-292.5	3.0		C	1958SIN
		582.50	l	SEL1	-2346.1	3.0	-275.4	3.0		DsubH	1958SIN
		582.50	g	SEL2	-2420.6	3.0	-200.9	3.0		DvH	1958SIN
		582.50	g	u	-2428	7	-194	7		R	2019DIK
		463.84	s		-2327.0	2.7	-294.4	2.7		R	2019DIK
		463.84	s		-2326.5	2.9	-295.0	3.0		C	1958SIN
		582.50	sg	SEL					91.60	s	2007VER
		463.84	sl	SEL					17.15	v	1991ACR

4.4. Recommended Heats of Combustion, Formation, and Transition

The following Table 4-9 contains the recommended values for heats of combustion, formation, and transition for many compounds, while the previous Table 4-8 contained all of the compiled values for these compounds. This table is divided into several sub-tables by primary compound class, *i.e.*, Small Molecules, Hydrocarbons, Aromatic Hydrocarbons, Oxidized Hydrocarbons, etc., and each of the sub-tables contain the compounds ordered by secondary compound class, *i.e.*, Alkanes, Alkenes, Alkadienes, Alkynes, etc., and then by molecular formula.

Table 4-9. Recommended Heats of Combustion, Formation, and Transition

Table 4-9(1). Small Molecules

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_f H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
Hydrogen/Oxygen													
H ₂	hydrogen	20.37	g	SEL	-285.826	0.000	0.000	0.000			def		def
O ₂	oxygen	90.19	g	SEL	0.000	0.000	0.000	0.000			def		def
H ₂ O ₂	hydrogen peroxide	426.40	l	SEL	-98.48	0.79	-187.35	0.79	51.89	0.79	TN		2021RUS
		426.40	g	SEL1	-150.368	0.068	-135.458	0.063			TN		2021RUS
		272.74	s	SEL2	-86.26	0.79	-199.56	0.79	12.22	0.03	DfusH		1955GIG
		272.74	sl	SEL					12.22	0.03	f		1951FOL
H ₂ O	water	373.12	l	SEL	0.000	0.037	-285.826	0.026	44.00	0.22	TN		2021RUS
		373.12	g	SEL1	-43.994	0.037	-241.832	0.026			TN		2021RUS
		273.153	s	SEL2	6.913	0.037	-292.739	0.026	6.91		TN		2021RUS
Nitrogen/Hydrogen													
N ₂	nitrogen	77.36	g	SEL	0.000	0.000	0.000	0.000			def		
H ₃ N	ammonia	239.82	g	SEL	-383.181	0.029	-45.558	0.029			TN		2021RUS
		239.82	l	SEL1	-363.32	0.11	-65.42	0.11			DvH		2021RUS
		239.82	lg	SEL					19.87	0.10	v		2019DIK
H ₄ N ₂	hydrazine	386.65	l	SEL	-622.34		50.69	0.18	-336.51	1.00			2021RUS
		386.65	g	SEL1	-669.22		97.6	1.0					2021RUS
Nitrogen/Oxygen													
ClH ₄ NO ₄	ammonium perchlorate	403.15	s(d)	SEL	-442.33	0.17	-295.77	0.17					1998CHA
Hydrogen Halides													
FH	hydrogen fluoride	292.7	g	SEL	na		-272.721	0.048			TN		2021RUS
		292.7	l	SEL1	na		-302.41	0.16			TN		2021RUS
	hydrogen fluoride (aq, 20 H ₂ O)	292.7	aq	SEL	na		-321.05	0.16			TN		2021RUS
ClH	hydrogen chloride	188.1	g	SEL	na		-92.173	0.006			TN		2021RUS
	hydrogen chloride (aq, 600 H ₂ O)	188.1	aq	SEL	na		-166.452	0.024			TN		2021RUS
BrH	hydrogen bromide	206.77	g	SEL	na		-36.29	0.16			R		1984COX
	hydrogen bromide (aq, 600 H ₂ O)	206.77	aq	SEL	na		-120.11	0.15			TN		2021RUS
Sulfur/Hydrogen													
H ₂ S	hydrogen sulfide	212.85	g	SEL	-867.21	0.5	-20.6	0.5			R		1984COX

		212.85	l	SEL1	-853.55	0.64	-34.27	0.64		DvH	1984COX
		212.85	lg	SEL					13.67 0.40	v	2019DIK
Sulfur/Oxygen											
O ₂ S	sulfur dioxide	263.14	l	SEL	na		-320.5	1.0	23.67	R	1982WAG
		263.14	g	SEL1	na		-296.83	0.20		R	1982WAG
H ₂ O ₄ S	sulfuric acid	283.46	l	SEL1	na		-814.0	1.0		R	1982WAG
		283.46	g	SEL2	na		-735.1	1.0		R	1998CHA
	sulfuric acid (aq, 115 H ₂ O)	283.46	aq	SEL	na		-887.81	0.25		R	1982WAG
Boron Compounds											
BH ₃ O ₃	boric acid	444.05	s	SEL	na		-1091.94	0.13		D	1950SMI
	boric acid (aq, 500 H ₂ O)	444.05	aq	SEL	na		-1069.89	0.13		H ₂ O	1950SMI
B ₂ H ₆	diborane	180.59	g	SEL	-2175.76	1.17	36.0	1.2		D	1965GUN
		180.59	l	SEL1	-2174.75	1.17	35.0	1.2		DvH	1965GUN
		180.59	lg	SEL			36.8		1.01	v	1959PAR
B ₂ O ₃	diboron trioxide	723.15	s	SEL	na		-1279.9	2.1			1959JOH
		2130	g	SEL1	na		-836.0	4.2			1963HIL
		2130	l	SEL2	na		-1259.3	1.3			1961EVA

Table 4-9(2). Hydrocarbons

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_f H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
Carbon													
C	diamond	4300	s	SEL1	-395.347	0.070	1.871	0.070			C		1966HAW
	graphite	4800	s	SEL	-393.476	0.025	0.000	0.025			def		1966HAW
		5000	g	SEL1	-1110.359	0.047	716.883	0.045			TN		2021RUS
Alkanes													
CH ₄	methane	111.67	g	SEL	-890.602	0.050	-74.526	0.049			TN		2021RUS
C ₂ H ₆	ethane	184.57	g	SEL	-1560.39	0.13	-84.04	0.13			TN		2021RUS
		184.57	l	SEL1	-1555.58	1.03	-88.85	1.03			DvH		2021RUS
		184.57	g	SEL	-1560.34	0.99	-84.09	0.99	4.81	0.29	v		2019DIK
C ₃ H ₈	propane	231.04	g	SEL	-2218.66	0.16	-105.07	0.16			TN		2021RUS
		231.04	l	SEL1	-2203.31	0.26	-120.42	0.26			DvH		2021RUS
		231.04	lg	SEL					15.4		v		1945PRO
C ₄ H ₁₀	isobutane	261.40	g	SEL	-2868.41	0.30	-134.62	0.30			TN		2021RUS
		261.40	l	SEL1	-2848.95	0.37	-154.08	0.37			DvH		2021RUS
		261.40	l	SEL	-2848.6	1.7	-154.4	1.7	19.46	0.21	v		2019DIK
	n-butane	272.66	g	SEL	-2877.28	0.24	-125.75	0.24			TN		2021RUS
		272.66	l	SEL1	-2854.88	0.69	-148.15	0.69			DvH		2021RUS
		272.66	lg	SEL					22.40		v		1971WIL
C ₅ H ₁₂	isopentane	300.98	l	SEL	-3503.95	0.43	-178.39	0.43			TN		2021RUS
		300.98	g	SEL1	-3529.17	0.43	-153.17	0.43			TN		2021RUS
		300.98	lg	SEL					25.22		v		1985MAJ
	neopentane	282.65	l	SEL	-3492.56	0.39	-189.78	0.39	22.39	0.30	TN		2021RUS
		282.65	g	SEL1	-3514.95	0.39	-167.39	0.39			TN		2021RUS
	n-pentane	309.21	l	SEL	-3509.35	0.43	-172.99	0.43	26.74	0.20	TN		2021RUS
		309.21	g	SEL1	-3536.09	0.31	-146.25	0.31			TN		2021RUS
C ₆ H ₁₄	3-methylpentane	336.43	l	SEL	-4159.90	0.88	-201.74	0.96	30.30	0.10	C		1945PRO
		336.43	g	SEL1	-4190.2	1.0	-171.4	1.0			DvH		1945PRO
		336.43	lg	SEL					30.27	0.10	v		1947OSB
	isohexane	333.42	l	SEL	-4157.65	0.83	-203.99	0.83	30.11	0.20	TN		2021RUS
		333.42	g	SEL1	-4187.76	0.84	-173.88	0.84			TN		2021RUS
	neohexane	322.89	l	SEL	-4148.50	0.88	-213.14	0.96	27.68	0.20	C		1945PRO
		322.89	g	SEL1	-4176.18	0.96	-185.46	0.96			DvH		1945PRO
		322.89	lg	SEL					27.68		v		1947OSB
	n-hexane	341.87	l	SEL	-4163.04	0.34	-198.60	0.34	31.71	0.20	TN		2021RUS

C ₇ H ₁₆	n-heptane	341.87	g	SEL1	-4194.75	0.39	-166.89	0.39	36.65	0.20	TN	2021RUS
		371.55	l	SEL	-4816.83	0.46	-224.11	0.46			TN	2021RUS
		371.55	g	SEL1	-4853.48	0.46	-187.46	0.46			TN	2021RUS
C ₈ H ₁₈	n-octane	398.79	l	SEL	-5470.63	0.55	-249.61	0.55	41.53	0.20	TN	2021RUS
		398.79	g	SEL1	-5512.16	0.54	-208.08	0.54			TN	2021RUS
C ₉ H ₂₀	n-nonane	423.91	l	SEL	-6124.85	0.50	-274.69	0.50	46.44	0.20	TN	2021RUS
		423.91	g	SEL1	-6171.29	0.51	-228.25	0.51			TN	2021RUS
C ₁₀ H ₂₂	n-decane	447.27	l	SEL	-6778.29	0.80	-300.56	0.80	51.39	0.20	TN	2021RUS
		447.27	g	SEL1	-6829.68	0.91	-249.17	0.91			TN	2021RUS
C ₁₆ H ₃₄	n-hexadecane	559.94	l	SEL	-10699.1	1.8	-455.6	2.0	81.40	0.40	C	1955FRA
		559.94	g	SEL1	-10780.5	1.8	-374.2	2.0			DvH	1955FRA
		559.94	lg	SEL							v	1995CHI
Alkenes												
C ₂ H ₄	ethene	169.38	g	SEL	-1410.92	0.12	52.32	0.12	8.44	0.13	TN	2021RUS
		169.38	l	SEL1	-1402.48	0.17	43.88	0.17			DvH	2021RUS
		169.38	l	SEL	-1402.7	2.7	44.1	2.7			v	2019DIK
C ₃ H ₆	propene	225.53	g	SEL	-2057.79	0.20	19.88	0.19	14.22	0.27	TN	2021RUS
		225.53	l	SEL1	-2043.57	0.33	5.66	0.33			DvH	2021RUS
		225.53	l	SEL	-2042.3	2.5	4.4	2.5			v	2019DIK
C ₄ H ₈	1-butene	266.83	g	SEL	-2717.30	0.36	0.09	0.36	20.1		TN	2021RUS
		266.83	l	SEL1	-2697.20	0.45	-20.01	0.45			DvH	2021RUS
		266.83	lg	SEL							v	1971WIL
	2-butene, (E)-	274.01	g	SEL	-2706.07	0.38	-11.14	0.38	21.3		TN	2021RUS
		274.01	l	SEL1	-2684.77	0.41	-32.44	0.41			DvH	2021RUS
		274.01	lg	SEL							v	1971WIL
	2-butene, (Z)-	276.87	g	SEL	-2710.16	0.40	-7.05	0.40	22.1		TN	2021RUS
		276.87	l	SEL1	-2688.06	0.59	-29.15	0.59			DvH	2021RUS
		276.87	lg	SEL							v	1971WIL
	isobutene	266.05	g	SEL	-2700.20	0.39	-17.01	0.39	20.60		TN	2021RUS
		266.05	l	SEL1	-2679.60	0.50	-37.61	0.49			DvH	2021RUS
		266.05	lg	SEL							v	1971WIL
C ₅ H ₁₀	1-pentene	303.12	l	SEL	-3349.72	0.64	-46.79	0.64	25.5		C	1979GOO
		303.12	g	SEL1	-3375.22	0.67	-21.29	0.67			DvH	1979GOO
		303.12	lg	SEL							v	1971WIL
	2-methyl-1-butene	304.30	l	SEL	-3335.74	0.84	-60.77	0.84	25.9		C	1979GOO
		304.30	g	SEL1	-3361.64	0.86	-34.87	0.86			DvH	1979GOO
		304.30	lg	SEL							v	1971WIL
	2-pentene, (E)-	309.51	l	SEL	-3338.71	0.76	-57.80	0.76			C	1979GOO

		309.51	g	SEL1	-3365.4	0.8	-31.1	0.8		DvH	1979GOO
		309.51	lg	SEL					26.7	v	1971WIL
	2-pentene, (Z)-	310.09	l	SEL	-3343.21	0.62	-53.30	0.62		C	1979GOO
		310.09	g	SEL1	-3370.0	0.7	-26.5	0.7		DvH	1979GOO
		310.09	lg	SEL					26.8	v	1971WIL
	3-methyl-1-butene	293.21	g	SEL	-3369.00	0.68	-27.51	0.68		DvH	1979GOO
		293.21	l	SEL1	-3345.10	0.62	-51.41	0.62		C	1979GOO
		293.21	lg	SEL					23.9	v	1971WIL
C ₆ H ₁₂	1-hexene	336.64	l	SEL	-4003.1	1.2	-72.7	1.2		H2	1984MOL
		336.64	g	SEL1	-4033.7	1.2	-42.1	1.2		DvH	1984MOL
		336.64	lg	SEL					30.60	v	1971WIL
C ₇ H ₁₄	1-heptene	366.71	l	SEL	-4657.41	0.63	-97.70	0.63		H2O	1984WIB
		366.71	g	SEL1	-4693.1	0.7	-62.0	0.7		DvH	1984WIB
		366.71	lg	SEL					35.7	v	1971WIL
C ₈ H ₁₆	1-octene	394.43	l	SEL	-5312.9	1.2	-121.5	1.2		C	1961ROC
		394.43	g	SEL1	-5353.5	1.2	-80.9	1.2		DvH	1961ROC
		394.43	lg	SEL					40.60	v	1995CHI
Alkadienes											
C ₃ H ₄	allene	238.65	g	SEL	-1941.86	0.24	189.78	0.24		TN	2021RUS
		238.65	l	SEL1	-1925.30	0.69	173.22	0.69		DvH	2021RUS
		238.65	l	SEL	-1931.3	4.8	179.3	4.8	16.56 0.65	v	2019DIK
C ₄ H ₆	1,2-butadiene	284.06	g	SEL	-2593.79	0.46	162.41	0.46	23.68 0.30	TN	2021RUS
		284.06	l	SEL1	-2570.11	0.66	138.73	0.66		TN	2021RUS
	1,3-butadiene	268.44	g	SEL	-2542.28	0.33	110.90	0.33		TN	2021RUS
		268.44	l	SEL1	-2521.18	0.43	89.80	0.43		DvH	2021RUS
		268.44	lg	SEL					21.1	v	1971WIL
C ₅ H ₈	1,2-pentadiene	318.80	l	SEL	-3222.90	0.69	112.22	0.69		DvH	1955FRA
		318.80	g	SEL1	-3251.60	0.67	140.92	0.67		C	1955FRA
		318.80	lg	SEL					28.7	v	1971WIL
	1,3-pentadiene, (E)-	315.32	l	SEL	-3158.90	0.78	48.22	0.78		DvH	1955FRA
		315.32	g	SEL1	-3186.70	0.67	76.02	0.67		C	1955FRA
		315.32	lg	SEL					27.8	v	1971WIL
	1,3-pentadiene, (Z)-	317.18	l	SEL	-3165.3	1.4	54.6	1.4		DvH	1955FRA
		317.18	g	SEL1	-3193.60	0.92	82.92	0.92		C	1955FRA
		317.18	lg	SEL					28.30 0.40	v	1971WIL
	1,4-pentadiene	299.12	g	SEL	-3217.2	1.3	106.5	1.3		C	1955FRA
		299.12	l	SEL1	-3192.0	1.3	81.3	1.3	25.20 0.20	DvH	1955FRA
		299.12	lg	SEL					25.2	v	1971WIL

2-methyl-1,3-butadiene		307.19	l	SEL	-3159.50	0.72	48.82	0.72			DvH	2021RUS
		307.19	g	SEL1	-3186.28	0.67	75.60	0.67			TN	2021RUS
		307.19	lg	SEL					26.78	0.25	v	1971ROG
Alkynes												
C ₂ H ₂	acetylene	188.41	g	SEL	-1301.06	0.13	228.28	0.13			TN	2021RUS
		188.41	l	SEL1	-1293.7	4.4	220.9	4.4			DvH	2021RUS
		188.41	l	SEL	-1291.8	4.7	219.0	4.7	7.4	4.4	v	2019DIK
C ₃ H ₄	propyne	249.80	g	SEL	-1937.72	0.22	185.64	0.22			TN	2021RUS
		249.80	l	SEL1	-1918.14	0.56	166.06	0.56			DvH	2021RUS
		249.80	l	SEL	-1918.5	3.8	166.5	3.8	19.58	0.51	v	2019DIK
C ₄ H ₆	1-butyne	281.18	g	SEL	-2598.06	0.67	166.68	0.67			TN	2021RUS
		281.18	l	SEL1	-2574.36	0.71	142.98	0.71			DvH	2021RUS
		281.18	lg	SEL					23.7		v	1971WIL
2-butyne		300.12	l	SEL	-2551.15	0.58	119.77	0.57			DvH	2021RUS
		300.12	g	SEL1	-2577.85	0.49	146.47	0.49			TN	2021RUS
		300.12	lg	SEL					26.7		v	1971WIL
C ₅ H ₈	1-pentyne	313.51	l	SEL	-3226.6	2.3	115.9	2.3			DvH	1945WAG
		313.51	g	SEL1	-3255.0	2.1	144.3	2.1			E	1945WAG
		313.51	lg	SEL					28.4		v	1971WIL
2-pentyne		329.27	l	SEL	-3208.8	2.2	98.1	2.2			DvH	1945WAG
		329.27	g	SEL1	-3239.6	2.1	128.9	2.1			E	1945WAG
		329.27	lg	SEL					30.8		v	1971WIL
C ₆ H ₁₀	1-hexyne	344.53	l	SEL	-3879.3	1.6	89.4	1.6			DvH	1979ROG
		344.53	g	SEL1	-3912.3	1.2	122.3	1.2			H2	1979ROG
		344.53	l	SEL	-3881.3	2.1	91.4	2.1	32.94	0.29	v	2019DIK
2-hexyne		357.59	l	SEL	-3861.9	2.5	71.9	2.5			DvH	1979ROG
		357.59	g	SEL1	-3897.7	2.4	107.7	2.4			H2	1979ROG
		357.59	lg	SEL					35.8		v	2007BOU
3-hexyne		354.58	l	SEL	-3860.2	1.9	70.2	1.9			DvH	1979ROG
		354.58	g	SEL1	-3895.4	1.9	105.4	1.9			H2	1979ROG
		354.58	l	SEL	-3864.8	2.6	74.9	2.6	35.21	0.29	v	2019DIK
C ₄ H ₄	butenyne	279.18	g	SEL	-2434.97	0.58	289.41	0.58			TN	2021RUS
		279.18	l	SEL1	-2411.8	2.1	266.2	2.1	23.21	0.40	TN	2021RUS
Alkenynes												
C ₅ H ₆	3-penten-1-yne	317.05	l	SEL	-3050.7	3.6	225.9	3.6	27.58	1.50	R	2019DIK
		317.05	g	SEL1	-3078.3	3.3	253.4	3.3			R	2019DIK
3-penten-1-yne, (E)-		323.65	l	SEL	-3050.7	3.2	225.9	3.2	28.35	1.54	R	2019DIK
		323.65	g	SEL1	-3079.1	2.8	254.2	2.8			R	2019DIK

3-penten-1-yne, (Z)-	317.15	l	SEL	-3050.2	2.9	225.4	2.9	27.28	0.77	R	2019DIK
	317.15	g	SEL1	-3077.5	2.8	252.6	2.8			R	2019DIK
Alkyldiynes											
C ₄ H ₂ butadiyne	283.65	g	SEL	-2319.73	0.58	460.00	0.58			TN	2021RUS
	283.65	l	SEL1	-2296.49	1.17	436.76	1.16			DvH	2021RUS
	283.65	l	SEL	-2297.1	3.1	437.4	3.1	23.24	1.01	v	2019DIK
Cycloalkanes											
C ₃ H ₆ cyclopropane	240.27	g	SEL	-2091.75	0.40	53.84	0.40			TN	2021RUS
	240.27	l	SEL1	-2073.63	0.53	35.72	0.53	18.12	0.20	TN	2021RUS
C ₄ H ₈ cyclobutane	285.64	g	SEL	-2745.02	0.39	27.81	0.39			TN	2021RUS
	285.64	l	SEL1	-2721.14	0.40	3.93	0.40	23.88	0.30	TN	2021RUS
C ₅ H ₁₀ cyclopentane	322.40	l	SEL	-3291.40	0.43	-105.11	0.42	28.69	0.20	TN	2021RUS
	322.40	g	SEL1	-3320.09	0.43	-76.42	0.42			TN	2021RUS
methylcyclobutane	310.00	l	SEL	-3339.5	1.3	-57.1	1.3			DvH	1950HUM
	310.00	g	SEL1	-3352.0	1.3	-44.5	1.3			C	1950HUM
	310.00	l	SEL	-3365.4	2.9	-31.1	2.9	25.92	1.30	v	2019DIK
C ₆ H ₁₂ cyclohexane	353.87	l	SEL	-3919.73	0.30	-156.08	0.29	33.14	0.15	TN	2021RUS
	353.87	g	SEL1	-3952.87	0.31	-122.94	0.30			TN	2021RUS
methylcyclopentane	344.96	l	SEL	-3937.74	0.37	-138.07	0.36	31.78	0.15	TN	2021RUS
	344.96	g	SEL1	-3969.52	0.38	-106.29	0.37			TN	2021RUS
C ₇ H ₁₄ methylcyclohexane	374.12	l	SEL	-4565.29	0.96	-189.82	1.05	35.40	0.20	C	1946PRO
	374.12	g	SEL1	-4600.69	1.08	-154.42	1.08			DvH	1946PRO
C ₁₀ H ₂₀ 1,3-diethylcyclohexane	445.70	l	SELu	-6515.9	6.2	-277.1	6.2	46.83	1.55	R	2019DIK
	445.70	g	SEL1u	-6562.7	6.0	-230.3	6.0			R	2019DIK
Cycloalkenes											
C ₄ H ₆ cyclobutene	275.74	g	SEL	-2591.98	0.71	160.60	0.71			TN	2021RUS
	275.74	l	SEL1	-2569.73	0.74	138.35	0.74			DvH	2021RUS
	275.74	l	SEL	-2565.8	1.7	134.4	1.7	22.26	0.21	v	2019DIK
C ₅ H ₈ cyclopentene	317.39	l	SEL	-3117.56	0.52	6.88	0.52	28.29	0.30	TN	2021RUS
	317.39	g	SEL1	-3145.85	0.47	35.17	0.47			TN	2021RUS
methylenecyclobutane	315.00	l	SEL	-3204.7	0.5	94.0	0.5	27.70	0.42	C	1974GOO
	315.00	g	SEL1	-3232.40	0.65	121.72	0.65			C	1974GOO
C ₆ H ₁₀ cyclohexene	356.10	l	SEL	-3752.00	0.32	-37.99	0.31	33.56	0.53	TN	2021RUS
	356.10	g	SEL1	-3785.56	0.33	-4.43	0.32			TN	2021RUS
Bicycloalkanes											
C ₁₀ H ₁₂ tetralin	480.56	l	SEL	-5621.5	1.0	-28.2	1.0			C	1976GOO
	480.56	g	SEL1	-5676.5	1.4	26.8	1.4			DvH	1976GOO
	480.56	lg	SEL					55.00	1.00	v	1971BOY

C ₁₀ H ₁₈	cis-decalin	468.93	l	SEL	-6287.7	1.3	-219.5	1.3	50.2	2.1	C	1941DAV
		468.93	g	SEL1	-6337.9	2.5	-169.3	2.5			DvH	1941DAV
		468.93	lg	SEL					50.2	2.1	v	1960SPE
	trans-decalin	460.42	l	SEL	-6278.9	1.3	-228.3	1.3			C	1941DAV
		460.42	g	SEL1	-6327.4		-179.8				DvH	1941DAV
		460.42	lg	SEL					48.5	2.1	v	1960SPE
C ₁₂ H ₂₂	1,1'-bicyclohexyl	511.10	l	SEL	-7593.0	1.2	-272.8	1.2			C	1976GOO
		511.10	g	SEL1	-7651.0	0.3	-214.8	1.2			DvH	1976GOO
		511.10	lg	SEL					57.98		v	1985MAJ
C ₁₀ H ₁₂	tetralin	480.56	l	SEL	-5621.5	0.9	-28.2	1.0			C	1976GOO
		480.56	g	SEL1	-5676.54	1.05	26.82	1.04			DvH	1976GOO
		480.56	lg	SEL					55.00	0.30	v	1971BOY

Table 4-9(3). Aromatic Hydrocarbons

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_{\text{c}}H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{f}}H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}}H$ (kJ mol ⁻¹)	unc	M	R	Reference
Alkylbenzenes													
C ₆ H ₆	benzene	353.22	l	SEL	-3267.51	0.23	49.18	0.22	33.94	0.15	TN		2021RUS
		353.22	g	SEL1	-3301.45	0.23	83.12	0.22			TN		2021RUS
		278.70	s	SEL2	-3256.83	0.92	38.50	0.92	10.50	0.20	DfusH		2008ROU
		278.70	sl	SEL					10.50	0.20	f		2008ROU
C ₇ H ₈	toluene	383.75	l	SEL	-3909.71	0.34	12.07	0.33	38.04	0.25	TN		2021RUS
		383.75	g	SEL1	-3947.7		50.1				TN		2021RUS
		178.20	s	SEL2	-3899.8	1.5	2.1	1.5	9.90	1.10	R		2008ROU
C ₈ H ₁₀	ethylbenzene	408.96	l	SEL	-4564.41	0.55	-12.53	0.55	42.26	0.40	TN		2021RUS
		408.96	g	SEL1	-4606.67	0.55	29.73	0.55			TN		2021RUS
	m-xylene	412.25	l	SEL	-4551.86	0.63	-25.08	0.63	42.60	0.30	C		1946PRO
		412.25	g	SEL1	-4594.46	0.69	17.52	0.69			DvH		1946PRO
	o-xylene	417.56	l	SEL	-4552.9	1.0	-24.1	1.0	43.40	0.30	C		1945PRO
		417.56	g	SEL1	-4596.26	1.04	19.32	1.04			DvH		1945PRO
	p-xylene	411.47	l	SEL	-4552.86	0.92	-24.08	0.92	42.30	0.30	C		1946PRO
		411.47	g	SEL1	-4595.16	0.97	18.22	0.97			C(l)		1946PRO
C ₉ H ₁₂	isopropylbenzene	425.54	l	SEL	-5215.44	0.96	-40.80	0.97			C		1945PRO
		425.54	g	SEL1	-5260.54	1.09	4.30	1.09			DvH		1945PRO
		425.54	lg	SEL					45.14		v		1947OSB
C ₁₀ H ₁₄	m-isopropyltoluene	448.35	l	SEL	-5857.3	1.0	-78.2	1.0			C		1973GOO
		448.35	g	SEL1	-5906.9	1.3	-28.7	1.3			DvH		1973GOO
		448.35	l	SEL	-5856.3	1.7	-79.2	1.7	49.55	0.88	v		2019DIK
	o-isopropyltoluene	451.50	l	SEL	-5862.62	0.84	-72.92	0.84			C		1973GOO
		451.50	g	SEL1	-5912.26	1.19	-23.28	1.19			DvH		1973GOO
		451.50	l	SEL	-5861.7	1.6	-73.9	1.6	49.64	0.84	v		2019DIK
	p-isopropyltoluene	450.27	l	SEL	-5857.9	1.0	-77.6	1.0			C		1973GOO
		450.27	g	SEL1	-5907.28	1.11	-28.26	1.10			DvH		1973GOO
		450.27	l	SEL	-5856.9	1.6	-78.7	1.6	49.39	0.47	v		2019DIK
Alkenylbenzenes													

C ₈ H ₈	C ₈ H ₈	styrene	418.45	l	SEL	-4394.87	0.84	103.76	0.84	43.50	0.40	C	1945PRO
			418.45	g	SEL1	-4438.35	0.92	147.24	0.92			C	1945PRO
Alkyl naphthalenes													
C ₁₀ H ₈	naphthalene	353.65	s	SEL	-5154.7	3.4	78.0	1.5	16.90	0.70	R		2008ROU
		353.65	l	SEL1	-5173.0	1.7	94.9	1.7			DvH		2008ROU
		491.11	g	SEL2	-5228.7	1.5	150.6	1.5			R		2008ROU
		491.11	lg	SEL					55.40	1.40	v		2008ROU
		353.65	sl	SEL					16.90	0.70	f		2008ROU
C ₁₁ H ₁₀	1-methylnaphthalene	517.84	l	SEL	-5814.0	1.7	56.7	1.7	60.10	0.80	C		1960SPE
		517.84	g	SEL1	-5874.1	1.9	116.8	1.9			DvH		1960SPE
		242.52	s	SEL2	-5807.1	1.7	49.7	1.7			DfusH		1960SPE
		517.84	lg	SEL					60.10	0.80	v		2003VER
		242.52	sl	SEL					6.95	1.00	f		1996DOM
	2-methylnaphthalene	307.69	s	SEL	-5802.7	1.5	45.3	1.5	12.13	1.00	C		1960SPE
		514.25	l	SEL1	-5814.8	2.2	57.5	2.4			DfusH		1960SPE
		514.25	g	SEL2	-5860.1	1.8	102.8	2.6			DvH		1960SPE
		514.25	lg	SEL					45.31	1.00	v		1957GLA
		307.69	sl	SEL					12.13	1.00	f		1996DOM

Table 4-9(4). Oxidized Hydrocarbons

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_f H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
Carbon Oxides													
CO ₂	carbon dioxide	194.69	g	SEL	0.000	0.021	-393.476	0.015			TN		2021RUS
CO	carbon monoxide	81.64	g	SEL	-282.952	0.030	-110.524	0.026			TN		2021RUS
Alkanols													
CH ₄ O	methanol	337.63	l	SEL	-726.52	0.16	-238.61	0.15	37.70	0.05	TN		2021RUS
		337.63	g	SEL1	-764.22	0.16	-200.91	0.15			TN		2021RUS
C ₂ H ₆ O	ethanol	351.57	l	SEL	-1366.92	0.23	-277.51	0.22	42.47		TN		2021RUS
		351.57	g	SEL1	-1409.39	0.23	-235.04	0.22			TN		2021RUS
C ₃ H ₈ O	1-propanol	370.26	l	SEL	-2021.03	0.25	-302.70	0.24	47.46		TN		2021RUS
		370.26	g	SEL1	-2068.49	0.26	-255.24	0.25			TN		2021RUS
	isopropanol	355.39	l	SEL	-2004.83	0.36	-318.90	0.36	45.44	0.10	TN		2021RUS
		355.39	g	SEL1	-2050.27	0.36	-273.46	0.36			TN		2021RUS
C ₄ H ₁₀ O	1-butanol	390.77	l	SEL	-2676.18	0.24	-326.85	0.25			C		1975MOS
		390.77	g	SEL1	-2728.68	0.39	-274.35	0.38			DvH		1975MOS
		390.77	lg	SEL					52.50		v		1995CHI
	isobutanol	380.90	l	SEL	-2669.64	0.59	-333.39	0.59			C		1965CHA
		380.90	g	SEL1	-2720.48	0.59	-282.56	0.59			DvH		1965CHA
		380.90	lg	SEL					50.84	0.08	v		1966WAD
	tert-butanol	355.50	l	SEL	-2643.95	0.79	-359.08	0.80	46.30		C		1960SKI
		355.50	g	SEL1	-2689.7	1.5	-313.4	1.5			DvH		1960SKI
		298.87	s	SEL2	-2636.7	5.1	-366.4	5.1	7.24		R		2019DIK
C ₅ H ₁₂ O	1-pentanol	410.90	l	SEL	-3330.91	0.28	-351.43	0.29			C		1975MOS
		410.90	g	SEL1	-3387.60	0.35	-294.74	0.34			DvH		1975MOS
		410.90	lg	SEL					56.69		v		1965CHA
	2-methyl-1-butanol	402.14	l	SEL	-3325.95	0.54	-356.39	0.55	54.56	1.26	C		1965CHA
	isopentanol	404.73	l	SEL	-3326.2	0.5	-356.1	0.5	55.02	1.26	C		1965CHA
		404.73	g	SEL1	-3381.2	1.4	-301.1	1.4			DvH		1965CHA

Glycols												
C ₂ H ₆ O ₂	1,2-ethanediol	470.73	l	SEL	-1189.19	0.37	-455.24	0.37	65.87	0.20	TN	2021RUS
		470.73	g	SEL1	-1255.06	0.40	-389.37	0.40			TN	2021RUS
C ₃ H ₈ O ₂	2-methoxyethanol	397.54	l	SEL	-1901.7	7.6	-422.0	7.6	45.1	2.7	H2O	1977GUT
		397.54	g	SEL1	-1947	8	-377	8			DvH	1977GUT
C ₃ H ₈ O ₃	glycerin	563.00	l	SEL	-1654.3	0.4	-669.4	0.4	91.70	0.90	C	1988BAS
		563.00	g	SEL1	-1746.00	0.99	-577.73	0.98			DvH	1988BAS
		563.00	s	SEL2	-1636.02	0.99	-687.71	0.98			DfusH	1988BAS
		563.00	lg	SEL					85.77		v	1962ROS
		291.05	sl	SEL					18.28		f	1991ACR
Sugars												
C ₆ H ₁₂ O ₆	alpha-D-glucose	415.0	s	SEL	-2805.0	1.3	-1270.8	1.3			C	1960PON
		706	g	SEL1	-2944	49	-1132	49			DsubH	1960PON
		415.0	sg	SEL					139	49	s	2019DIK
C ₆ H ₁₂ O ₆	fructose	415.0	s(d)	SEL	-2810.4	0.3	-1265.4	0.3			C	1939CLA
C ₁₂ H ₂₂ O ₁₁	sucrose	415.0	s(d)	SEL	-5643.4	1.8	-2222.4	1.8			C	1960PON
Alkyl Ethers												
C ₂ H ₆ O	dimethyl ether	248.37	g	SEL	-1460.41	0.40	-184.02	0.40			TN	2021RUS
		248.37	l	SEL1	-1441.9	1.1	-202.5	1.1			DvH	2021RUS
		248.37	lg	SEL					18.50		v	1976AMB
C ₃ H ₈ O	ethyl methyl ether	280.59	g	SEL	-2106.41	0.58	-217.32	0.58			DvH	2021RUS
		280.59	l	SEL1	-2082.5	1.2	-241.2	1.2			TN	2021RUS
		280.59	l	SEL	-2083.4	1.9	-240.3	1.9	23.90	1.10	v	2019DIK
C ₄ H ₁₀ O	diethyl ether	307.56	l	SEL	-2725.25	0.82	-277.78	0.82	27.37	0.25	TN	2021RUS
		307.56	g	SEL1	-2752.62	0.82	-250.41	0.82			TN	2021RUS
C ₆ H ₁₄ O	diisopropyl ether	341.49	l	SEL	-4010.4	1.3	-351.2	1.3			C	1965COL
		341.49	g	SEL1	-4043.1	1.4	-318.5	1.4			DvH	1965COL
		341.49	lg	SEL					32.70	0.50	v	2007EFI
Aldehydes												
CH ₂ O	formaldehyde	254.13	g	SEL	-570.088	0.096	-109.214	0.096			TN	2021RUS
		254.13	l	SEL1	-549.65	0.55	-129.65	0.55	20.44	0.54	TN	2021RUS
C ₂ H ₄ O	acetaldehyde	293.56	g	SEL	-1193.06	0.25	-165.54	0.25			TN	2021RUS

		293.56	l	SEL1	-1166.84	0.47	-191.76	0.47	26.22	0.40	TN	2021RUS
C ₃ H ₆ O	propanal	321.19	l	SEL	-1821.20	0.36	-216.71	0.35	29.94	0.25	TN	2021RUS
		321.19	g	SEL1	-1851.14	0.26	-186.77	0.25			TN	2021RUS
C ₄ H ₈ O	butanal	347.95	l	SEL	-2477.72	0.84	-239.49	0.84	33.68	0.42	TN	2021RUS
		347.95	g	SEL1	-2511.40	0.85	-205.81	0.85			TN	2021RUS
C ₃ H ₄ O	1-propenal	264.25	g	SEL	-1687.9		-64.2	1.1			TN	2021RUS
	2-propenal	325.40	l	SEL	-1656.4	1.4	-95.7	1.4	30.00	0.30	TN	2021RUS
		325.40	g	SEL1	-1686.4	1.2	-65.7	1.2			TN	2021RUS
Ketones												
C ₃ H ₆ O	acetone	329.22	l	SEL	-1789.79	0.35	-248.12	0.35	31.38	0.10	TN	2021RUS
		329.22	g	SEL1	-1821.17	0.35	-216.74	0.35			TN	2021RUS
C ₄ H ₈ O	butanone	352.74	l	SEL	-2443.83	0.79	-273.38	0.79	34.92	0.10	TN	2021RUS
		352.74	g	SEL1	-2478.75	0.79	-238.46	0.79			TN	2021RUS
Carboxylic Acids												
CH ₂ O ₂	formic acid	374.05	l	SEL	-254.59	0.20	-424.71	0.20	46.33		TN	2021RUS
		374.05	g	SEL1	-300.92	0.22	-378.38	0.22			TN	2021RUS
		281.47	s	SEL2	-241.9	1.0	-437.4	1.0	12.68	1.00	DfusH	2021RUS
		281.47	sl	SEL					12.68	1.00	f	1941STO
C ₂ H ₄ O ₂	acetic acid	391.20	l	SEL	-874.95	0.18	-483.65	0.18	50.91	0.30	TN	2021RUS
		391.20	g	SEL1	-925.86	0.38	-432.74	0.38			TN	2021RUS
		289.70	s	SEL2	-863.23	0.35	-495.37	0.35			DfusH	2021RUS
		289.70	sl	SEL					11.72		f	1996DOM
C ₃ H ₆ O	propanoic acid	413.99	l	SEL	-1528.31	0.18	-509.60	0.18	55.36	0.20	TN	2021RUS
		413.99	g	SEL1	-1583.67	0.28	-454.24	0.28			TN	2021RUS
C ₄ H ₈ O	n-butanoic acid	436.42	l	SEL	-2183.50	0.59	-533.70	0.59			C	1964LEB
		436.42	g	SEL1	-2241.73	0.43	-475.47	0.42			DvH	1964LEB
		436.42	lg	SEL					58.23	0.30	v	2000VER
Alkyl Esters												
C ₃ H ₆ O ₂	ethyl formate	327.29	l	SEL	-1611.8	1.2	-426.1	1.2	32.10	0.15	TN	2021RUS
		327.29	g	SEL1	-1643.9	1.2	-394.0	1.2			TN	2021RUS
C ₂ H ₄ O ₂	methyl formate	305.14	l	SEL	-970.10	0.86	-388.50	0.86	28.58	0.20	TN	2021RUS
		305.14	g	SEL1	-998.68	0.84	-359.92	0.84			TN	2021RUS

C ₄ H ₈ O ₂	ethyl acetate	350.21	l	SEL	-2237.35	0.46	-479.86	0.46	35.15	0.21	H2O	1991WIB2
		350.21	g	SEL1	-2272.49	0.51	-444.72	0.51			H2O	1991WIB2
C ₄ H ₆ O ₂	vinyl acetate	345.33	l	SEL	-2080.6	2.0	-350.8	2.0	37.15	0.84	H2	1980VIL
		345.33	g	SEL1	-2117.7	2.2	-313.6	2.2			DvH	1980VIL
C ₅ H ₈ O ₂	ethyl acrylate	372.59	l	SEL	-2740.1		-370.6				H2	1980VIL
		372.59	g	SEL1	-2779.3		-331.4				DvH	1980VIL
		372.59	lg	SEL					39.20		v	1975VIL
	methyl methacrylate	369.73	l	SEL	-2728.3	4.0	-382.4	4.0				1980VIL
		369.73	g	SEL1	-2768.3	4.0	-342.3	4.0				1980VIL
Oxacyclohydrocarbons												
C ₂ H ₄ O	oxirane	283.59	g	SEL	-1306.00	0.37	-52.60	0.37			TN	2021RUS
		283.59	l	SEL1	-1281.07	0.44	-77.53	0.44	24.93	0.24	TN	2021RUS
C ₄ H ₄ O	furan	304.51	l	SEL	-2083.5	0.6	-62.0	0.6	27.60	0.20	C	1952GUT
		304.51	g	SEL1	-2111.1	0.6	-34.4	0.6			DvH	1952GUT
C ₃ H ₆ O ₃	1,3,5-trioxane	335.65	s	SEL	-1515.70	0.30	-522.21	0.30			C	1969MAN
		387.20	l	SEL1	-1526.4	4.8	-511.5	4.8			DfusH	1969MAN
		387.20	g	SEL2	-1572.3	0.5	-465.6	0.5			DsubH	1969MAN
		335.65	s	SEL	-1514.7	0.5	-523.2	0.5	10.73	0.20	f	2019DIK
		387.20	sg	SEL					56.60	0.20	s	1969MAN
C ₄ H ₈ O ₂	1,3-dioxane	378.13	l	SEL	-2339.86	0.99	-377.35	0.99	39.09	0.05	C	1982BYS
		378.13	g	SEL1	-2378.95	0.99	-338.26	0.99			DvH	1982BYS
	1,4-dioxane	374.25	l	SEL	-2362.23	0.99	-354.98	0.99	38.64	0.05	C	1982BYS
		374.25	g	SEL1	-2362.28	0.99	-354.93	0.99			DvH	1982BYS
Furandione and D-Camphor												
C ₄ H ₂ O ₃	2,5-furandione	325.65	s	SEL	-1389.50	0.67	-470.23	0.67			C	1964WIL
		478.33	g	SEL1	-1458.5	2.1	-401.2	2.1			DvH	1964WIL
		478.33	l	SEL2	-1401.0	2.1	-458.8	2.1			DfusH	1964WIL
		478.33	l	SEL	-1400	10	-459	10	57.5	9.5	v	2019DIK
		325.65	s	SEL	-1389.8	1.1	-470.0	1.1	10.68		f	2019DIK
C ₁₀ H ₁₆ O	D-camphor	451.8	s	SEL	-5902.0	3.0	-319.4	3.0			C	1977STE
		493.00	g	SEL1	-5953.9	3.1	-267.5	3.1			DsubH	1977STE
		451.8	l	SEL2	-5908.8	3.2	-312.5	3.2			DvH	1977STE

		493.00	lg	SEL					6.82		v	1995CHI
		451.8	sg	SEL					51.88	0.84	s	1977STE
Aromatic Alcohols												
C ₆ H ₆ O	phenol	314.15	s	SEL	-3055.98	0.65	-162.35	0.65	69.01	0.30	TN	2021RUS
		454.91	g	SEL1	-3124.99	0.63	-93.34	0.63			TN	2021RUS
		454.91	l	SEL2	-3071.06	0.70	-147.28	0.70			DfusH	2021RUS
C ₇ H ₈ O	3-methylphenol	475.28	l	SEL	-3703.89	0.59	-193.75	0.59	61.71	1.05	C	1960AND
		475.28	g	SEL1	-3765.6	1.2	-132.0	1.2			DvH	1960AND
		285.37	s	SEL2	-3693.18	0.78	-204.46	0.78			DfusH	1960AND
		285.37	sl	SEL					10.71		f	1996DOM
	benzyl alcohol	478.50	l	SEL	-3736.9	1.3	-160.7	1.3			C	1954PAR
		478.50	g	SEL1	-3801.7	1.4	-95.9	1.4			DvH	1954PAR
		478.50	lg	SEL					64.80	0.60	v	1999VER
Aromatic Aldehydes and Acids												
C ₇ H ₆ O	benzaldehyde	451.15	l	SEL	-3525.01	1.10	-86.80	1.10	49.73	0.40	TN	2021RUS
		451.15	g	SEL1	-3574.74	0.91	-37.07	0.91			TN	2021RUS
C ₇ H ₆ O ₂	benzoic acid	395.50	s	SEL	-3227.01	0.18	-384.80	0.17			TN	2021RUS
		523.53	g	SEL1	-3317.62	0.21	-294.19	0.20	90.61	0.50	TN	2021RUS
		523.53	l	SEL2	-3238.72	0.21	-373.09	0.20			DvH	2021RUS
		523.53	lg	SEL					78.90		v	1995CHI
		395.50	sg	SEL					89.30	0.50	s	1990SIL

Table 4-9(5). Nitrogen-Substituted Hydrocarbons

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	unc	$\Delta_i H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}} H$ (kJ mol ⁻¹)	unc	M	R	Reference
Hydrogen Cyanide													
CHN	hydrogen cyanide	298.79	l	SEL	-639.08	0.30	102.69	0.30			DvH		2021RUS
		298.79	g	SEL1	-665.669	0.092	129.280	0.089			TN		2021RUS
		298.79	g	SEL	-665.4	2.7	129.0	2.7	26.59	0.29	v		2019DIK
Alkylamines													
CH ₅ N	methylanine	266.84	g	SEL	-1086.81	0.24	-21.23	0.24			TN		2021RUS
		266.84	l	SEL1	-1062.79	0.45	-45.25	0.45	24.02	0.30	TN		2021RUS
C ₂ H ₇ N	ethylanine	289.79	g	SEL	-1737.31	0.54	-50.03	0.54			TN		2021RUS
		289.79	l	SEL1	-1713.25	0.55	-74.09	0.55	24.06		TN		2021RUS
C ₃ H ₉ N	1-propylanine	320.38	l	SEL	-2365.13	0.41	-101.52	0.41	31.42	0.15	TN		2021RUS
		320.38	g	SEL1	-2396.55	0.42	-70.10	0.42			TN		2021RUS
	isopropylanine	304.92	l	SEL	-2354.37	0.52	-112.28	0.52	28.71	0.20	TN		2021RUS
		304.92	g	SEL1	-2383.08	0.53	-83.57	0.53			TN		2021RUS
C ₄ H ₁₁ N	n-butylanine	350.12	l	SEL	-3018.46	1.13	-127.48	1.13	32.64	1.26	C		1959EVA
		350.12	g	SEL1	-3054.17	1.13	-91.77	1.13			DvH		1959EVA
		350.12	lg	SEL					35.71	0.06	v		1969WAD
C ₂ H ₇ N	dimethylanine	280.02	g	SEL	-1769.79	0.42	-17.55	0.42			TN		2021RUS
		280.02	l	SEL1	-1743.91	0.37	-43.43	0.37	25.88	0.20	TN		2021RUS
C ₃ H ₉ N	methylethylanine	305.60	l	SEL	-2391.4	2.3	-75.2	2.3			TN		2021RUS
		305.60	g	SEL1	-2420.7	1.1	-45.9	1.1			TN		2021RUS
	trimethylanine	276.00	g	SEL	-2439.91	0.76	-26.74	0.76			TN		2021RUS
		276.00	l	SEL1	-2417.80	0.87	-48.85	0.87	22.11	0.10	TN		2021RUS
C ₆ H ₁₅ N	triethylanine	361.93	l	SEL	-4377.09	0.54	-127.46	0.55			C		1966LEB
		361.93	g	SEL1	-4412.49	0.59	-92.06	0.58			DvH		1966LEB
		361.93	lg	SEL					35.40	0.20	v		2009MOK
C ₆ H ₁₃ N	cyclohexylanine	406.98	l	SEL	-4071.3	1.3	-147.4	1.3			C		1979STE
		406.98	g	SEL1	-4113.8	1.3	-104.9	1.3			DvH		1979STE
		406.98	lg	SEL					42.80	0.10	v		1975BER

Alkyl nitriles												
C ₂ H ₃ N	ethanenitrile	355.05	l	SEL	-1256.29	0.22	40.60	0.22	33.45	0.21	TN	2021RUS
		355.05	g	SEL1	-1289.74	0.25	74.05	0.25			TN	2021RUS
C ₃ H ₅ N	propanenitrile	370.73	l	SEL	-1910.62	0.54	15.63	0.55	35.98		C	1971HAL
		370.73	g	SEL1	-1946.60	0.62	51.61	0.62			C	1971HAL
C ₄ H ₇ N	butanenitrile	390.77	l	SEL	-2568.68	0.92	-5.61	0.92	36.99		C	1959EVA
		390.77	g	SEL1	-2605.67	0.92	31.37	0.95			DvH	1959EVA
C ₃ H ₃ N	propenenitrile	351.13	l	SEL	-1756.40	0.42	147.23	0.42	32.64		C	1971HAL
		351.13	g	SEL1	-1789.04	0.72	179.87	0.72			DvH	1971HAL
Azahydrocarbons												
C ₂ N ₂	cyanogen	252.00	g	SEL	-1097.07	0.41	310.12	0.41			TN	2021RUS
		252.00	l	SEL1	-1076.13	0.42	289.18	0.42	20.94		TN	2021RUS
C ₂ H ₈ N ₂	1,1-dimethylhydrazine	335.57	l	SEL	-1978.7	3.6	48.4	3.6			C	1960DON
		335.57	g	SEL1	-2013.7	3.6	83.4	3.6			DvH	1960DON
		335.57	l	SEL					35.00	0.17	v	1952AST
C ₃ H ₆ N ₆	1,3,5-triazine-2,4,6-triamine	618	s	SEL	-1972.0	3.0	-65.9	3.0			C	1973SEL
C ₆ H ₇ N	aniline	457.30	l	SEL	-3393.06	0.84	31.81	0.84	55.77		C	1962HAT
		457.30	g	SEL1	-3448.27	0.88	87.03	0.88			DvH	1962HAT
	pyridine	388.36	l	SEL	-2782.4	1.5	100.2	1.5			C	1954COX
		388.36	g	SEL1	-2822.6	1.5	140.4	1.5	40.16	0.06	DvH	1996CHI
C ₈ H ₁₁ N	N,N-dimethylaniline	466.47	l	SEL	-4767.8	3.2	47.9	3.2	52.83	0.03	C	1982FUR
		466.47	g	SEL1	-4820.6	3.2	100.8	3.2			DvH	1982FUR
		275.65	s	SEL2	-4756.2	3.2	36.4	3.2			DfusH	1982FUR
		275.65	sl	SEL					11.56		f	1972AHN
C ₆ H ₁₅ NO ₃	triethanolamine	623.15	l	SEL	-3840.6	1.5	-664.0	1.5	105.9	2.2	C	1982MIN
		623.15	g	SEL1	-3946.3	2.7	-558.3	2.7			DvH	1982MIN
Alkyl Nitrates and Nitroalkanes												
CH ₃ NO ₃	methyl nitrate	338.00	g	SEL	-700.0	1.3	-122.2	1.3			E	1959RAY
CN ₄ O ₈	tetranitromethane	397.95	l	SEL	-431.8	1.7	38.3	1.7	43.93	0.42	C	1975LEB
		397.95	g	SEL1	-475.7	1.7	82.2	1.7			DvH	1975LEB
		287.05	s	SEL2	-395.4	4.3	1.9	4.3			DfusH	1975LEB
		287.05	sl	SEL					36.4	4.0	f(est)	2021CHE

CH ₃ NO ₂	nitromethane	374.34	l	SEL	-709.25	0.43	-112.97	0.43	38.26	0.10	TN	2021RUS
		374.34	g	SEL1	-747.51	0.46	-74.71	0.46			TN	2021RUS
C ₉ H ₆ N ₂ O ₂	2,4-diisocyanatotoluene	524.83	l	SEL	-4235.0		-163.7		49.5		C	1962STR
		524.83	g	SEL1	-4284.5		-114.2				DvH	1962STR
		293.65	s	SEL2	-4168.0		-230.8				est	1985CHR
		524.83	lg	SEL					49.5	5.2	v	2019DIK
C ₆ H ₅ NO ₂	nitrobenzene	483.81	l	SEL	-3082.0	1.4	6.6	1.4	55.00	0.05	TN	2021RUS
		483.81	g	SEL1	-3137.0	1.4	61.6	1.4			TN	2021RUS
C ₇ H ₅ N ₃ O ₆	trinitrotoluene	355.10	s	SEL	-3406.0	3.0	-62.9	3.0			C	1976ROU
		621.00	g	SEL1	-3519.2	3.4	50.3	3.4			DsubH	1976ROU
		621.00	l	SEL2	-3493.0	3.6	24.1	3.6			DvH	1976ROU
		621.00	lg	SEL					87.00	1.90	v	1978CUN
		355.10	sg	SEL					113.20	1.50	s	1979KUD

Table 4-9(6). Sulfur-Substituted Hydrocarbons

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_{\text{c}}H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{f}}H$ (kJ mol ⁻¹)	unc	$\Delta_{\text{trs}}H$ (kJ mol ⁻¹)	unc	M	R	Reference
Alkanethiols													
CH ₄ S	methanethiol	279.05	g	SEL	-1544.57	0.51	-22.55	0.51			DvH		1961GOO
		279.05	l	SEL1	-1520.8	0.5	-46.35	0.5	23.81	0.08	R	a	1961GOO
C ₂ H ₆ S	ethanethiol	308.19	l	SEL	-2173.17	0.42	-73.68	0.42			C	a	1957MCC
		308.19	g	SEL1	-2200.3	0.43	-46.1	0.43			DvH		1957MCC
Alkyl Sulfides													
CS ₂	carbon disulfide	318.35	l	SEL	-1687.2	0.5	89.71				C	a	1961GOO
		318.35	g	SEL1	-1714.5	0.5	117.1				DvH		1961GOO
C ₂ H ₆ S	dimethyl sulfide	310.42	l	SEL	-2181.45	0.33	-64.96				C	a	1957MCC
		310.42	g	SEL1	-2208.8	0.59	-37.57	0.59			DvH		1957MCC
C ₂ H ₆ OS	dimethyl sulfoxide	463.86	l	SEL	-2037.30	1.25	-203.40		52.90	0.83	C	g	1994MAS
		463.86	g	SEL1	-2095.9	1.5	-150.5	1.5			DvH		1994MAS

Table 4-9(7). Halogenated Hydrocarbons

Formula	Compound	T _{trs} (K)	Ph	SEL	Δ _c H (kJ mol ⁻¹)	unc	Δ _i H (kJ mol ⁻¹)	unc	Δ _{trs} H (kJ mol ⁻¹)	unc	M	R	Reference
Fluoromethanes													
CH ₃ F	fluoromethane	194.65	g	SEL	-764.86	0.28	-235.49	0.23	8.53	1.12	TN	k	2021RUS
		194.65	l	SEL1	-756.33	1.15	-244.02	1.14			DvH		2021RUS
		194.65	lg	SEL				v				2019DIK	
CH ₂ F ₂	difluoromethane	221.50	g	SEL	-584.82	0.40	-450.76	0.33	17.98	0.20	TN	k	2021RUS
		221.50	l	SEL1	-566.84	0.43	-468.74	0.36			TN		2021RUS
CHF ₃	trifluoromethane	188.75	g	SEL	-374.80		-696.00	0.41	8.46	0.30	TN	k	2021RUS
		188.75	l	SEL1	-366.34		-704.46	0.43			TN		2021RUS
CF ₄	tetrafluoromethane	145.56	g	SEL	-172.55		-933.47	0.25	11.81		TN	k	2021RUS
		145.56	l	SEL1	-160.74		-945.28	0.56			DvH		2021RUS
		145.56	lg	SEL							v		1969SMI
Fluoroethanes													
C ₂ H ₅ F	fluoroethane	236.05	g	SEL	-1407.52	0.40	-272.13	0.36	16.32	0.25	TN		2021RUS
		236.05	l	SEL1	-1391.21	0.47	-288.45	0.44			DvH		2021RUS
		236.05	lg	SEL							v		2019DIK
C ₂ H ₄ F ₂	1,1-difluoroethane	249.13	g	SEL	-1212.14	0.60	-502.74	0.55	21.13	0.50	DvH		2021RUS
		249.13	l	SEL1	-1193.66	0.88	-521.22	0.84			TN		2021RUS
	1,2-difluoroethane	293.00	g	SEL	-1267.04	0.84	-447.84	0.81			21.13	0.50	TN
293.00		l	SEL1	-1245.91	0.98	-468.97	0.95	DvH		2021RUS			
293.00		lg	SEL					v		2022BUR1			
C ₂ H ₃ F ₃	1,1,1-trifluoroethane	225.91	g	SEL	-1008.3	1.7	-748.5	1.7	13.43	0.97	C		1965KOL
		225.91	l	SEL1u	-988.2	2.0	-761.9	1.9			DvH		1965KOL
		225.91	lg	SEL							v		2019DIK
1,1,2-trifluoroethane	276.81	g	SEL	-1059	10	-691	10	24.28	0.45	H2		1965KOL	
	276.81	l	SEL1	-1035	10	-715	10			DvH		1965KOL	
	276.81	lg	SEL							v		2019DIK	
C ₂ H ₂ F ₄	1,1,1,2-tetrafluoroethane	247.08	g	SEL	-883.7	2.0	-901.6	2.0	18.10	0.16	CBS		1988BER
		247.08	l	SEL1	-865.6	2.0	-919.7	2.0			DvH		1988BER
		247.08	lg	SEL							v		2019DIK
1,1,2,2-tetrafluoroethane	253.23	g	SEL	-910.1	2.0	-875.2	2.0	19.79	1.13	CBS		1988BER	
	253.23	l	SEL1	-890.3	2.3	-895.0	2.3			G3B3		2016GOO	
	253.23	lg	SEL							v		2019DIK	
C ₂ HF ₅	pentafluoroethane	225.06	g	SEL	-701	8	-1120	8			G3B3		2016GOO
		225.06	l	SEL1	-687	8	-1133	8			DvH		2016GOO

C ₂ F ₆	hexafluoroethane	225.06	lg	SEL					13.44	0.16	v	2019DIK
		194.87	g	SEL	-513.33	-1342.44	0.97				TN	2021RUS
		194.87	l	SEL1	-506.33	-1349.44	1.09		7.00	0.50	DvH	2021RUS
		194.87	lg	SEL					7.00	0.50	v	2022BUR1
Fluoroethenes												
C ₂ H ₃ F	fluoroethene	201.65	g	SEL	-1251.41	0.48	-142.42	0.45			TN	2021RUS
		201.65	l	SEL1	-1240.61	0.90	-153.22	0.88			DvH	2021RUS
		201.65	l	SEL	-1244.0	3.3	-149.8	3.3	10.80	0.76	v	2019DIK
C ₂ H ₂ F ₂	1,1-difluoroethene	190.15	g	SEL	-1078.41	0.80	-350.64	0.77			TN	2021RUS
		190.15	l	SEL1	-1074.0	2.7	-355.1	2.7			DvH	2021RUS
		190.15	lg	SEL	-1090.5	4.3	-338.6	4.3	4.4	2.6	v	2019DIK
	1,2-difluoroethene, (E)-	231.15	g	SEL	-1123.0	0.9	-306.1	0.9			TN	2021RUS
		231.15	l	SEL1	-1108.05	1.39	-321.00	1.38			DvH	2021RUS
		231.15	lg	SEL					14.90	1.04	v	2019DIK
	1,2-difluoroethene, (Z)-		g	SEL	-1119.4	0.9	-309.7	0.9			TN	2021RUS
		253.15	l	SEL1	-1105.6	1.3	-323.4	1.3			DvH	2021RUS
		253.15	lg	SEL					13.79	0.94	v	2019DIK
C ₂ HF ₃	trifluoroethene	211.70	g	SEL	-966.2	1.6	-498.1	1.6			TN	2021RUS
		211.70	l	SEL1	-954.8	1.9	-509.5	1.8			DvH	2021RUS
		211.70	lg	SEL					11.37	0.92	v	2019DIK
C ₂ F ₄	tetrafluoroethene	197.51	g	SEL	-825.10	0.61	-674.40	0.52			TN	2021RUS
		197.51	l	SEL1	-819.00	1.44	-680.50	1.40			DvH	2021RUS
		197.51	lg	SEL	-844	23	-656	23	6.10	1.30	v	2019DIK
Chloromethanes												
CH ₃ Cl	chloromethane	249.00	g	SEL	-763.51	0.39	-82.24	0.39			TN	2021RUS
		249.00	l	SEL1	-743.61	0.25	-102.14	0.25	19.90	0.30	TN	2021RUS
CH ₂ Cl ₂	dichloromethane	312.92	g	SEL	-633.00	0.52	-93.38	0.52			TN	2021RUS
		312.92	l	SEL1	-603.99	0.54	-122.39	0.54	29.01	0.10	TN	2021RUS
CHCl ₃	trichloromethane	334.42	g	SEL	-504.94	0.50	-102.07	0.50			TN	2021RUS
		334.42	l	SEL1	-473.54	0.51	-133.47	0.51	31.40	0.10	TN	2021RUS
CCl ₄	tetrachloromethane	349.65	g	SEL	-392.02	0.43	-95.61	0.43			TN	2021RUS
		349.65	l	SEL1	-359.52	0.43	-128.11	0.43	32.5		TN	2021RUS
Chloroethanes												
C ₂ H ₅ Cl	chloroethane	285.40	g	SEL	-1413.57	0.26	-111.49	0.26			TN	2021RUS
		285.40	l	SEL1	-1389.14	0.40	-135.92	0.40	24.43	0.30	TN	2021RUS
C ₂ H ₄ Cl ₂	1,1-dichloroethane	330.42	g	SEL	-1272.61	0.44	-133.07	0.44			TN	2021RUS
		330.42	l	SEL1	-1241.81	0.48	-163.87	0.48	30.8		TN	2021RUS
	1,2-dichloroethane	356.70	l	SEL	-1238.5	3.5	-167.2	3.5	35.21	0.05	R	2002MAN

C ₂ H ₃ Cl ₃	1,1,1-trichloroethane	356.70	g	SEL1	-1273.7	3.5	-132.0	3.5	32.59	0.07	R	2002MAN
		347.22	l	SEL	-1109.1	2.0	-177.2	2.0			R	2002MAN
		347.22	g	SEL1	-1141.7	2.0	-144.6	2.0			R	2002MAN
	1,1,2-trichloroethane	386.78	l	SEL	-1098.0	4.0	-188.3	4.0	40.30	0.07	R	2002MAN
		386.78	g	SEL1	-1138.3	4.0	-148.0	4.0			R	2002MAN
C ₂ H ₂ Cl ₄	1,1,1,2-tetrachloroethane	403.33	l	SEL	-973.5	2.3	-193.4	2.3	41.10	0.50	R	2002MAN
		403.33	g	SEL1	-1014.6	2.4	-152.3	2.4			R	2002MAN
	1,1,2,2-tetrachloroethane	419.25	l	SEL	-964.5	3.5	-202.4	3.5	45.73	0.09	R	2002MAN
		419.25	g	SEL1	-1010.2	3.5	-156.7	3.5			R	2002MAN
C ₂ HCl ₅	pentachloroethane	433.45	l	SEL	-844.3	4.0	-203.3	4.0	47.40	1.50	R	2002MAN
		433.45	g	SEL1	-891.7	4.3	-155.9	4.3			R	2002MAN
C ₂ Cl ₆	hexachloroethane	462.96	l	SEL	-711.7	2.8	-216.5	2.8	67.1		TN	2021RUS
		462.96	g	SEL1	-778.8	1.3	-149.4	1.3			TN	2021RUS
Chloroethenes												
C ₂ H ₃ Cl	chloroethene	259.35	l	SEL	-1241.13	0.36	1.90	0.36	19.80	0.20	TN	2021RUS
		259.35	g	SEL1	-1260.93	0.30	21.70	0.30			TN	2021RUS
C ₂ H ₂ Cl ₂	1,1-dichloroethene	304.71	l	SEL	-1096.41	0.57	-23.45	0.57	26.75	0.15	TN	2021RUS
		304.71	g	SEL1	-1123.16	0.56	3.30	0.56			TN	2021RUS
	1,2-dichloroethene, (E)-	320.73	l	SEL	-1119.74	0.54	-0.12	0.54	29.50	1.00	DvH	2021RUS
		320.73	g	SEL1	-1149.24	1.14	29.38	1.14			TN	2021RUS
		320.73	lg	SEL							v	2002MAN
	1,2-dichloroethene, (Z)-	333.58	l	SEL	-1086.23	1.14	-33.63	1.14	29.50	1.00	TN	2021RUS
		333.58	g	SEL1	-1117.33	0.54	-2.53	0.54			TN	2021RUS
C ₂ HCl ₃	trichloroethene	360.50	l	SEL	-951.8	1.5	-48.7	1.5	34.60	0.15	TN	2021RUS
		360.50	g	SEL1	-986.4	1.5	-14.1	1.5			TN	2021RUS
C ₂ Cl ₄	tetrachloroethene	394.29	l	SEL	-818.2	1.1	-62.9	1.1	39.70	0.10	TN	2021RUS
		394.29	g	SEL1	-857.9	1.1	-23.2	1.1			TN	2021RUS
Chloroethynes												
C ₂ HCl	chloroacetylene	243.6	g	SEL	-1182.39	0.70	228.99	0.70	20.0	1.0	TN	2021RUS
		243.6	l	SEL1	-1162.4	1.2	209.0	1.2			DvH	2021RUS
		243.6	lg	SEL							v	2002MAN
C ₂ Cl ₂	dichloroacetylene	306.00	g	SEL	-1067.47	0.94	233.44	0.94	27.4	1.2	TN	2021RUS
		306.00	l	SEL1	-1040.1	1.5	206.0	1.5			DvH	2021RUS
		306.00	lg	SEL							v	2002MAN
Bromomethanes												
CH ₃ Br	bromomethane	276.70	g	SEL	-763.80	0.26	-35.61	0.26			TN	2021RUS
		276.70	l	SEL1	-740.47	0.27	-58.94	0.27			TN	2021RUS
CH ₂ Br ₂	dibromomethane	370.10	l	SEL	-601.3	1.1	-32.4	1.1			TN	2021RUS

CHBr ₃	tribromomethane	370.10	g	SEL1	-638.3	1.1	4.6	1.1	46.10	0.10	TN	2021RUS
		422.36	l	SEL	-471.5	1.3	3.5	1.3			TN	2021RUS
		422.36	g	SEL1	-517.6	1.3	49.6	1.3			TN	2021RUS
CBr ₄	tetrabromomethane	462.55	l	SEL	-351.6	1.4	49.3	1.4	54.50	0.70	TN	2021RUS
		462.55	g	SEL1	-406.1	1.3	103.8	1.3			TN	2021RUS
		462.55	sg	SEL							s	1984BIC
C ₂ H ₅ Br	bromoethane	311.54	l	SEL	-1387.66	0.26	-91.05	0.26			TN	2021RUS
		311.54	g	SEL1	-1415.65	0.25	-63.06	0.25			TN	2021RUS
Bromoethenes												
C ₂ H ₃ Br	bromoethene	288.60	g	SEL	-1266.79	0.57	73.90	0.57	33.76	1.74	TN	2021RUS
		288.60	l	SEL1	-1233.0	1.8	40.1	1.8			DvH	2021RUS
		288.60	lg	SEL							v	2019DIK
Chlorofluoromethanes												
CH ₂ ClF	chlorofluoromethane	264.00	g	SEL	-617.7	1.1	-263.3	1.1	18.62	0.96	TN	2021RUS
		264.00	l	SEL1	-599.1	1.5	-281.9	1.5			DvH	2021RUS
		264.00	lg	SEL							v	2019DIK
CHClF ₂	chlorodifluoromethane	232.35	g	SEL	-433.99	1.01	-482.21	0.98	15.70	0.24	TN	2021RUS
		232.35	l	SEL1	-418.29	1.05	-497.91	1.03			DvH	2021RUS
		232.35	lg	SEL							v	2019DIK
CHCl ₂ F	dichlorofluoromethane	282.05	g	SEL	-478.01	0.92	-283.59	0.91	23.64	1.18	TN	2021RUS
		282.05	l	SEL1	-454.4	1.5	-307.2	1.5			DvH	2021RUS
		282.05	lg	SEL							v	2019DIK
CClF ₃	chlorotrifluoromethane	191.75	g	SEL	-241.78	0.63	-709.65	0.57			TN	2021RUS
		191.75	l	SEL1	-225.28	0.70	-726.15	0.64			TN	2021RUS
CCl ₂ F ₂	dichlorodifluoromethane	243.35	g	SEL	-303.18	0.71	-493.65	0.67	16.80	0.23	TN	2021RUS
		243.35	l	SEL1	-286.38	0.73	-510.45	0.70			DvH	2021RUS
		243.35	lg	SEL							v	2019DIK
CCl ₃ F	trichlorofluoromethane	296.95	g	SEL	-353.80	0.80	-288.43	0.78			TN	2021RUS
		296.95	l	SEL1	-329.01	0.81	-313.22	0.80			TN	2021RUS
Chlorofluoroethenes												
C ₂ ClF ₃	chlorotrifluoroethene	244.80	g	SEL	-833.2	2.8	-511.7	4.7	17.2	3.8	DvH	1982ERA
		244.80	l	SEL1	-816.0	2.8	-528.9	2.8			C	1982ERA
Bromofluoromethanes												
CH ₂ BrF	bromofluoromethane	310.00	l	SEL	-591.3	5.2	-243.4	5.2	31.47	1.72	DvH	2021RUS
		310.00	g	SEL1	-622.7	4.9	-211.9	4.9			TN	2021RUS
		310.00	lg	SEL							v	2019DIK
CHBrF ₂	bromodifluoromethane	258.65	g	SEL	-445.5	1.2	-424.3	1.2			TN	2021RUS
		258.65	l	SEL1	-425.5	1.6	-444.4	1.5			DvH	2021RUS

CHBr ₂ F	dibromofluoromethane	258.65	lg	SEL	-456.9	5.2	-212.0	5.2	20.02	0.96	v	2019DIK
		338.05	l	SEL							DvH	2021RUS
		338.05	g	SEL1							TN	2021RUS
		338.05	lg	SEL							v	2021RUS
CBrF ₃	bromotrifluoromethane	215.35	g	SEL	-254.08	0.52	-651.00	0.44	32.73	1.81	TN	2019DIK
		215.35	l	SEL1	-242.31	0.53	-662.78	0.46			TN	2021RUS
CBr ₂ F ₂	dibromodifluoromethane	297.65	g	SEL	-322.1	1.3	-382.0	1.3	24.42	0.19	TN	2021RUS
		297.65	l	SEL1	-297.7	1.3	-406.4	1.3			DvH	2021RUS
		297.65	lg	SEL							v	2019DIK
CBr ₃ F	tribromofluoromethane	379.65	l	SEL	-324.4	3.0	-178.9	3.0	49.4	2.6	DvH	2021RUS
		379.65	g	SEL1	-373.7	1.5	-129.5	1.5			TN	2021RUS
		379.65	lg	SEL							v	2019DIK
Bromochloromethanes												
CH ₂ BrCl	bromochloromethane	341.21	l	SEL	-605.1	1.3	-74.9	1.3	32.75	0.18	DvH	2021RUS
		341.21	g	SEL1	-637.8	1.3	-42.2	1.3			TN	2021RUS
		341.21	lg	SEL							v	2019DIK
CHBrCl ₂	bromodichloromethane	362.65	l	SEL	-476.6	2.2	-84.1	2.2	35.01	1.84	DvH	2021RUS
		362.65	g	SEL1	-511.6	1.2	-49.1	1.2			TN	2021RUS
		362.65	lg	SEL							v	2019DIK
CHBr ₂ Cl	dibromochloromethane	393.00	l	SEL	-470.4	4.1	-44.0	4.1	44.0	2.3	DvH	2021RUS
		393.00	g	SEL1	-514.3	3.4	0.0	3.4			TN	2021RUS
		393.00	lg	SEL							v	2019DIK
CBrCl ₃	bromotrichloromethane	376.65	l	SEL	-363.8	1.6	-77.5	1.6	35.64	1.49	DvH	2021RUS
		376.65	g	SEL1	-399.40	0.59	-41.89	0.59			TN	2021RUS
		376.65	lg	SEL							v	2019DIK
CBr ₂ Cl ₂	dibromodichloromethane	423.35	l	SEL	-360.5	4.2	-34.4	4.2	-39.5	4.0	DvH	2021RUS
		423.35	g	SEL1	-400.0	1.3	5.1	1.3			TN	2021RUS
		423.35	lg	SEL							v	2019DIK
CBr ₃ Cl	tribromochloromethane	433.00	l	SEL	-355.0	2.9	6.4	2.9	48.1	2.5	DvH	2021RUS
		433.00	g	SEL1	-403.0	1.5	54.4	1.5			TN	2021RUS
		433.00	lg	SEL							v	2019DIK
Bromochlorofluoromethanes												
CHBrClF	bromochlorofluoromethane	312.60	l	SEL	-456.7	5.1	-258.5	5.1	27.6	1.52	DvH	2021RUS
		312.60	g	SEL1	-484.4	4.9	-230.9	4.9			TN	2021RUS
		312.60	lg	SEL							v	2019DIK
CBrClF ₂	bromochlorodifluoromethane	270.65	g	SEL	-310.6	4.9	-439.9	4.9	20.72	0.12	TN	2021RUS
		270.65	l	SEL1	-289.9	4.9	-460.6	4.9			DvH	2021RUS
		270.65	lg	SEL							v	2019DIK

CBrCl ₂ F	bromodichlorofluoromethane	325.15	l	SEL	-326.8	5.2	-269.1	5.2	30.78	1.70	DvH	2021RUS
		325.15	g	SEL1	-357.6	4.9	-238.3	4.9			TN	2021RUS
		325.15	lg	SEL							v	2019DIK
CBr ₂ ClF	dibromochlorofluoromethane	352.65	l	SEL	-327.2	5.2	-222.3	5.2	35.51	1.93	DvH	2021RUS
		352.65	g	SEL1	-362.7	4.9	-186.8	4.9			TN	2021RUS
		352.65	l	SEL							v	2019DIK
Carbonyl Halides												
CCl ₂ O	phosgene	280.71	g	SEL	-221.41	0.27	-219.14	0.27	22.84	0.39	TN	2021RUS
		280.71	l	SEL1	-198.58	0.47	-241.98	0.47			DvH	2021RUS
		280.71	l	SEL	-197.60	0.39	-242.96	0.39			v	2019DIK
Chlorophenols												
C ₆ HCl ₅ O	pentachlorophenol	463.84	s	SEL	-2329.0	2.9	-292.5	3.0	91.60	0.50	C	1958SIN
		582.50	l	SEL1	-2346.1	3.0	-275.4	3.0			DsubH	1958SIN
		582.50	g	SEL2	-2420.6	3.0	-200.9	3.0			DvH	1958SIN
		582.50	sg	SEL							s	2007VER
		463.84	sl	SEL							17.15	0.50

Table 4-10. Recommended Gross and Net Heats of Combustion

This table contains both recommended gross $\Delta_c H$ and net $\Delta_c H(\text{net})$ heats of combustion. Gross heats of combustion are where the products are at standard state (298.15 K and 1 bar) where water is in the liquid phase, while net heats of combustion are where the products are all gaseous. The recommended gross heats of combustion are the same as those provided in the previous Table 4-9. The compounds in this table are organized by primary and secondary compound class and then by molecular formula. **NOTE: This version corrects gross and net heats of combustion that were computed incorrectly in the original version for the following compounds: H₂S, propanoic acid, methanethiol, ethanethiol, carbon disulfide, 1,1-dichloroethene, and trichloroethene.**

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	$\Delta_c H(\text{net})$ (kJ mol ⁻¹)	<i>unc</i>	M	R	Reference
SMALL MOLECULES										
Hydrogen/Oxygen										
H ₂	hydrogen	20.37	g	SEL	-285.826	-241.832	0.000	def		def
O ₂	oxygen	90.19	g	SEL	0.000	0.000	0.000	def		def
H ₂ O ₂	hydrogen peroxide	426.40	l	SEL	-98.48	-54.49	0.79	TN		2021RUS
		426.40	g	SEL1	-150.368	-106.374	0.068	TN		2021RUS
		272.74	s	SEL2	-86.26	-42.27	0.79	DfusH		1955GIG
		373.12	l	SEL	0.000	43.994	0.037	TN		2021RUS
		373.12	g	SEL1	-43.994	0.000	0.037	TN		2021RUS
H ₂ O	water	273.153	s	SEL2	6.913	50.907	0.037	TN		2021RUS
Nitrogen/Hydrogen										
N ₂	nitrogen	77.36	g	SEL	0.000	0.000	0.000	def		
H ₃ N	ammonia	239.82	g	SEL	-383.181	-317.190	0.029	TN		2021RUS
		239.82	l	SEL1	-363.32	-297.33	0.11	DvH		2021RUS
ClH ₄ NO ₄	ammonium perchlorate	403.15	s(d)	SEL	-442.33	-280.07	0.17			1998CHA
H ₄ N ₂	hydrazine	386.65	l	SEL	-622.34	-534.35				2021RUS
		386.65	g	SEL1	-669.22	-581.23				2021RUS
Hydrogen Halides										
FH	hydrogen fluoride	292.7	g	SEL	na	na		TN		2021RUS
		292.7	l	SEL1	na	na		TN		2021RUS
ClH	hydrogen fluoride (aq, 20 H ₂ O)	292.7	aq	SEL	na	na		TN		2021RUS
	hydrogen chloride	188.1	g	SEL	na	na		TN		2021RUS
	hydrogen chloride (aq, 600 H ₂ O)	188.1	aq	SEL	na	na		TN		2021RUS
BrH	hydrogen bromide	206.77	g	SEL	na	na		R		1984COX
	hydrogen bromide (aq, 600 H ₂ O)	206.77	aq	SEL	na	na		TN		2021RUS
Sulfur/Hydrogen										

H ₂ S	hydrogen sulfide	212.85	g	SEL	-867.21	-518.04	0.5	R	1984COX
		212.85	l	SEL1	-853.55	-504.38	0.64	DvH	1984COX
O ₂ S	sulfur dioxide	263.14	l	SEL	na	na		R	1982WAG
Sulfur/Oxygen									
		263.14	g	SEL1	na	na		R	1982WAG
H ₂ O ₄ S	sulfuric acid	283.46	l	SEL1	na	na		R	1982WAG
		283.46	g	SEL2	na	na		R	1998CHA
	sulfuric acid (aq, 115 H ₂ O)	283.46	aq	SEL	na	na		R	1982WAG
Boron Compounds									
BH ₃ O ₃	boric acid	444.05	s	SEL	na	na		D	1950SMI
	boric acid (aq, 500 H ₂ O)	444.05	aq	SEL	na	na		H2O	1950SMI
B ₂ H ₆	diborane	180.59	g	SEL	-2175.76	-2043.79	1.17	D	1965GUN
		180.59	l	SEL1	-2174.75	-2042.78	1.17	DvH	1965GUN
B ₂ O ₃	diboron trioxide	723.15	s	SEL	na	na			1959JOH
		2130	g	SEL1	na	na			1963HIL
		2130	l	SEL2	na	na			1961EVA

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	$\Delta_c H(\text{net})$ (kJ mol ⁻¹)	<i>unc</i>	M	R	Reference
HYDROCARBONS										
Carbon										
C	diamond	4300	s	SEL1	-395.347	-395.347	0.070	C		1966HAW
	graphite	4800	s	SEL	-393.476	-393.476	0.025	def		1966HAW
		5000	g	SEL1	-1110.359	-1110.359	0.047	TN		2021RUS
Alkanes										
CH ₄	methane	111.67	g	SEL	-890.602	-802.614	0.050	TN		2021RUS
C ₂ H ₆	ethane	184.57	g	SEL	-1560.39	-1428.41	0.13	TN		2021RUS
		184.57	l	SEL1	-1555.58	-1423.60	1.03	DvH		2021RUS
C ₃ H ₈	propane	231.04	g	SEL	-2218.66	-2042.69	0.16	TN		2021RUS
		231.04	l	SEL1	-2203.31	-2027.34	0.26	DvH		2021RUS
C ₄ H ₁₀	isobutane	261.40	g	SEL	-2868.41	-2648.44	0.30	TN		2021RUS
		261.40	l	SEL1	-2848.95	-2628.98	0.37	DvH		2021RUS
	n-butane	272.66	g	SEL	-2877.28	-2657.31	0.24	TN		2021RUS
		272.66	l	SEL1	-2854.88	-2634.91	0.69	DvH		2021RUS
C ₅ H ₁₂	isopentane	300.98	l	SEL	-3503.95	-3239.98	0.43	TN		2021RUS
		300.98	g	SEL1	-3529.17	-3265.20	0.43	TN		2021RUS
	neopentane	282.65	l	SEL	-3492.56	-3228.59	0.39	TN		2021RUS
		282.65	g	SEL1	-3514.95	-3250.98	0.39	TN		2021RUS
	n-pentane	309.21	l	SEL	-3509.35	-3245.38	0.43	TN		2021RUS
		309.21	g	SEL1	-3536.09	-3272.12	0.31	TN		2021RUS
C ₆ H ₁₄	3-methylpentane	336.43	l	SEL	-4159.90	-3851.94	0.88	C		1945PRO
		336.43	g	SEL1	-4190.2	-3882.2	1.0	DvH		1945PRO
	isohexane	333.42	l	SEL	-4157.65	-3849.69	0.83	TN		2021RUS
		333.42	g	SEL1	-4187.76	-3879.80	0.84	TN		2021RUS
	neohexane	322.89	l	SEL	-4148.50	-3840.54	0.88	C		1945PRO
		322.89	g	SEL1	-4176.18	-3868.22	0.96	DvH		1945PRO
	n-hexane	341.87	l	SEL	-4163.04	-3855.08	0.34	TN		2021RUS
		341.87	g	SEL1	-4194.75	-3886.79	0.39	TN		2021RUS
C ₇ H ₁₆	n-heptane	371.55	l	SEL	-4816.83	-4464.88	0.46	TN		2021RUS
		371.55	g	SEL1	-4853.48	-4501.53	0.46	TN		2021RUS
C ₈ H ₁₈	n-octane	398.79	l	SEL	-5470.63	-5074.69	0.55	TN		2021RUS
		398.79	g	SEL1	-5512.16	-5116.22	0.54	TN		2021RUS
C ₉ H ₂₀	n-nonane	423.91	l	SEL	-6124.85	-5684.91	0.50	TN		2021RUS
		423.91	g	SEL1	-6171.29	-5731.35	0.51	TN		2021RUS
C ₁₀ H ₂₂	n-decane	447.27	l	SEL	-6778.29	-6294.35	0.80	TN		2021RUS

C ₁₆ H ₃₄	n-hexadecane	447.27	g	SEL1	-6829.68	-6345.74	0.91	TN	2021RUS
		559.94	l	SEL	-10699.1	-9951.2	1.8	C	1955FRA
		559.94	g	SEL1	-10780.5	-10032.6	1.8	DvH	1955FRA
Alkenes									
C ₂ H ₄	ethene	169.38	g	SEL	-1410.92	-1322.94	0.12	TN	2021RUS
		169.38	l	SEL1	-1402.48	-1314.50	0.17	DvH	2021RUS
C ₃ H ₆	propene	225.53	g	SEL	-2057.79	-1925.80	0.20	TN	2021RUS
		225.53	l	SEL1	-2043.57	-1911.58	0.33	DvH	2021RUS
C ₄ H ₈	1-butene	266.83	g	SEL	-2717.30	-2541.32	0.36	TN	2021RUS
		266.83	l	SEL1	-2697.20	-2521.22	0.45	DvH	2021RUS
	2-butene, (E)-	274.01	g	SEL	-2706.07	-2530.09	0.38	TN	2021RUS
		274.01	l	SEL1	-2684.77	-2508.79	0.41	DvH	2021RUS
	2-butene, (Z)-	276.87	g	SEL	-2710.16	-2534.18	0.40	TN	2021RUS
		276.87	l	SEL1	-2688.06	-2512.08	0.59	DvH	2021RUS
	isobutene	266.05	g	SEL	-2700.20	-2524.22	0.39	TN	2021RUS
		266.05	l	SEL1	-2679.60	-2503.62	0.50	DvH	2021RUS
C ₅ H ₈	2-methyl-1,3-butadiene	307.19	l	SEL	-3159.50	-2983.53	0.72	DvH	2021RUS
		307.19	g	SEL1	-3186.28	-3010.31	0.67	TN	2021RUS
C ₅ H ₁₀	1-pentene	303.12	l	SEL	-3349.72	-3129.75	0.64	C	1979GOO
		303.12	g	SEL1	-3375.22	-3155.25	0.67	DvH	1979GOO
	2-methyl-1-butene	304.30	l	SEL	-3335.74	-3115.77	0.84	C	1979GOO
		304.30	g	SEL1	-3361.64	-3141.67	0.86	DvH	1979GOO
	2-pentene, (E)-	309.51	l	SEL	-3338.71	-3118.74	0.76	C	1979GOO
		309.51	g	SEL1	-3365.4	-3145.4	0.8	DvH	1979GOO
	2-pentene, (Z)-	310.09	l	SEL	-3343.21	-3123.24	0.62	C	1979GOO
		310.09	g	SEL1	-3370.0	-3150.0	0.7	DvH	1979GOO
	3-methyl-1-butene	293.21	g	SEL	-3369.00	-3149.03	0.68	DvH	1979GOO
		293.21	l	SEL1	-3345.10	-3125.13	0.62	C	1979GOO
C ₆ H ₁₂	1-hexene	336.64	l	SEL	-4003.1	-3739.1	1.2	H2	1984MOL
		336.64	g	SEL1	-4033.7	-3769.7	1.2	DvH	1984MOL
C ₇ H ₁₄	1-heptene	366.71	l	SEL	-4657.41	-4349.46	0.63	H2O	1984WIB
		366.71	g	SEL1	-4693.1	-4385.2	0.7	DvH	1984WIB
C ₈ H ₁₆	1-octene	394.43	l	SEL	-5312.9	-4960.9	1.2	C	1961ROC
		394.43	g	SEL1	-5353.5	-5001.5	1.2	DvH	1961ROC
Alkadienes									
C ₃ H ₄	allene	238.65	g	SEL	-1941.86	-1853.87	0.24	TN	2021RUS
		238.65	l	SEL1	-1925.30	-1837.31	0.69	DvH	2021RUS
C ₄ H ₆	1,2-butadiene	284.06	g	SEL	-2593.79	-2461.81	0.46	TN	2021RUS

		284.06	l	SEL1	-2570.11	-2438.13	0.66	TN	2021RUS
	1,3-butadiene	268.44	g	SEL	-2542.28	-2410.30	0.33	TN	2021RUS
		268.44	l	SEL1	-2521.18	-2389.20	0.43	DvH	2021RUS
C ₅ H ₈	1,2-pentadiene	318.80	l	SEL	-3222.90	-3046.92	0.69	DvH	1955FRA
		318.80	g	SEL1	-3251.60	-3075.62	0.67	C	1955FRA
	1,3-pentadiene, (E)-	315.32	l	SEL	-3158.90	-2982.92	0.78	DvH	1955FRA
		315.32	g	SEL1	-3186.70	-3010.72	0.67	C	1955FRA
	1,3-pentadiene, (Z)-	317.18	l	SEL	-3165.3	-2989.3	1.4	DvH	1955FRA
		317.18	g	SEL1	-3193.60	-3017.62	0.92	C	1955FRA
	1,4-pentadiene	299.12	g	SEL	-3217.2	-3041.2	1.3	C	1955FRA
		299.12	l	SEL1	-3192.0	-3016.0	1.3	DvH	1955FRA

Alkynes									
C ₂ H ₂	acetylene	188.41	g	SEL	-1301.06	-1257.06	0.13	TN	2021RUS
		188.41	l	SEL1	-1293.7	-1249.7	4.4	DvH	2021RUS
C ₃ H ₄	propyne	249.80	g	SEL	-1937.72	-1849.73	0.22	TN	2021RUS
		249.80	l	SEL1	-1918.14	-1830.15	0.56	DvH	2021RUS
C ₄ H ₆	1-butyne	281.18	g	SEL	-2598.06	-2466.08	0.67	TN	2021RUS
		281.18	l	SEL1	-2574.36	-2442.38	0.71	DvH	2021RUS
	2-butyne	300.12	l	SEL	-2551.15	-2419.17	0.58	DvH	2021RUS
		300.12	g	SEL1	-2577.85	-2445.87	0.49	TN	2021RUS
C ₅ H ₈	1-pentyne	313.51	l	SEL	-3226.6	-3050.6	2.3	DvH	1945WAG
		313.51	g	SEL1	-3255.0	-3079.0	2.1	E	1945WAG
	2-pentyne	329.27	l	SEL	-3208.8	-3032.8	2.2	DvH	1945WAG
		329.27	g	SEL1	-3239.6	-3063.6	2.1	E	1945WAG
C ₆ H ₁₀	1-hexyne	344.53	l	SEL	-3879.3	-3659.4	1.6	DvH	1979ROG
		344.53	g	SEL1	-3912.3	-3692.3	1.2	H2	1979ROG
	2-hexyne	357.59	l	SEL	-3861.9	-3641.9	2.5	DvH	1979ROG
		357.59	g	SEL1	-3897.7	-3677.7	2.4	H2	1979ROG
	3-hexyne	354.58	l	SEL	-3860.2	-3640.2	1.9	DvH	1979ROG
		354.58	g	SEL1	-3895.4	-3675.4	1.9	H2	1979ROG
Alkenynes									
C ₄ H ₄	butenyne	279.18	g	SEL	-2434.97	-2346.98	0.58	TN	2021RUS
		279.18	l	SEL1	-2411.8	-2323.8	2.1	TN	2021RUS
C ₅ H ₆	3-penten-1-yne	317.05	l	SEL	-3050.7	-2918.7	3.6	R	2019DIK
		317.05	g	SEL1	-3078.3	-2946.3	3.3	R	2019DIK
	3-penten-1-yne, (E)-	323.65	l	SEL	-3050.7	-2918.8	3.2	R	2019DIK
		323.65	g	SEL1	-3079.1	-2947.1	2.8	R	2019DIK
	3-penten-1-yne, (Z)-	317.15	l	SEL	-3050.2	-2918.2	2.9	R	2019DIK
		317.15	g	SEL1	-3077.5	-2945.5	2.8	R	2019DIK
Alkadiynes									
C ₄ H ₂	butadiyne	283.65	g	SEL	-2319.73	-2275.74	0.58	TN	2021RUS
		283.65	l	SEL1	-2296.49	-2252.50	1.17	DvH	2021RUS
Cycloalkanes									
C ₃ H ₆	cyclopropane	240.27	g	SEL	-2091.75	-1959.76	0.40	TN	2021RUS
		240.27	l	SEL1	-2073.63	-1941.64	0.53	TN	2021RUS
C ₄ H ₈	cyclobutane	285.64	g	SEL	-2745.02	-2569.04	0.39	TN	2021RUS
		285.64	l	SEL1	-2721.14	-2545.16	0.40	TN	2021RUS
C ₅ H ₁₀	cyclopentane	322.40	l	SEL	-3291.40	-3071.43	0.43	TN	2021RUS
		322.40	g	SEL1	-3320.09	-3100.12	0.43	TN	2021RUS
	methylcyclobutane	310.00	l	SEL	-3339.5	-3119.5	1.3	DvH	1950HUM

C ₆ H ₁₂	cyclohexane	310.00	g	SEL1	-3352.0	-3132.0	1.3	C	1950HUM
		353.87	l	SEL	-3919.73	-3655.77	0.30	TN	2021RUS
		353.87	g	SEL1	-3952.87	-3688.91	0.31	TN	2021RUS
	methylcyclopentane	344.96	l	SEL	-3937.74	-3673.78	0.37	TN	2021RUS
		344.96	g	SEL1	-3969.52	-3705.56	0.38	TN	2021RUS
C ₇ H ₁₄	methylcyclohexane	374.12	l	SEL	-4565.29	-4257.33	0.96	C	1946PRO
		374.12	g	SEL1	-4600.69	-4292.73	1.08	DvH	1946PRO
C ₁₀ H ₂₀	1,3-diethylcyclohexane	445.70	l	SELu	-6515.9	-6076.0	6.2	R	2019DIK
		445.70	g	SEL1u	-6562.7	-6122.8	6.0	R	2019DIK
Cycloalkenes									
C ₄ H ₆	cyclobutene	275.74	g	SEL	-2591.98	-2460.00	0.71	TN	2021RUS
		275.74	l	SEL1	-2569.73	-2437.75	0.74	DvH	2021RUS
C ₅ H ₈	cyclopentene	317.39	l	SEL	-3117.56	-2941.59	0.52	TN	2021RUS
		317.39	g	SEL1	-3145.85	-2969.88	0.47	TN	2021RUS
	methylenecyclobutane	315.00	l	SEL	-3204.7	-3028.7	0.5	C	1974GOO
		315.00	g	SEL1	-3232.40	-3056.42	0.65	C	1974GOO
C ₆ H ₁₀	cyclohexene	356.10	l	SEL	-3752.00	-3532.03	0.32	TN	2021RUS
		356.10	g	SEL1	-3785.56	-3565.59	0.33	TN	2021RUS
Bicycloalkanes									
C ₁₀ H ₁₂	tetralin	480.56	l	SEL	-5621.5	-5357.6	1.0	C	1976GOO
		480.56	g	SEL1	-5676.5	-5412.6	1.4	DvH	1976GOO
C ₁₀ H ₁₈	cis-decalin	468.93	l	SEL	-6287.7	-5891.8	1.3	C	1941DAV
		468.93	g	SEL1	-6337.9	-5942.0	2.5	DvH	1941DAV
	trans-decalin	460.42	l	SEL	-6278.9	-5883.0	1.3	C	1941DAV
		460.42	g	SEL1	-6327.4	-5931.5		DvH	1941DAV
C ₁₂ H ₂₂	1,1'-bicyclohexyl	511.10	l	SEL	-7593.0	-7109.0	1.2	C	1976GOO
		511.10	g	SEL1	-7651.0	-7167.0	0.3	DvH	1976GOO

Formula	Compound	T _{trs} (K)	Ph	SEL	Δ _c H (kJ mol ⁻¹)	Δ _c H(net) (kJ mol ⁻¹)	unc	M	R	Reference
Aromatic Hydrocarbons										
C ₆ H ₆	benzene	353.22	l	SEL	-3267.51	-3135.53	0.23	TN		2021RUS
		353.22	g	SEL1	-3301.45	-3169.47	0.23	TN		2021RUS
		278.70	s	SEL2	-3256.83	-3124.85	0.92	DfusH		2008ROU
C ₇ H ₈	toluene	383.75	l	SEL	-3909.71	-3733.73	0.34	TN		2021RUS
		383.75	g	SEL1	-3947.7	-3771.8		TN		2021RUS
		178.20	s	SEL2	-3899.8	-3723.8	1.5	R		2008ROU
C ₈ H ₁₀	ethylbenzene	408.96	l	SEL	-4564.41	-4344.44	0.55	TN		2021RUS
		408.96	g	SEL1	-4606.67	-4386.70	0.55	TN		2021RUS
	m-xylene	412.25	l	SEL	-4551.86	-4331.89	0.63	C		1946PRO
		412.25	g	SEL1	-4594.46	-4374.49	0.69	DvH		1946PRO
	o-xylene	417.56	l	SEL	-4552.9	-4332.9	1.0	C		1945PRO
		417.56	g	SEL1	-4596.26	-4376.29	1.04	DvH		1945PRO
	p-xylene	411.47	l	SEL	-4552.86	-4332.89	0.92	C		1946PRO
		411.47	g	SEL1	-4595.16	-4375.19	0.97	C(l)		1946PRO
	C ₉ H ₁₂	isopropylbenzene	425.54	l	SEL	-5215.44	-4951.48	0.96	C	
425.54			g	SEL1	-5260.54	-4996.58	1.09	DvH		1945PRO
C ₁₀ H ₁₄	m-isopropyltoluene	448.35	l	SEL	-5857.3	-5549.3	1.0	C		1973GOO
		448.35	g	SEL1	-5906.9	-5598.9	1.3	DvH		1973GOO
	o-isopropyltoluene	451.50	l	SEL	-5862.62	-5554.66	0.84	C		1973GOO
		451.50	g	SEL1	-5912.26	-5604.30	1.19	DvH		1973GOO
	p-isopropyltoluene	450.27	l	SEL	-5857.9	-5549.9	1.0	C		1973GOO
		450.27	g	SEL1	-5907.28	-5599.32	1.11	DvH		1973GOO
C ₈ H ₈	styrene	418.45	l	SEL	-4395.42	-4219.45	0.48	TN		2021RUS
		418.45	g	SEL1	-4439.45	-4263.48	0.48	TN		2021RUS
C ₁₀ H ₈	naphthalene	353.65	s	SEL	-5154.7	-4978.7	3.4	R		2008ROU
		353.65	l	SEL1	-5173.0	-4997.0	1.7	DvH		2008ROU
		491.11	g	SEL2	-5228.7	-5052.7	1.5	R		2008ROU
C ₁₁ H ₁₀	1-methylnaphthalene	517.84	l	SEL	-5814.0	-5594.1	1.7	C		1960SPE
		517.84	g	SEL1	-5874.1	-5654.2	1.9	DvH		1960SPE
		242.52	s	SEL2	-5807.1	-5587.1	1.7	DfusH		1960SPE
	2-methylnaphthalene	307.69	s	SEL	-5802.7	-5582.7	1.5	C		1960SPE
		514.25	l	SEL1	-5814.8	-5594.9	2.2	DfusH		1960SPE
		514.25	g	SEL2	-5860.1	-5640.2	1.8	DvH		1960SPE

Formula	Compound	T _{trs} (K)	Ph	SEL	Δ _c H (kJ mol ⁻¹)	Δ _c H(net) (kJ mol ⁻¹)	unc	M	R	Reference
OXIDIZED HYDROCARBONS										
Carbon Oxides										
CO ₂	carbon dioxide	194.69	g	SEL	0.000	0.000	0.021	TN		2021RUS
CO	carbon monoxide	81.64	g	SEL	-282.952	-282.952	0.030	TN		2021RUS
Alkanols										
CH ₄ O	methanol	337.63	l	SEL	-726.52	-638.53	0.16	TN		2021RUS
		337.63	g	SEL1	-764.22	-676.23	0.16	TN		2021RUS
C ₂ H ₆ O	ethanol	351.57	l	SEL	-1366.92	-1234.94	0.23	TN		2021RUS
		351.57	g	SEL1	-1409.39	-1277.41	0.23	TN		2021RUS
C ₃ H ₈ O	1-propanol	370.26	l	SEL	-2021.03	-1845.06	0.25	TN		2021RUS
		370.26	g	SEL1	-2068.49	-1892.52	0.26	TN		2021RUS
	isopropanol	355.39	l	SEL	-2004.83	-1828.86	0.36	TN		2021RUS
C ₄ H ₁₀ O	1-butanol	355.39	g	SEL1	-2050.27	-1874.30	0.36	TN		2021RUS
		390.77	l	SEL	-2676.18	-2456.21	0.24	C		1975MOS
	390.77	g	SEL1	-2728.68	-2508.71	0.39	DvH		1975MOS	
	isobutanol	380.90	l	SEL	-2669.64	-2449.67	0.59	C		1965CHA
	380.90	g	SEL1	-2720.48	-2500.51	0.59	DvH		1965CHA	
	tert-butanol	355.50	l	SEL	-2643.95	-2423.98	0.79	C		1960SKI
C ₅ H ₁₂ O	1-pentanol	355.50	g	SEL1	-2689.7	-2469.7	1.5	DvH		1960SKI
		298.87	s	SEL2	-2636.7	-2416.7	5.1	R		2019DIK
	410.90	l	SEL	-3330.91	-3066.95	0.28	C		1975MOS	
	410.90	g	SEL1	-3387.60	-3123.63	0.35	DvH		1975MOS	
	2-methyl-1-butanol	402.14	l	SEL	-3325.95	-3061.98	0.54	C		1965CHA
	isopentanol	404.73	l	SEL	-3326.2	-3062.2	0.5	C		1965CHA
404.73	g	SEL1	-3381.2	-3117.3	1.4	DvH		1965CHA		
Glycols										
C ₂ H ₆ O ₂	1,2-ethanediol	470.73	l	SEL	-1189.19	-1057.21	0.37	TN		2021RUS
		470.73	g	SEL1	-1255.06	-1123.08	0.40	TN		2021RUS
C ₃ H ₈ O ₂	2-methoxyethanol	397.54	l	SEL	-1901.7	-1725.7	7.6	H2O		1977GUT
		397.54	g	SEL1	-1947	-1771	8	DvH		1977GUT
C ₃ H ₈ O ₃	glycerin	563.00	l	SEL	-1654.3	-1478.3	0.4	C		1988BAS
		563.00	g	SEL1	-1746.00	-1570.02	0.99	DvH		1988BAS
		563.00	s	SEL2	-1636.02	-1460.04	0.99	DfusH		1988BAS
Sugars										
C ₆ H ₁₂ O ₆	alpha-D-glucose	415.0	s	SEL	-2805.0	-2541.0	1.3	C		1960PON
		706	g	SEL1	-2944	-2680	49	DsubH		1960PON

C ₆ H ₁₂ O ₆	fructose	415.0	s(d)	SEL	-2810.4	-2546.4	0.3	C	1939CLA
C ₁₂ H ₂₂ O ₁₁	sucrose	415.0	s(d)	SEL	-5643.4	-5159.5	1.8	C	1960PON
Alkyl Ethers									
C ₂ H ₆ O	dimethyl ether	248.37	g	SEL	-1460.41	-1328.43	0.40	TN	2021RUS
		248.37	l	SEL1	-1441.9	-1309.9	1.1	DvH	2021RUS
C ₃ H ₈ O	ethyl methyl ether	280.59	g	SEL	-2106.41	-1930.44	0.58	DvH	2021RUS
		280.59	l	SEL1	-2082.5	-1906.5	1.2	TN	2021RUS
C ₄ H ₁₀ O	diethyl ether	307.56	l	SEL	-2725.25	-2505.28	0.82	TN	2021RUS
		307.56	g	SEL1	-2752.62	-2532.65	0.82	TN	2021RUS
C ₆ H ₁₄ O	diisopropyl ether	341.49	l	SEL	-4010.4	-3702.4	1.3	C	1965COL
		341.49	g	SEL1	-4043.1	-3735.1	1.4	DvH	1965COL
Aldehydes									
CH ₂ O	formaldehyde	254.13	g	SEL	-570.088	-526.094	0.096	TN	2021RUS
		254.13	l	SEL1	-549.65	-505.65	0.55	TN	2021RUS
C ₂ H ₄ O	acetaldehyde	293.56	g	SEL	-1193.06	-1105.08	0.25	TN	2021RUS
		293.56	l	SEL1	-1166.84	-1078.86	0.47	TN	2021RUS
C ₃ H ₆ O	propanal	321.19	l	SEL	-1821.20	-1689.21	0.36	TN	2021RUS
		321.19	g	SEL1	-1851.14	-1719.15	0.26	TN	2021RUS
C ₄ H ₈ O	butanal	347.95	l	SEL	-2477.72	-2301.74	0.84	TN	2021RUS
		347.95	g	SEL1	-2511.40	-2335.42	0.85	TN	2021RUS
C ₃ H ₄ O	1-propenal	264.25	g	SEL	-1687.9	-1599.9		TN	2021RUS
	2-propenal	325.40	l	SEL	-1656.4	-1568.4	1.4	TN	2021RUS
		325.40	g	SEL1	-1686.4	-1598.4	1.2	TN	2021RUS
Ketones									
C ₃ H ₆ O	acetone	329.22	l	SEL	-1789.79	-1657.80	0.35	TN	2021RUS
		329.22	g	SEL1	-1821.17	-1689.18	0.35	TN	2021RUS
C ₄ H ₈ O	butanone	352.74	l	SEL	-2443.83	-2267.85	0.79	TN	2021RUS
		352.74	g	SEL1	-2478.75	-2302.77	0.79	TN	2021RUS
Carboxylic Acids									
CH ₂ O ₂	formic acid	374.05	l	SEL	-254.59	-210.60	0.20	TN	2021RUS
		374.05	g	SEL1	-300.92	-256.93	0.22	TN	2021RUS
		281.47	s	SEL2	-241.9	-197.9	1.0	DfusH	2021RUS
C ₂ H ₄ O ₂	acetic acid	391.20	l	SEL	-874.95	-786.97	0.18	TN	2021RUS
		391.20	g	SEL1	-925.86	-837.88	0.38	TN	2021RUS
		289.70	s	SEL2	-863.23	-775.25	0.35	DfusH	2021RUS
C ₃ H ₆ O	propanoic acid	413.99	l	SEL	-1528.31	-1396.32	0.18	TN	2021RUS
		413.99	g	SEL1	-1583.67	-1451.68	0.28	TN	2021RUS
C ₄ H ₈ O	n-butanoic acid	436.42	l	SEL	-2183.50	-2007.53	0.59	C	1964LEB

		436.42	g	SEL1	-2241.73	-2065.76	0.43	DvH	1964LEB
Alkyl Esters									
C ₃ H ₆ O ₂	ethyl formate	327.29	l	SEL	-1611.8	-1479.8	1.2	TN	2021RUS
		327.29	g	SEL1	-1643.9	-1511.9	1.2	TN	2021RUS
C ₂ H ₄ O ₂	methyl formate	305.14	l	SEL	-970.10	-882.12	0.86	TN	2021RUS
		305.14	g	SEL1	-998.68	-910.70	0.84	TN	2021RUS
C ₄ H ₈ O ₂	ethyl acetate	350.21	l	SEL	-2237.35	-2061.37	0.46	H2O	1991WIB2
		350.21	g	SEL1	-2272.49	-2096.51	0.51	H2O	1991WIB2
C ₄ H ₆ O ₂	vinyl acetate	345.33	l	SEL	-2080.6	-1948.6	2.0	H2	1980VIL
		345.33	g	SEL1	-2117.7	-1985.8	2.2	DvH	1980VIL
C ₅ H ₈ O ₂	ethyl acrylate	372.59	l	SEL	-2740.1	-2564.1		H2	1980VIL
		372.59	g	SEL1	-2779.3	-2603.3		DvH	1980VIL
	methyl methacrylate	369.73	l	SEL	-2728.3	-2552.3	4.0		1980VIL
		369.73	g	SEL1	-2768.3	-2592.4	4.0		1980VIL
Oxacyclohydrocarbons									
C ₂ H ₄ O	oxirane	283.59	g	SEL	-1306.00	-1218.02	0.37	TN	2021RUS
		283.59	l	SEL1	-1281.07	-1193.09	0.44	TN	2021RUS
C ₄ H ₄ O	furan	304.51	l	SEL	-2083.5	-1995.5	0.6	C	1952GUT
		304.51	g	SEL1	-2111.1	-2023.1	0.6	DvH	1952GUT
C ₃ H ₆ O ₃	1,3,5-trioxane	335.65	s	SEL	-1515.70	-1383.72	0.30	C	1969MAN
		387.20	l	SEL1	-1526.4	-1394.4	4.8	DfusH	1969MAN
		387.20	g	SEL2	-1572.3	-1440.3	0.5	DsubH	1969MAN
C ₄ H ₈ O ₂	1,3-dioxane	378.13	l	SEL	-2339.86	-2163.88	0.99	C	1982BYS
		378.13	g	SEL1	-2378.95	-2202.97	0.99	DvH	1982BYS
	1,4-dioxane	374.25	l	SEL	-2362.23	-2186.25	0.99	C	1982BYS
		374.25	g	SEL1	-2362.28	-2186.30	0.99	DvH	1982BYS
Furandione and D-Camphor									
C ₄ H ₂ O ₃	2,5-Furandione	325.65	s	SEL	-1389.50	-1345.51	0.67	C	1964WIL
		478.33	g	SEL1	-1458.5	-1414.5	2.1	DvH	1964WIL
		478.33	l	SEL2	-1401.0	-1357.0	2.1	DfusH	1964WIL
C ₁₀ H ₁₆ O	D-camphor	451.8	s	SEL	-5902.0	-5550.0	3.0	C	1977STE
		493.00	g	SEL1	-5953.9	-5601.9	3.1	DsubH	1977STE
		451.8	l	SEL2	-5908.8	-5556.9	3.2	DvH	1977STE
Aromatic Alcohols									
C ₆ H ₆ O	phenol	314.15	s	SEL	-3055.98	-2924.00	0.65	TN	2021RUS
		454.91	g	SEL1	-3124.99	-2993.01	0.63	TN	2021RUS
		454.91	l	SEL2	-3071.06	-2939.08	0.70	DfusH	2021RUS
C ₇ H ₈ O	3-methylphenol	475.28	l	SEL	-3703.89	-3527.91	0.59	C	1960AND

		475.28	g	SEL1	-3765.6	-3589.6	1.2	DvH	1960AND
		285.37	s	SEL2	-3693.18	-3517.20	0.78	DfusH	1960AND
	benzyl alcohol	478.50	l	SEL	-3736.9	-3561.0	1.3	C	1954PAR
		478.50	g	SEL1	-3801.7	-3625.8	1.4	DvH	1954PAR
Aromatic Aldehydes and Acids									
C ₇ H ₆ O	benzaldehyde	451.15	l	SEL	-3525.01	-3393.03	1.10	TN	2021RUS
		451.15	g	SEL1	-3574.74	-3442.76	0.91	TN	2021RUS
C ₇ H ₆ O ₂	benzoic acid	395.50	s	SEL	-3227.01	-3095.03	0.18	TN	2021RUS
		523.53	g	SEL1	-3317.62	-3185.64	0.21	TN	2021RUS
		523.53	l	SEL2	-3238.72	-3106.74	0.21	DvH	2021RUS

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	$\Delta_c H(\text{net})$ (kJ mol ⁻¹)	unc	M	R	Reference
NITROGEN-SUBSTITUTED HYDROCARBONS										
Hydrogen Cyanide										
CHN	hydrogen cyanide	298.79	l	SEL	-639.08	-617.08	0.30	DvH		2021RUS
		298.79	g	SEL1	-665.669	-643.672	0.092	TN		2021RUS
Alkylamines										
CH ₅ N	methylamine	266.84	g	SEL	-1086.81	-976.83	0.24	TN		2021RUS
		266.84	l	SEL1	-1062.79	-952.81	0.45	TN		2021RUS
C ₂ H ₇ N	ethylamine	289.79	g	SEL	-1737.31	-1583.33	0.54	TN		2021RUS
		289.79	l	SEL1	-1713.25	-1559.27	0.55	TN		2021RUS
C ₃ H ₉ N	1-propylamine	320.38	l	SEL	-2365.13	-2167.15	0.41	TN		2021RUS
		320.38	g	SEL1	-2396.55	-2198.57	0.42	TN		2021RUS
	isopropylamine	304.92	l	SEL	-2354.37	-2156.39	0.52	TN		2021RUS
		304.92	g	SEL1	-2383.08	-2185.10	0.53	TN		2021RUS
C ₄ H ₁₁ N	n-butylamine	350.12	l	SEL	-3018.46	-2776.50	1.13	C		1959EVA
		350.12	g	SEL1	-3054.17	-2812.21	1.13	DvH		1959EVA
C ₂ H ₇ N	dimethylamine	280.02	g	SEL	-1769.79	-1615.81	0.42	TN		2021RUS
		280.02	l	SEL1	-1743.91	-1589.93	0.37	TN		2021RUS
C ₃ H ₉ N	methylethylamine	305.60	l	SEL	-2391.4	-2193.5	2.3	TN		2021RUS
		305.60	g	SEL1	-2420.7	-2222.8	1.1	TN		2021RUS
	trimethylamine	276.00	g	SEL	-2439.91	-2241.93	0.76	TN		2021RUS
		276.00	l	SEL1	-2417.80	-2219.82	0.87	TN		2021RUS
C ₆ H ₁₅ N	triethylamine	361.93	l	SEL	-4377.09	-4047.14	0.54	C		1966LEB
		361.93	g	SEL1	-4412.49	-4082.54	0.59	DvH		1966LEB
C ₆ H ₁₃ N	cyclohexylamine	406.98	l	SEL	-4071.3	-3785.3	1.3	C		1979STE
		406.98	g	SEL1	-4113.8	-3827.9	1.3	DvH		1979STE
Alkanenitriles										
C ₂ H ₃ N	ethanenitrile	355.05	l	SEL	-1256.29	-1190.30	0.22	TN		2021RUS
		355.05	g	SEL1	-1289.74	-1223.75	0.25	TN		2021RUS
C ₃ H ₅ N	propanenitrile	370.73	l	SEL	-1910.62	-1800.64	0.54	C		1971HAL
		370.73	g	SEL1	-1946.60	-1836.62	0.62	C		1971HAL
C ₄ H ₇ N	butanenitrile	390.77	l	SEL	-2568.68	-2414.70	0.92	C		1959EVA
		390.77	g	SEL1	-2605.67	-2451.69	0.92	DvH		1959EVA
C ₃ H ₃ N	propenenitrile	351.13	l	SEL	-1756.40	-1690.41	0.42	C		1971HAL
		351.13	g	SEL1	-1789.04	-1723.05	0.72	DvH		1971HAL
Azahydrocarbons										
C ₂ N ₂	cyanogen	252.00	g	SEL	-1097.07	-1097.07	0.41	TN		2021RUS

C ₂ H ₈ N ₂	1,1-dimethylhydrazine	252.00	l	SEL1	-1076.13	-1076.13	0.42	TN	2021RUS
		335.57	l	SEL	-1978.7	-1802.7	3.6	C	1960DON
		335.57	g	SEL1	-2013.7	-1837.7	3.6	DvH	1960DON
C ₃ H ₆ N ₆	1,3,5-triazine-2,4,6-triamine	618	s	SEL	-1972.0	-1840.0	3.0	C	1973SEL
C ₆ H ₇ N	aniline	457.30	l	SEL	-3393.06	-3239.08	0.84	C	1962HAT
		457.30	g	SEL1	-3448.27	-3294.30	0.88	DvH	1962HAT
C ₈ H ₁₁ N	N,N-dimethylaniline	466.47	l	SEL	-4767.8	-4525.8	3.2	C	1982FUR
		466.47	g	SEL1	-4820.6	-4578.7	3.2	DvH	1982FUR
		275.65	s	SEL2	-4756.2	-4514.3	3.2	DfusH	1982FUR
C ₆ H ₁₅ NO ₃	triethanolamine	623.15	l	SEL	-3840.6	-3510.6	1.5	C	1982MIN
		623.15	g	SEL1	-3946.3	-3616.3	2.7	DvH	1982MIN
Alkyl Nitrates and Nitroalkanes									
CH ₃ NO ₃	methyl nitrate	338.00	g	SEL	-700.0	-634.1	1.3	E	1959RAY
CN ₄ O ₈	tetranitromethane	397.95	l	SEL	-431.8	-431.8	1.7	C	1975LEB
		397.95	g	SEL1	-475.7	-475.7	1.7	DvH	1975LEB
		287.05	s	SEL2	-395.4	-395.4	4.3	DfusH	1975LEB
CH ₃ NO ₂	nitromethane	374.34	l	SEL	-709.25	-643.25	0.43	TN	2021RUS
		374.34	g	SEL1	-747.51	-681.51	0.46	TN	2021RUS
C ₉ H ₆ N ₂ O ₂	2,4-diisocyanatotoluene	524.83	l	SEL	-4235.0	-4103.1		C	1962STR
		524.83	g	SEL1	-4284.5	-4152.5		DvH	1962STR
		293.65	s	SEL2	-4168.0	-4036.0		est	1985CHR
C ₆ H ₅ NO ₂	nitrobenzene	483.81	l	SEL	-3082.0	-2972.0	1.4	TN	2021RUS
		483.81	g	SEL1	-3137.0	-3027.0	1.4	TN	2021RUS
C ₇ H ₅ N ₃ O ₆	trinitrotoluene	355.10	s	SEL	-3406.0	-3296.0	3.0	C	1976ROU
		621.00	g	SEL1	-3519.2	-3409.2	3.4	DsubH	1976ROU
		621.00	l	SEL2	-3493.0	-3383.0	3.6	DvH	1976ROU

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_c H$ (kJ mol ⁻¹)	$\Delta_c H(\text{net})$ (kJ mol ⁻¹)	unc	M	R	Reference
SULFUR-SUBSTITUTED HYDROCARBONS										
Akanethiols										
CH ₄ S	methanethiol	279.05	g	SEL	-1544.57	-1151.40	0.51	DvH		1961GOO
		279.05	l	SEL1	-1520.8	-1127.60	0.5	R	a	1961GOO
C ₂ H ₆ S	ethanethiol	308.19	l	SEL	-2173.17	-1736.02	0.42	C	a	1957MCC
		308.19	g	SEL1	-2200.3	-1763.32	0.43	DvH		1957MCC
Alkyl Sulfides										
CS ₂	carbon disulfide	318.35	l	SEL	-1687.2	-1076.81	0.5	C	a	1961GOO
		318.35	g	SEL1	-1714.5	-1104.46	0.5	DvH		1961GOO
C ₂ H ₆ S	dimethyl sulfide	310.42	l	SEL	-2181.45	-1656.31	0.33	C	a	1957MCC
		310.42	g	SEL1	-2208.8	-1683.70	0.59	DvH		1957MCC
C ₂ H ₆ OS	dimethyl sulfoxide	463.86	l	SEL	-2037.30	-1512.16	1.25	C	g	1994MAS
		463.86	g	SEL1	-2095.9	-1570.8	1.5	DvH		1994MAS

Formula	Compound	T_{trs} (K)	Ph	SEL	$\Delta_{\text{c}}H$ (kJ mol ⁻¹)	$\Delta_{\text{c}}H(\text{net})$ (kJ mol ⁻¹)	<i>unc</i>	M	R	Reference
HALOGENATED HYDROCARBONS										
Fluoromethanes										
CH ₃ F	fluoromethane	194.65	g	SEL	-764.86	-672.54	0.28	TN	k	2021RUS
		194.65	l	SEL1	-756.33	-664.01	1.15	DvH		2021RUS
CH ₂ F ₂	difluoromethane	221.50	g	SEL	-584.82	-488.16	0.40	TN	k	2021RUS
		221.50	l	SEL1	-566.84	-470.18	0.43	TN		2021RUS
CHF ₃	trifluoromethane	188.75	g	SEL	-374.80	-273.81		TN	k	2021RUS
		188.75	l	SEL1	-366.34	-265.35		TN		2021RUS
CF ₄	tetrafluoromethane	145.56	g	SEL	-172.55	-67.23		TN	k	2021RUS
		145.56	l	SEL1	-160.74	-55.42		DvH		2021RUS
Fluoroethanes										
C ₂ H ₅ F	fluoroethane	236.05	g	SEL	-1407.52	-1271.21	0.40	TN		2021RUS
		236.05	l	SEL1	-1391.21	-1254.89	0.47	DvH		2021RUS
C ₂ H ₄ F ₂	1,1-difluoroethane	249.13	g	SEL	-1212.14	-1071.49	0.60	DvH		2021RUS
		249.13	l	SEL1	-1193.66	-1053.01	0.88	TN		2021RUS
	1,2-difluoroethane	293.00	g	SEL	-1267.04	-1126.39	0.84	TN		2021RUS
		293.00	l	SEL1	-1245.91	-1105.26	0.98	DvH		2021RUS
C ₂ H ₃ F ₃	1,1,1-trifluoroethane	225.91	g	SEL	-1008.3	-863.4	1.7	C		1965KOL
		225.91	l	SEL1u	-988.2	-843.2	2.0	DvH		1965KOL
	1,1,2-trifluoroethane	276.81	g	SEL	-1059	-914	10	H2		1965KOL
		276.81	l	SEL1	-1035	-890	10	DvH		1965KOL
C ₂ H ₂ F ₄	1,1,1,2-tetrafluoroethane	247.08	g	SEL	-883.7	-734.4	2.0	CBS		1988BER
		247.08	l	SEL1	-865.6	-716.3	2.0	DvH		1988BER
	1,1,2,2-tetrafluoroethane	253.23	g	SEL	-910.1	-760.8	2.0	CBS		1988BER
		253.23	l	SEL1	-890.3	-741.0	2.3	G3B3		2016GOO
C ₂ HF ₅	pentafluoroethane	225.06	g	SEL	-701	-547	8	G3B3		2016GOO
		225.06	l	SEL1	-687	-533	8	DvH		2016GOO
C ₂ F ₆	hexafluoroethane	194.87	g	SEL	-513.33	-355.34		TN		2021RUS
		194.87	l	SEL1	-506.33	-348.34		DvH		2021RUS
Fluoroethenes										
C ₂ H ₃ F	fluoroethene	201.65	g	SEL	-1251.41	-1159.09	0.48	TN		2021RUS
		201.65	l	SEL1	-1240.61	-1148.29	0.90	DvH		2021RUS
C ₂ H ₂ F ₂	1,1-difluoroethene	190.15	g	SEL	-1078.41	-981.75	0.80	TN		2021RUS
		190.15	l	SEL1	-1074.0	-977.3	2.7	DvH		2021RUS
	1,2-difluoroethene, (<i>E</i>)-	231.15	g	SEL	-1123.0	-1026.3	0.9	TN		2021RUS
		231.15	l	SEL1	-1108.05	-1011.39	1.39	DvH		2021RUS

	1,2-difluoroethene, (Z)-	253.15	g	SEL	-1119.4	-1022.7	0.9	TN	2021RUS
		253.15	l	SEL1	-1105.6	-1008.9	1.3	DvH	2021RUS
C ₂ HF ₃	trifluoroethene	211.70	g	SEL	-966.2	-865.2	1.6	TN	2021RUS
		211.70	l	SEL1	-954.8	-853.8	1.9	DvH	2021RUS
C ₂ F ₄	tetrafluoroethene	197.51	g	SEL	-825.10	-719.77	0.61	TN	2021RUS
		197.51	l	SEL1	-819.00	-713.67	1.44	DvH	2021RUS
Chloromethanes									
CH ₃ Cl	chloromethane	249.00	g	SEL	-763.51	-645.24	0.39	TN	2021RUS
		249.00	l	SEL1	-743.61	-625.34	0.25	TN	2021RUS
CH ₂ Cl ₂	dichloromethane	312.92	g	SEL	-633.00	-484.44	0.52	TN	2021RUS
		312.92	l	SEL1	-603.99	-455.43	0.54	TN	2021RUS
CHCl ₃	trichloromethane	334.42	g	SEL	-504.94	-326.09	0.50	TN	2021RUS
		334.42	l	SEL1	-473.54	-294.69	0.51	TN	2021RUS
CCl ₄	tetrachloromethane	349.65	g	SEL	-392.02	-182.89	0.43	TN	2021RUS
		349.65	l	SEL1	-359.52	-150.39	0.43	TN	2021RUS
Chloroethanes									
C ₂ H ₅ Cl	chloroethane	285.40	g	SEL	-1413.57	-1251.30	0.26	TN	2021RUS
		285.40	l	SEL1	-1389.14	-1226.87	0.40	TN	2021RUS
C ₂ H ₄ Cl ₂	1,1-dichloroethane	330.42	g	SEL	-1272.61	-1080.06	0.44	TN	2021RUS
		330.42	l	SEL1	-1241.81	-1049.26	0.48	TN	2021RUS
	1,2-dichloroethane	356.70	l	SEL	-1238.5	-1045.9	3.5	R	2002MAN
		356.70	g	SEL1	-1273.7	-1081.1	3.5	R	2002MAN
C ₂ H ₃ Cl ₃	1,1,1-trichloroethane	347.22	l	SEL	-1109.1	-886.3	2.0	R	2002MAN
		347.22	g	SEL1	-1141.7	-918.9	2.0	R	2002MAN
	1,1,2-trichloroethane	386.78	l	SEL	-1098.0	-875.2	4.0	R	2002MAN
		386.78	g	SEL1	-1138.3	-915.5	4.0	R	2002MAN
C ₂ H ₂ Cl ₄	1,1,1,2-tetrachloroethane	403.33	l	SEL	-973.5	-720.4	2.3	R	2002MAN
		403.33	g	SEL1	-1014.6	-761.5	2.4	R	2002MAN
	1,1,2,2-tetrachloroethane	419.25	l	SEL	-964.5	-711.4	3.5	R	2002MAN
		419.25	g	SEL1	-1010.2	-757.1	3.5	R	2002MAN
C ₂ HCl ₅	pentachloroethane	433.45	l	SEL	-844.3	-560.9	4.0	R	2002MAN
		433.45	g	SEL1	-891.7	-608.3	4.3	R	2002MAN
C ₂ Cl ₆	hexachloroethane	462.96	l	SEL	-711.7	-398.0	2.8	TN	2021RUS
		462.96	g	SEL1	-778.8	-465.1	1.3	TN	2021RUS
Chloroethenes									
C ₂ H ₃ Cl	chloroethene	259.35	l	SEL	-1241.13	-1122.86	0.36	TN	2021RUS
		259.35	g	SEL1	-1260.93	-1142.66	0.30	TN	2021RUS
C ₂ H ₂ Cl ₂	1,1-dichloroethene	304.71	l	SEL	-1096.41	-947.85	0.57	TN	2021RUS

1,2-dichloroethene, (E)-	304.71	g	SEL1	-1123.16	-974.60	0.57	TN	2021RUS
	320.73	l	SEL	-1119.74	-971.18	0.54	DvH	2021RUS
	320.73	g	SEL1	-1149.24	-1000.68	1.14	TN	2021RUS
1,2-dichloroethene, (Z)-	333.58	l	SEL	-1086.23	-937.67	1.14	TN	2021RUS
	333.58	g	SEL1	-1117.33	-968.77	0.54	TN	2021RUS
C ₂ HCl ₃ trichloroethene	360.50	l	SEL	-951.8	-772.9	1.5	TN	2021RUS
	360.50	g	SEL1	-986.4	-807.5	1.5	TN	2021RUS
C ₂ Cl ₄ tetrachloroethene	394.29	l	SEL	-818.2	-609.1	1.1	TN	2021RUS
	394.29	g	SEL1	-857.9	-648.8	1.1	TN	2021RUS
Chloroethynes								
C ₂ HCl chloroacetylene	243.6	g	SEL	-1182.39	-1108.12	0.70	TN	2021RUS
	243.6	l	SEL1	-1162.4	-1088.1	1.2	DvH	2021RUS
C ₂ Cl ₂ dichloroacetylene	306.0	g	SEL	-1067.47		0.94	TN	2021RUS
	306.0	l	SEL1	-1040.1		1.5	DvH	2021RUS
Bromomethanes								
CH ₃ Br bromomethane	276.70	g	SEL	-763.80	-635.15	0.26	TN	2021RUS
	276.70	l	SEL1	-740.47	-611.82	0.27	TN	2021RUS
CH ₂ Br ₂ dibromomethane	370.10	l	SEL	-601.3	-432.0	1.1	TN	2021RUS
	370.10	g	SEL1	-638.3	-469.0	1.1	TN	2021RUS
CHBr ₃ tribromomethane	422.36	l	SEL	-471.5	-261.5	1.3	TN	2021RUS
	422.36	g	SEL1	-517.6	-307.6	1.3	TN	2021RUS
CBr ₄ tetrabromomethane	462.55	l	SEL	-351.6	-100.9	1.4	TN	2021RUS
	462.55	g	SEL1	-406.1	-155.4	1.3	TN	2021RUS
Bromoethanes								
C ₂ H ₅ Br bromoethane	311.54	l	SEL	-1387.66	-1215.02	0.26	TN	2021RUS
	311.54	g	SEL1	-1415.65	-1243.01	0.25	TN	2021RUS
Bromoethenes								
C ₂ H ₃ Br bromoethene	288.60	g	SEL	-1266.79	-1138.13	0.57	TN	2021RUS
	288.60	l	SEL1	-1233.0	-1104.4	1.8	DvH	2021RUS
Chlorofluoromethanes								
CH ₂ ClF chlorofluoromethane	264.00	g	SEL	-617.7	-495.1	1.1	TN	2021RUS
	264.00	l	SEL1	-599.1	-476.5	1.5	DvH	2021RUS
CHClF ₂ chlorodifluoromethane	232.35	g	SEL	-433.99	-307.05	1.01	TN	2021RUS
	232.35	l	SEL1	-418.29	-291.35	1.05	DvH	2021RUS
CHCl ₂ F dichlorofluoromethane	282.05	g	SEL	-478.01	-325.12	0.92	TN	2021RUS
	282.05	l	SEL1	-454.4	-301.5	1.5	DvH	2021RUS
CClF ₃ chlorotrifluoromethane	191.75	g	SEL	-241.78	-110.50	0.63	TN	2021RUS
	191.75	l	SEL1	-225.28	-94.00	0.70	TN	2021RUS

CCl ₂ F ₂	dichlorodifluoromethane	243.35	g	SEL	-303.18	-145.95	0.71	TN	2021RUS
		243.35	l	SEL1	-286.38	-129.15	0.73	DvH	2021RUS
CCl ₃ F	trichlorofluoromethane	296.95	g	SEL	-353.80	-170.62	0.80	TN	2021RUS
		296.95	l	SEL1	-329.01	-145.83	0.81	TN	2021RUS
Chlorofluoroethenes									
C ₂ ClF ₃	chlorotrifluoroethene	244.80	g	SEL	-833.2	-701.9	2.8	DvH	1982ERA
		244.80	l	SEL1	-816.0	-684.7	2.8	C	1982ERA
Bromofluoromethanes									
CH ₂ BrF	bromofluoromethane	310.00	l	SEL	-591.3	-458.3	5.2	DvH	2021RUS
		310.00	g	SEL1	-622.7	-489.7	4.9	TN	2021RUS
CHBrF ₂	bromodifluoromethane	258.65	g	SEL	-445.5	-308.2	1.2	TN	2021RUS
		258.65	l	SEL1	-425.5	-288.2	1.6	DvH	2021RUS
CHBr ₂ F	dibromofluoromethane	338.05	l	SEL	-456.9	-283.2	5.2	DvH	2021RUS
		338.05	g	SEL1	-489.6	-316.0	4.9	TN	2021RUS
CBrF ₃	bromotrifluoromethane	215.35	g	SEL	-254.08	-112.43	0.52	TN	2021RUS
		215.35	l	SEL1	-242.31	-100.65	0.53	TN	2021RUS
CBr ₂ F ₂	dibromodifluoromethane	297.65	g	SEL	-322.1	-144.2	1.3	TN	2021RUS
		297.65	l	SEL1	-297.7	-119.7	1.3	DvH	2021RUS
CBr ₃ F	tribromofluoromethane	379.65	l	SEL	-324.4	-110.0	3.0	DvH	2021RUS
		379.65	g	SEL1	-373.7	-159.4	1.5	TN	2021RUS
Bromochloromethanes									
CH ₂ BrCl	bromochloromethane	341.21	l	SEL	-605.1	-446.2	1.3	DvH	2021RUS
		341.21	g	SEL1	-637.8	-478.9	1.3	TN	2021RUS
CHBrCl ₂	bromodichloromethane	362.65	l	SEL	-476.6	-287.3	2.2	DvH	2021RUS
		362.65	g	SEL1	-511.6	-322.3	1.2	TN	2021RUS
CHBr ₂ Cl	dibromochloromethane	393.00	l	SEL	-470.4	-270.8	4.1	DvH	2021RUS
		393.00	g	SEL1	-514.3	-314.7	3.4	TN	2021RUS
CBrCl ₃	bromotrichloromethane	376.65	l	SEL	-363.8	-144.3	1.6	DvH	2021RUS
		376.65	g	SEL1	-399.40	-179.89	0.59	TN	2021RUS
CBr ₂ Cl ₂	dibromodichloromethane	423.35	l	SEL	-360.5	-130.6	4.2	DvH	2021RUS
		423.35	g	SEL1	-400.0	-170.2	1.3	TN	2021RUS
CBr ₃ Cl	tribromochloromethane	433.00	l	SEL	-355.0	-114.7	2.9	DvH	2021RUS
		433.00	g	SEL1	-403.0	-162.7	1.5	TN	2021RUS
Bromochlorofluoromethanes									
CHBrClF	bromochlorofluoromethane	312.60	l	SEL	-456.7	-293.5	5.1	DvH	2021RUS
		312.60	g	SEL1	-484.4	-321.1	4.9	TN	2021RUS
CBrClF ₂	bromochlorodifluoromethane	270.65	g	SEL	-310.6	-143.0	4.9	TN	2021RUS
		270.65	l	SEL1	-289.9	-122.3	4.9	DvH	2021RUS

CBrCl ₂ F	bromodichlorofluoromethane	325.15	l	SEL	-326.8	-133.2	5.2	DvH	2021RUS
		325.15	g	SEL1	-357.6	-164.0	4.9	TN	2021RUS
CBr ₂ ClF	dibromochlorofluoromethane	352.65	l	SEL	-327.2	-123.3	5.2	DvH	2021RUS
		352.65	g	SEL1	-362.7	-158.8	4.9	TN	2021RUS
Carbonyl Halides									
CCl ₂ O	phosgene	280.71	g	SEL	-221.41	-116.85	0.27	TN	2021RUS
		280.71	l	SEL1	-198.58	-94.01	0.47	DvH	2021RUS
Pentachlorophenol									
C ₆ HCl ₅ O	pentachlorophenol	463.84	s	SEL	-2329.0	-2045.6	2.9	C	1958SIN
		582.50	l	SEL1	-2346.1	-2062.7	3.0	DsubH	1958SIN
		582.50	g	SEL2	-2420.6	-2137.2	3.0	DvH	1958SIN

4.5. References for Heats of Combustion, Formation, and Transition

Reference	Source
[1901BER]	Berthelot, M.; Chimie Organique. - Nouvelles recherches sur l'isomerie des ethers sulfocyaniques. Compt. Rend., 1901, 132, 57-58
[1901ZUB]	Zubova, P.; Data about heat of combustion of compound cycle structure. Zh. Fiz. Khim., 1901, 33, 708-722
[1906SCH]	Schmidlin, M.J.; Recherches chimiques et thermochimiques sur la constitution des rosanilines. Ann. Chim. Phys., 1906, 1, 195-256
[1907LEM]	Lemoult, M.P.; Recherches theoriques et experimentales sur les chaleurs de combustion et de formation des composes organiques. Ann. Chim. Phys., 1907, 12, 395-432
[1909LEM]	Lemoult, M.P.; Jungfleisch, M.E.; Thermochimie. - Comparaisons entre les nitriles et les carbylamines. Compt. Rend., 1909, 148, 1602-1604
[1910AUW]	Auwers, K.; Roth, W.A.; Eisenlohr, F., III. Verbrennungswarmen von Terpenen und Styrolen, Justus Liebigs Ann. Chem., 1910, 267-290
[1910MUL]	Muller, J.-A.; Sur les chaleurs de combustion et les poids specifiques des methylamines. Ann. Chim. Phys., 1910, 20, 116-130
[1910RIC]	Richards, T.W.; Jesse, R.H., Jr.; The heats of combustion of the octanes and xylenes. J. Am. Chem. Soc., 1910, 32, 268-298
[1910SWI]	Swientoslawski, W.; Thermochemische Untersuchungen der organischen Verbindungen. Dritte Mitteilung. Stickstoffhaltige Verbindungen. Z. Phys. Chem., 1910, 72, 49-83
[1911EME]	Emery, A.G.; Benedict, F.G.; The heat of combustion of compounds of physiological importance. Am. J. Physiol., 1911, 28, 301-307
[1911LEM]	Lemoult, M.P., Thermochimie. - Recherches sur les derives du styrolene; rectification de quelques erreurs experimentales, Compt. Rend., 1911, 152, 1402-1404
[1911WRE]	Wrede, F., Uber die Bestimmung von Verbrennungswarmen mittels der kalorimetrischen Bombe unter Benutzung des Platinwiderstandsthermometers, Z. Phys. Chem. (Leipzig), 1911, 81-94
[1913ROT]	Roth, W.A.; Ostling, G.J., Thermochemische Untersuchungen in der alicyclischen Reihe, Chem. Ber., 1913, 46, 309-327
[1914MOU]	Moureu, C.; Andre, E., Thermochimie des composes acetyleniques, Ann. Chim. Phys., 1914, 1, 113-145
[1915RIC]	Richards, T.W.; Barry, F.; The heats of combustion of aromatic hydrocarbons and hexamethylene. J. Am. Chem. Soc., 1915, 37, 993-1020
[1918GUI]	Guinchant, M.J.; Etude sur la fonction acide dans les derives metheniques et methiniques. Ann. Chem., 1918, 10, 30-84
[1920FAJ]	Fajans, K.; Die Energie der Atombindungen im Diamanten und in aliphatischen Kohlenwasserstoffen. Chem. Ber., 1920, 53, 643-665
[1920MOU]	Moureu, C.; Boutaric, A.; Some physico-chemical constants of acrylic acid. J. Chim. Phys., 1920, 18, 348-352
[1920RIC]	Richards, T.W.; Davis, H.S.; The heats of combustion of benzene, toluene, aliphatic alcohols, cyclohexanol, and other carbon compounds. J. Am. Chem. Soc., 1920, 42, 1599-1617

- [1921GAR] Garner, W.E.; Abernethy, C.L.; Heats of combustion and formation of nitro-compounds. Part I. - Benzene, toluene, phenol and methylaniline series. Proc. Roy. Soc. London A, 1921, 99, 213-235
- [1922AUW] Auwers, K.v.; Kolligs, H., Oberparapräparate verschiedener Herkunft von Cymol und Äthylbenzol, Ber. Chem. Ges., 1922, 3872-3879
- [1923MOR] F. S. Mortimer and R. V. Murphy, Ind. Eng. Chem. 15, 1140 (1923)
- [1925BAR] Barker, M.F.; Calorific value and constitution. J. Phys. Chem., 1925, 29, 1345-1363
- [1925PAR] Parks, G.S.; Thermal data on organic compounds I. The heat capacities and free energies of methyl, ethyl and normal-butyl alcohols. J. Am. Chem. Soc., 1925, 47, 338-345
- [1925ROT] Roth, W.A.; Lasse, R.; Verbrennungswärme der Dekahydro-naphthaline und der Dekalone. Justus Liebigs Ann. Chem., 1925, 441, 48-53
- [1925WAR] Wartenberg, H.; Lerner-Steinberg; Heat of formation of formaldehyde. Z. Angew. Chem., 1925, 38, 591-592
- [1926HUC] Huckel, W.; Mentzel, R.; Zur Stereochemie bicyclischer Ringsteme II. Die Stereoisomerie des Dekahydronaphthalins und seiner Derivate II. Stereoisomere «beta»-substituierte Dekaline. Ann., 1926, 451, 109-132
- [1926MAT] Mathews, J.H.; The accurate measurement of heats of vaporization of liquids. J. Am. Chem. Soc., 1926, 48, 562-576
- [1927NAG] Nagornov, N.N.; Ann. Inst. Anal. Phys. Chem. (Leningrad), 1927, 3, 562
- [1928VER] Verkade, P.E.; Coops, J., Jr.; Further remarks on the method proposed by Swietoslawski for the correction of the older thermochemical data. Recl. Trav. Chim. Pays-Bas, 1928, 47, 701-708
- [1929HER] Herz, W.; Lorentz, E.; Physikalisch-chemische Untersuchungen an Dioxan. Z. Phys. Chem., 1929, 140, 406-422
- [1929KEL] Kelley, K.K.; The heat capacities of ethyl and hexyl alcohols from 16°K. to 298°K. and the corresponding entropies and free energies and free energies. J. Am. Chem. Soc., 1929, 51, 779-781
- [1929LAN] Landrieu, P.; Baylocq, F.; Johnson, J.R.; Etude thermochimique dans la série furanique. Bull. Soc. Chim. France, 1929, 45, 36-49
- [1929MAT] Matheson, G. L.; Maas, O.; The Properties of Pure Hydrogen Peroxide. VI, J. Am. Chem. Soc., 1929, 51, 674-687
- [1929ROT] Roth, W.A.; Müller, Fr.; Die Zersetzungswärme der Stickstoffwasserstoffsäure. Chem. Ber., 1929, 62, 1188-1194
- [1929SWI] Swietoslawski, M.W.; Bobinska, M.J., O ciepłoty spalania kamfory, azobenzenu i hydrazobenzenu - Sur la chaleur de combustion du camphre, de l'hydrazobenzène, Roczn. Chem., 1929, 9, 723-730
- [1930RIN] Rinkenbach, W.H., The heats of combustion and formation of aromatic nitro compounds, J. Am. Chem. Soc., 1930, 52, 115-120
- [1930ROT] W. Roth, R. Grau, and A. Z. Meichsner, Beiträge zur Thermochemie des Schwefels. III, Z. Anorg. Allg. Chem., 1930, 193, 161-175
- [1931BUR] Burriel, F., 7. Estudio físico-químico de algunos compuestos orgánicos sólidos a la temperatura ordinaria, que se proponen como patrones de temperatura, An. R. Soc. Esp. Fis. Quím., 1931, 29, 89-125
- [1931KEF] Keffler, L.J.P., Recherches calorimétriques sur les substances étalons, J. Chim. Phys., 1931, 28, 457-469
- [1931ROS] Rossini, F.D.; The heats of combustion of methane and carbon monoxide. J. Res. Nat. Bur. Stand., 1931, 6, 37-49

- [1932MIL] Milone, M.; Rossignoli, P., Sul calore di combustione di alcune miscele di composti organici, Gazz. Chim. Ital., 1932, 62, 644-655
- [1932ROS] Rossini, F.D.; The heats of combustion of methyl and ethyl alcohols. J. Res. Nat. Bur. Stand., 1932, 8, 119-139
- [1932ROT] Roth, W.A.; Banse, H., Die verbrennungs- und bildungswarme von kohlenoxyd und methan, Arch. Eisenhutten., 1932, 6, 43-46
- [1932SHC] Shchukarev, A.N.; Shchukareva, L.A., Heat standards, Russ. J. Phys. Chem., 1932, 3, 169-174
- [1933BAN] Banse, H.; Parks, G.S., Thermal data on organic compounds. XII. The heats of combustion of nine hydrocarbons, J. Am. Chem. Soc., 1933, 55, 3223-3227
- [1933FEN] R. W. Fenning and F. T. Cotton, A Bomb Calorimeter Determination of the Heats of Formation of Nitrous Oxide and Carbon Dioxide, Proc. Roy. Soc. London A, 1933, 141, 17-28
- [1933HEI] Heim, G.; Vapor tensions and latent heat of vaporization of some normal nitriles, Bull. Soc. Chim. Belg., 1933, 42, 467-482
- [1933MCM] McMorris, J.; Badger, R.M.; The heat of combustion, entropy and free energy of cyanogen gas. J. Am. Chem. Soc., 1933, 55, 1952-1957
- [1933ROT] Roth, W.A.; Meyer, I.; Einige physikalisch-chemische konstanten des dioxans. Z. Electrochem., 1933, 39, 35-37
- [1933WAR] Wartenberg, H.V.; Schutza, H.; Die verbrennungswarme des cyans. Z. Phys. Chem., 1933, 164, 386-388
- [1934HIR] Hirsbrunner, H., Uber das gleichgewicht der thermischen dissoziation der salicylsaure, Helv. Chim. Acta, 1934, 17, 477-504
- [1934ROS] Rossini, F.D.; Calorimetric determination of the heats of combustion of ethane, propane, normal butane, and normal pentane. J. Res. Nat. Bur. Stand., 1934, 12, 735-750
- [1935BRU] Brull, L.; Sui calori di combustione di alcuni derivati del bifenil. Gazz. Chim. Ital., 1935, 65, 19-28
- [1935HUF] Huffman, H.M.; Ellis, E.L., Thermal Data. II. The heats of combustion of L-cysteine, of L-cystine, β -thiolactic acid and β,β' -dithiodilactic acid, J. Am. Chem. Soc., 1935, 57, 41-46
- [1935SCH] Schjanberg, E.; Die Verbrennungswarmen und die Refraktionsdaten einiger chloresubstituierter Fettsauren und Ester. Z. Phys. Chem. Abt. A, 1935, 172, 197-233
- [1936ROT] Roth, W.A.; Pahlke, H., Sekundare eichsubstanz fur verbrennungscalorimeter fur gase und dampfe. Die verbrennungswarme von isopentandampf, Angew. Chem., 1936, 49, 618-619
- [1937AST] Aston, J.G.; Siller, C.W.; Messerly, G.H.; Heat capacities and entropies of organic compounds. III. Methylamine from 11.5°K. to the boiling point. Heat of vaporization and vapor pressure. The entropy from molecular data. J. Am. Chem. Soc., 1937, 59, 1743
- [1937GAL] Gallaughier, A.F.; Hibbert, H.; Studies on reactions relating to carbohydrates and polysaccharides. LV. Vapor pressures of the polyethylene glycols and their derivatives. J. Am. Chem. Soc., 1937, 59, 2521-2525
- [1937GAR] Gardner, George S.; Brewer, J. Edward; Vapor Pressure of Commerical High-Boiling Organic Solvents. Ind. Eng. Chem., 1937, 29, 179-181
- [1937HUC] Huckel, W.; Kamenz, E.; Gross, A.; Tappe, W.; Zur kenntnis der waldenschen umkehrung. Ann. Chim., 1937, 533, 16438

- [1937JES] Jessup, R.S., Heats of combustion of the liquid normal paraffin hydrocarbons from hexane to dodecane, J. Res. Nat. Bur. Stand., 1937, 18, 114-128
- [1937MEN] G. D. Mendenhall, D. M. Golden, and S. W. Benson, J. Phys. Chem., 1973, 77, 2707
- [1937MOU] Moureu, H.; Dode, M.; Chaleurs de formation de l'oxyde d'ethylene, de l'ethanediol et de quelques homologues. Bull. Soc. Chim. France, 1937, 4, 637-647
- [1937ROS] Rossini, F.d.; Knowlton, J.W., Calorimetric determination of the heats of combustion of ethylene and propylene, J. Res. Nat. Bur. Stand., 1937, 19, 249-262
- [1938CON] J. B. Conn, G. B. Kistiakowsky, and E. A. Smith, J. Am. Chem. Soc., 1938, 60, 2764
- [1938CRE] Crenshaw, J.L.; Cope, A.C.; Finkelstein, N.; Rogan, R.; The dioxanates of the mercuric halides. J. Am. Chem. Soc., 1938, 60, 2308-2323
- [1938EFT] E. Efring, Förbränningsvärmen för några alifatska kolväten, Thesis, Lund University, Stockholm (1938)
- [1938EG] Egan, C.J.; Kemp, J.D., Methyl bromide. The heat capacity, vapor pressure, heats of transition, fusion and vaporization. Entropy and density of the gas, J. Am. Chem. Soc., 1938, 60, 2097-2101
- [1938JES] Jessup, R.S., Heat of combustion of isoprene, J. Res. Nat. Bur. Stand., 1938, 20, 589-597
- [1938LUS] Luschinsky, W.V., Über verbrennungswärmen von mono- und polystyrol, Z. Phys. Chem., 1938, 182, 384-388
- [1939AST] Aston, J.G.; Eidinoff, M.L.; Forster, W.S.; The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of dimethylamine. J. Am. Chem. Soc., 1939, 61, 1539
- [1939BAD] Badoche, M.M., Determinations des chaleurs de combustion de derives nitres de la serie benzenique, Bull. Soc. Chim. France, 1939, 6, 570-579
- [1939BUR] Burlot, E., Etude relative a la calorimetrie des explosifs, Mem. Poudres, 1939, 29, 226-260
- [1939CLA] Clarke, T.H.; Stegeman, G., Heats of combustion of some mono- and disaccharides, J. Am. Chem. Soc., 1939, 61, 1726-1730
- [1939HUG] Hughes, A. M.; Corruccini, R. J.; Gilbert, E. C.; Studies on Hydrazine: The Heat of Formation of Hydrazine and of Hydrazine Hydrate, J. Am. Chem. Soc., 1939, 61, 2639-2642
- [1939KNO] Knowlton, J.W.; Rossini, F.D., Heats of combustion of tetramethylmethane and 2-methylbutane, J. Res. Nat. Bur. Stand., 1939, 22, 415-424
- [1939MAT] Matsui, M.; Abe, T., Arbeitsweise eines adiabatischen calormeters, Bull. Tokyo Univ. Eng., 1939, 8, 339
- [1939MOO] Moore, G.E.; Parks, G.S.; New thermodynamic data for the cyclohexane-methylcyclopentane isomerization. J. Am. Chem. Soc., 1939, 61, 2561-2562
- [1939PAT] Patnode, Winton.; Scheiber, W.J., The Density, Thermal Expansion, Vapor Pressure, and Refractive Index of Styrene, and the Density and Thermal Expansion of Polystyrene, J. Am. Chem. Soc., 1939, 61, 3449-3451
- [1939RIC] Richardson, J.W.; Parks, G.S., Thermal data on organic compounds. XIX. Modern combustion data for some non-volatile compounds containing carbon, hydrogen and oxygen, J. Am. Chem. Soc., 1939, 61, 3543-3546
- [1939ROS] Rossini, F. D.; Heat and free energy of formation of water and of carbon monoxide, J. Res. Nat. Bur. Stand. 22, 407 (1939)

- [1940MES] Messerly, G.H.; Kennedy, R.M.; The heat capacity and entropy, heats of fusion and vaporization and the vapor pressure of n-pentane. J. Am. Chem. Soc., 1940, 62, 2988-2991
- [1940MOO] Moore, G.E.; Renquist, M.L.; Parks, G.S.; Thermal data on organic compounds. XX. Modern combustion data for two methylnonanes, methyl ethyl ketone, thiophene and six cycloparaffins. J. Am. Chem. Soc., 1940, 62, 1505-1507
- [1941BAD] Badoche, M., No 19. - Chaleurs de combustion du phenol, du-m-cresol et del leurs ethers; Bull. Soc. Chim. Fr., 1941, 8, 212-220
- [1941DAV] Davies, G.F.; Gilbert, E.C., Heats of combustion and formation of the nine isomeric heptanes in the liquid state, J. Am. Chem. Soc., 1941, 63, 2730-2732
- [1941JON] Jones, W.H.; Starr, C.E., Jr., Determination of the heat of combustion of gasolines, Ind. Eng. Chem., 1941, 13, 287-290
- [1941LIS] Lister, M.W., Heats of organic reactions. X. Heats of bromination of cyclic olefins, J. Am. Chem. Soc., 1941, 63, 143-149
- [1941MIL] Miles, C.B.; Hunt, H., Heats of combustion. I. The heat of combustion of acetone, J. Phys. Chem., 1941, 45, 1346-1359
- [1941STO] Stout, J.W.; Fisher, Leon H.; The Entropy of Formic Acid. The Heat Capacity from 15 to 300°K. Heats of Fusion and Vaporization, J. Chem. Phys., 1941, 9, 163
- [1942AND] Anderson, C.M.; Gilbert, E.C.; The apparent energy of the N-N bond as calculated from heats of combustion. J. Am. Chem. Soc., 1942, 64, 2369-2372
- [1942CRO] Crog, R.S.; Hunt, H.; Heats of combustion. II. The heats of combustion of ethyl methyl ketone and ethylene oxide. J. Phys. Chem., 1942, 46, 1162-1163
- [1942DEL] Delepine, M.; Badoche, M.; Thermochemie de l'aldehyde formique, de l'hexamethylene-tetramine et de ses derive. C. R. Acad. Sci. Paris, 1942, 214, 777-780
- [1942JES] Jessup, R.S., Heat of combustion of benzoic acid, with special reference to the standardization of bomb calorimeters, J. Res. Nat. Bur. Stand., 1942, 29, 247-270
- [1942JUN] Jung, G.; Dahmlos, J.; Thermochemische untersuchungen zur frage der drehbarkeit um C-C-bindunge. Z. Phys. Chem., 1942, 190, 230-240
- [1942SCH] Schumann, S.C.; Aston, J.G.; Sagenkahn, M., The heat capacity and entropy, heats of fusion and vaporization and the vapor pressures of isopentane, J. Am. Chem. Soc., 1942, 64, 1039-1043
- [1943AST] Aston, J.G.; Fink, H.L.; Schumann, S.C., The heat capacity and entropy, heats of transition, fusion and vaporization and the vapor pressures of cyclopentane. Evidence for a non-planar structure, J. Am. Chem. Soc., 1943, 65, 341-346
- [1943LEM] Lemons, J. F.; Felsing, W.A.; The Heats of Vaporization of Some Hexanes, J. Am. Chem. Soc., 1943, 65, 46-48
- [1943WAL] Walker, J.F.; Carlisle, P.J., Trioxane, Chem. Eng. News, 1943, 21, 1250-1251
- [1944PRO] Prosen, E.J.; Rossini, F.D.; Heats of combustion of eight normal paraffin hydrocarbons in the liquid state. J. Res. Nat. Bur. Stand., 1944, 33, 255-272
- [1944ROT] Roth, W.A.; Isecke, K.; Die verbrennungs- und bildungswarme des tetranitromethans. Chem. Ber., 1944, 77, 537-539
- [1945DAV] Davis, H.S.; Wiedeman, O.F.; Physical properties of acrylonitrile. Ind. Chim. Belge, 1945, 37, 482-485

- [1945PRO] Prosen, E.J.; Rossini, F.D., Heats of combustion and formation of the paraffin hydrocarbons at 25° C, J. Res. Nat. Bur. Stand., 1945, 34, 263-267
- [1945PRO2] Prosen, E.J.; Gilmont, R.; Rossini, F.D.; Heats of combustion of benzene, toluene, ethylbenzene, o-xylene, m-xylene, p-xylene, n-propylbenzene, and styrene. J. Res. Nat. Bur. Stand., 1945, 34, 65-70
- [1945WAG] Wagman, D.D.; Kilpatrick, J.E.; Pitzer, K.S.; Rossini, F.D., Heats, equilibrium constants, and free energies of formation of the acetylene hydrocarbons through the pentynes, to 1,500° K, J. Res. Nat. Bur. Stand., 1945, 35, 467-496
- [1945WIL] Willingham, C.B.; Taylor, W.J.; Pignocco, J.M.; Rossini, F.D., Vapor pressures and boiling points of some paraffin, alkylcyclopentane, alkylcyclohexane, and alkylbenzene hydrocarbons, J. Res. Nat. Bur. Stand., 1945, 35, 3, 219-17
- [1946COO] Coops, J.; Mulder, D.; Dienske, J.W.; Smittenberg, J., The heats of combustion of a number of hydrocarbons, Rec. Trav. Chim. Pays/Bas, 1946, 65, 128
- [1946DOU] Douglas, T.B.; Heats of formation of liquid methyl sulfoxide and crystalline methyl sulfone at 18°. J. Am. Chem. Soc., 1946, 68, 1072-1076
- [1946JES] Jessup, R.S., Note on the density and heat of combustion of benzoic acid, J. Res. Nat. Bur. Stand., 1946, 36, 421-423
- [1946PAR] Parks, G.S.; West, T.J.; Naylor, B.F.; Fujii, P.S.; McClaine, L.A.; Thermal data on organic compounds. XXIII. Modern combustion data for fourteen hydrocarbons and five polyhydroxy alcohols. J. Am. Chem. Soc., 1946, 68, 2524-2527
- [1946PIT] Pitzer, K.S.; Guttman, L.; Westrum, E.F., Jr., The heat capacity, heats of fusion and vaporization, vapor pressure, entropy, vibration frequencies and barrier to internal rotation of styrene, J. Am. Chem. Soc., 1946, 68, 2209-22
- [1946PRO] Prosen, E.J.; Johnson, W.H.; Rossini, F.D.; Heats of formation and combustion of the normal alkylcyclopentanes and cyclohexanes and the increment per CH₂ group for several homologous series of hydrocarbons. J. Res. Nat. Bur. Stand., 1946, 37, 51-56
- [1947COO] Coops, J.; Mulder, D.; Dienske, J.W.; Smittenberg, J.; Researches on heat of combustion IV. Technique for the determination of the heats of combustion of volatile liquids. Recl. Trav. Chim. Pays-Bas, 1947, 66, 153-160
- [1947JON] Jones, W.M.; Giauque, W.F.; The entropy of nitromethane. Heat capacity of solid and liquid. Vapor pressure, heats of fusion and vaporization. J. Am. Chem. Soc., 1947, 69, 983-987
- [1947KET] Ketelaar, J.A.A.; Van Velden, P.F.; Zalm, P., Vapour pressures and heats of vaporization of cis- and trans-1,2-dichloroethene, Rec. Trav. Chim. Pays/Bas, 1947, 66, 27
- [1947MCC] McClaine, L.A.; Thermodynamic data for some compounds containing carbon, hydrogen and oxygen. Ph.D. Thesis, Stanford University, 1947
- [1947OSB] Osborne, N.S.; Ginnings, D.C.; Measurements of heat of vaporization and heat capacity of a number of hydrocarbons. J. Res. Nat. Bur. Stand., 1947, 39, 453-477
- [1947ROB] Roberts, D.E.; Walton, W.W.; Jessup, R.S., Heats of combustion and solution of liquid styrene and solid polystyrene, and the heat of polymerization of styrene, J. Polym. Sci. Part A, 1947, 2, 420-431
- [1947SIM] Simonetta, M.; Il problema termico nella catena di reazioni tra ossido di etilene ed alcool metilico. Chimi. Ind. (Milan), 1947, 29, 37-39

- [1947SPI] Spitzer, R.; Huffman, H.M.; The heats of combustion of cyclopentane, cyclohexane, cycloheptane and cyclooctane. J. Am. Chem. Soc., 1947, 69, 211-213
- [1948DOU] Douglas, T.B.; Vapor pressure of methyl sulfoxide from 20 to 50°. Calculation of the heat of vaporization. J. Am. Chem. Soc., 1948, 70, 2001-2020
- [1948REP] Reppe, W.; Schlichting, O.; Klager, K.; Toepel, T., Cyclisierende Polymerisation von Acetylen I Über Cyclooctatetraen, Justus Liebigs Ann. Chem., 1948, 1-93
- [1949DRE] Dreisbach, R.R.; Shrader, S.A.; Vapor Pressure--Temperature Data on Some Organic Compounds. Ind. Eng. Chem., 1949, 41, 2879-2880
- [1949EPS] Epstein, M.B.; Pitzer, K.S.; Rossini, F.D.; Heats, equilibrium constants, and free energies of formation of cyclopentene and cyclohexene. J. Res. Nat. Bur. Stand., 1949, 42, 379-382
- [1949GUE] Guerin, M.H.; Marthe, M.; Bastick, J.; Adam-Gironne, J.; Sur la chaleur de combustion du sulfure de carbon. Compt. Rend., 1949, 228, 87-89
- [1949HOL] Holcomb, D.E.; Dorsey, C.L., Jr.; Thermodynamic properties of nitroparaffins. Ind. Eng. Chem., 1949, 41, 2788-2792
- [1949KNO] Knowlton, J.W.; Rossini, F.D.; Heats of combustion and formation of cyclopropane. J. Res. Nat. Bur. Stand., 1949, 43, 113-115
- [1949PAR] Parks, G.S.; Moore, G.E., Vapor pressure and other thermodynamic data for n-hexadecane and n-dodecylcyclohexane near room temperature, J. Chem. Phys., 1949, 17, 1151
- [1949PRO] Prosen, E.J.; Maron, F.W.; Rossini, F.D., Heat of isomerization of the two butadienes, J. Res. Nat. Bur. Stand., 1949, 42, 269-275
- [1949SCO] Scott, D.W.; Waddington, G.; Smith, J.C.; Huffman, H.M.; Thermodynamic properties of three isomeric pentenes. J. Am. Chem. Soc., 1949, 71, 2767-2773
- [1949WAD] Waddington, G.; Smith, J.C.; Scott, D.W.; Huffman, H.M., Experimental vapor heat capacities and heats of vaporization of 2-methylpentane, 3-methylpentane and 2,3-dimethylbutane, J. Am. Chem. Soc., 1949, 71, 3902-3906
- [1950AST] Aston, J.G.; Mastrangelo, S.V.R.; Moessen, G.W.; The thermodynamics of 1-butyne from calorimetric and spectroscopic data. J. Am. Chem. Soc., 1950, 72, 5287-5291
- [1950COO] Coops, J.; Kaarsemaker, S. J.; Heat of combustion of cyclobutane. Recl. Trav. Chim. Pays-Bas, 1950, 69, 1364
- [1950DAI] F. S. Dainton and K. J. Ivin, Trans. Faraday Soc., 1950, 46, 295
- [1950EDW] Edwards, G., The vapour pressure of 2,4,6-trinitrotoluene, Trans. Faraday Soc., 1950, 46, 423
- [1950HOC] Hock, I.H.; Knaue, G.; Autoxydation von kohlenwasserstoffen, XIV. Mitteil. Über die energetische stellung organischer hydroperoxyde. Chem. Ber., 1950, 84, 1
- [1950HUM] Humphrey, G.I.; Spitzer, R.; Bond hybridization in the non-tetrahedral carbon atom. The heats of combustion of spiro-pentane and methylcyclobutane. J. Chem. Phys., 1950, 18, 902
- [1950JES] Jessup, R. S.; Prosen, E. J.; Heats of Combustion and Formation of Cellulose and Nitrocellulose (Cellulose Nitrate), J. Res. Nat. Bur. Stand., 1950, 44, 387-393

- [1950LAC] Lacher, J.R.; Walden, C.H.; Lea, K.R.; Park, J.D., Vapor phase heats of hydrobromination of cyclopropane and propylene, J. Am. Chem. Soc., 1950, 72, 331-333
- [1950PAR] Parks, G.S.; Mosley, J.R.; Peterson, P.V., Jr.; Heats of combustion and formation of some organic compounds containing oxygen. J. Chem. Phys., 1950, 18, 152
- [1950SMI] Smisko, J. Mason, L. S.; The Heats of Solution and Dilution of Orthoboric Acid at 25°, J. Am. Chem. Soc. 1950, 72, 3679-3680
- [1951CAS] Casey, D.W.H.; Fordham, S.; An all-glass calorimeter, and the heat of combustion of ethyl chloride. J. Chem. Soc., 1951, 2513-2516
- [1951COL] Cole, L.G.; Gilbert, E.C.; The heats of combustion of some nitrogen compounds and the apparent energy of the N-N bond. J. Am. Chem. Soc., 1951, 73, 5423-5427
- [1951FOL] Foley, W. T.; Giguere, P. A.; Hydrogen peroxide and its analogues. IV. Some thermal properties of hydrogen peroxide, Can. J. Chem. 29, 895-903 (1951)
- [1951KNO] Knowlton, J.W.; Prosen, E.J.; Heat of combustion and formation of cyanogen. J. Res. Nat. Bur. Stand., 1951, 46, 489-495
- [1951PRO] Prosen, E.J.; Maron, F.W.; Rossini, F.D., Heats of combustion, formation, and isomerization of ten C4 hydrocarbons, J. Res. Nat. Bur. Stand., 1951, 46, 106-112
- [1951SCO] Scott, D.W.; McCullough, J.P.; Williamson, K.D.; Waddington, G.; Rotational isomerism and thermodynamic functions of 2-methylbutane and 2,3-dimethylbutane. Vapor heat capacity and heat of vaporization of 2-methylbutane. J. Am. Chem. Soc., 1951, 73, 1707-1717
- [1951SUL] Sullivan, J.H.; Davidson, N.; The kinetics of the forward and reverse reactions for the vapor phase thermal bromination of chloroform. J. Chem. Phys., 1951, 19, 143-148
- [1951WIS] Wise, C.H.; Serijan, K.T.; Goodman, I.A., NACA Technical Report 1003, National Advisory Committee for Aeronautics, Washington, D.C., 1951
- [1952AST] Aston, J.G.; Rock, E.J.; Isserow, S.; The heats of combustion of the methyl substituted hydrazines and some observations on the burning of volatile liquids. J. Am. Chem. Soc., 1952, 74, 2484
- [1952GUT] Guthrie, G.B., Jr.; Scott, D.W.; Hubbard, W.N.; Katz, C.; McCullough, J.P.; Gross, M.E.; Williamson, K.D.; Waddington, G.; Thermodynamic properties of furan. J. Am. Chem. Soc., 1952, 74, 4662-46
- [1952KAA] Kaarsemaker, S.; Coops, J.; Thermal quantities of some cycloparaffins. Part III. Results of measurements. Rec. Trav. Chim. Pays/Bas, 1952, 71, 261
- [1952PAR] Parks, G.S.; Manchester, K.E.; The heats of solution of erythritol, mannitol and dulcitol; combustion values for liquid polyhydroxy alcohols. J. Am. Chem. Soc., 1952, 74, 3435
- [1952VRI] Vriens, G.N.; Hill, A.G.; Equilibria of several reactions of aromatic amines. Ind. Eng. Chem., 1952, 44, 2732
- [1953COO] Coops, J.; Mulder, D.; Dienske, J.W.; Smittenberg, J.; Thermochemical investigations on arylethanes I. Heats of combustion of phenylethanes. Rec. Trav. Chim. Pays/Bas, 1953, 72, 785
- [1953KAR] Karo, W.; McLaughlin, R.L.; Hipsher, H.F.; Dicyclic hydrocarbons. VI. 1,2,3,4-Tetrahydronaphthalene and 1-alkyl-1,2,3,4-tetrahydronaphthalenes. J. Am. Chem. Soc., 1953, 75, 3233-3235

- [1953LAN] Lane, M.R.; Linnett, J.W.; Oswin, H.G.; A study of the $\text{C}_2\text{H}_4 + \text{HCl} = \text{C}_2\text{H}_5\text{Cl}$ and $\text{C}_2\text{H}_4 + \text{HBr} = \text{C}_2\text{H}_5\text{Br}$ equilibria. *Proc. Roy. Soc. London A*, 1953, 216, 361-374
- [1954DAV] Davies, M.; Jones, J.I., The sublimation pressures and heats of sublimation of some carboxylic acids, *Trans. Faraday Soc.*, 1954, 50, 1042
- [1954DUU] H. C. Duus, E. I. du Pont de Nemours and Co., Wilmington, Del., presented at the meeting of American Chemical Society in New York on September 15, 1954
- [1954COX] Cox, J.D.; Challoner, A.R.; Meetham, A.R., The heats of combustion of pyridine and certain of its derivatives, *J. Chem. Soc.*, 1954, 265-271.
- [1954HIP] Hipsher, H.F.; Wise, P.H.; Dicyclic hydrocarbons. VIII. 1-Alkyl-naphthalenes and some of their tetrahydro derivatives. *J. Am. Chem. Soc.*, 1954, 76, 1747-1748
- [1954KIR] Kirkbridge, F.W., Davidson, F.G., Heats of formation of gaseous fluoro- and fluorochlorocarbons, *Nature*, 1954, 174, 79-80
- [1954MCC] McCullough, J.P.; Scott, D.W.; Pennington, R.E.; Hossenlopp, I.A.; Waddington, G.; Nitromethane. *J. Am. Chem. Soc.*, 1954, 76, 4791
- [1954PAR] Parks, G.S.; Manchester, K.E.; Vaughan, L.M., Heats of combustion and formation of some alcohols, phenols, and ketones, *J. Chem. Phys.*, 1954, 22, 2089-2090
- [1954PUS] Pushin, N.A.; Heats of combustion and heats of formation of isomeric organic compounds. *Bull. Soc. Chim. Belgrade*, 1954, 19, 531-547
- [1955CAM] D. L. Camin and F. O. Rossini, Physical Properties of Fourteen API Research Hydrocarbons, C9 to C15, *J. Phys. Chem.*, 1955, 59, 1173
- [1955CHA] Challoner, A.R.; Gundry, H.A.; Meetham, A.R., An electrically calibrated bomb calorimeter, *Phil. Trans. Roy. Soc. London*, 1955, 247, 553-580
- [1955DUU] Duus, H. C. Thermochemical Studies on Fluorocarbons. Heat of Formation of CF_4 , C_2F_4 , C_3F_6 , C_2F_4 Dimer, and C_2F_4 Polymer. *Ind. Eng. Chem.* 1955, 47, 1445
- [1955FRA] Fraser, F.M.; Prosen, E.J., Heats of combustion of liquid n-hexadecane, 1-hexadecene, n-decylbenzene, n-decylcyclohexane, n-decylcyclopentane, and the variation of heat of combustion with chain length, *J. Res. Nat. Bur. Stand.*, 1955, 55, 329-333
- [1955GIG] Giguere, P. A.; B. G. Morissette, A. W. Olmos, and O. Knop, Hydrogen peroxide and its analogues. 7. Calorimetric properties of the systems $\text{H}_2\text{O}-\text{H}_2\text{O}_2$ and $\text{D}_2\text{O}-\text{D}_2\text{O}_2$, *Can. J. Chem.*, 1955, 33, 804
- [1955HOW] Howlett, K.E.; The use of equilibrium constants to calculate thermodynamic quantities. Part II. *J. Chem. Soc.*, 1955, 1784-1789
- [1955JES] Jessup, R.S., McCoskey, R.E., Nelson, R.A., *J. Am. Chem. Soc.*, 1955, 77, 244
- [1955SAL] Salley, D.J.; Gray, J.B., Heats of combustion of some organic nitrogen compounds, *J. Am. Chem. Soc.*, 1951, 73, 5925
- [1955SCO] Scott DW; Good WD; Waddington G Heat of Formation of Tetrafluoromethane from Combustion Calorimetry of Polytetrafluoroethylene. *J. Am. Chem. Soc* 1955, 77, 245-246

- [1955TAF] Taft, R.W., Jr.; Riesz, P.; Thermodynamic properties for the system isobutene-t-butyl alcohol. J. Am. Chem. Soc., 1955, 77, 902-904
- [1955WAR]
[1956BRE] Wartenberg, H. v.; Schiefer, J. Bildungswärmen von Fluor-Chlor-Kohlenstoff-Verbindungen. Z. Anorg. Chem. 1955, 278, 326
Brennan, D.; Ubbelohde, A.R.; A thermochemical evaluation of bond strengths in some carbon compounds. Part IV. Bond-strength differences based on the reaction. J. Chem. Soc., 1956, 3011-3016
- [1956CAM] Camin, D.L.; Rossini, F.D.; Physical properties of the 17 isomeric hexenes of the API research series. J. Phys. Chem., 1956, 60, 1446
- [1956COO] Coops, J.; Adriaanse, N.; Van Nes, K., The absolute heat of combustion of benzoic acid, Rec. Trav. Chim. Pays/Bas, 1956, 75, 237-253
- [1956GOO] Good, W.D.; Scott, D.W.; Waddington, G.; Combustion calorimetry of organic fluorine compounds by a rotating-bomb method. J. Phys. Chem., 1956, 60, 1080-1089
- [1956HAN] Handrick, G.R., Report of the study of pure explosive compounds. Part IV. Calculation of heat of combustion of organic compounds from structural features and calculation of power of high explosives, Rpt. C-58247 for the Office of the Chief of Ordnance, contract DA-19-020-ORD-47 by the Arthur D. Little, Inc., Cambridge, MA, 1956, 467-573
- [1956HAV] Tavernier, P.; Lamouroux, M., Determinations calorimetriques relatives a vingt-six substances organiques, Mem. Poudres, 1956, 38, 65-88
- [1956KIR] Kirkbride, F.W.; The heats of chlorination of some hydrocarbons and their chloro-derivatives. J. Appl. Chem., 1956, 6, 11-21
- [1956LAC] Lacher, J. R.; Kianpour, A.; Oetting, F.; Park, J. D. Reaction Calorimetry. The Hydrogenation of Organic Fluorides and Chlorides. Trans. Faraday Soc. 1956, 52, 1500
- [1956NEU] Neugebauer, C.A., Margrave, J.L., The Heats of Formation of Tetrafluoroethylene, Tetrafluoromethane and 1,1-Difluoroethylene, J. Phys. Chem. 60, 1318 (1956)
- [1956YOU] Young, J.A.; Keith, J.E.; Stehle, P.; Dzombak, W.C.; Hunt, H., Heats of combustion of some organic nitrogen compounds, Ind. Eng. Chem., 1956, 48, 1375-1378
- [1957AND] Andon, R.J.L.; Cox, J.D.; Herington, E.F.G.; Martin, J.F.; The second virial coefficients of pyridine and benzene, and certain of their methyl homologues. Trans. Faraday Soc., 1957, 53, 1074
- [1957FLI] Flitcroft, T.; Skinner, H.A.; Whiting, M.C., Heats of hydrogenation Part 1.-Dodeca-3:9 and -5:7 Diynes, Trans. Faraday Soc., 1957, 53, 784-790
- [1957GLA] Glaser, F.; Ruland, H.; Untersuchungen über dampfdruckkurven und kritische daten einiger technisch wichtiger organischer substanzen. Chem. Ing. Techn., 1957, 29, 772
- [1957GRA] Gray, P.; Pratt, M.W.T.; The latent heats of vaporization of the alkyl nitrates. J. Chem. Soc., 1957, 2163
- [1957LAC] Lacher, J.R.; Kianpour, A.; Montgomery, P.; Knedler, H.; Park, J.D.; Reaction heats of organic halogen compounds. IX. The catalytic hydrogenation of vinyl and perfluorovinyl bromide. J. Phys. Chem., 1957, 61, 1125-1126
- [1957MCC] J. P. McCullough, H. L. Finke, J. F. Messerly, T. C. Kinceloe, and G. Waddington, J. Phys. Chem. 61, 1105, 1957

- [1957MUR] Murrin, J.W.; Goldhagen, S.; Determination of the C-O bond energy from the heats of combustion of four aliphatic ethers. NAVORD Report No. 5491, U.S. Naval Powder Factory Res. & Dev. Dept., Indian Head, MD, 1957
- [1957NEU] Neugebauer, C.A.; Margrave, J.L.; Heats of formation of the fluoromethanes and fluoroethylenes. Univ. Wisc. Tech. Rept., 1-45, 1957
- [1957PEN] Pennington, R.E.; Kobe, K.A., The thermodynamic properties of acetone, J. Am. Chem. Soc., 1957, 79, 300-305
- [1957SCH] Scheer, M. D. J. Phys. Chem. 1957, 61, 1184-1188
- [1957SKU] Skuratov, S.M.; Kozina, M.L.; Shteher, S.M.; Varushenko, R.M., The heats of combustion of several purified compounds, Thermochem. Bull. (Moscow State Univ.), March 1957, 36-37
- [1958BID] Biddiscombe, D.P.; Martin, J.F.; Vapour pressures of phenol and the cresols. Trans. Faraday Soc., 1958, 54, 1316
- [1958CAS] Cass, R.C.; Fletcher, S.E.; Mortimer, C.T.; Quincey, P.G.; Springall, H.D.; Heats of combustion and molecular structure. Part IV. Aliphatic nitroalkanes and nitric esters. J. Chem. Soc., 1958, 958
- [1958GUN] Gundry, H.A.; Meetham, A.R., Comparison of calorimeters for evaluating the heat of combustion of benzoic acid, Trans. Faraday Soc., 1958, 54, 665-670
- [1958NEU] Neugebauer, C.A.; Margrave, J.L.; The heats of formation of CHF₃ and CH₂F₂. J. Phys. Chem., 1958, 62, 1043-1048
- [1958PRO] Prosen, E. J.; Johnson, W. H.; Pergiel, F. Y.; Heats of Formation of Diborane and Pentaborane, J. Res. Nat. Bur. Stand., 1958, 61, 247-250
- [1958SIN] G. C. Sinke and D. R. Stull, J. Phys. Chem. 62, 397 (1958)
- [1959EVA] Evans, F.W.; Skinner, H.A.; The heat of combustion of acetic acid. Trans. Faraday Soc., 1959, 55, 260-261
- [1959FLE] Fletcher, S.E.; Mortimer, C.T.; Springall, H.D.; Heats of combustion and molecular structure. Part VII. 1. J. Chem. Soc., 580-584, 1959
- [1959HIL] Hildenbrand, D.L.; McDonald, R.A.; Kramer, W.R.; Stull, D.R.; Thermodynamic and spectroscopic study of vinylidene chloride. I. Thermodynamic properties of the solid, liquid, and ideal gas. J. Chem. Phys., 1959, 30, 930-934
- [1959HIL1] Hildenbrand, D.L.; McDonald, R.A.; The heat of vaporization and vapor pressure of carbon tetrachloride. J. Phys. Chem., 1959, 63, 1525
- [1959JOH] Johnson, W. H.; Miller, R. G.; Prosen, E. J.; Heat of Formation of Boron Trichloride, J. Res. Nat. Bur. Stand., 1959, 62, 213-217
- [1959MCC] McCullough, J.P.; Pennington, R.E.; Smith, J.C.; Hossenlopp, I.A.; Waddington, G.; Thermodynamics of cyclopentane, methylcyclopentane and cis-1,3-dimethylcyclopentane. J. Am. Chem. Soc., 1959, 81, 5880-5883
- [1959PAR] Paridon, L. J.; MacWood, G. E.; Hu, J.-H.; The heat of vaporization of diborane, J. Phys. Chem., 1959, 63, 1998-1999
- [1959PRO] Prosen, E. J.; Johnson, W. H.; Pergiel, F. Y.; J. Res. Nat. Bur. Stand., 1959, 62, 43-47
- [1959RAY] Ray, J.D.; Ogg, R.A., Jr.; The heat of formation of methyl nitrate. J. Phys. Chem., 1959, 63, 1522-1523
- [1959SIN] Sinke, G.C.; The heat of formation of formic acid. J. Phys. Chem., 1959, 63, 2063
- [1959SKI] Skinner, H.A.; Snelson, A., Heats of hydrogenation Part 3., Trans. Faraday Soc., 1959, 55, 405-407

- [1960AND] Andon, R.J.L.; Biddiscombe, D.P.; Cox, J.D.; Handley, R.; Harrop, D.; Herington, E.F.G.; Martin, J.F.; Thermodynamic properties of organic oxygen compounds. Part I. Preparation and physical properties of pure phenol, cresols, and xylenols. *J. Chem. Soc.*, 1960, 5246
- [1960COR] Corral, L.B., Investigaciones termoquimicas sobre los acidos toluicos y dimetilbenzoicos, *Rev. R. Acad. Cienc.*, 1960, 54, 365-403
- [1960DON] Donovan, T.M.; Shomate, C.H.; McBride, W.R.; The heat of combustion of tetramethyltetrazene and 1,1-dimethylhydrazine. *J. Phys. Chem.*, 1960, 64, 281-282
- [1960GOO] Good, W. D.; Lacina, J. L.; McCullough, J. P.; Sulfuric acid - heat of formation of aqueous solutions by rotating-bomb calorimetry, *J. Am. Chem. Soc.* 1960, 82, 5589
- [1960GRE] Green, J.H.S.; Revision of the values of the heats of formation of normal alcohols. *Chem. Ind. (London)*, 1960, 1215
- [1960GUN] S. R. Gunn and L. G. Green, The heats of hydrolysis of diborane and boron trichloride, *J. Phys. Chem.*, 64, 61-63 (1960)
- [1960ISS] Issoire, J.; Long, C.; Etude de la thermodynamique chimique de la reaction de formation des methylamines. *Bull. Soc. Chim. France*, 2004-2012 (1960)
- [1960LI] Li, S.; Gao, P.; Yu, X.; Li, X.; He, X., Certification of national research center for certified reference materials (NRCCRM, Beijing) thermochemical standard benzoic acid, *J. Chem. Thermodyn.*, 1990, 22, 319-320
- [1960NIC] Nicholson, G.R.; 478. The heats of combustion of butanal and heptanal, *J. Chem. Soc.*, 1960, 2377
- [1960PON] Ponomarev, V.V.; Migarskaya, L.B., Heats of combustion of some amino-acids, *Russ. J. Phys. Chem.*, 1960, 34, 1182-1183
- [1960SCO] Scott, D.W.; Berg, W.T.; McCullough, J.P.; Chemical thermodynamic properties of methylcyclopentane and 1-cis-3-dimethylcyclopentane. *J. Phys. Chem.*, 1960, 64, 906
- [1960SKI] Skinner, H.A.; Snelson, A.; The heats of combustion of the four isomeric butyl alcohols. *Trans. Faraday Soc.*, 1960, 56, 1776-1783
- [1960SPE] Speros, D.M.; Rossini, F.D.; Heats of combustion and formation of naphthalene, the two methylnaphthalenes, cis and trans decahydronaphthalene and related compounds. *J. Phys. Chem.*, 1960, 64, 1723-1727
- [1960TJE] Tjebbes, J.; Heats of combustion of butanal and some related compounds. *Acta Chem. Scand.*, 1960, 14, 180-188
- [1960VOR] Vorob'ev, A.F.; Skuratov, S.M.; Standard enthalpies of formation of CF₄. *Zh. Neorg. Khim.*, 1960, 5, 1398-1401
- [1960WAD] Wadso, I.; An adiabatic calorimeter for measuring heats of vaporisation at 25° C. *Acta Chem. Scand.*, 1960, 14, 566-572
- [1961BAI] Baibuz, V.F., Explosion method and heat of formation of CF₄, CFCl, CFCI₃, and CCl₄, *Dokl. Phys. Chem. (Engl. Transl.)*, 1961, 140, 786-788
- [1961COX] Cox, J.D.; The heats of combustion of phenol and the three cresols. *Pure Appl. Chem.*, 1961, 2, 125-128
- [1961EVA] Evans, W. H., Thermochemistry and thermodynamic functions of some boron compounds in "Preliminary report on the thermodynamic properties of selected light-elements and some related compounds," ed(s) Douglas, T. B. and Beckett, C. W., *Natl. Bur. Stand. Report No. 7093*, 1961, pgs. 1-135
- [1961GOO] Good, W.D.; Lacina, J.L.; McCullough, J.P.; Methanethiol and carbon disulfide. *J. Phys. Chem.*, 1961, 65, 2229-2231

- [1961GRE] Green, J.H.S.; Thermodynamic properties of organic oxygen compounds. Part 5.-Ethyl alcohol. Trans. Faraday Soc., 1961, 57, 2132
- [1961LAB] Labbauf, A.; Rossini, F.D.; Heats of combustion, formation, and hydrogenation of 14 selected cyclomonoolefin hydrocarbons. J. Phys. Chem., 1961, 65, 476-480
- [1961ROC] Rockenfeller, J.D.; Rossini, F.D., Heats of combustion, isomerization, and formation of selected C7, C8, and C10 monoolefin hydrocarbons, J. Phys. Chem., 1961, 65, 267-272
- [1961SNE] Snelson, A.; Skinner, H.A.; Heats of combustion. Trans. Faraday Soc., 1961, 57, 2125-2131
- [1962BAI] Baibuz and Medvedev, Tr., Gos. Inst. Prikl. Khim., 1962, 49, 84
- [1962GOL] Gollis, M.H.; Belenyessy, L.I.; Gudzinowicz, B.J.; Koch, S.D.; Smith, J.O.; Wineman, R.J.; Evaluation of pure hydrocarbons as jet fuels. J. Chem. Eng. Data, 1962, 7, 331-316
- [1962HAT] Hatton, W.E.; Hildenbrand, D.L.; Sinke, G.C.; Stull, D.R.; Chemical thermodynamic properties of aniline. J. Chem. Eng. Data, 1962, 7, 229-231
- [1962KOL] Kolesov, V.P.; Martynov, A.M.; Shtekher, S.M.; Skuratov, S.M.; Standard enthalpies of formation of 1,1-difluoroethylene and of trifluoroethylene. Russ. J. Phys. Chem., 1962, 36, 1118-1120
- [1962KOL2] Kolesov, V. P.; Zenkov, I. D.; Skuratov, S. M. The Standard Enthalpy of Formation of Tetrafluoroethylene. Russ. J. Phys. Chem., 1962, 36, 45
- [1962LAC] Lacher, J.R.; Gottlieb, H.B.; Park, J.D.; Reaction heats of organic compounds. Part 2.-Heat of addition of hydrogen chloride to acetylene. Trans. Faraday Soc., 1962, 58, 2348-2351
- [1962ROS] Ross, G.R.; Heideger, W.J.; Vapor pressure of glycerol. J. Chem. Eng. Data, 1962, 7, 505-507
- [1962STR] Strepikheev, Yu. A.; Baranov, Yu.I.; Burmistrova, O.A.; Determination of the heats of combustion and the heat capacities of several mono- and di-isocyanates. Izv. Vyssh. Uchebn. Zaved. Khim. Khim. Tekhnol., 1962, 5, 387-390
- [1962TJE] Tjebbes, J.; Heats of combustion of propanal and 2-methyl propanal. Acta Chem. Scand., 1962, 16, 953-857
- [1962WAD] Waddington, G.; Smith, J.C.; Williamson, K.D.; Scott, D.W.; Carbon Disulfide as a reference substance for vapor-flow calorimetry. J. Phys. Chem., 1962, 66, 1074
- [1963BUL] Bull, S.S.; Seregrenaja, I.I.; Tsherbakora, P.R.; Isothermic Equilibrium of Liquid-Steam in System Water-Acetoaldehyde, Khim. Prom. (Moscow), 1963, 7, 507-509
- [1963GAR] Gardner, D.M.; Grigger, J.C.; Heat of formation of tetranitromethane by combustion calorimetry. J. Chem. Eng. Data, 1963, 8, 73-74
- [1963HAL] Hales, J.L.; Cox, J.D.; Lees, E.B.; Thermodynamic properties of organic oxygen compounds. Part 10.-Measurement of vapour heat capacities and latent heats of vaporization of isopropyl alcohol. Trans. Faraday Soc., 1963, 59, 1544
- [1963HIL] Hildenbrand, D. L.; Hall, W. F.; Potter, N. D. J., Chem. Phys. 1963, 39, 296-301
- [1963KOL] Kolesov, V.P.; Zenkov, I.D.; Skuratov, S.M.; Standard enthalpies of formation of chlorotrifluoromethane and dichlorodifluoromethane. Russ. J. Phys. Chem., 1963, 37, 378-379
- [1963MAN] Mansson, M.; Sunner, S.; The heat of formation of sulphuric acid, Acta Chem. Scand., 1963, 17, 723-727

- [1963MCC] McCurdy, K.G.; Laidler, K.J.; Heats of vaporization of a series of aliphatic alcohols. *Can. J. Chem.*, 1963, 41, 1867-1871
- [1963MIL] Miller, G.A., Vapor pressure of naphthalene, *J. Chem. Eng. Data*, 1963, 8, 69
- [1963PIL] Pilcher, G.; Skinner, H.A.; Pell, A.S.; Pope, A.E.; Measurements of heats of combustion by flame calorimetry. Part 1.-Diethyl ether, ethyl vinyl ether and divinyl ether. *Trans. Faraday Soc.*, 1963, 59, 316-330
- [1963PUY] Puyo, J.; Balesdent, D.; Niclaude, M.; Dzierzynski, M.; Etude analytique et thermodynamique de la pyrolyse de l'hexachloroethane en phase gazeuse. *Compt. Rend.*, 1963, 256, 3471-3473
- [1964LEB] Lebedeva, N.D.; Heats of combustion of monocarboxylic acids. *Russ. J. Phys. Chem.*, 1964, 38, 1435-1437
- [1964PIL] Pilcher, G.; Pell, A.S.; Coleman, D.J.; Measurements of heats of combustion by flame calorimetry. Part 2-Dimethyl ether, methyl ethyl ether, methyl n-propyl ether, methyl isopropyl ether. *Trans. Faraday Soc.*, 1964, 60, 499-505
- [1964SIN] Sinke, G.C.; Oetting, F.L.; The chemical thermodynamic properties of methyl ethyl ketone. *J. Phys. Chem.*, 1964, 68, 1354-1358
- [1964WIL] Wilhoit, R.C.; Shiao, D., Thermochemistry of biologically important compounds. Heats of combustion of solid organic acids., *J. Chem. Eng. Data*, 1964, 9, 595-599
- [1965ASH] Ashcroft, S.J.; Carson, A.S.; Carter, W.; Laye, P.G.; Thermochemistry of reductions caused by lithium aluminium hydride. Part 3. The C-halogen bond dissociation energies in ethyl iodine and ethyl bromide. *Trans. Faraday Soc.*, 1965, 61, 225-229
- [1965BUC] Buckley, E.; Herington, E.F.G.; Equilibria in some secondary alcohol + hydrogen + ketone systems. *Trans. Faraday Soc.*, 1965, 61, 1618-1625
- [1965CHA] Chao, J.; Rossini, F.D.; Heats of combustion, formation, and isomerization of nineteen alkanols. *J. Chem. Eng. Data*, 1965, 10, 374-379
- [1965COL] Colomina, M.; Pell, A.S.; Skinner, H.A.; Coleman, D.J.; Heats of combustion of four dialkylethers. *Trans. Faraday Soc.*, 1965, 61, 2641
- [1965COX] Cox JD; Gundry HA; Head AJ Thermodynamic Properties of Fluorine Compounds. Part 4. – Enthalpy of Formation of Carbon Tetrafluoride. *Trans. Faraday Soc* 1965, 61, 1594–1600
- [1965DOM] Domalski, E.S., Armstrong, G.T., Heat of Formation of Aluminum Fluoride by Direct Combination of the Elements, *J. Res. Natl. Bur. Std.* 69A, 137-147 (1965)
- [1965EDW] Edwards, J.W.; Small, P.A.; Kinetics of the pyrolysis of chlorodifluoromethane. *Ind. Eng. Chem. Fundam.*, 1965, 4, 396-400
- [1965FOW] P. Fowell, J. R. Lacher, and J. D. Park, *Trans. Faraday Soc.* 61, 1324 (1965)
- [1965GUN] Gunn, S. R.; Heats of Reaction of Boron Trifluoride with HF-3.75H₂O and of Diborane with Trimethylamine. Correlation of Thermochemical Data for Some Boron Compound, *J. Phys. Chem.* 69, 1010-1015 (1965)
- [1965KOL] Kolesov, V.P.; Martynov, A.M.; Skuratov, S.M.; Standard enthalpy of formation of 1,1,1-trifluoroethane. *Russ. J. Phys. Chem.*, 1965, 39, 223-225
- [1965PEL] Pell, A.S.; Pilcher, G.; Measurements of heats of combustion by flame calorimetry. Part 3.-Ethylene oxide, trimethylene oxide, tetrahydrofuran and tetrahydropy. *Trans. Faraday Soc.*, 1965, 61, 71-77

- [1966ADA] Adams, G.P.; Carson, A.S.; Laye, P.G.; Thermochemistry of reductions caused by lithium aluminium hydride. Part 4.-Heat of formation of methyl bromide. *Trans. Faraday Soc.*, 1966, 62, 1447-1449
- [1966COL] Coleman, D.J.; Pilcher, G., Heats of combustion of biphenyl, bibenzyl, naphthalene, anthracene, and phenanthrene, *Trans. Faraday Soc.*, 1966, 62, 821-827
- [1966EGG] Egger, K.W.; Benson, S.W., Nitric oxide and iodine catalyzed isomerization of olefins. VI. Thermodynamic data from equilibrium studies of the geometrical and positional isomerization of n-pentenes, *J. Am. Chem. Soc.*, 1966, 88, 236-240
- [1966HAW] Hawtin, P.; Lewis, J. B.; Moul, N.; Phillips, R. H.; The Heats of Combustion of Graphite, Diamond and Some Non-Graphitic Carbons, *Phil. Trans. Roy. Soc. A*, 1966, 261, 67-95
- [1966LEB] Lebedeva, N.D.; Heats of combustion and formation of aliphatic tertiary amine homologues. *Russ. J. Phys. Chem.*, 1966, 40, 1465-1467
- [1966SIN] Sinke, G.C.; The heat of reaction of nitrogen trifluoride and hexafluoroethane. *J. Phys. Chem.*, 1966, 70, 1326-1327
- [1966WAD] Wadso, I., A heat of vaporization calorimeter for work at 25°C and for small amounts of substances, *Acta Chem. Scand.*, 1966, 20, 536
- [1967BUC] Buckley, E.; Cox, J. D.; Chemical Equilibria. Part 2. Dehydrogenation of Propanol and Butanol, *Trans. Faraday Soc.* 63, 895-901 (1967)
- [1967COO] Coomber, J.W.; Whittle, E.; Bond dissociation energies from equilibrium studies. Part 1.-D(CF₃-Br), D(C₂F₅-Br) and D(n-C₃F₇-Br). *Trans. Faraday Soc.*, 1967, 63, 608-619
- [1967DOM] Domalski, E.S.; Armstrong, G.T.; The heats of combustion of polytetrafluoroethylene (teflon) and graphite in elemental fluorine. *J. Res. Nat. Bur. Stand.*, 1967, 71, 105-118
- [1967LAC] Lacher, J.R.; Amador, A.; Park, J.D.; Reaction heats of organic compounds. Part 5.-Heats of hydrogenation of dichloromethane, 1,1- and 1,2-dichloroethane and 1,2-dichloropropane. *Trans. Faraday Soc.*, 1967, 63, 1608-1611
- [1967LOR] Lord, A.; Goy, C.A.; Pritchard, H.O.; The heats of formation of trifluoromethyl chloride and bromide. *J. Phys. Chem.*, 1967, 71, 2705-2707
- [1967PIH] Pihlaja, K.; Heikkila, J.; The configurations of isomeric 2,2,4,6-tetramethyl-1,3-dioxanes. *Acta Chem. Scand.*, 1967, 21, 2430-2434
- [1967PIL] Pilcher, G.; Chadwick, J.D.M., Measurements of heats of combustion by flame calorimetry. Part 4.-n-Pentane, isopentane, neopentane, *Trans. Faraday Soc.*, 1967, 63, 2357-2361
- [1967SMI] Smith, N.K.; Good, W.D.; Enthalpies of combustion and formation of propylamine, isopropylamine, and tert-butylamine. *J. Chem. Eng. Data*, 1967, 12, 572-574
- [1967WOO] Wood, J.L.; Lagow, R.J.; Margrave, J.L.; The heat of combustion of teflon in fluoride. A check on the heat of formation of carbon tetrafluoride. *J. Chem. Eng. Data*, 1967, 12, 255-256
- [1968FOW] L. Fowler, N. W. Trump, and C. E. Vogler, *J. Chem. Eng. Data* 13, 209, 1968
- [1968GRE] Greenberg, E.; Hubbard, W.N.; Fluorine bomb calorimetry. XXIII. The enthalpy of formation of carbon tetrafluoride. *J. Phys. Chem.*, 1968, 72, 222-227

- [1968KIN] King, R. C.; Armstrong, G. T.; Constant pressure flame calorimetry with fluorine. 2. Heat of formation of oxygen difluoride, J. Res. Nat. Bur. Stand. Sect. A 72, 113 (1968)
- [1968KOL] Kolesov, V.P.; Shtekher, S.N.; Martynov, A.M.; Skuratov, S.M.; Standard enthalpy of formation of 1,1-difluoroethane. Russ. J. Phys. Chem., 1968, 42, 975-976
- [1968PIH] Pihlaja, K.; Heikkil, J.; Heats of combustion. Diethyl ether and 1,1-diethoxyethane. Acta Chem. Scand., 1968, 22, 2731-2732
- [1968WAD] Wadso, I.; Heats of vaporization of organic compounds II. Chlorides, bromides, and iodides. Acta Chem. Scand., 1968, 22, 2438
- [1968WIB] Wiberg, K.B.; Fenoglio, R.A., Heats of formation of C₄H₆ hydrocarbons, J. Am. Chem. Soc., 1968, 90, 3395-3397
- [1969CHA] Chastel, R.; Steckel, F.; Tachoire, H., Proceedings of the First International Conference on Calorimetry, Thermodynamics Polish Scientific, Warsaw, 1969
- [1969FUR] Furuyama, S.; Golden, D.M.; Benson, S.W., Thermochemistry of the gas phase equilibria $i\text{-C}_3\text{H}_7\text{I} = \text{C}_3\text{H}_6 + \text{HI}$, $n\text{-C}_3\text{H}_7\text{I} = i\text{-C}_3\text{H}_7\text{I}$, and $\text{C}_3\text{H}_6 + 2\text{HI} = \text{C}_3\text{H}_8 + \text{I}_2$, J. Chem. Thermodyn., 1969, 1, 363-375
- [1969GOO] Good, W.D.; Smith, N.K., Enthalpies of combustion of toluene, benzene, cyclohexane, cyclohexene, methylcyclopentane, 1-methylcyclopentene, and n-hexane, J. Chem. Eng. Data, 1969, 14, 102-106
- [1969GUN] Gundry, H.A.; Harrop, D.; Head, A.J.; Lewis, G.B.; Thermodynamic properties of organic oxygen compounds. 21. Enthalpies of combustion of benzoic acid, pentan-1-ol, octan-1-ol, and hexadecan-1-ol. J. Chem. Thermodyn., 1969, 1, 321-332
- [1969HU] Hu, A.T.; Sinke, G.C.; Combustion calorimetry of some chlorinated organic compounds. J. Chem. Thermodyn., 1969, 1, 507-513
- [1969LOR] Lord, A.; Pritchard, H.O.; Thermodynamics of the reaction between carbon dioxide and carbon tetrachloride. J. Chem. Thermodyn., 1969, 1, 495-498
- [1969MAN] Mansson, M.; Morawetz, E.; Nakase, Y.; Sunner, S.; The enthalpies of formation of trioxane and tetroxane. Acta Chem. Scand., 1969, 23, 56-60
- [1969MOS] Mosselman, C.; Dekker, H.; A new method for handling volatile and hygroscopic substances in bomb calorimetry enthalpy of formation of liquid butanol. Recl. Trav. Chim. Pays-Bas, 257-261, 1969
- [1969SMI] Smith, J.H.; Pace, E.L., The thermodynamic properties of carbon tetrafluoride from 12°K to its boiling point. The significance of the parameter { ν }, J. Phys. Chem., 1969, 73, 4232-4236
- [1969STU] D. R. Stull, E. F. Westrum, Jr., and G. C. Sinke, The Chemical Thermodynamics of Organic Compounds, Wiley, New York, (1969)
- [1969WAD] Wadso, I.; Enthalpies of vaporization of organic compounds. Acta Chem. Scand., 1969, 23, 2061-2064
- [1970BRE] Bretschneider, E.; Rogers, D.W., A new microcalorimeter: heats of hydrogenation of four monoolefins, Mikrochim. Acta, 1970, 482-490
- [1970BUT] Butwill, M.E.; Rockenfeller, J.D.; Heats of combustion and formation of ethyl acetate and isopropyl acetate. Thermochim. Acta, 1970, 1, 289-295
- [1970COX] Cox, J.D.; Pilcher, G.; Thermochemistry of Organic and Organometallic Compounds. Academic Press, 1970, New York, 1-636

- [1970DEK] H. Dekker and C. Mosselman, Enthalpy of Formation of Liquid 1-Propanol by an Improved Combustion Technique, *Rec. Trav. Chim. Pays-Bas* 89, 1276-1280 (1970)
- [1970FLE] Fletcher, R.A.; Pilcher, G.; Measurements of heats of combustion by flame calorimetry. *Trans. Faraday Soc.*, 1970, 66, 794-799
- [1970FUR] Furuyama, S.; Golden, D.M.; Benson, S.W.; Thermochemistry of cyclopentene and cyclopentadiene from studies of gas-phase equilibria. *J. Chem. Thermodyn.*, 1970, 2, 161-169
- [1970GOO] Good, W.D.; The enthalpies of combustion and formation of the isomeric pentanes. *J. Chem. Thermodyn.*, 1970, 2, 237-244
- [1970HOW] Howard, P.B.; Wadso, I.; Enthalpies of vaporization of organic compounds IV. Alkyl Nitriles. *Acta Chem. Scand.*, 1970, 24, 145
- [1970KOL] Kolesov, V.P.; Papina, T.S.; Standard enthalpy of formation of vinyl fluoride. *Russ. J. Phys. Chem.*, 1970, 44, 611-613
- [1970KON] Konicek, J.; Wadso, I.; Enthalpies of vaporization of organic compounds. VII. Some carboxylic acids. *Acta Chem. Scand.*, 1970, 24, 2612-26
- [1970LIN] Lin, David C.K.; Silberberg, I. Harold; McKetta, John J., Volumetric behavior, vapor pressures, and critical properties of cyclopropane, *J. Chem. Eng. Data*, 1970, 15, 4, 483-492
- [1970LOR] Lord, A.; Pritchard, H.O.; Thermodynamics of phosgene formation from carbon monoxide and chlorine. *J. Chem. Thermodyn.*, 1970, 2, 187-191
- [1970MOI] Moiseev, V.D.; Antonova, N.D.; *Zh. Fiz. Khim.*, 1970, 44, 2912
- [1970WAL] Walker, L.C.; Sinke, G.C.; Perettie, D.J.; Janz, G.J.; Enthalpy of formation of trifluoroacetonitrile. *J. Am. Chem. Soc.*, 1970, 92, 4525-4526
- [1971BOY] Boyd, R.H.; Sanwal, S.N.; Shary-Tehrany, S.; McNally, D.; The thermochemistry, thermodynamic functions, and molecular structures of some cyclic hydrocarbons. *J. Phys. Chem.*, 1971, 75, 1264-1271
- [1971FLE] Fletcher, R.A.; Pilcher, G.; Measurements of heats of combustion by flame calorimetry. Part 7.-Chloromethane, chloroethane, 1-chloropropane, 2-chloropropane. *Trans. Faraday Soc.*, 1971, 67, 3191-3201
- [1971HAL] Hall, H.K., Jr.; Baldt, J.H.; Thermochemistry of strained-ring bridgehead nitriles and esters. *J. Am. Chem. Soc.*, 1971, 93, 140-145
- [1971HAY] Hayes, C.W.; Bomb calorimetric studies on normal alkan-1-ols, stereoregular polymethylmethacrylates, «alpha»-olefinic polymers, trioxane and oxygenated polymers. *Texas A&M Diss. Abs.*, College Station, TX, 1971, 31, 5903-5904
- [1971KNO] Knobel, Y.K.; Miroshnichenko, E.A.; Lebedev, Y.A.; Heats of combustion of nitromethane and dinitromethane. *Bull. Acad. Sci. USSR, Div. Chem. Sci.*, 425-428 (1971)
- [1971KUS] Kusano, K.; Wadso, I.; Enthalpy of vaporization of some organic substances at 25.0°C and test of calorimeter. *Bull. Chem. Soc. Jpn.*, 1971, 44, 1705-17
- [1971KUS2] Kusano, Kazuhito; Wadsö, Ingemar; Enthalpies of Vaporization of Organic Compounds. VIII. Alkoxyethanols, *Acta Chem. Scand.*, 1971, 25, 219-224
- [1971LEB] Lebedeva, N.D.; Katin, Y.A.; Akhmedova, G.Y.; Standard enthalpy of formation of nitrobenzene. *Russ. J. Phys. Chem.*, 1971, 45, 1192-1193

- [1971LEN] Lenchitz, C.; Velicky, R.W.; Silvestro, G.; Schlosberg, L.P.; Thermodynamic properties of several nitrotoluenes. J. Chem. Thermodyn., 1971, 3, 689-692
- [1971MAN] Mansson, M.; Ringner, B.; Sunner, S.; The enthalpies of combustion and formation of some simple chloroalkanes and chloralkenes 1,1,1-trichloroethane and 1,1-dichloroethene. J. Chem. Thermodyn., 1971, 3, 547-551
- [1971MOR] Morawetz, E.; A triplet calorimeter for the measurement of vaporization enthalpies in the pressure range 10^2 to 10^6 mmHg at 25 °C, Chem. Scr., 1971, 1, 103-111
- [1971PAR] Parsons, G.H.; Rochester, C.H.; Wood, C.E.C., Effect of 4-substitution on the thermodynamics of hydration of phenol and the phenoxide anion, J. Chem. Soc. B, 533 (1971)
- [1971POL] Polak, J.; Benson, G.C.; Enthalpies of vaporization of some aliphatic alcohols. J. Chem. Thermodyn., 1971, 3, 235-242
- [1971ROG] Rogers, F.E.; Thermochemistry of the Diels-Alder reaction. I. Enthalpy of addition of isoprene to tetracyanoethylene. J. Phys. Chem., 1971, 75, 1734-1737
- [1971SCO] Scott, D.W.; 1-Aminopropane, 2-aminopropane, and 2-methyl-2-aminopropane. Vibrational assignments, conformational analyses, and chemical thermodynamic properties. J. Chem. Thermodyn., 1971, 3, 843-852
- [1971SOL] Solly, R.K.; Benson, S.W.; Thermochemistry of the reaction of benzaldehyde with iodine. The enthalpy of formation of benzaldehyde and benzoyl iodide. J. Chem. Thermodyn., 1971, 3, 203-209
- [1971WIL] R. C. Wilhort and B. J. Zwolinski, Handbook on vapor pressure and heats of vaporization of hydrocarbons and related compounds, Texas A Research Foundation. College Station, Texas (1971)
- [1972AHN] Ahmed, A.M.I.; Eades, R.G., Proton relaxation in solid aniline and some methyl derivatives, J. Chem. Soc., Faraday Trans. 2, 1972, 68, 2017
- [1972BAR] Baroody, E.E.; Carpenter, G.A.; Heats of formation of propellant compounds, Rpt. Naval Ordnance Systems Command Task No. 331-003/067-1/UR2402-001 for Naval Ordnance Station, Indian Head, MD
- [1972CON] Connett, J.E.; Chemical equilibria. 5. Measurement of equilibrium constants for the dehydrogenation of propanol by a vapour flow technique. J. Chem. Thermodyn., 1972, 4, 233-237
- [1972DAV] Davies, J. V.; Pritchard, H. O.; The enthalpy of formation of phosgene, J. Chem. Thermodyn., 1972, 4, 23-29
- [1972GAR] Gardner, P.J.; Hussain, K.S.; The standard enthalpies of formation of some aliphatic diols. J. Chem. Thermodyn., 1972, 4, 819-827
- [1972GOO] Good, W.D., The enthalpies of combustion and formation of n-octane and 2,2,3,3-tetramethylbutane, J. Chem. Thermodyn., 1972, 4, 709-714
- [1972HU] Hu, A.T.; Sinke, G.C.; Mintz, M.J.; The enthalpy of formation of 1,1,1-trichloroethane from enthalpy of combustion and equilibrium studies. J. Chem. Thermodyn., 1972, 4, 239-245
- [1972IRV] Irving, R.J.; The standard enthalpy of sublimation of naphthalene. J. Chem. Thermodyn., 1972, 4, 793-794
- [1972LAY] Laynez, J.; Wadso, I.; Enthalpies of vaporization of organic compounds. IX. Some halogen substituted hydrocarbons and esters. Acta Chem. Scand., 1972, 26, 3148-3152

- [1972LEB] Lebedeva, N.D.; Katin, Yu. A.; Heats of combustion of certain monosubstituted benzenes. Russ. J. Phys. Chem., 1972, 46, 1088
- [1972MOR] Morawetz, E.; Enthalpies of vaporization of n-alkanes from C12 to C20. J. Chem. Thermodyn., 1972, 4, 139-144
- [1972PIT] Pittam, D.A.; Pilcher, G.; Measurements of heats of combustion by flame calorimetry. Part 8.-Methane, ethane, propane, n-butane and 2-methylpropane, J. Chem. Soc. Faraday Trans. 1, 1972, 68, 2224-2229
- [1972SAB] Beech, Graham; Lintonbon, Roger M.; The measurement of sublimation enthalpies by differential scanning calorimetry. Thermochimica Acta, 1971, 2, 86-88
- [1972WAL] Walker, L.C.; The enthalpy of decomposition of CF₃NF₂(g) to CF₄(g), N₂(g), and F₂(g). J. Chem. Thermodyn., 1972, 4, 219-223
- [1972WIE] Wiedemann, H.G.; Applications of thermogravimetry for vapor pressure determination, Thermochim. Acta, 1972, 355-366
- [1973ALF] Alfassi, Z.B.; Golden, D.M.; Benson, S.W.; The thermochemistry of the isomerization of 3-halopropenes (allyl halides) to 1-halopropenes; entropy and enthalpy of formation contribution of the Cd-(H)(X) group. J. Chem. Thermodyn., 1973, 5, 411-420
- [1973DEL] Delafontaine, J.; Sabbah, R.; Laffitte, M.; Mesures microcalorimetriques des enthalpies de reference de formation (a l'etat condense) de composes volatils - mise au point de la technique. Z. Phys. Chem. (Neue Folge), 1973, 84, 157-172
- [1973FER] Ferguson, K.C.; Okafo, E.N.; Whittle, E.; Bond dissociation energies from equilibrium studies Part 4.-The equilibrium Br₂ + CH₄ = HBr + CH₃Br. Determination of D(CH₃-Br) and DHf°(CH₃Br,g). J. Chem. Soc. Faraday Trans. 1, 1973, 69, 295-301
- [1973GOO] Good, W.D.; The enthalpies of combustion and formation of n-butylbenzene, the diethylbenzenes, the methyl-n-propylbenzenes, and the methyl-iso-propylbenzenes. J. Chem. Thermodyn., 1973, 5, 707-714
- [1973JOH] Johnson, G. K.; Smith, P. N.; Hubbard, W. N.; The Enthalpies of Solution and Neutralization of HF(l); Entahlpies of Dilution and Derived Thermodynamic Properties of HF(aq). J. Chem. Thermodyn., 1973, 5, 793-809
- [1973KON] Konicek, J.; Design and testing of a vaporisation calorimeter. Enthalpies of vaporisation of some alkyl cyanides. Acta Chem. Scand., 1973, 27, 1496-15
- [1973LEB] P. Verevkin, V. N. Emel'yanenko, V. Diky, and O. V. Dorofeeva, Enthalpies of Formation of Nitromethane and Nitrobenzene: New Experiments vs. Quantum Chemical Calculations, J. Chem. Thermodyn. 73, 163-170 (2014)
- [1973MAL] Malaspina, L.; Gigli, R.; Bardi, G.; Microcalorimetric determination of the enthalpy of sublimation of benzoic acid and anthracene, J. Chem. Phys., 1973, 59, 387-394
- [1973MEN] G. D. Mendenhall, D. M. Golden, and S. W. Benson, J. Phys. Chem. 77, 2707 (1973)
- [1973SEL] Selivanov, V.D.; Karlik, V.M.; Zagranicnyi, V.I.; Standard enthalpies of formation ammeline, ammelide, melem, and melon, Russ. J. Phys. Chem., 1973, 47, 272
- [1973SVO] Svoboda, V.; Veselý, F.; Holub, R.; Pick, J.; Enthalpy data of liquids. II. The dependence of heats of vaporization of methanol, propanol, butanol, cyclohexane, cyclohexene, and benzene on temperature, Collect. Czech. Chem. Commun., 1973, 38, 3539-3543

- [1974AMB] Sinke, G.C.; A method for measurement of vapor pressures of organic compounds below 0.1 Torr Naphthalene as a reference substance. *J. Chem. Thermodyn.*, 1974, 6, 311-316
- [1974CHA] J. Chao, A. S. Rodgers, R. C. Wilhoit, and B. J. Zwolinski, *J. Phys. Chem. Ref. Data* 3, 141 (1974)
- [1974GOO] Good, W.D.; Moore, R.T.; Osborn, A.G.; Douslin, D.R.; The enthalpies of formation of ethylcyclobutane, methylenecyclobutane, and 1,1-dimethylcyclopropane. *J. Chem. Thermodyn.*, 1974, 6, 303-310
- [1974GUT] Guthrie, J.P.; Hydration of carboxamides. Evaluation of the free energy change for addition of water to acetamide and formamide derivatives. *J. Am. Chem. Soc.*, 1974, 96, 3608-3615
- [1974HIN] Hine, J.; Klueppet, A.W.; Structural effects on rates and equilibria. XVIII. Thermodynamic stability of ortho esters. *J. Am. Chem. Soc.*, 1974, 96, 2924-2929
- [1974OKA] Okafo, E.N.; Whittle, E.; Bond dissociation energies from equilibrium studies. Part 5.-The equilibria $\text{Br}_2 + \text{CH}_2\text{F}_2 = \text{HBr} + \text{CHF}_2\text{Br}$ and $\text{Br}_2 + \text{CH}_3\text{F} = \text{HBr} + \text{CH}_2\text{FBr}$. Determination of $D(\text{CHF}_2\text{-Br})$ and $DH^\circ_f(\text{CHF}_2\text{Br}, g)$. *Trans. Faraday Soc.*, 1974, 70, 1366-1375
- [1974ROG] Rogers, D.W.; Skanupong, S.; Heats of hydrogenation of sixteen terminal monoolefins. The alternating effect. *J. Phys. Chem.*, 1974, 78, 2569-2572
- [1974ROZ] Rozhnov, A.M.; Levanova, S.V.; Dvornikova, G.A.; Sadovnikova, S.L.; Equilibrium of isomerization of dichloroethylenes. *J. Appl. Chem. USSR*, 1974, 47, 667-669
- [1974SAB] Sabbah, R.; Chastel, R.; Laffitte, M.; Thermochemical study of methyl naphthalenes. *Thermochim. Acta*, 1974, 10, 353-358
- [1974SIN] Sinke, G.C., A method for measurement of vapor pressures of organic compounds below 0.1 Torr Naphthalene as a reference substance, *J. Chem. Thermodyn.*, 1974, 6, 311-316
- [1974WU] Wu, E.; Rodgers, A.S.; Thermochemistry of gas-phase equilibrium $\text{CF}_3\text{CH}_3 + \text{I}_2 = \text{CF}_3\text{CHI} + \text{HI}$. The carbon-hydrogen bond dissociation energy in 1,1,1-trifluoroethane and the heat of formation of the 2,2,2-trifluoroethyl radical. *J. Phys. Chem.*, 1974, 78, 2315-2317
- [1975AMB] Ambrose, D.; Ellender, J.H.; Lees, E.B.; Sprake, C.H.S.; Townsend, R.; Thermodynamic properties of organic oxygen compounds XXXVIII. Vapour pressures of some aliphatic ketones, *J. Chem. Thermodyn.*, 1975, 7, 453-472
- [1975BER] Bergström, Stina; Olofsson, Gerd; Thermodynamic quantities for the solution and protonation of four C6-amines in water over a wide temperature range. *J. Solution Chem.*, 1975, 4, 535-554
- [1975CHU] Chun, Kil W.; Drummond, Joseph C.; Smith, William H.; Davison, Richard R.; Vapor-liquid equilibrium of binary mixtures of butyl alcohols with triethylamine, 1975, *J. Chem. Eng. Data*, 20, 58-61
- [1975CON] Connett, J.E.; Chemical equilibria 6. Measurement of equilibrium constants for the dehydrogenation of 2-methylpropan-1-ol by a vapour-flow technique. *J. Chem. Thermodyn.*, 1975, 7, 1159-1162
- [1975HOR] Hörner, Ch.; Höpfner, A.; Schmeiser, B., Der Dampfdruck-Isotopie-Effekt von deuterierten n-Pentanen, *Berichte der Bunsengesellschaft für physikalische Chemie*, 1975, 79, 2, 222-225
- [1975KUD] Kudchadker, S.A.; Kudchadker, A.P.; Thermodynamic properties of oxygen compounds. III. Benzaldehyde and furfural (2-furaldehyde). *Ber. Bunsenges. Phys. Chem.*, 1975, 79, 432-437

- [1975KUS] Kusano, K.; Saito, Y.; Preprints 33rd Ann. Meeting Chem. Soc. Japan, 123 (1975)
- [1975LEB] Lebedev, V.P.; Miroshnichenko, E.A.; Matyushin, Yu.N.; Larionov, V.P.; Romanov, V.S.; Bukolov, Yu.E.; Denisov, G.M.; Balepin, A.A.; Lebedev, Yu.A., Semimicrocalorimetric determination of gas-phase enthalpies of formation of compounds containing nitrogen and oxygen. II. calorimetric determination of enthalpies of formation [of Tetranitromethane], Russ. J. Phys. Chem., 1975, 49, 1133-1136
- [1975LEV] Levanova, S.V.; Treger, Yu.A.; Velichko, S.M.; Rozhnov, A.M.; Bshneva, L.I.; Talanov, A.I.; Equilibrium of the vinylidene chloride-methylchloroform system. Zh. Prikl. Khim. (Leningrad), 1975, 42, 480-481
- [1975MOS] Mosselman, C.; Dekker, H., Enthalpies of formation of n-alkan-1-ols, J. Chem. Soc. Faraday Trans. 1, 1975, 417-424
- [1975ROG] Rogers, D.W.; Papadimetriou, P.M.; Siddiqui, N.A., An improved hydrogen microcalorimeter for use with large molecules, Mikrochim. Acta, 1975, 2, 389-400
- [1975VIL] Vilcu, R.; Perisanu, S.; Ciocazanu, I.; Heats of formation in the ideal gas state of some substances containing carbon, hydrogen and oxygen. Conf. Int. Thermodyn. Chim. C.R. 4th, 1975, 1, 105-112
- [1976AMB] Ambrose, D.; Ellender, J.H.; Sprake, C.H.S.; Townsend, R.; Thermodynamic properties of organic oxygen compounds XLIII. Vapour pressures of some ethers, J. Chem. Thermodyn., 1976, 8, 165-178
- [1976CHA] Chao, J.; Zwolinski, B.J.; Ideal gas thermodynamic properties of propanone and 2-butanone. J. Phys. Chem. Ref. Data, 1976, 5, 319-328
- [1976GOO] Good, W.D.; The enthalpies of formation of five isomeric heptenes. J. Chem. Thermodyn., 1976, 8, 67-71
- [1976HIN] Hine, J.; Arata, K.; Keto-Enol tautomerism. II. The calorimetric determination of the equilibrium constants for keto-enol tautomerism for cyclohexanone. Bull. Chem. Soc. Jpn., 1976, 49, 3089-3092
- [1976LBR] S. V. Levanova, L. I. Bushneva, R. M. Rodova, A. M. Rozhnov, and Yu. A. Treger, Russ. J. Phys. Chem., Engl. Transl. 50, 1616 (1976)
- [1976LEV] S. V. Levanova, Y. A. Treger, S. M. Velichko, and A. M. Rozhnov, Russ. J. Phys. Chem., Engl. Transl. 50, 1148 (1976)
- [1976ROU] Rouse, P.E., Jr., Enthalpies of formation and calculated detonation properties of some thermally stable explosives, J. Chem. Eng. Data, 1976, 21, 16-20
- [1977AMM] Ammar, M.M.; El Sayed, N.; Morsi, S.E.; El Azmirly, A., The enthalpies of combustion and formation of p-tertiary butyl phenol and naphthalene, Egypt. J. Phys., 1977, 8, 111-118
- [1977GUT] Guthrie, J.P.; Cyclization of glycol monoesters to give hemiorthoesters. Can. J. Chem., 1977, 55, 3562-3574
- [1977MAN] Mansson, M.; Sellers, P.; Stridh, G.; Sunner, S., Enthalpies of vaporization of some 1-substituted n-alkanes, J. Chem. Thermodyn., 1977, 9, 91
- [1977STE] Steele, W.V.; The standard enthalpies of formation of bicyclic compounds. III. 1,7,7-trimethylbicyclo[2.2.1]heptan-2-one. J. Chem. Thermodyn., 1977, 9, 311-314
- [1978CUN] Pingel, Norbert; Poser, Uwe; Würflinger, Albert; Dielectric measurements at high pressures and low temperatures. Part 6. Dielectric and thermodynamic properties of cyclohexanol. J. Chem. Soc., Faraday Trans. 1, 1984, 80, 3221

- [1978FEN] Fenwick, J.O.; Harrop, D.; Head, A.J.; Thermodynamic properties of organic oxygen compounds. 46. Enthalpies of formation of ethyl acetate and 1-hexanoic acid. *J. Chem. Thermodyn.*, 1978, 10, 687-690
- [1978FUC] Fuchs, R.; Peacock, L.A.; Heats of vaporization of monoalkylcyclohexanes by the gas chromatography-calorimetry method. *Can. J. Chem.*, 1978, 56, 2493-24
- [1978GUN] Gundry, H.A.; Head, A.J., Thermodynamic properties of aliphatic halogen compounds 4. The enthalpy of combustion of 1,1,1,2-tetrachloroethane, *J. Chem. Thermodyn.*, 1978, 10, 195-200
- [1978MON] Montgomery, R.L.; Rossini, F.D.; Mansson, M.; Enthalpies of combustion, vaporization, and formation of phenylbenzene, cyclohexylbenzene, and cyclohexylcyclohexane; enthalpy of hydrogenation of certain aromatic systems. *J. Chem. Eng. Data*, 1978, 23, 125-129
- [1978SCH] Shultz, M. M.; V.L. Stolyarova, V. L.; Semenov, G. A. *Fiz. Khim. Stekla* 1978, 4, 653-660
- [1978VIL] Vilcu, R.; Perisanu, S.; Ciocazanu, I., The thermochemical properties of some organic anhydrides, *Bull. Inst. Politeh. Gheorghe Gheorghiu-Dej, Bucur., Ser. Chim. Metal.* 40, 9 (1978)
- [1979BUS] L. I. Bushneva, S. V. Levanova, R. M. Rodova, A. M. Rozhnov, Yu. A. Treger, and V. S. Popov, *Zhur. Prikl. Khim.* 52, 445 (1979)
- [1979GOO] Good, W.D., The enthalpies of combustion and formation of the isomeric pentanes, *J. Chem. Thermodyn.*, 1970, 2, 237-244
- [1979JAK] Jakli, G.; Tzias, P.; Van Hook, W.A., The vapor pressure isotope effect of perdeuteroheptane from 5 to 80°C, *Acta Chim. Acad. Sci. Hung.*, 1979, 99, 2, 121
- [1979KUD] Kudchadker, Shanti A.; Kudchadker, Arvind P.; Zwolinski, Bruno J., Chemical thermodynamic properties of anthracene and phenanthrene, *J. Chem. Thermodyn.*, 1979, 11, 1051-1059
- [1979LEV] S. V. Levanova, L. I. Bushneva, R. M. Rodova, A. M. Rozhnov, Yu. A. Treger, and S. A. Aprelkin, *Zhur. Prikl. Khim.* 52, 1513 (1979)
- [1979MAJ] Majer, Vladimír; Svoboda, Václav; Hála, Slavoj; Pick, Jirí; Temperature dependence of heats of vaporization of saturated hydrocarbons C5-C8; Experimental data and an estimation method, *Collect. Czech. Chem. Commun.*, 1979, 44, 637-651
- [1979MAJ2] Majer, Vladimír; Svoboda, Václav; Koubek, Josef; Pick, Jirí; Temperature dependence of heats of vaporization, saturated vapour pressures and cohesive energies for a group of amines, *Collect. Czech. Chem. Commun.*, 1979, 44, 3521-3528
- [1979PRI] Price, S.J.W.; Sapiaro, H.J.; Determination of $\Delta H^\circ_{f298}(\text{C}_6\text{F}_{10}, \text{g})$ and $\Delta H^\circ_{f298}(\text{C}_6\text{F}_{12}, \text{g})$ from studies of the combustion of decafluorocyclohexene and dodecafluorocyclohexene in oxygen and calculation of the resonance energy of hexafluorobenzene. *Can. J. Chem.*, 1979, 57, 685-688
- [1979ROG] Rogers, D.W.; Dagdagan, O.A.; Allinger, N.L., Heats of hydrogenation and formation of linear alkynes and a molecular mechanics interpretation, *J. Am. Chem. Soc.*, 1979, 101, 671-676
- [1979STE] Steele, W.V.; The standard enthalpies of formation of cyclohexylamine and cyclohexylamine hydrochloride. *J. Chem. Thermodyn.*, 1979, 11, 1185-1188
- [1979SUN] Sunner, S.; Svensson, Ch.; Zelepuga, A.S.; Enthalpies of vaporization at 298.15 K for some 2-alkanones and methyl alkanoates. *J. Chem. Thermodyn.*, 1979, 11, 491-495
- [1980AND] R. J. L. Andon and J. E. Connett, *Thermochim. Acta* 42, 241, 1980

- [1980HUG] Hugh Davies, R.; Finch, Arthur; Gardner, P.J.; The standard enthalpy of formation of liquid and gaseous ethylchloroformate (C₃H₅O₂Cl). *J. Chem. Thermodyn.*, 1980, 12, 291-296
- [1980MAJ] Majer, V.; Wagner, Z.; Svoboda, V.; Cadek, V.; Enthalpies of vaporization and cohesive energies for a group of aliphatic ethers. *J. Chem. Thermodyn.*, 1980, 12, 387-391
- [1980SVO] Svoboda, Václav; Uchytlová, Vera; Majer, Vladimír; Pick, Jirí; Heats of vaporization of alkyl esters of formic, acetic and propionic acids, *Collect. Czech. Chem. Commun.*, 1980, 45, 3233-3240
- [1980VIL] Vilcu, R.; Perisanu, S.; The ideal gas state enthalpies of formation of some monomers. *Rev. Roum. Chim.*, 1980, 25, 619-624
- [1981AMB] Ambrose, D.; Hall, D.J.; Thermodynamic properties of organic oxygen compounds L. The vapour pressures of 1,2-ethanediol (ethylene glycol) and bis(2-hydroxyethyl) ether (diethylene glycol), *J. Chem. Thermodyn.*, 1981, 13, 61-66
- [1981DEK] C. G. De Kruif, T. Kuipers, J. C. Van Miltenburg, R. C. F. Schaake, and G. J. Stevens, *J. Chem. Thermodyn.* 13, 108 (1981)
- [1981HAL] Murata, S.; Sakiyama, M.; Seki, S.; Construction and testing of a sublimation calorimetric system using a Calvet microcalorimeter. *J. Chem. Thermodyn.*, 1982, 14, 707-721
- [1981HOS] Hossenlopp, I.A.; Scott, D.W., Vapor heat capacities and enthalpies of vaporization of five alkane hydrocarbons, *J. Chem. Thermodyn.*, 1981, 13, 415-421
- [1981JOO] Joo, Hyoek-Jong; Arlt, Wolfgang, Vapor-liquid equilibrium for the binary systems ethylene glycol-n-amyl alcohol and ethylene glycol-isoamyl alcohol, *J. Chem. Eng. Data*, 1981, 26, 2, 138-140
- [1981SHI] Shimizu, M.; Saito, Y.; Kusano, K.; Preprints 17th Conf. Chem. Thermodyn. Thermal Anal., 50 (1981)
- [1981WIB] Wiberg, K.B.; Wasserman, D.J., Enthalpies of hydration of alkenes. 1. The n-hexenes, *J. Am. Chem. Soc.*, 1981, 103, 6563-6566
- [1982ALL] Allinger, N.L.; Dodziuk, H.; Rogers, D.W.; Naik, S.N.; Heats of hydrogenation and formation of some 5-membered ring compounds by molecular mechanics calculations and direct measurements. *Tetrahedron*, 1982, 38, 1593-1597
- [1982BYS] Bystrm, K.; Mansson, M., Enthalpies of formation of some cyclic 1,3- and 1,4-Di- and poly-ethers: Thermochemical strain in the -O-C-O and -O-C-C-O- groups, *J. Chem. Soc. Perkin Trans. 2*, 1982, 565
- [1982ERA] Erastov, P.A.; Kolesov, V.P.; Enthalpy of formation of trifluorochloroethene. *J. Chem. Thermodyn.*, 1982, 14, 103-106
- [1982FUC] Fuchs, Richard; Peacock, L. Alan; Stephenson, W. Kirk, Enthalpies of interaction of polar and nonpolar molecules with aromatic solvents, *Can. J. Chem.*, 1982, 60, 15, 1953-1958
- [1982FUR] Furukawa, J.; Sakiyama, M.; Seki, S.; Saito, Y.; Kusano, K.; Standard enthalpies of combustion and vaporization of N,N-dimethylaniline. *Bull. Chem. Soc. Jpn.*, 1982, 55, 3329-3330
- [1982GUT] Gutner, N.M.; Masalitinova, T.N.; Mon'yakova, O.N.; Lebedeva, N.D., Heats of combustion and formation of some cyclic compounds, *Termodin. Organ. Soedin.*, 1982, 46-49
- [1982JOC] Jochems, R.; Dekker, H.; Mosselman, C.; Somsen, G., The use of the LKB 8721-3 vaporization calorimeter to measure enthalpies of sublimation. The enthalpies of sublimation of bicyclo[2.2.1]hept-2-ene (norbornene), bicyclo[2.2.1]heptane (norbornane), and tricyclo[3.3.1.1^{3,7}]decane (adamantane), *J. Chem. Thermodyn.*, 1982, 14, 395-398

- [1982MIN] Minadakis, C.; Sabbah, R.; Thermodynamique de composés azotés. VIII. Etude thermochimique de la diéthanolamine et de la triéthanolamine et de la réaction de réduction de l'eau par celle-ci. *Thermochim. Acta*, 1982, 55, 147-159
- [1982MOS] Mosselman, C.; Dekker, H., Enthalpies of formation of n-alkan-1-ols, *J. Chem. Soc. Faraday Trans. 1*, 1975, 417-424
- [1982MUR] Murata, S.; Sakiyama, M.; Seki, S., Construction and testing of a sublimation calorimetric system using a Calvet microcalorimeter, *J. Chem. Thermodyn.*, 1982, 14, 8
- [1982PAP] Papina, T.S.; Kolesov, V.P.; Golovanova, Yu. G.; The standard enthalpy of formation of bromoform. *Russ. J. Phys. Chem.*, 1982, 56, 1666-1668
- [1982WAG] Wagman, D. D.; Evans, W. H.; Parker, V. B.; Schumm, R. H.; Halow, I.; Bailey, S. M.; Churney, K. L.; Nuttall, R. L.; The NBS tables of chemical thermodynamic properties. Selected values for inorganic and C1 and C2 organic substances in SI units, *J. Phys. Chem. Ref. Data* 1982, 11, Suppl. 2
- [1983AN] An, X.; Mansson, M.; Enthalpies of combustion and formation of acetonitrile. *J. Chem. Thermodyn.*, 1983, 15, 287-293
- [1983EKE] Van Ekeren, P.J.; Jacobs, M.H.G.; Offringa, J.C.A.; De Kruif, C.G., Vapour-pressure measurements on trans-diphenylethene and naphthalene using a spinning-rotor friction gauge, *J. Chem. Thermodyn.*, 1983, 15, 5, 409-417
- [1983HOL] Holdiness, M.R., Heats of atomization and resonance energy of some ortho-substituted benzoic acids, *Thermochim. Acta*, 1983, 71, 257-263
- [1983SCH] Schmeling, T.; Strey, K.; Equilibrium Vapor Pressure Measurements for the n-Alcohols in the Temperature Range from -30 °C to + 30 °C, *Ber. Bunsenges. Phys. Chem.*, 1983, 87, 871-874
- [1983UCH] Uchytlova, V.; Majer, V.; Svoboda, V.; Hynek, V.; Enthalpies of vaporization and cohesive energies for seven aliphatic ketones. *J. Chem. Thermodyn.*, 1983, 15, 853-858
- [1983WIT] De Wit, H.G.M.; Van Miltenburg, J.C.; De Kruif, C.G., Thermodynamic properties of molecular organic crystals containing nitrogen, oxygen, and sulphur 1. Vapour pressures and enthalpies of sublimation, *J. Chem. Thermodyn.*, 1983, 15, 7, 651-663
- [1984BIC] Bickerton, J.; Minas Da Piedade, M.E.; Pilcher, G.; Enthalpy of formation of tetrabromomethane by rotating-bomb calorimetry. *J. Chem. Thermodyn.*, 1984, 16, 661-668
- [1984BOU] Boublik, T.; Fried, V.; Hala, E.; *The Vapour Pressures of Pure Substances*, Elsevier, Amsterdam, 1984
- [1984COX] Cox, J.D.; Wagman, D.D.; Medvedev, V.A., *CODATA Key Values for Thermodynamics*, Hemisphere Publishing Corp., New York, 1984
- [1984LAW] Lawrence, A.H.; Elias, L.; Authier-Martin, M.; Determination of amphetamine, cocaine, and heroin vapour pressures using a dynamic gas blending system and gas chromatographic analysis. *Can. J. Chem.*, 1984, 62, 1886-1888
- [1984MAJ] Majer, V.; Svoboda, V.; Hynek, V.; On the enthalpy of vaporization of isomeric butanols. *J. Chem. Thermodyn.*, 1984, 16, 1059-1066
- [1984MOL] Molnar, A.; Rachford, R.; Smith, G.V.; Liu, R., Heats of hydrogenation by a simple and rapid flow calorimetric method, *Appl. Catal.*, 1984, 9, 219-223

- [1984PIL] Pilcher, G.; Bickerton, J.; Ribeiro Da Silva, M.M.D.C.C., Certification of B.D.H. thermochemical standard benzoic acid, *J. Chem. Thermodyn.*, 1984, 16, 401-402
- [1984PLA] Platonov, V.A.; Simulin, Yu. N.; Experimental determination of the standard enthalpies of formation of polychlorobenzenes. II. Standard enthalpies of formation of dichlorobenzenes. *Russ. J. Phys. Chem.*, 1984, 58, 1630-1632
- [1984WIB] Wiberg, K.B.; Wasserman, D.J.; Martin, E., Enthalpies of hydration of alkenes. 2. The n-heptenes and n-pentenes, *J. Phys. Chem.*, 1984, 88, 3684-3688
- [1985CHR] U.S. Coast Guard, Department of Transportation. CHRIS - Hazardous Chemical Data. Volume II. Washington, D.C.: U.S. Government Printing Office, 1984-5
- [1985KUS] Kusano, Kazuhito; Micro conduction calorimeters to measure enthalpies of vaporization, *Thermochimica Acta*, 1985, 88, 109-120
- [1985LIA] Lias, S.G.; Karpas, Z.; Liebman, J.F.; Halomethylenes. *J. Am. Chem. Soc.*, 1985, 107, 6089-6096
- [1985MAJ] Majer, V.; Svoboda, V., Enthalpies of Vaporization of Organic Compounds: A Critical Review and Data Compilation, Blackwell Scientific Publications, Oxford, 1985
- [1985PAP] Papina, T.S.; Kolesov, V.P.; Standard enthalpy of formation of trichloroethylene. *Russ. J. Phys. Chem.*, 1985, 59, 1289-1291
- [1987AZA] Azandegbe, E.C.; Analisis, 1987, 15, 370
- [1987MAR] Markovnik, V.S.; Shvarko, O.V.; Al'khimovich, V.M.; Sachek, A.I., Temperature dependence of vapor pressure, of heats of formation, combustion, and vaporization of 3-methyl-1-butanol, *Termodin. Org. Soedin.*, 1987, 68-70
- [1987PAP] Papina, T.S.; Kolesov, V.P.; Standard enthalpy of formation of 1,1,2-trichloroethane. *Russ. J. Phys. Chem.*, 1987, 61, 1170-1172
- [1987SAB] Sabbah, R. Antipine, I.; *Bull. Soc. Chim. Fr*, 3, 392, 1987
- [1987STE] Stephenson, Richard M.; Malanowski, Stanislaw; Handbook of the Thermodynamics of Organic Compounds, New York, Elsevier, 1987. doi:10.1007/978-94-009-3173-2
- [1988BAS] Bastos, M.; Nilsson, S-O.; Ribeiro Da Silva, M.D.M.C.; Ribeiro Da Silva, M.A.V.; Wadso, I.; Thermodynamic properties of glycerol enthalpies of combustion and vaporization and the heat capacity at 298.15 K. Enthalpies of solution in water at 288.15, 298.15, and 308.15 K. *J. Chem. Thermodyn.*, 1988, 20, 1353-1359
- [1988BER] Berry, R. J.; Ehlers, C. J.; Burgess, D. R., Jr.; Zachariah, M. R.; Nyden, M. R.; Schwartz, M.; Halon thermochemistry: Ab initio calculations of the enthalpies of formation of fluoroethanes, *J. Mol. Struct. THEOCHEM*, 1998, 422, 89-98
- [1988KNA] Knauth, P.; Sabbah, R.; Energetics of intra- and intermolecular bonds in «omega»-alkanediols. I. Calorimetric investigation of the enthalpies of vaporization at 298.15 K. *Bull. Soc. Chim. Fr.*, 1988, 5, 834
- [1988LIT] Litorja, M.; Rusci, B.; and B. Rusci, "A photoionization study of the hydroperoxyl radical, HO₂, and hydrogen peroxide," H₂O₂. *J. Electr. Spectr. Rel. Phenom.* 97, 131 (1998)
- [1988MES] Messerly, J.F.; Finke, H.L.; Good, W.D.; Gammon, B.E.; Condensed-phase heat capacities and derived thermodynamic properties for 1,4-dimethylbenzene, 1,2-diphenylethane, and 2,3-dimethylnaphthalene. *J. Chem. Thermodynam.*, 1988, 20, 485-501

- [1988RUS] B. Ruscic, C. A. Mayhew, and J. Berkowitz, Photoionization studies of (BH₃)_n (n=1,2), J. Chem. Phys. 88, 5580-5593 (1988)
- [1988TOR] Torres-Gomez, L.A.; Barreiro-Rodriguez, G.; Galarza-Mondragon, A., A new method for the measurement of enthalpies of sublimation using differential scanning calorimetry, Thermochim. Acta, 1988, 124, 229-233
- [1989AN] An, Xuwu; Hu, Hui; Enthalpies of Vaporization of Some Multichloro-Alkanes, Acta Phys. Chim. Sin., 5, 565-571 (1989)
- [1989CHI] Chirico, R.D.; Nguyen, A.; Steele, W.V.; Strube, M.M.; Vapor pressure of n-alkanes revisited. New high-precision vapor pressure data on n-decane, n-eicosane, and n-octacosane. J. Chem. Eng. Data, 1989, 34, 149-156
- [1989DAU] Daubert, T.E., R.P. Danner. Physical and Thermodynamic Properties of Pure Chemicals Data Compilation. Washington, D.C.: Taylor and Francis, 1989
- [1989GUR] Gurvich, L.V.; Veyts, I. V.; Alcock, C. B., Thermodynamic Properties of Individual Substances, Fourth Edition, Hemisphere Pub. Co., New York, 1989
- [1989VOR] Voronkov, M.G.; Klyuchnikov, V.A.; Kolabin, S.N.; Shvets, G.N.; Varusin, P.I.; Deryagina, E.N.; Korchevin, N.A.; Tsvetnitskaya, S.I., Thermochemical properties of diorganyl chalcogenides and dichalcogenides RMnR (M = S, Se, Te; n = 1, 2), Dokl. Phys. Chem., 1989, 307, 650-653
- [1990AMB] Ambrose, D.; Ghiassee, N.B.; Vapour pressures, critical temperatures, and critical pressures of benzyl alcohol, octan-2-ol, and 2-ethylhexan-1-ol, J. Chem. Thermodyn., 1990, 22, 307-311
- [1990GLA] Gladii, S.L.; Starchevskii, M.K.; Pazderskii, Yu. A.; Moiseev, I.I.; Chemical equilibrium in the methyl formate-water-methanol-formic acid system. J. Appl. Chem. USSR, 1990, 63, 106-111
- [1990KNA] Knauth, P.; Sabbah, R.; Energetics of intra- and intermolecular bonds in «omega»-alkanediols (II) Thermochemical study of 1,2-ethanediol, 1,3-propanediol, 1,4-butanediol, and 1,5-pentanediol at 298.15 K. Struct. Chem., 1990, 1, 43-46
- [1990SIL] Da Silva, Manuel A.V. Ribeiro; Monte, Manuel J.S., The construction, testing and use of a new Knudsen effusion apparatus, Thermochimica Acta, 1990, 171, 169-183
- [1990SMI] Smith, Norman O.; Vapor pressures of 1,4-dimethylbenzene, 1,4-di(methyl-d₃)benzene, and 1,4-dimethylbenzene-d₁₀ at 20-50.degree.C, J. Chem. Eng. Data, 1990, 35, 387-389
- [1991ACR] Acree, William E., Thermodynamic properties of organic compounds: enthalpy of fusion and melting point temperature compilation, Thermochimica Acta, 1991, 189, 1, 37-56
- [1991FRO] Frolova, E.I.; Petrov, A.A.; Chemical equilibrium in the system formic acid-ethanol-ethyl formate-water. J. Appl. Chem. USSR, 1991, 64, 2361-2365
- [1991ROT] Roth, W.R.; Adamczak, O.; Breuckmann, R.; Lennartz, H.-W.; Boese, R., Die Berechnung von Resonanzenergien; das MM2ERW-Kraftfeld, Chem. Ber., 1991, 124, 2499-2521
- [1991WIB] Wiberg, K.B.; Hao, S., Enthalpies of hydration of alkenes. 4. Formation of acyclic tert-alcohols, J. Org. Chem., 1991, 56, 5108-5110
- [1991WIB2] Wiberg, K.B.; Waldron, R.F.; Lactones. 2. Enthalpies of hydrolysis, reduction, and formation of the C₄-C₁₃ monocyclic lactones. Strain energies and conformations. J. Am. Chem. Soc., 1991, 113, 7697-7705

- [1991ZAH] Zaheeruddin, M.; Lodhi, Z.H.; Enthalpies of formation of some cyclic compounds. *Phys. Chem. (Peshawar Pak.)*, 1991, 10, 111-118
- [1992GRA] Gracia, M.; Sánchez, F.; Pérez, P.; Valero, J.; Gutiérrez Losa, C., Vapour pressures of (butan-1-ol + hexane) at temperatures between 283.10 K and 323.12 K, *J. Chem. Thermodyn.*, 1992, 24, 5, 463-471
- [1992KIE] Kiefer, J.H.; Sidhu, S.S.; Kern, R.D.; Xie, K.; Chen, H.; Harding, L.B., The homogeneous pyrolysis of acetylene II: the high temperature radical chain mechanism, *Combust. Sci. Technol.*, 1992, 82, 101-130
- [1992LUO] Luo, X.; P. R. Fleming, and T. R. Rizzo, Vibrational overtone spectroscopy of the $4\nu_{OH} + \nu_{OH'}$ combination level of HOOH via sequential local mode–local mode excitation, *J. Chem. Phys.* 96, 5659 (1992)
- [1993CHI] R. D. Chirico, S. E. Knipmeyer, A. Nguyen, and W. V. Steele, *J. Chem. Thermodyn.* 25, 1461, (1993)
- [1994CAR] Carson, A.S.; Laye, P.G.; Pedley, J.B.; Welsby, A.M.; Chickos, J.S.; Hosseini, S.; The enthalpies of formation of iodoethane, 1,2-diiodoethane, 1,3-diiodopropane, and 1,4-diiodobutane. *J. Chem. Thermodyn.*, 1994, 26, 1103-1109
- [1994MAS] Masuda, N.; Nagano, Y.; Sakiyama, M.; Standard molar enthalpy of formation of (CH₃)₂SO, dimethylsulfoxide, by combustion calorimetry. *J. Chem. Thermodyn.*, 1994, 26, 971-975
- [1994RUI] Terol, A.; Bouassab, A.; Pauvert, B.; Chevallet, P.; Cassanas, G.; Polymorphism and thermal behaviour of 3-para-toluy-1,2,2-trimethyl cyclopentane carboxylic acid. *Thermochimica Acta*, 1993, 225, 97-109
- [1994RUZ] Ruzicka, Kvetoslav; Majer, Vladimir, Simultaneous Treatment of Vapor Pressures and Related Thermal Data Between the Triple and Normal Boiling Temperatures for n-Alkanes C₅--C₂₀, *J. Phys. Chem. Ref. Data*, 1994, 23, 1
- [1994SPI] Lencka, Malgorzata; Measurements of the vapour pressures of pyridine, 2-methylpyridine, 2,4-dimethylpyridine, 2,6-dimethylpyridine, and 2,4,6-trimethylpyridine from 0.1 kPa to atmospheric pressure using a modified Swietoslawski ebulliometer. *J. Chem. Thermodyn.*, 1990, 22, 473-480
- [1995CHI] Chickos, James S.; Hosseini, Sarah; Hesse, Donald G., Determination of vaporization enthalpies of simple organic molecules by correlations of changes in gas chromatographic net retention times, *Thermochimica Acta*, 1995, 249, 41-62
- [1995DIO] Diogo, Hermínio P.; Santos, Rui C.; Nunes, Paulo M.; Minas da Piedade, Manuel E., Ebulliometric apparatus for the measurement of enthalpies of vaporization, *Thermochimica Acta*, 1995, 249, 113-120
- [1995PAP] Papina, T.S.; Pimenova, S.M.; Luk'yanova, V.A.; Kolesov, V.P., Standard enthalpies of formation of benzyl alcohol and α,α,α -trichlorotoluene, *Russ. J. Phys. Chem.*, 1995, 69, 1951.
- [1995WOL] Wolff, Hans; Landeck, Heiner; Frerichs, Heinz-Peter; Wolff, Erika; The association of normal and tertiary butylamine in mixtures with n-hexane according to isothermal vapour pressure measurements, *Fluid Phase Equilibria*, 1995, 109, 245-263
- [1996BAL] Balcan, M.; Arzik, S.; Altunata, T., The determination of the heats of combustion and the resonance energies of some substituted naphthalenes, *Thermochim. Acta*, 1996, 278, 49-56
- [1996BOU] G. Bouchoux, J.Y. Salpin, Thermokinetic Determination of Gas-Phase Basicities. Applications to Ketene, Methylketene, and Formaldimine, *J. Phys. Chem.*, 1996, 100, 16555.

- [1996CHI] Chirico, R. D.; Steele, W. V.; Nguyen, A.; Klots, T. D.; Knipmeyer, S. E.; "Thermodynamic properties of pyridine. I. Vapor pressures, high-temperature heat capacities, densities, critical properties, derived thermodynamic functions, vibrational assignment, and derivation of recommended values," *J. Chem. Thermodyn.* 28, 797-818 (1996).
- [1996DOM] Domalski, Eugene S.; Hearing, Elizabeth D., *Heat Capacities and Entropies of Organic Compounds in the Condensed Phase. Volume III*, *J. Phys. Chem. Ref. Data*, 1996, 25, 1
- [1996HUY] G. Huybrechts, M. Narmon, and B. van Mele, *Int. J. Chem. Kinet.* 28, 27 (1996)
- [1996SKO] Skorobogatov, G.A.; Dymov, B.P.; Nedozrelova, I.V.; Equilibrium and rate constants of the reactions $RBr \rightarrow R + Br$ and $Br + RBr \rightleftharpoons Br_2 + R$ ($R = CH_2Cl$, $CHCl_2$, CCl_3 , and $CBrF_2$) at 300-700°C. *Russ. J. Gen. Chem. (Engl. Transl.)*, 1996, 66, 1777-1785
- [1996STE] Steele, W.V.; Chirico, R.D.; Knipmeyer, S.E.; Nguyen, A.; Smith, N.K.; Tasker, I.R.; Thermodynamic properties and ideal-gas enthalpies of formation for cyclohexene, phthalan (2,5-dihydrobenzo-3,4-furan), isoxazole, octylamine, dioctylamine, trioctylamine, phenyl isocyanate, and 1,4,5,6-tetrahydropyrimidine. *J. Chem. Eng. Data*, 1996, 41, 1269-1284
- [1996VAR] Varushchenko, R.M.; Pashchenko, L.L.; Druzhinina, A.I.; Enthalpies of vaporization of some mono- and polycyclic hydrocarbons. *Zh. Fiz. Khim.*, 1996, 70, 228
- [1997AUB] C. Aubry, J.L. Holmes, J.K. Terlouw, *J. Phys. Chem. A* 101 (1997) 5958
- [1997STE] Steele, W.V.; Chirico, R.D.; Cowell, A.B.; Knipmeyer, S.E.; Nguyen, A.; Thermodynamic properties and ideal-gas enthalpies of formation for 2-aminoisobutyric acid (2-methylalanine), acetic acid, (4-methyl-3-penten-2-one), 4-methylpent-1-ene, 2,2'-bis(phenylthio)propane, and glycidyl phenyl ether (1,2-epoxy-3-phenoxypropane). *J. Chem. Eng. Data*, 1997, 42, 1052-1066
- [1998CHA] Chase, M.W., Jr., *NIST-JANAF Thermochemical Tables, Fourth Edition*, *J. Phys. Chem. Ref. Data*, Monograph 9, 1998
- [1998CHI] Chickos, James; Hesse, Donald; Hosseini, Sarah; Nichols, Gary; Webb, Paul; Sublimation enthalpies at 298.15 K using correlation gas chromatography and differential scanning calorimetry measurements, *Thermochimica Acta*, 1998, 313, 101-110
- [1998LIA] S.G. Lias, J.E. Bartmess, J.F. Liebman, J.L. Holmes, R.D. Levin, W.G. Mallard, *J. Phys. Chem. Ref. Data* 17 (1998) Suppl. 1
- [1999MOK] Mokbel, I.; Porcedda, S.; Guetachew, T.; Marongiu, B.; Jose, J.; Static measurements of the total vapor pressure of ethane-1,2-diol + water mixtures at temperatures from 258 K to 363 K, *ELDATA*, 1999, 5, 79
- [1999ORT] Ortega, Juan; Hernández, Pablo; Thermodynamic Study of Binary Mixtures Containing an Isobutylalkanol and an Alkyl (Ethyl to Butyl) Alkanoate (Methanoate to Butanoate), Contributing with Experimental Values of Excess Molar Enthalpies and Volumes, and Isobaric Vapor-Liquid Equilibria, *J. Chem. Eng. Data*, 1999, 44, 757-771
- [1999VER] Verevkin, Sergey P.; Strain Effects in Phenyl-Substituted Methanes. Geminal Interaction between Phenyl and the Electron-Releasing Substituent in Benzylamines and Benzyl Alcohols, *J. Chem. Eng. Data*, 1999, 44, 1245-1251
- [2000NIC] Nichols, Gary; Orf, Jennifer; Reiter, Sarah M.; Chickos, James; Gokel, George W., The vaporization enthalpies of some crown and polyethers by correlation gas chromatography, *Thermochimica Acta*, 2000, 346, 1-2, 15-28
- [2000VER] Verevkin, S.P., Measurement and Prediction of the Monocarboxylic Acids Thermochemical Properties, *J. Chem. Eng. Data*, 2000, 45, 5, 953-960

- [2001KIY] Kiyobayashi, Tetsu; Minas da Piedade, Manuel E., The standard molar enthalpy of sublimation of η^5 -bis-pentamethylcyclopentadienyl iron measured with an electrically calibrated vacuum-drop sublimation microcalorimetric apparatus, *J. Chem. Thermodyn.*, 2001, 33, 11-21
- [2001RIJ] A. R. Aguilar, E. Orozco-Guareño, and M. M. Herrera, *J. Chem. Thermodyn.* 33, 1405 (2001)
- [2002CHI] Chirico, R.D.; Knipmeyer, S.E.; Steele, W.V., Heat capacities, enthalpy increments, and derived thermodynamic functions for naphthalene between the temperatures 5K and 440K, *J. Chem. Thermodyn.*, 2002, 34, 1873-1884
- [2002DAL] A. Dale, C. Lythall, J. Aucott, and C. Sayer, High Precision Calorimetry to Determine the Enthalpy of Combustion of Methane, *Thermochim. Acta* 382, 47-54 (2002)
- [2002LEI] Lei, Ying Duan; Chankalal, Raymond; Chan, Anita; Wania, Frank, Supercooled Liquid Vapor Pressures of the Polycyclic Aromatic Hydrocarbons, *J. Chem. Eng. Data*, 2002, 47, 801-806
- [2002MAN] Manion, J.A.; Evaluated Enthalpies of Formation of the Stable Closed Shell C1 and C2 Chlorinated Hydrocarbons. *J. Phys. Chem. Ref. Data*, 2002, 31, 123-172
- [2002ROO] van Roon, André; Parsons, John R.; Govers, Harrie A.J., Gas chromatographic determination of vapour pressure and related thermodynamic properties of monoterpenes and biogenically related compounds, *Journal of Chromatography A*, 2002, 955, 1, 105-115
- [2002YU] Yu. I. Alexandrov, E. N. Korchagina, and A. G. Chunovkina, *Measurement Techniques* 45, 268-273 (2002). New Generalization of Data on the Heat of Combustion of High-Purity Methane.
- [2003DER] Derawi, Samer O.; Michelsen, Michael L.; Kontogeorgis, Georgios M.; Stenby, Erling H.; Application of the CPA equation of state to glycol/hydrocarbons liquid-liquid equilibria, *Fluid Phase Equilibria*, 2003, 209, 163-184
- [2003VER] Verevkin, Sergey P., Vapor pressures and enthalpies of vaporization of a series of 1- and 2-halogenated naphthalenes, *J. Chem. Thermodyn.*, 2003, 35, 1237-1251
- [2004ANT] Antosik, Maria; Galka, Maria; Malanowski, Stanislaw K., Vapor-Liquid Equilibrium for Acetonitrile + Propanenitrile and 1-Pentanamine + 1-Methoxy-2-propanol «8224», *J. Chem. Eng. Data*, 2004, 49, 11-17
- [2004CHI] Chickos, James S.; Hanshaw, William; Vapor Pressures and Vaporization Enthalpies of the n -Alkanes from C21 to C30 at T = 298.15 K by Correlation Gas Chromatography. *J. Chem. Eng. Data*, 2004, 49, 77-85
- [2004LUB] Lubomska, Monika; Malanowski, Stanislaw K.; Vapor-Liquid Equilibrium for Benzene + 2-Methylpentane and Allyl Alcohol + 1-Propanol, *J. Chem. Eng. Data*, 2004, 49, 1488-1493
- [2004NAS] Nasirzadeh, Karamat; Zimin, Denys; Neueder, Roland; Kunz, Werner; Vapor-Pressure Measurements of Liquid Solutions at Different Temperatures. *J. Chem. Eng. Data*, 2004, 49, 607.
- [2004VER] Verevkin, Sergey P., Determination of vapor pressures and enthalpies of vaporization of 1,2-alkanediols, *Fluid Phase Equilibria*, 2004, 224, 23-29
- [2005HOS] Hoskovec, Michal; Grygarová, Dana; Cvacka, Josef; Streinz, Ludvík; Zima, Jirí; Verevkin, Sergey P.; Koutek, Bohumír; Determining the vapour pressures of plant volatiles from gas chromatographic retention data. *Journal of Chromatography A*, 2005, 1083, 161-172

- [2005VAS] Vasiltsova, Tatiana V.; Verevkin, Sergey P.; Bich, Eckard; Heintz, Andreas; Bogel-Lukasik, Rafal; Domanska, Urszula; Thermodynamic Properties of Mixtures Containing Ionic Liquids. Activity Coefficients of Ethers and Alcohols in 1-Methyl-3-Ethyl-Imidazolium Bis(Trifluoromethyl-sulfonyl) Imide Using the Transpiration Method. *J. Chem. Eng. Data*, 2005, 50, 142-148
- [2006HAF] Ribeiro da Silva, Manuel A.V.; Monte, Manuel J.S.; Huinink, Jan; Vapour pressures and standard molar enthalpies of sublimation of beryllium(II) and aluminium(III) benzoylacetates. *Journal of Alloys and Compounds*, 1995, 224, 181-183
- [2006NAS] Vasiltsova, Tatiana V.; Verevkin, Sergey P.; Bich, Eckard; Heintz, Andreas; Bogel-Lukasik, Rafal; Domanska, Urszula; Thermodynamic Properties of Mixtures Containing Ionic Liquids. Activity Coefficients of Ethers and Alcohols in 1-Methyl-3-Ethyl-Imidazolium Bis(Trifluoromethyl-sulfonyl) Imide Using the Transpiration Method. *J. Chem. Eng. Data*, 2005, 50, 142-148
- [2006REI] Reistel, R.; Wagner, W.; A New Equation of State for Ice Ih, *J. Phys. Chem. Ref. Data* 35, 1021-1047 (2006)
- [2006RIB] Ribeiro da Silva, Manuel A.V.; Monte, Manuel J.S.; Santos, Luís M.N.B.F., The design, construction, and testing of a new Knudsen effusion apparatus, *J. Chem. Thermodyn.*, 2006, 38, 778-787
- [2007BOU] Boukais-Belaribi, Ghénima; Belaribi, Brahim Farid; Lohmann, Juergen; Jose, Jacques, Isothermal vapour--liquid equilibria and excess molar enthalpies of hex-2-yne+methyl butyl ether and hex-3-yne+dibutyl ether mixtures, *Fluid Phase Equilibria*, 2007, 262, 180-186
- [2007EFI] Efimova, A.A.; Pashchenko, L.L.; Varushchenko, R.M.; Krasnyh, E.; Levanova, S.V.; The thermodynamics of vaporization of ethyl tert-butyl ether, isobutyl tert-butyl ether, and di-isopropyl ether, *J. Chem. Thermodyn.* 39, 142-147 (2007)
- [2007EME] Emel'yanenko, Vladimir N.; Dabrowska, Aldona; Hertel, Marcus O.; Scheuren, Hans; Sommer, Karl; Vapor Pressures, Enthalpies of Vaporization, and Limiting Activity Coefficients in Water at 100 °C of 2-Furaldehyde, Benzaldehyde, Phenylethanal, and 2-Phenylethanol, *J. Chem. Eng. Data*, 2007, 52, 468-471
- [2007MAL] Malanowski, Stanislaw K.; Vapor-liquid equilibrium for benzaldehyde with 1-methylethylbenzene and for 2-methyl-propan-2-ol with 2,4,4-trimethyl-1-pentene, *J. Chem. Eng. Data*, 2007, 52, 239
- [2007PAS] Pashchenko, L.L.; Kuznetsova, T.S., The enthalpies of vaporization of some hydrocarbons with three-membered rings, *Russ. J. Phys. Chem.*, 2007, 81, 11, 1738-1742
- [2007RIC] Richard, Laurence S.; Bernardes, Carlos E.S.; Diogo, Hermínio P.; Leal, João P.; Minas da Piedade, Manuel E.; Energetics of Cresols and of Methylphenoxyl Radicals, *J. Phys. Chem. A* 111, 8741-8748 (2007)
- [2007VAR] Varushchenko, R.M.; Druzhinina, A.I.; Kuramshina, G.M.; Dorofeeva, O.V.; Thermodynamics of vaporization of some freons and halogenated ethanes and propanes, *Fluid Phase Equilibria* 256, 112-122 (2007)
- [2007VER] Verevkin, Sergey P.; Emel'yanenko, Vladimir N.; Klamt, Andreas, Thermochemistry of Chlorobenzenes and Chlorophenols: Ambient Temperature Vapor Pressures and Enthalpies of Phase Transitions, *J. Chem. Eng. Data*, 2007, 52, 499-510
- [2008FRE] Freedman, Andrew; Kebabian, Paul L.; Li, Ziman; Robinson, Wade A.; Wormhoudt, Joda C., Apparatus for determination of vapor pressures at ambient temperatures employing a Knudsen effusion cell and quartz crystal microbalance, *Meas. Sci. Technol.*, 2008, 19, 125102

- [2008ROU] Roux, M.V.; Temprado, M.; Chickos, J.S.; Nagano, Y., Critically Evaluated Thermochemical Properties of Polycyclic Aromatic Hydrocarbons, J. Phys. Chem. Ref. Data, 2008, 37, 1855-1996
- [2008ZHA] Zhao, Hui; Unhannanant, Patamaporn; Hanshaw, William; Chickos, James S., Enthalpies of Vaporization and Vapor Pressures of Some Deuterated Hydrocarbons. Liquid-Vapor Pressure Isotope Effects, J. Chem. Eng. Data, 2008, 53, 1545-1556
- [2009MOK] Mokbel, Ilham; Razzouk, Antonio; Sawaya, Terufat; Jose, Jacques; Experimental Vapor Pressures of 2-Phenylethylamine, Benzylamine, Triethylamine, and cis -2,6-Dimethylpiperidine in the Range between 0.2 Pa and 75 kPa, J. Chem. Eng. Data, 2009, 54, 819-822
- [2011JAC] Jacobson, N. S.; Myers, D. L.; High-temperature vaporization of B₂O₃(l) under reducing conditions, J. Phys. Chem. B, 2011, 115, 1253-13260
- [2012KRO] Kenneth Kroenlein, Chris D. Muzny, Andrei F. Kazakov, Vladimir Diky, Robert D. Chirico, Joseph W. Magee, Ilmutdin Abdulagatov and Michael Frenkel. NIST/TRC Web Thermo Tables (WTT) NIST Standard Reference Subscription Database 3 - Professional Edition Version 2-2012-1-Pro. <https://wtt-pro.nist.gov/wtt-pro/> (accessed April 27, 2002)
- [2016GOO] Goos, E.; Burcat, A.; Ruscic, B.; Extended Third Millennium Thermodynamic Database for Combustion and Air-Pollution Use with updates from Active Thermochemical Tables (October 5, 2016). <http://garfield.chem.elte.hu/Burcat/BURCAT.THR> (accessed January 21, 2022)
- [2019DIK] Diky, V.; Chirico, R. D.; Frenkel, M.; Bazyleva, A.; Magee, J. W.; Paulechka, E.; Kazakov, A.; Lemmon, E. W.; Muzny, C. D.; Smolyanitsky, A. Y.; Townsend, S.; Kroenlein, K.; ThermoData Engine (TDE), Vewrs. 10.4.2, NIST Standard Reference Database 103b, (2019)
- [2019PAU] Paulechka, E.; Kazakov, A.; Critical Evaluation of the Enthalpies of Formation for Fluorinated Compounds Using Experimental Data and High-Level Ab Initio Calculations, J. Chem. Eng. Data, 2019, 64, 4863-4874
- [2019PRO] Prosen, E.J.; Maron, F.W.; Rossini, F.D., Heats of combustion, formation, and isomerization of ten C₄ hydrocarbons, J. Res. Nat. Bur. Stand., 1951, 46, 106-112
- [2021CHE] <https://www.chemeo.com/>
- [2021RUS] Ruscic, B.; Bross, D. H.; Active Thermochemical Tables (ATcT), ver. 1.122r (2021). <http://ATcT.anl.gov> (accessed December 24, 2021)
- [2022BUR1] this work estimated from $\Delta_v H(T)$ correlation

5. Heat Capacities

5.1. Overview

In this work, we compiled, evaluated, and recommended gas-phase $C_p(g)$, liquid-phase $C_p(l)$, and solid-phase $C_p(s)$ heat capacities at standard state ($T = 298.15$ K) for hydrocarbons, substituted hydrocarbons, and a few other compounds of physical, chemical, and engineering interest. We provide about 650 values for heat capacities for about 310 compounds from about 90 source references. The substituted hydrocarbons include different classes of oxygen-, nitrogen-, sulfur-, and halogen-substituted hydrocarbons. The compounds are organized by compound class – for example, Hydrocarbons: Alkanes, Alkenes; Oxidized Hydrocarbons: Alkanols, Aldehydes; etc. The values were taken from the literature and the NIST Chemistry Webbook.[2021LIN] We checked these values against those provided in the NIST TRC Thermodata Engine (TDE) [2021DIK], and where not readily available in the literature, we utilized values from TDE. The full references are given in Section 5.4.

We provide uncertainties from the original where they were readily available. For many of the uncertainties reported in the literature, they are standard uncertainties of 1σ . For our recommended values, we provide expanded uncertainties U . In cases where uncertainties in heat capacities which were not provided (or recorded), we utilized expanded uncertainties (2σ) of 3 % for gas- and liquid-phase heat capacities, and 0.5 % for solid-phase heat capacities. These estimated expanded uncertainties were chosen based on our analysis of the cases where uncertainties were available. We found for gas-phase heat capacities that about 40% of the uncertainties were less than 1 %, about 30 % were (1 to 3) %, and about 30 % were (3 to 7) %. For liquid-phase heat capacities, we found that about 60 % were under 1 %, 15 % were (1 to 3) %, and 25 % were (3 to 6) %. For solid-phase heat capacities, we found that about 70 % were under 0.5 %, and about 30 % were (0.5 to 1.2) %. Uncertainties of 1.0 % for gas-phase heat capacities $C_p(g)$ from TDE are generic estimates from TDE where experimental values were not available.

In the Table 5-1, heat capacities in the gas-, liquid-, and solid-phases are given by $C_p(g)$, $C_p(l)$, and $C_p(s)$, respectively. The heat capacities are all at $T = 298.15$ K. The gas-phase heat capacities are for ideal gases at standard state, while the liquid-phase heat capacities are at saturation pressures, and the solid-phase heat capacities are either at saturation pressures or at 1 atm (101.325 kPa). For ice, $H_2O(s)$, the solid-phase heat capacity is at the melting point 273.15 K. The column “SEL” indicates all the selected value for all phases SEL, gas-phase SEL(g), liquid-phase SEL(l), or solid-phase SEL(s) heat capacities are from the indicated source, while those labeled SEL(g), SEL(l), or SEL(s) are from the indicated source. Table 5-1 provides all of the data along with reference sources, while Table 5-2 just provides the recommended values.

5.2. Heat Capacities (compilation)

This Table 5-1 contains all the heat capacities that were compiled, while the next Table 5-2 provides only the selected recommended values for heat capacities in the gas, liquid, and solid phases.

Table 5-1. Heat capacities in gas, liquid, and solid phases

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{pg})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{pl})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{ps})$	Reference
SMALL MOLECULES									
Hydrogen/Oxygen									
H ₂	hydrogen	SEL	28.84	0.10					1998CHA
			28.83	0.29					2019DIK
			28.68	0.06					1979BER
O ₂	oxygen	SEL	29.39	0.10					1998CHA
			29.37	0.29					2019DIK
H ₂ O	water	SEL	33.59	0.07	75.33	0.01			2018LEM
		SEL(s)					37.77	0.38	2006FEI
			33.58	0.340	75.3	0.5	39.80	1.30	2019DIK
					75.23	0.08			1939OSB
							37.85		1936GIA
H ₂ O ₂	hydrogen peroxide	SEL(g)	43.07						1998CHA
		SEL(s)	42.36	0.42			59.10	0.20	2019DIK
Nitrogen/Hydrogen									
N ₂	nitrogen	SEL	29.09	0.12					1995ESP
			29.12	0.29					2019DIK
			29.12						1998CHA
			29.11	0.02					1992EWI
H ₃ N	ammonia	SEL(g)	35.64	0.04					2014SOU
		SEL(l)	35.54	0.36	81.4	0.8			2018LEM
			35.65	0.36	80.7	0.7			2019DIK
			35.64	0.20					1998CHA
H ₄ N ₂	hydrazine	SEL	50.81	0.20	98.81	0.02			1998CHA
			49.14	0.49	98.67	0.39			2019DIK

HN ₃	hydrogen azide	SEL	44.22	0.35				1989GUR
			44.16	0.44				2019DIK
Nitrogen/Oxygen								
N ₂ O	nitrous oxide	SEL	38.60	0.20				1998CHA
			38.54	0.39				2019DIK
ClH ₄ NO ₄	ammonium perchlorate	SEL			128.07	0.33		1969WES
Sulfur/Hydrogen								
H ₂ S	hydrogen sulfide	SEL(g)	34.20	0.20				1998CHA
		SEL(l)	34.12	0.34	75.8	3.2		2019DIK
Boron/Hydrogen								
B ₂ H ₆	diborane	SEL	58.07					1998CHA

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{p,g})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{p,l})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{p,s})$	Reference
HYDROCARBONS									
Carbon									
C	carbon (gas)	SEL	20.79	0.00					definition 2019DIK 1998CHA
			20.88	0.21					
			20.82	0.20					
	graphite	SEL					8.43	0.10	1980TAY 2019DIK 1973BUT
							8.52	0.01	
							8.62		
Alkanes									
CH ₄	methane	SEL	35.69	0.20					1989GUR 2019DIK
			35.66	0.36					
C ₂ H ₆	ethane	SEL(g)	52.49	0.20					1989GUR 2019DIK
		SEL(l)	52.50	0.53	160.6	1.9			
C ₃ H ₈	propane	SEL(g)	73.60	0.20					1973CHA 2019DIK 1940KIS
		SEL(l)	73.62	0.74	117.41	0.86			
			72.67	0.07					
C ₄ H ₁₀	isobutane	SEL(g)	96.65	0.20					1975CHE 2019DIK
		SEL(l)	96.69	0.97	140.69	0.29			
	<i>n</i> -butane	SEL(g)	98.49	0.20					1975CHE 2019DIK
		SEL(l)	98.57	0.99	140.50	0.87			
C ₅ H ₁₂	isopentane	SEL(g)	118.90	0.40					1974SCO 2019DIK
		SEL(l)	118.89	1.19	164.84	0.73			
	neopentane	SEL(g)	120.83	0.25					1974SCO 2019DIK
		SEL(l)	120.93	1.21	169.2	6.3			
	<i>n</i> -pentane	SEL(g)	120.00	0.10					1974SCO 1967MES
		SEL(l)			167.19				
	2,3-dimethylbutane	SEL(g)	120.10	1.20	167.14	0.53			2019DIK 1974SCO
		SEL(l)	139.40	0.70					
C ₆ H ₁₄	3-methylpentane		139.39	1.39	189.02				1989OHN 2019DIK 1974SCO
			189.23	0.46					
		SEL(g)	140.10	0.40					
	Isohexane	SEL(l)			191.16				1989OHN 2019DIK 1974SCO
			140.11	1.40	190.67	0.35			
		SEL(g)	142.2	0.2					
	neohehexane	SEL(l)			193.88				1989OHN 2019DIK 1974SCO
			142.21	1.42	193.88	0.69			
		SEL(g)	141.5	0.30					

		SEL(l)			189.67		1989OHN
			141.47	1.41	189.31	0.55	2019DIK
	<i>n</i> -hexane	SEL(g)	142.6	0.20			1974SCO
		SEL(l)			195.52		1989OHN
			142.60	1.43	196.12	0.49	2019DIK
C ₇ H ₁₆	isoheptane	SEL(g)	164.50	0.20			1974SCO
		SEL(l)			222.92	0.45	1961HUF
			164.49	1.64	222.77	0.46	2019DIK
	neoheptane	SEL(g)	166.70	0.40			1974SCO
		SEL(l)			221.12	0.42	1961HUF
			166.74	1.67	221.13	0.53	2019DIK
	<i>n</i> -heptane	SEL(g)	165.20	0.30			1974SCO
		SEL(l)			224.73	0.12	1988SHI
					224.64		1988AND
			165.19	1.65	224.82	0.26	2019DIK
C ₈ H ₁₈	2,2,4-trimethylpentane	SEL(g)	188.40	0.40			1974SCO
		SEL(l)			238.87	1.1	1976FOR
			188.40	1.88	239.12	1.21	2019DIK
	2-methylheptane	SEL(g)	187.2	0.2			1974SCO
		SEL(l)			252.0	0.4	1971MES
			187.21	1.87	251.98	0.25	2019DIK
	<i>n</i> -octane	SEL(g)	187.8	0.4			1974SCO
		SEL(l)			254.18	1.0	1985LAI2
			187.75	1.88	254.30	0.65	2019DIK
C ₉ H ₂₀	<i>n</i> -nonane	SEL(g)	210.4	0.5			1974SCO
		SEL(l)			284.39	0.5	1954FIN
			210.4	2.1	283.73	1.30	2019DIK
C ₁₀ H ₂₂	<i>n</i> -decane	SEL(g)	233.1	0.6	314.23	0.43	1974SCO
		SEL(l)			314.2	1.4	1984ROU
			233.0	2.3	314.23	0.43	2019DIK
C ₁₆ H ₃₄	<i>n</i> -hexadecane	SEL(g)	369.2	3.7	501.14	0.70	2019DIK
		SEL(l)			499.7	1.0	1981GRO
Alkenes							
C ₂ H ₄	ethane	SEL	43.29	0.30			1941BUR
			42.88	0.43			2019DIK
C ₃ H ₆	propene	SEL(g)	63.79	0.13			1974BIE
		SEL(l)			102.00		1983CHA
			64.36	0.64	100.82	1.42	2019DIK
C ₄ H ₈	1-butene	SEL(g)	85.65	0.86	128.71	0.50	2019DIK

		SEL(l)			128.96	0.40	1991TAK
					118.00		1983CHA
	2-butene, (E)-	SEL(g)	87.78	0.40			1940KIS2
		SEL(l)	87.74	0.88	130.0	2.2	2019DIK
	2-butene, (Z)-	SEL(g)	81.13	0.16	126.15	0.50	1944SCO
		SEL(l)			127.00	0.50	1983CHA
			80.21	0.80	127.01	1.18	2019DIK
	isobutene	SEL	88.13	0.88	133.6	4.2	2019DIK
C ₅ H ₁₀	1-pentene	SEL(g)	108.16	1.08	155.04	0.51	2019DIK
		SEL(l)			154.87	0.30	1990MES
					154.00		1983CHA
	2-methyl-1-butene	SEL(g)	109.98	1.10	157.41	0.76	2019DIK
		SEL(l)			157.3	0.8	1983CHA
	2-methyl-2-butene	SEL(g)	104.99	1.05	152.65	0.73	2019DIK
		SEL(l)			152.80		1983CHA
	2-pentene, (E)-	SEL	108.89	1.09	156.97	0.63	2019DIK
	2-pentene, (Z)-	SEL	98.79	0.99	151.58	0.65	2019DIK
	3-methyl-1-butene	SEL	114.48	1.14	156.16	0.68	2019DIK
C ₆ H ₁₂	1-hexene	SEL	130.82	1.31	183.08	0.55	2019DIK
C ₇ H ₁₄	1-heptene	SEL	153.46	1.53	211.86	0.49	2019DIK
C ₈ H ₁₆	1-octene	SEL	176.17	1.76	241.30	0.43	2019DIK
Alkadienes							
C ₃ H ₄	allene	SEL(g)	59.07	0.24			1940KIS
		SEL(l)	59.06	0.59	102.0	3.7	2019DIK
C ₄ H ₆	1,2-butadiene	SEL	79.50	0.80	123.83	0.86	2019DIK
	1,3-butadiene	SEL	79.64	0.80	123.66	0.48	2019DIK
C ₅ H ₈	1,2-pentadiene	SEL	100.98	1.01	150.84	1.25	2019DIK
	1,3-pentadiene	SEL	98.9	6.3	148.6	4.9	2019DIK
	1,4-pentadiene	SEL	98.34	0.98	146.99	1.23	2019DIK
	2-methyl-1,3-butadiene	SEL	102.76	1.03	150.85	1.37	2019DIK
Alkynes							
C ₂ H ₂	acetylene	SEL	44.04	0.30			1989GUR
			43.99	0.44			2019DIK
C ₃ H ₄	propyne	SEL	60.7	0.6	92.5	5	2018LEM
			60.79				1940KIS
			60.75	0.61	109.4	4.1	2019DIK
C ₄ H ₆	1-butyne	SEL	81.4	0.8	135.2	1.1	2019DIK
C ₄ H ₆	2-butyne	SEL	78.1	0.8	125.3	1.3	2019DIK
C ₄ H ₆	2-butyne				125.1		1941YOS

C ₅ H ₈	1-pentyne	SEL	102.67	1.03	143.3	4.0	2019DIK
	2-pentyne	SEL	99.72	1.00	129.6	3.5	2019DIK
C ₆ H ₁₀	1-hexyne	SEL	128.94	1.29	184.8	5.1	2019DIK
	2-hexyne	SEL	119.65		170.8	4.8	2019DIK
	3-hexyne	SEL	119.51	1.20	179.3	5.0	2019DIK
Alkenynes							
C ₄ H ₄	butenyne	SEL(g)	73.18	1.0			1949STA
		SEL(l)	73.20	0.73	118.4	3.9	2019DIK
C ₅ H ₆	3-penten-1-yne	SEL	96.3	5.9	139.5	4.7	2019DIK
Alkadiynes							
C ₄ H ₂	butadiyne	SEL(g)	73.70	1.00			1991DOR
		SEL(l)	73.66	0.74	117.0	3.6	2019DIK
Cycloalkanes							
C ₃ H ₆	cyclopropane	SEL(g)	55.60	1.00			1986DOR2
		SEL(l)	55.58	0.56	96.37	1.42	2019DIK
			55.91				1940KIS
C ₄ H ₈	cyclobutane	SEL(g)	70.60	1.50			1986DOR2
		SEL(l)	70.66	0.71	109.89	0.75	2019DIK
C ₅ H ₁₀	cyclopentane	SEL(g)	82.80	2.00			1986DOR2
		SEL(l)			126.74	0.2	1985TAN
					126.87		1946DOU
			83.01	0.83	127.33	0.66	2019DIK
	methylcyclobutane	SEL	98.79	0.99	145.7	4.9	2019DIK
C ₆ H ₁₂	cyclohexane	SEL(g)	105.30	2.00			1986DOR2
		SEL(l)			156.90		1989LAI
			106.33	1.06	156.20	0.38	2019DIK
	methylcyclopentane	SEL(g)	109.79	1.0			1960SCO
		SEL(l)			158.67	0.2	1946DOU
			109.51	1.10	158.81	0.65	2019DIK
C ₇ H ₁₄	methylcyclohexane	SEL(g)	136.13	1.36	184.75	0.46	2019DIK
		SEL(l)			184.96	0.3	1985TAN
			136.14	1.36	185.44	1.95	2018LEM
					184.2		1946DOU
C ₁₀ H ₂₀	1,3-diethylcyclohexane	SEL	204	13	270	11	2019DIK
Cycloalkenes							
C ₃ H ₄	cyclopropene	SEL(g)	52.90	1.00			1986DOR2
		SEL(l)	52.88	0.53	95.8	3.7	2019DIK
C ₄ H ₆	cyclobutene	SEL(g)	64.40	2.00			1986DOR2
		SEL(l)	64.45	0.64	101.0	3.4	2019DIK

C ₅ H ₈	cyclopentene	SEL(g)	81.3	2.00			1986DOR2
		SEL(l)			122.38		1948HUF
			81.38	0.81	122.60	0.64	2019DIK
	methylenecyclobutane	SEL	87.36	0.87	131.56	0.46	2019DIK
C ₆ H ₁₀	cyclohexene	SEL(g)	101.5	3.0			1986DOR2
		SEL(l)			148.80		1991STE
			101.58	1.02	150.66	0.83	2019DIK
					149.16		1948HUF
Cycloalkadienes							
C ₄ H ₄	cyclobutadiene	SEL(g)	61.0	2.5			1986DOR2
		SEL(l)	60.78	0.61	112.1	3.7	2019DIK
C ₅ H ₆	cyclopentadiene	SEL(g)	75.4	2.00			1986DOR2
		SEL(l)	75.41	0.75	119.9	3.4	2019DIK
C ₆ H ₈	1,3-cyclohexadiene	SEL(g)	94.2	2.5			1986DOR2
		SEL(l)	94.73	0.95	143.7	2.4	2019DIK
	1,4-cyclohexadiene	SEL(g)	94.1	3.0			1986DOR2
		SEL(l)	95.03	0.95	145.6	2.6	2019DIK
Bicycloalkanes							
C ₁₀ H ₁₈	cis-decalin	SEL(g)	168.60	1.00			1988DOR
		SEL(l)			229.17		1989OHN
	trans-decalin	SEL(g)	168.57	1.69	228.52	0.40	2019DIK
		SEL(l)			232.08		1989OHN
C ₁₂ H ₂₂	cyclohexylcyclohexane	SEL(g)	218	15	286.96	0.19	2019DIK
		SEL(l)			286.92	0.2	1998CHI
Alkylbenzenes							
C ₆ H ₆	benzene	SEL(g)	82.2	0.20			2014RUZ
		SEL(l)			135.69	0.30	1993GRO
			82.54	0.83	135.62	0.80	2019DIK
			81.54	0.41	135.43	0.67	2018LEM
					135.71		1985TAN
					135.23		1955STA
C ₇ H ₈	toluene	SEL(g)	103.7	0.40			1985DRA
		SEL(l)			157.09	0.30	1993GRO
			103.80	0.52	157.25	0.32	2019DIK
			103.79	0.52	156.75	0.39	2018LEM
C ₈ H ₁₀	1,2-dimethylbenzene	SEL(g)	132.5	0.40			1985DRA
		SEL(l)			187.65	0.3	1979FOR
			132.43	0.33	187.79	0.28	2019DIK
			131.34	0.33	187.36	0.24	2018LEM

	1,3-dimethylbenzene	SEL(g) SEL(l)	125.80	0.40	184.63			1985DRA	
			125.57	0.32	181.69	0.23		1993GRO	
			125.89	0.63	181.68	0.32		2018LEM	
					181.55			2019DIK	
								1979FOR	
	1,4-dimethylbenzene	SEL(g) SEL(l)	125.6	0.40	181.94	0.3		1985DRA	
			126.21	0.32	182.15	0.40		1979FOR	
			126.23	0.32	182.26	0.45		2019DIK	
			126.5	0.5	183.6	0.4		2018LEM	
								1943PIT	
	ethylbenzene	SEL(g) SEL(l)	127.5	0.2				1978MIL	
					185.87	0.20		1997CHI	
					185.57	0.3		1979FOR	
			127.54	0.32	185.77	0.17		2019DIK	
			127.53	0.32	185.35	0.46		2018LEM	
C ₉ H ₁₂	isopropylbenzene	SEL	159.75	1.60	215.21	0.52		2019DIK	
C ₁₀ H ₁₄	2-isopropyltoluene	SEL	180	12	251.4	8.3		2019DIK	
	3-isopropyltoluene	SEL	180	12	250.4	8.0		2019DIK	
	4-isopropyltoluene	SEL	180	12	238.5	2.8		2019DIK	
Alkenylbenzenes									
C ₈ H ₈	ethenylbenzene	SEL(g) SEL(l)	120.27	1.20	181.3 182.60	2.1		2019DIK 1961WAR	
Benzocycloalkanes									
C ₉ H ₁₀	benzocyclopentane	SEL(g) SEL(l)	130.7 130.73	1.00 1.31	189.66	0.21		1986DOR 2019DIK	
C ₁₀ H ₁₂	benzocyclohexane	SEL(g) SEL(l)	150.9	2.00	217.44 217.43			1988DOR 1957MCC 2019DIK	
Naphthalenes									
C ₁₀ H ₈	naphthalene	SEL(g) SEL(l) SEL(s)	131.94	1.32	196.06		165.69 165.69	0.46 0.5	2019DIK 1993CHI 1957MCC
C ₁₁ H ₁₀	1-methylnaphthalene	SEL(g) SEL(l)	159.59	1.60	224.34 224.39	0.65		2019DIK 1957MCC	
	2-methylnaphthalene	SEL(g) SEL(l)	159.83	1.60	195.98	0.6		2019DIK 1957MCC	
Polycyclic Aromatics									
C ₁₂ H ₈	acenaphthylene	SEL(g) SEL(s)	154.80 154.79	1.00 1.55			183	15	1986DOR 2019DIK

	biphenylene	SEL	159.35	1.59				2019DIK	
C ₁₂ H ₁₀	acenaphthene	SEL(g)	162.20	1.00				1986DOR	
		SEL(s)				190.37	0.50	1977FIN	
			162.17	1.62		190.38	0.45	2019DIK	
C ₁₃ H ₁₀	fluorene	SEL(g)	173.10	1.00				1989DOR	
		SEL(s)				203.13		1977FIN	
			173.12	1.73		203.14	0.50	2019DIK	
C ₁₄ H ₈	pyracclene	SEL	172.50	1.50				1986DOR	
			172.56	1.73				2019DIK	
C ₁₄ H ₁₀	anthracene	SEL(g)	184.70	1.00				1988DOR	
		SEL(s)				211.70	1.00	1980RAD	
			184.78	1.85		210.50	0.32	2019DIK	
	phenanthrene	SEL(g)	185.70	1.86				2019DIK	
		SEL(s)				220.63	0.54	2019DIK	
						220.62		1977FIN	
C ₁₄ H ₁₂	pyracene	SEL	187.20	1.50				1986DOR	
			187.22	1.87				2019DIK	
C ₁₆ H ₁₀	aceanthrylene	SEL(g)	203.00	1.50				1986DOR	
		SEL(l)	203.0	2.0	334	18		2019DIK	
			fluoranthene	SEL(g)	203.60	2.00			
SEL(s)							230.25		1971WON
	206.3			2.1			230.25	0.35	2019DIK
	pyrene	SEL(g)	204.20					1988DOR	
		SEL(s)					229.36		1980SMI
			201.6	2.0			229.64	0.34	2019DIK
						229.70		1971WON	
C ₁₈ H ₁₂	naphthacene	SEL(g)	233.40	1.50				1988DOR	
		SEL(s)					236.60		1980WON
			233.1	2.3			259.37	3.09	2019DIK
	triphenylene	SEL(g)	222.1	2.2				2019DIK	
		SEL(s)					259.20	0.33	2019DIK
							229.70		1971WON
C ₂₀ H ₁₂	perylene	SEL(g)	255.4	2.4				1988DOR	
		SEL(s)	na				274.93	0.46	1980WON
C ₂₄ H ₁₂	coronene	SEL(g)	286.6	5.0				1988DOR	
		SEL(s)					313.80	1.00	1980WON
				277.9	2.8			313.81	1.21

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{pg})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{pl})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{ps})$	Reference
Carbon Oxides									
CO	carbon monoxide	SEL	29.14	0.15					1998CHA
CO ₂	carbon dioxide	SEL	37.14	0.16	284.63	1.06			2018LEM
			37.12	0.16					2019DIK
			37.12	0.20					1998CHA
			37.40	1.47					1964SEN
			37.44	0.42					1955HIL
Alkanols									
CH ₄ O	methanol	SEL(g) SEL(l)	44.04	0.03					1986CHA
					81.08	0.20			1985WIL
			44.08	0.44	81.16	0.26			2019DIK
			44.0	0.4	81.3	0.8			2018LEM
					80.35				1988OKA
					80.24				1988AND
					81.56	0.4			1986TAN
C ₂ H ₆ O	ethanol	SEL(g) SEL(l)	65.20	0.14					1986CHA
					112.25	0.30			1985WIL
			65.30	0.65	112.35	0.47			2019DIK
			65.2	0.7	112.2	1.1			2018LEM
					111.53				1988AND
					112.68				1986TAN
C ₃ H ₈ O	1-propanol	SEL(g) SEL(l)	85.56	0.14					1986CHA
					143.74	0.4			1985WIL
					143.96	0.30			1986TAN
			85.77	0.86	143.74	0.23			2019DIK
	isopropanol	SEL(g) SEL(l)	89.32	0.15					1986CHA
					155.0	0.3			1985WIL
			89.51	0.90	154.92	0.34			2019DIK
C ₄ H ₁₀ O	1-butanol	SEL(g) SEL(l)	108.03	0.25					1986CHA
					177.06	0.35			1985WIL
					176.67	0.50			1986GAT
			108.35	1.08	177.17	0.62			2019DIK
					176.86				1988AND
					176.69				1986TAN
	2-butanol	SEL(g) SEL(l)	112.74	0.17					1986CHA
					196.89	0.25			1985WIL
					198.03				1988OKA

	isobutanol	SEL(g) SEL(l)	112.95 113	1.13 7	196.81 181.20 181.16 181.05	0.84 0.28 0.4			2019DIK 2019DIK 1985WIL 1988OKA
	tert-butanol	SEL(g) SEL(l,s)	113.63 113.93	0.21 1.14	219.08 221.88 215.37	0.4 145.9 146.1	2.0 2.0		1986CHA 1985WIL 2019DIK 1988OKA 1988CAC 1963OET
C ₅ H ₁₂ O	1-pentanol	SEL(g) SEL(l)	130.75	1.31	208.31 208.19	0.41 0.30			2019DIK 1986TAN
	2-methyl-1-butanol	SEL	136.2	8.6	213.77	1.84			2019DIK
	isopentanol	SEL	136.2	8.6	209.75	1.66			2019DIK
C ₆ H ₁₄ O	n-hexanol	SEL(g) SEL(l)	153.39	1.53	242.00 240.57 241.32	1.25			2019DIK 1986TAN 1988AND
Glycols									
C ₂ H ₆ O ₂	1,2-ethanediol	SEL(g) SEL(l)	82.7	0.13	149.3	0.8			1986CHA 1985WIL
			82.67	0.83	148.75	0.61			2019DIK
C ₃ H ₈ O ₂	2-methoxyethanol	SEL	106.8	6.4	175.28	1.41			2019DIK
C ₃ H ₈ O ₃	glycerin	SEL(g) SEL(l)	124.0	7.6	220.57 206.3	0.79 1.0			2019DIK 1985WIL
Sugars									
C ₆ H ₁₂ O ₆	alpha-D-glucose	SEL					219.19	0.42	2019DIK
	fructose	SEL					213.15	0.67	2019DIK
C ₁₂ H ₂₂ O ₁₁	sucrose	SEL					424.30	0.82	2019DIK
Cycloalkanols									
C ₅ H ₁₀ O	cyclopentanol	SEL(g) SEL(l)	113.12	1.13	182.37 184.14	0.85			2019DIK 1956PAR
C ₆ H ₁₂ O	cyclohexanol	SEL	127.65 132.70	1.28	209.38 214.06 209.99	1.63	192.31	0.30	2019DIK 1990MAY 1988KAB 1988CAC
Ethers									
C ₂ H ₆ O	dimethyl ether	SEL(g) SEL(l)	65.57 65.60 65.54	0.08 0.66	112.66	1.47			1986CHA 2019DIK 1940KIS

C ₄ H ₁₀ O	diethyl ether	SEL(g) SEL(l)	119.46 119.62	0.15 1.20	172.4 172.37 172.50	0.2 0.44	1986CHA 1985WIL 2019DIK 1971COU
C ₆ H ₁₄ O	diisopropyl ether	SEL(g) SEL(l)	158.31	1.58	216.42 216.74	0.70	2019DIK 1993GRO
Aldehydes							
CH ₂ O	formaldehyde	SEL(g) SEL(l)	35.39	0.02	82.6	3.0	1986CHA 2019DIK
C ₂ H ₄ O	acetaldehyde	SEL(g) SEL(l)	55.32	0.08	89.05 100.8	4.2	1986CHA 1988LEB 2012KRO
C ₃ H ₆ O	propanal	SEL(g) SEL(l)	80.73	0.10	137.2	1.4	1986CHA 1980FUC
C ₄ H ₈ O	butanal	SEL(g) SEL(l)	103.36	0.1	163.7	1.0	1986CHA 1985WIL
Alkenals							
C ₃ H ₄ O	1-propenal	SEL	71.46	0.72			2019DIK
C ₃ H ₄ O	2-propenal	SEL	71.28	0.71	118.7	3.3	2019DIK
Ketones							
C ₃ H ₆ O	acetone	SEL(g) SEL(l)	75.02 74.52	0.11 0.75	126.81 125.45 126.6 123.80 128.24	0.67 1.00 1.5	1986CHA 2019DIK 1991MAL 1985WIL 1985COS 1955STA
C ₄ H ₈ O	butanone	SEL(g) SEL(l)	101.68 103.27	0.14 1.03	158.40 158.74 158.90 158.41	0.30 0.41 0.4	1986CHA 1985COS 2019DIK 1985WIL 1956PAR
Carboxylic Acids							
CH ₂ O ₂	formic acid	SEL(g) SEL(l)	45.68 45.70	0.07 0.46	99.51 99.77 99.04	0.2 0.90	1986CHA 1985WIL 2019DIK 1941STO
C ₂ H ₄ O ₂	acetic acid	SEL(g) SEL(l)	63.44 63.46	0.11 0.64	123.10 123.09	0.30 0.38	1978CHA 1982MAR 2019DIK

C ₃ H ₆ O	propanoic acid	SEL(g) SEL(l)	87.6	5.4	123.6 152.81 152.80 152.7	2.0 0.20 2.0	1985WIL 2019DIK 1982MAR 1985WIL
C ₄ H ₈ O	<i>n</i> -butanoic acid	SEL	110.8	6.9	177.67 178.6 177.70 178.00	0.19 1.4	2019DIK 1985WIL 1982MAR 1971KON
C ₅ H ₁₀ O ₂	<i>n</i> -pentanoic acid	SEL(g) SEL(l)	134.1	8.4	207.8 210.33 197.00	2.1	2019DIK 1965MCD 1971KON
Esters							
C ₂ H ₄ O ₂	methyl formate	SEL(g) SEL(l)	64.38	0.09	119.70		1986CHA 1979FUC
C ₃ H ₆ O ₂	ethyl formate	SEL(g) SEL(l)	66.61 88.64	0.67 0.89	119.4 143.7 144.30	3.5 2.5	2019DIK 1979FUC
	methyl acetate	SEL(g) SEL(l)	86.03	0.12	141.34		1986CHA 1988PIN
			86.04	0.86	143.09 140.60	0.42	2019DIK 1985COS
C ₄ H ₈ O ₂	ethyl acetate	SEL(g) SEL(l)	113.66	1.14	168.84 168.94 170.6 169.60 167.40	0.62 0.5	2019DIK 1988PIN 1985WIL 1985COS 1979FUC
	methyl propanoate	SEL(g) SEL(l)	116.58	1.17	173.96 174.20 171.21	0.68	2019DIK 1979FUC 1988PIN
	<i>n</i> -propyl formate	SEL(g) SEL(l)	117.6	7.1	171.91 172.10	1.88	2019DIK 1988PIN
C ₅ H ₁₀ O ₂	methyl butanoate	SEL(g) SEL(l)	130.7	8.3	197.96 200.80	0.25	2019DIK 1979FUC
Alkenyl Esters							
C ₄ H ₆ O ₂	ethenyl acetate	SEL	99.00	0.99	163.7	4.5	2019DIK
C ₅ H ₈ O ₂	ethyl propenoate	SEL	121.2	7.7	184.0	5.2	2019DIK
	methyl 2-methylpropenoate	SEL	122.1	7.7	187.3	5.9	2019DIK
Oxacycloalkanes							
C ₂ H ₄ O	oxacyclopropane	SEL(g)	47.0	1.00			1992DOR

C ₃ H ₆ O	methyloxacyclopropane	SEL(l)	47.87	0.48	89.56	0.68		2019DIK
		SEL(g)	72.55	0.12				1986CHA
		SEL(l)			120.41	0.35		1985WIL
	oxacyclobutane	SEL(g)	61.50	2.00				1992DOR
		SEL(l)	62.03	1.41	114.5	3.2		2019DIK
C ₄ H ₈ O	oxacyclopentane	SEL(g)	76.25	0.16				1986CH
		SEL(l)			124.1	0.2		1985WIL
			76.57	0.77	124.15	0.64		2019DIK
					124.10			1985COS
Oxacycloalkenes								
C ₂ H ₂ O	oxacyclopropene	SEL	54.2	3.0				1992DOR
C ₄ H ₄ O	oxacyclopentadiene	SEL(g)	65.40	0.29				1986CHA
		SEL(l)			114.56			1952GUT
			65.40	1.50				1992DOR
			65.43	0.65	114.61	0.35		2019DIK
C ₄ H ₆ O	1-oxa-3-cyclopentene	SEL(g)	75.60	0.12				1986CHA
		SEL(l)	75.78	0.76	126.3	4.5		2019DIK
			73.21	0.11				1992DOR
Multioxacycloalkanes								
CH ₂ O ₂	1,2-dioxacyclopropane	SEL(g)	43.9	3.0				1992DOR
C ₂ H ₄ O ₂	1,2-dioxacyclobutane	SEL(g)	77.0	4.0				1992DOR
C ₂ H ₄ O ₃	1,2,3-trioxacyclopentane	SEL(g)	78.0	4.5				1992DOR
C ₃ H ₆ O ₂	1,3-dioxacyclopentane	SEL(g)	71.0	4.0				1992DOR
		SEL(l)	73.9	4.9	121.71	1.19		2019DIK
C ₃ H ₆ O ₃	1,3,5-trioxacyclohexane	SEL	82.36	0.82			113.08	2019DIK
C ₄ H ₈ O ₂	1,3-dioxane	SEL	89.54	0.90	140.3	2.4		2019DIK
	1,4-dioxane	SEL	92.25	0.92	150.56	0.66		2019DIK
					149.65			1993GRO
					153.6	2.5		1985WIL
Furandione and D-Camphor								
C ₄ H ₂ O ₃	2,5-furandione	SEL	84.6	5.3			119.24	2019DIK
C ₁₀ H ₁₆ O	D-camphor	SEL	191	13			246.29	2019DIK
Aromatic Alcohols								
C ₆ H ₆ O	phenol	SEL(g)	103.22					1978KUD
		SEL(s)					127.21	1975NIC
			103.22	1.03			127.45	2019DIK
C ₇ H ₈ O	2-methylphenol	SEL(g)	127.30					1978KUD
		SEL(s)					154.56	1967AND
			127.30	1.27			154.56	2019DIK

3-methylphenol	SEL(g)	124.68						1978KUD
	SEL(l)			224.93	1.30			1967AND
4-methylphenol		124.77	1.25	223.3	2.0			2019DIK
	SEL(g)	124.97	0.50					1978KUD
	SEL(s)					150.25	1.00	1967AND
		125.10	1.25			150.25	1.16	2019DIK
benzyl alcohol						221.03		1975NIC
	SEL(g)	126.8	8.1	215.9	2.4			2019DIK
	SEL(l)			215.94	1.00			1975NIC
Aromatic Aldehydes and Acids								
C ₇ H ₆ O	benzaldehyde	SEL(g)	111.70	1.00				1975AMB
		SEL(l)			172.00			1975AMB2
C ₇ H ₆ O ₂	benzoic acid		111.68	1.12	171.99	0.37		2019DIK
		SEL(g)	124.0	7.9			146.74	1.42
		SEL(s)					147.78	1.00
			103.47					1993KAJ
								1969STU

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{pg})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{pl})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{ps})$	Reference
NITROGEN-SUBSTITUTED HYDROCARBONS									
Hydrogen Cyanides									
CHN	hydrogen cyanide	SEL	35.86	0.36	72.95	1.43			2019DIK
Alkyl Amines									
CH ₅ N	methylamine	SEL	50.08	0.50	104.8	2.7			2019DIK
C ₂ H ₇ N	ethylamine	SEL	71.65	0.72					2019DIK
C ₃ H ₉ N	1-propylamine	SEL(g)	91.25	0.91	164.35	0.92			2019DIK
		SEL(l)			162.51	1.00			1972FIN
					160.00				1971KON
	isopropylamine	SEL(g)	97.62	0.98	163.84	0.63			2019DIK
		SEL(l)			163.85				1972FIN
					164.00				1971KON
C ₄ H ₁₁ N	isobutylamine	SEL(g)	117.1	7.5					2019DIK
		SEL(l)			194.00				1971KON
	<i>n</i> -butylamine	SEL(g)	118.52	1.19	188.1	4.5			2019DIK
		SEL(l)			188.00				1971KON
	<i>tert</i> -butylamine	SEL(g)	121.00	1.21	192.06	0.63			2019DIK
		SEL(l)			190.00				1971KON
C ₂ H ₇ N	dimethylamine	SEL	70.45	0.70	140.5	2.6			2019DIK
C ₃ H ₉ N	methylethylamine	SEL	87.6	5.7					2019DIK
C ₄ H ₁₁ N	diethylamine	SEL(g)	115.77	1.16	178.09	0.85			2019DIK
		SEL(l)			178.10				1985COS
C ₃ H ₉ N	trimethylamine	SEL	91.80	0.92	138.15	0.99			2019DIK
C ₆ H ₁₅ N	<i>n</i> -hexylamine	SEL(g)	164	10	250.5	2.4			2019DIK
		SEL(l)			252.00				1971KON
		SEL(g)	160.98	1.61	221.3	2.3			2019DIK
	triethylamine	SEL(l)			216.43				1993GRO
C ₆ H ₁₃ N	cyclohexylamine	SEL	132.0	8.7	201.2	8.9			2019DIK
C ₆ H ₁₂ N ₄	methenamine	SEL(s)					152.31	0.55	2019DIK
Alkanenitriles									
C ₂ H ₃ N	ethanenitrile	SEL	52.25	0.52	91.22	0.37			2019DIK
C ₃ H ₅ N	propanenitrile	SEL	73.77	0.74	120.26	0.89			2019DIK
C ₄ H ₇ N	butanenitrile	SEL	94.51	0.95	133.50	1.88			2019DIK
C ₃ H ₃ N	propenenitrile	SEL	63.93	0.64	108.48	0.49			2019DIK
Cyanogens									
C ₂ N ₂	cyanogen	SEL	56.92	0.57	113.9	4.7			2019DIK
Hydrazines									

C ₂ H ₈ N ₂	1,1-dimethylhydrazine	SEL	89.7	5.7	164.06	0.67		2019DIK
Aromatic Amines								
C ₃ H ₆ N ₆	melamine	SEL	137.8	8.5			155.16 0.75	2019DIK
C ₆ H ₇ N	aniline	SEL(g)	108.14	1.08	192.42	0.82		2019DIK
		SEL(l)			191.01	1.20		1975NIC
C ₈ H ₁₁ N	<i>N,N</i> -dimethylaniline	SEL	151.2	9.8	211.9	5.6		2019DIK
Ureas								
CH ₄ N ₂ O	urea	SEL	77.55	0.78			92.79 0.32	2019DIK
Oxyamines								
C ₆ H ₁₅ NO ₃	triethanolamine	SEL	213	13	361	21		2019DIK
Alkyl Nitrates and Nitroalkanes								
CH ₃ NO ₃	methyl nitrate	SEL(g)	76.49	0.77	152.1	7.1		2019DIK
		SEL(l)			157.19			1953GRA
C ₃ H ₅ N ₃ O ₉	nitroglycerin	SEL	234.5	2.6				2019DIK
CN ₄ O ₈	tetranitromethane	SEL	171.6	5.4	207	11		2019DIK
CHN ₃ O ₆	trinitromethane	SEL	131.5	3.8	182	11		2019DIK
CH ₃ NO ₂	nitromethane	SEL(g)	57.27	0.57	106.25	0.78		2019DIK
		SEL(l)			105.98			1947JON
Isocyano Aromatics								
C ₉ H ₆ N ₂ O ₂	2,4-diisocyanatotoluene	SEL			287.80			2019DIK
Nitro Aromatics								
C ₆ H ₃ N ₃ O ₆	trinitrobenzene	SEL(g)	197.48	1.97			222.67 5.81	2019DIK
		SEL(s)					214.60	1980RAD
C ₆ H ₅ NO ₂	nitrobenzene	SEL(g)	119.57	1.20	182.1	2.2		2019DIK
		SEL(l)			181.13			1985LAI
C ₇ H ₅ N ₃ O ₆	trinitrotoluene	SEL(g)	213.3	5.6			236.80 12.30	2019DIK
		SEL(s)					243.30	1991YIN

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{pg})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{pl})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{ps})$	Reference
SULFUR-SUBSTITUTED HYDROCARBONS									
Alkanethiols									
CH ₄ S	methanethiol	SEL	50.31	0.50	90.47	0.63			2019DIK
C ₂ H ₆ S	ethanethiol	SEL(g) SEL(l)	73.02	0.73	117.82 117.99	0.24			2019DIK 1952MCC
Alkyl Sulfides									
CS ₂	carbon disulfide	SEL(g) SEL(l)	45.51	0.46	78.14 78.99	0.62 0.80			2019DIK 1955STA
C ₂ H ₆ S	dimethyl sulfide	SEL	74.08	0.74	118.28	0.84			2019DIK
C ₂ H ₆ OS	dimethyl sulfoxide	SEL(g) SEL(l)	89.43	0.89	150.97 149.39	1.05			2019DIK 1993GRO

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{pg})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{pl})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{ps})$	Reference
HALOGENATED HYDROCARBONS									
Fluoromethanes									
CH ₃ F	fluoromethane	SEL(g) SEL(l)	37.49		106.26 105.89	1.09 0.91			1998CHA 1998HO 2019DIK 1965BAR
CH ₂ F ₂	difluoromethane	SEL(g) SEL(l)	42.83		97.26 97.33	0.80 0.42			1998CHA 1996LUD 2019DIK 1965BAR
CHF ₃	trifluoromethane	SEL	51.04 51.01 50.99 51.38	0.30 0.51 0.51					1962VAL 2019DIK 1998CHA 1965BAR
CF ₄	tetrafluoromethane	SEL	61.02 61.04 61.07 61.55	0.61 0.03 0.60					1998CHA 2019DIK 1989EWI 1965BAR
Fluoroethanes									
C ₂ H ₅ F	fluoroethane	SEL	59.62 59.65	0.60 0.60	108.9 112.2	5 5.3			2018LEM 2019DIK
C ₂ H ₄ F ₂	1,1-difluoroethane	SEL	67.48 68.49	0.67 0.69	118.9 118.38	1.2 0.21			2018LEM 2019DIK
C ₂ H ₄ F ₂	1,2-difluoroethane	SEL	67.77	0.68	217.0	7.2			2019DIK
C ₂ H ₃ F ₃	1,1,1-trifluoroethane	SEL	78.08	0.16	140.24	0.70			2018LEM
	1,1,2-trifluoroethane	SEL	77.42	0.77	137.5	4.6			2019DIK
C ₂ H ₂ F ₄	1,1,1,2-tetrafluoroethane	SEL	85.03 86.69	0.21 0.87	145.36 143.55	0.73 0.74			2018LEM 2019DIK
	1,1,2,2-tetrafluoroethane	SEL	90.35	0.90	147.7	5.8			2019DIK
C ₂ HF ₅	pentafluoroethane	SEL	94.44	0.94	166.0	6.7			2019DIK
C ₂ F ₆	hexafluoroethane	SEL	106.61	1.07					2019DIK
Fluoroethenes									
C ₂ H ₃ F	fluoroethene	SEL	50.43	0.50	99.1	6.7			2019DIK
C ₂ H ₂ F ₂	1,1-difluoroethene	SEL	59.08	0.59					2019DIK
	1,2-difluoroethene, (E)-	SEL	60.26	0.60	129.9	8.0			2019DIK
	1,2-difluoroethene, (Z)-	SEL	58.35	0.58	103.5	5.5			2019DIK
C ₂ HF ₃	trifluoroethene	SEL	69.20	0.69	145	11			2019DIK

C ₂ F ₄	tetrafluoroethene	SEL	80.42	0.80			2019DIK
Chloromethanes							
CH ₃ Cl	chloromethane	SEL(g) SEL(l)	40.74 42.07	0.41 1.69	81.44 81.70 81.20	0.72 0.50	2019DIK 1940AWB 1964SEN 1924SHO
CH ₂ Cl ₂	dichloromethane	SEL(g) SEL(l)	50.85 50.90 51.60 52.6	0.51 0.48 2.1	98.6 100.13 97.97 102.27 99.50 100.0 100.59	2.4 0.91 1.96 0.71 0.36 3.2 1.00	1998CHA 2015CHO 2019DIK 2008TJA 1978MOS 1965BAR 1964SEN 1943MCG 1940RIE 1937PERE
CHCl ₃	trichloromethane	SEL(g) SEL(l)	65.33 65.36 65.87	0.65 0.61	113.20 114.43 114.25 114.18	0.47	1998CHA 1993SHE 2019DIK 1993GRO 1965BAR 1955STA
CCl ₄	tetrachloromethane	SEL(g) SEL(l)	82.65 83.46	0.84	131.34 131.74 131.30 132.59 131.69 132.7	0.41 0.46 8.0	1998CHA 1982TAN 2019DIK 1993SHE 1955STA 1944HIC 1937STU
Chloroethanes							
C ₂ H ₅ Cl	chloroethane	SEL	62.65	0.63	105.31	0.81	2019DIK
C ₂ H ₄ Cl ₂	1,1-dichloroethane	SEL	76.33	0.76	126.44	0.63	2019DIK
	1,2-dichloroethane	SEL	77.32	0.77	129.19	0.52	2019DIK
C ₂ H ₃ Cl ₃	1,1,1-trichloroethane	SEL	92.33	0.92	144.37	0.81	2019DIK
	1,1,2-trichloroethane	SEL	84.94	0.85	141.4	5.1	2019DIK
C ₂ H ₂ Cl ₄	1,1,1,2-tetrachloroethane	SEL	102.88	1.03	157.9	6.1	2019DIK
	1,1,2,2-tetrachloroethane	SEL(g) SEL(l)	98.97	0.99	165.62 168.00	0.83	2019DIK 1989WIL
C ₂ HCl ₅	pentachloroethane	SEL	118.07	1.18	174.3	7.2	2019DIK

C ₂ Cl ₆	hexachloroethane	SEL	136.78	1.37			198.24	1.64	2019DIK
Chloroethenes									
C ₂ H ₃ Cl	chloroethene	SEL	53.60	0.54	96.2	3.3			2019DIK
C ₂ H ₂ Cl ₂	1,1-dichloroethene	SEL	66.93	0.67	110.80	0.67			2019DIK
	1,2-dichloroethene, (E)-	SEL	66.55	0.67	113.83	1.18			2019DIK
	1,2-dichloroethene, (Z)-	SEL	65.00	0.65	112.85	1.12			2019DIK
C ₂ HCl ₃	trichloroethene	SEL	79.75	1.00	128.70	1.13			2019DIK
C ₂ Cl ₄	tetrachloroethene	SEL(g) SEL(l)	95.51						1993GRO 1989WIL 2019DIK
			94.88	0.95	151.92	1.94			
Chloroalkynes									
C ₂ HCl	chloroethyne	SEL	51.6	2.8	135.5	4.4			2019DIK
C ₂ Cl ₂	dichloroethyne	SEL	61.1	3.2	185.1	8.0			2019DIK
Chloroaromatics									
C ₆ H ₅ Cl	chlorobenzene	SEL(g) SEL(l)	97.99	0.98	150.87 152.10	1.14			2019DIK 1993SHE
Bromoalkanes									
CH ₃ Br	bromomethane	SEL	42.45	0.43	81.59	1.21			2019DIK
CH ₂ Br ₂	dibromomethane	SEL(g) SEL(l)	54.59	0.55	104.48 104.1 105.15	0.88 3.1 0.41			2019DIK 1993SHE 1957HAR
CHBr ₃	tribromomethane	SEL	70.99	0.71	121.5 135.10	6.3			2019DIK 1932TRE
CBr ₄	tetrabromomethane	SEL(g) SEL(s)	90.71				145.90 127.78	0.51	1998CHA 1984BIC 2019DIK
C ₂ H ₅ Br	bromoethane	SEL(g) SEL(l)	64.27	0.64	107.38 105.80	0.81			2019DIK 1993SHE
Bromoethenes									
C ₂ H ₃ Br	bromoethene	SEL(g) SEL(l)	55.49	0.56					2019DIK 2012KRO
					114.3	3.7			
Iodoalkanes									
CH ₃ I	iodomethane	SEL(g) SEL(l)	44.08	0.44	82.58 82.75 82.00	0.61			2019DIK 1993CAR 1993SHE
CH ₂ I ₂	diiodomethane	SEL(g) SEL(l)	57.75	0.58	112.6 112.80 133.81	5.8			2019DIK 1993SHE 1993CAR
CHI ₃	triiodomethane	SEL(g)	75.06	0.75					2019DIK

Cl ₄	tetraiodomethane	SEL(I) SEL(g) SEL(l)	95.78	0.96	157.50 259.80		1993CAR 2019DIK 1993CAR
Chlorofluoroalkanes							
CH ₂ ClF	chlorofluoromethane	SEL(g) SEL(l)	47.29 47.08	0.46 0.47			1965BAR 2019DIK
CHCl ₂ F	dichlorofluoromethane	SEL(g) SEL(l)	61.33 60.95	0.64 0.61	73.2 107.5	2.4 3.8	1965BAR 2019DIK
CHClF ₂	chlorodifluoromethane	SEL(g) SEL(l)	56.00	0.22			1995ESP
					107.2	3.2	1940BEN
			55.87	0.56	108.5	2.5	2019DIK
			56.35	0.06			1970ERN
			56.17	0.60			1965BAR
CCl ₂ F ₂	dichlorodifluoromethane	SEL(g) SEL(l)	73.92 72.30	0.21 0.72			1991BUF
			73.66	0.47	119.9	4.8	2019DIK
			72.94	0.73			1969SHU
CCl ₃ F	trichlorofluoromethane	SEL(g) SEL(l)	78.09	0.78	121.67 121.55	0.83 0.46	1965BAR
			74.64	0.76			1941OSB
					123.6	3.6	1965BAR
CClF ₃	chlorotrifluoromethane	SEL	66.87	0.67			1940BEN
							2019DIK
Chlorofluoroethenes							
C ₂ ClF ₃	chlorotrifluoroethene	SEL	83.94	0.84	137.0	2.7	2019DIK
Bromofluoroalkanes							
CH ₂ BrF	bromofluoromethane	SEL	49.14	0.49	114.3	3.8	2019DIK
CHBrF ₂	bromodifluoromethane	SEL	58.78	0.59	106.5	3.8	2019DIK
CBr ₂ F ₂	dibromodifluoromethane	SEL	77.02	0.77	118.6	3.6	2019DIK
CBr ₃ F	tribromofluoromethane	SEL	84.37	0.84	169.7	6.4	2019DIK
CBrF ₃	bromotrifluoromethane	SEL	69.26	0.69	125.24	0.27	2019DIK
Bromochloroalkanes							
CH ₂ BrCl	bromochloromethane	SEL(g) SEL(l)	52.76	0.53	103.90 104.1	1.22 2.6	2019DIK
CHBr ₂ Cl	dibromochloromethane	SEL	69.15	0.69	129.3	5.2	2015CHO
CHBrCl ₂	bromodichloromethane	SEL	67.25	0.67	115.9	3.6	2019DIK
CBr ₂ Cl ₂	dibromodichloromethane	SEL	87.15	0.87	140.43	1.08	2019DIK
CBrCl ₃	bromotrichloromethane	SEL	85.38	0.85	135.29	0.56	2019DIK
Bromochlorofluoroalkanes							
CHBrClF	bromochlorofluoromethane	SEL	62.85	0.63	103.9	3.6	2019DIK

CBr ₂ ClF	dibromochlorofluoromethane	SEL	82.33	0.82	130.0	4.8	2019DIK
CBrCl ₂ F	bromodichlorofluoromethane	SEL	80.06	0.80	126.4	4.4	2019DIK
CBrClF ₂	bromochlorodifluoromethane	SEL	74.66	0.75	118.8	3.9	2019DIK
Carbonyl Halides							
CCl ₂ O	phosgene	SEL(g)	57.69	0.58			1998CHA
		SEL(l)	57.69	0.58	104.08	0.97	2019DIK
Chloroaromatic Alcohols							
C ₆ HCl ₅ O	pentachlorophenol	SEL(g)	189.80	1.90			2019DIK
		SEL(l)			201.96		1958HIL

Formula	Compound	SEL	$C_p(g)$ $J mol^{-1} K^{-1}$	$u(C_{pg})$	$C_p(l)$ $J mol^{-1} K^{-1}$	$u(C_{pl})$	$C_p(s)$ $J mol^{-1} K^{-1}$	$u(C_{ps})$	Reference
Siloxanes									
$C_6H_{18}OSi_2$	hexamethyldisiloxane	SEL			311.38	0.82			2019DIK
$C_8H_{24}O_4Si_4$	octamethylcyclotetrasiloxane	SEL			495.9	9.5			2019DIK

5.3. Recommended Heat Capacities

This Table 5-2 provides the selected recommended values for heat capacities in the gas, liquid, and solid phases. All the heat capacities that were compiled are provided previously in Table 5-1.

Table 5-2. Recommended heat capacities in gas, liquid, and solid phases

Table 5-2(1). Small Molecules

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_p g)$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_p l)$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_p s)$	Reference
Hydrogen/Oxygen									
H ₂	hydrogen	SEL	28.83	0.29					2019DIK
O ₂	oxygen	SEL	29.39						1998CHA
H ₂ O	water	SEL(g,l) SEL(s)	33.59	0.07	75.33	0.01	37.77 59.10	0.38 0.20	2018LEM 2006FEI 2019DIK
H ₂ O ₂	hydrogen peroxide	SEL	42.36	0.42					
Nitrogen/Hydrogen/Oxygen									
N ₂	nitrogen	SEL	29.09	0.12					1995ESP
H ₃ N	ammonia	SEL(g) SEL(l)	35.64	0.20					1998CHA 2018LEM
H ₄ N ₂	hydrazine	SEL	49.14	0.49	81.4 98.67	0.8 0.39			2019DIK
HN ₃	hydrogen azide	SEL	44.22	0.35					1989GUR
N ₂ O	nitrous oxide	SEL	38.54	0.39					2019DIK
ClH ₄ NO ₄	ammonium perchlorate	SEL					128.07	0.33	1969WES
Sulfur/Hydrogen									
H ₂ S	hydrogen sulfide	SEL(g) SEL(l)	34.20	0.20					1998CHA 2019DIK
					75.8	3.2			
Boron/Hydrogen									
B ₂ H ₆	diborane	SEL	58.07						1998CHA

Table 5-2(2). Hydrocarbons

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{p,g})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{p,l})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{p,s})$	Reference
Carbon									
C	carbon (gas)	SEL	20.79	0.00					definition
	graphite	SEL					8.43		1980TAY
Alkanes									
CH ₄	methane	SEL	35.69	0.20					1989GUR
C ₂ H ₆	ethane	SEL(g)	52.49	0.20					1989GUR
		SEL(l)			160.6	1.9			2019DIK
C ₃ H ₈	propane	SEL(g)	73.60	0.20					1973CHA
		SEL(l)			117.41	0.86			2019DIK
C ₄ H ₁₀	isobutane	SEL(g)	96.65	0.20					1975CHE
		SEL(l)			140.69	0.29			2019DIK
	<i>n</i> -butane	SEL(g)	98.49	0.20					1975CHE
		SEL(l)			140.50	0.87			2019DIK
C ₅ H ₁₂	isopentane	SEL(g)	118.90	0.40					1974SCO
		SEL(l)			164.84	0.73			2019DIK
	neopentane	SEL(g)	120.83	0.25					1974SCO
		SEL(l)			169.2	6.3			2019DIK
	<i>n</i> -pentane	SEL(g)	120.00	0.10					1974SCO
		SEL(l)			167.19				1967MES
C ₆ H ₁₄	2,3-dimethylbutane	SEL(g)	139.40	0.70					1974SCO
		SEL(l)			189.02				1989OHN
	3-methylpentane	SEL(g)	140.10	0.40					1974SCO
		SEL(l)			191.16				1989OHN
	isohexane	SEL(g)	142.2	0.2					1974SCO
		SEL(l)			193.88				1989OHN
	neohexane	SEL(g)	141.5	0.30					1974SCO
		SEL(l)			189.67				1989OHN
	<i>n</i> -hexane	SEL(g)	142.6	0.20					1974SCO
		SEL(l)			195.52				1989OHN
C ₇ H ₁₆	isoheptane	SEL(g)	164.50	0.20					1974SCO
		SEL(l)			222.92	0.45			1961HUF
	neoheptane	SEL(g)	166.70	0.40					1974SCO
		SEL(l)			221.12	0.42			1961HUF
	<i>n</i> -heptane	SEL(g)	165.20	0.30					1974SCO

C ₈ H ₁₈	2,2,4-trimethylpentane	SEL(l) SEL(g) SEL(l)	188.40 0.40	224.73 0.12		1988SHI 1974SCO 1976FOR
	2-methylheptane	SEL(g) SEL(l)	187.2 0.2	238.87 1.1		1974SCO 1971MES
	<i>n</i> -octane	SEL(g) SEL(l)	187.8 0.4	252.0 0.4		1974SCO 1985LAI2
C ₉ H ₂₀	<i>n</i> -nonane	SEL(g) SEL(l)	210.4 0.5	254.18 1.0		1974SCO 1954FIN
C ₁₀ H ₂₂	<i>n</i> -decane	SEL SEL(l)	233.1 0.6	284.39 0.5		1974SCO 1984ROU
C ₁₆ H ₃₄	<i>n</i> -hexadecane	SEL(g) SEL(l)	369.2 3.7	314.2 1.4		2019DIK 1981GRO
Alkenes						
C ₂ H ₄	ethene	SEL	43.29 0.30			1941BUR
C ₃ H ₆	propene	SEL(g) SEL(l)	63.79 0.13	102.00		1974BIE 1983CHA
C ₄ H ₈	1-butene	SEL(g) SEL(l)	85.65 0.86	128.96 0.40		2019DIK 1991TAK
	2-butene, (<i>E</i>)-	SEL(g) SEL(l)	87.78 0.40	130.0 2.2		1940KIS2 2019DIK
	2-butene, (<i>Z</i>)-	SEL(g) SEL(l)	81.13 0.16	127.00 0.50		1944SCO 1983CHA
	isobutene	SEL	88.13 0.88	133.6 4.2		2019DIK
C ₅ H ₁₀	1-pentene	SEL(g) SEL(l)	108.16 1.08	154.87 0.30		2019DIK 1990MES
	2-methyl-1-butene	SEL(g) SEL(l)	109.98 1.10	157.3 0.8		2019DIK 1983CHA
	2-methyl-2-butene	SEL(g) SEL(l)	104.99 1.05	152.8 0.7		2019DIK 1983CHA
	2-pentene, (<i>E</i>)-	SEL	108.89 1.09	156.97 0.63		2019DIK
	2-pentene, (<i>Z</i>)-	SEL	98.79 0.99	151.58 0.65		2019DIK
	3-methyl-1-butene	SEL	114.48 1.14	156.16 0.68		2019DIK
C ₆ H ₁₂	1-hexene	SEL	130.82 1.31	183.08 0.55		2019DIK
C ₇ H ₁₄	1-heptene	SEL	153.46 1.53	211.86 0.49		2019DIK
C ₈ H ₁₆	1-octene	SEL	176.17 1.76	241.30 0.43		2019DIK
Alkadienes						
C ₃ H ₄	allene	SEL(g) SEL(l)	59.07 0.24 59.06 0.59	102.0 3.7		1940KIS 2019DIK

C ₄ H ₆	1,2-butadiene	SEL	79.50	0.80	123.83	0.86	2019DIK
	1,3-butadiene	SEL	79.64	0.80	123.66	0.48	2019DIK
C ₅ H ₈	1,2-pentadiene	SEL	100.98	1.01	150.84	1.25	2019DIK
	1,3-pentadiene	SEL	98.9	6.3	148.6	4.9	2019DIK
	1,4-pentadiene	SEL	98.34	0.98	146.99	1.23	2019DIK
	2-methyl-1,3-butadiene	SEL	102.76	1.03	150.85	1.37	2019DIK
Alkynes							
C ₂ H ₂	acetylene	SEL	44.04	0.30			1989GUR
C ₃ H ₄	propyne	SEL(g) SEL(l)	60.79		109.4	4.1	1940KIS 2019DIK
C ₄ H ₆	1-butyne	SEL	81.37	0.81	135.24	1.05	2019DIK
C ₅ H ₈	1-pentyne	SEL	102.67	1.03	143.3	4.0	2019DIK
	2-pentyne	SEL	99.72	1.00	129.6	3.5	2019DIK
C ₆ H ₁₀	1-hexyne	SEL	128.94	1.29	184.8	5.1	2019DIK
	2-hexyne	SEL	119.65		170.8	4.8	2019DIK
	3-hexyne	SEL	119.5	1.2	179.3	5.0	2019DIK
C ₄ H ₄	butenyne	SEL	73.20	0.73	118.4	3.9	2019DIK
C ₅ H ₆	3-penten-1-yne	SEL	96.3	5.9	139.5	4.7	2019DIK
C ₄ H ₂	butadiyne	SEL(g) SEL(l)	73.7	1.0	117.0	3.6	1991DOR 2019DIK
Cycloalkanes							
C ₃ H ₆	cyclopropane	SEL(g) SEL(l)	55.6	1.0	96.37	1.42	1986DOR2 2019DIK
C ₄ H ₈	cyclobutane	SEL(g) SEL(l)	70.60	1.50	109.89	0.75	1986DOR2 2019DIK
C ₅ H ₁₀	cyclopentane	SEL(g) SEL(l)	82.80	2.00	126.74		1986DOR2 1985TAN
	methylcyclobutane	SEL	98.79	0.99	145.7	4.9	2019DIK
C ₆ H ₁₂	cyclohexane	SEL(g) SEL(l)	105.30	2.00	156.90		1986DOR2 1989LAI
	methylcyclopentane	SEL	109.51	1.10	158.81	0.65	2019DIK
C ₇ H ₁₄	methylcyclohexane	SEL(g) SEL(l)	136.13	1.36	184.96	0.5	2019DIK 1985TAN
C ₁₀ H ₂₀	1,3-diethylcyclohexane	SEL	204	13	270	11	2019DIK
Cyclopropanes							
C ₃ H ₄	cyclopropene	SEL(g) SEL(l)	52.90	1.00	95.8	3.7	1986DOR2 2019DIK
C ₄ H ₆	cyclobutene	SEL(g) SEL(l)	64.40	2.00	101.0	3.4	1986DOR2 2019DIK

C ₅ H ₈	cyclopentene	SEL(g) SEL(l)	81.3	2.00				1986DOR2 1948HUF
	methylenecyclobutane	SEL	87.36	0.87	122.38	131.56	0.46	2019DIK
C ₆ H ₁₀	cyclohexene	SEL(g) SEL(l)	101.5	3.0				1986DOR2 1991STE
					148.80			
Cycloalkadienes								
C ₄ H ₄	cyclobutadiene	SEL(g) SEL(l)	61.0	2.5			3.7	1986DOR2 2019DIK
C ₅ H ₆	cyclopentadiene	SEL(g) SEL(l)	75.4	2.00				1986DOR2 2019DIK
					119.9		3.4	
C ₆ H ₈	1,3-cyclohexadiene	SEL(g) SEL(l)	94.2	2.5				1986DOR2 2019DIK
					143.7		2.4	
	1,4-cyclohexadiene	SEL(g) SEL(l)	94.1	3.0				1986DOR2 2019DIK
					145.6		2.6	
Bicycloalkanes								
C ₁₀ H ₁₈	cis-decalin	SEL(g) SEL(l)	168.60	1.00				1988DOR 1989OHN
					229.17			
	trans-decalin	SEL(g) SEL(l)	168.57	1.69				2019DIK 1989OHN
					232.08			
C ₁₂ H ₂₂	cyclohexylcyclohexane	SEL	218	15	286.96		0.19	2019DIK

Table 5-2(3). Aromatic Hydrocarbons

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{pg})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{pl})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{ps})$	Reference
Alkyl Benzenes									
C ₆ H ₆	benzene	SEL(g)	82.2	0.20					2014RUZ
		SEL(l)			135.69	0.30			1993GRO
C ₇ H ₈	toluene	SEL(g)	103.7	0.40					1985DRA
		SEL(l)			157.09	0.30			1993GRO
C ₈ H ₁₀	1,2-dimethylbenzene	SEL(g)	132.5	0.40					1985DRA
		SEL(l)			187.65	0.3			1979FOR
	1,3-dimethylbenzene	SEL(g)	125.80	0.40					1985DRA
		SEL(l)			184.63				1993GRO
	1,4-dimethylbenzene	SEL(g)	125.6	0.40					1985DRA
		SEL(l)			181.94	0.3			1979FOR
	ethylbenzene	SEL(g)	127.5	0.2					1978MIL
		SEL(l)			185.87	0.20			1997CHI
C ₉ H ₁₂	isopropylbenzene	SEL	159.75	1.60	215.21	0.52			2019DIK
C ₁₀ H ₁₄	2-isopropyltoluene	SEL	180	12	251.4	8.3			2019DIK
	3-isopropyltoluene	SEL	180	12	250.4	8.0			2019DIK
	4-isopropyltoluene	SEL	180	12	238.5	2.8			2019DIK
Alkenylbenzenes									
C ₈ H ₈	ethenylbenzene	SEL(g)	120.27	1.20					2019DIK
		SEL(l)			182.60				1961WAR
Benzocycloalkanes									
C ₉ H ₁₀	benzocyclopentane	SEL(g)	130.7	1.00					1986DOR
		SEL(l)			189.66	0.21			2019DIK
C ₁₀ H ₁₂	benzocyclohexane	SEL(g)	150.9	2.00					1988DOR
		SEL(l)			217.44				1957MCC
Naphthalenes									
C ₁₀ H ₈	naphthalene	SEL(g)	131.94	1.32			165.69	0.46	2019DIK
		SEL(l)			196.06				1993CHI
		SEL(s)					165.69		1957MCC
C ₁₁ H ₁₀	1-methylnaphthalene	SEL(g)	159.59	1.60					2019DIK
		SEL(l)			224.39				1957MCC
	2-methylnaphthalene	SEL(g)	159.83	1.60					2019DIK
		SEL(l)			195.98	0.6			1957MCC
Polycyclic Aromatic									

C ₁₂ H ₈	acenaphthylene	SEL(g)	154.80	1.00			1986DOR
		SEL(s)	154.79	1.55	183	15	2019DIK
	biphenylene	SEL	159.35	1.59			2019DIK
C ₁₂ H ₁₀	acenaphthene	SEL(g)	162.20	1.00			1986DOR
		SEL(s)			190.37	0.50	1977FIN
C ₁₃ H ₁₀	fluorene	SEL(g)	173.10	1.00			1989DOR
		SEL(s)			203.13		1977FIN
C ₁₄ H ₈	pyracylene	SEL	172.50	1.50			1986DOR
C ₁₄ H ₁₀	anthracene	SEL(g)	184.70	1.00			1988DOR
		SEL(s)			211.70	1.00	1980RAD
	phenanthrene	SEL(g)	185.70	1.86	220.63	0.54	2019DIK
		SEL(s)			220.62		1977FIN
C ₁₄ H ₁₂	pyracene	SEL	187.20	1.50			1986DOR
C ₁₆ H ₁₀	aceanthrylene	SEL(g)	203.00	1.50			1986DOR
		SEL(l)			334	18	2019DIK
	fluoranthene	SEL(g)	203.60	2.00			1989DOR
		SEL(s)			230.25		1971WON
	pyrene	SEL(g)	204.20				1988DOR
		SEL(s)			229.36		1980SMI
C ₁₈ H ₁₂	naphthacene	SEL(g)	233.40	1.50			1988DOR
		SEL(s)			236.60		1980WON
	triphenylene	SEL(g)	222.1	2.2	259.20	0.33	2019DIK
		SEL(s)			229.70		1971WON
C ₂₀ H ₁₂	perylene	SEL(g)	255.4	2.4			1988DOR
		SEL(s)			274.93	0.46	1980WON
C ₂₄ H ₁₂	coronene	SEL(g)	286.6	5.0			1988DOR
		SEL(s)			313.80	1.00	1980WON

Table 5-2(4). Oxidized Hydrocarbons

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{pg})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{pl})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{ps})$	Reference
Carbon Oxides									
CO	carbon monoxide	SEL	29.14	0.15					1998CHA
CO ₂	carbon dioxide	SEL	37.14	0.16	284.63	1.06			2018LEM
Alkanols									
CH ₄ O	methanol	SEL(g) SEL(l)	44.04	0.03	81.08	0.20			1986CHA 1985WIL
C ₂ H ₆ O	ethanol	SEL(g) SEL(l)	65.20	0.14	112.25	0.30			1986CHA 1985WIL
C ₃ H ₈ O	1-propanol	SEL(g) SEL(l)	85.56	0.14	143.74	0.4			1986CHA 1985WIL
	isopropanol	SEL(g) SEL(l)	89.32	0.15	155.0	0.3			1986CHA 1985WIL
C ₄ H ₁₀ O	1-butanol	SEL(g) SEL(l)	108.03	0.25	177.06	0.35			1986CHA 1985WIL
	2-butanol	SEL(g) SEL(l)	112.74	0.17	196.89	0.25			1986CHA 1985WIL
	isobutanol	SEL(g) SEL(l)	113	7	181.16	0.4			2019DIK 1985WIL
	<i>tert</i> -butanol	SEL(g) SEL(l,s)	113.63	0.21	219.08	0.4	145.9	2.0	1986CHA 1985WIL
C ₅ H ₁₂ O	1-pentanol	SEL(g) SEL(l)	130.75	1.31	208.19	0.30			2019DIK 1986TAN
	2-methyl-1-butanol	SEL	136.2	8.6	213.77	1.84			2019DIK
	isopentanol	SEL	136.2	8.6	209.75	1.66			2019DIK
C ₆ H ₁₄ O	<i>n</i> -hexanol	SEL(g) SEL(l)	153.39	1.53	240.57	1.5			2019DIK 1986TAN
Glycols									
C ₂ H ₆ O ₂	1,2-ethanediol	SEL(g) SEL(l)	82.7	0.13	149.3	0.8			1986CHA 1985WIL
C ₃ H ₈ O ₂	2-methoxyethanol	SEL	106.8	6.4	175.28	1.41			2019DIK
C ₃ H ₈ O ₃	glycerin	SEL(g) SEL(l)	124.0	7.6	206.3	1.0			2019DIK 1985WIL
Sugars									
C ₆ H ₁₂ O ₆	alpha- <i>D</i> -glucose	SEL					219.19	0.42	2019DIK

C ₁₂ H ₂₂ O ₁₁	fructose	SEL						213.15	0.67	2019DIK
	sucrose	SEL						424.30	0.82	2019DIK
Cycloalkanols										
C ₅ H ₁₀ O	cyclopentanol	SEL(g)	113.12	1.13						2019DIK
		SEL(l)			184.14					1956PAR
C ₆ H ₁₂ O	cyclohexanol	SEL	127.65	1.28	209.38	1.63	192.31	0.30		2019DIK
Alkoxyalkanols										
C ₃ H ₈ O ₂	2-methoxyethanol	SEL	106.8	6.4	175.28	1.41				2019DIK
Alkyl Ethers										
C ₂ H ₆ O	dimethyl ether	SEL(g)	65.57	0.08						1986CHA
		SEL(l)			112.7	1.5				2019DIK
C ₄ H ₁₀ O	diethyl ether	SEL(g)	119.46	0.15						1986CHA
		SEL(l)			172.4	0.2				1985WIL
C ₆ H ₁₄ O	diisopropyl ether	SEL(g)	158.3	1.6						2019DIK
		SEL(l)			216.7	0.7				1993GRO
Aldehydes										
CH ₂ O	formaldehyde	SEL(g)	35.39	0.02						1986CHA
		SEL(l)			82.6	3.0				2019DIK
C ₂ H ₄ O	acetaldehyde	SEL(g)	55.32	0.08						1986CHA
		SEL(l)			89.05					1988LEB
C ₃ H ₆ O	propanal	SEL(g)	80.73	0.10						1986CHA
		SEL(l)			137.20	1.40				1980FUC
C ₄ H ₈ O	butanal	SEL(g)	103.36	0.1						1986CHA
		SEL(l)			163.7	1.0				1985WIL
Alkenals										
C ₃ H ₄ O	1-propenal	SEL	71.46	0.72						2019DIK
C ₃ H ₄ O	2-propenal	SEL	71.28	0.71	118.7	3.3				2019DIK
Ketones										
C ₃ H ₆ O	acetone	SEL(g)	75.02	0.11						1986CHA
		SEL(l)			126.8	0.7				2019DIK
C ₄ H ₈ O	butanone	SEL(g)	101.68	0.14						1986CHA
		SEL(l)			158.40	0.30				1985COS
Carboxylic Acids										
CH ₂ O ₂	formic acid	SEL(g)	45.68	0.07						1986CHA
		SEL(l)			99.51	0.2				1985WIL
C ₂ H ₄ O ₂	acetic acid	SEL(g)	63.44	0.11						1978CHA
		SEL(l)			123.10	0.30				1982MAR
C ₃ H ₆ O	propanoic acid	SEL(g)	87.6	5.4						2019DIK
		SEL(l)			152.80	0.2				1982MAR

C ₄ H ₈ O	<i>n</i> -butanoic acid	SEL	110.8	6.9	177.67	0.19	2019DIK
C ₅ H ₁₀ O ₂	<i>n</i> -pentanoic acid	SEL(g) SEL(l)	134.1	8.4	210.33	2.0	2019DIK 1965MCD
Alkyl Esters							
C ₂ H ₄ O ₂	methyl formate	SEL(g) SEL(l)	64.38	0.09	119.70		1986CHA 1979FUC
C ₃ H ₆ O ₂	ethyl formate	SEL(g) SEL(l)	88.64	0.89	144.30		2019DIK 1979FUC
	methyl acetate	SEL(g) SEL(l)	86.03	0.12	141.34		1986CHA 1988PIN
C ₄ H ₈ O ₂	ethyl acetate	SEL(g) SEL(l)	113.66	1.14	168.94	0.6	2019DIK 1988PIN
	methyl propanoate	SEL(g) SEL(l)	116.58	1.17	174.20	0.7	2019DIK 1979FUC
	<i>n</i> -propyl formate	SEL(g) SEL(l)	117.6	7.1	172.10	1.5	2019DIK 1988PIN
C ₅ H ₁₀ O ₂	methyl butanoate	SEL(g) SEL(l)	130.7	8.3	200.80	0.6	2019DIK 1979FUC
Alkenyl Esters							
C ₄ H ₆ O ₂	ethenyl acetate	SEL	99.00	0.99	163.7	4.5	2019DIK
C ₅ H ₈ O ₂	ethyl propenoate	SEL	121.2	7.7	184.0	5.2	2019DIK
	methyl 2-methylpropenoate	SEL	122.1	7.7	187.3	5.9	2019DIK
Oxacycloalkanes							
C ₂ H ₄ O	oxacyclopropane	SEL(g) SEL(l)	47.0	1.00	89.56	0.68	1992DOR 2019DIK
C ₃ H ₆ O	methyloxacyclopropane	SEL(g) SEL(l)	72.55	0.12	120.41	0.35	1986CHA 1985WIL
	oxacyclobutane	SEL(g) SEL(l)	61.50	2.00	114.5	3.2	1992DOR 2019DIK
C ₄ H ₈ O	oxacyclopentane	SEL(g) SEL(l)	76.57	0.77	124.10	0.6	2019DIK 1985COS
Oxacycloalkenes							
C ₂ H ₂ O	oxacyclopropene	SEL	54.2	3.0			1992DOR
C ₄ H ₄ O	oxacyclopentadiene	SEL(g) SEL(l)	65.40	0.29	114.56		1986CHA 1952GUT
C ₄ H ₆ O	1-oxa-3-cyclopentene	SEL(g) SEL(l)	73.21	0.11	126.3	4.5	1992DOR 2019DIK
Multioxacycloalkanes							
CH ₂ O ₂	1,2-dioxacyclopropane	SEL(g)	43.9	3.0			1992DOR

C ₂ H ₄ O ₂	1,2-dioxacyclobutane	SEL(g)	77.0	4.0				1992DOR
C ₂ H ₄ O ₃	1,2,3-trioxacyclopentane	SEL(g)	78.0	4.5				1992DOR
C ₃ H ₆ O ₂	1,3-dioxacyclopentane	SEL(g)	71.0	4.0				1992DOR
		SEL(l)			121.71	1.19		2019DIK
C ₃ H ₆ O ₃	1,3,5-trioxacyclohexane	SEL	82.36	0.82			113.08	2019DIK
C ₄ H ₈ O ₂	1,3-dioxane	SEL	89.54	0.90	140.3	2.4		2019DIK
	1,4-dioxane	SEL(g)	92.25	0.92				2019DIK
		SEL(l)			149.65	0.7		1993GRO
Furandione and D-Camphor								
C ₄ H ₂ O ₃	2,5-furandione	SEL	84.6	5.3			119.24	2019DIK
C ₁₀ H ₁₆ O	D-camphor	SEL	191	13			246.29	2019DIK
Aromatic Alcohols								
C ₆ H ₆ O	phenol	SEL(g)	103.22					1978KUD
		SEL(s)					127.21	1975NIC
C ₇ H ₈ O	2-methylphenol	SEL(g)	127.30					1978KUD
		SEL(s)					154.56	1967AND
	3-methylphenol	SEL(g)	124.68					1978KUD
		SEL(l)			224.93	1.30		1967AND
C ₇ H ₈ O	4-methylphenol	SEL(g)	124.97	0.50				1978KUD
		SEL(s)					150.25	1967AND
	benzyl alcohol	SEL(g)	126.8	8.1				2019DIK
		SEL(l)			215.94	1.00		1975NIC
Aromatic Aldehydes and Acids								
C ₇ H ₆ O	benzaldehyde	SEL(g)	111.70	1.00				1975AMB
		SEL(l)			172.00			1975AMB2
C ₇ H ₆ O ₂	benzoic acid	SEL(g)	124.0	7.9				2019DIK
		SEL(s)					147.78	1993KAJ

Table 5-2(5). Nitrogen-Substituted Hydrocarbons

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_p g)$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_p l)$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_p s)$	Reference
Hydrogen Cyanides									
CHN	hydrogen cyanide	SEL	35.86	0.36	72.95	1.43			2019DIK
Alkylamines									
CH ₅ N	methylamine	SEL	50.08	0.50	104.8	2.7			2019DIK
C ₂ H ₇ N	ethylamine	SEL(g)	71.65	0.72					2019DIK
C ₂ H ₇ N	ethylamine	SEL(l)			129.5	4.8			2012KRO
C ₃ H ₉ N	1-propylamine	SEL(g)	91.25	0.91					2019DIK
		SEL(l)			162.51	1.00			1972FIN
	isopropylamine	SEL(g)	97.62	0.98					2019DIK
		SEL(l)							1972FIN
C ₄ H ₁₁ N	isobutylamine	SEL(g)	117.1	7.5					2019DIK
		SEL(l)			194.00				1971KON
	<i>n</i> -butylamine	SEL(g)	118.52	1.19					2019DIK
		SEL(l)			188.00	4.0			1971KON
	<i>tert</i> -butylamine	SEL(g)	121.00	1.21					2019DIK
		SEL(l)			190.00	1.0			1971KON
C ₂ H ₇ N	dimethylamine	SEL	70.45	0.70	140.5	2.6			2019DIK
C ₃ H ₉ N	methylethylamine	SEL	87.6	5.7					2019DIK
C ₄ H ₁₁ N	diethylamine	SEL(g)	115.77	1.16					2019DIK
		SEL(l)			178.10	0.8			1985COS
C ₃ H ₉ N	trimethylamine	SEL	91.80	0.92	138.15	0.99			2019DIK
C ₆ H ₁₅ N	<i>n</i> -hexylamine	SEL(g)	164	10					2019DIK
		SEL(l)			252.00	2.0			1971KON
	triethylamine	SEL(g)	160.98	1.61					2019DIK
		SEL(l)			216.43	2.0			1993GRO
Cycloalkylamines									
C ₆ H ₁₃ N	cyclohexylamine	SEL	132.0	8.7	201.2	8.9			2019DIK
Methenamines									
C ₆ H ₁₂ N ₄	methenamine	SEL(s)					152.31	0.55	2019DIK
Alkanenitriles									
C ₂ H ₃ N	ethanenitrile	SEL	52.25	0.52	91.22	0.37			2019DIK
C ₃ H ₅ N	propanenitrile	SEL	73.77	0.74	120.26	0.89			2019DIK
C ₄ H ₇ N	butanenitrile	SEL	94.51	0.95	133.50	1.88			2019DIK
Alkenenitriles									

C ₃ H ₃ N	propenenitrile	SEL	63.93	0.64	108.48	0.49		2019DIK
Cyanogens								
C ₂ N ₂	cyanogen	SEL	56.92	0.57	113.9	4.7		2019DIK
Alkylhydrazines								
C ₂ H ₈ N ₂	1,1-dimethylhydrazine	SEL	89.7	5.7	164.06	0.67		2019DIK
Aromatic Amines								
C ₆ H ₇ N	aniline	SEL(g) SEL(l)	108.14	1.08		191.01	1.20	2019DIK 1975NIC
C ₈ H ₁₁ N	<i>N,N</i> -dimethylaniline	SEL	151.2	9.8	211.9	5.6		2019DIK
Melamines								
C ₃ H ₆ N ₆	melamine	SEL	137.8	8.5			155.16	0.75
Oxidized Amines								
CH ₄ N ₂ O	urea	SEL	77.55	0.78			92.79	0.32
C ₆ H ₁₅ NO ₃	triethanolamine	SEL	213	13	361	21		2019DIK
Alkyl Nitrates and Nitroalkanes								
CH ₃ NO ₃	methyl nitrate	SEL(g) SEL(l)	76.49	0.77		157.19		2019DIK 1953GRA
C ₃ H ₅ N ₃ O ₉	nitroglycerin	SEL	234.5	2.6				2019DIK
CN ₄ O ₈	tetranitromethane	SEL	171.6	5.4	207	11		2019DIK
CHN ₃ O ₆	trinitromethane	SEL	131.5	3.8	182	11		2019DIK
CH ₃ NO ₂	nitromethane	SEL(g) SEL(l)	57.27	0.57	106.25	0.78		2019DIK 1947JON
Isocyanates								
C ₉ H ₆ N ₂ O ₂	2,4-diisocyanatotoluene	SEL			287.80			2019DIK
Nitro Aromatics								
C ₆ H ₃ N ₃ O ₆	trinitrobenzene	SEL(g) SEL(s)	197.48	1.97			215	5
C ₆ H ₅ NO ₂	nitrobenzene	SEL(g) SEL(l)	119.57	1.20				
C ₇ H ₅ N ₃ O ₆	trinitrotoluene	SEL(g) SEL(s)	213.3	5.6	181.13	2.0		
							243	10

Table 5-2(6). Sulfur-Substituted Hydrocarbons

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{pg})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{pl})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{ps})$	Reference
Alkanethiols									
CH ₄ S	methanethiol	SEL	50.31	0.50	90.47	0.63			2019DIK
C ₂ H ₆ S	ethanethiol	SEL(g) SEL(l)	73.02	0.73					2019DIK 1952MCC
					117.99	0.2			
Alkyl Sulfides									
CS ₂	carbon disulfide	SEL(g) SEL(l)	45.51	0.46					2019DIK 1955STA
					78.99	0.80			
C ₂ H ₆ S	dimethyl sulfide	SEL	74.08	0.74	118.28	0.84			2019DIK
Alkyl Sulfoxides									
C ₂ H ₆ OS	dimethyl sulfoxide	SEL(g) SEL(l)	89.43	0.89					2019DIK 1993GRO
					149.39	1.0			

Table 5-2(7). Halogenated Hydrocarbons

Formula	Compound	SEL	$C_p(g)$ J mol ⁻¹ K ⁻¹	$u(C_{pg})$	$C_p(l)$ J mol ⁻¹ K ⁻¹	$u(C_{pl})$	$C_p(s)$ J mol ⁻¹ K ⁻¹	$u(C_{ps})$	Reference
Fluoromethanes									
CH ₃ F	fluoromethane	SEL(g) SEL(l)	37.49		106.26	1.09			1998CHA 1998HO
CH ₂ F ₂	difluoromethane	SEL(g) SEL(l)	42.83		97.26	0.80			1998CHA 1996LUD
CHF ₃	trifluoromethane	SEL	51.04	0.30					1962VAL
CF ₄	tetrafluoromethane	SEL	61.02						1998CHA
Fluoroethanes									
C ₂ H ₅ F	fluoroethane	SEL	59.62	0.60	108.9	2.2			2018LEM
C ₂ H ₄ F ₂	1,1-difluoroethane	SEL	67.48	0.67	118.9	1.2			2018LEM
C ₂ H ₄ F ₂	1,2-difluoroethane	SEL	67.77	0.68	217.0	7.2			2019DIK
C ₂ H ₃ F ₃	1,1,1-trifluoroethane	SEL	78.08	0.16	140.24	0.70			2018LEM
	1,1,2-trifluoroethane	SEL	77.42	0.77	137.5	4.6			2019DIK
C ₂ H ₂ F ₄	1,1,1,2-tetrafluoroethane	SEL	85.03	0.21	145.36	0.73			2019DIK
	1,1,2,2-tetrafluoroethane	SEL	90.35	0.90	147.7	5.8			2019DIK
C ₂ HF ₅	pentafluoroethane	SEL	94.44	0.94	166.0	6.7			2019DIK
C ₂ F ₆	hexafluoroethane	SEL	106.61	1.07					2019DIK
Fluoroalkenes									
C ₂ H ₃ F	fluoroethene	SEL	50.43	0.50	99.1	6.7			2019DIK
C ₂ H ₂ F ₂	1,1-difluoroethene	SEL	59.08	0.59					2019DIK
	1,2-difluoroethene, (E)-	SEL	60.26	0.60	129.9	8.0			2019DIK
	1,2-difluoroethene, (Z)-	SEL	58.35	0.58	103.5	5.5			2019DIK
C ₂ HF ₃	trifluoroethene	SEL	69.20	0.69	145	11			2019DIK
C ₂ F ₄	tetrafluoroethene	SEL	80.42	0.80					2019DIK
Chloromethanes									
CH ₃ Cl	chloromethane	SEL(g) SEL(l)	40.74	0.41	81.70	0.50			2019DIK 1940AWB
CH ₂ Cl ₂	dichloromethane	SEL(g) SEL(l)	50.85		98.6	2.4			1998CHA 2015CHO
CHCl ₃	trichloromethane	SEL(g) SEL(l)	65.33		113.20				1998CHA 1993SHE
CCl ₄	tetrachloromethane	SEL(g) SEL(l)	82.65		131.34				1998CHA 1982TAN
Chloroethanes									

C ₂ H ₅ Cl	chloroethane	SEL	62.65	0.63	105.31	0.81		2019DIK
C ₂ H ₄ Cl ₂	1,1-dichloroethane	SEL	76.33	0.76	126.44	0.63		2019DIK
	1,2-dichloroethane	SEL	77.32	0.77	129.19	0.52		2019DIK
C ₂ H ₃ Cl ₃	1,1,1-trichloroethane	SEL	92.33	0.92	144.37	0.81		2019DIK
	1,1,2-trichloroethane	SEL	84.94	0.85	141.4	5.1		2019DIK
C ₂ H ₂ Cl ₄	1,1,1,2-tetrachloroethane	SEL	102.88	1.03	157.9	6.1		2019DIK
	1,1,2,2-tetrachloroethane	SEL(g)	98.97	0.99				2019DIK
		SEL(l)			168.00	1.5		1989WIL
C ₂ HCl ₅	pentachloroethane	SEL	118.07	1.18	174.3	7.2		2019DIK
C ₂ Cl ₆	hexachloroethane	SEL	136.78	1.37			198.24 1.64	2019DIK
Chloroalkenes								
C ₂ H ₃ Cl	chloroethene	SEL	53.60	0.54	96.2	3.3		2019DIK
C ₂ H ₂ Cl ₂	1,1-dichloroethene	SEL	66.93	0.67	110.80	0.67		2019DIK
	1,2-dichloroethene, (E)-	SEL	66.55	0.67	113.83	1.18		2019DIK
	1,2-dichloroethene, (Z)-	SEL	65.00	0.65	112.85	1.12		2019DIK
C ₂ HCl ₃	trichloroethene	SEL	79.75	1.00	128.70	1.13		2019DIK
C ₂ Cl ₄	tetrachloroethene	SEL(g)	95.51					1993GRO
		SEL(l)			147.16			1989WIL
Chloroalkynes								
C ₂ HCl	chloroethyne	SEL	51.6	2.8	135.5	4.4		2019DIK
C ₂ Cl ₂	dichloroethyne	SEL	61.1	3.2	185.1	8.0		2019DIK
Chloroaromatics								
C ₆ H ₅ Cl	chlorobenzene	SEL(g)	97.99	0.98				2019DIK
		SEL(l)			152.10	1.1		1993SHE
Bromoalkanes								
CH ₃ Br	bromomethane	SEL	42.45	0.43	81.59	1.21		2019DIK
CH ₂ Br ₂	dibromomethane	SEL(g)	54.59	0.55	104.48	0.88		2019DIK
		SEL(l)			104.1	3.1		1993SHE
CHBr ₃	tribromomethane	SEL	70.99	0.71	121.5	6.3		2019DIK
CBr ₄	tetrabromomethane	SEL(g)	90.71					1998CHA
		SEL(s)					145.90	1984BIC
C ₂ H ₅ Br	bromoethane	SEL(g)	64.27	0.64				2019DIK
		SEL(l)			105.80	1.0		1993SHE
Bromoalkenes								
C ₂ H ₃ Br	bromoethene	SEL(g)	55.49	0.56				2019DIK
		SEL(l)			114.3	3.7		2012KRO
Iodoalkanes								
CH ₃ I	iodomethane	SEL(g)	44.08	0.44	82.58	0.61		2019DIK
		SEL(l)			82.75			1993CAR

CH ₂ I ₂	diiodomethane	SEL(g) SEL(l)	57.75	0.58	112.80	6.0	2019DIK 1993SHE
CHI ₃	triiodomethane	SEL(g) SEL(l)	75.06	0.75	157.50		2019DIK 1993CAR
CI ₄	tetraiodomethane	SEL(g) SEL(l)	95.78	0.96	259.80		2019DIK 1993CAR
Chlorofluoroalkanes							
CH ₂ ClF	chlorofluoromethane	SEL(g) SEL(l)	47.29	0.46	73.2	2.4	1965BAR 2019DIK
CHCl ₂ F	dichlorofluoromethane	SEL(g) SEL(l)	61.33	0.64	107.5	3.8	1965BAR 2019DIK
CHClF ₂	chlorodifluoromethane	SEL(g) SEL(l)	56.00	0.22	107.2	3.2	1995ESP 1940BEN
CCl ₂ F ₂	dichlorodifluoromethane	SEL(g) SEL(l)	73.92	0.21	119.9	4.8	1991BUF 2019DIK
CCl ₃ F	trichlorofluoromethane	SEL(g) SEL(l)	78.09	0.78	121.55	0.46	2019DIK 1941OSB
CClF ₃	chlorotrifluoromethane	SEL	66.87	0.67			2019DIK
Chlorofluoroalkenes							
C ₂ ClF ₃	chlorotrifluoroethene	SEL	83.94	0.84	137.0	2.7	2019DIK
Bromofluoroalkanes							
CH ₂ BrF	bromofluoromethane	SEL	49.14	0.49	114.3	3.8	2019DIK
CHBrF ₂	bromodifluoromethane	SEL	58.78	0.59	106.5	3.8	2019DIK
CBr ₂ F ₂	dibromodifluoromethane	SEL	77.02	0.77	118.6	3.6	2019DIK
CBr ₃ F	tribromofluoromethane	SEL	84.37	0.84	169.7	6.4	2019DIK
CBrF ₃	bromotrifluoromethane	SEL	69.26	0.69	125.24	0.27	2019DIK
Bromochloroalkanes							
CH ₂ BrCl	bromochloromethane	SEL(g) SEL(l)	52.76	0.53	104.1	1.5	2019DIK 2015CHO
CHBr ₂ Cl	dibromochloromethane	SEL	69.15	0.69	129.3	5.2	2019DIK
CHBrCl ₂	bromodichloromethane	SEL	67.25	0.67	115.9	3.6	2019DIK
CBr ₂ Cl ₂	dibromodichloromethane	SEL	87.15	0.87	140.43	1.08	2019DIK
CBrCl ₃	bromotrichloromethane	SEL	85.38	0.85	135.29	0.56	2019DIK
Bromochlorofluoroalkanes							
CHBrClF	bromochlorofluoromethane	SEL	62.85	0.63	103.9	3.6	2019DIK
CBr ₂ ClF	dibromochlorofluoromethane	SEL	82.33	0.82	130.0	4.8	2019DIK
CBrCl ₂ F	bromodichlorofluoromethane	SEL	80.06	0.80	126.4	4.4	2019DIK
CBrClF ₂	bromochlorodifluoromethane	SEL	74.66	0.75	118.8	3.9	2019DIK
Carbonyl Halides							

CCl ₂ O	phosgene	SEL(g) SEL(l)	57.69	0.58	104.08	0.97	1998CHA 2019DIK
Chlorophenols							
C ₆ HCl ₅ O	pentachlorophenol	SEL(g) SEL(l)	189.80	1.90	201.96		2019DIK 1958HIL
Siloxanes							
C ₆ H ₁₈ OSi ₂	hexamethyldisiloxane	SEL			311.38	0.82	2019DIK
C ₈ H ₂₄ O ₄ Si ₄	octamethylcyclotetrasiloxane	SEL			495.9	9.5	2019DIK

5.4. References for Heat Capacities

Reference	Source
[1924SHO]	Shorthose, D. N.; The Thermal Properties of Methyl Chloride. Food Invest. Board, 1924, Spec. Rep. No. 19, 1-18
[1930WIE]	Wiebe, R.; Hubbard, K.H.; Brevoort, M.J., The heat capacity of saturated liquid ethane from the boiling point to the critical temperature and heat fusion of the solid, J. Am. Chem. Soc., 1930, 52, 611-622. https://doi.org/10.1021/ja01365a026
[1932TRE]	Trew, V.C.G., Physical properties of mixtures of acetone and bromoform, Trans. Faraday Soc., 1932, 28, 509-514. https://doi.org/10.1039/tf9322800509
[1936GIA]	Giauque, W. F.; Stout, J. W.; The Entropy of Water and the Third Law of Thermodynamics. The Heat Capacity of Ice from 15 to 273°K, J. Am. Chem. Soc. 58, 1144 (1936).
[1937PER]	Perlick, A.; Calorimetric investigations on dichloromethane, difluoromonochloroethane and tetrafluorodichloroethane. Bull. Inst. Int. Froid, 1937, 18, A-1
[1937STU]	Stull, D. R.; A Semi-micro Calorimeter for Measuring Heat Capacities at Low Temp.. J. Am. Chem. Soc., 1937, 59, 2726-2733
[1939OSB]	Osborn, N. S.; Stimson, H. F.; Ginnings, D. C.; Measurements of heat capacity and heat of vaporization of water in the range 0 degrees to 100 degrees C, J. Res. Nat. Bur. Std. 23, 197 (1939).
[1940AWB]	Awberg, J.H.; Griffiths, E., The specific heat of liquid methyl chloride, Proc. Phys. Soc. (London), 1940, 52, 770-776 https://doi.org/10.1088/0959-5309/52/6/306
[1940BEN]	Benning, A. F.; McHarness, R. C.; Markwood; Smith, J. O.; Thermodynamic Properties of Fluorochloromethanes and -ethanes, Heat Capacity of the Liquid and Vapor of Three Fluorochloromethanes and Trifluorotrichloroethane. Ind. Eng. Chem., 1940, 32, 976
[1940KIS]	Kistiakowsky G.B., Gaseous heat capacities. II, J. Chem. Phys., 1940, 8, 610-618. https://doi.org/10.1063/1.1750722
[1940KIS2]	Kistiakowsky G.B., Gaseous heat capacities. III, J. Chem. Phys., 1940, 8, 618-622. https://doi.org/10.1063/1.1750723
[1940RIE]	Riedel, L.; The Determination of the Specific Heat of Ethyl Chloride and Methylene Chloride in the Liquid State. Z. Gesamte Kalte-Ind., 1940, 47, 87
[1941BUR]	Burcik E.J., The vibrational energy levels and specific heat of ethylene, J. Chem. Phys., 1941, 9, 118-119. https://doi.org/10.1063/1.1750816
[1941OSB]	Osborne, D.W.; Garner, C.S.; Doescher, R.N.; Yost, D.M., The heat capacity, entropy, heats of fusion and vaporization and vapor pressure of fluorotrichloromethane, J. Am. Chem. Soc., 1941, 63, 3496-3499. https://doi.org/10.1021/ja01857a069
[1941STO]	Stout, J.W.; Fisher, L.H., The entropy of formic acid. The heat capacity from 15 to 300K. Heats of fusion and vaporization, J. Chem. Phys., 1941, 9, 163-168. https://doi.org/10.1063/1.1750869

- [1941YOS] Yost, D.M.; Osborne, D.W.; Garner, C.S., The heat capacity, entropy, and heats of transition, fusion, and vaporization of dimethylacetylene. Free rotation in the dimethylacetylene molecule, *J. Am. Chem. Soc.*, 1941, 63, 3492-3496
- [1943MCG] McGovern, E. W.; Chlorohydrocarbon Solvents. *Ind. Eng. Chem.*, 1943, 35, 1230-9
- [1943PIT] Pitzer K.S., The thermodynamics and molecular structure of benzene and its methyl derivatives, *J. Am. Chem. Soc.*, 1943, 65, 803-829
- [1944HIC] Hicks, J. F. G.; Hooley, J. G.; Stephenson, C. C.; The Heat Capacity of Carbon Tetrachloride from 15 to 300 K. The Heats of Transition and Fusion. The Entropy from Thermal Measurements Compared with the Entropy from Molecular Data. *J. Am. Chem. Soc.*, 1944, 66, 1064-1067
- [1944SCO] Scott R.B., Thermodynamic properties of cis-2-butene from 15 to 1500 K, *J. Res. Nat. Bur. Stand.*, 1944, 33, 1-20
<https://doi.org/10.6028/jres.033.001>
- [1946DOU] Douslin, D.R.; Huffman, H.M., The heat capacities, heats of transition, heats of fusion and entropies of cyclopentane, methylcyclopentane and methylcyclohexane, *J. Am. Chem. Soc.*, 1946, 68, 173-176.
- [1947JON] Jones, W.M.; Giaque, W.F., The entropy of nitromethane. Heat capacity of solid and liquid. Vapor pressure, heats of fusion and vaporization, *J. Am. Chem. Soc.*, 1947, 69, 983-987 <https://doi.org/10.1021/ja01197a001>
- [1948HUF] Huffman, H.M.; Eaton, M.; Oliver, G.D., The heat capacities, heats of transition, heats of fusion and entropies of cyclopentene and cyclohexene, *J. Am. Chem. Soc.*, 1948, 70, 2911-2914. <https://doi.org/10.1021/ja01189a025>
- [1949COL] Coleman C.F., The heat capacity of organic vapors. V. Acetaldehyde, *J. Am. Chem. Soc.*, 1949, 71, 2839-2841
<https://doi.org/10.1021/ja01176a072>
- [1949STA] Stamm R.F., Fundamental vibrational frequencies and thermodynamic functions for vinylacetylene, revised thermodynamic functions for hydrogen cyanide, and thermodynamics of two reactions involved in the synthesis of acrylonitrile, *J. Chem. Phys.*, 1949, 17, 104-105.
- [1952GUT] Guthrie, G.B., Jr.; Scott, D.W.; Hubbard, W.N.; Katz, C.; McCullough, J.P.; Gross, M.E.; Williamson, K.D.; Waddington, G., Thermodynamic properties of furan, *J. Am. Chem. Soc.*, 1952, 74, 4662-4669
<https://doi.org/10.1021/ja01138a063>
- [1952MCC] McCullough, J.P.; Scott, D.W.; Finke, H.L.; Gross, M.E.; Williamson, K.D.; Pennington, R.E.; Waddington, G.; Huffman, H.M., Ethanethiol (ethyl mercaptan): thermodynamic properties in the solid, liquid and vapor states. Thermodynamic functions to 1000K, *J. Am. Chem. Soc.*, 1952, 74, 2801-2804 <https://doi.org/10.1021/ja01131a029>
- [1953GRA] Gray, P.; Smith, P.L., The thermodynamic properties of methyl nitrate, *J. Chem. Soc.*, 1953, 1953, 2380-2385.
<https://doi.org/10.1039/jr9530002380>
- [1954FIN] Finke, H.L.; Gross, M.E.; Waddington, G.; Huffman, H.M., Low-temperature thermal data for the nine normal paraffin hydrocarbons from octane to hexadecane, *J. Am. Chem. Soc.*, 1954, 76, 333-341.

- [1955HIL] Hilsenrath, J.; Tables of Thermal Properties of Gases Comprising Tables of Thermodynamic and Transport Properties of Air, Argon, Carbon Dioxide, Carbon Monoxide Hydrogen, Nitrogen, Oxygen, and Steam, NBS Circular 564, 1955
- [1955STA] Staveley, L.A.K.; Tupman, W.I.; Hart, K.R., Some thermodynamic properties of the systems benzene + ethylene dichloride, benzene + carbon tetrachloride, acetone + chloroform, and acetone + carbon disulphide, *Trans. Faraday Soc.*, 1955, 51, 323-342. <https://doi.org/10.1039/tf9555100323>
- [1956PAR] Parks, G.S.; Kennedy, W.D.; Gates, R.R.; Mosley, J.R.; Moore, G.E.; Renquist, M.L., Thermal data on organic compounds. XXVI. Some heat capacity, entropy and free energy data for seven compounds containing oxygen. *J. Am. Chem. Soc.*, 1956, 78, 56-59 <https://doi.org/10.1021/ja01582a015>
- [1957HAR] Harrison, D.; Moelwyn-Hughes, E. A.; The heat capacities of certain liquids. *Proc. R. Soc. London, Ser. A*, 1957, 239, 230-46
- [1957MCC] McCullough, J.P.; Finke, H.L.; Messerly, J.F.; Kincheloe, T.C.; Waddington, G., The low temperature thermodynamic properties of naphthalene, 1-methylnaphthalene, 2-methylnaphthalene, 1,2,3,4-tetrahydronaphthalene, trans-decahydronaphthalene and cis-decahydronaphthalene, *J. Phys. Chem.*, 1957, 61, 1105-1116. <https://doi.org/10.1021/j150554a016>
- [1958HIL] Hildenbrand, D.L.; Kramer, W.R.; Stull, D.R., The heat capacities of hexachlorobenzene and pentachlorophenol from 15 to 300K, *J. Phys. Chem.*, 1958, 62, 958-959. <https://doi.org/10.1021/j150566a014>
- [1960SCO] Scott D.W., Chemical thermodynamic properties of methylcyclopentane and 1-cis-3-dimethylcyclopentane, *J. Phys. Chem.*, 1960, 64, 906-908.
- [1961HUF] Huffman, H.M.; Gross, M.E.; Scott, D.W.; McCullough, I.P., Low temperature thermodynamic properties of six isomeric heptanes, *J. Phys. Chem.*, 1961, 65, 495-503
- [1961WAR] Warfield, R.W.; Petree, M.C., Thermodynamic properties of polystyrene and styrene, *J. Polymer Sci.*, 1961, 55, 497-505. <https://doi.org/10.1002/pol.1961.1205516208>
- [1962VAL] Valentine, R. H.; Brodale, G. E.; Giauque, W. F.; Trifluoromethane: entropy, low temp. heat capacity, heats of fusion and vaporization, and vapor pressure. *J. Phys. Chem.*, 1962, 66, 392
- [1963OET] Oetting F.L., The heat capacity and entropy of 2-methyl-2-propanol from 15 to 330 K, *J. Phys. Chem.*, 1963, 67, 2757-2761. <https://doi.org/10.1021/j100806a059>
- [1964SEN] Senftleben, H.; New values of thermal conductivity and specific heat at different temperatures for a series of gases. *Z. Angew. Phys.*, 1964, 17, 86-7
- [1965BAR] Barho, W.; Molar specific heats of fluorine-chlorine derivatives of methane in the ideal gas state.. *Kaeltetech.-Klim.*, 1965, 17, 219
- [1965MCD] McDougall, L.A.; Kilpatrick, J.E., Entropy and related thermodynamic properties of n-valeric acid, *J. Chem. Phys.*, 1965, 42, 2307-2310. <https://doi.org/10.1063/1.1696293>

- [1967AND] Andon, R.J.L.; Counsell, J.F.; Lees, E.B.; Martin, J.F.; Mash, C.J., Thermodynamic properties of organic oxygen compounds. Part 17. Low-temperature heat capacity and entropy of the cresols, *Trans. Faraday Soc.*, 1967, 63, 1115-1121 <https://doi.org/10.1039/TF9676301115>
- [1967MES] Messerly, J.F.; Guthrie, G.B.; Todd, S.S.; Finke, H.L., Low-temperature thermal data for n-pentane, n-heptadecane, and n-octadecane, *J. Chem. Eng. Data*, 1967, 12, 338-346. <https://doi.org/10.1021/je60034a014>
- [1969SHU] Shumskaya, A. I.; Gruzdev, V. A.; Isobar specific heat of Freon vapors. *Teplofiz. Svoistva Freonov*, 1969, Pg. 35-42, Gruzdev, V. A., Ed., Nauka: Novosibirsk
- [1969STU] Stull D.R., Jr., *The Chemical Thermodynamics of Organic Compounds*. Wiley, New York, 1969.
- [1969WES] Westrum, E. F. J.; Justice, B. H.; Molecular Freedom of the Ammonium Ion. Heat Capacity and Thermodynamic Properties of Ammonium Perchlorate from 5deg-350deg K. *J. Chem. Phys.*, 1969, 50, 5083-5087 <https://doi.org/10.1063/1.1671022>
- [1970ERN] Ernst, G.; Busser, J.; Ideal and real state heat capacities C_p of C_3H_8 , $i-C_4H_{10}$, C_2F_5Cl , CH_2ClCF_3 , $CF_2ClCFCl_2$, and CHF_2Cl . *J. Chem. Thermodyn.*, 1970, 2, 787-791
- [1971COU] Counsell, J.F.; Lee, D.A.; Martin, J.F., Thermodynamic properties of organic oxygen compounds. Part XXVI. Diethyl ether, *J. Chem. Soc. A*, 1971, 313-316 <https://doi.org/10.1039/j19710000313>
- [1971KON] Konicek, J.; Wadso, I., Thermochemical properties of some carboxylic acids, amines and N-substituted amides in aqueous solution, *Acta Chem. Scand.*, 1971, 25, 1541-1551. <https://doi.org/10.3891/acta.chem.scand.25-1541>
- [1971MES] Messerly, J.F.; Finke, H.L., Low-temperature thermal properties of 2-methylheptane and 2-methyldecane: the thermodynamic properties of the 2-methylalkanes, *J. Chem. Thermodynam.*, 1971, 3, 675-687.
- [1971WON] Wong, W-K.; Westrum, E.F., Jr., Thermodynamics of polynuclear aromatic molecules. I. Heat capacities and enthalpies of fusion of pyrene, flouranthene, and triphenylene, *J. Chem. Thermodynam.*, 1971, 3, 105-124. [https://doi.org/10.1016/S0021-9614\(71\)80071-X](https://doi.org/10.1016/S0021-9614(71)80071-X)
- [1972FIN] Finke, H.L.; Messerly, J.F.; Todd, S.S., Thermodynamic properties of acrylonitrile, 1-aminopropane, 2-aminopropane, and 2-methyl-2-aminopropane, *J. Chem. Thermodynam.*, 1972, 4, 359-374. [https://doi.org/10.1016/0021-9614\(72\)90019-5](https://doi.org/10.1016/0021-9614(72)90019-5)
- [1973BUT] Butland, A.T.D.; Maddison, R.J., The specific heat of graphite: an evaluation of measurements, *J. Nucl. Mater.*, 1973, 49, 45-56. [https://doi.org/10.1016/0022-3115\(73\)90060-3](https://doi.org/10.1016/0022-3115(73)90060-3)
- [1973CHA] Chao J., Ideal gas thermodynamic properties of ethane and propane, *J. Phys. Chem. Ref. Data*, 1973, 2, 427-438 <https://doi.org/10.1063/1.3253123>
- [1974BIE] Bier K., Thermodynamic properties of propylene from calorimetric measurements, *J. Chem. Thermodyn.*, 1974, 6, 1039-1052 [https://doi.org/10.1016/0021-9614\(74\)90066-4](https://doi.org/10.1016/0021-9614(74)90066-4)

- [1974SCO] Scott D.W., Chemical Thermodynamic Properties of Hydrocarbons and Related Substances. Properties of the Alkane Hydrocarbons, C1 through C10 in the Ideal Gas State from 0 to 1500 K. U.S. Bureau of Mines, Bulletin 666, 1974
- [1975AMB] Ambrose D., Thermodynamic properties of organic oxygen compounds. 42. Physical and thermodynamic properties of benzaldehyde, J. Chem. Thermodyn., 1975, 7, 1143-1157 [https://doi.org/10.1016/0021-9614\(75\)90035-X](https://doi.org/10.1016/0021-9614(75)90035-X)
- [1975AMB2] Ambrose, D.; Connett, J.E.; Green, J.H.S.; Hales, J.L.; Head, A.J.; Martin, J.F., Thermodynamic properties of organic oxygen compounds. 42. Physical and thermodynamic properties of benzaldehyde, J. Chem. Thermodyn., 1975, 7, 1143-1157 [https://doi.org/10.1016/0021-9614\(75\)90035-X](https://doi.org/10.1016/0021-9614(75)90035-X)
- [1975CHE] Chen S.S., Ideal gas thermodynamic properties and isomerization of n-butane and isobutane, J. Phys. Chem. Ref. Data, 1975, 4, 859-869. <https://doi.org/10.1063/1.555526>
- [1975NIC] Nichols, N.; Wads, I., Thermochemistry of solutions of biochemical model compounds. 3. Some benzene derivatives in aqueous solution, J. Chem. Thermodynam., 1975, 7, 329-336. [https://doi.org/10.1016/0021-9614\(75\)90169-X](https://doi.org/10.1016/0021-9614(75)90169-X)
- [1976FOR] Fortier, J.-L.; Benson, G.C., Excess heat capacities of binary liquid mixtures determined with a Picker flow calorimeter, J. Chem. Thermodynam., 1976, 8, 411-423.
- [1977FIN] Finke, H.L.; Messerly, J.F.; Lee, S.H.; Osborn, A.G.; Douslin, D.R., Comprehensive thermodynamic studies of seven aromatic hydrocarbons, J. Chem. Thermodyn., 1977, 9, 937-956 [https://doi.org/10.1016/0021-9614\(77\)90215-4](https://doi.org/10.1016/0021-9614(77)90215-4)
- [1977KOR] Korkhov, A.D.; Vasil'ev, I.A., Heat capacity and thermodynamic functions of propanal at low temperatures, Termodin. Org. Soedin., 1977, (6), 34-37.
- [1978CHA] Chao J., Ideal gas thermodynamic properties of methanoic and ethanoic acids, J. Phys. Chem. Ref. Data, 1978, 7, 363-377. <https://doi.org/10.1063/1.555571>
- [1978KUD] Kudchadker S.A., Ideal gas thermodynamic properties of phenol and cresols, J. Phys. Chem. Ref. Data, 1978, 7, 417-423 <https://doi.org/10.1063/1.555573>
- [1978MIL] Miller, A.; Scott, D. W.; Chemical Thermodynamic Properties of Ethylbenzene, J. Chem. Phys., 1978, 68, 1317.
- [1978MOS] Moseeva, E. M.; Rabinovich, I. B.; Busygina, G. I.; Safonov, V. A.; Ovchinnikov, E. Yu.; Thermodynamic proerties of methylene chloride. Termodin. Org. Soedin., 1978, 1, 8-11
- [1979BER] Berryman, James M.; DeVaney, Will E.; Eakin, Bert E.; Bailey, Nancy L.; Enthalpy Measurements on Synthetic Gas Systems: Hydrogen-Methane, Hydrogen-Carbon Monoxide. GPA, 1979, Research Report No. 37
- [1979FOR] Fortier, J.-L.; Benson, G.C., Heat capacities of some binary aromatic hydrocarbon mixtures containing benzene or toluene, J. Chem. Eng. Data, 1979, 24(1), 34-37. <https://doi.org/10.1021/jc60080a015>
- [1979FUC] Fuchs, R., Heat capacities of some liquid aliphatic, alicyclic, and aromatic esters at 298.15 K, J. Chem. Thermodyn., 1979, 11, 959-961. [https://doi.org/10.1016/0021-9614\(79\)90044-2](https://doi.org/10.1016/0021-9614(79)90044-2)
- [1980FUC] Fuchs, R.; Heat Capacities of Liquid Ketones and Aldehydes at 298 K, Can. J. Chem., 1980, 58, 2305 <https://doi.org/10.1139/v80-370>

- [1980RAD] Radomska, M.; Radomski, R., Calorimetric studies of binary systems of 1,3,5-trinitrobenzene with naphthalene, anthracene, and carbazole. I. Phase transitions and heat capacities of the pure components and charge-transfer complexes, *Thermochim. Acta*, 1980, 40, 405-414. [https://doi.org/10.1016/0040-6031\(80\)80082-7](https://doi.org/10.1016/0040-6031(80)80082-7)
- [1980SMI] Smith, N.K.; Stewart, R.C., Jr.; Osborn, A.G.; Scott, D.W., Pyrene: vapor pressure, enthalpy of combustion, and chemical thermodynamic properties, *J. Chem. Thermodyn.*, 1980, 12, 919-926 [https://doi.org/10.1016/0021-9614\(80\)90132-9](https://doi.org/10.1016/0021-9614(80)90132-9)
- [1980TAY] Taylor, R.E.; Groot, H., Thermophysical properties of POCO graphite, *High Temp.-High Press.*, 1980, 12, 147-160
- [1980WON] Wong, W.K.; Westrum, E.F., Jr., Thermodynamics of polynuclear aromatic molecules. II. Low temperature thermal properties of perylene, coronene, and naphthacene, *Mol. Cryst. Liq. Cryst.*, 1980, 61, 207-228 <https://doi.org/10.1080/00268948008081342>
- [1981GRO] Grolier, J.P.E.; Inglese, A.; Roux, A.H.; Wilhelm, E., Thermodynamics of (1-chloronaphthalene + n-alkane): excess enthalpies, excess volumes and excess heat capacities, *Ber. Bunsenges. Phys. Chem.*, 1981, 85, 768-772
- [1982MAR] Martin, J.F.; Andon, R.J.L., Thermodynamic properties of organic oxygen compounds. Part LII. Molar heat capacity of ethanoic, propanoic, and butanoic acids, *J. Chem. Thermodynam.*, 1982, 14, 679-688 [https://doi.org/10.1016/0021-9614\(82\)90083-0](https://doi.org/10.1016/0021-9614(82)90083-0)
- [1982TAN] Tanaka, R., Determination of excess heat capacities of (benzene + tetrachloromethane and + cyclohexane) between 293.15 and 303.15 K by use of a Picker flow calorimeter, *J. Chem. Thermodynam.*, 1982, 14, 259-268 [https://doi.org/10.1016/0021-9614\(82\)90016-7](https://doi.org/10.1016/0021-9614(82)90016-7)
- [1983CHA] Chao, J.; Hall, K.R.; Yao, J.M., Thermodynamic properties of simple alkenes, *Thermochim. Acta*, 1983, 64(3), 285-303. [https://doi.org/10.1016/0040-6031\(83\)85005-9](https://doi.org/10.1016/0040-6031(83)85005-9)
- [1984BIC] Bickerton, J.; Minas Da Piedade, M.E.; Pilcher, G., Enthalpy of formation of tetrabromomethane by rotating-bomb calorimetry, *J. Chem. Thermodyn.*, 1984, 16, 661-668 [https://doi.org/10.1016/0021-9614\(84\)90046-6](https://doi.org/10.1016/0021-9614(84)90046-6)
- [1984ROU] Roux, A.H.; Grolier, J.-P.E.; Inglese, A.; Wilhelm, E., Excess molar enthalpies, excess molar heat capacities and excess molar volumes of (fluorobenzene + an n-alkane), *Ber. Bunsenges. Phys. Chem.*, 1984, 88, 986-992.
- [1985COS] Costas, M.; Patterson, D., Heat capacities of water + organic-solvent mixtures, *J. Chem. Soc., Faraday Trans. 1*, 1985, 81, 2381-2398 <https://doi.org/10.1039/f19858102381>
- [1985DRA] Draeger, J.A., The methylbenzenes II. Fundamental vibrational shifts, statistical thermodynamic functions, and properties of formation, *J. Chem. Thermodyn.*, 1985, 17, 263-275. [https://doi.org/10.1016/0021-9614\(85\)90020-5](https://doi.org/10.1016/0021-9614(85)90020-5)
- [1985LAI] Lainez, A.; Rodrigo, M.; Roux, A.H.; Grolier, J.-P.E.; Wilhelm, E., Relations between structure and thermodynamic properties. Heat capacities of polar substances (nitrobenzene and benzonitrile) in alkane solutions, *Calorim. Anal. Therm.*, 1985, 16, 153-158.

- [1985LAI2] Lainez, A.; Grolier, J.-P.E.; Wilhelm, E., Excess molar heat capacity and excess molar volume of 1,6-dichlorohexane + n-octane, *Thermochim. Acta*, 1985, 91, 243-248.
- [1985TAN] Tanaka, R., Excess heat capacities for mixtures of benzene with cyclopentane, methylcyclohexane, and cyclooctane at 298.15 K, *J. Chem. Eng. Data*, 1985, 30, 267-269. <https://doi.org/10.1021/je00041a009>
- [1985WIL] Wilhoit, R. C.; Chao, J.; Hall, K. R.; Thermodynamic Properties of Key Organic Oxygen Compounds in the Carbon Range C1 to C4. Part 1. Properties of Condensed Phases, *J. Phys. Chem. Ref. Data* 14, 1 (1985).
- [1986CHA] Chao J., Thermodynamic properties of key organic oxygen compounds in the carbon range C1 to C4. Part 2. Ideal gas properties, *J. Phys. Chem. Ref. Data*, 1986, 15, 1369-1436 <https://doi.org/10.1063/1.555769>
- [1986DOR] Dorofeeva O.V., On calculation of thermodynamic properties of polycyclic aromatic hydrocarbons, *Thermochim. Acta*, 1986, 102, 59-66. [https://doi.org/10.1016/0040-6031\(86\)85313-8](https://doi.org/10.1016/0040-6031(86)85313-8)
- [1986DOR2] Dorofeeva O.V., Thermodynamic properties of twenty-one monocyclic hydrocarbons, *J. Phys. Chem. Ref. Data*, 1986, 15, 437-464 <https://doi.org/10.1063/1.555773>
- [1986GAT] Gates, J.A.; Wood, R.H.; Cobos, J.C.; Casanova, C.; Roux, A.H.; Roux-Desgranges, G.; Grolier, J.-P.E., Densities and heat capacities of 1-butanol + n-decane from 298 K to 400 K, *Fluid Phase Equilib.*, 1986, 27, 137-151 [https://doi.org/10.1016/0378-3812\(86\)87046-7](https://doi.org/10.1016/0378-3812(86)87046-7)
- [1986TAN] Tanaka, R.; Toyama, S.; Murakami, S., Heat capacities of $\{xC_nH_{2n+1}OH+(1-x)C_7H_{16}\}$ for $n = 1$ to 6 at 298.15 K, *J. Chem. Thermodynam.*, 1986, 18, 63-73 [https://doi.org/10.1016/0021-9614\(86\)90044-3](https://doi.org/10.1016/0021-9614(86)90044-3)
- [1988AND] Andreoli-Ball, L.; Patterson, D.; Costas, M.; Caceres-Alonso, M., Heat capacity and corresponding states in alkan-1-ol-n-alkane systems, *J. Chem. Soc., Faraday Trans. 1*, 1988, 84(11), 3991-4012. <https://doi.org/10.1039/f19888403991>
- [1988CAC] Caceres-Alonso, M.; Costas, M.; Andreoli-Ball, L.; Patterson, D., Steric effects on the self-association of branched and cyclic alcohols in inert solvents. Apparent heat capacities of secondary and tertiary alcohols in hydrocarbons, *Can. J. Chem.*, 1988, 66, 989-998. <https://doi.org/10.1139/v88-165>
- [1988DOR] Dorofeeva O.V., Thermodynamic Properties of Polycyclic Aromatic Hydrocarbons in the Gaseous Phase. Institute for High Temperatures, USSR Academy of Sciences, Preprint No.1-238 (in Russian), Moscow, 1988
- [1988KAB] Kabo G.J., Thermodynamic properties of cyclohexanol and cyclohexanone, *J. Chem. Thermodyn.*, 1988, 20, 429-437. [https://doi.org/10.1016/0021-9614\(88\)90179-6](https://doi.org/10.1016/0021-9614(88)90179-6)
- [1988LEB] Lebedev, B.V.; Vasil'ev, V.G., Thermodynamics of ethanal at 0-300 K, *Zhur. Fiz. Khim.*, 1988, 62, 3099-3102.
- [1988OKA] Okano, T.; Ogawa, H.; Murakami, S., Molar excess volumes, isentropic compressions, and isobaric heat capacities of methanol-isomeric butanol systems at 298.15 K, *Can. J. Chem.*, 1988, 66, 713-717 <https://doi.org/10.1139/v88-124>
- [1988PIN] Pintos, M.; Bravo, R.; Baluja, M.C.; Paz Andrade, M.I.; Roux-Desgranges, G.; Grolier, J.-P.E., *Can. J. Chem.*, 1988, 1179 <https://doi.org/10.1139/v88-193>

- [1988SHI] Shiohama, Y.; Ogawa, H.; Murakami, S.; Fujihara, I., Excess molar isobaric heat capacities and isentropic compressibilities of (cis- or trans-decalin + benzene or toluene or iso-octane or n-heptane) at 298.15 K, *J. Chem. Thermodynam.*, 1988, 20, 1183-1189.
- [1989DOR] Dorofeeva O.V., Thermodynamic Properties of Gaseous Polycyclic Aromatic Hydrocarbons Containing Five-Membered Rings. Institute for High Temperatures, USSR Academy of Sciences, Preprint No.1-263 (in Russian), Moscow, 1989.
- [1989EWI] Ewing, M. B.; Trusler, J. P. M.; Speeds of sound in tetrafluoromethane between 175 and 300 K measured with a spherical resonator. *J. Chem. Phys.*, 1989, 90, 1106-15
- [1989GUR] Gurvich, L.V.; Veyts, I.V.; Alcock, C.B., Thermodynamic Properties of Individual Substances, 4th ed.; Vols. 1 and 2, Hemisphere, New York, 1989
- [1989LAI] Lainez, A.; Rodrigo, M.M.; Wilhelm, E.; Grolier, J.-P.E., Excess volumes and excess heat capacities of some mixtures with trans,trans,cis-1,5,9-cyclododecatriene at 298.15K, *J. Chem. Eng. Data*, 1989, 34, 332-335
<https://doi.org/10.1021/je00057a021>
- [1989OHN] Ohnishi, K.; Fujihara, I.; Murakami, S., Thermodynamic properties of decalins mixed with hexane isomers at 298.15K. 1. Excess enthalpies and excess isobaric heat capacities, *Fluid Phase Equilib.*, 1989, 46, 59-72.
[https://doi.org/10.1016/0378-3812\(89\)80275-4](https://doi.org/10.1016/0378-3812(89)80275-4)
- [1989WIL] Wilhelm, E.; Lainez, A.; Grolier, J.-P.E., Thermodynamics of (a halogenated ethane or ethene + an n-alkane). VE and CpE of mixtures containing either 1,1,2,2-tetrachloroethane or tetrachloroethene, *Fluid Phase Equilib.*, 1989, 49, 233-250. [https://doi.org/10.1016/0378-3812\(89\)80018-4](https://doi.org/10.1016/0378-3812(89)80018-4)
- [1990MAY] Mayer, J.; Rachwalska, M.; Sciesinska, E.; Sciesinski, J., On the polymorphism of solid cyclohexanol by adiabatic calorimetry and far infrared methods, *J. Phys.(Paris)*, 1990, 51(9), 857-867.
<https://doi.org/10.1051/jphys:01990005109085700>
- [1990MES] Messerly, J.F.; Todd, S.S.; Finke, H.L.; Lee-Bechtold, S.H.; Guthrie, G.B.; Steele, W.V.; Chirico, R.D., Heat capacities of pent-1-ene (10K to 320K), cis-hex-2-ene (10K to 330K), non-1-ene (10K to 400K), and hexadec-1-ene (10K to 400K), *J. Chem. Thermodynam.*, 1990, 22, 1107-1128 [https://doi.org/10.1016/0021-9614\(90\)90159-N](https://doi.org/10.1016/0021-9614(90)90159-N)
- [1991BUF] Buffington, R. M.; Fleischer, J.; Thermodynamic properties of dichlorodifluoromethane, a new refrigerant specific heat of liquid and vapor and latent heat of vaporization. *Ind. Eng. Chem.*, 1931, 23, 1290
- [1991DOR] Dorofeeva O.V., Thermodynamic properties of linear carbon chain molecules with conjugated triple bonds. Part I. Polyacetylenes, H(CC)nH (n=2-6), cyanopolyacetylenes, H(CC)nCN (n=1-5), and dicyanopolyacetylenes, NC(CC)nCN (n=1-4), *Thermochim. Acta*, 1991, 178, 273-286. [https://doi.org/10.1016/0040-6031\(91\)80319-E](https://doi.org/10.1016/0040-6031(91)80319-E)
- [1991MAL] Malhotra, R.; Woolf, L.A., Thermodynamic properties of propanone (acetone) at temperatures from 278 K to 323 K and pressures up to 400 Mpa, *J. Chem. Thermodynam.*, 1991, 23, 867-876. [https://doi.org/10.1016/S0021-9614\(05\)80282-4](https://doi.org/10.1016/S0021-9614(05)80282-4)

- [1991STE] Steele, W.V.; Chirico, R.D.; Knipmeyer, S.E.; Nguyen, A.; Tasker, I.R., Determination of ideal gas enthalpies of formation for key compounds the 1991 project results, DIPPR Project, 1993, 871, NIPER-716
<https://doi.org/10.2172/10107077>
- [1991TAK] Takeda, K.; Yamamuro, O.; Suga, H., Thermodynamic study of 1-butene. Exothermic and endothermic enthalpy relaxations near the glass transition, *J. Phys. Chem. Solids*, 1991, 22, 607-615. [https://doi.org/10.1016/0022-3697\(91\)90155-S](https://doi.org/10.1016/0022-3697(91)90155-S)
- [1991YIN] Yin, C.; Ziru, L.; Ganghe, W.; Chengyun, W., et al., Determination of heat capacity of explosives and related materials by DSC, *Proc. Int. Pyrotech. Semin. 17th(Vol. 1)*, 1991, 515-521
- [1992DOR] Dorofeeva O.V., Ideal gas thermodynamic properties of oxygen heterocyclic compounds. Part 1. Three-membered, four-membered and five-membered rings, *Thermochim. Acta*, 1992, 194, 9-46 [https://doi.org/10.1016/0040-6031\(92\)80002-E](https://doi.org/10.1016/0040-6031(92)80002-E)
- [1992EWI] Ewing, M. B.; Trusler, J. P. M.; Second acoustic virial coefficients of nitrogen between 80 and 373 K. *Physica A (Amsterdam)*, 1992, 184, 415-36
- [1993CAR] Carson, A.S.; Laye, P.G.; Pedley, J.B.; Welsby, A.M., The enthalpies of formation iodomethane, diiodomethane, triiodomethane, and tetraiodomethane by rotating combustion calorimetry, *J. Chem. Thermodyn.*, 1993, 25, 261-269 <https://doi.org/10.1006/jcht.1993.1025>
- [1993CHI] Chirico, R.D.; Knipmeyer, S.E.; Nguyen, A.; Steele, W.V., The thermodynamic properties to the temperature 700 K of naphthalene and of 2,7-dimethylnaphthalene, *J. Chem. Thermodyn.*, 1993, 25, 1461-1494.
<https://doi.org/10.1006/jcht.1993.1148>
- [1993GRO] Grolier, J.-P.E.; Roux-Desgranges, G.; Berkane, M.; Jimenez, E.; Wilhelm, E., Heat capacities and densities of mixtures of very polar substances 2. Mixtures containing N,N-dimethylformamide, *J. Chem. Thermodynam.*, 1993, 25(1), 41-50 <https://doi.org/10.1006/jcht.1993.1005>
- [1993KAJ] Kaji, K.; Tochigi, K.; Misawa, Y.; Suzuki, T., An adiabatic calorimeter for samples of mass less than 0.1 g and heat capacity measurements on benzoic acid at temperatures from 19 K to 312 K, *J. Chem. Thermodynam.*, 1993, 25(6), 699-709 <https://doi.org/10.1006/jcht.1993.1068>
- [1993SHE] Shehatta, I., Heat capacity at constant pressure of some halogen compounds, *Thermochim. Acta*, 1993, 213, 1-10
[https://doi.org/10.1016/0040-6031\(93\)80001-Q](https://doi.org/10.1016/0040-6031(93)80001-Q)
- [1995ESP] Esper, G.; Lemming, W.; Beckermann, W.; Kohler, F.; Acoustic determination of ideal gas heat capacity and second virial coefficient of small hydrocarbons. *Fluid Phase Equilib.*, 1995, 105, 173-192
- [1996LUD] Luddecke, T. O.; Magee, J. W.; Molar Heat Capacity at Constant Volume of Difluoromethane (R32) and Pentafluoroethane (R125) from the Triple-Point Temperature to 345 K at Pressures to 35 MPa. *Int. J. Thermophys.*, 1996, 17, 823-849
- [1997CHI] Chirico, R. D.; Knipmeyer, S. E.; Nguyen, A.; Steele, W. V.; "Thermodynamic Equilibria in Xylene Isomerization. 4. The thermodynamic Properties of Ethylbenzene," *J. Chem. Eng. Data*, 1997, 42, 772-783

- [1998CHA] Chase, M.W., Jr., NIST-JANAF Thermochemical Tables, Fourth Edition, J. Phys. Chem. Ref. Data, Monograph 9, 1998
- [1998CHI] Chirico, R. D.; Cowell, A. B.; Good, W. D.; Klots, T. D.; Knipmeyer, S. E.; Nguyen, A.; Rau, A. P.; Reynolds, J. W.; Smith, N. K.; Steele, W. V.; Heat capacities, enthalpy increments, phase transitions, and derived thermodynamic functions for the condensed phases of bicyclohexyl between the temperatures 6 K and 440 K, J. Chem. Thermodyn., 1998, 30, 1423-1439
- [1998HO] Holcomb, C.D.; Magee, J. W.; Scott, J. L.; Outcalt, S. L.; Haynes, W. M.; Selected Thermodynamic Properties for Mixtures of R-32 (difluoromethane), R-125 (pentafluoroethane), R-134a (1,1,1,2-tetrafluoroethane), R-143a (1,1,1-trifluoroethane), R-41 (fluoromethane), R-290 (propane), and R-744 (carbon dioxide),. NIST, 1998, Selected Thermodynamic Properties for Mixtures of R-32 R-125 R-134A R-143A R-41 R-290 and R-744 No. Technical Note 1397, Boulder CO USA
- [2006FEI] Feistel, R.; Wagner, W.; A New Equation of State for H₂O Ice Ih, J. Phys. Chem. Ref. Data, 2006, 35, 1021.
- [2008TJA] Tjahjono, M.; Garland, M.; Deuterium isotope effect on molar heat capacities and apparent molar heat capacities in dilute aqueous solutions: A multi-channel heat-flow microcalorimeter study. J. Chem. Thermodyn., 2008, 40, 1600-1606
- [2012KRO] Kroenlein, K., Muzny, C., Kazakov, A., Diky, V., Chirico, R., Magee, J., Abdulagatov, I. and Frenkel, M. (2012), NIST/TRC Web Thermo Tables (WTT) NIST Standard Reference Subscription Database 3 - Professional Edition Version 2, NIST/TRC Web Thermo Tables (WTT) - Professional Edition, [online], <http://wtt-pro.nist.gov/> (Accessed August 10, 2021)
- [2014RUZ] Ruzicka, K.; Fulem, M.; Cervinka, C.; Recommended sublimation pressure and enthalpy of benzene, J. Chem. Thermodyn., 2014, 68, 40-47
- [2014SOU] Sousa-Silva, C.; Hesketh, N.; Yurchenko, S. N.; Hill, C.; Tennyson, J.; High temperature partition functions and thermodynamic data for ammonia and phosphine, J. Quant. Spec. Radiat. Transf. 142, 66 (2014).
- [2015CHO] Chorazewski, M.; Troncoso, J.; Jacquemin, J.; Thermodynamic Properties of Dichloromethane, Bromochloromethane, and Dibromomethane under Elevated Pressure: Experimental Results and SAFT-VR Mie Predictions. Ind. Eng. Chem. Res., 2015, 54, 720-730
- [2018LEM] Lemmon, E. W.; Bell, I. H.; Huber, M. L.; McLinden, M. O.; Reference Fluid Thermodynamic and Transport Properties (REFPROP), NIST Standard Reference Database 23, Vers. 10.0, 2018
- [2019DIK] Diky, V.; Chirico, R. D.; Frenkel, M.; Bazyleva, A.; Magee, J. W.; Paulechka, E.; Kazakov, A.; Lemmon, E. W.; Muzny, C. D.; Smolyanitsky, A. Y.; Townsend, S.; Kroenlein, K.; ThermoData Engine (TDE), Vewrs. 10.4.2, NIST Standard Reference Database 103b, (2019).
- [2021LIN] P.J. Linstrom and W.G. Mallard, Eds., NIST Chemistry WebBook, NIST Standard Reference Database Number 69, National Institute of Standards and Technology, Gaithersburg MD, 20899, <https://doi.org/10.18434/T4D303>, (accessed July 22, 2021).